



Red Hat Enterprise Linux 7

Installation Guide

Installing Red Hat Enterprise Linux 7 on all architectures

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Abstract

This manual explains how to boot the Red Hat Enterprise Linux 7 installation program (Anaconda) and how to install Red Hat Enterprise Linux 7 on AMD64 and Intel 64 systems, 64-bit ARM systems, 64-bit IBM Power Systems servers, and IBM Z servers. It also covers advanced installation methods such as Kickstart installations, PXE installations, and installations over VNC. Finally, it describes common post-installation tasks and explains how to troubleshoot installation problems. Information on installing Red Hat Enterprise Linux Atomic Host can be found in the Red Hat Enterprise Linux Atomic Host Installation and Configuration Guide.

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CHAPTER 1. GETTING STARTED

You can install Red Hat Enterprise Linux with an installation utility called **Anaconda**. Most users can simply follow the procedure outlined in [Section 4.1, “Interactive Installation”](#) to install Red Hat Enterprise Linux using the graphical interface in **Anaconda**.

Users with advanced requirements can also use the graphical interface to configure many aspects of the installation, and install Red Hat Enterprise Linux on a wide variety of systems. On systems without a local interface, installation can be accessed entirely remotely. Installation can also be automated by using a **Kickstart** file, and performed with no interaction at all.

1.1. GRAPHICAL INSTALLATION

The Red Hat Enterprise Linux installer, **Anaconda**, provides a simple graphical method to install Red Hat Enterprise Linux. The graphical installation interface has a built-in help system which can guide you through most installations, even if you have never installed Linux before. However, **Anaconda** can also be used to configure advanced installation options if required.

Anaconda is different from most other operating system installation programs due to its parallel nature. Most installers follow a linear path; you must choose your language first, then you configure networking, and so on. There is usually only one way to proceed at any given time.

In the graphical interface in **Anaconda** you are at first only required to select your language and locale, and then you are presented with a central screen, where you can configure most aspects of the installation in any order you like. While certain parts require others to be completed before configuration – for example, when installing from a network location, you must configure networking before you can select which packages to install – most options in **Anaconda** can be configured in any order. If a background task, such as network initialization or disk detection, is blocking configuration of a certain option, you can configure unrelated options while waiting for it to complete.

Additional differences appear in certain screens; notably the custom partition process is very different from other Linux distributions. These differences are described in each screen's subsection.

Some screens will be automatically configured depending on your hardware and the type of media you used to start the installation. You can still change the detected settings in any screen. Screens which have not been automatically configured, and therefore require your attention before you begin the installation, are marked by an exclamation mark. You cannot start the actual installation process before you finish configuring these settings.

Installation can also be performed in text mode, however certain options, notably including custom partitioning, are unavailable. See [Section 8.3, “Installing in Text Mode”](#), or if using an IBM Power system or IBM Z, see [Section 13.3, “Installing in Text Mode”](#), or [Section 18.4, “Installing in Text Mode”](#), respectively, for more information.

1.2. REMOTE INSTALLATION

You can use the graphical interface remotely to install Red Hat Enterprise Linux. For headless systems, **Connect Mode** can be used to perform a graphical installation completely remotely. For systems with a display and keyboard, but without the capacity to run the graphical interface, **Direct Mode** can instead be used to facilitate setup. For more information, see the [Chapter 25, Using VNC](#).

1.3. AUTOMATED INSTALLATION

Anaconda installations can be automated through the use of a **Kickstart** file. **Kickstart** files can be used to configure any aspect of installation, allowing installation without user interaction, and can be used to easily automate installation of multiple instances of Red Hat Enterprise Linux.

In most situations, you can simply follow the procedure outlined in [Section 4.2, “Automatic Installation”](#) to create and configure a **Kickstart** file, which can be used to perform an arbitrary number of non-interactive installations of Red Hat Enterprise Linux.

Kickstart files can be automatically created based on choices made using the graphical interface, through the online [Kickstart Generator](#) tool, or written from scratch using any text editor. For more information, see [Section 27.2.1, “Creating a Kickstart File”](#).

Kickstart files can be easily maintained and updated using various utilities in Red Hat Enterprise Linux. For more information, see [Section 27.2.2, “Maintaining the Kickstart File”](#).

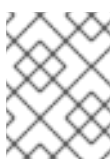
CHAPTER 2. DOWNLOADING RED HAT ENTERPRISE LINUX

If you have a Red Hat subscription, you can download *ISO image files* of the Red Hat Enterprise Linux 7 installation DVD from the Red Hat Customer Portal. If you do not have a subscription, either purchase one or obtain a free evaluation subscription from the Software & Download Center at <https://access.redhat.com/downloads/>.

There are two basic types of installation media available for the AMD64 and Intel 64 (x86_64), ARM (Aarch64), and IBM Power Systems (ppc64) architectures:

Binary DVD

A full installation image that boots the installation program and performs the entire installation without additional package repositories.



NOTE

Binary DVDs are also available for IBM Z. They can be used to boot the installation program using a SCSI DVD drive or as installation sources.

Boot.iso

A minimal boot image that boots the installation program but requires access to additional package repositories. Red Hat does not provide the repository; you must create it using the full installation ISO image.



NOTE

Supplementary DVD images containing additional packages, such as the IBM Java Runtime Environment and additional virtualization drivers may be available, but they are beyond the scope of this document.

If you have a subscription or evaluation subscription, follow these steps to obtain the Red Hat Enterprise Linux 7 ISO image files:

Procedure 2.1. Downloading Red Hat Enterprise Linux ISO Images

1. Visit the Customer Portal at <https://access.redhat.com/home>. If you are not logged in, click **LOG IN** on the right side of the page. Enter your account credentials when prompted.
2. Click **DOWNLOADS** at the top of the page.
3. Click **Red Hat Enterprise Linux**.
4. Ensure that you select the appropriate **Product Variant** and **Architecture** for your installation target. By default, **Red Hat Enterprise Linux Server** and **x86_64** are selected. If you are not sure which variant best suits your needs, see <http://www.redhat.com/en/technologies/linux-platforms/enterprise-linux>. Additionally, a list of packages available for every variant is available in the [Red Hat Enterprise Linux 7 Package Manifest](#).
5. A list of available downloads is displayed; most notably, a minimal **Boot ISO** image and a full installation **Binary DVD** ISO image. These files are described above. Additional images can be available, such as preconfigured virtual machine images, which are beyond the scope of this document.

6. Choose the image file that you want to use. You have two ways to download it from the Customer Portal:
 - Click its name to begin downloading it to your computer using your web browser.
 - Right-click the name and then click **Copy Link Location** or a similar menu item, the exact wording of which depends on the browser that you are using. This action copies the URL of the file to your clipboard, which allows you to use an alternative application to download the file to your computer. This approach is especially useful if your Internet connection is unstable: in that case, your browser might fail to download the whole file, and an attempt to resume the interrupted download process fails because the download link contains an authentication key which is only valid for a short time. Specialized applications such as **curl** can, however, be used to resume interrupted download attempts from the Customer Portal, which means that you need not download the whole file again and thus you save your time and bandwidth consumption.

Procedure 2.2. Using curl to Download Installation Media

1. Make sure the curl package is installed by running the following command as root:

```
# yum install curl
```

If your Linux distribution does not use **yum**, or if you do not use Linux at all, download the most appropriate software package from the [curl web site](#).

2. Open a terminal window, enter a suitable directory, and type the following command:

```
$ curl -o filename.iso 'copied_link_location'
```

Replace *filename.iso* with the ISO image name as displayed in the Customer Portal, such as **rhel-server-7.0-x86_64-dvd.iso**. This is important because the download link in the Customer Portal contains extra characters which **curl** would otherwise use in the downloaded file name, too. Then, keep the single quotation mark in front of the next parameter, and replace *copied_link_location* with the link that you have copied from the Customer Portal; copy it again if you copied the commands above in the meantime. Note that in Linux, you can paste the content of the clipboard into the terminal window by middle-clicking anywhere in the window, or by pressing **Shift+Insert**. Finally, use another single quotation mark after the last parameter, and press **Enter** to run the command and start transferring the ISO image. The single quotation marks prevent the command line interpreter from misinterpreting any special characters that might be included in the download link.

Example 2.1. Downloading an ISO image with curl

The following is an example of a **curl** command line:

```
$ curl -o rhel-server-7.0-x86_64-dvd.iso  
'https://access.cdn.redhat.com/content/origin/files/sha256/85/85a...46c/rhel-  
server-7.0-x86_64-dvd.iso?_auth_=141...7bf'
```

Note that the actual download link is much longer because it contains complicated identifiers.

3. If your Internet connection does drop before the transfer is complete, refresh the

download page in the Customer Portal; log in again if necessary. Copy the new download link, use the same basic **curl** command line parameters as earlier but be sure to use the new download link, and add **-C -** to instruct **curl** to automatically determine where it should continue based on the size of the already downloaded file.

Example 2.2. Resuming an interrupted download attempt

The following is an example of a **curl** command line that you use if you have only partially downloaded the ISO image of your choice:

```
$ curl -o rhel-server-7.0-x86_64-dvd.iso
'https://access.cdn.redhat.com/content/origin/files/sha256/85/85a...46c/rhel-
server-7.0-x86_64-dvd.iso?_auth_=141...963' -C -
```

7. Optionally, you can use a checksum utility such as **sha256sum** to verify the integrity of the image file after the download finishes. All downloads on the Download Red Hat Enterprise Linux page are provided with their checksums for reference:

```
$ sha256sum rhel-server-7.0-x86_64-dvd.iso
85a...46c rhel-server-7.0-x86_64-dvd.iso
```

Similar tools are available for [Microsoft Windows](#) and [Mac OS X](#). You can also use the installation program to verify the media when starting the installation; see [Section 23.2.2, “Verifying Boot Media”](#) for details.

After you have downloaded an ISO image file from the Customer Portal, you can:

- Burn it to a CD or DVD as described in [Section 3.1, “Making an Installation CD or DVD”](#).
- Use it to create a bootable USB drive; see [Section 3.2, “Making Installation USB Media”](#).
- Place it on a server to prepare for a network installation. For specific directions, see [Section 3.3.3, “Installation Source on a Network”](#).
- Place it on a hard drive to use the drive as an installation source. For specific instructions, see [Section 3.3.2, “Installation Source on a Hard Drive”](#).
- Use it to prepare a *Preboot Execution Environment* (PXE) server, which allows you to boot the installation system over a network. See [Chapter 24, Preparing for a Network Installation](#) for instructions.

CHAPTER 3. MAKING MEDIA

This chapter describes how to use ISO image files obtained by following the steps in [Chapter 2, Downloading Red Hat Enterprise Linux](#) to create bootable physical media, such as a DVD or a USB flash drive. You can then use these media to boot the installation program and start the installation. These steps only apply if you plan to install Red Hat Enterprise Linux on a 64-bit AMD, Intel, or ARM system, or an IBM Power Systems server using physical boot media. For information about installing Red Hat Enterprise Linux on an IBM Z server, see [Chapter 16, Booting the Installation on IBM Z](#). For instructions on how to set up a *Preboot Execution Environment* (PXE) server to perform a PXE-based installation over a network, see [Chapter 24, Preparing for a Network Installation](#).



NOTE

By default, the **inst.stage2=** boot option is used on the installation media and set to a specific label (for example, **inst.stage2=hd:LABEL=RHEL7x20Server.x86_64**). If you modify the default label of the file system containing the runtime image, or if using a customized procedure to boot the installation system, you must ensure this option is set to the correct value. See [Specifying the Installation Source](#) for details.

3.1. MAKING AN INSTALLATION CD OR DVD

You can make an installation CD or DVD using burning software on your computer and a CD/DVD burner. The exact series of steps that produces an optical disc from an ISO image file varies greatly from computer to computer, depending on the operating system and disc burning software installed. Consult your burning software's documentation for the exact steps needed to burn a CD or DVD from an ISO image file.



NOTE

It is possible to use optical discs (CDs and DVDs) to create both minimal boot media and full installation media. However, it is important to note that due to the large size of the full installation ISO image (between 4 and 4.5 GB), only a DVD can be used to create a full installation disc. Minimal boot ISO is roughly 300 MB, allowing it to be burned to either a CD or a DVD.

Make sure that your disc burning software is capable of burning discs from image files. Although this is true of most disc burning software, exceptions exist. In particular, note that the disc burning feature built into Windows XP and Windows Vista cannot burn DVDs; and that earlier Windows operating systems did not have any disc burning capability installed by default at all. Therefore, if your computer has a Windows operating system prior to Windows 7 installed on it, you need a separate piece of software for this task. Examples of popular disc burning software for Windows that you might already have on your computer include **Nero Burning ROM** and **Roxio Creator**. Most widely used disc burning software for Linux, such as **Brasero** and **K3b**, also has the built-in ability to burn discs from ISO image files.

On some computers, the option to burn a disc from an ISO file is integrated into a context menu in the file browser. For example, when you right-click an ISO file on a computer with a Linux or UNIX operating system which runs the **GNOME** desktop, the **Nautilus** file browser presents you with the option to **Write to disk**.

3.2. MAKING INSTALLATION USB MEDIA

You can use a USB drive or an SD card instead of a CD or DVD to create bootable media for installing Red Hat Enterprise Linux on 64-bit AMD, Intel, or ARM systems. The exact procedure varies depending

on whether you want to perform it on a Linux or Windows system. You can create minimal boot media and full installation media using the same procedure; the only limitation is the capacity of the USB drive - it must have enough space to fit the entire image, which means roughly 450 MB for minimal boot media and 4.8 GB for full installation media.

3.2.1. Making Installation USB Media on Linux

The following procedure assumes you are using a Linux system and that you have downloaded an appropriate ISO image as described in [Chapter 2, Downloading Red Hat Enterprise Linux](#). On most Linux distributions, it will work without the need for installing any additional packages.



WARNING

This procedure is destructive. Any data on the USB flash drive will be destroyed with no warning. Make sure that you specify the correct drive, and make sure that this drive does not contain any data you want to preserve.

Many Linux distributions provide their own tools for creating live USB media: `liveusb-creator` on Fedora, `usb-creator` on Ubuntu, and others. Describing these tools is beyond the scope of this book; the following procedure will work on most Linux systems.

Procedure 3.1. Making USB Media on Linux

1. Connect a USB flash drive to the system and execute the **`dmesg`** command. A log detailing all recent events will be displayed. At the bottom of this log, you will see a set of messages caused by the USB flash drive you just connected. It will look like a set of lines similar to the following:

```
[ 170.171135] sd 5:0:0:0: [sdb] Attached SCSI removable disk
```

Note the name of the connected device - in the above example, it is **`sdb`**.

2. Log in as **`root`**:

```
$ su -
```

Provide your root password when prompted.

3. Make sure that the device is not mounted. First, use the **`findmnt device`** command and the device name you found in the earlier steps. For example, if the device name is **`sdb`**, use the following command:

```
# findmnt /dev/sdb
```

If the command displays no output, you can proceed with the next step. However, if the command does provide output, it means that the device was automatically mounted and you must unmount it before proceeding. A sample output will look similar to the following:

```
# findmnt /dev/sdb
TARGET SOURCE FSTYPE OPTIONS
/mnt/iso /dev/sdb iso9660 ro,relatime
```

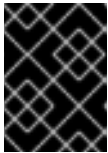
Note the **TARGET** column. Next, use the **umount *target*** command to unmount the device:

```
# umount /mnt/iso
```

4. Use the **dd** command to write the installation ISO image directly to the USB device:

```
# dd if=/image_directory/image.iso of=/dev/device bs=blocksize
```

Replace */image_directory/image.iso* with the full path to the ISO image file you downloaded, *device* with the device name as reported by the **dmesg** command earlier, and *blocksize* with a reasonable block size (for example, **512k**) to speed up the writing process. The **bs** parameter is optional, but it can speed up the process considerably.



IMPORTANT

Make sure to specify the output as the device name (for example, **/dev/sda**), not as a name of a *partition* on the device (for example, **/dev/sda1**).

For example, if the ISO image is located in **/home/testuser/Downloads/rhel-server-7-x86_64-boot.iso** and the detected device name is **sdb**, the command will look like the following:

```
# dd if=/home/testuser/Downloads/rhel-server-7-x86_64-boot.iso of=/dev/sdb bs=512k
```

5. Wait for **dd** to finish writing the image to the device. Note that no progress bar is displayed; the data transfer is finished when the **#** prompt appears again. After the prompt is displayed, log out from the **root** account and unplug the USB drive.

The USB drive is now ready to be used as a boot device. You can continue with [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) on AMD, Intel, and ARM systems, or [Chapter 12, Booting the Installation on IBM Power Systems](#) on IBM Power Systems servers.



NOTE

Non-virtualized installations (known as "bare metal" installations) on IBM Power Systems servers require that the **inst.stage2=** boot option is specified. Refer to [Section 23.1, "Configuring the Installation System at the Boot Menu"](#) for information about the **inst.stage2=** boot option.

3.2.2. Making Installation USB Media on Windows

The procedure of creating bootable USB media on Windows depends on which tool you use. There are many different utilities which allow you to write an ISO image to a USB drive. Red Hat recommends using the **Fedora Media Writer**, available for download at <https://github.com/FedoraQt/MediaWriter/releases>.

**NOTE**

Fedora Media Writer is a community product and is not supported by Red Hat. Any issues with the tool can be reported at <https://github.com/FedoraQt/MediaWriter/issues>.

**IMPORTANT**

Transferring the ISO image file to the USB drive using Windows Explorer or a similar file manager will not work – you will not be able to boot from the device.

Procedure 3.2. Making USB Media on Windows

1. Download and install **Fedora Media Writer**.
2. Download the Red Hat Enterprise Linux ISO image you want to use to create the media. (See [Chapter 2, Downloading Red Hat Enterprise Linux](#) for instructions on obtaining ISO images.)
3. Plug in the USB drive you will be using to create bootable media.
4. Open **Fedora Media Writer**.
5. In the main window, click **Custom Image** and select the downloaded Red Hat Enterprise Linux ISO image.
6. From the drop-down menu, select the drive you want to use. If the drive does not appear, verify that the USB drive is connected and restart **Fedora Media Writer**.
7. Click **Write to disk**. The boot media creation process will begin. Do not unplug the drive until the operation completes. Depending on the size of the ISO image and the write speed of the USB drive, writing the image can take several minutes.

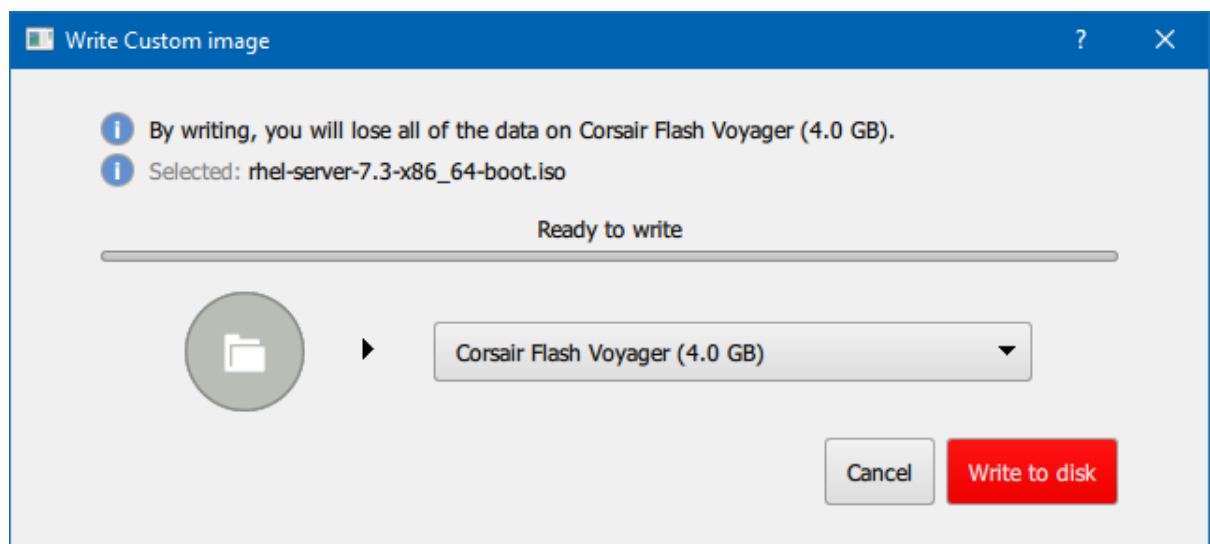


Figure 3.1. Fedora Media Writer

8. When the creation process finishes and the **Complete!** message appears, unmount the USB drive using the **Safely remove hardware** icon in the system's notification area.

The USB drive is now ready to be used as a boot device. You can continue with [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) on AMD, Intel, and ARM systems, or [Chapter 12, Booting the Installation on IBM Power Systems](#) on IBM Power Systems servers.

3.2.3. Making Installation USB Media on Mac OS X

This procedure involves using the **dd** command line tool to write the installation image to a USB flash drive. Note that some steps involve use of the **sudo** command, which is only available when logged in with an administrator account that requires a password.



WARNING

All data on the USB flash drive will be deleted by this procedure.

Procedure 3.3. Making USB Media on Mac OS X

1. Connect a USB flash drive to the system and identify the device path with the **diskutil list** command. The device path has the format of **/dev/disknumber**, where *number* is the number of the disk. The disks are numbered starting at zero (0). Disk 0 is likely to be the OS X recovery disk, and Disk 1 is likely to be your main OS X installation. In the following example, it is **disk2**:

```
$ diskutil list
/dev/disk0
#:              TYPE NAME              SIZE      IDENTIFIER
0:      GUID_partition_scheme          *500.3 GB   disk0
1:              EFI EFI                209.7 MB   disk0s1
2:      Apple_CoreStorage              400.0 GB   disk0s2
3:      Apple_Boot Recovery HD         650.0 MB   disk0s3
4:      Apple_CoreStorage              98.8 GB    disk0s4
5:      Apple_Boot Recovery HD         650.0 MB   disk0s5
/dev/disk1
#:              TYPE NAME              SIZE      IDENTIFIER
0:      Apple_HFS YosemiteHD          *399.6 GB   disk1
Logical Volume on disk0s1
8A142795-8036-48DF-9FC5-84506DFBB7B2
Unlocked Encrypted
/dev/disk2
#:              TYPE NAME              SIZE      IDENTIFIER
0:      FDisk_partition_scheme         *8.0 GB     disk2
1:      Windows_NTFS SanDisk USB       8.0 GB     disk2s1
```

To identify your USB flash drive, compare the **NAME**, **TYPE** and **SIZE** columns to what you know about your flash drive. For example, the **NAME** should be the same as the title of the flash drive icon in the **Finder**. You can also compare these values to those in the flash drive's information panel; right-click on the drive icon and select **Get Info**.

2. Use the **diskutil unmountDisk** command to unmount the flash drive's filesystem volumes:

```
$ diskutil unmountDisk /dev/disknumber
Unmount of all volumes on disknumber was successful
```

When you do this, the icon for the flash drive disappears from your desktop. If it does not, you might have identified the wrong disk. If you attempt to unmount the system disk accidentally, you get a **failed to unmount** error.

3. Use the **dd** command as a parameter of the **sudo** command to write the ISO image to the flash drive:

```
$ sudo dd if=/path/to/image.iso of=/dev/rdisknumber bs=1m>
```



NOTE

Mac OS X provides both a block (**/dev/disk***) and character device (**/dev/rdisk***) file for each storage device. Writing an image to the **/dev/rdisknumber** character device is faster than to the **/dev/disknumber** block device.

Example 3.1. Writing an ISO Image to a Disk

To write the **/Users/user_name/Downloads/rhel-server-7-x86_64-boot.iso** file to the **/dev/rdisk2** device:

```
$ sudo dd if=/Users/user_name/Downloads/rhel-server-7-x86_64-boot.iso of=/dev/rdisk2
```

4. Wait for the command to finish. Note that no progress bar is displayed; however, to check the status of the operation while it is still running, press **Ctrl+t** in the terminal:

```
load: 1.02 cmd: dd 3668 uninterruptible 0.00u 1.91s
112+0 records in
111+0 records out
116391936 bytes transferred in 114.834860 secs (1013559 bytes/sec)
```

5. The speed of the data transfer depends on the speed of your USB ports and the flash drive. After the prompt is displayed again, the data transfer is finished. You can then unplug the flash drive.

The flash drive is now ready to be used as a boot device. You can continue with [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) on AMD64 and Intel 64 systems or [Chapter 12, Booting the Installation on IBM Power Systems](#) on IBM Power Systems servers.

3.3. PREPARING INSTALLATION SOURCES

As explained in [Chapter 2, Downloading Red Hat Enterprise Linux](#), two basic types of media are available for Red Hat Enterprise Linux: a minimal boot image and a full installation image (also known as a binary DVD). If you downloaded the binary DVD and created a boot DVD-ROM or USB drive from it, you can proceed with the installation immediately, as this image contains everything you need to install the system.

However, if you use the minimal boot image, you must also configure an additional source of the installation. This is because the minimal boot image only contains the installation program itself and tools needed to boot your system and start the installation; it does not include the software packages to be installed on your system.

The full installation DVD ISO image can be used as the source for the installation. If your system will require additional software not provided by Red Hat, you should configure additional repositories and install these packages *after* the installation is finished. For information about configuring additional **Yum** repositories on an installed system, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

The installation source can be any of the following:

- **DVD:** You can burn the binary DVD ISO image onto a DVD and configure the installation program to install packages from this disk.
- **Hard drive:** You can place the binary DVD ISO image on a hard drive and install packages from it.
- **Network location:** You can copy the binary DVD ISO image or the *installation tree* (extracted contents of the binary DVD ISO image) to a network location accessible from the installation system and perform the installation over the network using the following protocols:
 - **NFS:** The binary DVD ISO image is placed into a *Network File System* (NFS) share.
 - **HTTPS, HTTP or FTP:** The installation tree is placed on a network location accessible over **HTTP, HTTPS, or FTP**.

When booting the installation from minimal boot media, you must always configure an additional installation source. When booting the installation from the full binary DVD, it is also possible to configure another installation source, but it is not necessary – the binary DVD ISO image itself contains all packages you need to install the system, and the installation program will automatically configure the binary DVD as the source.

You can specify an installation source in any of the following ways:

- In the installation program's graphical interface: After the graphical installation begins and you select your preferred language, the **Installation Summary** screen will appear. Navigate to the **Installation Source** screen and select the source you want to configure. For details, see:
 - [Section 8.11, “Installation Source”](#) for 64-bit AMD, Intel, and ARM systems
 - [Section 13.12, “Installation Source”](#) for IBM Power Systems servers
 - [Section 18.12, “Installation Source”](#) for IBM Z
- Using a boot option: You can specify custom boot options to configure the installation program before it starts. One of these options allows you to specify the installation source to be used. See the **inst.repo=** option in [Section 23.1, “Configuring the Installation System at the Boot Menu”](#) for details.
- Using a Kickstart file: You can use the **install** command in a Kickstart file and specify an installation source. See [Section 27.3.1, “Kickstart Commands and Options”](#) for details on the **install** Kickstart command, and [Chapter 27, Kickstart Installations](#) for information about Kickstart installations in general.

3.3.1. Installation Source on a DVD

You can burn the binary DVD ISO image onto a DVD and configure the installation program to install packages from this disk while booting the installation from another drive (for example, a minimal boot ISO on a USB flash drive). This procedure is the same as creating bootable optical media – see [Section 3.1, “Making an Installation CD or DVD”](#) for more information.

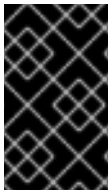
When using a DVD as an installation source, make sure the DVD is in the drive when the installation begins. The **Anaconda** installation program is not able to detect media inserted after the installation begins.

3.3.2. Installation Source on a Hard Drive

Hard drive installations use an ISO image of the binary installation DVD. To use a hard drive as the installation source, transfer the binary DVD ISO image to the drive and connect it to the installation system. Then, boot the **Anaconda** installation program.

You can use any type of hard drive accessible to the installation program, including USB flash drives. The binary ISO image can be in any directory of the hard drive, and it can have any name; however, if the ISO image is not in the top-level directory of the drive, or if there is more than one image in the top-level directory of the drive, you will be required to specify the image to be used. This can be done using a boot option, an entry in a Kickstart file, or manually in the **Installation Source** screen during a graphical installation.

A limitation of using a hard drive as the installation source is that the binary DVD ISO image on the hard drive must be on a partition with a file system which **Anaconda** can mount. These file systems are **xfs**, **ext2**, **ext3**, **ext4**, and **vfat (FAT32)**. Note that on Microsoft Windows systems, the default file system used when formatting hard drives is **NTFS**, and the **exFAT** file system is also available; however, neither of these file systems can be mounted during the installation. If you are creating a hard drive or a USB drive to be used as an installation source on Microsoft Windows, make sure to format the drive as **FAT32**.



IMPORTANT

The **FAT32** file system does not support files larger than 4 GiB. Some Red Hat Enterprise Linux 7 installation media can be larger than that, which means you cannot copy them to a drive with this file system.

When using a hard drive or a USB flash drive as an installation source, make sure it is connected to the system when the installation begins. The installation program is not able to detect media inserted after the installation begins.

3.3.3. Installation Source on a Network

Placing the installation source on a network has the advantage of allowing you to install multiple systems from a single source, without having to connect and disconnect any physical media. Network-based installations can be especially useful when used together with a TFTP server, which allows you to boot the installation program from the network as well. This approach completely eliminates the need for creating physical media, allowing easy deployment of Red Hat Enterprise Linux on multiple systems at the same time. For further information about setting up a TFTP server, see [Chapter 24, Preparing for a Network Installation](#).

3.3.3.1. Installation Source on an NFS Server

The **NFS** installation method uses an ISO image of the Red Hat Enterprise Linux binary DVD placed in a **Network File System** server's *exported directory*, which the installation system must be able to read. To perform an NFS-based installation, you will need another running system which will act as the NFS host.

For more information about NFS servers, see the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

The following procedure is only meant as a basic outline of the process. The precise steps you must take to set up an NFS server will vary based on the system's architecture, operating system, package manager, service manager, and other factors. On Red Hat Enterprise Linux 7 systems, the procedure

can be followed exactly as documented. For procedures describing the installation source creation process on earlier releases of Red Hat Enterprise Linux, see the appropriate *Installation Guide* for that release.

Procedure 3.4. Preparing for Installation Using NFS

1. Install the `nfs-utils` package by running the following command as **root**:

```
# yum install nfs-utils
```

2. Copy the full Red Hat Enterprise Linux 7 binary DVD ISO image to a suitable directory on the NFS server. For example, you can create directory `/rhel7-install/` for this purpose and save the ISO image here.
3. Open the `/etc/exports` file using a text editor and add a line with the following syntax:

```
/exported_directory/ clients
```

Replace `/exported_directory/` with the full path to the directory holding the ISO image. Instead of `clients`, use the host name or IP address of the computer which is to be installed from this NFS server, the subnetwork from which all computers are to have access the ISO image, or the asterisk sign (*) if you want to allow any computer with network access to the NFS server to use the ISO image. See the **exports(5)** man page for detailed information about the format of this field.

The following is a basic configuration which makes the `/rhel7-install/` directory available as read-only to all clients:

```
/rhel7-install *
```

4. Save the `/etc/exports` file after finishing the configuration and exit the text editor.
5. Start the **nfs** service:

```
# systemctl start nfs.service
```

If the service was already running before you changed the `/etc/exports` file, enter the following command instead, in order for the running NFS server to reload its configuration:

```
# systemctl reload nfs.service
```

After completing the procedure above, the ISO image is accessible over **NFS** and ready to be used as an installation source.

When configuring the installation source before or during the installation, use **nfs:** as the protocol, the server's host name or IP address, the colon sign (:), and the directory holding the ISO image. For example, if the server's host name is **myserver.example.com** and you have saved the ISO image in `/rhel7-install/`, specify **nfs:myserver.example.com:/rhel7-install/** as the installation source.

3.3.3.2. Installation Source on an HTTP, HTTPS or FTP Server

This installation method allows for a network-based installation using an installation tree, which is a directory containing extracted contents of the binary DVD ISO image and a valid **.treeinfo** file. The installation source is accessed over **HTTP**, **HTTPS**, or **FTP**.

For more information about HTTP and FTP servers, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

The following procedure is only meant as a basic outline of the process. The precise steps you must take to set up an FTP server will vary based on the system's architecture, operating system, package manager, service manager, and other factors. On Red Hat Enterprise Linux 7 systems, the procedure can be followed exactly as documented. For procedures describing the installation source creation process on earlier releases of Red Hat Enterprise Linux, see the appropriate *Installation Guide* for that release.

Procedure 3.5. Preparing Installation Using HTTP or HTTPS

1. Install the `httpd` package by running the following command as **root**:

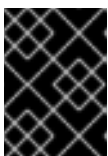
```
# yum install httpd
```

An **HTTPS** server needs additional configuration. For detailed information, see section [Setting Up an SSL Server](#) in the Red Hat Enterprise Linux 7 System Administrator's Guide. However, **HTTPS** is not necessary in most cases, because no sensitive data is sent between the installation source and the installer, and **HTTP** is sufficient.



WARNING

If your **Apache** web server configuration enables SSL security, make sure to only enable the **TLSv1** protocol, and disable **SSLv2** and **SSLv3**. This is due to the POODLE SSL vulnerability (CVE-2014-3566). See <https://access.redhat.com/solutions/1232413> for details.



IMPORTANT

If you decide to use **HTTPS** and the server is using a self-signed certificate, you must boot the installer with the **noverifyssl** option.

2. Copy the full Red Hat Enterprise Linux 7 binary DVD ISO image to the HTTP(S) server.
3. Mount the binary DVD ISO image, using the **mount** command, to a suitable directory:

```
# mount -o loop,ro -t iso9660 /image_directory/image.iso /mount_point/
```

Replace `/image_directory/image.iso` with the path to the binary DVD ISO image, and `/mount_point/` with the path to the directory in which you want the content of the ISO image to appear. For example, you can create directory `/mnt/rhel7-install/` for this purpose and use that as the parameter of the **mount** command.

4. Copy the files from the mounted image to the HTTP server root.

—

```
# cp -r /mnt/rhel7-install/ /var/www/html/
```

This command creates the **/var/www/html/rhel7-install/** directory with the content of the image.

5. Start the **httpd** service:

```
# systemctl start httpd.service
```

After completing the procedure above, the installation tree is accessible and ready to be used as the installation source.

When configuring the installation source before or during the installation, use **http://** or **https://** as the protocol, the server's host name or IP address, and the directory in which you have stored the files from the ISO image, relative to the HTTP server root. For example, if you are using **HTTP**, the server's host name is **myserver.example.com**, and you have copied the files from the image to **/var/www/html/rhel7-install/**, specify **http://myserver.example.com/rhel7-install/** as the installation source.

Procedure 3.6. Preparing for Installation Using FTP

1. Install the **vsftpd** package by running the following command as **root**:

```
# yum install vsftpd
```

2. Optionally, open the **/etc/vsftpd/vsftpd.conf** configuration file in a text editor, and edit any options you want to change. For available options, see the **vsftpd.conf(5)** man page. The rest of this procedure assumes that default options are used; notably, to follow the rest of the procedure, anonymous users of the FTP server must be permitted to download files.



WARNING

If you configured SSL/TLS security in your **vsftpd.conf** file, make sure to only enable the **TLSv1** protocol, and disable **SSLv2** and **SSLv3**. This is due to the POODLE SSL vulnerability (CVE-2014-3566). See <https://access.redhat.com/solutions/1234773> for details.

3. Copy the full Red Hat Enterprise Linux 7 binary DVD ISO image to the FTP server.
4. Mount the binary DVD ISO image, using the **mount** command, to a suitable directory:

```
# mount -o loop,ro -t iso9660 /image_directory/image.iso /mount_point
```

Replace **/image_directory/image.iso** with the path to the binary DVD ISO image, and **/mount_point** with the path to the directory in which you want the content of the ISO image to appear. For example, you can create directory **/mnt/rhel7-install/** for this purpose and use that as the parameter of the **mount** command.

5. Copy the files from the mounted image to the FTP server root:

```
# cp -r /mnt/rhel7-install/ /var/ftp/
```

This command creates the **/var/ftp/rhel7-install/** directory with the content of the image.

6. Start the **vsftpd** service:

```
# systemctl start vsftpd.service
```

If the service was already running before you changed the **/etc/vsftpd/vsftpd.conf** file, restart it to ensure the edited file is loaded. To restart, execute the following command:

```
# systemctl restart vsftpd.service
```

After completing the procedure above, the installation tree is accessible and ready to be used as the installation source.

When configuring the installation source before or during the installation, use **ftp://** as the protocol, the server's host name or IP address, and the directory in which you have stored the files from the ISO image, relative to the FTP server root. For example, if the server's host name is **myserver.example.com** and you have copied the files from the image to **/var/ftp/rhel7-install/**, specify **ftp://myserver.example.com/rhel7-install/** as the installation source.

3.3.3.3. Firewall Considerations for Network-based Installations

When using a network-based installation source, make sure that your firewall allows the server you are installing to access the remote installation source. The following table shows which ports must be open for each type of network-based installation

Table 3.1. Ports Used by Network Protocols

Protocol used	Ports to open
FTP	21
HTTP	80
HTTPS	443
NFS	2049, 111, 20048
TFTP	69

For information about opening specific firewall ports, see the [Red Hat Enterprise Linux 7 Security Guide](#).

PART I. AMD64, INTEL 64, AND ARM 64 - INSTALLATION AND BOOTING

This part of the *Red Hat Enterprise Linux Installation Guide* discusses the installation of Red Hat Enterprise Linux 7 on 64-bit AMD, Intel, and ARM systems as well as some basic troubleshooting. For advanced installation options, see [Part IV, "Advanced Installation Options"](#).

CHAPTER 4. QUICK INSTALLATION GUIDE

4.1. INTERACTIVE INSTALLATION

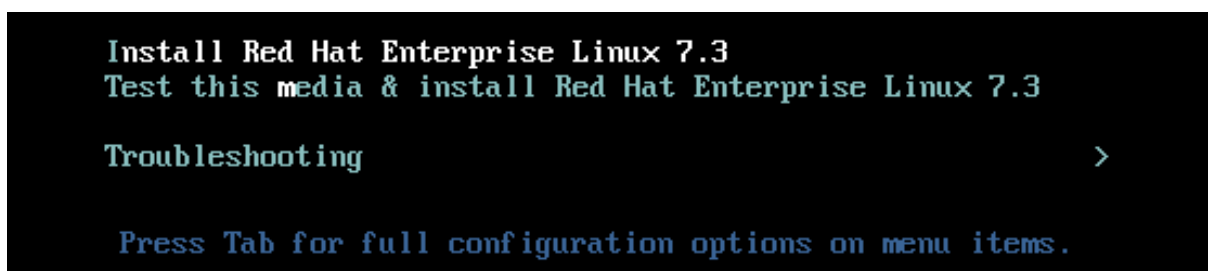
This section describes the simple procedure to install and register Red Hat Enterprise Linux after you have created and booted from an installation USB drive.

Prerequisites: Create an installation USB drive and boot it. For details, see:

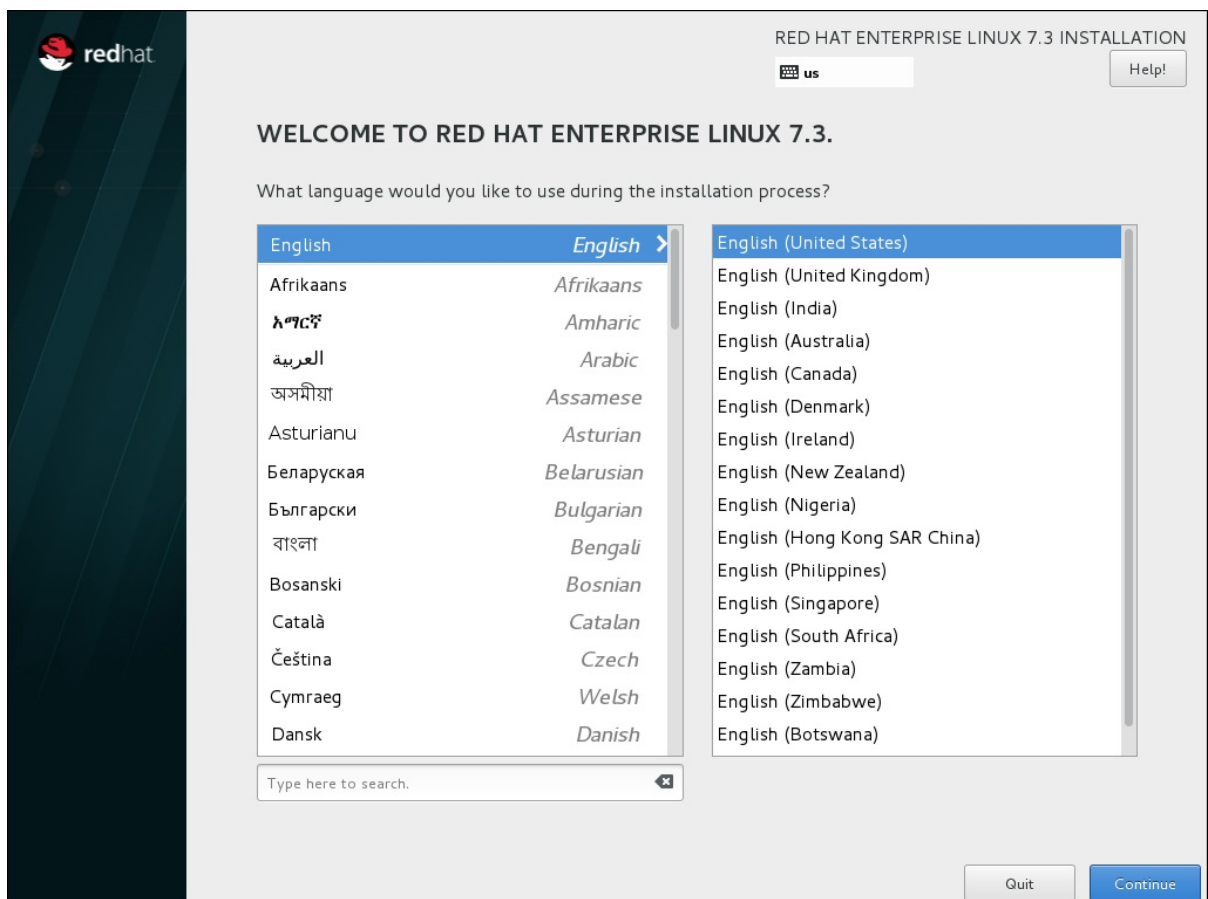
- [Chapter 3, Making Media](#).
- [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#).

After booting the installation USB drive:

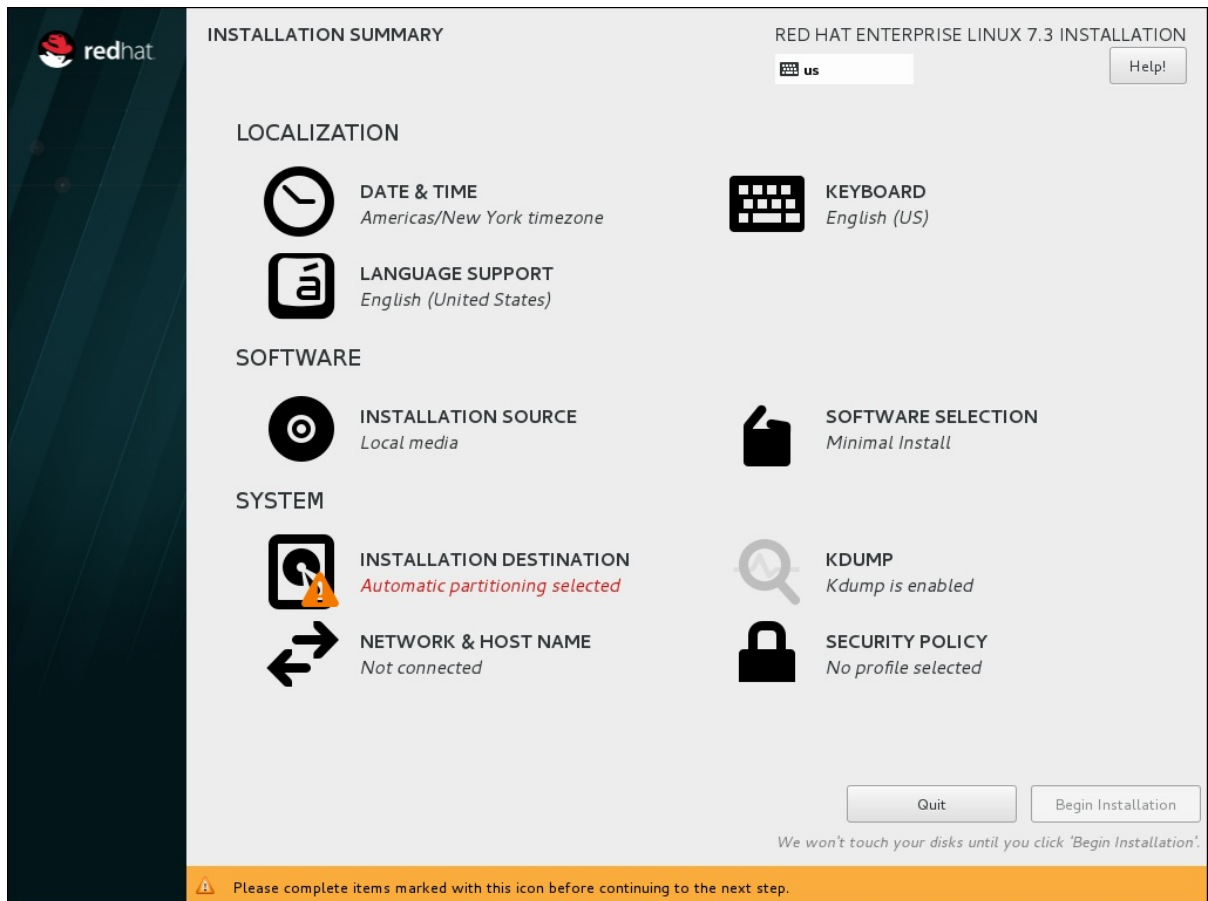
1. Select **Install Red Hat Enterprise Linux** in the boot menu and press **Enter**.



2. After **Anaconda**, the Red Hat Enterprise Linux installer, started, select your language and region, and click **Continue**.



3. The **Installation Summary** is the central screen to set configuration options:



You can display and modify the individual options in any order. If a configuration option was automatically configured correctly, no further action is required. However, if items are marked with an exclamation point icon, you must complete the configuration for these items before you can begin the installation.



NOTE

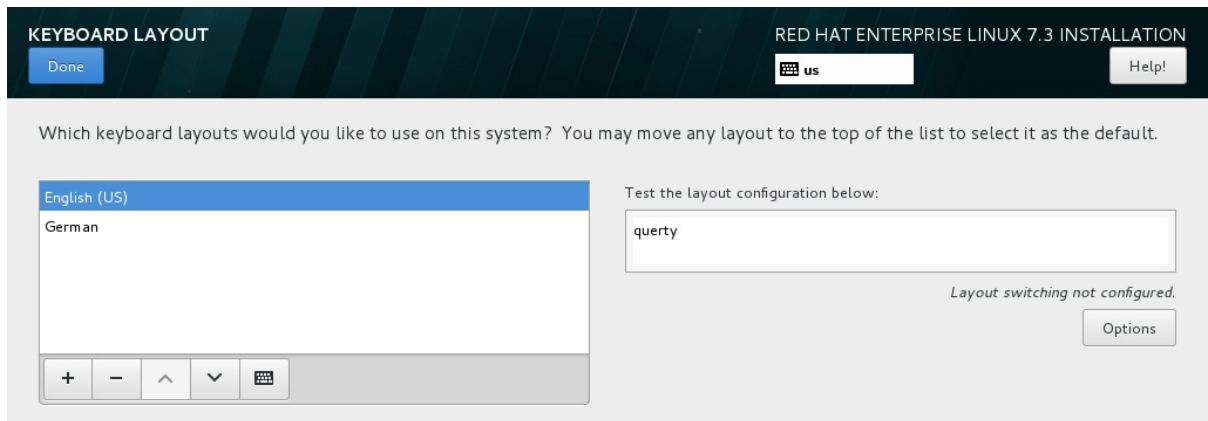
Nothing will be written to the disk until you click the **Begin Installation** button.

4. Select **Date & Time**:

- a. Set your region and the nearest city in your time zone.
- b. Click **Done** to return to the **Installation Summary**.

5. Select **Keyboard Layout**:

- a. Use the **+** and **-** buttons to add and remove keyboard layouts.
- b. If you enable multiple keyboard layouts, move your preferred layout to the top of the list using the **↑** button to set it as default.
- c. Click **Done** to return to the **Installation Summary**.

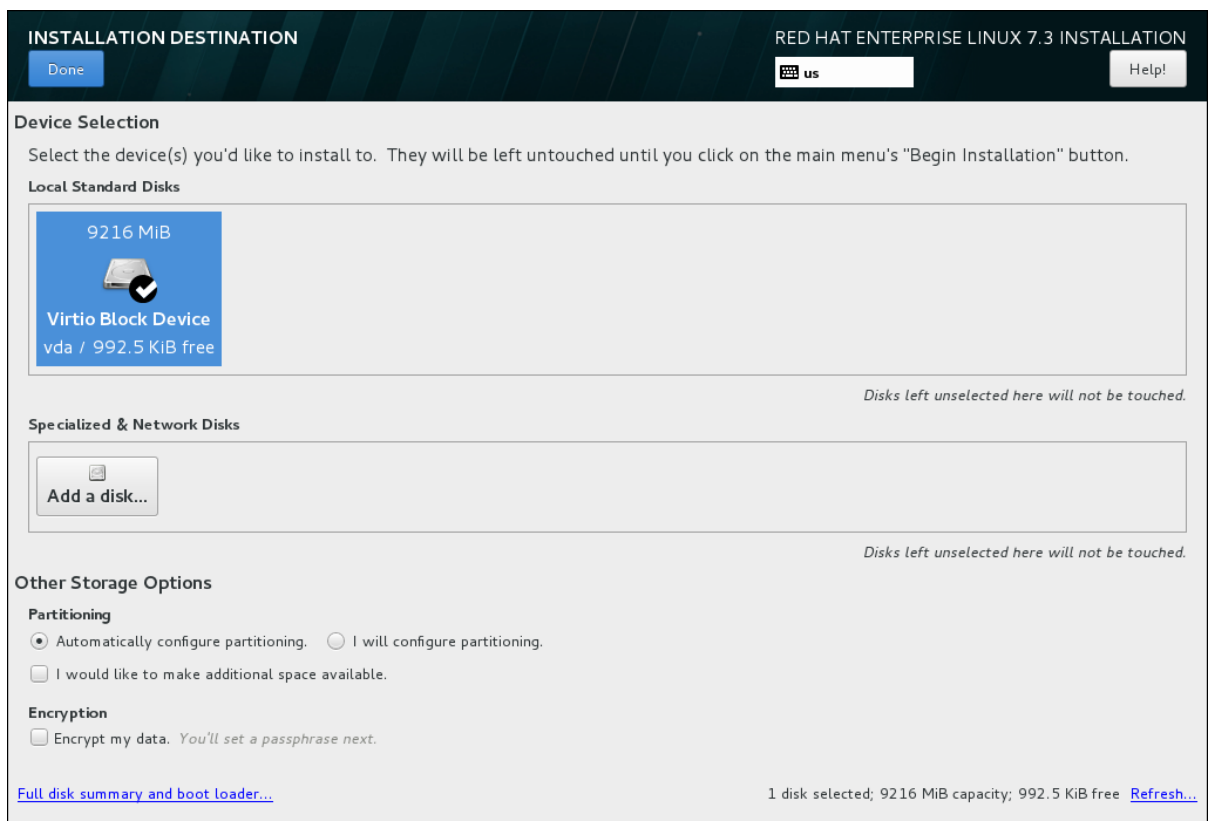


6. Select **Installation Destination**:

- a. Select the target disk. A check mark is displayed next to the selected target.

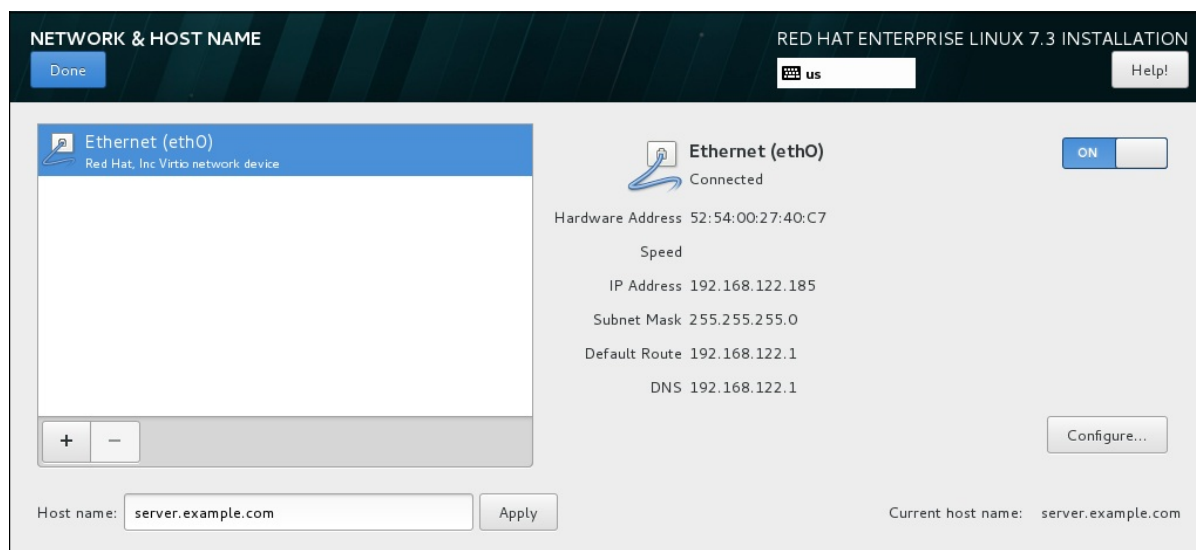
The selected disk is partitioned automatically.

- b. Click **Done** to return to the **Installation Summary**.



7. Select **Network and Hostname**:

- a. Click the **Ethernet** sliding switch in the top right corner to enable the network configuration.
- b. Optional, select the device and click **Configure** to update the network interface configuration.
- c. Click **Done** to return to the **Installation Summary**.



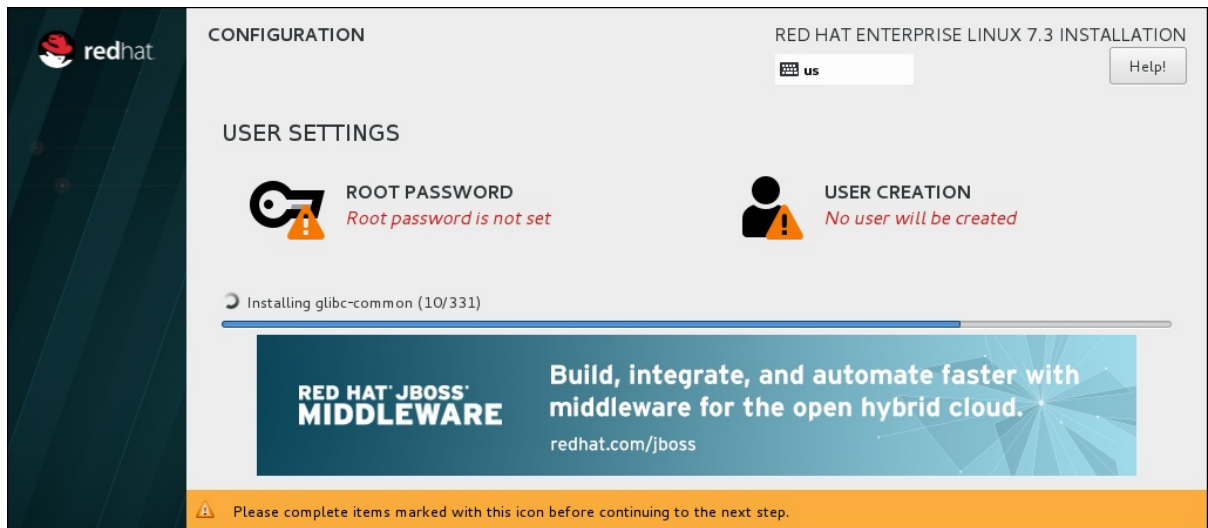
NOTE

Anaconda applies the network settings immediately. They are used during the setup and after the installation.

- On the **Installation Summary** screen, click **Begin Installation**.



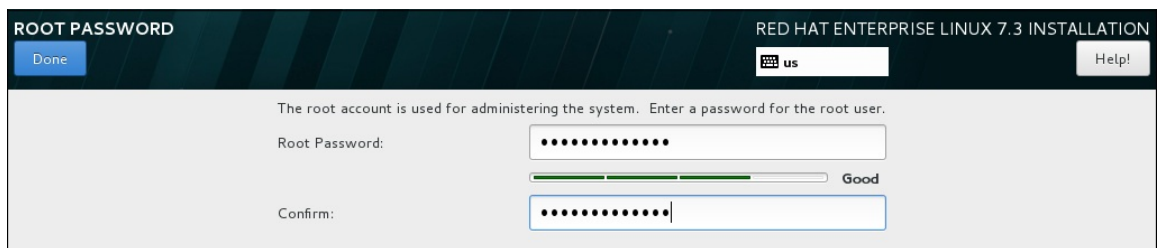
- The installation starts and the **Configuration** screen is displayed:



During the installation:

a. Select **Root Password**:

- i. Enter the password for the **root** user and confirm it.
- ii. Click **Done** to return to the **Configuration** screen.



b. Select **User Creation**:

- i. Enter the user's full name.
- ii. Optionally, update the automatically generated user name.
- iii. Set the password and confirm it.
- iv. Optionally, check the **Make this user administrator** check box. This will add the user to the **wheel** group and allow this account to use **sudo** without any further configuration.
- v. Click **Done** to return to the **Configuration** screen.

c. Wait until the installation completes and click **Reboot**.

10. After the installed system has been started:

- If you installed the server using the **Server with GUI** base environment, the **Initial Setup** application is started automatically:

1. Accept the license agreement.
2. Register the system.

For details, see [Chapter 30, Initial Setup](#).

- If you selected any other base environment during the installation:

1. Log into the system as the **root** user.
2. Register the system and automatically attach a subscription:

```
# subscription-manager register --auto-attach \
--username=user_name --password=password
```

4.2. AUTOMATIC INSTALLATION

This section describes a simple procedure on how to add a Kickstart file to the installation USB drive, which automatically installs and registers Red Hat Enterprise Linux. You can use this procedure to deploy Red Hat Enterprise Linux on multiple machines.

Generating the USB Boot Media

1. Record an installation in a Kickstart file:
 - a. Manually install Red Hat Enterprise Linux once. For details see [Section 4.1, “Interactive Installation”](#).
 - b. Boot the installed system. During the installation, **Anaconda** created a Kickstart file with the settings in the **/root/anaconda-ks.cfg** file.
2. Download the Red Hat Enterprise Linux installation DVD ISO file to the **/tmp/** directory.
3. Mount the installation ISO file to the **/mnt/** directory. For example:

```
# mount -o loop /tmp/rhel-server-7.3-x86_64-dvd.iso /mnt/
```

4. Create a working directory and copy the DVD content to it. For example:

```
# mkdir /root/rhel-install/
# shopt -s dotglob
# cp -avRf /mnt/* /root/rhel-install/
```

5. Unmount the ISO file:

```
# umount /mnt/
```

6. Copy the Kickstart file generated during the installation to the working directory:

```
# cp /root/anaconda-ks.cfg /root/rhel-install/
```

7. To register Red Hat Enterprise Linux after the installation automatically and attach a subscription, append the following to the **/root/rhel-install/anaconda-ks.cfg** file:

```
%post
subscription-manager register --auto-attach --username=user_name --password=password
%end
```

8. Display the installation DVD volume name:

```
# isoinfo -d -i rhel-server-7.3-x86_64-dvd.iso | grep "Volume id" | \
sed -e 's/Volume id: //' -e 's/ \x20/g'
RHEL-7.3\x20Server.x86_64
```

9. Add a new menu entry to the boot **/root/rhel-install/isolinux/isolinux.cfg** file that uses the Kickstart file. For example:

```
#####
label kickstart
menu label ^Kickstart Installation of RHEL7.3
kernel vmlinuz

append initrd=initrd.img inst.stage2=hd:LABEL=RHEL-7.3\x20Server.x86_64
inst.ks=hd:LABEL=RHEL-7.3\x20Server.x86_64:/anaconda-ks.cfg
#####
```



NOTE

Set the **inst.stage2=hd:LABEL=** and **inst.ks=hd:LABEL=** options to the DVD volume name retrieved in the previous step.

10. Before you create the **/root/rhel-ks.iso** file from the working directory, execute the following steps for a **USB UEFI boot** or for a **CDROM UEFI boot**

- For a **USB UEFI boot**, follow the steps:

1. Mount the volume:

```
# mount /root/rhel-install/images/efiboot.img /mnt/
```

2. Edit the file **/mnt/EFI/BOOT/grub.cfg**:

3. Add a new menu entry:

```
#####
'Kickstart Installation of RHEL-7.3' --class fedora --class gnu-linux --class gnu --class
os {
    linuxefi /images/pxeboot/vmlinuz inst.stage2=hd:LABEL=RHEL-
7.3\x20Server.x86_64 inst.ks=hd:LABEL=RHEL-7.3\x20Server.x86_64:/anaconda-
ks.cfg
    initrdefi /images/pxeboot/initrd.img
}
#####
```

4. Unmount the volume:

```
# umount /mnt
```

- For a **CDROM UEFI boot**, follow the steps:

1. Edit the file **/root/rhel-install/EFI/BOOT/grub.cfg**:

2. Add a new menu entry to the file:

```
#####
'Kickstart Installation of RHEL-7.3' --class fedora --class gnu-linux --class gnu --class
os {
    linuxefi /images/pxeboot/vmlinuz inst.stage2=hd:LABEL=RHEL-
7.3\x20Server.x86_64 inst.ks=hd:LABEL=RHEL-7.3\x20Server.x86_64:/anaconda-
ks.cfg
    initrdefi /images/pxeboot/initrd.img
}
#####
```

11. Create the **/root/rhel-ks.iso** file from the working directory:

```
# mkisofs -untranslated-filenames -volid "RHEL-7.3 Server.x86_64" -J -joliet-long -rational-
rock -translation-table -input-charset utf-8 -b isolinux/isolinux.bin -c isolinux/boot.cat -no-
emul-boot -boot-load-size 4 -boot-info-table -eltorito-alt-boot -e images/efiboot.img -no-emul-
boot -o /root/rhel-ks.iso -graft-points /root/rhel-install/
```



NOTE

Set the **-V** option to the DVD volume name retrieved in an earlier step and replace **\x20** in the string with a space.

12. Make the ISO image created by the command ``mkisofs`` bootable:


```
# isohybrid --uefi /root/rhel-ks.iso
```

13. Create an installation USB drive. For details, see [Section 3.2.1, “Making Installation USB Media on Linux”](#).

Install Red Hat Enterprise Linux Using the Kickstart File

1. Boot the installation USB drive. See [Chapter 7, *Booting the Installation on 64-bit AMD, Intel, and ARM systems*](#).
2. Select the entry with the Kickstart configuration that you created in [Section 4.2, “Automatic Installation”](#).

CHAPTER 5. PLANNING FOR INSTALLATION ON 64-BIT AMD, INTEL, AND ARM SYSTEMS

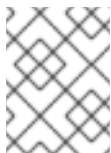
This chapter outlines the decisions and preparations you will need to make when deciding how to proceed with the installation.

5.1. UPGRADE OR INSTALL?

There are two procedures available for upgrading your current system to the next major version of Red Hat Enterprise Linux. To decide which procedure is the right one for your system, read the following descriptions:

Clean Install

A clean install is performed by backing up all data from the system, formatting disk partitions, performing an installation of Red Hat Enterprise Linux from installation media, and then restoring any user data.



NOTE

This is the recommended method for upgrading between major versions of Red Hat Enterprise Linux.

In-Place Upgrade

An in-place upgrade is a way of upgrading your system without removing the older version first. The procedure requires installing the migration utilities available for your system and running them as any other software. In Red Hat Enterprise Linux, the **Preupgrade Assistant** assesses your current system and identifies potential problems you might encounter during or after the upgrade. It also performs minor fixes and modifications to the system. The **Red Hat Upgrade Tool** utility downloads the packages and performs the actual upgrade. An in-place upgrade requires a lot of troubleshooting and planning and should only be done if there is no other choice. For more information on the **Preupgrade Assistant**, see [Chapter 29, Upgrading Your Current System](#).



WARNING

Never perform an in-place upgrade on a production system without first testing it on a cloned backup copy of the system.

5.2. IS YOUR HARDWARE COMPATIBLE?

Red Hat Enterprise Linux 7 should be compatible with most hardware in systems that were factory built within the last two years. Hardware compatibility is a particularly important concern if you have an older or custom-built system. Because hardware specifications change almost daily, it is recommended that all systems be checked for compatibility.

The most recent list of supported hardware can be found in the *Red Hat Hardware Compatibility List*, available online at <https://access.redhat.com/ecosystem/search/#/category/Server>. Also see [Red Hat Enterprise Linux technology capabilities and limits](#) for general information about system requirements.

5.3. SUPPORTED INSTALLATION TARGETS

An installation target is a storage device that will store Red Hat Enterprise Linux and boot the system. Red Hat Enterprise Linux supports the following installation targets for AMD, Intel, and ARM systems:

- Storage connected by a standard internal interface, such as SCSI, SATA, or SAS
- BIOS/firmware RAID devices
- NVDIMM devices in sector mode on the Intel64 and AMD64 architectures, supported by the **nd_pmem** driver.
- Fibre Channel Host Bus Adapters and multipath devices. Some can require vendor-provided drivers.
- Xen block devices on Intel processors in Xen virtual machines.
- VirtIO block devices on Intel processors in KVM virtual machines.

Red Hat does not support installation to USB drives or SD memory cards. For information about the support for third-party virtualization technologies, see the *Red Hat Hardware Compatibility List*, available online at <https://hardware.redhat.com>.

5.4. SYSTEM SPECIFICATIONS LIST

The installation program automatically detects and installs your computer's hardware and you do not usually need to supply the installation program with any specific details about your system. However, when performing certain types of installation, it is important to know specific details about your hardware. For this reason, it is recommended that you record the following system specifications for reference during the installation, depending on your installation type.

- If you plan to use a customized partition layout, record:
 - The model numbers, sizes, types, and interfaces of the hard drives attached to the system. For example, Seagate ST3320613AS 320 GB on SATA0, Western Digital WD7500AAKS 750 GB on SATA1. This will allow you to identify specific hard drives during the partitioning process.
- If you are installing Red Hat Enterprise Linux as an additional operating system on an existing system, record:
 - Information about the partitions used on the system. This information can include file system types, device node names, file system labels, and sizes. This will allow you to identify specific partitions during the partitioning process. Remember that different operating systems identify partitions and drives differently, therefore even if the other operating system is a Unix operating system, the device names can be reported by Red Hat Enterprise Linux differently. This information can usually be found by executing the equivalent of the **mount** command and **blkid** command and in the **/etc/fstab** file.

If you have other operating systems already installed, the Red Hat Enterprise Linux 7 installation program attempts to automatically detect and configure to boot them. You can manually configure any additional operating systems if they are not detected properly. For

more information, see [Section 8.14.1, “Boot Loader Installation”](#).

- If you plan to install from an image on a local hard drive:
 - The hard drive and directory that holds the image.
- If you plan to install from a network location:
 - The make and model numbers of the network adapters on your system. For example, Netgear GA311. This will allow you to identify adapters when manually configuring the network.
 - IP, DHCP, and BOOTP addresses
 - Netmask
 - Gateway IP address
 - One or more name server IP addresses (DNS)
 - The location of the installation source on an FTP server, HTTP (web) server, HTTPS (web) server, or NFS server.

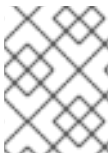
If any of these networking requirements or terms are unfamiliar to you, contact your network administrator for assistance.

- If you plan to install on an iSCSI target:
 - The location of the iSCSI target. Depending on your network, you might also need a CHAP user name and password, and perhaps a reverse CHAP user name and password.
- If your computer is part of a domain:
 - You should verify that the domain name will be supplied by the DHCP server. If not, you will need to input the domain name manually during installation.

5.5. DISK SPACE AND MEMORY REQUIREMENTS

Red Hat Enterprise Linux, like most modern operating systems, uses *disk partitions*. When you install Red Hat Enterprise Linux, you might have to work with disk partitions. For more information, see [Appendix A, *An Introduction to Disk Partitions*](#).

If you have other operating systems installed on your system, the disk space they use must be separate from the disk space used by Red Hat Enterprise Linux.



NOTE

For AMD64/Intel 64 and ARM systems, at least two partitions (*/* and **swap**) must be dedicated to Red Hat Enterprise Linux.

To install Red Hat Enterprise Linux, you must have a minimum of 10 GiB of space in either unpartitioned disk space or in partitions which can be deleted. For more information on partition and disk space recommendations, see the recommended partitioning sizes discussed in [Section 8.14.4.4, “Recommended Partitioning Scheme”](#).

Red Hat Enterprise Linux requires minimum the following amount of RAM:

Installation type	Minimum required RAM
Local media installation (USB, DVD)	768 MiB
NFS network installation	768 MiB
HTTP, HTTPS, or FTP network installation	1.5 GiB

**NOTE**

It may be possible to perform the installation with less memory than listed in this section. However, the exact requirements depend heavily on your environment and exact installation path, and they also change with each new release. Determining the absolute minimum required RAM for your specific use case therefore requires you to test various configurations, and periodically re-test with each new release.

Installing Red Hat Enterprise Linux using a Kickstart file has the same minimum RAM requirements as a manual installation. However, if you use a Kickstart file that runs commands which require additional memory or write data to the RAM disk, additional RAM might be necessary.

For more information about the minimum requirements and technology limits of Red Hat Enterprise Linux 7, see the [Red Hat Enterprise Linux technology capabilities and limits](#) article on the Red Hat Customer Portal.

5.6. RAID AND OTHER DISK DEVICES

Some storage technology requires special consideration when using Red Hat Enterprise Linux. Generally, it is important to understand how these technologies are configured, visible to Red Hat Enterprise Linux, and how support for them might have changed between major versions.

5.6.1. Hardware RAID

RAID (Redundant Array of Independent Disks) allows a group, or array, of drives to act as a single device. Configure any RAID functions provided by the mainboard of your computer, or attached controller cards, before you begin the installation process. Each active RAID array appears as one drive within Red Hat Enterprise Linux.

5.6.2. Software RAID

On systems with more than one hard drive, you can use the Red Hat Enterprise Linux installation program to operate several of the drives as a Linux software RAID array. With a software RAID array, RAID functions are controlled by the operating system rather than dedicated hardware. These functions are explained in detail in [Section 8.14.4, “Manual Partitioning”](#).

**NOTE**

When a pre-existing RAID array's member devices are all unpartitioned disks/drives, the installer will treat the array itself as a disk and will not provide a way to remove the array.

5.6.3. USB Disks

You can connect and configure external USB storage after installation. Most such devices are recognized by the kernel and available for use at that time.

Some USB drives might not be recognized by the installation program. If configuration of these disks at installation time is not vital, disconnect them to avoid potential problems.

5.6.4. NVDIMM devices

To use a Non-Volatile Dual In-line Memory Module (NVDIMM) device as storage, the following conditions must be satisfied:

- Version of Red Hat Enterprise Linux is 7.6 or later.
- The architecture of the system is Intel 64 or AMD64.
- The device is configured to sector mode. Anaconda can reconfigure NVDIMM devices to this mode.
- The device must be supported by the **nd_pmem** driver.

Bootting from a NVDIMM device is possible under the following additional conditions:

- The system uses UEFI.
- The device must be supported by firmware available on the system, or by a UEFI driver. The UEFI driver may be loaded from an option ROM of the device itself.
- The device must be made available under a namespace.

To take advantage of the high performance of NVDIMM devices during booting, place the **/boot** and **/boot/efi** directories on the device. See [Section 8.14.4, “Manual Partitioning”](#) for more information. Note that the Execute-in-place (XIP) feature of NVDIMM devices is not supported during booting and the kernel is loaded into conventional memory.

5.6.5. Considerations for Intel BIOS RAID Sets

Red Hat Enterprise Linux 7 uses **mdraid** for installation onto Intel BIOS RAID sets. These sets are detected automatically during the boot process and their device node paths can change from boot to boot. For this reason, local modifications to **/etc/fstab**, **/etc/crypttab** or other configuration files which refer to devices by their device node paths might not work in Red Hat Enterprise Linux 7. Therefore, you should replace device node paths (such as **/dev/sda**) with file system labels or device UUIDs instead. You can find the file system labels and device UUIDs using the **blkid** command.

5.6.6. Considerations for Intel BIOS iSCSI Remote Boot

If you are installing using Intel iSCSI Remote Boot, all attached iSCSI storage devices must be disabled, otherwise the installation will succeed but the installed system will not boot.

5.7. CHOOSE AN INSTALLATION BOOT METHOD

You can use several methods to boot the Red Hat Enterprise Linux 7 installation program. The method you choose depends upon your installation media.

Your system's firmware (BIOS or UEFI) settings might need to be changed to allow booting from removable media such as a DVD or a USB flash drive. See [Section 7.1.1, “Booting from Physical Media”](#) for information.



NOTE

Installation media must remain mounted throughout installation, including during execution of the **%post** section of a kickstart file.

Full installation DVD or USB drive

You can create bootable media from the full installation DVD ISO image. In this case, a single DVD or USB drive can be used to complete the entire installation – it will serve both as a boot device and as an installation source for installing software packages. See [Chapter 3, Making Media](#) for instructions on how to make a full installation DVD or USB drive.

Minimal boot CD, DVD or USB Flash Drive

A minimal boot CD, DVD or USB flash drive is created using a small ISO image, which only contains data necessary to boot the system and start the installation. If you use this boot media, you will need an additional installation source from which packages will be installed. See [Section 3.2, “Making Installation USB Media”](#) for instructions on making boot CDs, DVDs and USB flash drives.

PXE Server

A *preboot execution environment* (PXE) server allows the installation program to boot over the network. After you boot the system, you complete the installation from a different installation source, such as a local hard drive or a location on a network. For more information on PXE servers, see [Chapter 24, Preparing for a Network Installation](#).

5.8. AUTOMATING THE INSTALLATION WITH KICKSTART

Red Hat Enterprise Linux 7 offers a way to partially or fully automate the installation process using a *Kickstart file*. Kickstart files contain answers to all questions normally asked by the installation program, such as what time zone do you want the system to use, how should the drives be partitioned or which packages should be installed. Providing a prepared Kickstart file at the beginning of the installation therefore allows you to perform the entire installation (or parts of it) automatically, without need for any intervention from the user. This is especially useful when deploying Red Hat Enterprise Linux on a large number of systems at once.

In addition to allowing you to automate the installation, Kickstart files also provide more options regarding software selection. When installing Red Hat Enterprise Linux manually using the graphical installation interface, your software selection is limited to pre-defined environments and add-ons. A Kickstart file allows you to install or remove individual packages as well.

For instructions about creating a Kickstart file and using it to automate the installation, see [Chapter 27, Kickstart Installations](#).

5.9. USING A BETA RELEASE WITH UEFI SECURE BOOT



NOTE

This section *only* concerns Beta releases of Red Hat Enterprise Linux 7.

The UEFI Secure Boot technology requires that the operating system kernel must be signed with a recognized private key in order to be able to boot. In every beta release of Red Hat Enterprise Linux 7, the kernel is signed with a Red Hat Beta-specific private key, which is different from the more common Red Hat key used to sign kernels in a General Availability (non-Beta) releases.

The Beta private key will likely not be recognized by your hardware, which means that any Beta release of Red Hat Enterprise Linux 7 will not be able to boot. In order to use a Beta release with UEFI Secure Boot enabled, you need to add the Red Hat Beta public key to your system using the *Machine Owner Key* (MOK) facility.

The procedure to add the Red Hat Beta key to your system is below.

Procedure 5.1. Adding a Custom Private Key for UEFI Secure Boot

1. First, disable UEFI Secure Boot on the system, and install Red Hat Enterprise Linux 7 normally.
2. After the installation finishes, the system will reboot. Secure Boot should still be disabled at this point. Reboot the system, log in and, if applicable, go through the Initial Setup screens as described in [Chapter 30, Initial Setup](#).
3. After finishing the first boot and going through Initial Setup, install the `kernel-doc` package if not installed already:

```
# yum install kernel-doc
```

This package provides a certificate file which contains the Red Hat CA public Beta key, located in `/usr/share/doc/kernel-keys/kernel-version/kernel-signing-ca.cer`, where *kernel-version* is the kernel version string without the platform architecture suffix - for example, **3.10.0-686.el7**.

4. Execute the following commands to enroll the public key into the system Machine Owner Key (MOK) list:

```
# kr=$(uname -r)
# mokutil --import /usr/share/doc/kernel-keys/${kr%.$(uname -p)}/kernel-signing-ca.cer
```

Enter a password of your choosing when prompted.



NOTE

Make sure to remember the password. It is required to finish this procedure as well as to remove the imported key when it is no longer needed.

5. Reboot the system again. During startup you will be prompted to confirm that you want to complete the pending key enrollment request. Select yes, and provide the password which you set earlier using the **mokutil** command in the previous step. The system will reboot again after you do so, and the key will be imported into the system firmware. You can turn on Secure Boot on this or any subsequent reboot.



WARNING

Remove the imported Beta public key when you no longer need it.

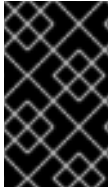
If you install a final (General Availability) release of Red Hat Enterprise Linux 7, or when you install a different operating system, you should remove the imported key. If you have *only* imported this public key, you can use the following command to reset the MOK:

```
# mokutil --reset
```

After the next reboot, the firmware will prompt you for a confirmation and the password you created when importing the key. The key will be removed from the MOK after providing the correct password, and the system will revert to its original state.

CHAPTER 6. UPDATING DRIVERS DURING INSTALLATION ON AMD64 AND INTEL 64 SYSTEMS

In most cases, Red Hat Enterprise Linux already includes drivers for the devices that make up your system. However, if your system contains hardware that has been released very recently, drivers for this hardware might not yet be included. Sometimes, a driver update that provides support for a new device might be available from Red Hat or your hardware vendor on a *driver disc* that contains *RPM packages*. Typically, the driver disc is available for download as an *ISO image file*.



IMPORTANT

Driver updates should only be performed if a missing driver prevents you to complete the installation successfully. The drivers included in the kernel should always be preferred over drivers provided by other means.

Often, you do not need the new hardware during the installation process. For example, if you use a DVD to install to a local hard drive, the installation will succeed even if drivers for your network card are not available. In such a situation, complete the installation and add support for the new hardware afterward – see [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for details of adding this support.

In other situations, you might want to add drivers for a device during the installation process to support a particular configuration. For example, you might want to install drivers for a network device or a storage adapter card to give the installation program access to the storage devices that your system uses. You can use a driver disc to add this support during installation in one of two ways:

1. place the ISO image file of the driver disc in a location accessible to the installation program, on a local hard drive, on a USB flash drive, or on a CD or DVD.
2. create a driver disc by extracting the image file onto a CD or a DVD, or a USB flash drive. See the instructions for making installation discs in [Section 3.1, “Making an Installation CD or DVD”](#) for more information on burning ISO image files to a CD or DVD, and [Section 3.2, “Making Installation USB Media”](#) for instructions on writing ISO images to USB drives.

If Red Hat, your hardware vendor, or a trusted third party told you that you will require a driver update during the installation process, choose a method to supply the update from the methods described in this chapter and test it before beginning the installation. Conversely, do not perform a driver update during installation unless you are certain that your system requires it. The presence of a driver on a system for which it was not intended can complicate support.



WARNING

Driver update disks sometimes disable conflicting kernel drivers, where necessary. In rare cases, unloading a kernel module in this way can cause installation errors.

6.1. LIMITATIONS OF DRIVER UPDATES DURING INSTALLATION

On UEFI systems with the Secure Boot technology enabled, all drivers being loaded must be signed with a valid certificate, otherwise the system will refuse them. All drivers provided by Red Hat are signed by one of Red Hat's private keys and authenticated by the corresponding Red Hat public key in the kernel.

If you load any other drivers (ones not provided on the Red Hat Enterprise Linux installation DVD), you must make sure that they are signed as well.

More information about signing custom drivers can be found in the Working with Kernel Modules chapter in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

6.2. PREPARING FOR A DRIVER UPDATE DURING INSTALLATION

If a driver update is necessary and available for your hardware, Red Hat, your hardware vendor, or another trusted third party will typically provide it in the form of an image file in ISO format. Once you obtain the ISO image, you must decide on the method you want to use to perform the driver update.

The available methods are:

Automatic driver update

When starting the installation, the **Anaconda** installation program will attempt to detect all attached storage devices. If there is a storage device labeled **OEMDRV** present when the installation begins, **Anaconda** will always treat it like a driver update disc and attempt to load drivers present on it.

Assisted driver update

You can specify the **inst.dd** boot option when starting the installation. If you use this option without any parameters, **Anaconda** will display a list of all storage devices connected to the system, and it will prompt you to select a device which contains a driver update.

Manual driver update

You can specify the **inst.dd=location** boot option when starting the installation, where *location* is the path to a driver update disc or ISO image. When you specify this option, **Anaconda** will attempt to load any driver updates it finds at the specified location. With manual driver updates, you can specify either locally available storage devices, or a network location (an **HTTP**, **HTTPS** or **FTP** server).



NOTE

You can also use both **inst.dd=location** and **inst.dd** at the same time. However, what **Anaconda** does in this case depends on the type of *location* that you use. If it is a device, **Anaconda** prompts you to select drivers to update from the specified device and then it offers you additional devices. If *location* is a network location, **Anaconda** first prompts you to select a device containing a driver update and then it lets you update drivers from the specified network location.

If you want to use the automatic driver update method, you must create a storage device labeled **OEMDRV**, and it must be physically connected to the installation system. To use the assisted method, you can use any local storage device any label other than **OEMDRV**. To use the manual method, you can use any local storage with a different label, or a network location accessible from the installation system.



IMPORTANT

Make sure to initialize the network using the **ip=** option when loading a driver update from a network location. See [Section 23.1, “Configuring the Installation System at the Boot Menu”](#) for details.

6.2.1. Preparing to Use a Driver Update Image File on Local Storage

If you use a local storage device to provide the ISO file, such as a hard drive or USB flash drive, you can make the installation program to recognize it automatically by properly labeling the device. Only if it is not possible, install the update manually as described below.

- In order for the installation program to automatically recognize the driver disk, the volume label of the storage device must be **OEMDRV**. Also, you will need to extract the contents of the ISO image file to the root directory of the storage device rather than copy the ISO image itself. See [Section 6.3.1, “Automatic Driver Update”](#). Note that installation of a driver from a device labeled **OEMDRV** is always recommended and preferable to the manual installation.
- For manual installation, simply copy the ISO image, as a single file, onto the storage device. You can rename the file if you find it helpful but you must not change the file name extension, which must remain **.iso**, for example **dd.iso**. See [Section 6.3.3, “Manual Driver Update”](#) to learn how to select the driver update manually during installation.

6.2.2. Preparing a Driver Disc

You can create a driver update disc on a CD or DVD. See [Section 3.1, “Making an Installation CD or DVD”](#) to learn more about burning discs from image files.

After you burn a driver update disc CD or DVD, verify that the disc was created successfully by inserting it into your system and browsing to it using the file manager. You should see a single file named **rhdd3**, which is a signature file that contains the driver disc's description, and a directory named **rpms**, which contains the RPM packages with the actual drivers for various architectures.

If you see only a single file ending in **.iso**, then you have not created the disc correctly and should try again. Ensure that you choose an option similar to **Burn from Image** if you use a Linux desktop other than **GNOME**, or if you use a different operating system.

6.3. PERFORMING A DRIVER UPDATE DURING INSTALLATION

At the very beginning of the installation process, you can perform a driver update in the following ways:

- let the installation program automatically find and offer a driver update for installation,
- let the installation program prompt you to locate a driver update,
- manually specify a path to a driver update image or an RPM package.

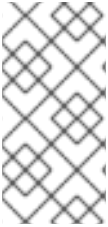


IMPORTANT

Always make sure to put your driver update discs on a standard disk partition. Advanced storage, such as RAID or LVM volumes, might not be accessible during the early stage of the installation when you perform driver updates.

6.3.1. Automatic Driver Update

To have the installation program automatically recognize a driver update disc, connect a block device with the **OEMDRV** volume label to your computer before starting the installation process.



NOTE

Starting with Red Hat Enterprise Linux 7.2, you can also use the **OEMDRV** block device to automatically load a Kickstart file. This file must be named **ks.cfg** and placed in the root of the device to be loaded. See [Chapter 27, Kickstart Installations](#) for more information about Kickstart installations.

When the installation begins, the installation program detects all available storage connected to the system. If it finds a storage device labeled **OEMDRV**, it will treat it as a driver update disc and attempt to load driver updates from this device. You will be prompted to select which drivers to load:

```
DD: Checking devices /dev/sr1
DD: Checking device /dev/sr1
DD: Processing DD repo /media/DD//rpms/x86_64 on /dev/sr1

Page 1 of 1
Select drivers to install
  1) [ ] /media/DD//rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue:
```

Figure 6.1. Selecting a Driver

Use number keys to toggle selection on individual drivers. When ready, press **c** to install the selected drivers and proceed to the **Anaconda** graphical user interface.

6.3.2. Assisted Driver Update

It is always recommended to have a block device with the **OEMDRV** volume label available to install a driver during installation. However, if no such device is detected and the **inst.dd** option was specified at the boot command line, the installation program lets you find the driver disk in interactive mode. In the first step, select a local disk partition from the list for **Anaconda** to scan for ISO files. Then, select one of the detected ISO files. Finally, select one or more available drivers. The image below demonstrates the process in the text user interface with individual steps highlighted.

```

Starting Driver Update Disk UI on tty1...
DD: Checking devices

Page 1 of 1
Driver disk device selection
  DEVICE      TYPE  LABEL      UUID
  1) vda1      ext2  HOME       8c9d0c6e-4fea-4910-9bac-6609bc8ff847
  2) vda2      xfs   DD_PART    9dcc606d-a9ca-41d1-98b5-e9411769e37f
  3) vdb1      ext4  DD_PART    dd69ffa5-c72e-4b61-ae39-0197d6960fc3

# to select, 'n'-next page, 'p'-previous page or 'c'-continue: 3
[ 97.268612] EXT4-fs (vdb1): mounted filesystem without journal. Opts: (null)

Page 1 of 1
Choose driver disk ISO file
  1) dd.iso

# to select, 'n'-next page, 'p'-previous page or 'c'-continue: 1
DD: Checking device /media/DD-search/dd.iso
[ 112.233480] loop: module loaded
DD: Processing DD repo /media/DD/rpms/x86_64 on /media/DD-search/dd.iso

Page 1 of 1
Select drivers to install
  1) [ ] /media/DD/rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue: 1

Page 1 of 1
Select drivers to install
  1) [x] /media/DD/rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue: _

```

Figure 6.2. Selecting a Driver Interactively

**NOTE**

If you extracted your ISO image file and burned it on a CD or DVD but the media does not have the **OEMDRV** volume label, either use the **inst.dd** option with no arguments and use the menu to select the device, or use the following boot option for the installation program to scan the media for drivers:

```
inst.dd=/dev/sr0
```

Hit number keys to toggle selection on individual drivers. When ready, press **c** to install the selected drivers and proceed to the **Anaconda** graphical user interface.

6.3.3. Manual Driver Update

For manual driver installation, prepare an ISO image file containing your drivers to an accessible location, such a USB flash drive or a web server, and connect it to your computer. At the welcome screen, hit **Tab** to display the boot command line and append the **inst.dd=location** to it, where *location* is a path to the driver update disc:

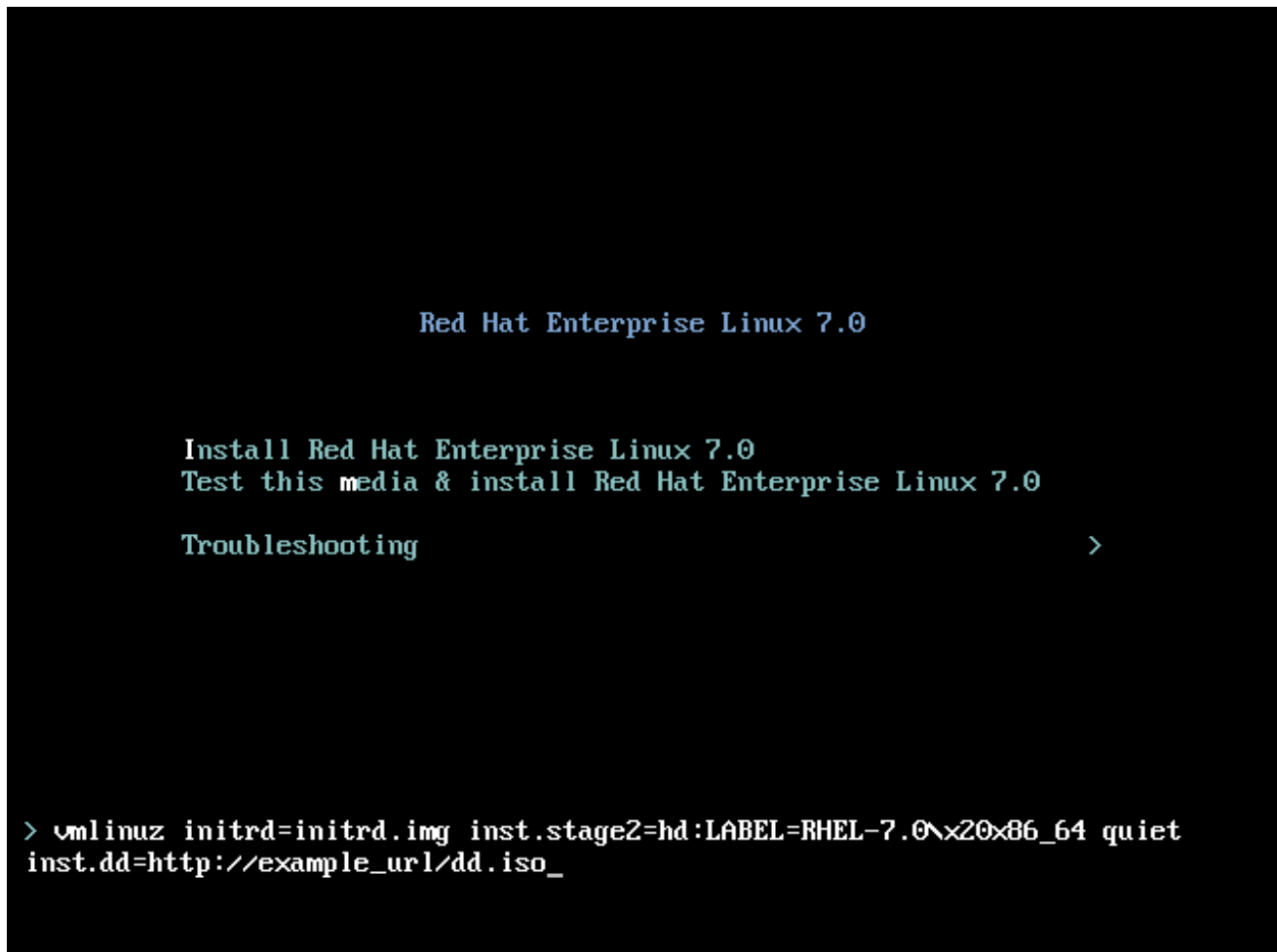


Figure 6.3. Specifying a Path to a Driver Update

Typically, the image file is located on a web server (for example, *http://server.example.com/dd.iso*) or on a USB flash drive (for example, */dev/sdb1*). It is also possible to specify an RPM package containing the driver update (for example *http://server.example.com/dd.rpm*).

When ready, hit **Enter** to execute the boot command. Then, your selected drivers will be loaded and the installation process will proceed normally.

6.3.4. Blacklisting a Driver

A malfunctioning driver can prevent a system from booting normally during installation. When this happens, you can disable (or blacklist) the driver by customizing the boot command line. At the boot menu, display the boot command line by hitting the **Tab** key. Then, append the **modprobe.blacklist=driver_name** option to it. Replace *driver_name* with names of a driver or drivers you want to disable, for example:

```
modprobe.blacklist=ahci
```

Note that the drivers blacklisted during installation using the **modprobe.blacklist=** boot option will remain disabled on the installed system and appear in the */etc/modprobe.d/anaconda-blacklist.conf* file. See [Chapter 23, Boot Options](#) for more information about blacklisting drivers and other boot options.

CHAPTER 7. BOOTING THE INSTALLATION ON 64-BIT AMD, INTEL, AND ARM SYSTEMS

You can install Red Hat Enterprise Linux from the ISO images stored on hard disk, or from a network using **NFS**, **FTP**, **HTTP**, or **HTTPS** methods. Booting and installing from the full installation DVD is the easiest method to get started with. Other methods require some additional setup but provide different advantages that might suit your needs better. For example, when installing Red Hat Enterprise Linux on a large number of computers at the same time, the best approach is booting from a PXE server and installing from a source in a shared network location.

The following table summarizes the different boot methods and recommended installation methods to use with each:

Table 7.1. Boot Methods and Installation Sources

Boot method	Installation source
Full installation media (DVD or USB)	The boot media itself
Minimal boot media (CD or USB)	Full installation DVD ISO image or the installation tree extracted from this image, placed in a network location or on a hard drive
Network boot (PXE)	Full installation DVD ISO image or the installation tree extracted from this image, placed in a network location

To create a boot CD-ROM or to prepare your USB flash drive for booting or installation, see [Section 3.2, “Making Installation USB Media”](#).

This chapter covers the following topics:

- [Section 7.1.1, “Booting from Physical Media”](#) describes how to boot the installation program using physical media (Red Hat Enterprise Linux DVD, Boot CD-ROM, USB flash drive).
- [Section 7.1.2, “Booting from the Network Using PXE”](#) describes how to boot the installation program using PXE.
- [Section 7.2, “The Boot Menu”](#) contains information on the boot menu.

7.1. STARTING THE INSTALLATION PROGRAM

To start, first make sure that you have all necessary resources for the installation. If you have already read through [Chapter 5, Planning for Installation on 64-bit AMD, Intel, and ARM Systems](#), and followed the instructions, you should be ready to start the installation process. When you have verified that you are ready to begin, boot the installation program using the Red Hat Enterprise Linux DVD or any boot media that you have created.



IMPORTANT

Excessive input (for example, clicking the mouse repeatedly) during the boot sequence can cause the installer to ignore keyboard input later in the installation process.

**NOTE**

Occasionally, some hardware components require a *driver update* during the installation. A driver update adds support for hardware that is not otherwise supported by the installation program. See [Chapter 6, *Updating Drivers During Installation on AMD64 and Intel 64 Systems*](#) for more information.

7.1.1. Booting from Physical Media

To start the installation program from a Red Hat Enterprise Linux DVD or from minimal boot media, follow this procedure:

Procedure 7.1. Booting the Installation from Physical Media

1. Disconnect any drives which you do not need for the installation. See [Section 5.6.3, “USB Disks”](#) for more information.
2. Power on your computer system.
3. Insert the media in your computer.
4. Power off your computer with the boot media still inside.
5. Power on your computer system. Note that you might need to press a specific key or combination of keys to boot from the media or configure your system's *Basic Input/Output System* (BIOS) to boot from the media. For more information, see the documentation that came with your system.

After a short delay, the boot screen appears, which contains information on a variety of boot options. Installation program automatically begins if you take no action within the first minute. For a description of the options available on this screen, see [Section 7.2, “The Boot Menu”](#).

7.1.2. Booting from the Network Using PXE

To boot with PXE, you need a properly configured TFTP server, and a network interface in your computer that supports PXE. For information on how to configure a PXE server, see [Chapter 24, *Preparing for a Network Installation*](#).

Configure the computer to boot from the network interface. This option is in the BIOS, and can be labeled **Network Boot** or **Boot Services**. Also, ensure that the BIOS is configured to boot first from the correct network interface. Some BIOS systems specify the network interface as a possible boot device, but do not support the PXE standard. See your hardware's documentation for more information. Once you properly enable PXE booting, the computer can boot the Red Hat Enterprise Linux installation system without any other media.

Follow the procedure below to boot the installation program from a PXE server. Note that this procedure requires the use of a physical network connection, for example Ethernet. It will not work with a wireless connection.

Procedure 7.2. Booting the Installation from the Network Using PXE

1. Ensure that the network cable is attached. The link indicator light on the network socket should be lit, even if the computer is not switched on.
2. Switch on the computer.

3. Depending on your hardware, some network setup and diagnostic information can be displayed before your computer connects to a PXE server. Once it connects, a menu is displayed according to the configuration of the PXE server. Press the number key that corresponds to the desired option. If you are not sure of which option to select, ask your server administrator.

At this point, the installation program starts successfully and the boot screen appears, which contains information on a variety of boot options. Installation program automatically begins if you take no action within the first minute. For a description of the options available on this screen, see [Section 7.2, “The Boot Menu”](#).

7.2. THE BOOT MENU

Once your system has completed loading the boot media, a boot menu is displayed using **GRUB2** (**GR**and **Un**ified **B**ootloader, version 2). The boot menu provides several options in addition to launching the installation program. If no key is pressed within 60 seconds, the default boot option (the one highlighted in white) will be run. To choose the default, either wait for the timer to run out or press **Enter**.

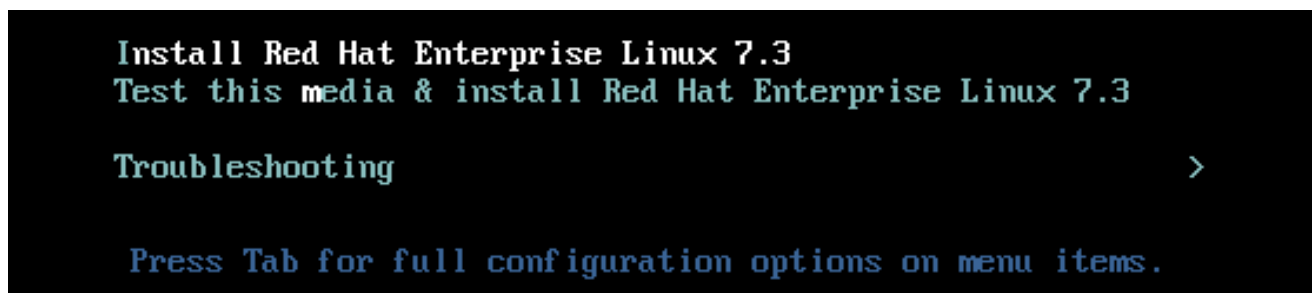


Figure 7.1. The Boot Screen

To select a different option than the default, use the arrow keys on your keyboard, and press **Enter** when the correct option is highlighted.

To customize the boot options for a particular menu entry:

- On BIOS-based systems, the preferred way is to press the **Tab** key and add custom boot options to the command line. You can also access the **boot:** prompt by pressing the **Esc** key but no required boot options will be preset in it. In that case, you must always specify the **linux** option before using any other boot options.
- On UEFI-based systems, press the **e** key and add custom boot options to the command line. When ready press **Ctrl+X** to boot the modified option.

See [Chapter 23, Boot Options](#) for more information about additional boot options.

The boot menu options are:

Install Red Hat Enterprise Linux 7.0

Choose this option to install Red Hat Enterprise Linux onto your computer system using the graphical installation program.

Test this media & install Red Hat Enterprise Linux 7.0

This option is the default. Prior to starting the installation program, a utility is launched to check the integrity of the installation media.

Troubleshooting >

This item is a separate menu containing options that help resolve various installation issues. When highlighted, press **Enter** to display its contents.

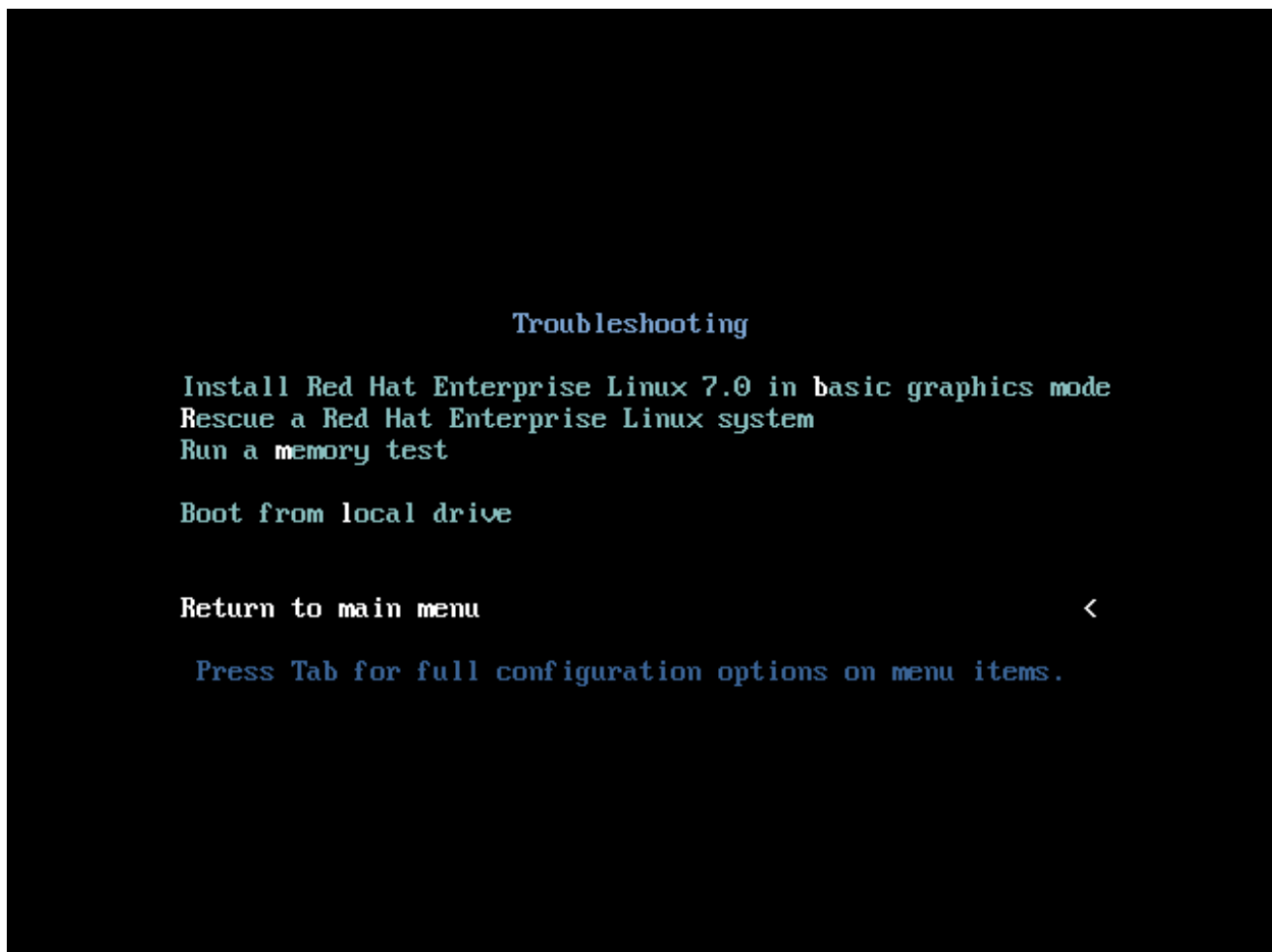


Figure 7.2. The Troubleshooting Menu

Install Red Hat Enterprise Linux 7.0 in basic graphics mode

This option allows you to install Red Hat Enterprise Linux in graphical mode even if the installation program is unable to load the correct driver for your video card. If your screen appears distorted or goes blank when using the **Install Red Hat Enterprise Linux 7.0** option, restart your computer and try this option instead.

Rescue a Red Hat Enterprise Linux system

Choose this option to repair a problem with your installed Red Hat Enterprise Linux system that prevents you from booting normally. The rescue environment contains utility programs that allow you fix a wide variety of these problems.

Run a memory test

This option runs a memory test on your system. For more information, see [Section 23.2.1, “Loading the Memory \(RAM\) Testing Mode”](#).

Boot from local drive

This option boots the system from the first installed disk. If you booted this disc accidentally, use this option to boot from the hard disk immediately without starting the installation program.

CHAPTER 8. INSTALLING USING ANACONDA

This chapter provides step-by-step instructions for installing Red Hat Enterprise Linux using the **Anaconda** installer. The bulk of this chapter describes installation using the graphical user interface. A text mode is also available for systems with no graphical display, but this mode is limited in certain aspects (for example, custom partitioning is not possible in text mode).

If your system does not have the ability to use the graphical mode, you can:

- Use Kickstart to automate the installation as described in [Chapter 27, Kickstart Installations](#)
- Perform the graphical installation remotely by connecting to the installation system from another computer with a graphical display using the VNC (Virtual Network Computing) protocol – see [Chapter 25, Using VNC](#)

8.1. INTRODUCTION TO ANACONDA

The Red Hat Enterprise Linux installer, **Anaconda**, is different from most other operating system installation programs due to its parallel nature. Most installers follow a fixed path: you must choose your language first, then you configure network, then installation type, then partitioning, and so on. There is usually only one way to proceed at any given time.

In **Anaconda** you are only required to select your language and locale first, and then you are presented with a central screen, where you can configure most aspects of the installation in any order you like. This does not apply to all parts of the installation process, however – for example, when installing from a network location, you must configure the network before you can select which packages to install.

Some screens will be automatically configured depending on your hardware and the type of media you used to start the installation. You can still change the detected settings in any screen. Screens which have not been automatically configured, and therefore require your attention before you begin the installation, are marked by an exclamation mark. You cannot start the actual installation process before you finish configuring these settings.

Additional differences appear in certain screens; notably the custom partitioning process is very different from other Linux distributions. These differences are described in each screen's subsection.

8.2. CONSOLES AND LOGGING DURING THE INSTALLATION

The following sections describe how to access logs and an interactive shell during the installation. This is useful when troubleshooting problems, but should not be necessary in most cases.

8.2.1. Accessing Consoles

The Red Hat Enterprise Linux installer uses the **tmux** terminal multiplexer to display and control several windows you can use in addition to the main interface. Each of these windows serves a different purpose – they display several different logs, which can be used to troubleshoot any issues during the installation, and one of the windows provides an interactive shell prompt with **root** privileges, unless this prompt was specifically disabled using a boot option or a Kickstart command.



NOTE

In general, there is no reason to leave the default graphical installation environment unless you need to diagnose an installation problem.

The terminal multiplexer is running in virtual console 1. To switch from the graphical installation environment to **tmux**, press **Ctrl+Alt+F1**. To go back to the main installation interface which runs in virtual console 6, press **Ctrl+Alt+F6**.



NOTE

If you choose text mode installation, you will start in virtual console 1 (**tmux**), and switching to console 6 will open a shell prompt instead of a graphical interface.

The console running **tmux** has 5 available windows; their contents are described in the table below, along with keyboard shortcuts used to access them. Note that the keyboard shortcuts are two-part: first press **Ctrl+b**, then release both keys, and press the number key for the window you want to use.

You can also use **Ctrl+b n** and **Ctrl+b p** to switch to the next or previous **tmux** window, respectively.

Table 8.1. Available tmux Windows

Shortcut	Contents
Ctrl+b 1	Main installation program window. Contains text-based prompts (during text mode installation or if you use VNC Direct Mode), and also some debugging information.
Ctrl+b 2	Interactive shell prompt with root privileges.
Ctrl+b 3	Installation log; displays messages stored in /tmp/anaconda.log .
Ctrl+b 4	Storage log; displays messages related storage devices from kernel and system services, stored in /tmp/storage.log .
Ctrl+b 5	Program log; displays messages from other system utilities, stored in /tmp/program.log .

In addition to displaying diagnostic information in **tmux** windows, **Anaconda** also generates several log files, which can be transferred from the installation system. These log files are described in [Table 9.1, “Log Files Generated During the Installation”](#), and directions for transferring them from the installation system are available in [Chapter 9, Troubleshooting Installation on 64-bit AMD, Intel, and ARM Systems](#).

8.2.2. Saving Screenshots

You can press **Shift+Print Screen** at any time during the graphical installation to capture the current screen. These screenshots are saved to **/tmp/anaconda-screenshots/**.

Additionally, you can use the **autostep --autoscreenshot** command in a Kickstart file to capture and save each step of the installation automatically. See [Section 27.3.1, “Kickstart Commands and Options”](#) for details.

8.3. INSTALLING IN TEXT MODE

Text mode installation offers an interactive, non-graphical interface for installing Red Hat Enterprise Linux. This can be useful on systems with no graphical capabilities; however, you should always consider the available alternatives before starting a text-based installation. Text mode is limited in the amount of choices you can make during the installation.



IMPORTANT

Red Hat recommends that you install Red Hat Enterprise Linux using the graphical interface. If you are installing Red Hat Enterprise Linux on a system that lacks a graphical display, consider performing the installation over a VNC connection – see [Chapter 25, Using VNC](#). The text mode installation program will prompt you to confirm the use of text mode if it detects that a VNC-based installation is possible.

If your system has a graphical display, but graphical installation fails, try booting with the **inst.xdriver=vesa** option – see [Chapter 23, Boot Options](#).

Alternatively, consider a Kickstart installation. See [Chapter 27, Kickstart Installations](#) for more information.

```

Installation

1) [!] Timezone settings          2) [x] Language settings
   (Timezone is not set.)        (English (United States))
3) [!] Software selection        4) [!] Installation source
   (Processing...)              (Processing...)
5) [x] Network settings          6) [!] Install Destination
   (Wired (eth0) connected)     (No disks selected)
7) [x] Kdump                     8) [!] Set root password
   (Kdump is enabled)           (Password is not set.)
9) [!] Create user
   (No user will be created)

Please make your choice from above ['q' to quit | 'b' to begin installation |
'r' to refresh]: _

```

Figure 8.1. Text Mode Installation

Installation in text mode follows a pattern similar to the graphical installation: There is no single fixed progression; you can configure many settings in any order you want using the main status screen. Screens which have already been configured, either automatically or by you, are marked as **[x]**, and screens which require your attention before the installation can begin are marked with **[!]**. Available commands are displayed below the list of available options.



NOTE

When related background tasks are being run, certain menu items can be temporarily unavailable or display the **Processing...** label. To refresh to the current status of text menu items, use the **r** option at the text mode prompt.

At the bottom of the screen in text mode, a green bar is displayed showing five menu options. These options represent different screens in the **tmux** terminal multiplexer; by default you start in screen 1, and you can use keyboard shortcuts to switch to other screens which contain logs and an interactive command prompt. For information about available screens and shortcuts to switch to them, see [Section 8.2.1, “Accessing Consoles”](#).

Limits of interactive text mode installation include:

- The installer will always use the English language and the US English keyboard layout. You can configure your language and keyboard settings, but these settings will only apply to the installed system, not to the installation.
- You cannot configure any advanced storage methods (LVM, software RAID, FCoE, zFCP and iSCSI).
- It is not possible to configure custom partitioning; you must use one of the automatic partitioning settings. You also cannot configure where the boot loader will be installed.
- You cannot select any package add-ons to be installed; they must be added after the installation finishes using the **Yum** package manager.

To start a text mode installation, boot the installation with the **inst.text** boot option used either at the boot command line in the boot menu, or in your PXE server configuration. See [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) for information about booting and using boot options.

8.4. INSTALLING IN THE GRAPHICAL USER INTERFACE

The graphical installation interface is the preferred method of manually installing Red Hat Enterprise Linux. It allows you full control over all available settings, including custom partitioning and advanced storage configuration, and it is also localized to many languages other than English, allowing you to perform the entire installation in a different language. The graphical mode is used by default when you boot the system from local media (a CD, DVD or a USB flash drive).

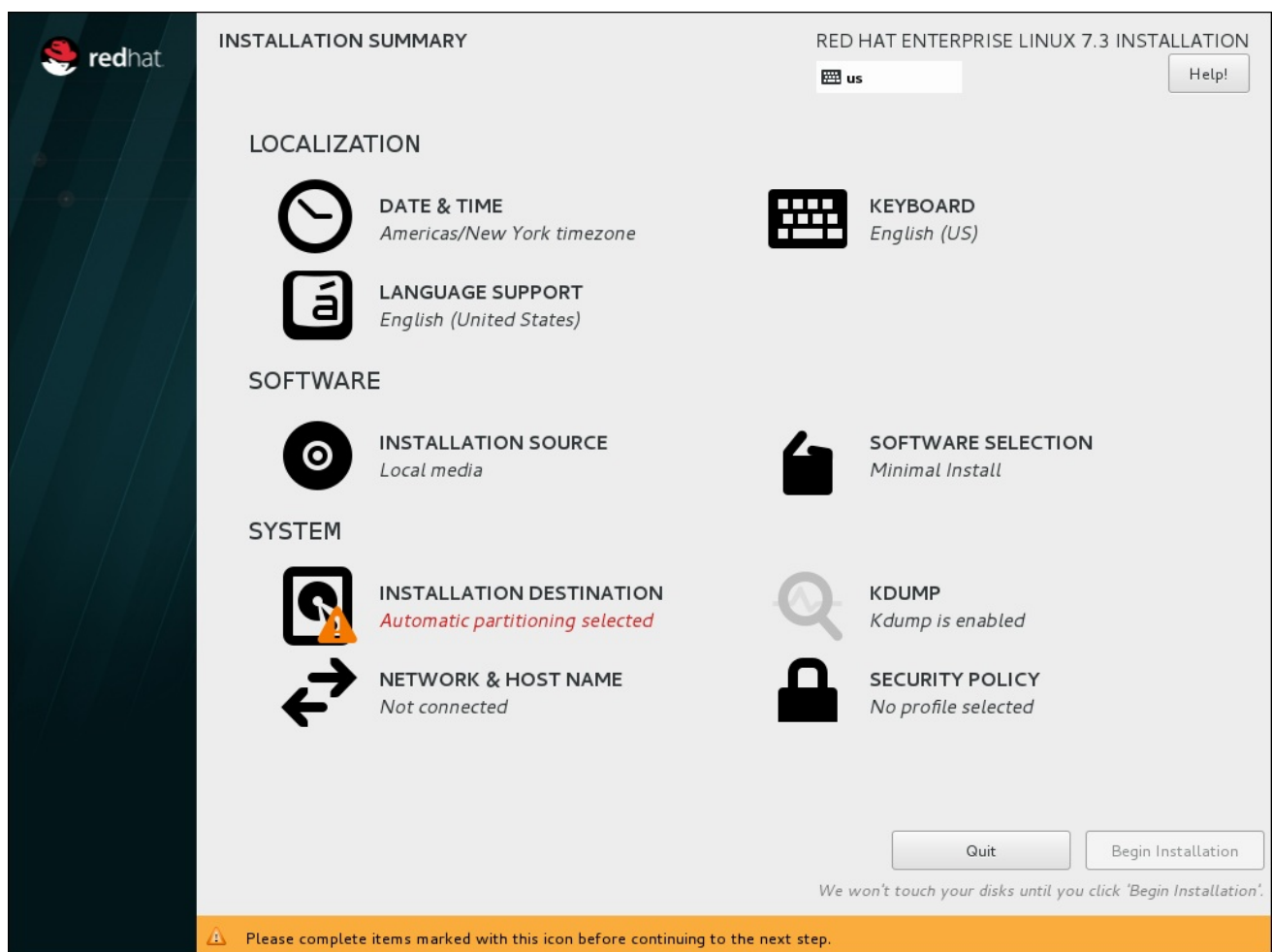


Figure 8.2. The Installation Summary Screen

The sections below discuss each screen available in the installation process. Note that due to the installer's parallel nature, most of the screens do not have to be completed in the order in which they are described here.

Each screen in the graphical interface contains a **Help** button. This button opens the **Yelp** help browser displaying the section of the *Red Hat Enterprise Linux Installation Guide* relevant to the current screen.

You can also control the graphical installer with your keyboard. Following table shows you the shortcuts you can use.

Table 8.2. Graphical installer keyboard shortcuts

Shortcut keys	Usage
Tab and Shift+Tab	Cycle through active control elements (buttons, check boxes, and so on.) on the current screen
Up and Down	Scroll through lists
Left and Right	Scroll through horizontal toolbars and table entries
Space and Enter	Select or remove a highlighted item from selection and expand and collapse drop-down menus

Additionally, elements in each screen can be toggled using their respective shortcuts. These shortcuts are highlighted (underlined> when you hold down the **Alt** key; to toggle that element, press **Alt+X**, where **X** is the highlighted letter.

Your current keyboard layout is displayed in the top right hand corner. Only one layout is configured by default; if you configure more than layout in the **Keyboard Layout** screen ([Section 8.9, "Keyboard Configuration"](#)), you can switch between them by clicking the layout indicator.

8.5. WELCOME SCREEN AND LANGUAGE SELECTION

The first screen of the installation program is the **Welcome to Red Hat Enterprise Linux** screen. Here you select the language that **Anaconda** will use for the rest of the installation. This selection will also become the default for the installed system, unless changed later. In the left panel, select your language of choice, for example **English**. Then you can select a locale specific to your region in the right panel, for example **English (United States)**.



NOTE

One language is pre-selected by default on top of the list. If network access is configured at this point (for example, if you booted from a network server instead of local media), the pre-selected language will be determined based on automatic location detection using the **GeoIP** module.

Alternatively, type your preferred language into the search box as shown below.

Once you have made your selection, click the **Continue** button to proceed to the **Installation Summary** screen.

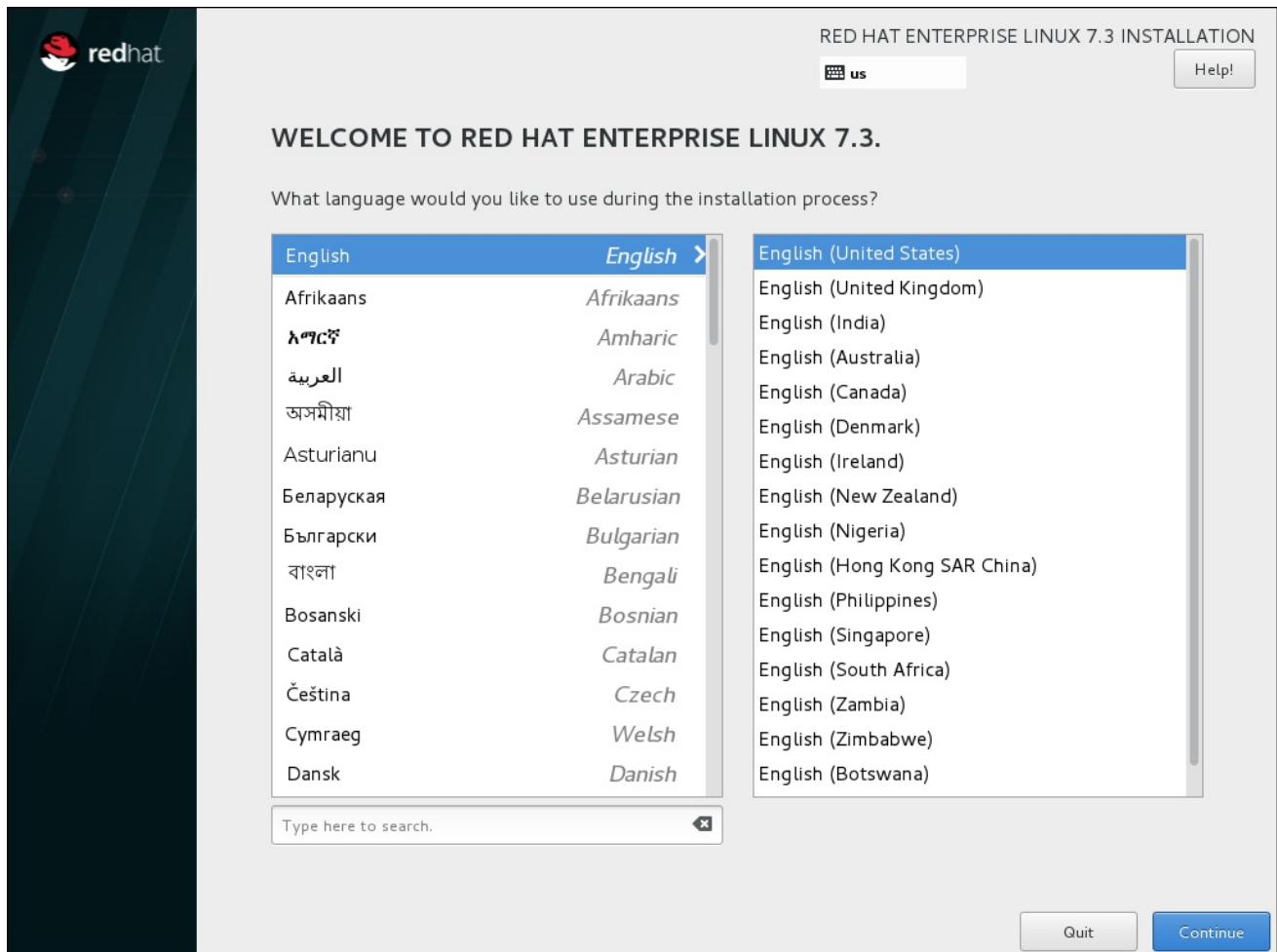


Figure 8.3. Language Configuration

After you click the **Continue** button, the unsupported hardware dialog may appear. This happens if you are using hardware that the kernel does not support.

8.6. THE INSTALLATION SUMMARY SCREEN

The **Installation Summary** screen is the central location for setting up an installation.

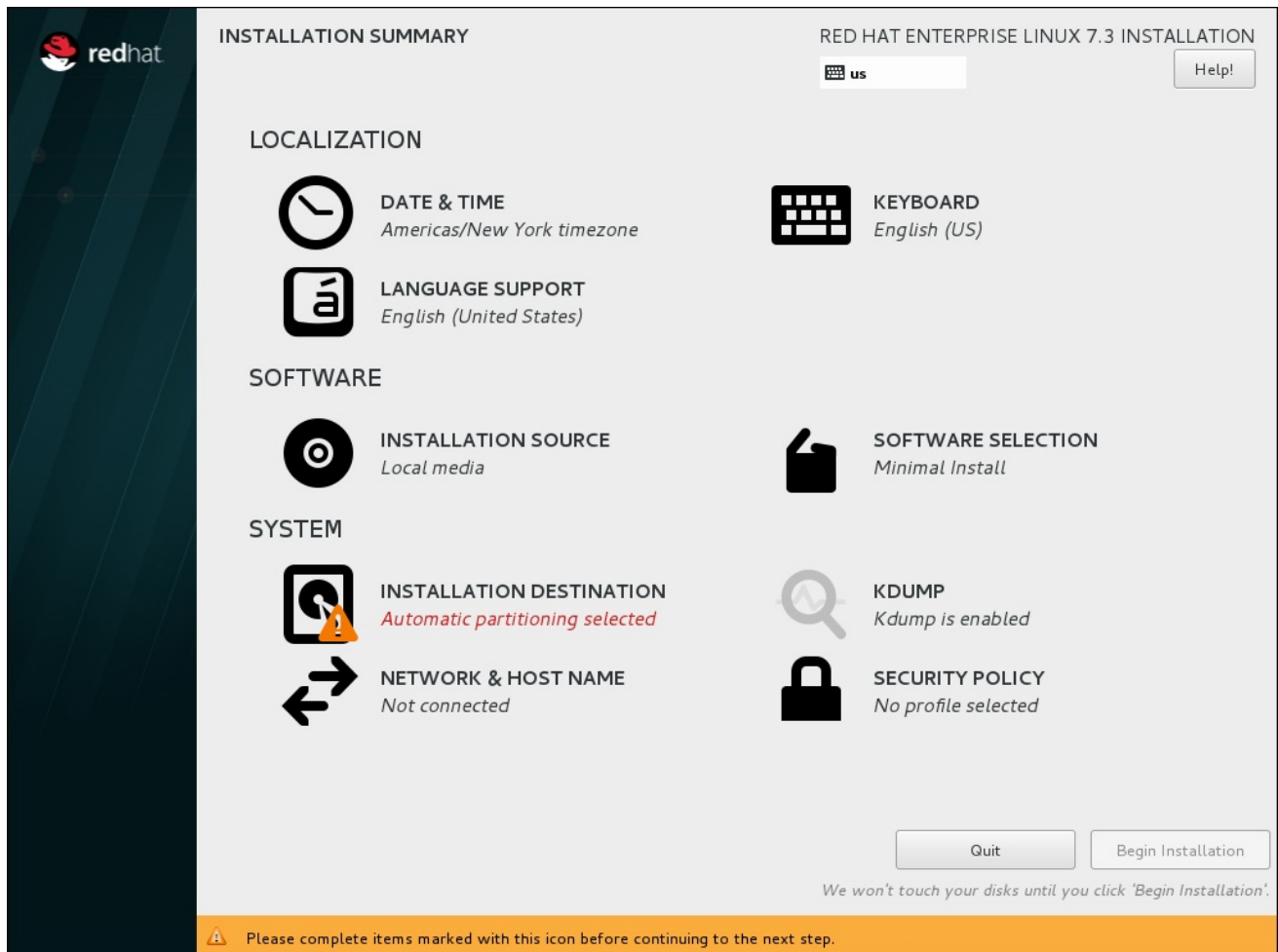


Figure 8.4. The Installation Summary Screen

Instead of directing you through consecutive screens, the Red Hat Enterprise Linux installation program allows you to configure your installation in the order you choose.

Use your mouse to select a menu item to configure a section of the installation. When you have completed configuring a section, or if you would like to complete that section later, click the **Done** button located in the upper left corner of the screen.

Only sections marked with a warning symbol are mandatory. A note at the bottom of the screen warns you that these sections must be completed before the installation can begin. The remaining sections are optional. Beneath each section's title, the current configuration is summarized. Using this you can determine whether you need to visit the section to configure it further.

Once all required sections are complete, click the **Begin Installation** button. Also see [Section 8.17, "Begin Installation"](#).

To cancel the installation, click the **Quit** button.



NOTE

When related background tasks are running, certain menu items might be temporarily unavailable.

If you used a Kickstart option or a boot command-line option to specify an installation repository on a network, but no network is available at the start of the installation, the installation program will display the configuration screen for you to set up a network connection prior to displaying the **Installation**

Summary screen.

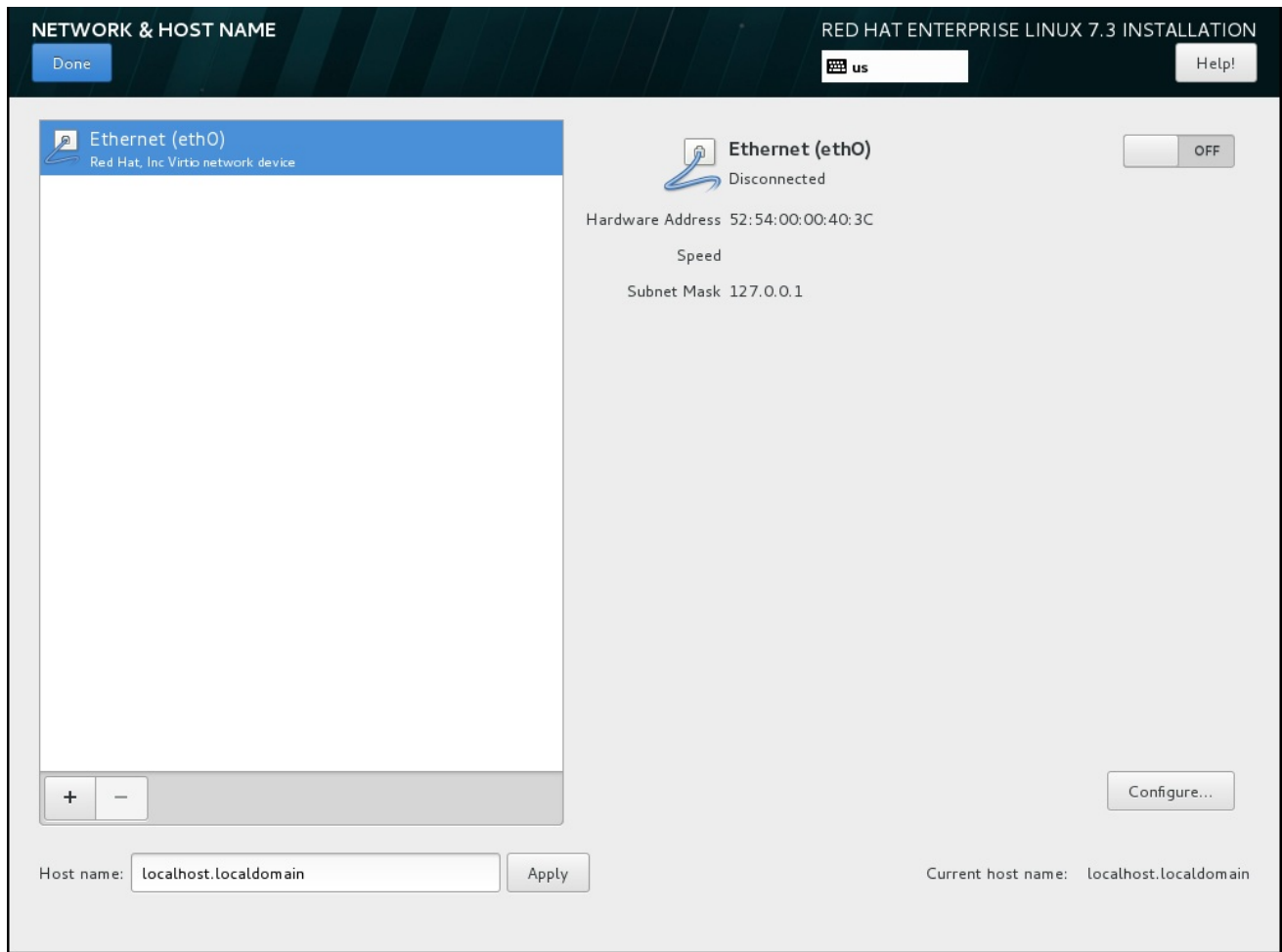


Figure 8.5. Network Configuration Screen When No Network Is Detected

You can skip this step if you are installing from an installation DVD or other locally accessible media, and you are certain you will not need network to finish the installation. However, network connectivity is necessary for network installations (see [Section 8.11, “Installation Source”](#)) or for setting up advanced storage devices (see [Section 8.15, “Storage Devices”](#)). For more details about configuring a network in the installation program, see [Section 8.12, “Network & Hostname”](#).

8.7. DATE & TIME

To configure time zone, date, and optionally settings for network time, select **Date & Time** at the **Installation Summary** screen.

There are three ways for you to select a time zone:

- Using your mouse, click on the interactive map to select a specific city. A red pin appears indicating your selection.
- You can also scroll through the **Region** and **City** drop-down menus at the top of the screen to select your time zone.
- Select **Etc** at the bottom of the **Region** drop-down menu, then select your time zone in the next menu adjusted to GMT/UTC, for example **GMT+1**.

If your city is not available on the map or in the drop-down menu, select the nearest major city in the same time zone. Alternatively you can use a Kickstart file, which will allow you to specify some additional

time zones which are not available in the graphical interface. See the **timezone** command in [timezone \(required\)](#) for details.



NOTE

The list of available cities and regions comes from the Time Zone Database (tzdata) public domain, which is maintained by the Internet Assigned Numbers Authority (IANA). Red Hat cannot add cities or regions into this database. You can find more information at the official website, available at <http://www.iana.org/time-zones>.

Specify a time zone even if you plan to use NTP (Network Time Protocol) to maintain the accuracy of the system clock.

If you are connected to the network, the **Network Time** switch will be enabled. To set the date and time using NTP, leave the **Network Time** switch in the **ON** position and click the configuration icon to select which NTP servers Red Hat Enterprise Linux should use. To set the date and time manually, move the switch to the **OFF** position. The system clock should use your time zone selection to display the correct date and time at the bottom of the screen. If they are still incorrect, adjust them manually.

Note that NTP servers might be unavailable at the time of installation. In such a case, enabling them will not set the time automatically. When the servers become available, the date and time will update.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your time zone configuration after you have completed the installation, visit the **Date & Time** section of the **Settings** dialog window.

8.8. LANGUAGE SUPPORT

To install support for additional locales and language dialects, select **Language Support** from the **Installation Summary** screen.

Use your mouse to select the language for which you would like to install support. In the left panel, select your language of choice, for example **Español**. Then you can select a locale specific to your region in the right panel, for example **Español (Costa Rica)**. You can select multiple languages and multiple locales. The selected languages are highlighted in bold in the left panel.

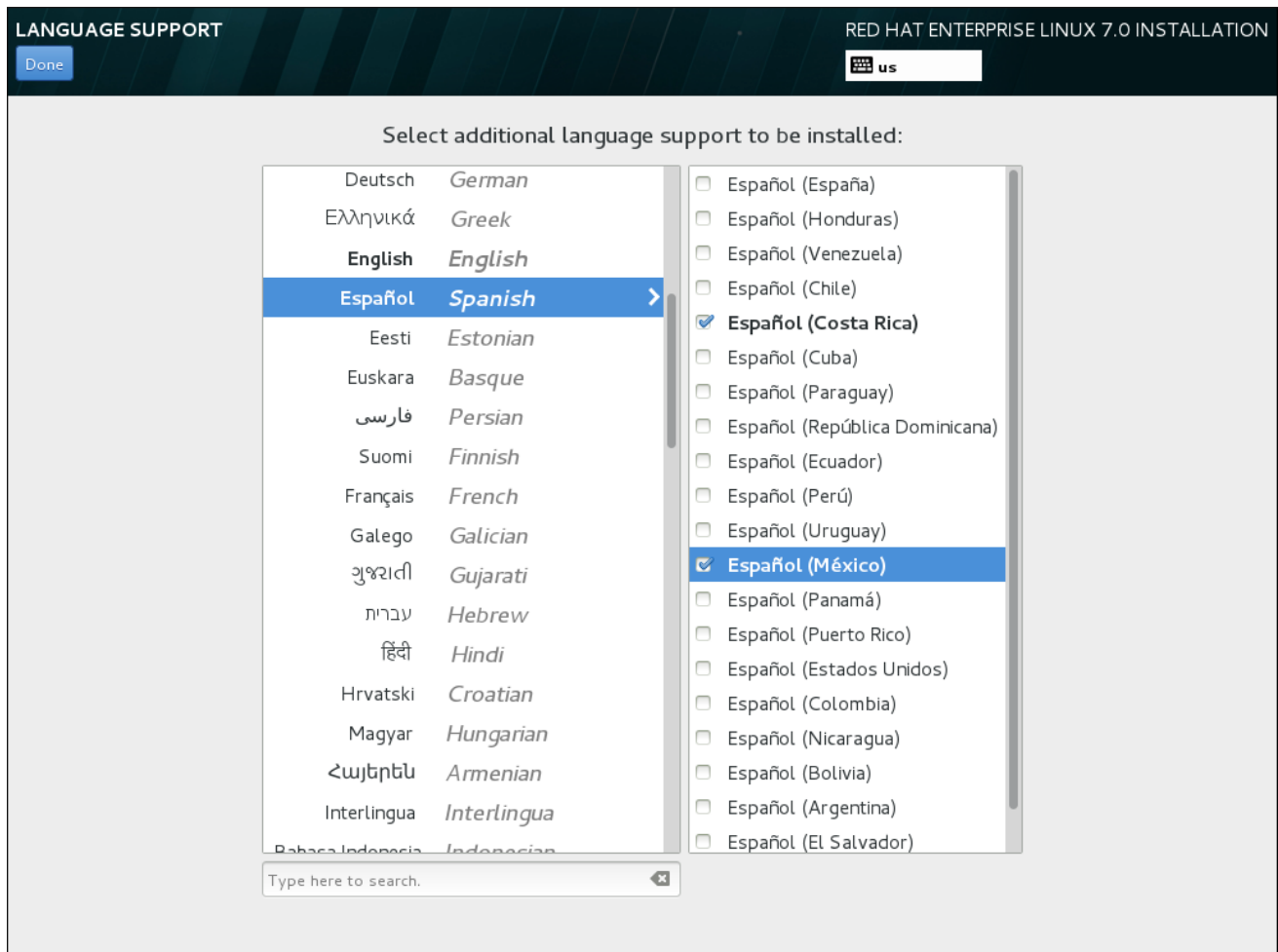


Figure 8.6. Configuring Language Support

Once you have made your selections, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your language support configuration after you have completed the installation, visit the **Region & Language** section of the **Settings** dialog window.

8.9. KEYBOARD CONFIGURATION

To add multiple keyboard layouts to your system, select **Keyboard** from the **Installation Summary** screen. Upon saving, the keyboard layouts are immediately available in the installation program and you can switch between them by using the keyboard icon located at all times in the upper right corner of the screen.

Initially, only the language you selected in the welcome screen is listed as the keyboard layout in the left pane. You can either replace the initial layout or add more layouts. However, if your language does not use ASCII characters, you might need to add a keyboard layout that does, to be able to properly set a password for an encrypted disk partition or the root user, among other things.

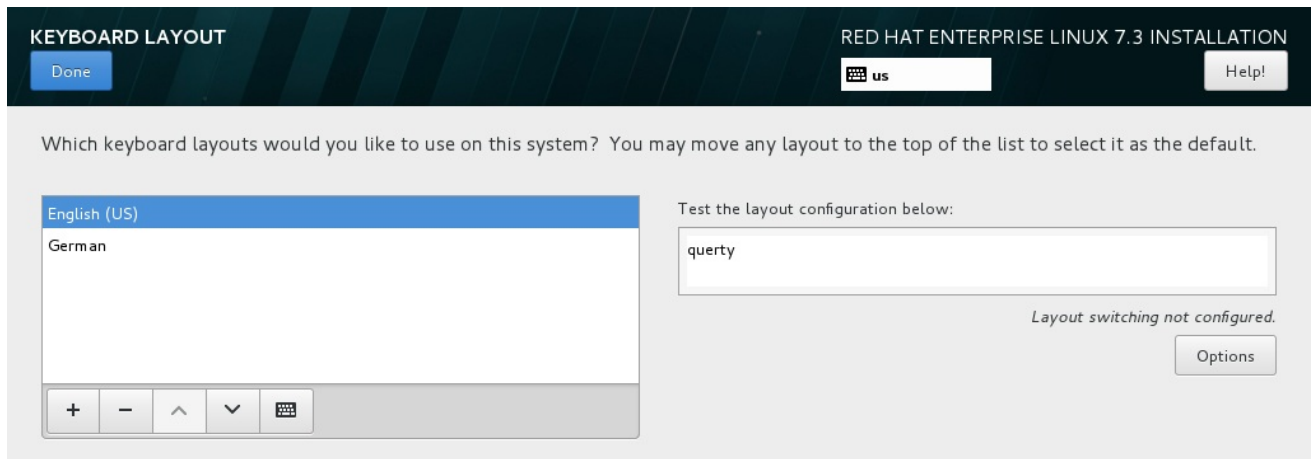
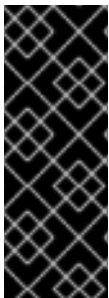


Figure 8.7. Keyboard Configuration

To add an additional layout, click the **+** button, select it from the list, and click **Add**. To delete a layout, select it and click the **-** button. Use the arrow buttons to arrange the layouts in order of preference. For a visual preview of the keyboard layout, select it and click the keyboard button.

To test a layout, use the mouse to click inside the text box on the right. Type some text to confirm that your selection functions correctly.

To test additional layouts, you can click the language selector at the top on the screen to switch them. However, it is recommended to set up a keyboard combination for switching layout. Click the **Options** button at the right to open the **Layout Switching Options** dialog and choose a combination from the list by selecting its check box. The combination will then be displayed above the **Options** button. This combination applies both during the installation and on the installed system, so you must configure a combination here in order to use one after installation. You can also select more than one combination to switch between layouts.



IMPORTANT

If you use a layout that cannot accept Latin characters, such as **Russian**, Red Hat recommends additionally adding the **English (United States)** layout and configuring a keyboard combination to switch between the two layouts. If you only select a layout without Latin characters, you might be unable to enter a valid root password and user credentials later in the installation process. This can prevent you from completing the installation.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your keyboard configuration after you have completed the installation, visit the **Keyboard** section of the **Settings** dialogue window.

8.10. SECURITY POLICY

The **Security Policy** spoke allows you to configure the installed system following restrictions and recommendations (*compliance policies*) defined by the Security Content Automation Protocol (SCAP) standard. This functionality is provided by an add-on which has been enabled by default since Red Hat

Enterprise Linux 7.2. When enabled, the packages necessary to provide this functionality will automatically be installed. However, by default, no policies are enforced, meaning that no checks are performed during or after installation unless specifically configured.

See the information about scanning the system for configuration compliance and vulnerabilities in the [Red Hat Enterprise Linux 7 Security Guide](#), which includes background information, practical examples, and additional resources.



IMPORTANT

Applying a security policy is not necessary on all systems. This screen should only be used when a specific policy is mandated by your organization rules or government regulations.

If you apply a security policy to the system, it will be installed using restrictions and recommendations defined in the selected profile. The `openscap-scanner` package will also be added to your package selection, providing a preinstalled tool for compliance and vulnerability scanning. After the installation finishes, the system will be automatically scanned to verify compliance. The results of this scan will be saved to the `/root/openscap_data` directory on the installed system.

Pre-defined policies which are available in this screen are provided by **SCAP Security Guide**. See the [OpenSCAP Portal](#) for links to detailed information about each available profile.

You can also load additional profiles from an HTTP, HTTPS or FTP server.

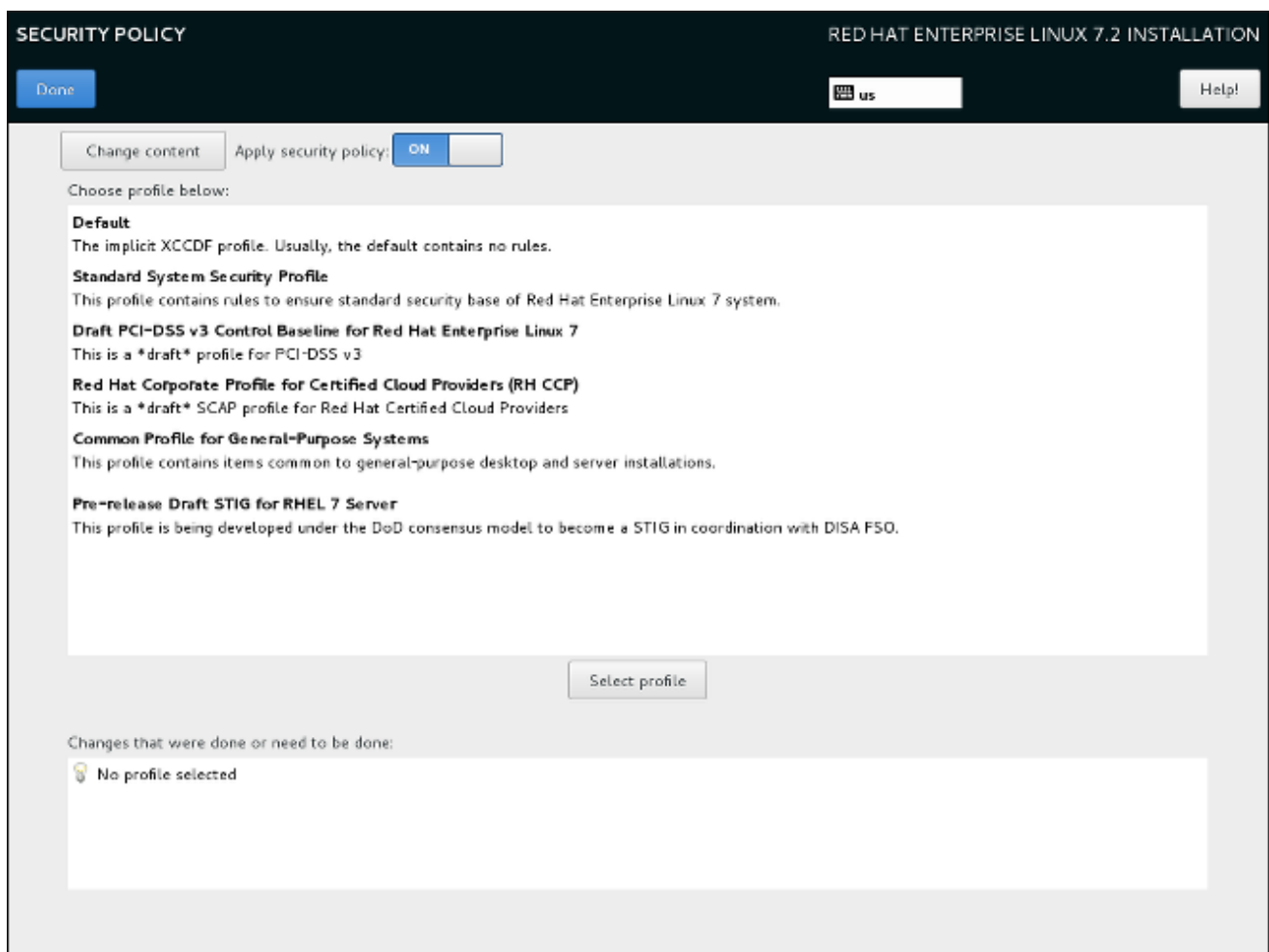


Figure 8.8. Security policy selection screen

To configure the use of security policies on the system, first enable configuration by setting the **Apply security policy** switch to **ON**. If the switch is in the **OFF** position, controls in the rest of this screen have no effect.

After enabling security policy configuration using the switch, select one of the profiles listed in the top window of the screen, and click the **Select profile** below. When a profile is selected, a green check mark will appear on the right side, and the bottom field will display whether any changes will be made before beginning the installation.



NOTE

None of the profiles available by default perform any changes before the installation begins. However, loading a custom profile as described below can require some pre-installation actions.

To use a custom profile, click the **Change content** button in the top left corner. This will open another screen where you can enter an URL of a valid security content. To go back to the default security content selection screen, click **Use SCAP Security Guide** in the top left corner.

Custom profiles can be loaded from an **HTTP**, **HTTPS** or **FTP** server. Use the full address of the content, including the protocol (such as **http://**). A network connection must be active (enabled in [Section 8.12, "Network & Hostname"](#)) before you can load a custom profile. The content type will be detected automatically by the installer.

After you select a profile, or if you want to leave the screen, click **Done** in the top left corner to return to [Section 8.6, "The Installation Summary Screen"](#).

8.11. INSTALLATION SOURCE

To specify a file or a location to install Red Hat Enterprise Linux from, select **Installation Source** from the **Installation Summary** screen. On this screen, you can choose between locally available installation media, such as a DVD or an ISO file, or a network location.

Figure 8.9. Installation Source Screen

Select one of the following options:

Auto-detected installation media

If you started the installation using the full installation DVD or USB drive, the installation program will detect it and display basic information under this option. Click the **Verify** button to ensure that the media is suitable for installation. This integrity test is the same as the one performed if you selected **Test this media & Install Red Hat Enterprise Linux** in the boot menu, or if you used the **rd.live.check** boot option.

ISO file

This option will appear if the installation program detected a partitioned hard drive with mountable file systems. Select this option, click the **Choose an ISO** button, and browse to the installation ISO file's location on your system. Then click **Verify** to ensure that the file is suitable for installation.

On the network

To specify a network location, select this option and choose from the following options in the drop-down menu:

- **http://**
- **https://**
- **ftp://**
- **nfs**

Using your selection as the start of the location URL, type the rest into the address box. If you choose NFS, another box will appear for you to specify any NFS mount options.



IMPORTANT

When selecting an NFS-based installation source, you must specify the address with a colon (:) character separating the host name from the path. For example:

server.example.com:/path/to/directory

To configure a proxy for an HTTP or HTTPS source, click the **Proxy setup** button. Check **Enable HTTP proxy** and type the URL into the **Proxy URL** box. If your proxy requires authentication, check **Use Authentication** and enter a user name and password. Click **Add**.

If your HTTP or HTTPS URL refers to a repository mirror list, mark the check box under the input field.

You can also specify additional repositories to gain access to more installation environments and software add-ons. See [Section 8.13, "Software Selection"](#) for more information.

To add a repository, click the **+** button. To delete a repository, click the **-** button. Click the arrow icon to revert to the previous list of repositories, that is, to replace current entries with those that were present at the time you entered the **Installation Source** screen. To activate or deactivate a repository, click the check box in the **Enabled** column at each entry in the list.

In the right part of the form, you can name your additional repository and configure it the same way as the primary repository on the network.

Once you have selected your installation source, click **Done** to return to the **Installation Summary** screen.

8.12. NETWORK & HOSTNAME

To configure essential networking features for your system, select **Network & Hostname** at the **Installation Summary** screen.

IMPORTANT

When the installation finishes and the system boots for the first time, any network interfaces which you configured during the installation will be activated. However, the installation does not prompt you to configure network interfaces on some common installation paths – for example, when you install Red Hat Enterprise Linux from a DVD to a local hard drive.

When you install Red Hat Enterprise Linux from a local installation source to a local storage device, be sure to configure at least one network interface manually if you require network access when the system boots for the first time. You will also need to set the connection to connect automatically after boot when editing the configuration.

Locally accessible interfaces are automatically detected by the installation program and cannot be manually added or deleted. The detected interfaces are listed in the left pane. Click an interface in the list to display more details about it on the right. To activate or deactivate a network interface, move the switch in the top right corner of the screen to either **ON** or **OFF**.

NOTE

There are several types of network device naming standards used to identify network devices with persistent names such as **em1** or **wl3sp0**. For information about these standards, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

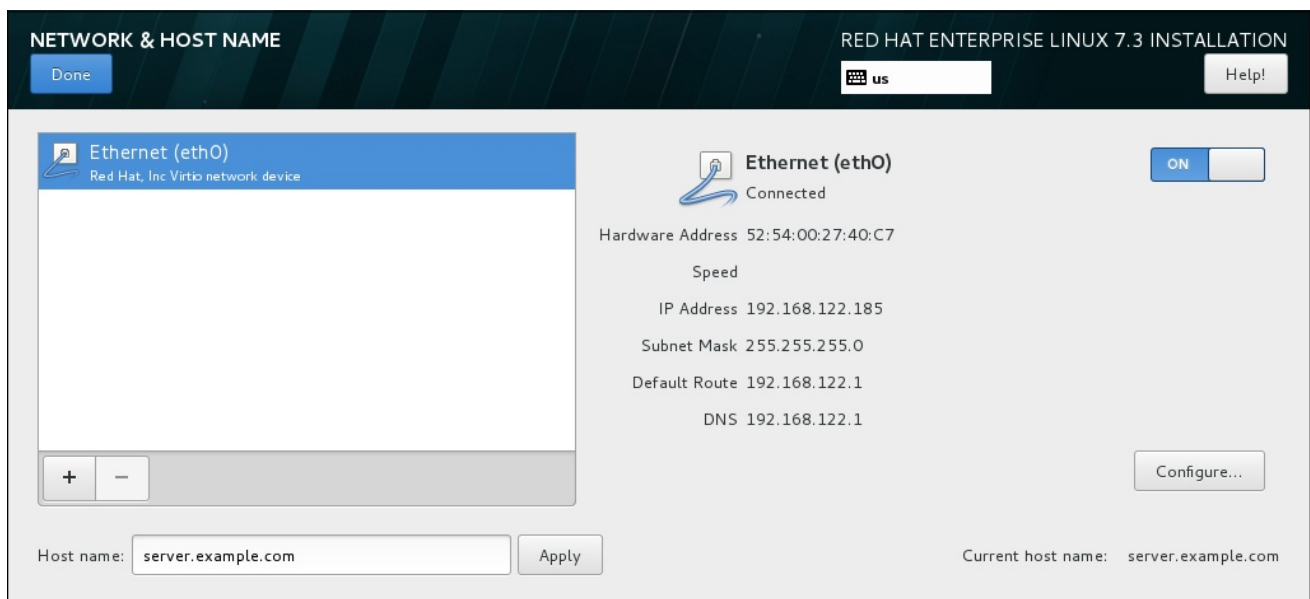


Figure 8.10. Network & Hostname Configuration Screen

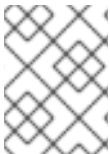
Below the list of connections, enter a host name for this computer in the **Hostname** input field. The host name can be either a *fully-qualified domain name* (FQDN) in the format *hostname.domainname* or a

short host name in the format *hostname*. Many networks have a *Dynamic Host Configuration Protocol* (DHCP) service that automatically supplies connected systems with a domain name. To allow the DHCP service to assign the domain name to this machine, only specify the short host name. The value **localhost.localdomain** means that no specific static host name for target system is configured, and the actual host name of installed system will be configured during process of network configuration (for example, by NetworkManager using DHCP or DNS).



IMPORTANT

If you want to manually assign the host name, make sure you do not use a domain name that is not delegated to you, as this can result in network resources becoming unavailable. For more information, see the recommended naming practices in the [Red Hat Enterprise Linux 7 Networking Guide](#).



NOTE

You can use the **Network** section of the system **Settings** dialog to change your network configuration after you have completed the installation.

Once you have finished network configuration, click **Done** to return to the **Installation Summary** screen.

8.12.1. Edit Network Connections

This section only details the most important settings for a typical wired connection used during installation. Many of the available options do not have to be changed in most installation scenarios and are not carried over to the installed system. Configuration of other types of network is broadly similar, although the specific configuration parameters are necessarily different. To learn more about network configuration after installation, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

To configure a network connection manually, click the **Configure** button in the lower right corner of the screen. A dialog appears that allows you to configure the selected connection. The configuration options presented depends on whether the connection is wired, wireless, mobile broadband, VPN, or DSL. If required, see the *Networking Guide* for more detailed information on network settings.

The most useful network configuration options to consider during installation are:

- Mark the **Automatically connect to this network when it is available** check box if you want to use the connection every time the system boots. You can use more than one connection that will connect automatically. This setting will carry over to the installed system.

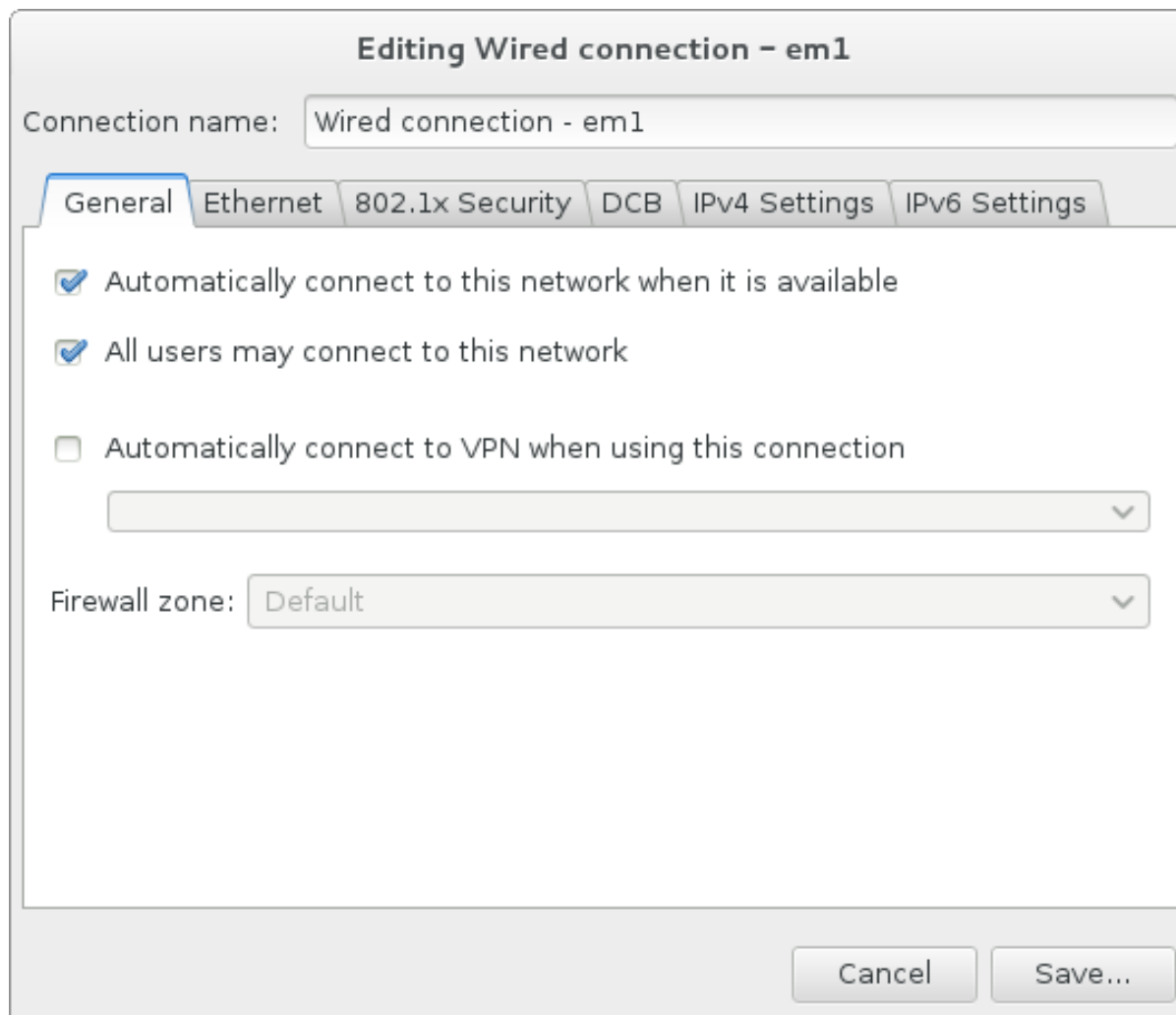


Figure 8.11. Network Auto-Connection Feature

- By default, IPv4 parameters are configured automatically by the DHCP service on the network. At the same time, the IPv6 configuration is set to the **Automatic** method. This combination is suitable for most installation scenarios and usually does not require any changes.

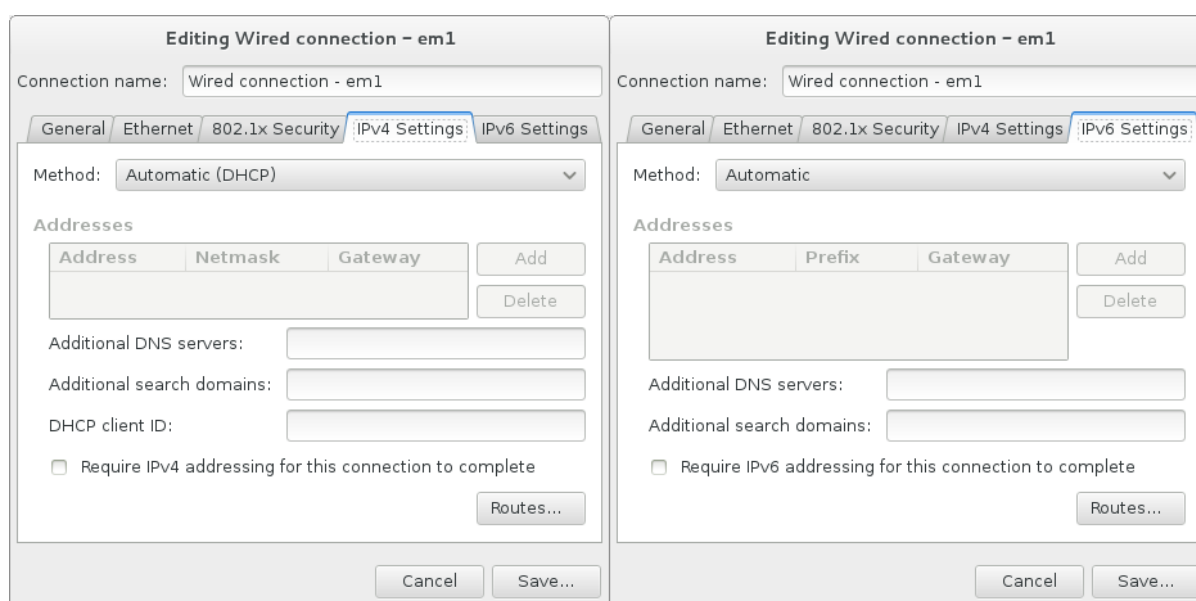


Figure 8.12. IP Protocol Settings

When you have finished editing network settings, click **Save** to save the new configuration. If you

reconfigured a device that was already active during installation, you must restart the device in order to use the new configuration in the installation environment. Use the **ON/OFF** switch on the **Network & Host Name** screen to restart the device.

8.12.2. Advanced Network Interfaces

Advanced network interfaces are also available for installation. This includes virtual local area networks (VLANs) and three methods to use aggregated links. Detailed description of these interfaces is beyond the scope of this document; read the [Red Hat Enterprise Linux 7 Networking Guide](#) for more information.

To create an advanced network interface, click the **+** button in the lower left corner of the **Network & Hostname** screen.

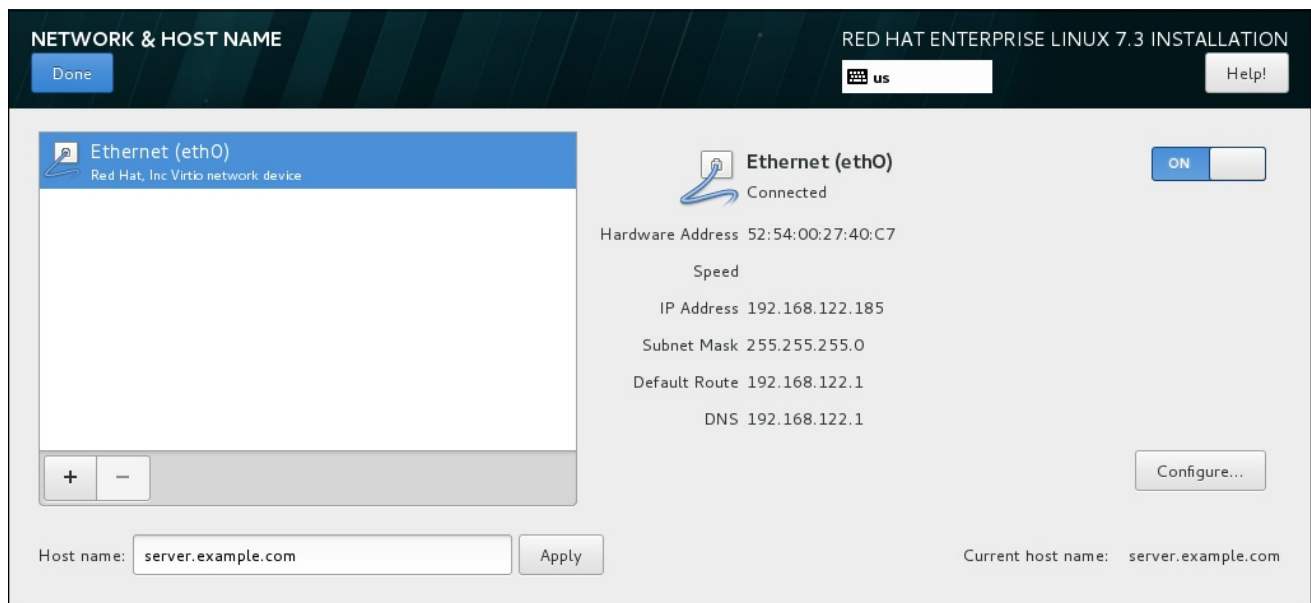


Figure 8.13. Network & Hostname Configuration Screen

A dialog appears with a drop-down menu with the following options:

- **Bond** - represents NIC (*Network Interface Controller*) Bonding, a method to bind multiple network interfaces together into a single, bonded, channel.
- **Bridge** - represents NIC Bridging, a method to connect multiple separate network into one aggregate network.
- **Team** - represents NIC Teaming, a new implementation to aggregate links, designed to provide a small kernel driver to implement the fast handling of packet flows, and various applications to do everything else in user space.
- **VLAN** - represents a method to create multiple distinct broadcast domains, which are mutually isolated.

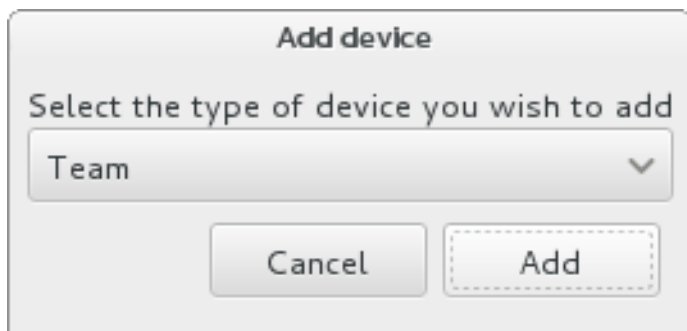


Figure 8.14. Advanced Network Interface Dialog



NOTE

Note that locally accessible interfaces, wired or wireless, are automatically detected by the installation program and cannot be manually added or deleted by using these controls.

Once you have selected an option and clicked the **Add** button, another dialog appears for you to configure the new interface. See the respective chapters in the [Red Hat Enterprise Linux 7 Networking Guide](#) for detailed instructions. To edit configuration on an existing advanced interface, click the **Configure** button in the lower right corner of the screen. You can also remove a manually-added interface by clicking the - button.

8.13. SOFTWARE SELECTION

To specify which packages will be installed, select **Software Selection** at the **Installation Summary** screen. The package groups are organized into *Base Environments*. These environments are pre-defined sets of packages with a specific purpose; for example, the **Virtualization Host** environment contains a set of software packages needed for running virtual machines on the system. Only one software environment can be selected at installation time.

For each environment, there are additional packages available in the form of *Add-ons*. Add-ons are presented in the right part of the screen and the list of them is refreshed when a new environment is selected. You can select multiple add-ons for your installation environment.

A horizontal line separates the list of add-ons into two areas:

- Add-ons listed *above* the horizontal line are specific to the environment you selected. If you select any add-ons in this part of the list and then select a different environment, your selection will be lost.
- Add-ons listed *below* the horizontal line are available for all environments. Selecting a different environment will not impact the selections made in this part of the list.

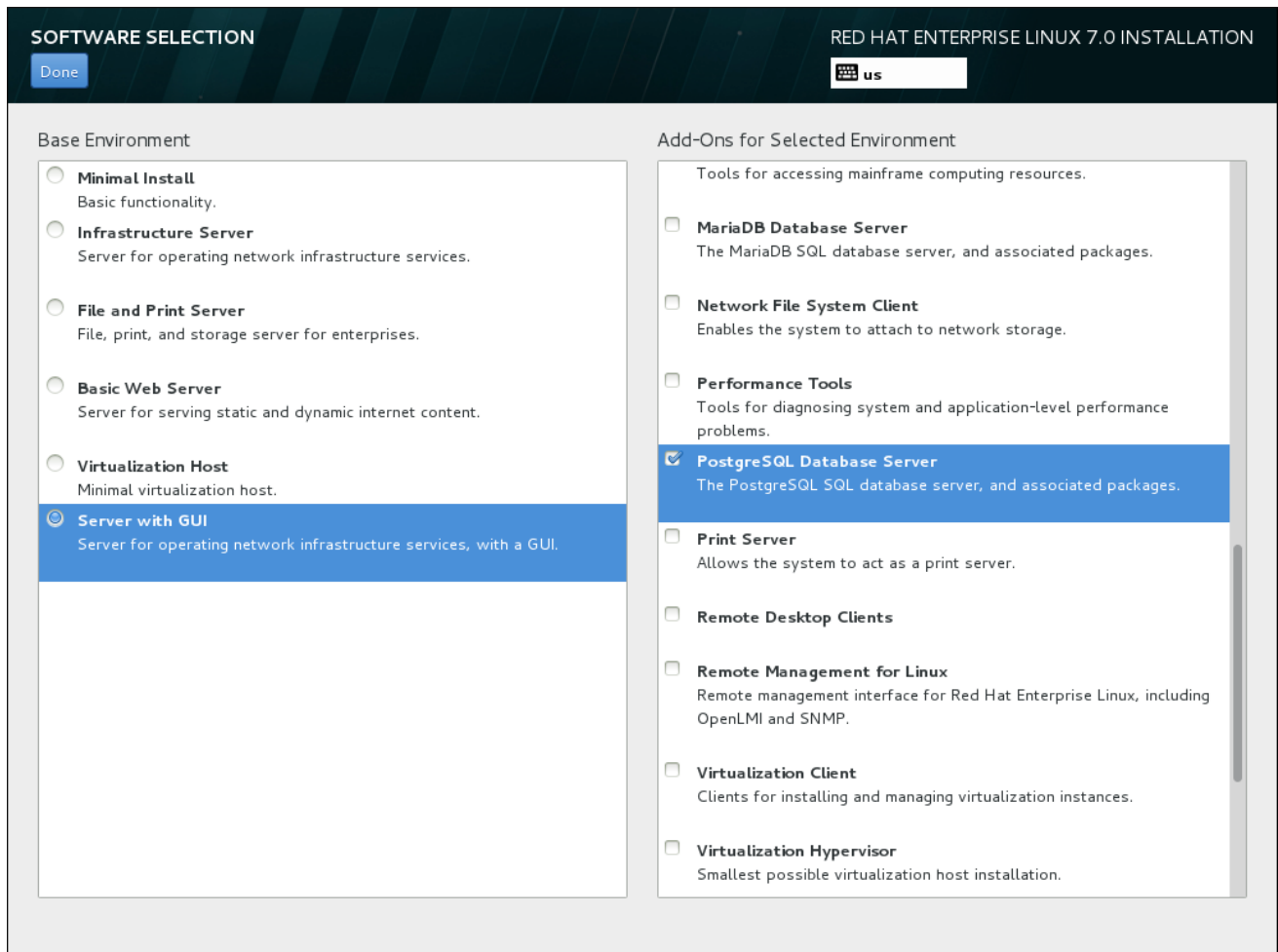


Figure 8.15. Example of a Software Selection for a Server Installation

The availability of base environments and add-ons depends on the variant of the installation ISO image which you are using as the installation source. For example, the **server** variant provides environments designed for servers, while the **workstation** variant has several choices for deployment as a developer workstation, and so on.

The installation program does not show which packages are contained in the available environments. To see which packages are contained in a specific environment or add-on, see the **repodata/*-comps-variant.architecture.xml** file on the Red Hat Enterprise Linux Installation DVD which you are using as the installation source. This file contains a structure describing available environments (marked by the **<environment>** tag) and add-ons (the **<group>** tag).



IMPORTANT

The pre-defined environments and add-ons allow you to customize your system, but in a manual installation, there is no way to select individual packages to install. If you are not sure what package should be installed, Red Hat recommends you to select the **Minimal Install** environment. **Minimal install** only installs a basic version of Red Hat Enterprise Linux with only a minimal amount of additional software. This will substantially reduce the chance of the system being affected by a vulnerability. After the system finishes installing and you log in for the first time, you can use the **Yum** package manager to install any additional software you need. For more details on **Minimal install**, see the [Installing the Minimum Amount of Packages Required](#) section of the Red Hat Enterprise Linux 7 Security Guide.

Alternatively, automating the installation with a Kickstart file allows for a much higher degree of control

over installed packages. You can specify environments, groups and individual packages in the **%packages** section of the Kickstart file. See [Section 27.3.2, “Package Selection”](#) for instructions on selecting packages to install in a Kickstart file, and [Chapter 27, Kickstart Installations](#) for general information about automating the installation with Kickstart.

Once you have selected an environment and add-ons to be installed, click **Done** to return to the **Installation Summary** screen.

8.13.1. Core Network Services

All Red Hat Enterprise Linux installations include the following network services:

- centralized logging through the **rsyslog** service
- email through SMTP (Simple Mail Transfer Protocol)
- network file sharing through NFS (Network File System)
- remote access through SSH (Secure SHell)
- resource advertising through mDNS (multicast DNS)

Some automated processes on your Red Hat Enterprise Linux system use the email service to send reports and messages to the system administrator. By default, the email, logging, and printing services do not accept connections from other systems.

You can configure your Red Hat Enterprise Linux system after installation to offer email, file sharing, logging, printing, and remote desktop access services. The SSH service is enabled by default. You can also use NFS to access files on other systems without enabling the NFS sharing service.

8.14. INSTALLATION DESTINATION

To select the disks and partition the storage space on which you will install Red Hat Enterprise Linux, select **Installation Destination** in the **Installation Summary** screen. If you are unfamiliar with disk partitions, see [Appendix A, An Introduction to Disk Partitions](#) for more information.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.



IMPORTANT

If you install Red Hat Enterprise Linux in text mode, you can only use the default partitioning schemes described in this section. You cannot add or remove partitions or file systems beyond those that the installation program automatically adds or removes.

IMPORTANT

Special cases

- If you have a RAID card, be aware that some BIOS types do not support booting from the RAID card. In such a case, the **/boot** partition must be created on a partition outside of the RAID array, such as on a separate hard drive. It is necessary to use an internal hard drive for partition creation with problematic RAID cards. A **/boot** partition is also necessary for software RAID setups. If you have chosen to automatically partition your system, you should manually edit your **/boot** partition; see [Section 8.14.4, “Manual Partitioning”](#) for more details.
- To configure the Red Hat Enterprise Linux boot loader to *chain load* from a different boot loader, you must specify the boot drive manually by clicking the **Full disk summary and bootloader** link from the **Installation Destination** screen. See [Section 8.14.1, “Boot Loader Installation”](#) for instructions on specifying a boot drive.
- When you install Red Hat Enterprise Linux on a system with both multipath and non-multipath storage devices, the automatic partitioning layout in the installation program might create volume groups that contain a mix of multipath and non-multipath devices. This defeats the purpose of multipath storage. We advise that you select only multipath or only non-multipath devices on the **Installation Destination** screen. Alternatively, proceed to manual partitioning.

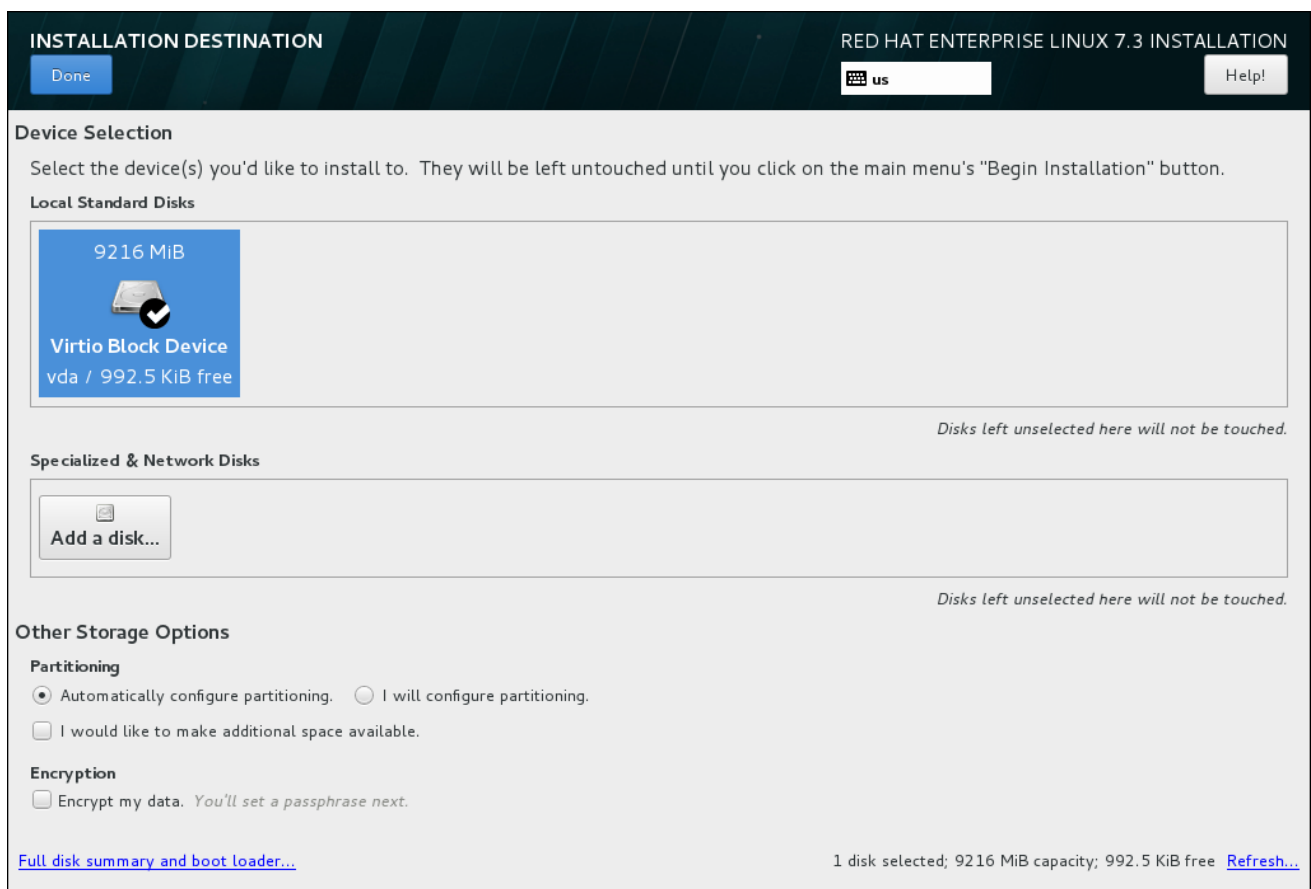


Figure 8.16. Storage Space Overview

On this screen, you can see storage devices available locally on your computer. You can also add additional specialized or network devices by clicking the **Add a disk** button. To learn more about these devices see [Section 8.15, “Storage Devices”](#).

Choose the disks to install Red Hat Enterprise Linux on by clicking their icons in the pane at the top of the screen. Each disk is marked with its label, size, and available space. Disks left unselected on the screen will not be touched once the installation begins.

Below the panes for storage devices is a form of additional controls labeled **Other Storage Options**:

- In the **Partitioning** section, you can select how storage devices are partitioned and how volumes are created. You can configure the partitions, manually or allow the installation program to do it automatically.

Automatic partitioning is recommended if you are doing a clean installation on previously unused storage or do not need to keep any data that might be present on the storage. To proceed this way, leave the default selection of the **Automatically configure partitioning** radio button, and the installation program will create the necessary partitions and volumes on the storage space for you.

For automatic partitioning, you can also select the **I would like to make additional space available** check box to choose how to reassign space from other file systems to this installation. After you click **Done**, there are two dialogs that can appear. If you selected automatic partitioning but there is not enough storage space to complete the installation using the recommended partitioning configuration, a dialog will appear:

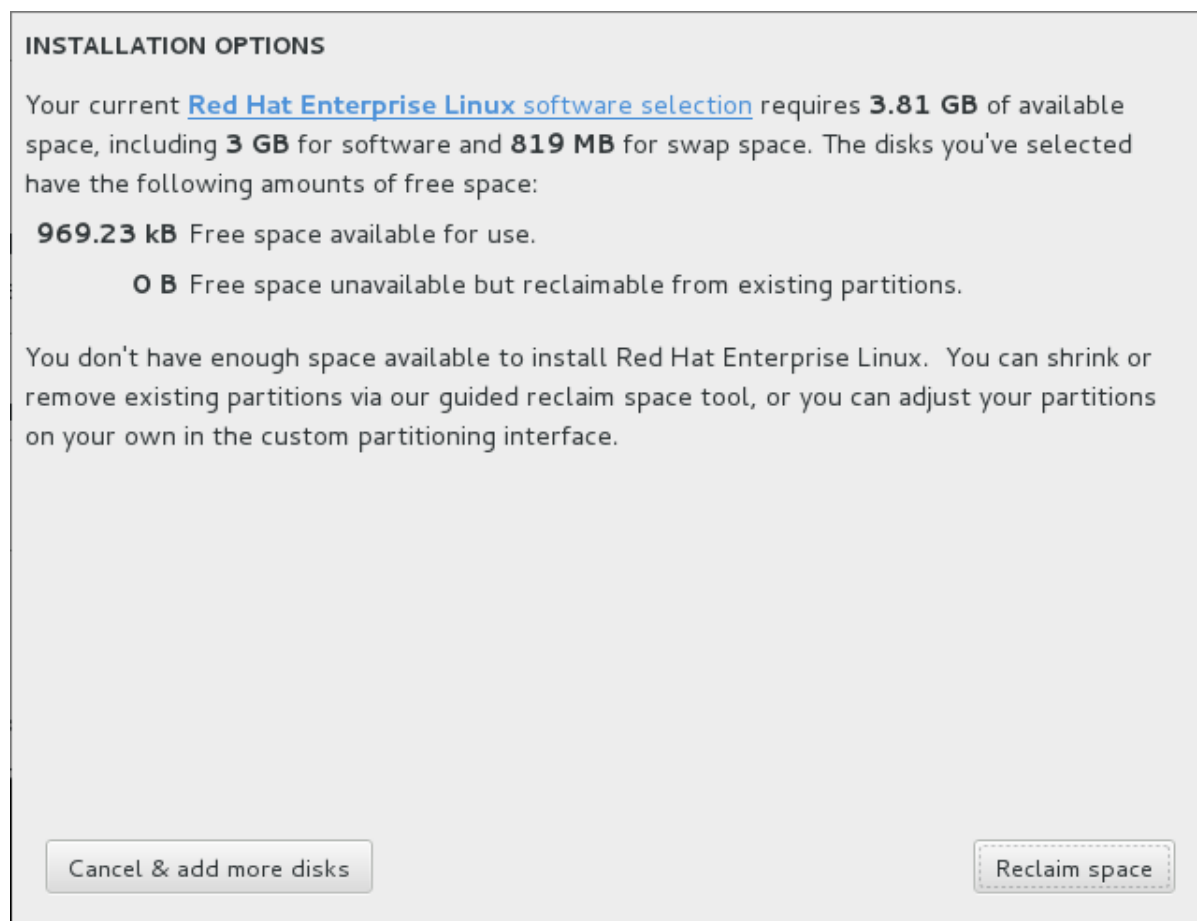


Figure 8.17. Installation Options Dialog with Option to Reclaim Space

You can click to the **Red Hat Enterprise Linux software selection** link. The link will navigate you to the **Software selection** section, where you can change what software you want to install, and free some additional storage space.

Alternatively, you can click **Cancel & add more disks** to return to the **Installation Destination** screen, where it is possible to add more storage devices, or to choose to configure partitioning

manually. Click **Reclaim space** to free some storage space from existing file systems. See [Section 8.14.3, “Reclaim Disk Space”](#) for details.

The second dialog appears if you cannot create enough free space. In this case, you have to add more disks on the initial storage screen or quit the installation.

If you select the **I will configure partitioning** radio button for manual setup, you will be brought to the **Manual Partitioning** screen after clicking **Done**. See [Section 8.14.4, “Manual Partitioning”](#) for details.

- In the **Encryption** section, you can select the **Encrypt my data** check box to encrypt all partitions except for the **/boot** partition. See the [Red Hat Enterprise Linux 7 Security Guide](#) for information on encryption.

At the bottom of the screen is the **Full disk summary and bootloader** button for you to configure a disk on which a boot loader will be installed.

See [Section 8.14.1, “Boot Loader Installation”](#) for more information.

Click the **Done** button once you have made your selections to either return to the **Installation Summary** screen or to proceed to the **Manual Partitioning** screen.

8.14.1. Boot Loader Installation

Red Hat Enterprise Linux uses GRUB2 (GRand Unified Bootloader version 2) as its boot loader. The boot loader is the first program that runs when the computer starts and is responsible for loading and transferring control to an operating system. GRUB2 can boot any compatible operating system and can also use *chain loading* to transfer control to other boot loaders for unsupported operating systems.



WARNING

Installing GRUB2 can overwrite your existing boot loader.

If you have other operating systems already installed, Red Hat Enterprise Linux attempts to automatically detect and configure GRUB2 to boot them. You can manually configure any additional operating systems if they are not detected properly.

To specify which device the boot loader should be installed on, click the **Full disk summary and bootloader** link at the bottom of the **Installation Destination** screen. The **Selected Disks** dialog will appear. If you are partitioning the drive manually, this dialog can be reached by clicking **Storage device/s selected** on the **Manual Partitioning** screen.

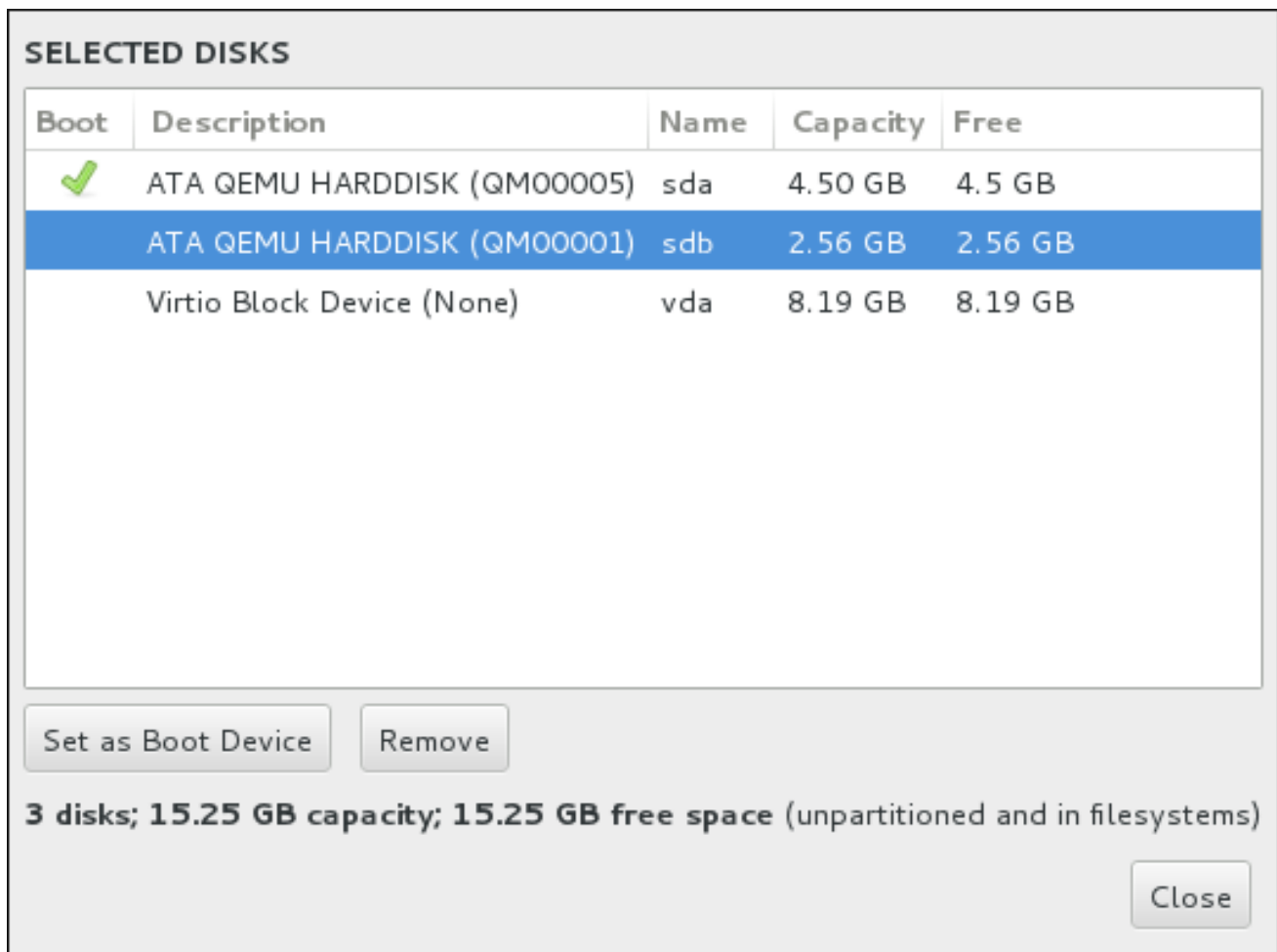


Figure 8.18. Summary of Selected Disks

In the **Boot** column, a green tick icon marks one of the devices as the intended boot device. To change the boot device, select a device from the list and click the **Set as Boot Device** button to install the boot loader there instead.

To decline installation of a new boot loader, select the marked device and click the **Do not install bootloader** button. This will remove the tick and ensure GRUB2 is not installed on any device.



WARNING

If you choose not to install a boot loader for any reason, you will not be able to boot the system directly, and you must use another boot method, such as a commercial boot loader application. Use this option only if you are sure you have another way to boot your system.

8.14.1.1. MBR and GPT Considerations

The installation program installs GRUB2 either in the *master boot record* (MBR) or the *GUID partition table* (GPT) of the device for the root file system. In order to determine which of these methods to use, the installation program considers the following variations:

BIOS systems, and UEFI systems in BIOS compatibility mode

If the disk is already formatted, the partitioning scheme is retained.

If the disk is not formatted, or the user erased all partitions from the disk, **Anaconda** will use:

- MBR if the disk has less than 2^{32} sectors. Most commonly, disks sectors are 512 bytes in size, in which case this would be equivalent to 2 TiB.
- GPT if the disk has 2^{32} sectors or more.



NOTE

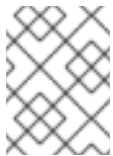
Append the **inst.gpt** option to the boot command line to override the default behavior and use GPT on a disk of less than 2^{32} sectors in size. Note that you cannot manually override **Anaconda** to use MBR on a disk which is 2^{32} sectors in size or larger.

You need to create a BIOS Boot (*biosboot*) partition to install on a BIOS system where the disk containing the boot loader uses GPT. The **biosboot** partition should be 1 MiB in size. However, you do *not* need the **biosboot** partition if the disk containing the boot loader uses MBR.

UEFI systems

Only GPT is allowed on UEFI systems. In order to install on a formatted disk with a MBR, you must first reformat it.

You need to create an EFI System Partition (**/boot/efi**), regardless of the partitioning scheme. The **/boot/efi** partition should be at least 50 MiB in size; its recommended size is 200 MiB.



NOTE

Neither the **biosboot** nor **efi** partition can reside on an LVM volume. Use standard physical partitions for them.

8.14.2. Encrypt Partitions

If you selected the **Encrypt my data** option, when you click to proceed to the next screen the installation program will prompt you for a passphrase with which to encrypt the partitions on the system.

Partitions are encrypted using the *Linux Unified Key Setup* – see the [Red Hat Enterprise Linux 7 Security Guide](#) for more information.

DISK ENCRYPTION PASSPHRASE

You have chosen to encrypt some of your data. You will need to create a passphrase that you will use to access your data when you start your computer.

Passphrase:


Strong

Confirm:



Warning: You won't be able to switch between keyboard layouts (from the default one) when you decrypt your disks after install.

Figure 8.19. Enter Passphrase for an Encrypted Partition

Choose a passphrase and type it into each of the two fields in the dialog box. Note that you need to use the same keyboard layout for setting up this passphrase that you will use to unlock partitions later. Use the language layout icon to ensure the correct layout is selected. You must provide this passphrase every time that the system boots. Press **Tab** while in the **Passphrase** input field to retype it. If the passphrase is too weak, a warning icon appears in the field and you will not be allowed to type in the second field. Hover your mouse cursor over the warning icon to learn how to improve the passphrase.



WARNING

If you lose this passphrase, any encrypted partitions and the data on them will become completely inaccessible. There is no way to recover a lost passphrase.

Note that if you perform a Kickstart installation, you can save encryption passphrases and create backup encryption passphrases during installation. See the [Red Hat Enterprise Linux 7 Security Guide](#) for more information about disk encryption.

8.14.3. Reclaim Disk Space

If there is insufficient space to install Red Hat Enterprise Linux on the disks selected in **Installation Destination** and you selected **Reclaim Space** at the **Installation Options** dialog, the **Reclaim Disk Space** dialog appears.

**WARNING**

Unless you select to shrink a partition, reclaiming space on a partition involves deleting all the data on it and you should always verify that any data you need to keep was backed up.

RECLAIM DISK SPACE

You can remove existing filesystems you no longer need to free up space for this installation. Removing a filesystem will permanently delete all of the data it contains.

Disk	Name	Filesystem	Reclaimable Space	Action
▼ 4.50 GB ATA QEMU HARDDISK	sda		2.92 GB total	Delete
└─ xfs	sda1	xfs	Not resizeable	
└─ rhel	sda2	lvmpv	Not resizeable	
└─ Free space			1.58 GB	
▼ 2.56 GB ATA QEMU HARDDISK	sdb		975 MB total	Preserve
└─ rhel	sdb1	lvmpv	Not resizeable	Preserve
└─ Free space			1.58 GB	Preserve
▼ 8.19 GB Virtio Block Device	vda		8.19 GB total	Preserve
└─ Free space			8.19 GB	

Preserve
Delete
Shrink

Delete all

3 disks; 12.08 GB reclaimable space (in filesystems)

Total selected space to reclaim: **4.50 GB**

Installation requires a total of **3.99 GB** for system data.

Cancel

Reclaim space

Figure 8.20. Reclaim Disk Space from Existing File Systems

The existing file systems Red Hat Enterprise Linux has detected are listed in a table as part of their respective disks. The **Reclaimable Space** column lists the space that could be reassigned to this installation. The **Action** column lists what action will be taken with the file system to reclaim space.

Beneath the table are four buttons:

- **Preserve** - leaves the file system untouched and no data will be deleted. This is the default action.

- **Delete** - removes the file system entirely. All the space it takes up on the disk will be made available for the installation.
- **Shrink** - recovers free space from the file system and makes it available for this installation. Use the slider to set a new size for the selected partition. Can only be used on resizable partitions where LVM or RAID is not used.
- **Delete all/Preserve all** - this button, located on the right, marks all file systems for deletion by default. Upon clicking, it changes the label and allows you to mark all file systems to be preserved again.

Select a file system or a whole disk in the table with your mouse and click one of the buttons. The label in the **Action** column will change to match your selection and the amount of **Total selected space to reclaim** displayed beneath the table will adjust accordingly. Beneath this value is the amount of space the installation requires based on the packages you have selected to install.

When enough space has been reclaimed for the installation to proceed, the **Reclaim Space** button will become available. Click this button to return to the Installation Summary screen and proceed with the installation.

8.14.4. Manual Partitioning

The **Manual Partitioning** screen is displayed when you click **Done** from Installation Destination if you selected the **I will configure partitioning** option. On this screen you configure your disk partitions and mount points. This defines the file system that Red Hat Enterprise Linux will be installed on.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.

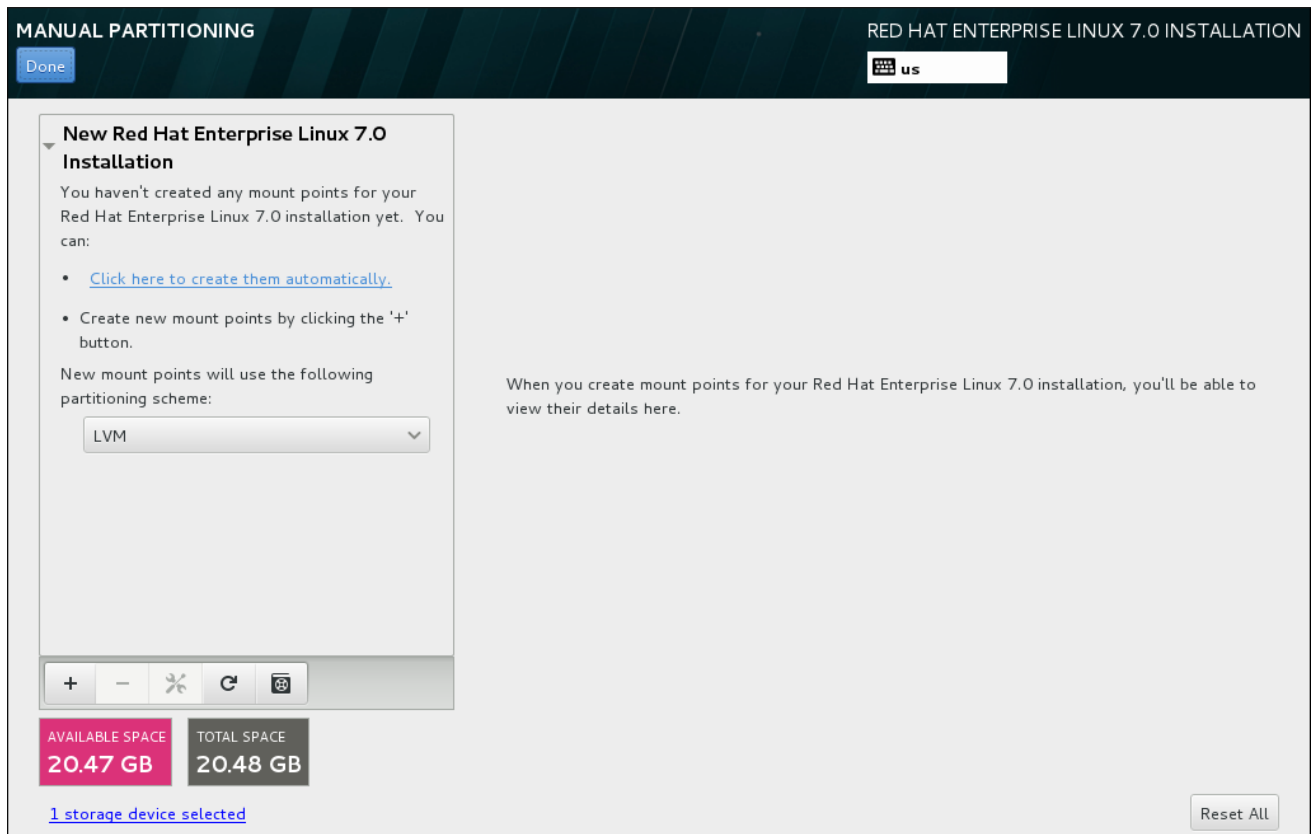


Figure 8.21. The Manual Partitioning Screen

The **Manual Partitioning** screen initially features a single pane on the left for the mount points. The pane is either empty except for information about creating mount points, or it displays existing mount points that the installation program has detected. These mount points are organized by detected operating system installations. Therefore, some file systems might be displayed multiple times if a partition is shared among several installations. The total space and available space on selected storage devices are displayed beneath this pane.

If your system contains existing file systems, ensure that enough space will be available for the installation. Use the - button to remove unneeded partitions.



NOTE

For recommendations and additional information about disk partitions, see [Appendix A, *An Introduction to Disk Partitions*](#) and [Section 8.14.4.4, “Recommended Partitioning Scheme”](#). At a bare minimum, you need an appropriately sized root partition, and usually a swap partition appropriate to the amount of RAM you have on your system.

8.14.4.1. Adding File Systems and Configuring Partitions

An installation of Red Hat Enterprise Linux requires a minimum of one partition but Red Hat recommends using at least the following partitions or volumes: **/**, **/home**, **/boot**, and **swap**. You can also create additional partitions and volumes as you require. See [Section 8.14.4.4, “Recommended Partitioning Scheme”](#) for further details.



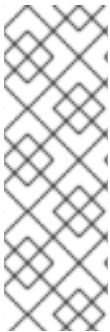
NOTE

If you have any specific requirements for some partitions (for example, requiring that a particular partition be on a specific disk) and less specific requirements for other partitions, create the partitions first which have more specific requirements.

Adding a file system is a two-step process. You first create a mount point in a certain partitioning scheme. The mount point appears in the left pane. Next, you can customize it using the options in the right pane, where you can change the mount point, capacity, the device type, file system type, label, and whether to encrypt or reformat the corresponding partition.

If you have no existing file systems and want the installation program to create the required file systems and their mount points for you, select your preferred partitioning scheme from the drop-down menu in the left pane (default for Red Hat Enterprise Linux is LVM), then click the link on top of the pane for creating mount points automatically. This will generate a **/boot** partition, a **/** (root) volume, and a swap volume proportionate to the size of the available storage. These are the recommended file systems for a typical installation, but you can add additional file systems and mount points if you need to.

Alternatively, create individual mount points using the **+** button at the bottom of the pane. The **Add a New Mount Point** dialog then opens. Either select one of the preset paths from the **Mount Point** drop-down menu or type your own; for example, select **/** for the root partition or **/boot** for the boot partition. Then enter the size of the file system in the **Desired Capacity** text field; for example, **2GiB**. If you leave the field empty or if you specify a size bigger than available space, all remaining free space is used instead. After entering these details, click the **Add mount point** button to create the partition.



NOTE

To avoid problems with space allocation, first create small partitions with known fixed sizes, such as **/boot**, and then create the rest of the partitions, letting the installation program allocate the remaining capacity to them.

Similarly, if you have multiple disks that the system is to reside on, they differ in size, and a particular partition must be created on the first disk detected by BIOS, be sure to start by creating such a partition.

For each new mount point you create manually, you can set its partitioning scheme from the drop-down menu located in the left pane. The available options are **Standard Partition**, **BTRFS**, **LVM**, and **LVM Thin Provisioning**. Note that the **/boot** partition will always be located on a standard partition, regardless of the value selected in this menu.

To change on which devices a single non-LVM mount point should be located, select the mount point and click the **Modify...** button in the right pane to open the **Configure Mount Point** dialog. Select one or more devices and click **Select**. After the dialog closes, note that you also need to confirm this setting by clicking the **Update Settings** button on the right side of the **Manual Partitioning** screen.

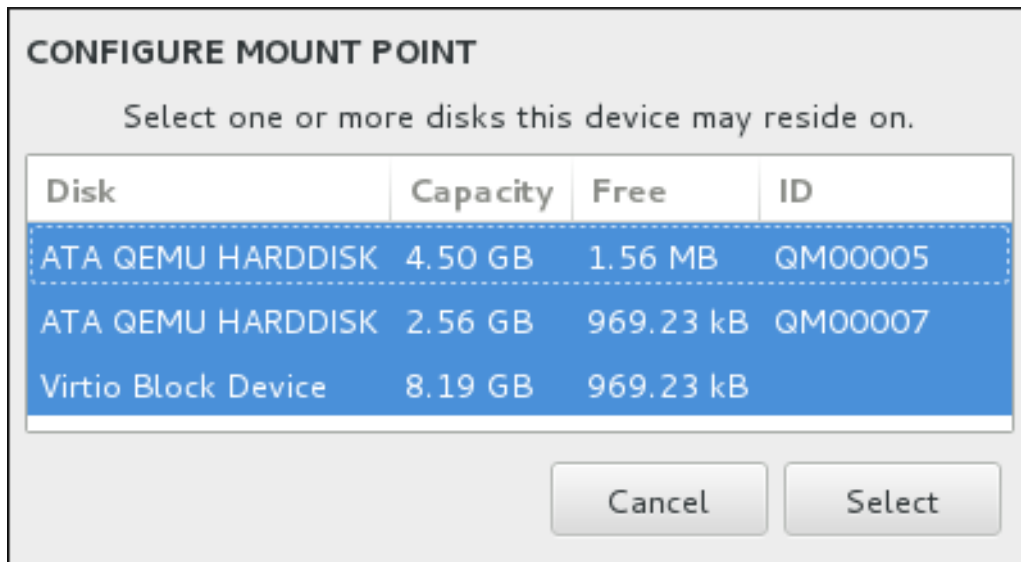


Figure 8.22. Configuring Mount Points

To refresh information about all local disks and partitions on them, click the **Rescan** button (with the circular arrow icon on it) in the toolbar. You only need to do this action after performing advanced partition configuration outside the installation program. Note that if you click the **Rescan Disks** button, all configuration changes you previously made in the installation program will be lost.

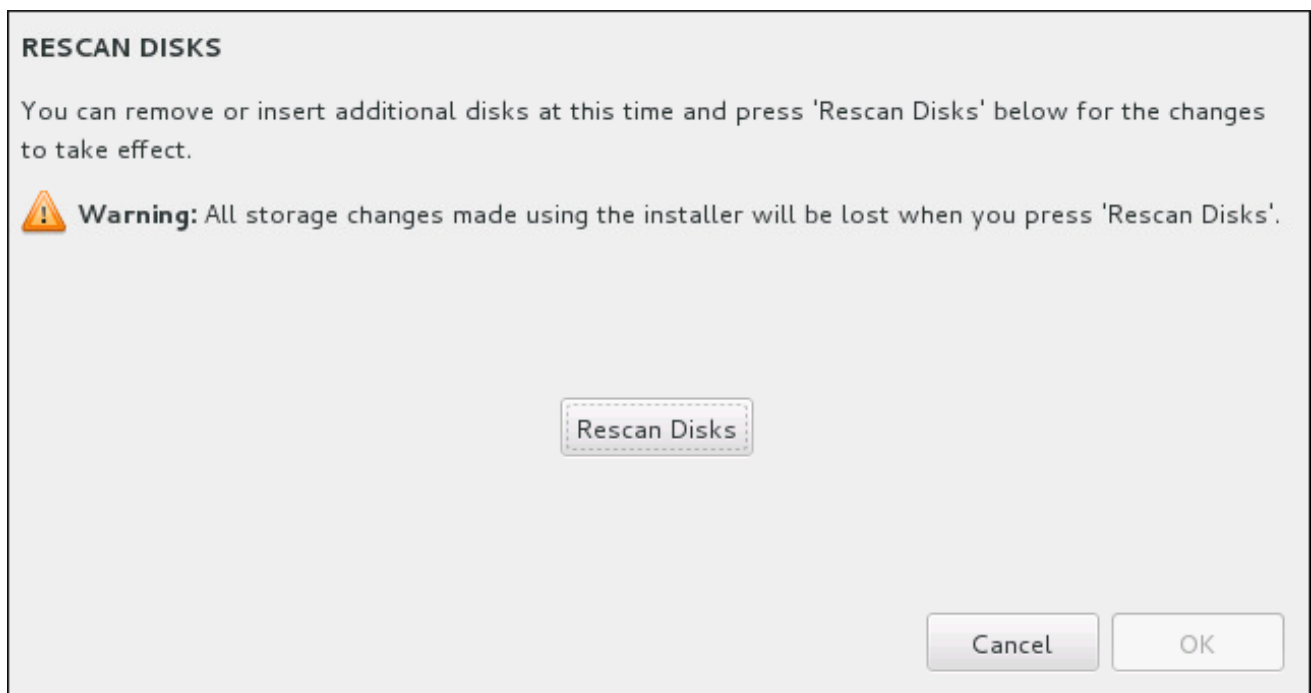


Figure 8.23. Rescanning Disks

At the bottom of the screen, a link states how many storage devices have been selected in **Installation Destination** (see [Section 8.14, "Installation Destination"](#)). Clicking on this link opens the **Selected Disks** dialog, where you review the information about the disks. See [Section 8.14.1, "Boot Loader Installation"](#) for more information.

To customize a partition or a volume, select its mount point in the left pane and the following customizable features then appear to the right:

MANUAL PARTITIONING RED HAT ENTERPRISE LINUX 7.1 INSTALLATION

Done us Help!

New Red Hat Enterprise Linux 7.1 Installation

DATA

SYSTEM

/boot 476 MiB

sdal

/ 4743.34 MiB >

rhel-root

swap 1020 MiB

rhel-swap

rhel-root

Mount Point: /

Device(s): ATA QEMU HARDDISK (sda) and 1 other

Desired Capacity: 4743.34 MiB

Device Type: LVM ☐ Encrypt

File System: xfs ☒ Reformat

Volume Group: rhel (0 B free)

Label:

Name: root

Update Settings

Note: The settings you make on this screen will not be applied until you click on the main menu's 'Begin Installation' button.

Reset All

AVAILABLE SPACE 2395 KiB TOTAL SPACE 6246.4 MiB

[2 storage devices selected](#)

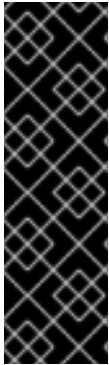
Figure 8.24. Customizing Partitions

- **Mount Point** – enter the file system's mount point. For example, if a file system should be the root file system, enter `/`; enter `/boot` for the `/boot` file system, and so on. For a swap file system, the mount point should not be set – setting the file system type to **swap** is sufficient.
- **Desired Capacity** – enter the desired size of the file system. You can use common size units such as KiB or GiB. The default is MiB if no other unit is specified.
- **Device type** – choose one of these types: **Standard Partition**, **LVM**, **RAID**, **LVM Thin Provisioning**, or **BTRFS**. Check the adjacent **Encrypt** box to encrypt the partition or volume. You will be prompted to set a password later. **RAID** is only available if two or more disks are selected for partitioning, and if you choose this type, you can also set the **RAID Level**. Similarly, if you select **LVM**, you can specify the **Volume Group**.
- **File system** – in the drop-down menu, select the appropriate file system type for this partition or volume. Check the adjacent **Reformat** box to format an existing partition, or leave it unchecked to retain your data. Note that newly created partitions and volumes must be reformatted, and the check box cannot be unchecked in this case.
- **Label** – assign a label to the partition. Labels are used for you to easily recognize and address individual partitions.
- **Name** – assign a name to an LVM or Btrfs volume. Note that standard partitions are named automatically when they are created and their name cannot be edited, such as `/home` being assigned the name `sda1`.

See [Section 8.14.4.1.1, “File System Types”](#) for more information about file system and device types.

Click the **Update Settings** button to save your changes and select another partition to customize. Note that the changes will not be applied until you actually start the installation from the Installation summary page. Click the **Reset All** button to discard all changes to all partitions and start over.

When all file systems and mount points have been created and customized, click the **Done** button. If you chose to encrypt any file system, you will now be prompted to create a passphrase. Then, a dialog appears, showing a summary of all actions related to storage that the installation program will take. This includes creating, resizing, or deleting partitions and file systems. You can review all the changes and click **Cancel & Return to Custom Partitioning** to go back. To confirm your changes, click **Accept Changes** to return to the Installation Summary page. To partition additional devices, select them in the **Installation Destination** screen, return to the **Manual Partitioning** screen, repeat the steps outlined in this section for the additional devices.



IMPORTANT

If **/usr** or **/var** is partitioned separately from the rest of the root volume, the boot process becomes much more complex because these directories contain components critical to it. In some situations, such as when these directories are placed on an iSCSI drive or an FCoE location, the system can either be unable to boot, or hang with a **Device is busy** error when powering off or rebooting.

This limitation only applies to **/usr** or **/var**, not to directories below them. For example, a separate partition for **/var/www** will work without issues.

8.14.4.1.1. File System Types

Red Hat Enterprise Linux allows you to create different device types and file systems. The following is a brief description of the different device types and file systems available, and how they can be used.

Device Types

- **standard partition** - A standard partition can contain a file system or swap space, or it can provide a container for software RAID or an LVM physical volume.
- **logical volume (LVM)** - Creating an LVM partition automatically generates an LVM logical volume. LVM can improve performance when using physical disks. For information on how to create a logical volume, see [Section 8.14.4.3, “Create LVM Logical Volume”](#). For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.
- **LVM thin provisioning** - Using thin provisioning, you can manage a storage pool of free space, known as a thin pool, which can be allocated to an arbitrary number of devices when needed by applications. The thin pool can be expanded dynamically when needed for cost-effective allocation of storage space. For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.



WARNING

Anaconda does not support overprovisioned LVM thin pools.



NOTE

The installer will automatically reserve 20% of any requested space for an LVM thin pool logical volume in the volume group containing it. This is a safety measure to ensure that you can extend either the metadata volume or the data volume of your thinly provisioned logical volume.

- **software RAID** - Creating two or more software RAID partitions allows you to create a RAID device. One RAID partition is assigned to each disk on the system. To create a RAID device, see [Section 8.14.4.2, “Create Software RAID”](#). For more information regarding RAID, see the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

File Systems

- **xfs** - XFS is a highly scalable, high-performance file system that supports file systems up to 16 EiB (approximately 16 billion GiB), files up to 8 EiB (approximately 8 billion GiB), and directory structures containing tens of millions of entries. XFS supports metadata journaling, which facilitates quicker crash recovery. The XFS file system can also be defragmented and resized while mounted and active. This file system is selected by default and is highly recommended. For information on how to translate common commands from previously used ext4 file system to XFS, see [Appendix F, Reference Table for ext4 and XFS Commands](#).

The maximum supported size of an XFS file system in Red Hat Enterprise Linux is currently 500 TiB.

- **ext4** - The ext4 file system is based on the ext3 file system and features a number of improvements. These include support for larger file systems and larger files, faster and more efficient allocation of disk space, no limit on the number of subdirectories within a directory, faster file system checking, and more robust journaling.

The maximum supported size of an ext4 file system in Red Hat Enterprise Linux is currently 50 TiB.

- **ext3** - The ext3 file system is based on the ext2 file system and has one main advantage - journaling. Using a journaling file system reduces time spent recovering a file system after a crash as there is no need to check the file system for metadata consistency by running the **fsck** utility every time a crash occurs.
- **ext2** - An ext2 file system supports standard Unix file types, including regular files, directories, or symbolic links. It provides the ability to assign long file names, up to 255 characters.
- **vfat** - The VFAT file system is a Linux file system that is compatible with Microsoft Windows long file names on the FAT file system.
- **swap** - Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- **BIOS Boot** - A very small partition required for booting a device with a GUID partition table (GPT) on a BIOS system. See [Section 8.14.1, “Boot Loader Installation”](#) for details.
- **EFI System Partition** - A small partition required for booting a device with a GUID partition table (GPT) on a UEFI system. See [Section 8.14.1, “Boot Loader Installation”](#) for details.

Each file system has different size limits for the file system itself as well as individual files contained within. For a list of maximum supported file and file system sizes, see the Red Hat Enterprise Linux technology capabilities and limits page, available on the Customer Portal at

<https://access.redhat.com/site/articles/rhel-limits>.

8.14.4.2. Create Software RAID

Redundant arrays of independent disks (RAIDs) are constructed from multiple storage devices that are arranged to provide increased performance and, in some configurations, greater fault tolerance. See below for a description of different kinds of RAIDs.

A RAID device is created in one step and disks are added or removed as necessary. One RAID partition per physical disk is allowed for each device, so the number of disks available to the installation program determines which levels of RAID device are available to you. For example, if your system has two hard drives, the installation program will not allow you to create a RAID10 device, which requires 4 separate partitions.

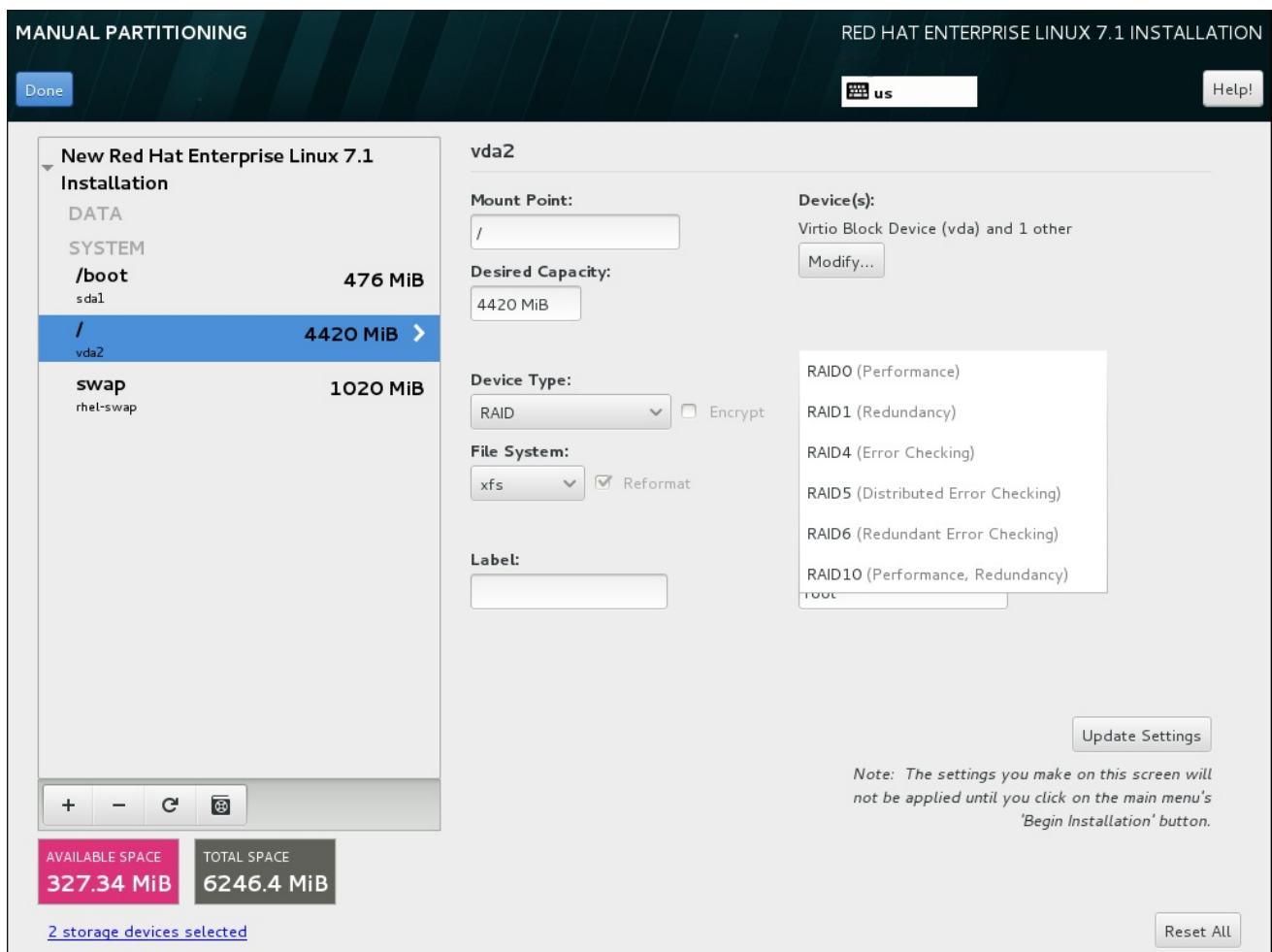


Figure 8.25. Creating a Software RAID Partition - the Device Type Menu Expanded

RAID configuration options are only visible if you have selected two or more disks for installation. At least two disks are required to create a RAID device.

To create a RAID device:

1. Create a mount point as described in [Section 8.14.4.1, “Adding File Systems and Configuring Partitions”](#). By configuring this mount point, you configure the RAID device.
2. Keeping the partition selected in the left pane, select the configuration button below the pane to open the **Configure Mount Point** dialog. Select which disks will be included in the RAID device and click **Select**.

3. Click the **Device Type** drop-down menu and select **RAID**.
4. Click the **File System** drop-down menu and select your preferred file system type (see [Section 8.14.4.1.1, "File System Types"](#)).
5. Click the **RAID Level** drop-down menu and select your preferred level of RAID.

The available RAID levels are:

RAID0 - Optimized performance (stripe)

Distributes data across multiple disks. Level 0 RAID offers increased performance over standard partitions, and can be used to pool the storage of multiple disks into one large virtual device. Note that Level 0 RAID offers no redundancy, and that the failure of one device in the array destroys data in the entire array. RAID 0 requires at least two RAID partitions.

RAID1 - Redundancy (mirror)

Mirrors all data on one disk onto one or more other disks. Additional devices in the array provide increasing levels of redundancy. RAID 1 requires at least two RAID partitions.

RAID4 - Error detection (parity)

Distributes data across multiple disks, and uses one disk in the array to store parity information that safeguards the array in case any disk within the array fails. Because all parity information is stored on one disk, access to this disk creates a bottleneck in the performance of the array. RAID 4 requires at least three RAID partitions.

RAID5 - Distributed error detection

Distributes data *and* parity information across multiple disks. Level 5 RAID therefore offers the performance advantages of distributing data across multiple disks, but does not share the performance bottleneck of level 4 RAID because the parity information is also distributed through the array. RAID 5 requires at least three RAID partitions.

RAID6 - Redundant

Level 6 RAID is similar to level 5 RAID, but instead of storing only one set of parity data, they store two sets. RAID 6 requires at least four RAID partitions.

RAID10 - Redundancy (mirror) and Optimized performance (stripe)

Level 10 RAID is a *nested RAID* or *hybrid RAID*. They are constructed by distributing data over mirrored sets of disks. For example, a level 10 RAID array constructed from four RAID partitions consists of two mirrored pairs of striped partitions. RAID 10 requires at least four RAID partitions.

6. Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.

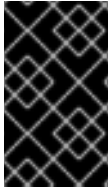
If fewer disks are included than the specified RAID level requires, a message will be displayed at the bottom of the window, informing you how many disks are actually required for your selected configuration.

8.14.4.3. Create LVM Logical Volume

Logical Volume Management (LVM) presents a simple logical view of underlying physical storage space,

such as hard drives or LUNs. Partitions on physical storage are represented as *physical volumes* that can be grouped together into *volume groups*. Each volume group can be divided into multiple *logical volumes*, each of which is analogous to a standard disk partition. Therefore, LVM logical volumes function as partitions that can span multiple physical disks.

To learn more about LVM, see [Appendix D, Understanding LVM](#) or read the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide. Note that LVM configuration is only available in the graphical installation program.



IMPORTANT

During text-mode installation, LVM configuration is not available. If you need to create an LVM configuration from scratch, press **Ctrl+Alt+F2** to use a different virtual console, and run the **lvm** command. To return to the text-mode installation, press **Ctrl+Alt+F1**.

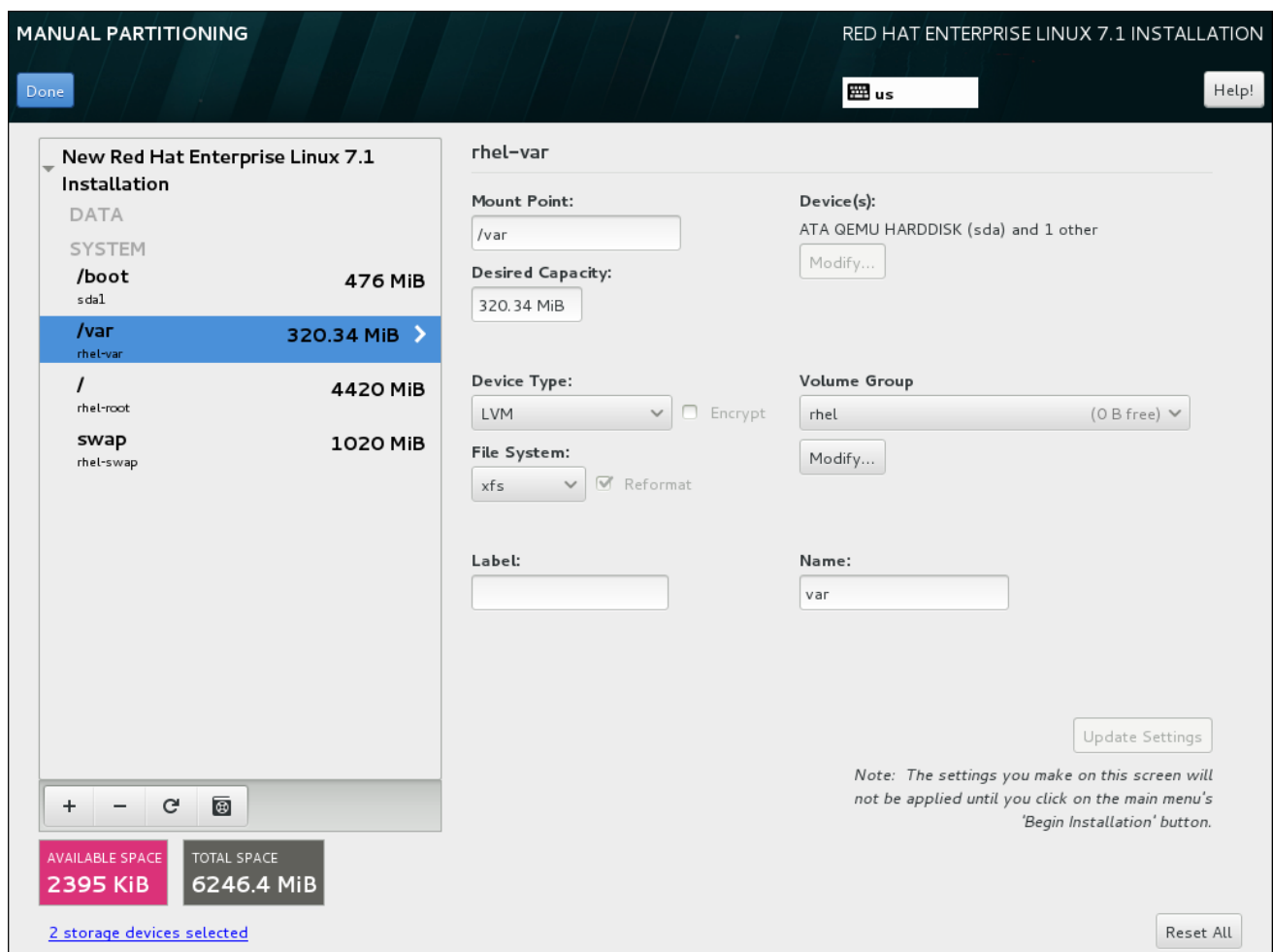


Figure 8.26. Configuring a Logical Volume

To create a logical volume and add it to a new or existing volume group:

1. Create a mount point for the LVM volume as described in [Section 8.14.4.1, “Adding File Systems and Configuring Partitions”](#).
2. Click the **Device Type** drop-down menu and select **LVM**. The **Volume Group** drop-down menu appears and displays the newly-created volume group name.
3. Optionally, either click the menu and select **Create a new volume group** or click **Modify** to configure the newly-created volume group, if you need to. Both the **Create a new volume**

group option and the **Modify** button lead to the **Configure Volume Group** dialog, where you can rename the logical volume group and select which disks will be included.



NOTE

The configuration dialog does not allow you to specify the size of the volume group's physical extents. The size will always be set to the default value of 4 MiB. If you want to create a volume group with different physical extents, create it manually by switching to an interactive shell and using the **vgcreate** command, or use a Kickstart file with the **volgroup --pesize=size** command.

CONFIGURE VOLUME GROUP

Please create a name for this volume group and select at least one disk below.

Name:

Disk	Capacity	Free	ID
Virtio Block Device	28.67 GB	969.23 kB	

RAID Level: ☐ Encrypt

Size policy:

Figure 8.27. Customizing an LVM Volume Group

The available RAID levels are the same as with actual RAID devices. See [Section 8.14.4.2, “Create Software RAID”](#) for more information. You can also mark the volume group for encryption and set the size policy for it. The available policy options are:

- **Automatic** – the size of the volume group is set automatically so that it is just large enough to contain the configured logical volumes. This is optimal if you do not need free space within the volume group.
- **As large as possible** – the volume group is created with maximum size, regardless of the size of the configured logical volumes it contains. This is optimal if you plan to keep most of your data on LVM and later need to increase the size of some existing logical volumes, or if you need to create additional logical volumes within this group.

- **Fixed** - with this option, you can set an exact size of the volume group. Any configured logical volumes must then fit within this fixed size. This is useful if you know exactly how large you would like the volume group to be.

Click **Save** when the group is configured.

4. Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.



WARNING

Placing the **/boot** partition on an LVM volume is not supported.

8.14.4.4. Recommended Partitioning Scheme

Red Hat recommends that you create separate file systems at the following mount points:

- **/boot**
- **/ (root)**
- **/home**
- **swap**

/boot partition - recommended size at least 1 GiB

The partition mounted on **/boot** contains the operating system kernel, which allows your system to boot Red Hat Enterprise Linux, along with files used during the bootstrap process. Due to the limitations of most firmwares, creating a small partition to hold these is recommended. In most scenarios, a 1 GiB boot partition is adequate. Unlike other mount points, using an LVM volume for **/boot** is not possible - **/boot** must be located on a separate disk partition.



WARNING

Normally, the **/boot** partition is created automatically by the installation program. However, if the **/ (root)** partition is larger than 2 TiB and (U)EFI is used for booting, you need to create a separate **/boot** partition that is smaller than 2 TiB to boot the machine successfully.



NOTE

If you have a RAID card, be aware that some BIOS types do not support booting from the RAID card. In such a case, the **/boot** partition must be created on a partition outside of the RAID array, such as on a separate hard drive.

root - recommended size of 10 GiB

This is where "/", or the root directory, is located. The root directory is the top-level of the directory structure. By default, all files are written to this file system unless a different file system is mounted in the path being written to (for example, **/boot** or **/home**).

While a 5 GiB root file system allows you to install a minimal installation, it is recommended to allocate at least 10 GiB so that you can install as many package groups as you want.

**IMPORTANT**

Do not confuse the / directory with the **/root** directory. The **/root** directory is the home directory of the root user. The **/root** directory is sometimes referred to as *slash root* to distinguish it from the root directory.

/home - recommended size at least 1 GiB

To store user data separately from system data, create a dedicated file system for the **/home** directory. This file system should be sized based on the amount of data that will be stored locally, number of users, and so on. This will enable you to upgrade or reinstall Red Hat Enterprise Linux without erasing user data files. If you select automatic partitioning, it is recommended to have at least 55GiB of disk space available for the installation, to ensure that the **/home** file system is created.

swap partition - recommended size at least 1 GB

Swap file systems support virtual memory; data is written to a swap file system when there is not enough RAM to store the data your system is processing. Swap size is a function of system memory workload, not total system memory and therefore is not equal to the total system memory size. Therefore, it is important to analyze what applications a system will be running and the load those applications will serve in order to determine the system memory workload. Application providers and developers should be able to provide some guidance.

When the system runs out of swap space, the kernel terminates processes as the system RAM memory is exhausted. Configuring too much swap space results in storage devices being allocated but idle and is a poor use of resources. Too much swap space can also hide memory leaks. The maximum size for a swap partition and other additional information can be found in the **mkswap(8)** manual page.

The following table provides the recommended size of a swap partition depending on the amount of RAM in your system and whether you want sufficient memory for your system to hibernate. If you let the installation program partition your system automatically, the swap partition size will be established using these guidelines. Automatic partitioning setup assumes hibernation is not in use. The maximum size of the swap partition is limited to 10% of the total size of the hard drive, and the installer cannot create swap partitions more than 128GB in size. If you want to set up enough swap space to allow for hibernation, or if you want to set the swap partition size to more than 10% of the system's storage space, or more than 128GB, you must edit the partitioning layout manually.

Table 8.3. Recommended System Swap Space

Amount of RAM in the system	Recommended swap space	Recommended swap space if allowing for hibernation
less than 2 GB	2 times the amount of RAM	3 times the amount of RAM

Amount of RAM in the system	Recommended swap space	Recommended swap space if allowing for hibernation
2 GB - 8 GB	Equal to the amount of RAM	2 times the amount of RAM
8 GB - 64 GB	4GB to 0.5 times the amount of RAM	1.5 times the amount of RAM
more than 64 GB	workload dependent (at least 4GB)	hibernation not recommended

At the border between each range listed above (for example, a system with 2 GB, 8 GB, or 64 GB of system RAM), discretion can be exercised with regard to chosen swap space and hibernation support. If your system resources allow for it, increasing the swap space can lead to better performance.

Distributing swap space over multiple storage devices – particularly on systems with fast drives, controllers and interfaces – also improves swap space performance.

Many systems have more partitions and volumes than the minimum listed above. Choose partitions based on your particular system needs. See [Section 8.14.4.4.1, “Advice on Partitions”](#) for more information.



NOTE

Only assign storage capacity to those partitions you require immediately. You can allocate free space at any time, to meet needs as they occur. To learn about a more flexible method for storage management, see [Appendix D, Understanding LVM](#).

If you are not sure how best to configure the partitions for your computer, accept the automatic default partition layout provided by the installation program.

8.14.4.4.1. Advice on Partitions

Optimal partition setup depends on the usage for the Linux system in question. Use these tips to decide how to configure your disk space.

- Consider encrypting any partitions that might contain sensitive data. Encryption prevents unauthorized people from accessing the data on the partitions, even if they have access to the physical storage device. In most cases, you should at least encrypt the **/home** partition.
- Each kernel installed on your system requires approximately 56 MB on the **/boot** partition.
 - 32 MB initramfs
 - 14 MB kdump initramfs
 - 3.5 MB system map
 - 6.6 MB vmlinuz

**NOTE**

For rescue mode, **initramfs** and **vmlinuz** require 80 MB.

The default partition size of 1 GiB for **/boot** should suffice for most common uses. However, it is recommended that you increase the size of this partition if you are planning on retaining multiple kernel releases or errata kernels.

- The **/var** directory holds content for a number of applications, including the **Apache** web server. It also is used to store downloaded update packages on a temporary basis. Ensure that the partition containing the **/var** directory has enough space to download pending updates and hold your other content.
- The **PackageKit** update software downloads updated packages to **/var/cache/yum/** by default. If you create a separate partition or volume for **/var**, ensure that it is at least 3GB in size to accommodate downloaded package updates.
- The **/usr** directory holds the majority of software content on a Red Hat Enterprise Linux system. For an installation of the default set of software, allocate at least 5 GB of space. If the system will be used as a software development workstation, allocate at least 10GB.
- If **/usr** or **/var** is partitioned separately from the rest of the root volume, the boot process becomes much more complex because these directories contain components critical to it. In some situations, such as when these directories are placed on an iSCSI drive or an FCoE location, the system can either be unable to boot, or it can hang with a **Device is busy** error when powering off or rebooting.

This limitation only applies to **/usr** or **/var**, not to directories below them. For example, a separate partition for **/var/www** will work without issues.

- Consider leaving a portion of the space in an LVM volume group unallocated. This unallocated space gives you flexibility if your space requirements change but you do not want to remove data from other partitions to reallocate storage. You can also select the **Thin provisioning** device type for the partition to have the unused space handled automatically by the volume.
- If you separate subdirectories into partitions, you can retain content in those subdirectories if you decide to install a new version of Red Hat Enterprise Linux over your current system. For instance, if you intend to run a **MySQL** database in **/var/lib/mysql/**, make a separate partition for that directory in case you need to reinstall later.
- On a BIOS system with its boot loader using GPT (GUID partition table), you need to create the **biosboot** partition of 1 MiB in size. See [Section 8.14.1, “Boot Loader Installation”](#) for more details.
- UEFI systems need to contain a small partition with a mount point of **/boot/efi/** containing an EFI System Partition file system. Its recommended size is 200 MiB, which is also the default value for automatic partitioning.

8.15. STORAGE DEVICES

You can install Red Hat Enterprise Linux on a large variety of storage devices. You can see basic, locally accessible, storage devices in the **Installation Destination** page, as described in [Section 8.14, “Installation Destination”](#). To add a specialized storage device, click the **Add a disk** button in the **Specialized & Network Disks** section of the screen.

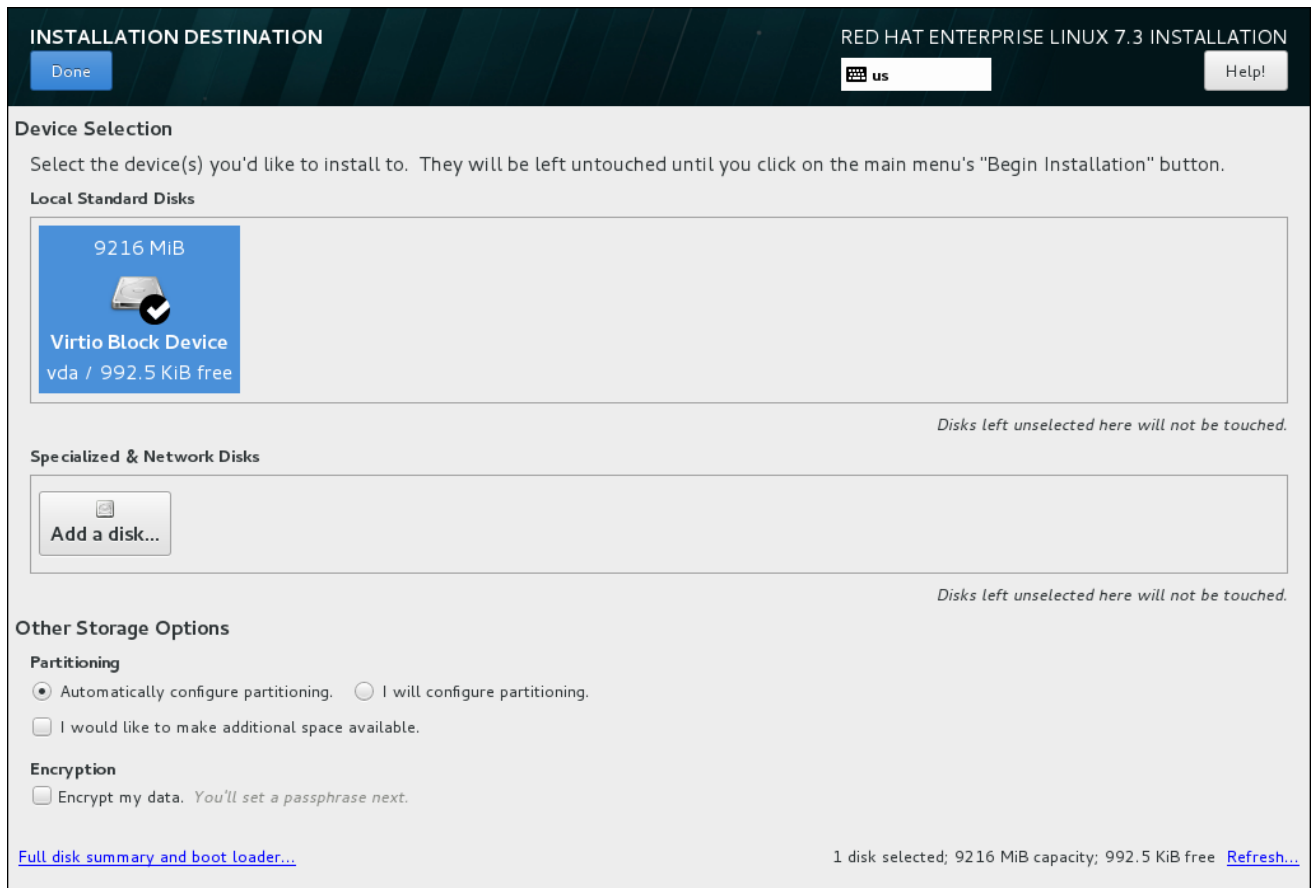


Figure 8.28. Storage Space Overview

**NOTE**

Monitoring of LVM and software RAID devices by the **dmeventd** daemon is not performed during installation.

8.15.1. The Storage Devices Selection Screen

The storage device selection screen displays all storage devices to which the **Anaconda** installation program has access.

The devices are grouped under the following tabs:

Multipath Devices

Storage devices accessible through more than one path, such as through multiple SCSI controllers or Fiber Channel ports on the same system.

The installation program only detects multipath storage devices with serial numbers that are 16 or 32 characters long.

Other SAN Devices

Devices available on a Storage Area Network (SAN).

NVDIMM Devices

Non-Volatile Dual In-line Memory Module (NVDIMM) storage devices on the machine.

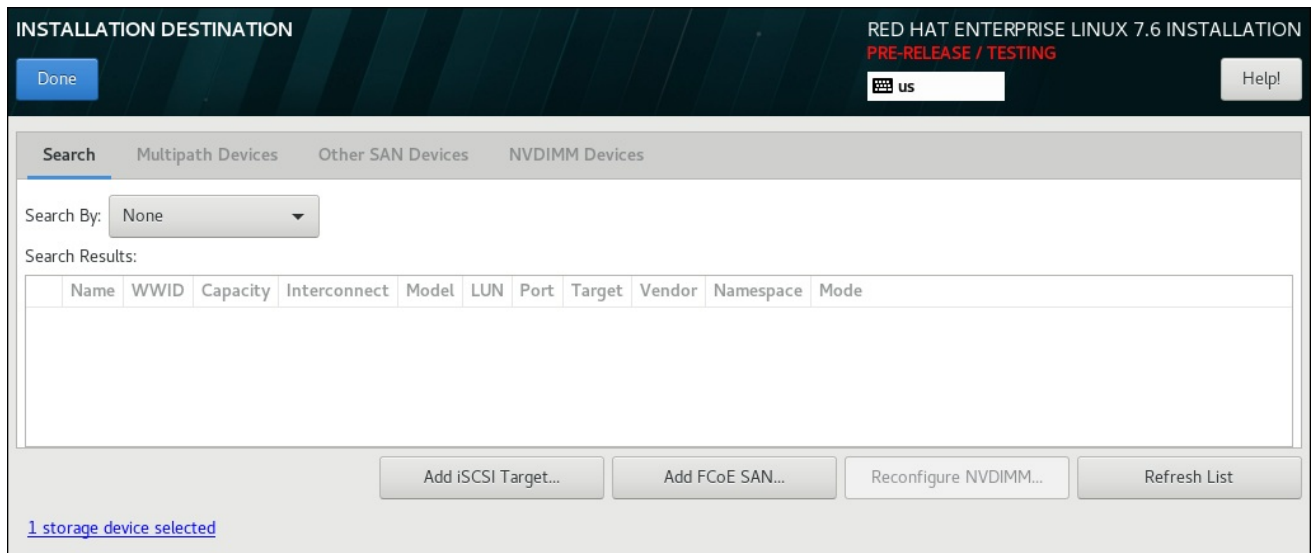


Figure 8.29. Tabbed Overview of Specialized Storage Devices

A set of buttons is available in the bottom right corner of the screen. Use these buttons to add additional storage devices. The available buttons are:

- **Add iSCSI Target** - use to attach iSCSI devices; continue with [Section 8.15.1.1, "Configuring iSCSI Parameters"](#)
- **Add FCoE SAN** - use to configure a Fibre Channel Over Internet storage device; continue with [Section 8.15.1.2, "Configuring FCoE Parameters"](#)
- **Reconfigure NVDIMM** - use to reconfigure a NVDIMM device into sector mode; continue with [Section 8.15.1.3, "Configuring NVDIMM Devices"](#)
- **Refresh List** - use to reload the list when a device is added after the installer starts.

The overview page also contains the **Search** tab that allows you to filter storage devices either by their *World Wide Identifier* (WWID) or by the port, target, or *logical unit number* (LUN) at which they are accessed.

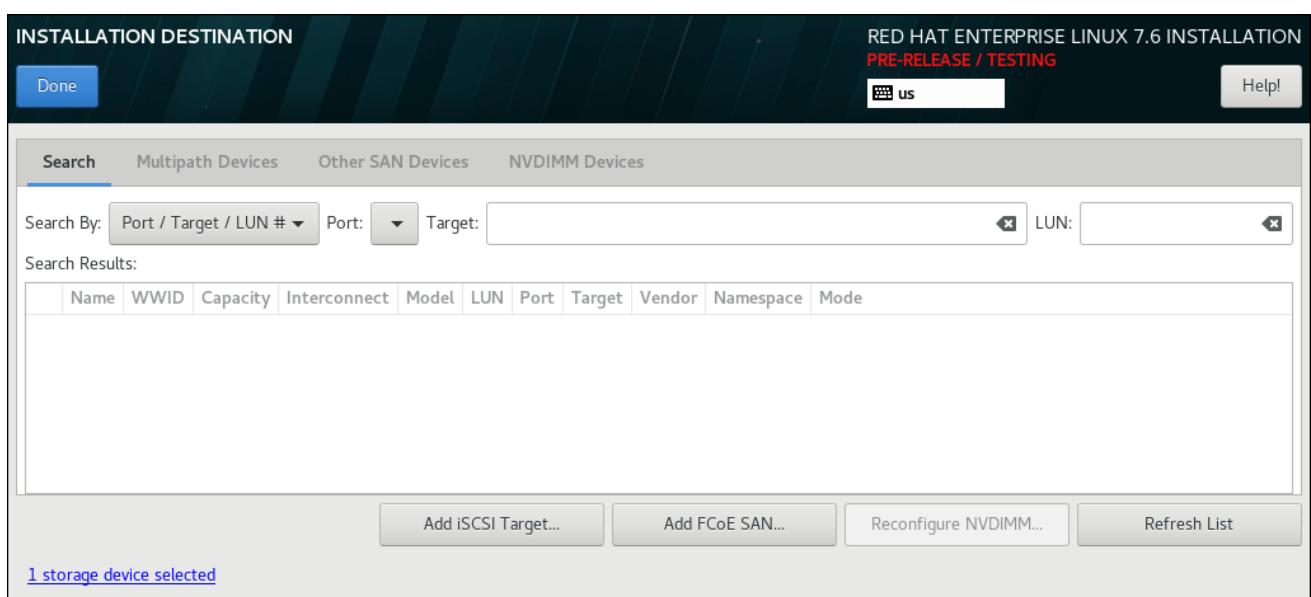
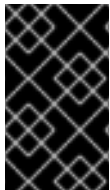


Figure 8.30. The Storage Devices Search Tab

The Search tab contains the **Search By** drop-down menu to select searching by port, target, LUN, or WWID. Searching by WWID or LUN requires additional values in the corresponding input text fields. Click the **Find** button to start the search.

Each device is presented on a separate row, with a check box to its left. Click the check box to make the device available during the installation process. Later in the installation process, you can choose to install Red Hat Enterprise Linux onto any of the devices selected here, and can choose to automatically mount any of the other devices selected here as part of the installed system.

Note that the devices that you select here are not automatically erased by the installation process. Selecting a device on this screen does not, in itself, place data stored on the device at risk. Also note that any devices that you do not select here to form part of the installed system can be added to the system after installation by modifying the **/etc/fstab** file.



IMPORTANT

Any storage devices that you do not select on this screen are hidden from **Anaconda** entirely. To *chain load* the Red Hat Enterprise Linux boot loader from a different boot loader, select all the devices presented in this screen.

When you have selected the storage devices to make available during installation, click **Done** to return to the Installation Destination screen.

8.15.1.1. Advanced Storage Options

To use an advanced storage device, you can configure a SCSI over TCP/IP (*iSCSI*) target, Fibre Channel over Ethernet (*FCoE*) Storage Area Network (*SAN*), or Non-Volatile Dual In-line Memory Module (*NVDIMM*) device by clicking the appropriate button in the lower right corner of the Installation Destination screen. For an introduction to *iSCSI*, see [Appendix B, *iSCSI Disks*](#).

INSTALLATION DESTINATION

RED HAT ENTERPRISE LINUX 7.6 INSTALLATION
PRE-RELEASE / TESTING

Done

Help!

Search Multipath Devices Other SAN Devices NVDIMM Devices

Search By: Port / Target / LUN # Port: Target: LUN:

Search Results:

Name	WWID	Capacity	Interconnect	Model	LUN	Port	Target	Vendor	Namespace	Mode

Add iSCSI Target... Add FCoE SAN... Reconfigure NVDIMM... Refresh List

1 storage device selected

Figure 8.31. Advanced Storage Options

8.15.1.1.1. Configuring iSCSI Parameters

When you click the **Add iSCSI target...** button, the **Add iSCSI Storage Target** dialog appears.

ADD iSCSI STORAGE TARGET

To use iSCSI disks, you must provide the address of your iSCSI target and the iSCSI initiator name you've configured for your host.

Target IP Address:

iSCSI Initiator Name:

 Example: iqn.2012-09.com.example:diskarrays-sn-a8675309

Discovery Authentication Type: CHAP pair and a reverse pair

CHAP Username:

CHAP Password:

Reverse CHAP Username:

Reverse CHAP Password:

☐ Bind targets to network interfaces

Figure 8.32. The iSCSI Discovery Details Dialog

To use iSCSI storage devices for the installation, **Anaconda** must be able to *discover* them as iSCSI targets and be able to create an iSCSI *session* to access them. Each of these steps might require a user name and password for *CHAP* (Challenge Handshake Authentication Protocol) authentication. Additionally, you can configure an iSCSI target to authenticate the iSCSI initiator on the system to which the target is attached (*reverse CHAP*), both for discovery and for the session. Used together, CHAP and reverse CHAP are called *mutual CHAP* or *two-way CHAP*. Mutual CHAP provides the greatest level of security for iSCSI connections, particularly if the user name and password are different for CHAP authentication and reverse CHAP authentication.



NOTE

Repeat the iSCSI discovery and iSCSI login steps as many times as necessary to add all required iSCSI storage. However, you cannot change the name of the iSCSI initiator after you attempt discovery for the first time. To change the iSCSI initiator name, you must restart the installation.

Procedure 8.1. iSCSI Discovery and Starting an iSCSI Session

Use the **Add iSCSI Storage Target** dialog to provide **Anaconda** with the information necessary to discover the iSCSI target.

1. Enter the IP address of the iSCSI target in the **Target IP Address** field.
2. Provide a name in the **iSCSI Initiator Name** field for the iSCSI initiator in *iSCSI qualified name* (IQN) format. A valid IQN entry contains:
 - the string **iqn.** (note the period)
 - a date code that specifies the year and month in which your organization's Internet domain or subdomain name was registered, represented as four digits for the year, a dash, and two digits for the month, followed by a period. For example, represent September 2010 as **2010-09.**
 - your organization's Internet domain or subdomain name, presented in reverse order with the top-level domain first. For example, represent the subdomain **storage.example.com** as **com.example.storage**
 - a colon followed by a string that uniquely identifies this particular iSCSI initiator within your domain or subdomain. For example, **:diskarrays-sn-a8675309**

A complete IQN can therefore look as follows: **iqn.2010-09.storage.example.com:diskarrays-sn-a8675309**. **Anaconda** prepopulates the **iSCSI Initiator Name** field with a name in this format to help you with the structure.

For more information on IQNs, see 3.2.6. *iSCSI Names* in *RFC 3720 - Internet Small Computer Systems Interface (iSCSI)* available from <http://tools.ietf.org/html/rfc3720#section-3.2.6> and 1. *iSCSI Names and Addresses* in *RFC 3721 - Internet Small Computer Systems Interface (iSCSI) Naming and Discovery* available from <http://tools.ietf.org/html/rfc3721#section-1>.

3. Use the **Discovery Authentication Type** drop-down menu to specify the type of authentication to use for iSCSI discovery. The following options are available:
 - no credentials
 - CHAP pair
 - CHAP pair and a reverse pair
4.
 - If you selected **CHAP pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** fields.
 - If you selected **CHAP pair and a reverse pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** field and the user name and password for the iSCSI initiator in the **Reverse CHAP Username** and **Reverse CHAP Password** fields.
5. Optionally check the box labeled **Bind targets to network interfaces**.
6. Click the **Start Discovery** button. **Anaconda** attempts to discover an iSCSI target based on the information that you provided. If discovery succeeds, the dialog displays a list of all iSCSI nodes discovered on the target.
7. Each node is presented with a check box beside it. Click the check boxes to select the nodes to use for installation.

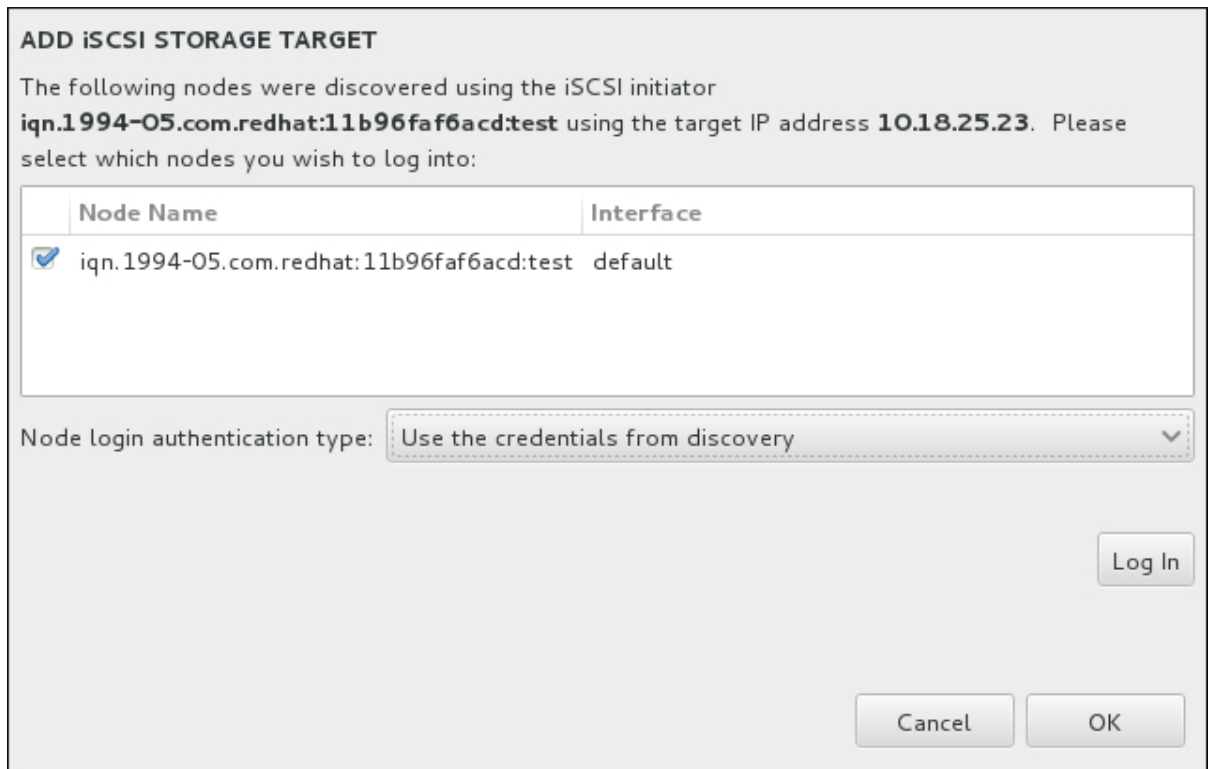


Figure 8.33. The Dialog of Discovered iSCSI Nodes

8. The **Node login authentication type** menu provides the same options as the **Discovery Authentication Type** menu described in step 3. However, if you needed credentials for discovery authentication, it is typical to use the same credentials to log into a discovered node. To do that, use the additional **Use the credentials from discovery** option from the menu. When the proper credentials have been provided, the **Log In** button becomes available.
9. Click **Log In** to initiate an iSCSI session.

8.15.1.1.2. Configuring FCoE Parameters

When you click the **Add FCoE SAN...** button, a dialog appears for you to configure network interfaces for discovering FCoE storage devices.

First, select a network interface that is connected to a FCoE switch in the **NIC** drop-down menu and click the **Add FCoE disk(s)** button to scan the network for SAN devices.

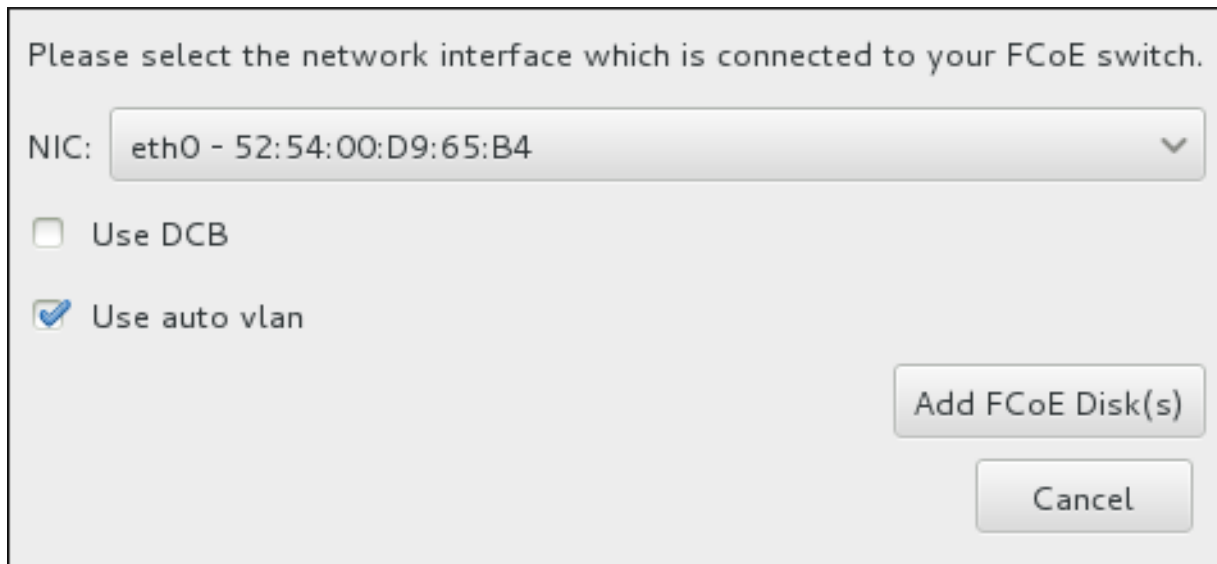


Figure 8.34. Configure FCoE Parameters

There are check boxes with additional options to consider:

Use DCB

Data Center Bridging (DCB) is a set of enhancements to the Ethernet protocols designed to increase the efficiency of Ethernet connections in storage networks and clusters. Enable or disable the installation program's awareness of DCB with the check box in this dialog. This option should only be enabled for network interfaces that require a host-based DCBX client. Configurations on interfaces that implement a hardware DCBX client should leave this check box empty.

Use auto vlan

Auto VLAN indicates whether VLAN discovery should be performed. If this box is checked, then the FIP (FCoE Initiation Protocol) VLAN discovery protocol will run on the Ethernet interface once the link configuration has been validated. If they are not already configured, network interfaces for any discovered FCoE VLANs will be automatically created and FCoE instances will be created on the VLAN interfaces. This option is enabled by default.

Discovered FCoE devices will be displayed under the **Other SAN Devices** tab in the Installation Destination screen.

8.15.1.1.3. Configuring NVDIMM Devices

Click the **Reconfigure NVDIMM** button to reconfigure the selected NVDIMM device into sector mode and use it as an installation destination. The **Sector size** drop-down list contains the supported sector sizes of **512** and **4096**.

From the **Sector size** drop-down list, select the sector size and click the **Start Reconfiguration** button.

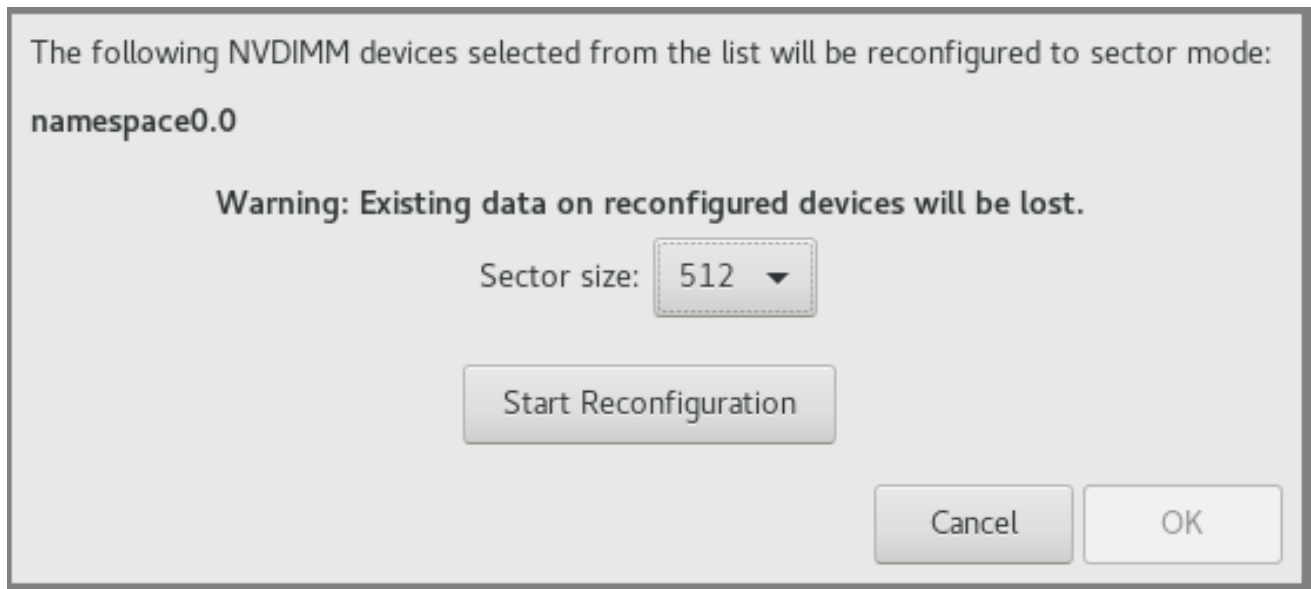
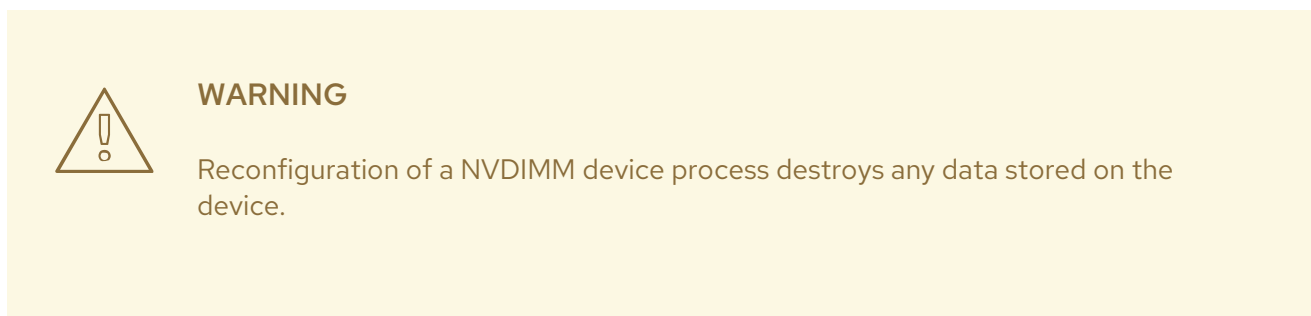


Figure 8.35. Reconfigure NVDIMM



When the device is configured, click the **OK** button to return to the **Installation Destination** window.

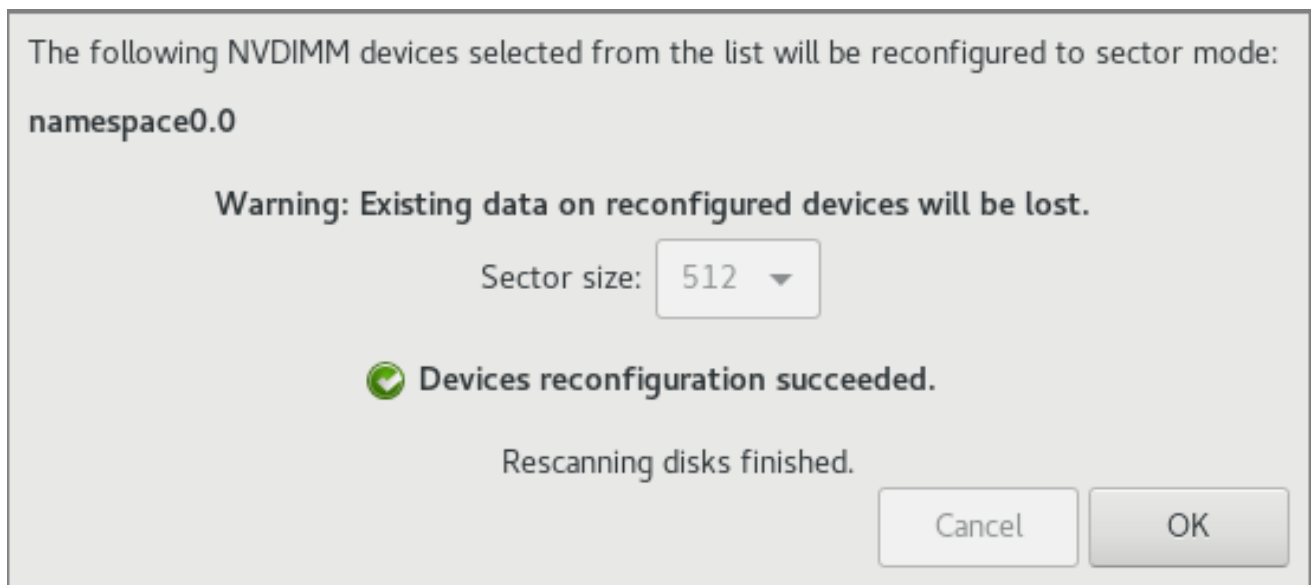


Figure 8.36. Successfully reconfigured NVDIMM

NVDIMM devices in sector mode are displayed under the **NVDIMM Devices** tab in the **Installation Destination** window and are available for installation.

8.16. KDUMP

Use this screen to select whether or not to use **Kdump** on this system. **Kdump** is a kernel crash dumping mechanism which, in the event of a system crash, captures information that can be invaluable in determining the cause of the crash.

Note that if you enable **Kdump**, you must reserve a certain amount of system memory for it. As a result, less memory is available for your processes.

If you do not want to use **Kdump** on this system, uncheck **Enable kdump**. Otherwise, set the amount of memory to reserve for **Kdump**. You can let the installer reserve a reasonable amount automatically, or you can set any amount manually. When you are satisfied with the settings, click **Done** to save the configuration and return to the previous screen.

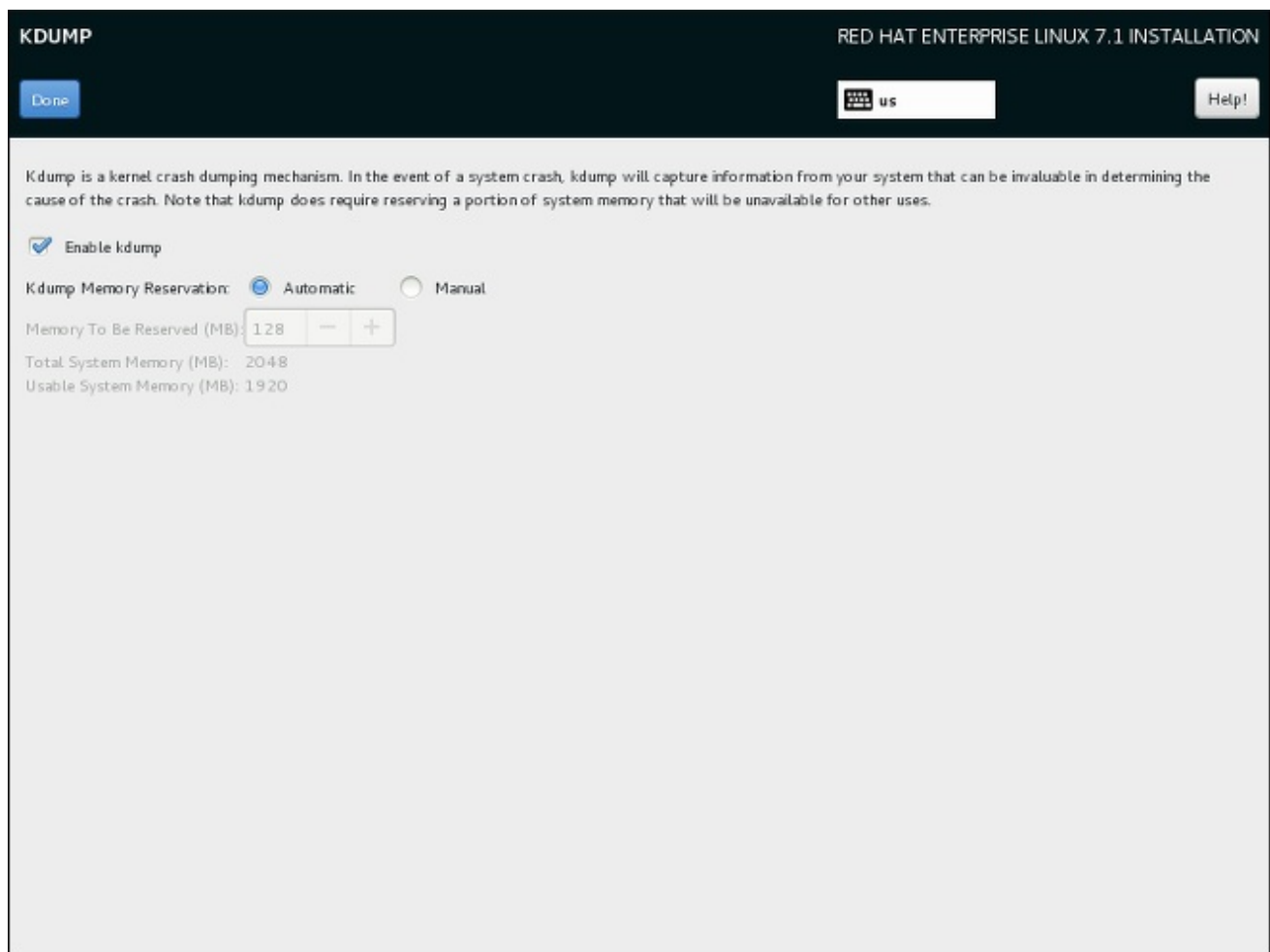


Figure 8.37. Kdump Enablement and Configuration

8.17. BEGIN INSTALLATION

When all required sections of the **Installation Summary** screen have been completed, the admonition at the bottom of the menu screen disappears and the **Begin Installation** button becomes available.

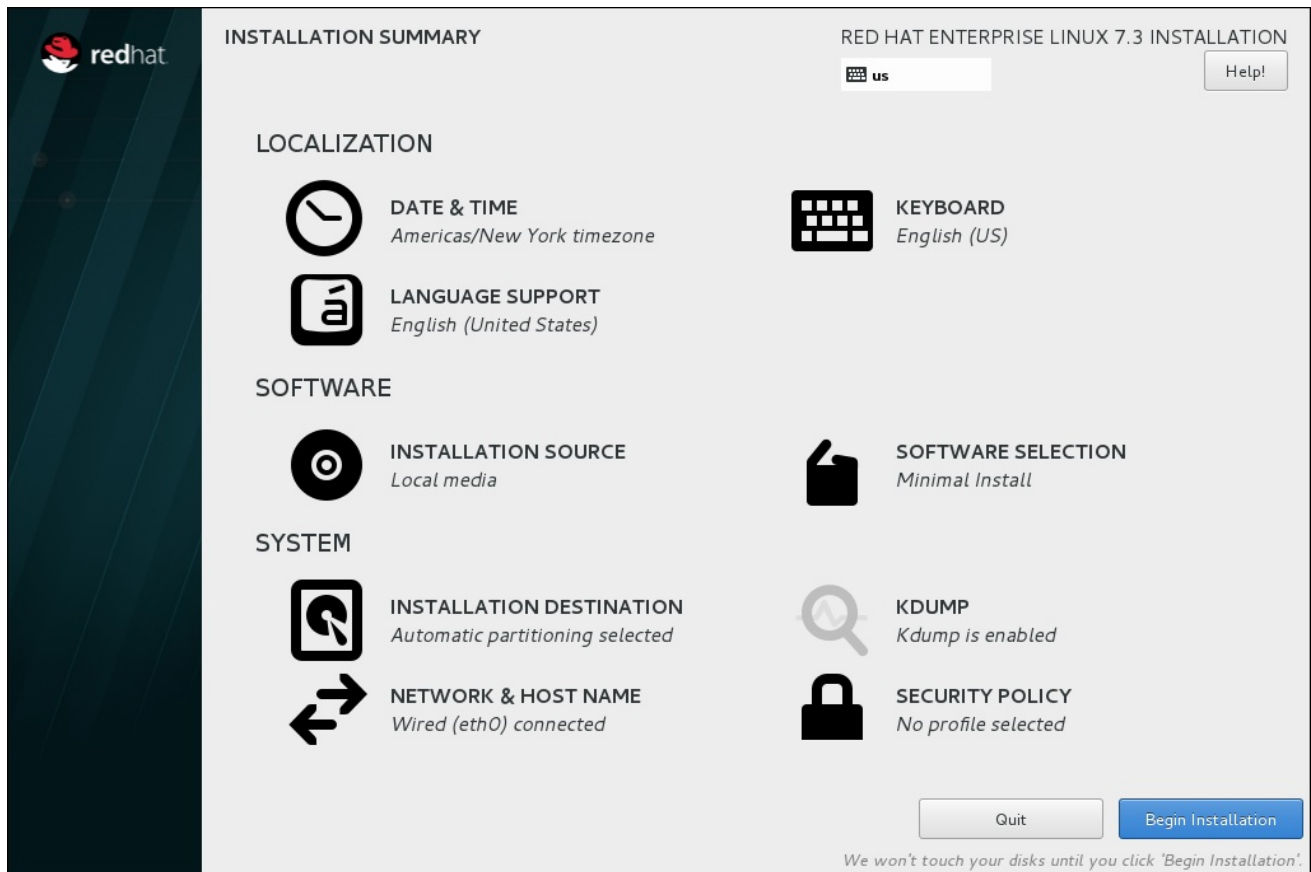


Figure 8.38. Ready to Install

**WARNING**

Up to this point in the installation process, no lasting changes have been made on your computer. When you click **Begin Installation**, the installation program will allocate space on your hard drive and start to transfer Red Hat Enterprise Linux into this space. Depending on the partitioning option that you chose, this process might include erasing data that already exists on your computer.

To revise any of the choices that you made up to this point, return to the relevant section of the **Installation Summary** screen. To cancel installation completely, click **Quit** or switch off your computer. To switch off most computers at this stage, press the power button and hold it down for a few seconds.

If you have finished customizing your installation and are certain that you want to proceed, click **Begin Installation**.

After you click **Begin Installation**, allow the installation process to complete. If the process is interrupted, for example, by you switching off or resetting the computer, or by a power outage, you will probably not be able to use your computer until you restart and complete the Red Hat Enterprise Linux installation process, or install a different operating system.

8.18. THE CONFIGURATION MENU AND PROGRESS SCREEN

Once you click **Begin Installation** at the **Installation Summary** screen, the progress screen appears. Red Hat Enterprise Linux reports the installation progress on the screen as it writes the selected packages to your system.

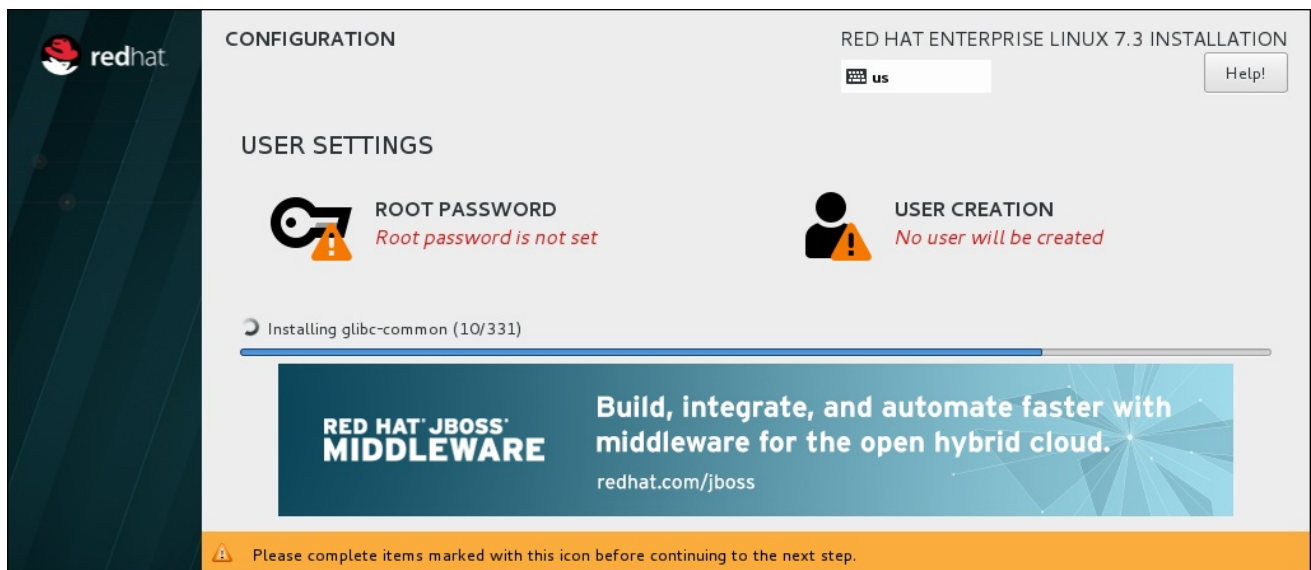


Figure 8.39. Installing Packages

For your reference, a complete log of your installation can be found in the `/var/log/anaconda/anaconda.packaging.log` file, once you reboot your system.

If you chose to encrypt one or more partitions during partitioning setup, a dialog window with a progress bar will be displayed during the early stage of the installation process. This window informs that the installer is attempting to gather enough entropy (random data) to ensure that the encryption is secure. This window will disappear after 256 bits of entropy are gathered, or after 10 minutes. You can speed up the gathering process by moving your mouse or randomly typing on the keyboard. After the window disappears, the installation process will continue.

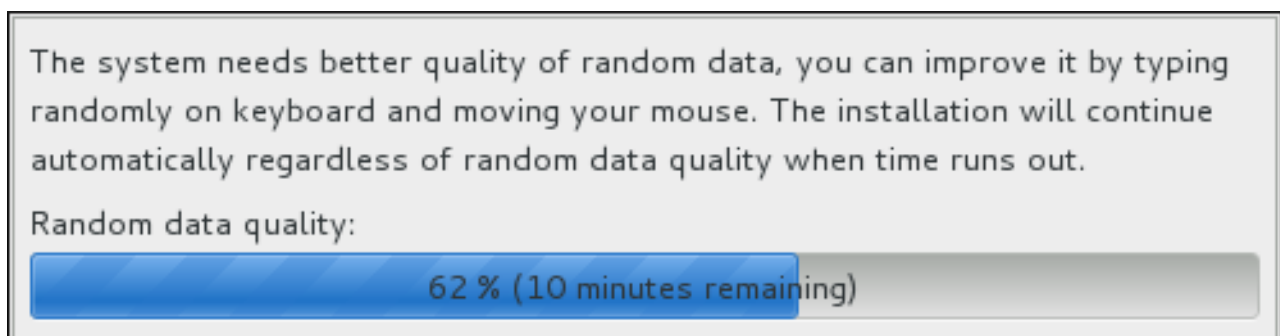


Figure 8.40. Gathering Entropy for Encryption

While the packages are being installed, more configuration is required. Above the installation progress bar are the **Root Password** and **User Creation** menu items.

The **Root Password** screen is used to configure the system's **root** account. This account can be used to perform critical system management and administration tasks. The same tasks can also be performed with a user account with the **wheel** group membership; if such an user account is created during installation, setting up a **root** password is not mandatory.

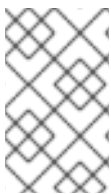
Creating a user account is optional and can be done after installation, but it is recommended to do it on this screen. A user account is used for normal work and to access the system. Best practice suggests that you always access the system through a user account, not the root account.

It is possible to disable access to the **Root Password** or **Create User** screens. To do so, use a Kickstart file which includes the **rootpw --lock** or **user --lock** commands. See [Section 27.3.1, “Kickstart Commands and Options”](#) for more information these commands.

8.18.1. Set the Root Password

Setting up a root account and password is an important step during your installation. The root account (also known as the superuser) is used to install packages, upgrade RPM packages, and perform most system maintenance. The root account gives you complete control over your system. For this reason, the root account is best used *only* to perform system maintenance or administration. See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about becoming root.

Figure 8.41. Root Password Screen



NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

Click the **Root Password** menu item and enter your new password into the **Root Password** field. Red Hat Enterprise Linux displays the characters as asterisks for security. Type the same password into the **Confirm** field to ensure it is set correctly. After you set the root password, click **Done** to return to the User Settings screen.

The following are the requirements and recommendations for creating a strong root password:

- *must* be at least eight characters long
- may contain numbers, letters (upper and lower case) and symbols
- is case-sensitive and should contain a mix of cases
- something you can remember but that is not easily guessed
- should not be a word, abbreviation, or number associated with you, your organization, or found in a dictionary (including foreign languages)
- should not be written down; if you must write it down keep it secure



NOTE

To change your root password after you have completed the installation, run the **passwd** command as **root**. If you forget the root password, see [Section 32.1.3, “Resetting the Root Password”](#) for instructions on how to use the rescue mode to set a new one.

8.18.2. Create a User Account

To create a regular (non-root) user account during the installation, click **User Settings** on the progress screen. The **Create User** screen appears, allowing you to set up the regular user account and configure its parameters. Though recommended to do during installation, this step is optional and can be performed after the installation is complete.



NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

To leave the user creation screen after you have entered it, without creating a user, leave all the fields empty and click **Done**.

Figure 8.42. User Account Configuration Screen

Enter the full name and the user name in their respective fields. Note that the system user name must be shorter than 32 characters and cannot contain spaces. It is highly recommended to set up a password for the new account.

When setting up a strong password even for a non-root user, follow the guidelines described in [Section 8.18.1, “Set the Root Password”](#).

Click the **Advanced** button to open a new dialog with additional settings.

ADVANCED USER CONFIGURATION

Home Directory

☒ Create a home directory for this user.

Home directory:

User and Group IDs

☐ Specify a user ID manually:
☐ Specify a group ID manually:

Group Membership

Add user to the following groups:

Example: wheel, my-team (1245), project-x (29935)

Tip: You may input a comma-separated list of group names and group IDs here.
Groups that do not already exist will be created; specify their GID in parentheses.

Figure 8.43. Advanced User Account Configuration

By default, each user gets a home directory corresponding to their user name. In most scenarios, there is no need to change this setting.

You can also manually define a system identification number for the new user and their default group by selecting the check boxes. The range for regular user IDs starts at the number **1000**. At the bottom of the dialog, you can enter the comma-separated list of additional groups, to which the new user shall belong. The new groups will be created in the system. To customize group IDs, specify the numbers in parenthesis.



NOTE

Consider setting IDs of regular users and their default groups at range starting at **5000** instead of **1000**. That is because the range reserved for system users and groups, **0-999**, might increase in the future and thus overlap with IDs of regular users.

For creating users with custom IDs using kickstart, see [user \(optional\)](#).

For changing the minimum UID and GID limits after the installation, which ensures that your chosen UID and GID ranges are applied automatically on user creation, see [the Users and Groups chapter of the System Administrator's Guide](#).

Once you have customized the user account, click **Save Changes** to return to the **User Settings** screen.

8.19. INSTALLATION COMPLETE

Congratulations! Your Red Hat Enterprise Linux installation is now complete!

Click the **Reboot** button to reboot your system and begin using Red Hat Enterprise Linux. Remember to remove any installation media if it is not ejected automatically upon reboot.

After your computer's normal power-up sequence has completed, Red Hat Enterprise Linux loads and starts. By default, the start process is hidden behind a graphical screen that displays a progress bar. Eventually, a GUI login screen (or if the X Window System is not installed, a **login:** prompt) appears.

If your system was installed with the X Window System during this installation process, the first time you start your Red Hat Enterprise Linux system, applications to set up your system are launched. These applications guide you through initial configuration of Red Hat Enterprise Linux and allow you to set your system time and date, register your machine with Red Hat Network, and more.

See [Chapter 30, *Initial Setup*](#) for information about the configuration process. For instructions on post-installation procedures, configuration and updates for Red Hat Enterprise Linux Atomic Host, see the [Getting Started with Red Hat Enterprise Linux Atomic Host](#) document.

CHAPTER 9. TROUBLESHOOTING INSTALLATION ON 64-BIT AMD, INTEL, AND ARM SYSTEMS

This chapter discusses some common installation problems and their solutions.

For debugging purposes, **Anaconda** logs installation actions into files in the `/tmp` directory. These files are listed in the following table.

Table 9.1. Log Files Generated During the Installation

Log file	Contents
<code>/tmp/anaconda.log</code>	general Anaconda messages
<code>/tmp/program.log</code>	all external programs run during the installation
<code>/tmp/storage.log</code>	extensive storage module information
<code>/tmp/packaging.log</code>	yum and rpm package installation messages
<code>/tmp/syslog</code>	hardware-related system messages

If the installation fails, the messages from these files are consolidated into `/tmp/anaconda-tb-identifier`, where *identifier* is a random string.

After successful installation, by default, these files will be copied to the installed system under the directory `/var/log/anaconda/`. However, if installation is unsuccessful, or if the **inst.nosave=all** or **inst.nosave=logs** options are used when booting the installation system, these logs will only exist in the installation program's RAM disk. This means they are not saved permanently and will be lost once the system is powered down. To store them permanently, copy those files to another system on the network by using **scp** on the system running the installation program, or copy them to a mounted storage device (such as a USB flash drive). Details on how to transfer the log files over the network are below. Note that if you use a USB flash drive or other removable media, you should make sure to back up any data on it before starting the procedure.

Procedure 9.1. Transferring Log Files Onto a USB Drive

1. On the system you are installing, press **Ctrl+Alt+F2** to access a shell prompt. You will be logged into a root account and you will have access to the installation program's temporary file system.
2. Connect a USB flash drive to the system and execute the **dmesg** command. A log detailing all recent events will be displayed. At the bottom of this log, you will see a set of messages caused by the USB flash drive you just connected. It will look like a set of lines similar to the following:

```
[ 170.171135] sd 5:0:0:0: [sdb] Attached SCSI removable disk
```

Note the name of the connected device - in the above example, it is **sdb**.

3. Go to the `/mnt` directory and once there, create new directory which will serve as the mount target for the USB drive. The name of the directory does not matter; this example uses the name **usb**.

```
# mkdir usb
```

4. Mount the USB flash drive onto the newly created directory. Note that in most cases, you do not want to mount the whole drive, but a partition on it. Therefore, do not use the name **sdb** – use the name of the partition you want to write the log files to. In this example, the name **sdb1** is used.

```
# mount /dev/sdb1 /mnt/usb
```

You can now verify that you mounted the correct device and partition by accessing it and listing its contents – the list should match what you expect to be on the drive.

```
# cd /mnt/usb
```

```
# ls
```

5. Copy the log files to the mounted device.

```
# cp /tmp/*log /mnt/usb
```

6. Unmount the USB flash drive. If you get an error message saying that the target is busy, change your working directory to outside the mount (for example, /).

```
# umount /mnt/usb
```

The log files from the installation are now saved on the USB flash drive.

Procedure 9.2. Transferring Log Files Over the Network

1. On the system you are installing, press **Ctrl+Alt+F2** to access a shell prompt. You will be logged into a root account and you will have access to the installation program's temporary file system.
2. Switch to the **/tmp** directory where the log files are located:

```
# cd /tmp
```

3. Copy the log files onto another system on the network using the **scp** command:

```
# scp *log user@address:path
```

Replace *user* with a valid user name on the target system, *address* with the target system's address or host name, and *path* with the path to the directory you want to save the log files into. For example, if you want to log in as **john** to a system with an IP address of **192.168.0.122** and place the log files into the **/home/john/logs/** directory on that system, the command will have the following form:

```
# scp *log john@192.168.0.122:/home/john/logs/
```

When connecting to the target system for the first time, the SSH client asks you to confirm that the fingerprint of the remote system is correct and that you want to continue:

The authenticity of host '192.168.0.122 (192.168.0.122)' can't be established.
ECDSA key fingerprint is a4:60:76:eb:b2:d0:aa:23:af:3d:59:5c:de:bb:c4:42.
Are you sure you want to continue connecting (yes/no)?

Type **yes** and press **Enter** to continue. Then, provide a valid password when prompted. The files will start transferring to the specified directory on the target system.

The log files from the installation are now permanently saved on the target system and available for review.

9.1. TROUBLE BEGINNING THE INSTALLATION

9.1.1. System Does Not Boot When UEFI Secure Boot Is Enabled

Beta releases of Red Hat Enterprise Linux 7 have their kernels signed with a special public key which is not recognized by standard UEFI Secure Boot implementations. This prevents the system from booting when the Secure Boot technology is enabled.

To fix this issue, you must disable UEFI Secure Boot, install the system, and then import the Beta public key using the *Machine Owner Key* facility. See [Section 5.9, “Using a Beta Release with UEFI Secure Boot”](#) for instructions.

9.1.2. Problems with Booting into the Graphical Installation

Systems with some video cards have trouble booting into the graphical installation program. If the installation program does not run using its default settings, it attempts to run in a lower resolution mode. If that still fails, the installation program attempts to run in text mode.

There are several possible solutions to display issues, most of which involve specifying custom boot options. For more information, see [Section 23.1, “Configuring the Installation System at the Boot Menu”](#).

Use the basic graphics mode

You can attempt to perform the installation using the basic graphics driver. To do this, either select **Troubleshooting > Install Red Hat Enterprise Linux in basic graphics mode** in the boot menu, or edit the installation program's boot options and append **inst.xdriver=vesa** at the end of the command line.

Specify the display resolution manually

If the installation program fails to detect your screen resolution, you can override the automatic detection and specify it manually. To do this, append the **inst.resolution=x** option at the boot menu, where x is your display's resolution (for example, **1024x768**).

Use an alternate video driver

You can also attempt to specify a custom video driver, overriding the installation program's automatic detection. To specify a driver, use the **inst.xdriver=x** option, where x is the device driver you want to use (for example, **nouveau**).



NOTE

If specifying a custom video driver solves your problem, you should report it as a bug at <https://bugzilla.redhat.com> under the **anaconda** component. **Anaconda** should be able to detect your hardware automatically and use the appropriate driver without your intervention.

Perform the installation using VNC

If the above options fail, you can use a separate system to access the graphical installation over the network, using the *Virtual Network Computing* (VNC) protocol. For details on installing using VNC, see [Chapter 25, Using VNC](#).

9.1.3. Serial Console Not Detected

In some cases, attempting to install in text mode using a serial console will result in no output on the console. This happens on systems which have a graphics card, but no monitor connected. If **Anaconda** detects a graphics card, it will attempt to use it for a display, even if no display is connected.

If you want to perform a text-based installation on a serial console, use the **inst.text** and **console=** boot options. See [Chapter 23, Boot Options](#) for more details.

9.2. TROUBLE DURING THE INSTALLATION

9.2.1. No Disks Detected

In the **Installation Destination** screen, the following error message can appear at the bottom: **No disks detected. Please shut down the computer, connect at least one disk, and restart to complete installation.**

The message indicates that **Anaconda** did not find any writable storage devices to install to. In that case, first make sure that your system does have at least one storage device attached.

If your system uses a hardware RAID controller, verify that the controller is properly configured and working. See your controller's documentation for instructions.

If you are installing into one or more iSCSI devices and there is no local storage present on the system, make sure that all required LUNs (*Logical Unit Numbers*) are being presented to the appropriate HBA (*Host Bus Adapter*). For additional information about iSCSI, see [Appendix B, iSCSI Disks](#).

If you made sure you have a connected and properly configured storage device and the message still appears after you reboot the system and start the installation again, it means that the installation program failed to detect the storage. In most cases this message appears when you attempt to install on an SCSI device which has not been recognized by the installation program.

In that case, you will have to perform a driver update before starting the installation. Check your hardware vendor's website to determine if a driver update is available that fixes your problem. For more general information on driver updates, see [Chapter 6, Updating Drivers During Installation on AMD64 and Intel 64 Systems](#).

You can also consult the *Red Hat Hardware Compatibility List*, available online at <https://hardware.redhat.com>.

9.2.2. Reporting Traceback Messages

If the graphical installation program encounters an error, it presents you with a crash reporting dialog box. You can then choose to send information about the problem you encountered to Red Hat. To send a crash report, you will need to enter your Customer Portal credentials. If you do not have a Customer Portal account, you can register at <https://www.redhat.com/wapps/ugc/register.html>. Automated crash reporting also requires a working network connection.

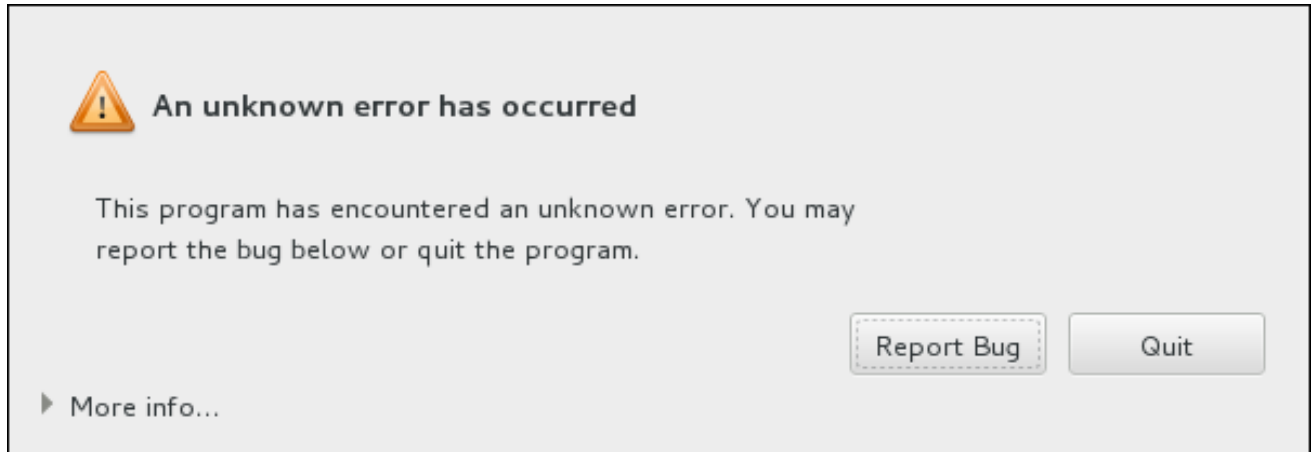


Figure 9.1. The Crash Reporting Dialog Box

When the dialog appears, select **Report Bug** to report the problem, or **Quit** to exit the installation.

Optionally, click **More Info** to display detailed output that can help determine the cause of the error. If you are familiar with debugging, click **Debug**. This will take you to virtual terminal **tty1**, where you can request more precise information that will enhance the bug report. To return to the graphical interface from **tty1**, use the **continue** command.

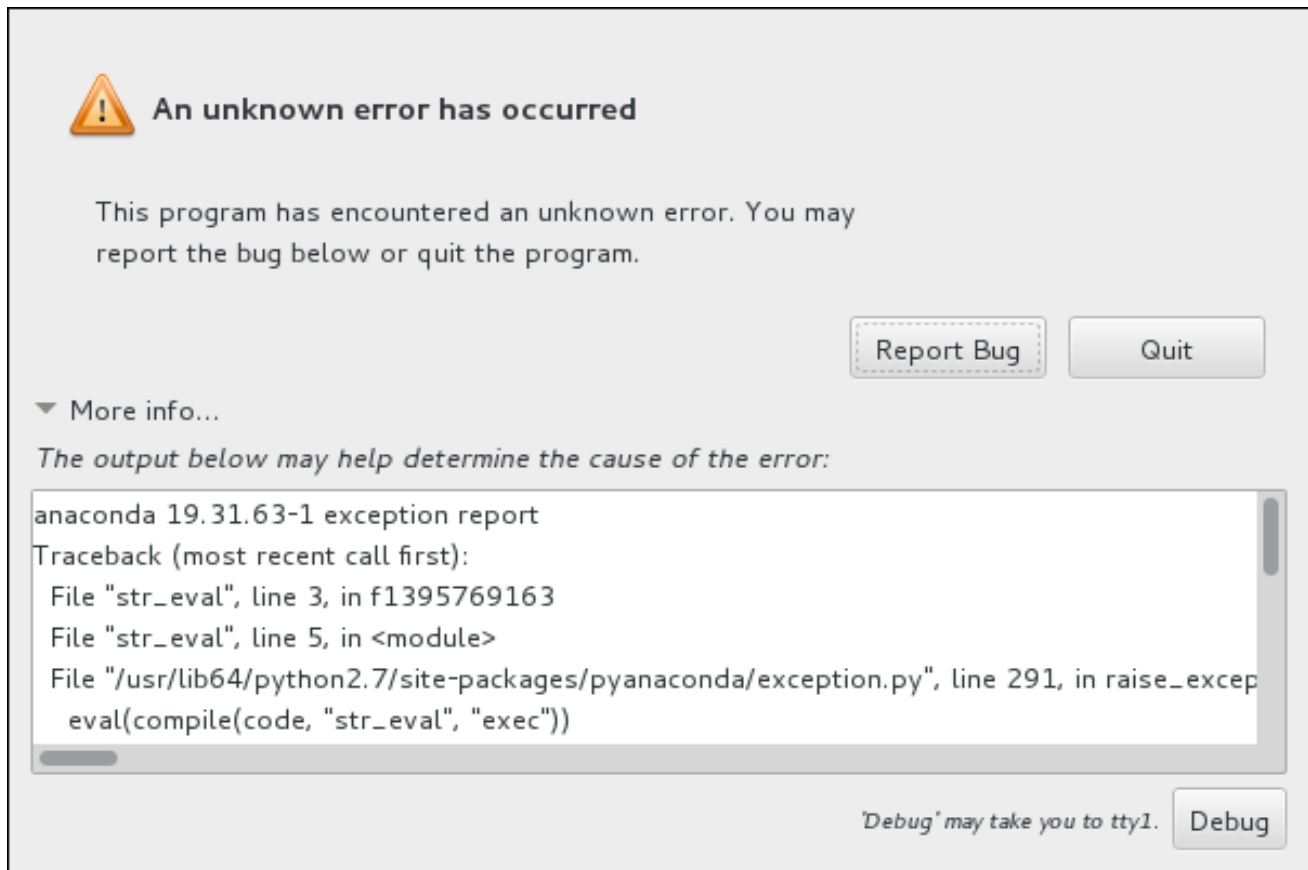


Figure 9.2. The Expanded Crash Reporting Dialog Box

If you want to report the bug to the customer portal, follow the procedure below.

Procedure 9.3. Reporting Errors to Red Hat Customer Support

1. In the menu that appears, select **Report a bug to Red Hat Customer Portal**.
2. To report the bug to Red Hat, you first need to provide your Customer Portal credentials. Click **Configure Red Hat Customer Support**.

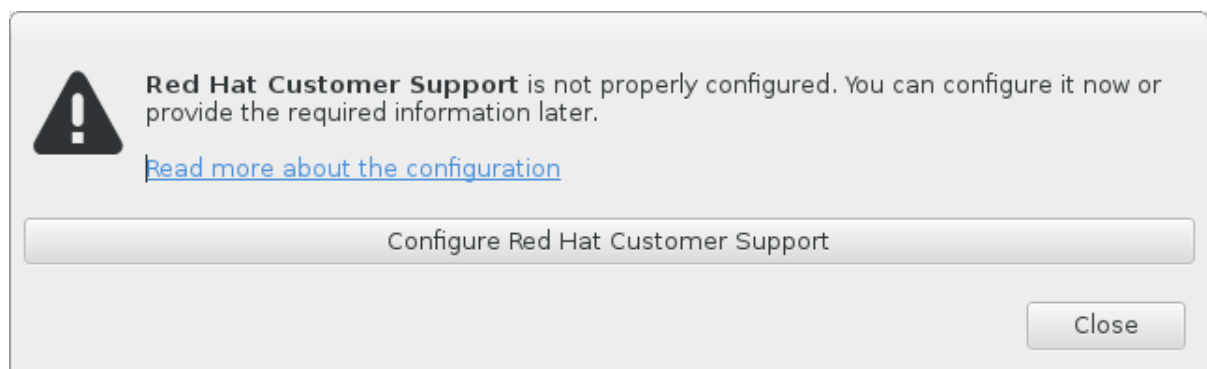
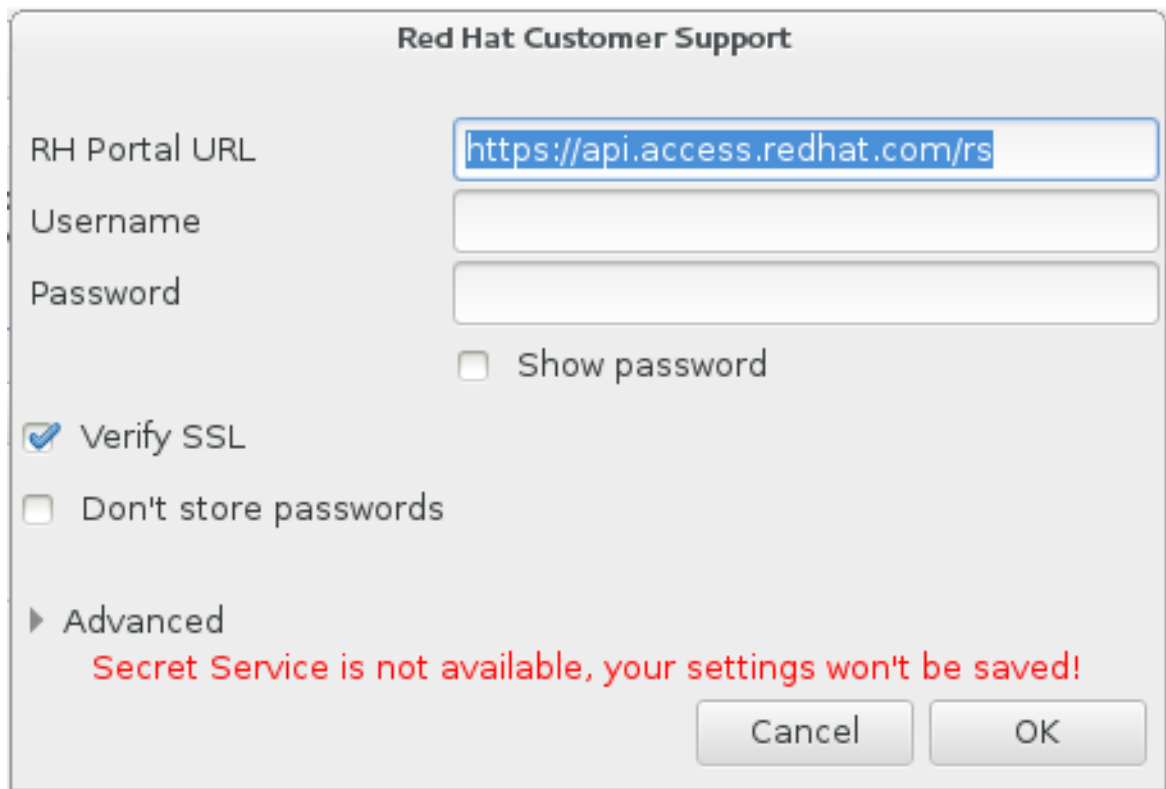


Figure 9.3. Customer Portal Credentials

3. A new window is now open, prompting you to enter your Customer Portal user name and password. Enter your Red Hat Customer Portal credentials.



The image shows a dialog box titled "Red Hat Customer Support". It contains several input fields and checkboxes. The "RH Portal URL" field is filled with "https://api.access.redhat.com/rs". The "Username" and "Password" fields are empty. There is a checkbox for "Show password" which is unchecked. Below these are two checkboxes: "Verify SSL" (checked) and "Don't store passwords" (unchecked). At the bottom left, there is a section titled "Advanced" with a right-pointing triangle icon. Below this section, a red error message reads: "Secret Service is not available, your settings won't be saved!". At the bottom right, there are two buttons: "Cancel" and "OK".

Figure 9.4. Configure Red Hat Customer Support

If your network settings require you to use a **HTTP** or **HTTPS** proxy, you can configure it by expanding the **Advanced** menu and entering the address of the proxy server.

When you put in all required credentials, click **OK** to proceed.

4. A new window appears, containing a text field. Write down any useful information and comments here. Describe how the error can be reproduced by explaining each step you took before the crash reporting dialog appeared. Provide as much relevant detail as possible, including any information you acquired when debugging. Be aware that the information you provide here can become publicly visible on the Customer Portal.

If you do not know what caused the error, check the box labeled **I don't know what caused this problem** at the bottom of the dialog.

Then, click **Forward**.

How did this problem happen (step-by-step)? How can it be reproduced? Any additional comments useful for diagnosing the problem? Please use English if possible.

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (stage1 on first disk). First disk is not used in partitioning section.

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the Kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to the second volume.

Your comments are not private. They may be included into publicly visible problem reports.

If you don't know how to describe it, you can [add a screencast](#)

☐ I don't know what caused this problem

Close Forward

Figure 9.5. Describe the Problem

- Next, review the information that will be sent to the Customer Portal. The explanation you provided is in the **comment** tab. Other tabs include such information as your system's host name and other details about the installation environment. You can remove any items you do not want sent to Red Hat, but be aware that providing less detail can affect the investigation of the issue.

Click **Forward** when you finish reviewing the information to be sent.

Please review the data before it gets reported. Depending on reporter chosen, it may end up publicly visible.

environ cmdline backtrace hostname **comment** reason

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the Kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to t

Close Forward

Figure 9.6. Review the Data to Be Sent

- Review the list of files that will be sent and included in the bug report as individual attachments.

These files provide system information that will assist the investigation. If you do not want to send certain files, uncheck the box next to each one. To provide additional files that can help find the problem, click **Attach a file**.

Once you have reviewed the files to be sent, check the box labeled **I have reviewed the data and agree with submitting it**. Then, click **Forward** to send the report and attachments to the Customer Portal.

Include	Name	Value
☒	anaconda.log	(click here to view/edit)
☒	packaging.log	(click here to view/edit)
☒	os_info	(click here to view/edit)
☒	environ	(click here to view/edit)
☒	last_occurrence	1395769224
☒	anaconda-tb	(click here to view/edit)
☒	version	7.0
☒	cmdline	/usr/bin/python /sbin/anaconda
☒	package	anaconda-19.31.63-1
☒	duphash	aeb4ed3992aab62d1cd03d3b1e0a89c79753f34da115a739be021d2873d0a131
☒	lsblk_output	(click here to view/edit)
☒	pkg_version	19.31.63
☒	product	Red Hat Enterprise Linux
☒	release	Red Hat Enterprise Linux Workstation release 7.0 Beta (Maipo)
☒	pkg_arch	x86_64

Attach a file

☐ I reviewed the data and agree with submitting it

Close Forward

Figure 9.7. Review the Files to Be Sent

- When the dialog reports that processing has finished, you can click **Show log** to view details of the reporting process or **Close** to return to the initial crash reporting dialog box. There, click **Quit** to exit the installation.

9.2.3. Creating Pre-installation Log Files

To debug installation problems you can set the **inst.debug** option to create log files from the environment before the installation starts. These log files contain, for example, the current storage configuration.

To set the option in the Red Hat Enterprise Linux installation boot menu:

- Select the **Install Red Hat Enterprise Linux 7.3** entry.
- Press the **Tab** key to edit the boot options.
- Append **inst.debug** to the options. For example:

```
> vmlinuz ... inst.debug
```

For further details, see [Chapter 23, Boot Options](#).

- Press **Enter** to start the setup.

The system stores the pre-installation log files in the `/tmp/pre-anaconda-logs/` directory before Anaconda starts. To access the log files:

1. Switch to the console. See [Section 8.2.1, “Accessing Consoles”](#).
2. Change into the `/tmp/pre-anaconda-logs/` directory:

```
# cd /tmp/pre-anaconda-logs/
```

9.3. PROBLEMS AFTER INSTALLATION

9.3.1. Are You Unable to Boot With Your RAID Card?

If you have performed an installation and cannot boot your system properly, you might need to reinstall and partition your system's storage differently.

Some BIOS types do not support booting from RAID cards. After you finish the installation and reboot the system for the first time, a text-based screen showing the boot loader prompt (for example, **grub>**) and a flashing cursor might be all that appears. If this is the case, you must repartition your system and move your **/boot** partition and the boot loader outside the RAID array. The **/boot** partition and the boot loader must be on the same drive.

Once these changes have been made, you should be able to finish your installation and boot the system properly. For more information about partitioning, see [Section 8.14, “Installation Destination”](#).

9.3.2. Trouble With the Graphical Boot Sequence

After you finish the installation and reboot your system for the first time, it is possible that the system stops responding during the graphical boot sequence, requiring a reset. In this case, the boot loader is displayed successfully, but selecting any entry and attempting to boot the system results in a halt. This usually means a problem with the graphical boot sequence; to solve this issue, you must disable graphical boot. To do this, temporarily alter the setting at boot time before changing it permanently.

Procedure 9.4. Disabling Graphical Boot Temporarily

1. Start your computer and wait until the boot loader menu appears. If you set your boot loader timeout period to 0, hold down the **Esc** key to access it.
2. When the boot loader menu appears, use your cursor keys to highlight the entry you want to boot and press the **e** key to edit this entry's options.
3. In the list of options, find the kernel line - that is, the line beginning with the keyword **linux** (or, in some cases, **linux16** or **linuxefi**). On this line, locate the **rhgb** option and delete it. The option might not be immediately visible; use the cursor keys to scroll up and down.
4. Press **F10** or **Ctrl+X** to boot your system with the edited options.

If the system started successfully, you can log in normally. Then you will need to disable the graphical boot permanently - otherwise you will have to perform the previous procedure every time the system boots. To permanently change boot options, do the following.

Procedure 9.5. Disabling Graphical Boot Permanently

1. Log in to the **root** account using the **su -** command:


```
$ su -
```

2. Use the **grubby** tool to find the default **GRUB2** kernel:

```
# grubby --default-kernel  
/boot/vmlinuz-3.10.0-229.4.2.el7.x86_64
```

3. Use the **grubby** tool to remove the **rhgb** boot option from the default kernel, identified in the last step, in your **GRUB2** configuration. For example:

```
# grubby --remove-args="rhgb" --update-kernel /boot/vmlinuz-3.10.0-229.4.2.el7.x86_64
```

After you finish this procedure, you can reboot your computer. Red Hat Enterprise Linux will not use the graphical boot sequence any more. If you want to enable graphical boot in the future, follow the same procedure, replacing the **--remove-args="rhgb"** parameter with the **--args="rhgb"** parameter. This will restore the **rhgb** boot option to the default kernel in your **GRUB2** configuration.

See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about working with the **GRUB2** boot loader.

9.3.3. Booting into a Graphical Environment

If you have installed the **X Window System** but are not seeing a graphical desktop environment once you log into your system, you can start it manually using the **startx** command. Note, however, that this is just a one-time fix and does not change the log in process for future log ins.

To set up your system so that you can log in at a graphical login screen, you must change the default **systemd** target to **graphical.target**. When you are finished, reboot the computer. You will be presented with a graphical login prompt after the system restarts.

Procedure 9.6. Setting Graphical Login as Default

1. Open a shell prompt. If you are in your user account, become root by typing the **su -** command.
2. Change the default target to **graphical.target**. To do this, execute the following command:

```
# systemctl set-default graphical.target
```

Graphical login is now enabled by default – you will be presented with a graphical login prompt after the next reboot. If you want to reverse this change and keep using the text-based login prompt, execute the following command as **root**:

```
# systemctl set-default multi-user.target
```

For more information about targets in **systemd**, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

9.3.4. No Graphical User Interface Present

If you are having trouble getting **X** (the **X Window System**) to start, it is possible that it has not been installed. Some of the preset base environments you can select during the installation, such as **Minimal install** or **Web Server**, do not include a graphical interface – it has to be installed manually.

If you want **X**, you can install the necessary packages afterwards. See the Knowledgebase article at <https://access.redhat.com/site/solutions/5238> for information on installing a graphical desktop environment.

9.3.5. X Server Crashing After User Logs In

If you are having trouble with the **X** server crashing when a user logs in, one or more of your file systems can be full or nearly full. To verify that this is the problem you are experiencing, execute the following command:

```
$ df -h
```

The output will help you diagnose which partition is full – in most cases, the problem will be on the **/home** partition. The following is a sample output of the **df** command:

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/vg_rhel-root	20G	6.0G	13G	32%	/
devtmpfs	1.8G	0	1.8G	0%	/dev
tmpfs	1.8G	2.7M	1.8G	1%	/dev/shm
tmpfs	1.8G	1012K	1.8G	1%	/run
tmpfs	1.8G	0	1.8G	0%	/sys/fs/cgroup
tmpfs	1.8G	2.6M	1.8G	1%	/tmp
/dev/sda1	976M	150M	760M	17%	/boot
/dev/dm-4	90G	90G	0	100%	/home

In the above example, you can see that the **/home** partition is full, which causes the crash. You can make some room on the partition by removing unneeded files. After you free up some disk space, start **X** using the **startx** command.

For additional information about **df** and an explanation of the options available (such as the **-h** option used in this example), see the **df(1)** man page.

9.3.6. Is Your RAM Not Being Recognized?

In some cases the kernel does not recognize all of your memory (RAM), which causes the system to use less memory than is installed. You can find out how much RAM is being utilized using the **free -m** command. If the displayed total amount of memory does not match your expectations, it is likely that at least one of your memory modules is faulty. On BIOS-based systems, you can use the **Memtest86+** utility to test your system's memory – see [Section 23.2.1, “Loading the Memory \(RAM\) Testing Mode”](#) for details.



NOTE

Some hardware configurations have a part of the system's RAM reserved and unavailable to the main system. Notably, laptop computers with integrated graphics cards will reserve some memory for the GPU. For example, a laptop with 4 GiB of RAM and an integrated Intel graphics card will show only roughly 3.7 GiB of available memory.

Additionally, the **kdump** crash kernel dumping mechanism, which is enabled by default on most Red Hat Enterprise Linux systems, reserves some memory for the secondary kernel used in case of the primary kernel crashing. This reserved memory will also not be displayed as available when using the **free** command. For details about **kdump** and its memory requirements, see the [Red Hat Enterprise Linux 7 Kernel Crash Dump Guide](#).

If you made sure that your memory does not have any issues, you can try and set the amount of memory manually using the **mem=** kernel option.

Procedure 9.7. Configuring the Memory Manually

1. Start your computer and wait until the boot loader menu appears. If you set your boot loader timeout period to 0, hold down the **Esc** key to access it.
2. When the boot loader menu appears, use your cursor keys to highlight the entry you want to boot and press the **e** key to edit this entry's options.
3. In the list of options, find the kernel line - that is, the line beginning with the keyword **linux** (or, in some cases, **linux16**). Append the following option to the end of this line:

```
mem=xxM
```

Replace *xx* with the amount of RAM you have in MiB.

4. Press **F10** or **Ctrl+X** to boot your system with the edited options.
5. Wait for the system to boot and log in. Then, open a command line and execute the **free -m** command again. If total amount of RAM displayed by the command matches your expectations, append the following to the line beginning with **GRUB_CMDLINE_LINUX** in the **/etc/default/grub** file to make the change permanent:

```
mem=xxM
```

Replace *xx* with the amount of RAM you have in MiB.

6. After you updated the file and saved it, refresh the boot loader configuration so that the change will take effect. Run the following command with root privileges:

```
# grub2-mkconfig --output=/boot/grub2/grub.cfg
```

In **/etc/default/grub**, the above example would look similar to the following:

```
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR="$(sed 's, release.*$,g' /etc/system-release)"
GRUB_DEFAULT=saved
GRUB_DISABLE_SUBMENU=true
GRUB_TERMINAL_OUTPUT="console"
GRUB_CMDLINE_LINUX="rd.lvm.lv=rhel/root vconsole.font=latarcyrheb-sun16 rd.lvm.lv=rhel/swap
$([ -x /usr/sbin/rhcrashkernel.param ] && /usr/sbin/rhcrashkernel-param || :) vconsole.keymap=us
rhgb quiet mem=1024M"
GRUB_DISABLE_RECOVERY="true"
```

See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about working with the **GRUB2** boot loader.

9.3.7. Is Your System Displaying Signal 11 Errors?

A signal 11 error, commonly known as a *segmentation fault*, means that a program accessed a memory location that was not assigned to it. A signal 11 error can occur due to a bug in one of the software programs that is installed, or faulty hardware.

If you receive a fatal signal 11 error during the installation, first make sure you are using the most recent installation images, and let **Anaconda** verify them to make sure they are not corrupted. Bad installation media (such as an improperly burned or scratched optical disk) are a common cause of signal 11 errors. Verifying the integrity of the installation media is recommended before every installation.

For information about obtaining the most recent installation media, see [Chapter 2, Downloading Red Hat Enterprise Linux](#). To perform a media check before the installation starts, append the **rd.live.check** boot option at the boot menu. See [Section 23.2.2, “Verifying Boot Media”](#) for details.

If you performed a media check without any errors and you still have issues with segmentation faults, it usually means that your system encountered a hardware error. In this case, the problem is most likely in the system's memory (RAM). This can be a problem even if you previously used a different operating system on the same computer without any errors. On BIOS-based systems, you can use the **Memtest86+** memory testing module included on the installation media to perform a thorough test of your system's memory. See [Section 23.2.1, “Loading the Memory \(RAM\) Testing Mode”](#) for details.

Other possible causes are beyond this document's scope. Consult your hardware manufacturer's documentation and also see the *Red Hat Hardware Compatibility List*, available online at <https://hardware.redhat.com>.

PART II. IBM POWER SYSTEMS - INSTALLATION AND BOOTING

This part of the *Red Hat Enterprise Linux Installation Guide* includes information about installation and basic post-installation troubleshooting for IBM Power Systems servers. IBM Power Systems servers include IBM PowerLinux servers and POWER7, POWER8, and POWER9 Power Systems servers running Linux. For advanced installation options, see [Part IV, “Advanced Installation Options”](#).



IMPORTANT

Previous releases of Red Hat Enterprise Linux supported 32-bit and 64-bit Power Systems servers (**ppc** and **ppc64**, respectively). Red Hat Enterprise Linux 7 supports only 64-bit Power Systems servers (**ppc64**).

CHAPTER 10. PLANNING FOR INSTALLATION ON IBM POWER SYSTEMS

This chapter outlines the decisions and preparations you will need to make when deciding how to proceed with the installation.

10.1. UPGRADE OR INSTALL?

While automated in-place upgrades are now supported, the support is currently limited to AMD64 and Intel 64 systems. If you have an existing installation of a previous release of Red Hat Enterprise Linux on an IBM Power Systems server, you must perform a clean install to migrate to Red Hat Enterprise Linux 7. A clean install is performed by backing up all data from the system, formatting disk partitions, performing an installation of Red Hat Enterprise Linux from installation media, and then restoring any user data.

10.2. IS YOUR HARDWARE COMPATIBLE?

Red Hat Enterprise Linux 7 (big endian) is compatible with IBM Power Systems servers which use the POWER7, POWER8, and POWER9 processor series. POWER6 processors and older are no longer supported.

Red Hat Enterprise Linux also offers a little endian variant for IBM Power Systems. This variant is currently compatible with POWER8 and POWER9 processors, and is supported as a KVM guest on Red Hat Enterprise Virtualization for Power, on PowerVM, and PowerNV (bare metal).

The most recent list of supported hardware can be found in the *Red Hat Hardware Compatibility List*, available online at <https://access.redhat.com/ecosystem/search/#/category/Server>. Also see [Red Hat Enterprise Linux technology capabilities and limits](#) for general information about system requirements.

10.3. IBM INSTALLATION TOOLS

IBM Installation Toolkit is an optional utility that speeds up the installation of Linux on IBM Power Systems and is especially helpful for those unfamiliar with Linux. You can use the **IBM Installation Toolkit** to: ^[1]

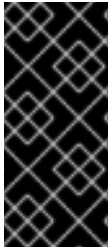
- Install and configure Linux on a non-virtualized IBM Power Systems server.
- Install and configure Linux on servers with previously-configured logical partitions (LPARs, also known as virtualized servers).
- Install IBM service and productivity tools on a new or previously installed Linux system. The IBM service and productivity tools include dynamic logical partition (DLPAR) utilities.
- Upgrade system firmware level on IBM Power Systems servers.
- Perform diagnostics or maintenance operations on previously installed systems.
- Migrate a LAMP server (software stack) and application data from a System x to a System p system. A LAMP server is a bundle of open source software. LAMP is an acronym for Linux, **Apache HTTP Server**, **MySQL** relational database, and the PHP (or sometimes Perl, or Python) language.

Documentation for the **IBM Installation Toolkit** for PowerLinux is available in the Linux Information Center at <http://publib.boulder.ibm.com/infocenter/lnxinfo/v3r0m0/topic/liaan/powerpack.htm>

PowerLinux service and productivity tools is an optional set of tools that include hardware service diagnostic aids, productivity tools, and installation aids for Linux operating systems on IBM servers based on POWER7, POWER6, POWER5, and POWER4 technology.

Documentation for the service and productivity tools is available in the Linux Information Center at <http://publib.boulder.ibm.com/infocenter/lnxinfo/v3r0m0/topic/liaau/liaauraskickoff.htm>

10.4. PREPARATION FOR IBM POWER SYSTEMS SERVERS



IMPORTANT

Ensure that the real-base boot parameter is set to **c00000**, otherwise you might see errors such as:

DEFAULT CATCH!, exception-handler=fff00300

IBM Power Systems servers offer many options for partitioning, virtual or native devices, and consoles.

If you are using a non-partitioned system, you do not need any pre-installation setup. For systems using the HVSI serial console, hook up your console to the T2 serial port.

If using a partitioned system the steps to create the partition and start the installation are largely the same. You should create the partition at the HMC and assign some CPU and memory resources, as well as SCSI and Ethernet resources, which can be either virtual or native. The HMC create partition wizard steps you through the creation.

For more information on creating the partition, see the *Partitioning for Linux with an HMC* PDF in the IBM Systems Hardware Information Center at:

http://publib.boulder.ibm.com/infocenter/powersys/v3r1m5/topic/iphbi_p5/iphbibook.pdf

If you are using virtual SCSI resources, rather than native SCSI, you must configure a 'link' to the virtual SCSI serving partition, and then configure the virtual SCSI serving partition itself. You create a 'link' between the virtual SCSI client and server slots using the HMC. You can configure a virtual SCSI server on either Virtual I/O Server (VIOS) or IBM i, depending on which model and options you have.

If you are installing using Intel iSCSI Remote Boot, all attached iSCSI storage devices must be disabled. Otherwise, the installation will succeed but the installed system will not boot.

For more information on using virtual devices, see the IBM Redbooks publication *Virtualizing an Infrastructure with System p and Linux* at: <http://publib-b.boulder.ibm.com/abstracts/sg247499.html>

Once you have your system configured, you need to Activate from the HMC or power it on. Depending on the type of installation, you need to configure SMS to correctly boot the system into the installation program.

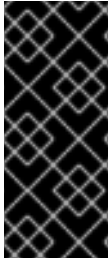
10.5. SUPPORTED INSTALLATION TARGETS

An installation target is a storage device that will store Red Hat Enterprise Linux and boot the system. Red Hat Enterprise Linux supports the following installation targets for AMD64 and Intel 64 systems:

- Storage connected by a standard internal interface, such as SCSI, SATA, or SAS
- Fibre Channel Host Bus Adapters and multipath devices. Some can require vendor-provided drivers.

- Virtualized installation on IBM Power Systems servers is also supported when using Virtual SCSI (vSCSI) adapters in virtual client LPARs

Red Hat does not support installation to USB drives or SD memory cards. For information about the support for third-party virtualization technologies, see the *Red Hat Hardware Compatibility List*, available online at <https://hardware.redhat.com>.



IMPORTANT

On IBM Power Systems servers, the eHEA module fails to initialize if 16GB *huge pages* are assigned to a system or partition and the kernel command line does not contain the huge page parameters. Therefore, when you perform a network installation through an IBM eHEA ethernet adapter, you cannot assign huge pages to the system or partition during the installation. Use *large pages* instead.

10.6. SYSTEM SPECIFICATIONS LIST

The installation program automatically detects and installs your computer's hardware and you do not usually need to supply the installation program with any specific details about your system. However, when performing certain types of installation, it is important to know specific details about your hardware. For this reason, it is recommended that you record the following system specifications for reference during the installation, depending on your installation type.

- If you plan to use a customized partition layout, record:
 - The model numbers, sizes, types, and interfaces of the hard drives attached to the system. For example, Seagate ST3320613AS 320 GB on SATA0, Western Digital WD7500AAKS 750 GB on SATA1. This will allow you to identify specific hard drives during the partitioning process.
- If you are installing Red Hat Enterprise Linux as an additional operating system on an existing system, record:
 - Information about the partitions used on the system. This information can include file system types, device node names, file system labels, and sizes. This will allow you to identify specific partitions during the partitioning process. Remember that different operating systems identify partitions and drives differently, therefore even if the other operating system is a Unix operating system, the device names can be reported by Red Hat Enterprise Linux differently. This information can usually be found by executing the equivalent of the **mount** command and **blkid** command and in the **/etc/fstab** file.

If you have other operating systems already installed, the Red Hat Enterprise Linux 7 installation program attempts to automatically detect and configure to boot them. You can manually configure any additional operating systems if they are not detected properly. For more information, see [Section 13.15.1, “Boot Loader Installation”](#).

- If you plan to install from an image on a local hard drive:
 - The hard drive and directory that holds the image.
- If you plan to install from a network location:
 - The make and model numbers of the network adapters on your system. For example, Netgear GA311. This will allow you to identify adapters when manually configuring the network.

- IP, DHCP, and BOOTP addresses
- Netmask
- Gateway IP address
- One or more name server IP addresses (DNS)
- The location of the installation source on an FTP server, HTTP (web) server, HTTPS (web) server, or NFS server.

If any of these networking requirements or terms are unfamiliar to you, contact your network administrator for assistance.

- If you plan to install on an iSCSI target:
 - The location of the iSCSI target. Depending on your network, you might also need a CHAP user name and password, and perhaps a reverse CHAP user name and password.
- If your computer is part of a domain:
 - You should verify that the domain name will be supplied by the DHCP server. If not, you will need to input the domain name manually during installation.

10.7. DISK SPACE AND MEMORY REQUIREMENTS

Red Hat Enterprise Linux, like most current operating systems, uses *disk partitions*. When you install Red Hat Enterprise Linux, you have to work with disk partitions. For more information about disk partitions, see [Appendix A, An Introduction to Disk Partitions](#).

The disk space used by Red Hat Enterprise Linux must be separate from the disk space used by other operating systems you might have installed on your system.



NOTE

For IBM Power Systems servers, at least three partitions (`/`, **swap** and a **PReP** boot partition) must be dedicated to Red Hat Enterprise Linux.

To install Red Hat Enterprise Linux you must have a minimum of 10 GB of space in either unpartitioned disk space or in partitions which can be deleted. For more information on partition and disk space recommendations, see the recommended partitioning sizes discussed in [Section 13.15.4.4, "Recommended Partitioning Scheme"](#).

Red Hat Enterprise Linux requires minimum the following amount of RAM:

Installation type	Minimum required RAM
Local media installation (USB, DVD)	1,280 MiB
NFS network installation	1,280 MiB
HTTP, HTTPS, or FTP network installation	1,664 MiB

Installing Red Hat Enterprise Linux using a Kickstart file has the same minimum RAM requirements as a manual installation. However, if you use a Kickstart file that runs commands which require additional memory or write data to the RAM disk, additional RAM might be necessary.

For more information about the minimum requirements and technology limits of Red Hat Enterprise Linux 7, see the [Red Hat Enterprise Linux technology capabilities and limits](#) article on the Red Hat Customer Portal.

10.8. RAID AND OTHER DISK DEVICES

Some storage technology requires special consideration when using Red Hat Enterprise Linux. Generally, it is important to understand how these technologies are configured, visible to Red Hat Enterprise Linux, and how support for them might have changed between major versions.

10.8.1. Hardware RAID

RAID (Redundant Array of Independent Disks) allows a group, or array, of drives to act as a single device. Configure any RAID functions provided by the mainboard of your computer, or attached controller cards, before you begin the installation process. Each active RAID array appears as one drive within Red Hat Enterprise Linux.

10.8.2. Software RAID

On systems with more than one hard drive, you can use the Red Hat Enterprise Linux installation program to operate several of the drives as a Linux software RAID array. With a software RAID array, RAID functions are controlled by the operating system rather than dedicated hardware. These functions are explained in detail in [Section 13.15.4, “Manual Partitioning”](#).



NOTE

When a pre-existing RAID array's member devices are all unpartitioned disks/drives, the installer will treat the array itself as a disk and will not provide a way to remove the array.

10.8.3. USB Disks

You can connect and configure external USB storage after installation. Most such devices are recognized by the kernel and available for use at that time.

Some USB drives might not be recognized by the installation program. If configuration of these disks at installation time is not vital, disconnect them to avoid potential problems.

10.9. CHOOSE AN INSTALLATION BOOT METHOD

You can use several methods to boot the Red Hat Enterprise Linux 7 installation program. The method you choose depends upon your installation media.



NOTE

Installation media must remain mounted throughout installation, including during execution of the **%post** section of a kickstart file.

Full installation DVD or USB drive

You can create bootable media from the full installation DVD ISO image. In this case, a single DVD or USB drive can be used to complete the entire installation – it will serve both as a boot device and as an installation source for installing software packages. See [Chapter 3, Making Media](#) for instructions on how to make a full installation DVD or USB drive.

Minimal boot CD, DVD or USB Flash Drive

A minimal boot CD, DVD or USB flash drive is created using a small ISO image, which only contains data necessary to boot the system and start the installation. If you use this boot media, you will need an additional installation source from which packages will be installed. See [Chapter 3, Making Media](#) for instructions on making boot CDs, DVDs and USB flash drives.

PXE Server

A *preboot execution environment* (PXE) server allows the installation program to boot over the network. After you boot the system, you complete the installation from a different installation source, such as a local hard drive or a location on a network. For more information on PXE servers, see [Chapter 24, Preparing for a Network Installation](#).

10.10. AUTOMATING THE INSTALLATION WITH KICKSTART

Red Hat Enterprise Linux 7 offers a way to partially or fully automate the installation process using a *Kickstart file*. Kickstart files contain answers to all questions normally asked by the installation program, such as what time zone do you want the system to use, how should the drives be partitioned or which packages should be installed. Providing a prepared Kickstart file at the beginning of the installation therefore allows you to perform the entire installation (or parts of it) automatically, without need for any intervention from the user. This is especially useful when deploying Red Hat Enterprise Linux on a large number of systems at once.

In addition to allowing you to automate the installation, Kickstart files also provide more options regarding software selection. When installing Red Hat Enterprise Linux manually using the graphical installation interface, your software selection is limited to pre-defined environments and add-ons. A Kickstart file allows you to install or remove individual packages as well.

For instructions about creating a Kickstart file and using it to automate the installation, see [Chapter 27, Kickstart Installations](#).

[1] Parts of this section were previously published at IBM's *Linux information for IBM systems* resource at http://publib.boulder.ibm.com/infocenter/lnxinfo/v3r0m0/index.jsp?topic=%2Fliaay%2Ftools_overview.htm

CHAPTER 11. UPDATING DRIVERS DURING INSTALLATION ON IBM POWER SYSTEMS

In most cases, Red Hat Enterprise Linux already includes drivers for the devices that make up your system. However, if your system contains hardware that has been released very recently, drivers for this hardware might not yet be included. Sometimes, a driver update that provides support for a new device might be available from Red Hat or your hardware vendor on a *driver disc* that contains *RPM packages*. Typically, the driver disc is available for download as an *ISO image file*.



IMPORTANT

Driver updates should only be performed if a missing driver prevents you to complete the installation successfully. The drivers included in the kernel should always be preferred over drivers provided by other means.

Often, you do not need the new hardware during the installation process. For example, if you use a DVD to install to a local hard drive, the installation will succeed even if drivers for your network card are not available. In such a situation, complete the installation and add support for the new hardware afterward - see [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for details of adding this support.

In other situations, you might want to add drivers for a device during the installation process to support a particular configuration. For example, you might want to install drivers for a network device or a storage adapter card to give the installation program access to the storage devices that your system uses. You can use a driver disc to add this support during installation in one of two ways:

1. place the ISO image file of the driver disc in a location accessible to the installation program, on a local hard drive, on a USB flash drive, or on a CD or DVD.
2. create a driver disc by extracting the image file onto a CD or a DVD, or a USB flash drive. See the instructions for making installation discs in [Section 3.1, "Making an Installation CD or DVD"](#) for more information on burning ISO image files to a CD or DVD, and [Section 3.2, "Making Installation USB Media"](#) for instructions on writing ISO images to USB drives.

If Red Hat, your hardware vendor, or a trusted third party told you that you will require a driver update during the installation process, choose a method to supply the update from the methods described in this chapter and test it before beginning the installation. Conversely, do not perform a driver update during installation unless you are certain that your system requires it. The presence of a driver on a system for which it was not intended can complicate support.



WARNING

Driver update disks sometimes disable conflicting kernel drivers, where necessary. In rare cases, unloading a kernel module in this way can cause installation errors.

11.1. PREPARING FOR A DRIVER UPDATE DURING INSTALLATION

If a driver update is necessary and available for your hardware, Red Hat, your hardware vendor, or another trusted third party will typically provide it in the form of an image file in ISO format. Once you obtain the ISO image, you must decide on the method you want to use to perform the driver update.

The available methods are:

Automatic driver update

When starting the installation, the **Anaconda** installation program will attempt to detect all attached storage devices. If there is a storage device labeled **OEMDRV** present when the installation begins, **Anaconda** will always treat it like a driver update disc and attempt to load drivers present on it.

Assisted driver update

You can specify the **inst.dd** boot option when starting the installation. If you use this option without any parameters, **Anaconda** will display a list of all storage devices connected to the system, and it will prompt you to select a device which contains a driver update.

Manual driver update

You can specify the **inst.dd=location** boot option when starting the installation, where *location* is the path to a driver update disc or ISO image. When you specify this option, **Anaconda** will attempt to load any driver updates it finds at the specified location. With manual driver updates, you can specify either locally available storage devices, or a network location (an **HTTP**, **HTTPS** or **FTP** server).



NOTE

You can also use both **inst.dd=location** and **inst.dd** at the same time. However, what **Anaconda** does in this case depends on the type of *location* that you use. If it is a device, **Anaconda** prompts you to select drivers to update from the specified device and then it offers you additional devices. If *location* is a network location, **Anaconda** first prompts you to select a device containing a driver update and then it lets you update drivers from the specified network location.

If you want to use the automatic driver update method, you must create a storage device labeled **OEMDRV**, and it must be physically connected to the installation system. To use the assisted method, you can use any local storage device any label other than **OEMDRV**. To use the manual method, you can use any local storage with a different label, or a network location accessible from the installation system.



IMPORTANT

Make sure to initialize the network using the **ip=** option when loading a driver update from a network location. See [Section 23.1, “Configuring the Installation System at the Boot Menu”](#) for details.

11.1.1. Preparing to Use a Driver Update Image File on Local Storage

If you use a local storage device to provide the ISO file, such as a hard drive or USB flash drive, you can make the installation program to recognize it automatically by properly labeling the device. Only if it is not possible, install the update manually as described below.

- In order for the installation program to automatically recognize the driver disk, the volume label of the storage device must be **OEMDRV**. Also, you will need to extract the contents of the ISO image file to the root directory of the storage device rather than copy the ISO image itself. See [Section 11.2.1, “Automatic Driver Update”](#). Note that installation of a driver from a device labeled **OEMDRV** is always recommended and preferable to the manual installation.
- For manual installation, simply copy the ISO image, as a single file, onto the storage device. You can rename the file if you find it helpful but you must not change the file name extension, which

must remain **.iso**, for example **dd.iso**. See [Section 11.2.2, “Assisted Driver Update”](#) to learn how to select the driver update manually during installation.

11.1.2. Preparing a Driver Disc

You can create a driver update disc on a CD or DVD. See [Section 3.1, “Making an Installation CD or DVD”](#) to learn more about burning discs from image files.

After you burn a driver update disc CD or DVD, verify that the disc was created successfully by inserting it into your system and browsing to it using the file manager. You should see a single file named **rhdd3**, which is a signature file that contains the driver disc's description, and a directory named **rpms**, which contains the RPM packages with the actual drivers for various architectures.

If you see only a single file ending in **.iso**, then you have not created the disc correctly and should try again. Ensure that you choose an option similar to **Burn from Image** if you use a Linux desktop other than **GNOME**, or if you use a different operating system.

11.2. PERFORMING A DRIVER UPDATE DURING INSTALLATION

At the very beginning of the installation process, you can perform a driver update in the following ways:

- let the installation program automatically find and offer a driver update for installation,
- let the installation program prompt you to locate a driver update,
- manually specify a path to a driver update image or an RPM package.



IMPORTANT

Always make sure to put your driver update discs on a standard disk partition. Advanced storage, such as RAID or LVM volumes, might not be accessible during the early stage of the installation when you perform driver updates.

11.2.1. Automatic Driver Update

To have the installation program automatically recognize a driver update disc, connect a block device with the **OEMDRV** volume label to your computer before starting the installation process.



NOTE

Starting with Red Hat Enterprise Linux 7.2, you can also use the **OEMDRV** block device to automatically load a Kickstart file. This file must be named **ks.cfg** and placed in the root of the device to be loaded. See [Chapter 27, Kickstart Installations](#) for more information about Kickstart installations.

When the installation begins, the installation program detects all available storage connected to the system. If it finds a storage device labeled **OEMDRV**, it will treat it as a driver update disc and attempt to load driver updates from this device. You will be prompted to select which drivers to load:

```
DD: Checking devices /dev/sr1
DD: Checking device /dev/sr1
DD: Processing DD repo /media/DD//rpms/x86_64 on /dev/sr1

Page 1 of 1
Select drivers to install
  1) [ ] /media/DD//rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue:
```

Figure 11.1. Selecting a Driver

Use number keys to toggle selection on individual drivers. When ready, press **c** to install the selected drivers and proceed to the **Anaconda** graphical user interface.

11.2.2. Assisted Driver Update

It is always recommended to have a block device with the **OEMDRV** volume label available to install a driver during installation. However, if no such device is detected and the **inst.dd** option was specified at the boot command line, the installation program lets you find the driver disk in interactive mode. In the first step, select a local disk partition from the list for **Anaconda** to scan for ISO files. Then, select one of the detected ISO files. Finally, select one or more available drivers. The image below demonstrates the process in the text user interface with individual steps highlighted.

```

Starting Driver Update Disk UI on tty1...
DD: Checking devices

Page 1 of 1
Driver disk device selection
  DEVICE      TYPE  LABEL      UUID
  1) vda1      ext2  HOME       8c9d0c6e-4fea-4910-9bac-6609bc8ff847
  2) vda2      xfs   DD_PART    9dcc606d-a9ca-41d1-98b5-e9411769e37f
  3) vdb1      ext4  DD_PART    dd69ffa5-c72e-4b61-ae39-0197d6960fc3

# to select, 'n'-next page, 'p'-previous page or 'c'-continue: 3
[ 97.268612] EXT4-fs (vdb1): mounted filesystem without journal. Opts: (null)

Page 1 of 1
Choose driver disk ISO file
  1) dd.iso

# to select, 'n'-next page, 'p'-previous page or 'c'-continue: 1
DD: Checking device /media/DD-search/dd.iso
[ 112.233480] loop: module loaded
DD: Processing DD repo /media/DD/rpms/x86_64 on /media/DD-search/dd.iso

Page 1 of 1
Select drivers to install
  1) [ ] /media/DD/rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue: 1

Page 1 of 1
Select drivers to install
  1) [x] /media/DD/rpms/x86_64/kmod_e10.rpm

# to toggle selection, 'n'-next page, 'p'-previous page or 'c'-continue: _

```

Figure 11.2. Selecting a Driver Interactively

**NOTE**

If you extracted your ISO image file and burned it on a CD or DVD but the media does not have the **OEMDRV** volume label, either use the **inst.dd** option with no arguments and use the menu to select the device, or use the following boot option for the installation program to scan the media for drivers:

```
inst.dd=/dev/sr0
```

Hit number keys to toggle selection on individual drivers. When ready, press **c** to install the selected drivers and proceed to the **Anaconda** graphical user interface.

11.2.3. Manual Driver Update

For manual driver installation, prepare an ISO image file containing your drivers to an accessible location, such a USB flash drive or a web server, and connect it to your computer. At the welcome screen, hit **Tab** to display the boot command line and append the **inst.dd=location** to it, where *location* is a path to the driver update disc:

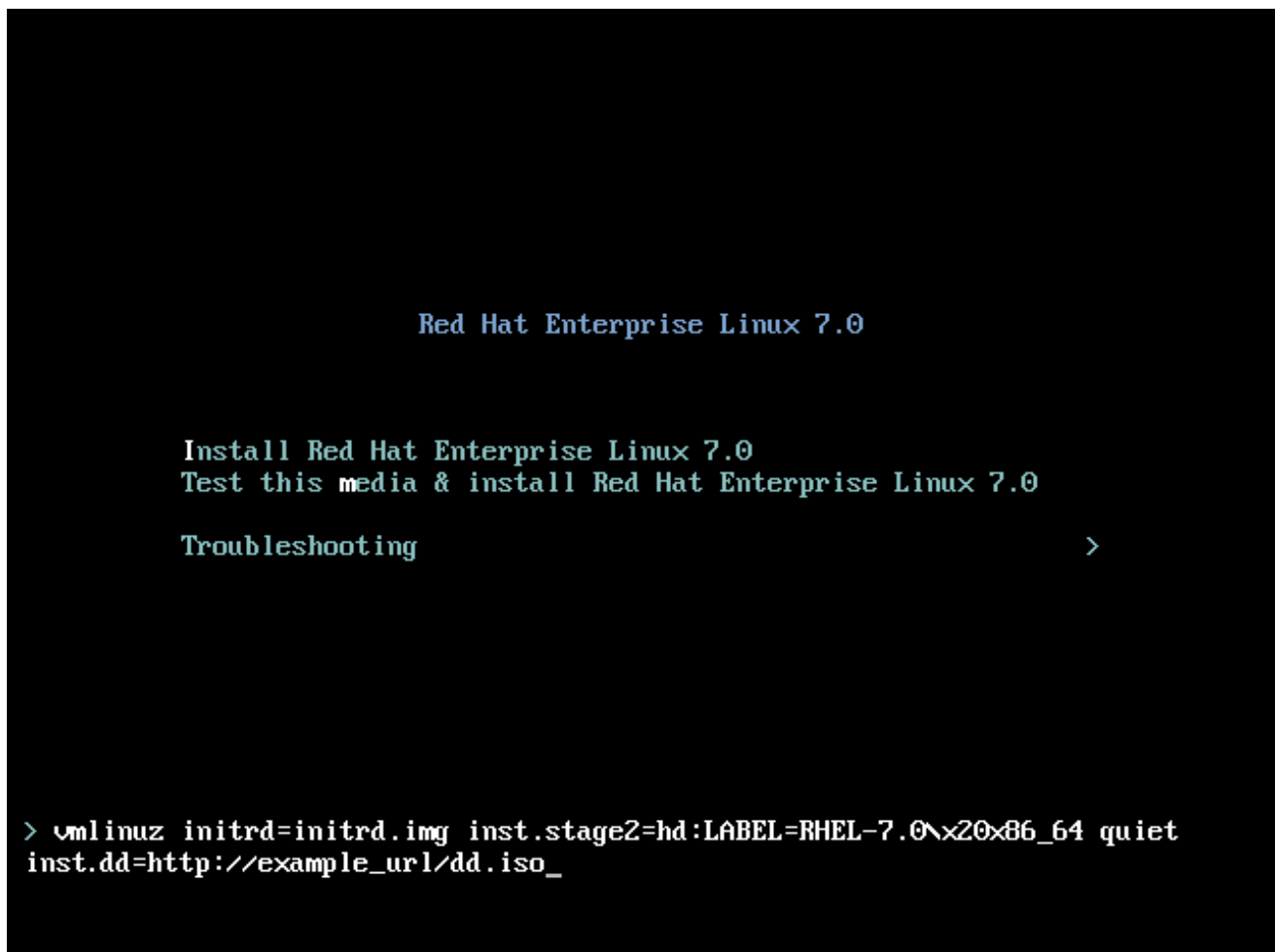


Figure 11.3. Specifying a Path to a Driver Update

Typically, the image file is located on a web server (for example, *http://server.example.com/dd.iso*) or on a USB flash drive (for example, */dev/sdb1*). It is also possible to specify an RPM package containing the driver update (for example *http://server.example.com/dd.rpm*).

When ready, hit **Enter** to execute the boot command. Then, your selected drivers will be loaded and the installation process will proceed normally

11.2.4. Blacklisting a Driver

A malfunctioning driver can prevent a system from booting normally during installation. When this happens, you can disable (or blacklist) the driver by customizing the boot command line. At the boot menu, display the boot command line by hitting the **Tab** key. Then, append the **modprobe.blacklist=driver_name** option to it. Replace *driver_name* with names of a driver or drivers you want to disable, for example:

```
modprobe.blacklist=ahci
```

Note that the drivers blacklisted during installation using the **modprobe.blacklist=** boot option will remain disabled on the installed system and appear in the */etc/modprobe.d/anaconda-blacklist.conf* file. See [Chapter 23, Boot Options](#) for more information about blacklisting drivers and other boot options.

CHAPTER 12. BOOTING THE INSTALLATION ON IBM POWER SYSTEMS

To boot an IBM Power Systems server from a DVD, you must specify the install boot device in the **System Management Services** (SMS) menu.

To enter the **System Management Services** GUI, press the **1** key during the boot process when you hear the chime sound. This brings up a graphical interface similar to the one described in this section.

On a text console, press **1** when the self test is displaying the banner along with the tested components:

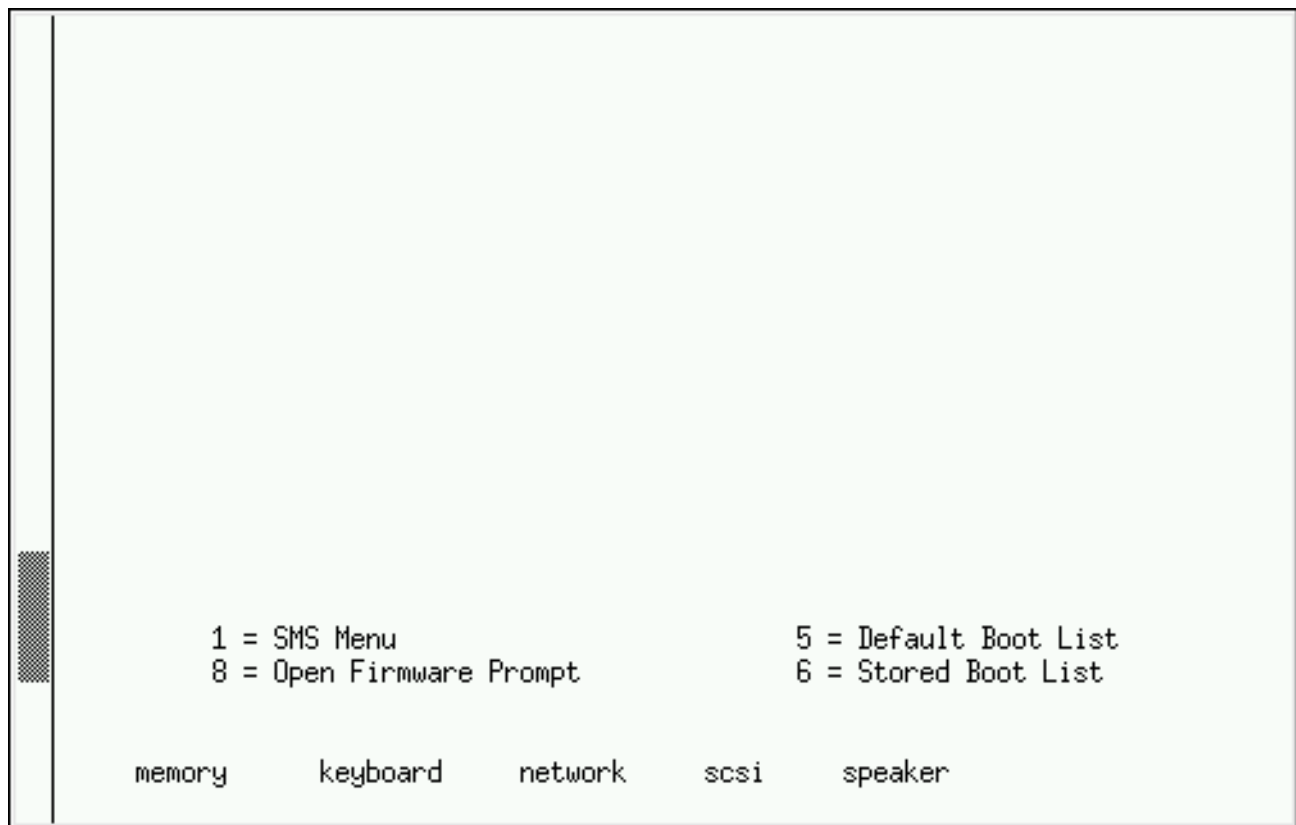


Figure 12.1. The SMS Console

Once in the SMS menu, select the option for **Select Boot Options**. In that menu, specify **Select Install or Boot a Device**. There, select **CD/DVD**, and then the bus type (in most cases SCSI). If you are uncertain, you can select to view all devices. This scans all available buses for boot devices, including network adapters and hard drives.

Finally, select the device containing the installation DVD. The boot menu will now load.



IMPORTANT

Because IBM Power Systems servers primarily use text consoles, **Anaconda** will not automatically start a graphical installation. However, the graphical installation program offers more features and customization and is recommended if your system has a graphical display.

To start a graphical installation, pass the **inst.vnc** boot option (see [Enabling Remote Access](#)).

12.1. THE BOOT MENU

Once your system has completed loading the boot media, a boot menu is displayed using **GRUB2** (**GR**and **Unified Bootloader**, version 2). The boot menu provides several options in addition to launching the installation program. If no key is pressed within 60 seconds, the default boot option (the one highlighted in white) will be run. To choose the default, either wait for the timer to run out or press **Enter**.

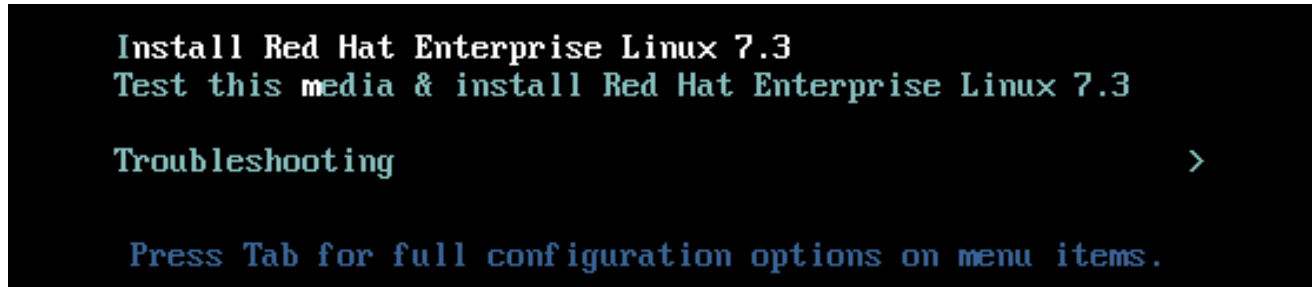


Figure 12.2. The Boot Screen

To select a different option than the default, use the arrow keys on your keyboard, and press **Enter** when the correct option is highlighted.

To customize the boot options for a particular menu entry, press the **e** key and add custom boot options to the command line. When ready press **Ctrl+X** to boot the modified option.

See [Chapter 23, Boot Options](#) for more information about additional boot options.

The boot menu options are:

Install Red Hat Enterprise Linux 7.0

Choose this option to install Red Hat Enterprise Linux onto your computer system using the graphical installation program.

Test this media & install Red Hat Enterprise Linux 7.0

This option is the default. Prior to starting the installation program, a utility is launched to check the integrity of the installation media.

Troubleshooting >

This item is a separate menu containing options that help resolve various installation issues. When highlighted, press **Enter** to display its contents.

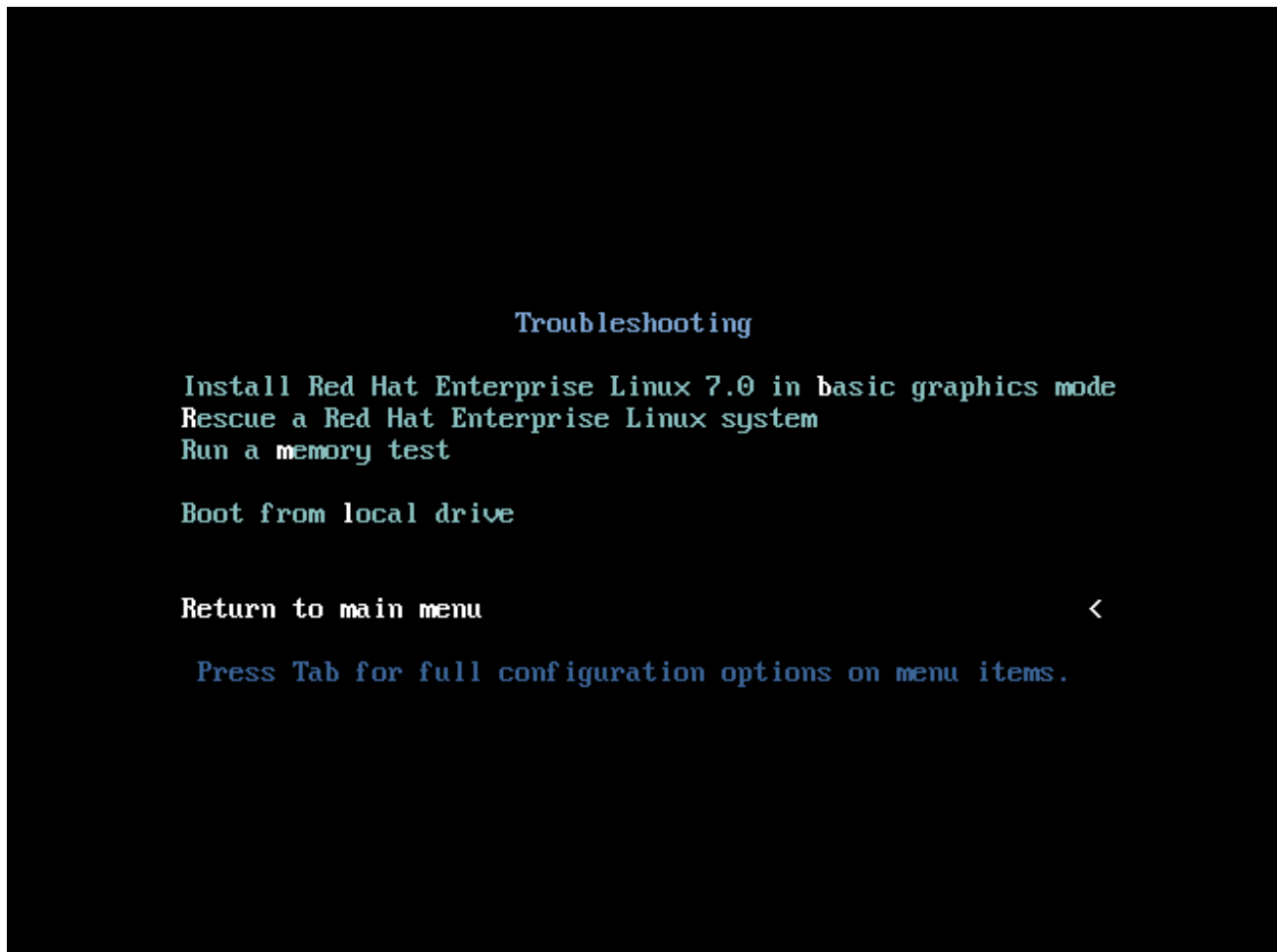


Figure 12.3. The Troubleshooting Menu

Install Red Hat Enterprise Linux 7.0 in basic graphics mode

This option allows you to install Red Hat Enterprise Linux in graphical mode even if the installation program is unable to load the correct driver for your video card. If your screen appears distorted or goes blank when using the **Install Red Hat Enterprise Linux 7.0** option, restart your computer and try this option instead.

Rescue a Red Hat Enterprise Linux system

Choose this option to repair a problem with your installed Red Hat Enterprise Linux system that prevents you from booting normally. The rescue environment contains utility programs that allow you fix a wide variety of these problems.

Run a memory test

This option runs a memory test on your system. For more information, see [Section 23.2.1, “Loading the Memory \(RAM\) Testing Mode”](#).

Boot from local drive

This option boots the system from the first installed disk. If you booted this disc accidentally, use this option to boot from the hard disk immediately without starting the installation program.

12.2. INSTALLING FROM A DIFFERENT SOURCE

You can install Red Hat Enterprise Linux from the ISO images stored on hard disk, or from a network using NFS, FTP, HTTP, or HTTPS methods. Experienced users frequently use one of these methods because it is often faster to read data from a hard disk or network server than from a DVD.

The following table summarizes the different boot methods and recommended installation methods to use with each:

Table 12.1. Boot Methods and Installation Sources

Boot method	Installation source
Full installation media (DVD)	The boot media itself
Minimal boot media (CD or DVD)	Full installation DVD ISO image or the installation tree extracted from this image, placed in a network location or on a hard drive
Network boot	Full installation DVD ISO image or the installation tree extracted from this image, placed in a network location

12.3. BOOTING FROM THE NETWORK USING AN INSTALLATION SERVER

For network boot, you need a properly configured server, and a network interface in your computer that can support an installation server. For information on how to configure an installation server, see [Section 24.1.3, “Configuring Network Boot for IBM Power Systems Using GRUB2”](#).

Configure the computer to boot from the network interface by selecting **Select Boot Options** in the SMS menu, then **Select Boot/Install Device**. Finally, select your network device from the list of available devices.

Once you properly configure booting from an installation server, the computer can boot the Red Hat Enterprise Linux installation system without any other media.

To boot a computer from a server:

Procedure 12.1. How to Start the Installation Program from the Network

1. Ensure that the network cable is attached. The link indicator light on the network socket should be lit, even if the computer is not switched on.
2. Switch on the computer.
3. Networking setup and diagnostic information usually appears before your computer connects to the server, although this depends on the hardware in use. Then you will see a menu with options specifying how the network boot server is setup. Press the number key that corresponds to the desired option. In case you are not sure which option you should select, ask your server administrator.

If your system does not boot from the network installation server, ensure that the SMS is configured to boot first from the correct network interface. See your hardware's documentation for more information.



IMPORTANT

Use the **vmlinuz** and **initrd.img** images to boot your system over a network. You cannot use the **ppc64.img** image to boot over a network; the file is too large for TFTP.

CHAPTER 13. INSTALLING USING ANACONDA

This chapter provides step-by-step instructions for installing Red Hat Enterprise Linux using the **Anaconda** installer. The bulk of this chapter describes installation using the graphical user interface. A text mode is also available for systems with no graphical display, but this mode is limited in certain aspects (for example, custom partitioning is not possible in text mode).

If your system does not have the ability to use the graphical mode, you can:

- Use Kickstart to automate the installation as described in [Chapter 27, Kickstart Installations](#)
- Perform the graphical installation remotely by connecting to the installation system from another computer with a graphical display using the VNC (Virtual Network Computing) protocol – see [Chapter 25, Using VNC](#)

13.1. INTRODUCTION TO ANACONDA

The Red Hat Enterprise Linux installer, **Anaconda**, is different from most other operating system installation programs due to its parallel nature. Most installers follow a fixed path: you must choose your language first, then you configure network, then installation type, then partitioning, and so on. There is usually only one way to proceed at any given time.

In **Anaconda** you are only required to select your language and locale first, and then you are presented with a central screen, where you can configure most aspects of the installation in any order you like. This does not apply to all parts of the installation process, however – for example, when installing from a network location, you must configure the network before you can select which packages to install.

Some screens will be automatically configured depending on your hardware and the type of media you used to start the installation. You can still change the detected settings in any screen. Screens which have not been automatically configured, and therefore require your attention before you begin the installation, are marked by an exclamation mark. You cannot start the actual installation process before you finish configuring these settings.

Additional differences appear in certain screens; notably the custom partitioning process is very different from other Linux distributions. These differences are described in each screen's subsection.

13.2. CONSOLES AND LOGGING DURING THE INSTALLATION

The following sections describe how to access logs and an interactive shell during the installation. This is useful when troubleshooting problems, but should not be necessary in most cases.

13.2.1. Accessing Consoles

The Red Hat Enterprise Linux installer uses the **tmux** terminal multiplexer to display and control several windows you can use in addition to the main interface. Each of these windows serves a different purpose – they display several different logs, which can be used to troubleshoot any issues during the installation, and one of the windows provides an interactive shell prompt with **root** privileges, unless this prompt was specifically disabled using a boot option or a Kickstart command.



NOTE

In general, there is no reason to leave the default graphical installation environment unless you need to diagnose an installation problem.

The terminal multiplexer is running in virtual console 1. To switch from the graphical installation environment to **tmux**, press **Ctrl+Alt+F1**. To go back to the main installation interface which runs in virtual console 6, press **Ctrl+Alt+F6**.



NOTE

If you choose text mode installation, you will start in virtual console 1 (**tmux**), and switching to console 6 will open a shell prompt instead of a graphical interface.

The console running **tmux** has 5 available windows; their contents are described in the table below, along with keyboard shortcuts used to access them. Note that the keyboard shortcuts are two-part: first press **Ctrl+b**, then release both keys, and press the number key for the window you want to use.

You can also use **Ctrl+b n** and **Ctrl+b p** to switch to the next or previous **tmux** window, respectively.

Table 13.1. Available tmux Windows

Shortcut	Contents
Ctrl+b 1	Main installation program window. Contains text-based prompts (during text mode installation or if you use VNC Direct Mode), and also some debugging information.
Ctrl+b 2	Interactive shell prompt with root privileges.
Ctrl+b 3	Installation log; displays messages stored in /tmp/anaconda.log .
Ctrl+b 4	Storage log; displays messages related storage devices from kernel and system services, stored in /tmp/storage.log .
Ctrl+b 5	Program log; displays messages from other system utilities, stored in /tmp/program.log .

In addition to displaying diagnostic information in **tmux** windows, **Anaconda** also generates several log files, which can be transferred from the installation system. These log files are described in [Table 14.1, “Log Files Generated During the Installation”](#), and directions for transferring them from the installation system are available in [Chapter 14, Troubleshooting Installation on IBM Power Systems](#).

13.2.2. Saving Screenshots

You can press **Shift+Print Screen** at any time during the graphical installation to capture the current screen. These screenshots are saved to **/tmp/anaconda-screenshots/**.

Additionally, you can use the **autostep --autoscreenshot** command in a Kickstart file to capture and save each step of the installation automatically. See [Section 27.3.1, “Kickstart Commands and Options”](#) for details.

13.3. INSTALLING IN TEXT MODE

Text mode installation offers an interactive, non-graphical interface for installing Red Hat Enterprise Linux. This can be useful on systems with no graphical capabilities; however, you should always consider the available alternatives before starting a text-based installation. Text mode is limited in the amount of choices you can make during the installation.



IMPORTANT

Red Hat recommends that you install Red Hat Enterprise Linux using the graphical interface. If you are installing Red Hat Enterprise Linux on a system that lacks a graphical display, consider performing the installation over a VNC connection – see [Chapter 25, Using VNC](#). The text mode installation program will prompt you to confirm the use of text mode if it detects that a VNC-based installation is possible.

If your system has a graphical display, but graphical installation fails, try booting with the **inst.xdriver=vesa** option – see [Chapter 23, Boot Options](#).

Alternatively, consider a Kickstart installation. See [Chapter 27, Kickstart Installations](#) for more information.

```

Installation

1) [!] Timezone settings          2) [x] Language settings
   (Timezone is not set.)        (English (United States))
3) [!] Software selection        4) [!] Installation source
   (Processing...)              (Processing...)
5) [x] Network settings          6) [!] Install Destination
   (Wired (eth0) connected)     (No disks selected)
7) [x] Kdump                     8) [!] Set root password
   (Kdump is enabled)           (Password is not set.)
9) [!] Create user
   (No user will be created)

Please make your choice from above ['q' to quit | 'b' to begin installation |
'r' to refresh]: _

```

Figure 13.1. Text Mode Installation

Installation in text mode follows a pattern similar to the graphical installation: There is no single fixed progression; you can configure many settings in any order you want using the main status screen. Screens which have already been configured, either automatically or by you, are marked as **[x]**, and screens which require your attention before the installation can begin are marked with **[!]**. Available commands are displayed below the list of available options.



NOTE

When related background tasks are being run, certain menu items can be temporarily unavailable or display the **Processing...** label. To refresh to the current status of text menu items, use the **r** option at the text mode prompt.

At the bottom of the screen in text mode, a green bar is displayed showing five menu options. These options represent different screens in the **tmux** terminal multiplexer; by default you start in screen 1, and you can use keyboard shortcuts to switch to other screens which contain logs and an interactive command prompt. For information about available screens and shortcuts to switch to them, see [Section 13.2.1, “Accessing Consoles”](#).

Limits of interactive text mode installation include:

- The installer will always use the English language and the US English keyboard layout. You can configure your language and keyboard settings, but these settings will only apply to the installed system, not to the installation.
- You cannot configure any advanced storage methods (LVM, software RAID, FCoE, zFCP and iSCSI).
- It is not possible to configure custom partitioning; you must use one of the automatic partitioning settings. You also cannot configure where the boot loader will be installed.
- You cannot select any package add-ons to be installed; they must be added after the installation finishes using the **Yum** package manager.

To start a text mode installation, boot the installation with the **inst.text** boot option used either at the boot command line in the boot menu, or in your PXE server configuration. See [Chapter 12, Booting the Installation on IBM Power Systems](#) for information about booting and using boot options.

13.4. USING THE HMC VTERM

The HMC vterm is the console for any partitioned IBM Power system. Open the console by right-clicking on the partition on the HMC, and then selecting **Open Terminal Window**. Only a single vterm can be connected to the console at a time and there is no console access for partitioned system besides the vterm. This often is referred to as a *virtual console*, but is different from the virtual consoles in [Section 13.2.1, “Accessing Consoles”](#).

13.5. INSTALLING IN THE GRAPHICAL USER INTERFACE

The graphical installation interface is the preferred method of manually installing Red Hat Enterprise Linux. It allows you full control over all available settings, including custom partitioning and advanced storage configuration, and it is also localized to many languages other than English, allowing you to perform the entire installation in a different language. The graphical mode is used by default when you boot the system from local media (a CD, DVD or a USB flash drive).

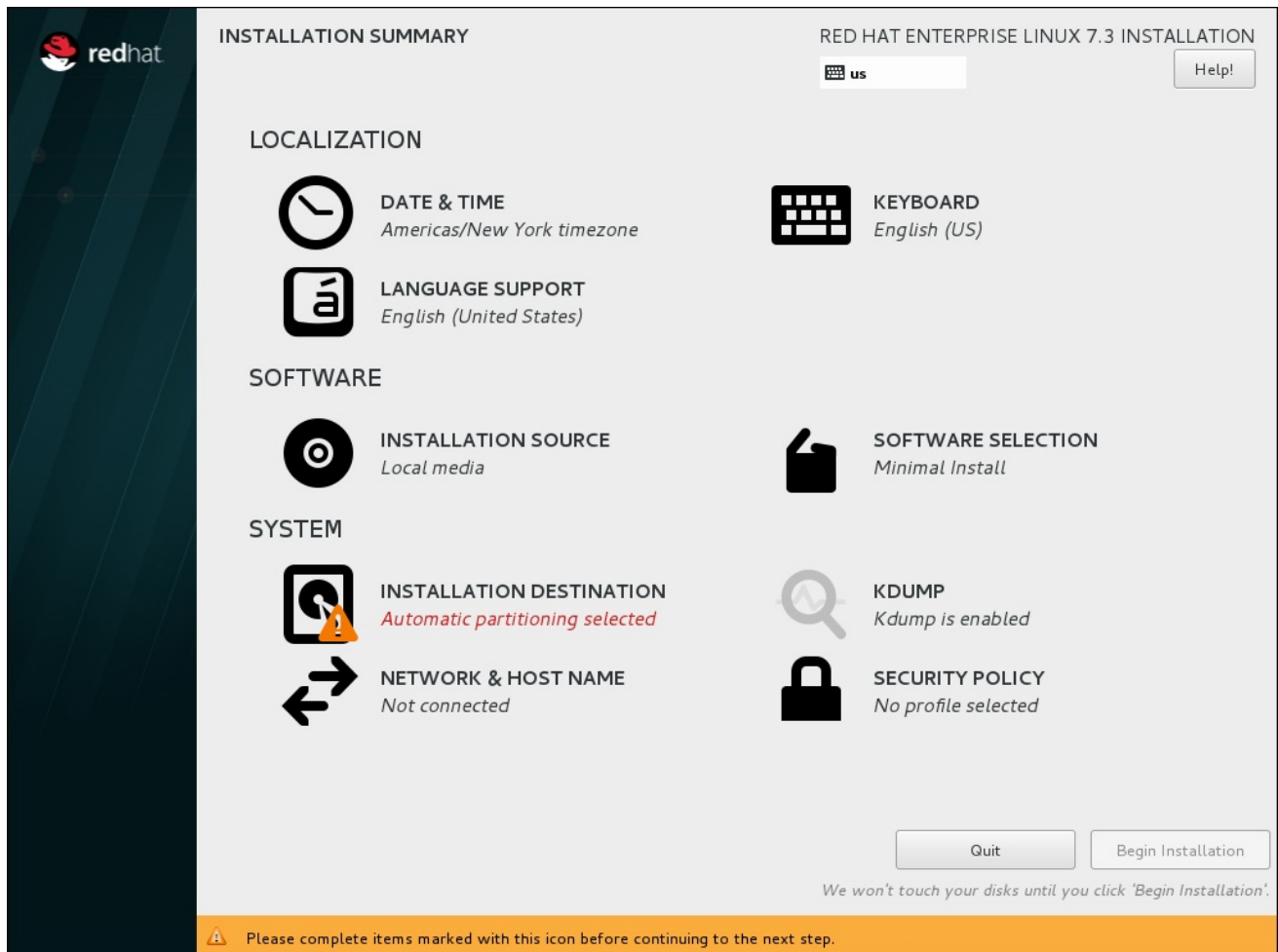


Figure 13.2. The Installation Summary Screen

The sections below discuss each screen available in the installation process. Note that due to the installer's parallel nature, most of the screens do not have to be completed in the order in which they are described here.

Each screen in the graphical interface contains a **Help** button. This button opens the **Yelp** help browser displaying the section of the *Red Hat Enterprise Linux Installation Guide* relevant to the current screen.

You can also control the graphical installer with your keyboard. Following table shows you the shortcuts you can use.

Table 13.2. Graphical installer keyboard shortcuts

Shortcut keys	Usage
Tab and Shift+Tab	Cycle through active control elements (buttons, check boxes, and so on.) on the current screen
Up and Down	Scroll through lists
Left and Right	Scroll through horizontal toolbars and table entries
Space and Enter	Select or remove a highlighted item from selection and expand and collapse drop-down menus

Additionally, elements in each screen can be toggled using their respective shortcuts. These shortcuts are highlighted (underlined) when you hold down the **Alt** key; to toggle that element, press **Alt+X**, where **X** is the highlighted letter.

Your current keyboard layout is displayed in the top right hand corner. Only one layout is configured by default; if you configure more than layout in the **Keyboard Layout** screen ([Section 13.10, “Keyboard Configuration”](#)), you can switch between them by clicking the layout indicator.

13.6. WELCOME SCREEN AND LANGUAGE SELECTION

The first screen of the installation program is the **Welcome to Red Hat Enterprise Linux** screen. Here you select the language that **Anaconda** will use for the rest of the installation. This selection will also become the default for the installed system, unless changed later. In the left panel, select your language of choice, for example **English**. Then you can select a locale specific to your region in the right panel, for example **English (United States)**.



NOTE

One language is pre-selected by default on top of the list. If network access is configured at this point (for example, if you booted from a network server instead of local media), the pre-selected language will be determined based on automatic location detection using the **GeoIP** module.

Alternatively, type your preferred language into the search box as shown below.

Once you have made your selection, click the **Continue** button to proceed to the **Installation Summary** screen.

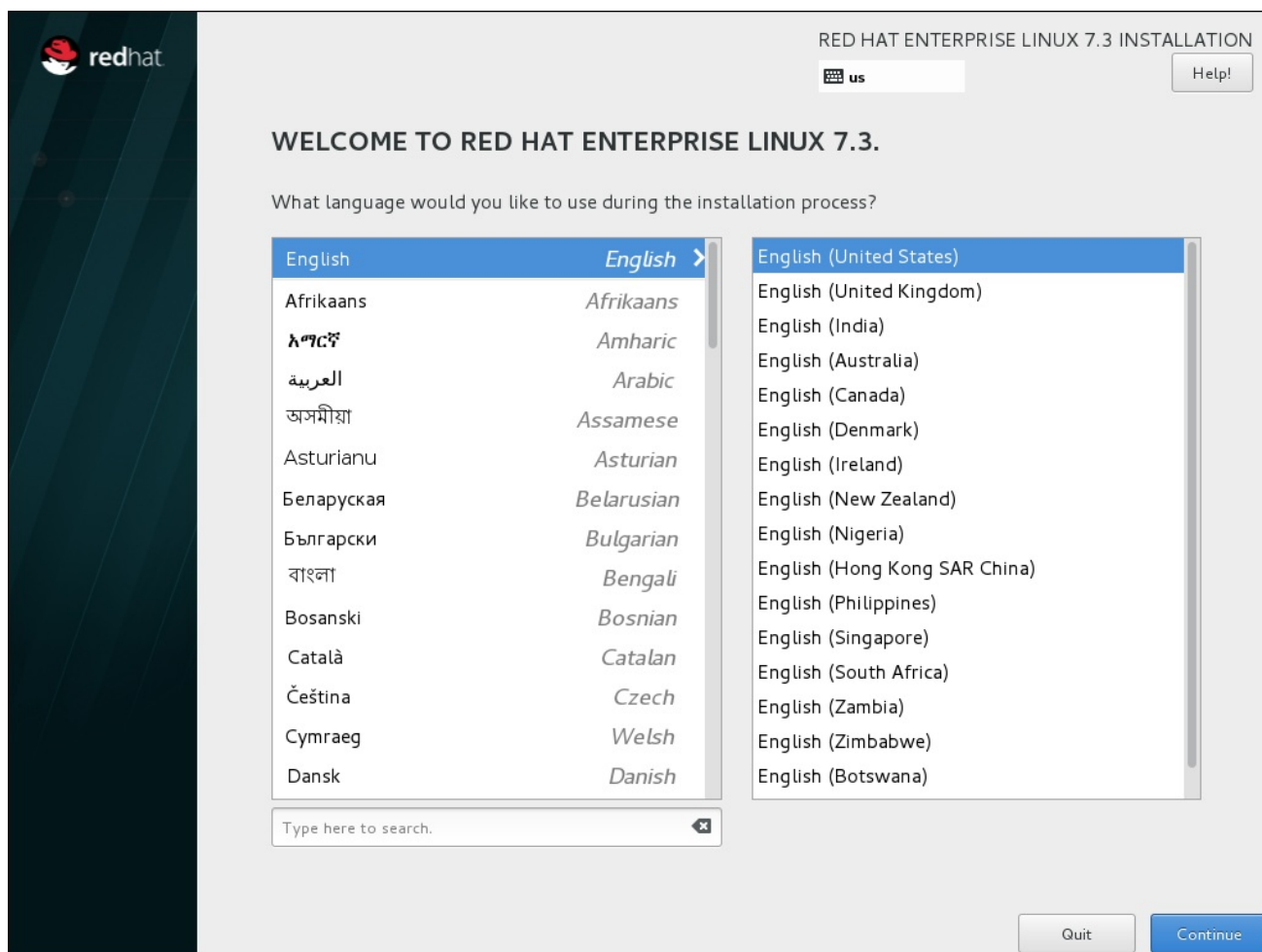


Figure 13.3. Language Configuration

After you click the **Continue** button, the unsupported hardware dialog may appear. This happens if you are using hardware that the kernel does not support.

13.7. THE INSTALLATION SUMMARY SCREEN

The **Installation Summary** screen is the central location for setting up an installation.

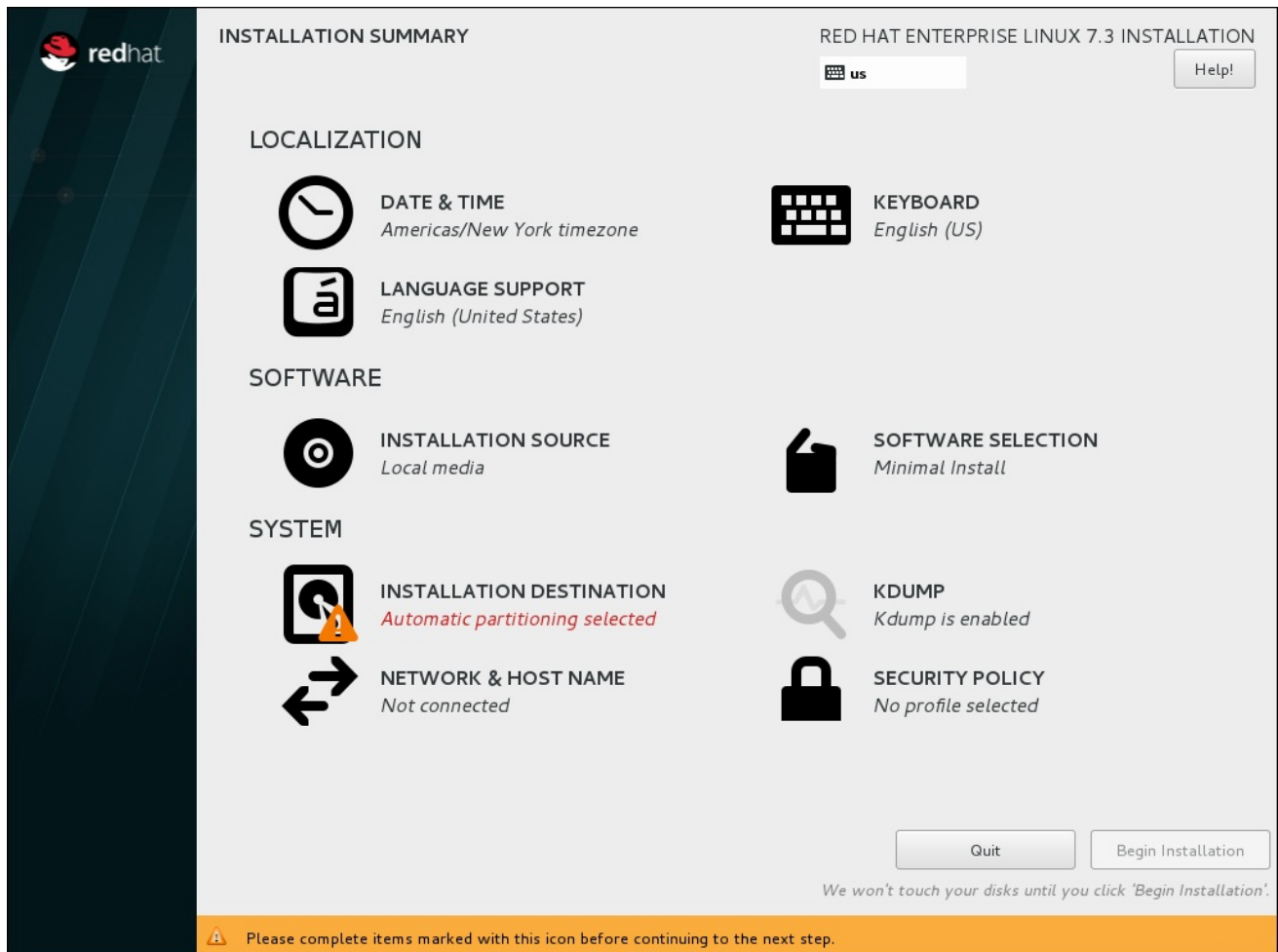


Figure 13.4. The Installation Summary Screen

Instead of directing you through consecutive screens, the Red Hat Enterprise Linux installation program allows you to configure your installation in the order you choose.

Use your mouse to select a menu item to configure a section of the installation. When you have completed configuring a section, or if you would like to complete that section later, click the **Done** button located in the upper left corner of the screen.

Only sections marked with a warning symbol are mandatory. A note at the bottom of the screen warns you that these sections must be completed before the installation can begin. The remaining sections are optional. Beneath each section's title, the current configuration is summarized. Using this you can determine whether you need to visit the section to configure it further.

Once all required sections are complete, click the **Begin Installation** button. Also see [Section 13.18, "Begin Installation"](#).

To cancel the installation, click the **Quit** button.



NOTE

When related background tasks are running, certain menu items might be temporarily unavailable.

If you used a Kickstart option or a boot command-line option to specify an installation repository on a network, but no network is available at the start of the installation, the installation program will display the configuration screen for you to set up a network connection prior to displaying the **Installation**

Summary screen.

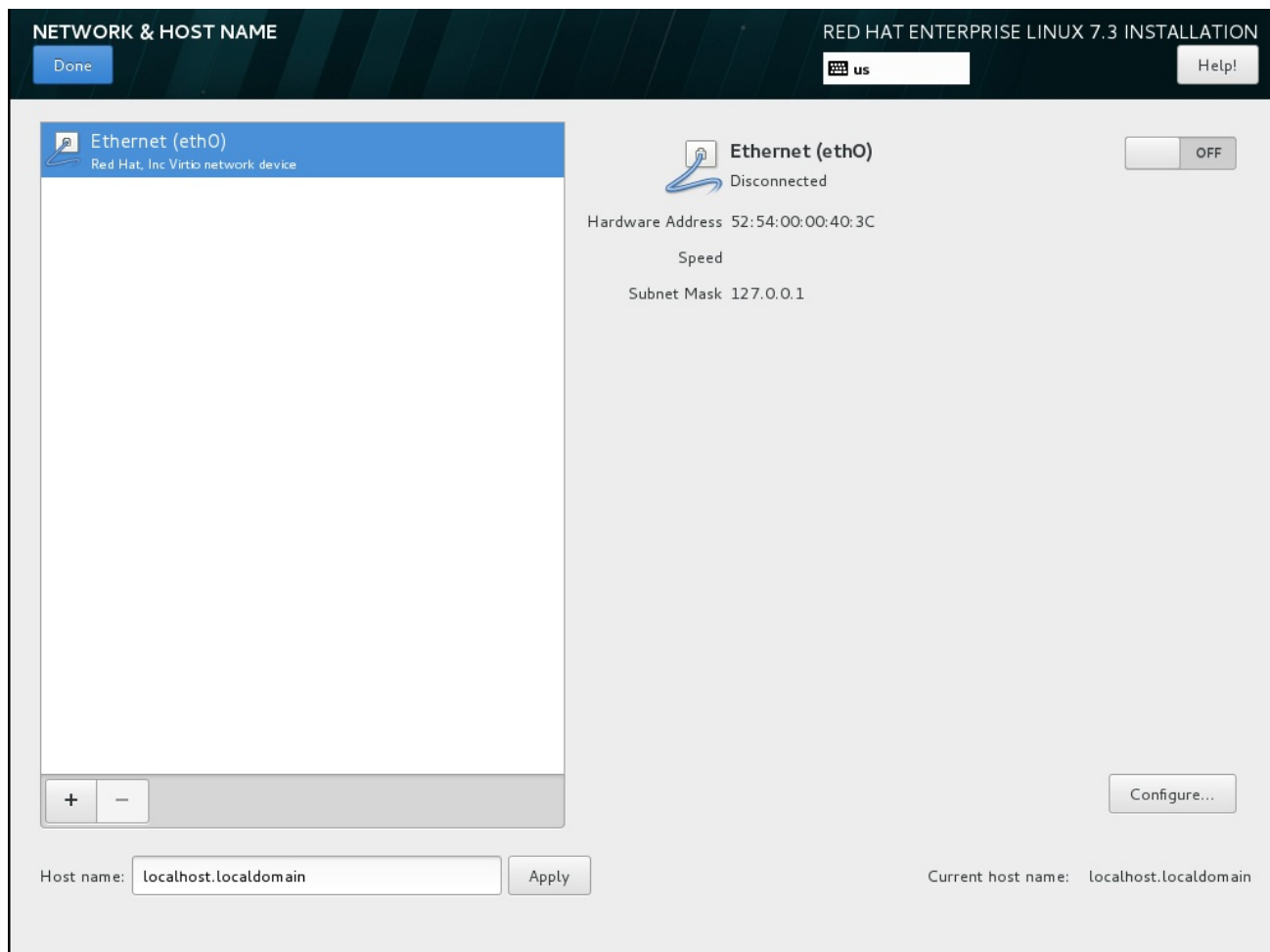


Figure 13.5. Network Configuration Screen When No Network Is Detected

You can skip this step if you are installing from an installation DVD or other locally accessible media, and you are certain you will not need network to finish the installation. However, network connectivity is necessary for network installations (see [Section 8.11, “Installation Source”](#)) or for setting up advanced storage devices (see [Section 8.15, “Storage Devices”](#)). For more details about configuring a network in the installation program, see [Section 8.12, “Network & Hostname”](#).

13.8. DATE & TIME

To configure time zone, date, and optionally settings for network time, select **Date & Time** at the **Installation Summary** screen.

There are three ways for you to select a time zone:

- Using your mouse, click on the interactive map to select a specific city. A red pin appears indicating your selection.
- You can also scroll through the **Region** and **City** drop-down menus at the top of the screen to select your time zone.
- Select **Etc** at the bottom of the **Region** drop-down menu, then select your time zone in the next menu adjusted to GMT/UTC, for example **GMT+1**.

If your city is not available on the map or in the drop-down menu, select the nearest major city in the same time zone. Alternatively you can use a Kickstart file, which will allow you to specify some additional

time zones which are not available in the graphical interface. See the **timezone** command in [timezone \(required\)](#) for details.



NOTE

The list of available cities and regions comes from the Time Zone Database (tzdata) public domain, which is maintained by the Internet Assigned Numbers Authority (IANA). Red Hat cannot add cities or regions into this database. You can find more information at the official website, available at <http://www.iana.org/time-zones>.

Specify a time zone even if you plan to use NTP (Network Time Protocol) to maintain the accuracy of the system clock.

If you are connected to the network, the **Network Time** switch will be enabled. To set the date and time using NTP, leave the **Network Time** switch in the **ON** position and click the configuration icon to select which NTP servers Red Hat Enterprise Linux should use. To set the date and time manually, move the switch to the **OFF** position. The system clock should use your time zone selection to display the correct date and time at the bottom of the screen. If they are still incorrect, adjust them manually.

Note that NTP servers might be unavailable at the time of installation. In such a case, enabling them will not set the time automatically. When the servers become available, the date and time will update.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your time zone configuration after you have completed the installation, visit the **Date & Time** section of the **Settings** dialog window.

13.9. LANGUAGE SUPPORT

To install support for additional locales and language dialects, select **Language Support** from the **Installation Summary** screen.

Use your mouse to select the language for which you would like to install support. In the left panel, select your language of choice, for example **Español**. Then you can select a locale specific to your region in the right panel, for example **Español (Costa Rica)**. You can select multiple languages and multiple locales. The selected languages are highlighted in bold in the left panel.

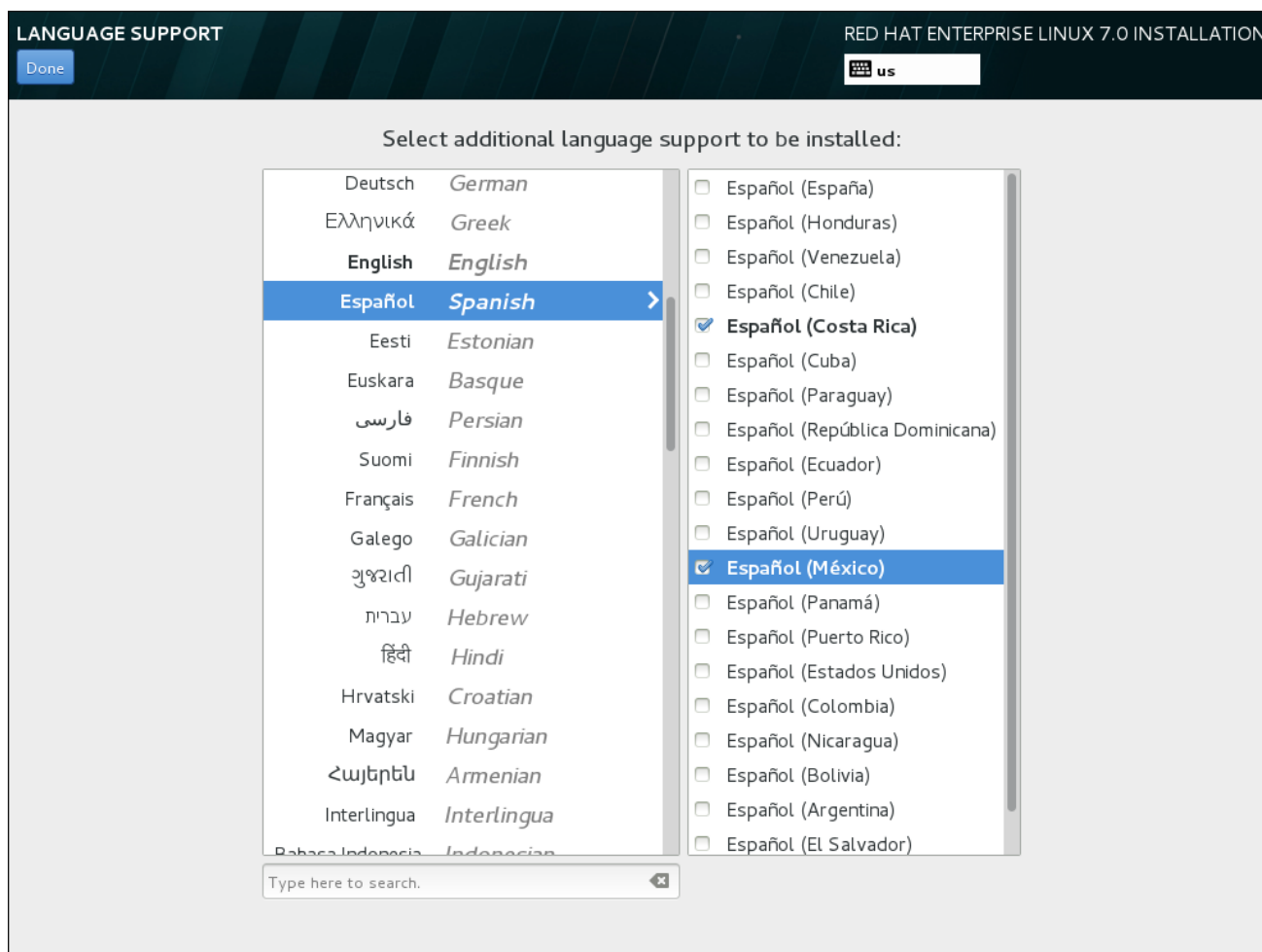


Figure 13.6. Configuring Language Support

Once you have made your selections, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your language support configuration after you have completed the installation, visit the **Region & Language** section of the **Settings** dialog window.

13.10. KEYBOARD CONFIGURATION

To add multiple keyboard layouts to your system, select **Keyboard** from the **Installation Summary** screen. Upon saving, the keyboard layouts are immediately available in the installation program and you can switch between them by using the keyboard icon located at all times in the upper right corner of the screen.

Initially, only the language you selected in the welcome screen is listed as the keyboard layout in the left pane. You can either replace the initial layout or add more layouts. However, if your language does not use ASCII characters, you might need to add a keyboard layout that does, to be able to properly set a password for an encrypted disk partition or the root user, among other things.

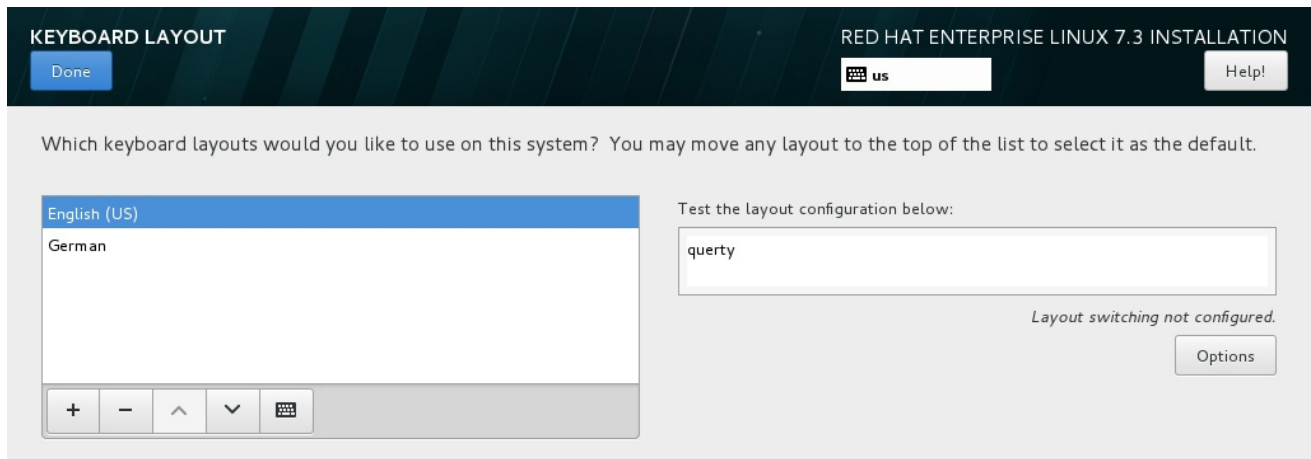


Figure 13.7. Keyboard Configuration

To add an additional layout, click the **+** button, select it from the list, and click **Add**. To delete a layout, select it and click the **-** button. Use the arrow buttons to arrange the layouts in order of preference. For a visual preview of the keyboard layout, select it and click the keyboard button.

To test a layout, use the mouse to click inside the text box on the right. Type some text to confirm that your selection functions correctly.

To test additional layouts, you can click the language selector at the top on the screen to switch them. However, it is recommended to set up a keyboard combination for switching layout. Click the **Options** button at the right to open the **Layout Switching Options** dialog and choose a combination from the list by selecting its check box. The combination will then be displayed above the **Options** button. This combination applies both during the installation and on the installed system, so you must configure a combination here in order to use one after installation. You can also select more than one combination to switch between layouts.



IMPORTANT

If you use a layout that cannot accept Latin characters, such as **Russian**, Red Hat recommends additionally adding the **English (United States)** layout and configuring a keyboard combination to switch between the two layouts. If you only select a layout without Latin characters, you might be unable to enter a valid root password and user credentials later in the installation process. This can prevent you from completing the installation.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your keyboard configuration after you have completed the installation, visit the **Keyboard** section of the **Settings** dialogue window.

13.11. SECURITY POLICY

The **Security Policy** spoke allows you to configure the installed system following restrictions and recommendations (*compliance policies*) defined by the Security Content Automation Protocol (SCAP) standard. This functionality is provided by an add-on which has been enabled by default since Red Hat

Enterprise Linux 7.2. When enabled, the packages necessary to provide this functionality will automatically be installed. However, by default, no policies are enforced, meaning that no checks are performed during or after installation unless specifically configured.

The [Red Hat Enterprise Linux 7 Security Guide](#) provides detailed information about security compliance including background information, practical examples, and additional resources.



IMPORTANT

Applying a security policy is not necessary on all systems. This screen should only be used when a specific policy is mandated by your organization rules or government regulations.

If you apply a security policy to the system, it will be installed using restrictions and recommendations defined in the selected profile. The `openscap-scanner` package will also be added to your package selection, providing a preinstalled tool for compliance and vulnerability scanning. After the installation finishes, the system will be automatically scanned to verify compliance. The results of this scan will be saved to the `/root/openscap_data` directory on the installed system.

Pre-defined policies which are available in this screen are provided by **SCAP Security Guide**. See the [OpenSCAP Portal](#) for links to detailed information about each available profile.

You can also load additional profiles from an HTTP, HTTPS or FTP server.

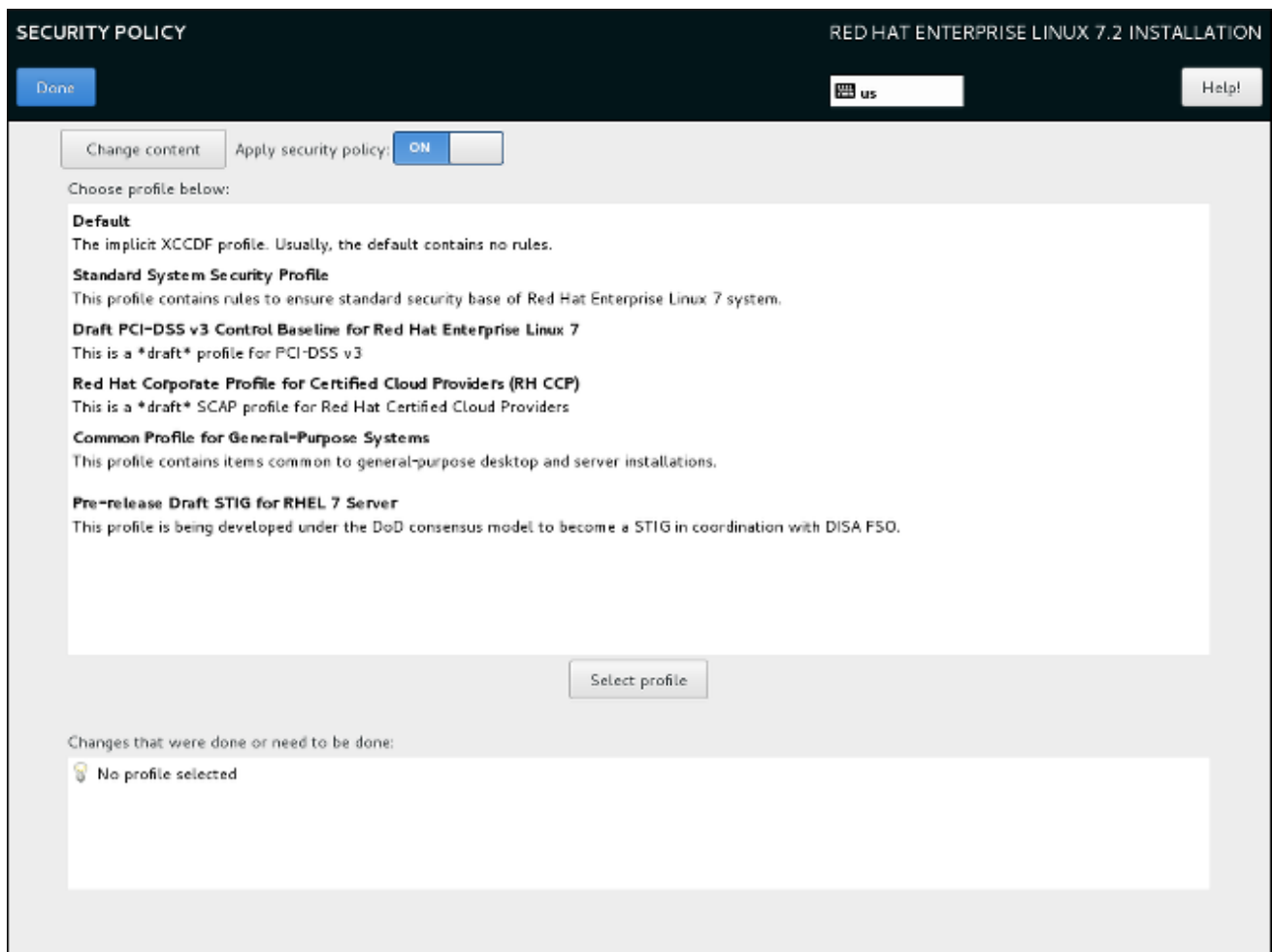


Figure 13.8. Security policy selection screen

To configure the use of security policies on the system, first enable configuration by setting the **Apply security policy** switch to **ON**. If the switch is in the **OFF** position, controls in the rest of this screen have no effect.

After enabling security policy configuration using the switch, select one of the profiles listed in the top window of the screen, and click the **Select profile** below. When a profile is selected, a green check mark will appear on the right side, and the bottom field will display whether any changes will be made before beginning the installation.



NOTE

None of the profiles available by default perform any changes before the installation begins. However, loading a custom profile as described below can require some pre-installation actions.

To use a custom profile, click the **Change content** button in the top left corner. This will open another screen where you can enter an URL of a valid security content. To go back to the default security content selection screen, click **Use SCAP Security Guide** in the top left corner.

Custom profiles can be loaded from an **HTTP**, **HTTPS** or **FTP** server. Use the full address of the content, including the protocol (such as **http://**). A network connection must be active (enabled in [Section 13.13, "Network & Hostname"](#)) before you can load a custom profile. The content type will be detected automatically by the installer.

After you select a profile, or if you want to leave the screen, click **Done** in the top left corner to return to [Section 13.7, "The Installation Summary Screen"](#).

13.12. INSTALLATION SOURCE

To specify a file or a location to install Red Hat Enterprise Linux from, select **Installation Source** from the **Installation Summary** screen. On this screen, you can choose between locally available installation media, such as a DVD or an ISO file, or a network location.

Figure 13.9. Installation Source Screen

Select one of the following options:

Auto-detected installation media

If you started the installation using the full installation DVD or USB drive, the installation program will detect it and display basic information under this option. Click the **Verify** button to ensure that the media is suitable for installation. This integrity test is the same as the one performed if you selected **Test this media & Install Red Hat Enterprise Linux** in the boot menu, or if you used the **rd.live.check** boot option.

ISO file

This option will appear if the installation program detected a partitioned hard drive with mountable file systems. Select this option, click the **Choose an ISO** button, and browse to the installation ISO file's location on your system. Then click **Verify** to ensure that the file is suitable for installation.

On the network

To specify a network location, select this option and choose from the following options in the drop-down menu:

- **http://**
- **https://**
- **ftp://**
- **nfs**

Using your selection as the start of the location URL, type the rest into the address box. If you choose NFS, another box will appear for you to specify any NFS mount options.



IMPORTANT

When selecting an NFS-based installation source, you must specify the address with a colon (:) character separating the host name from the path. For example:

server.example.com:/path/to/directory

To configure a proxy for an HTTP or HTTPS source, click the **Proxy setup** button. Check **Enable HTTP proxy** and type the URL into the **Proxy URL** box. If your proxy requires authentication, check **Use Authentication** and enter a user name and password. Click **Add**.

If your HTTP or HTTPS URL refers to a repository mirror list, mark the check box under the input field.

You can also specify additional repositories to gain access to more installation environments and software add-ons. See [Section 13.14, "Software Selection"](#) for more information.

To add a repository, click the **+** button. To delete a repository, click the **-** button. Click the arrow icon to revert to the previous list of repositories, that is, to replace current entries with those that were present at the time you entered the **Installation Source** screen. To activate or deactivate a repository, click the check box in the **Enabled** column at each entry in the list.

In the right part of the form, you can name your additional repository and configure it the same way as the primary repository on the network.

Once you have selected your installation source, click **Done** to return to the **Installation Summary** screen.

13.13. NETWORK & HOSTNAME

To configure essential networking features for your system, select **Network & Hostname** at the **Installation Summary** screen.

IMPORTANT

When the installation finishes and the system boots for the first time, any network interfaces which you configured during the installation will be activated. However, the installation does not prompt you to configure network interfaces on some common installation paths – for example, when you install Red Hat Enterprise Linux from a DVD to a local hard drive.

When you install Red Hat Enterprise Linux from a local installation source to a local storage device, be sure to configure at least one network interface manually if you require network access when the system boots for the first time. You will also need to set the connection to connect automatically after boot when editing the configuration.

Locally accessible interfaces are automatically detected by the installation program and cannot be manually added or deleted. The detected interfaces are listed in the left pane. Click an interface in the list to display more details about it on the right. To activate or deactivate a network interface, move the switch in the top right corner of the screen to either **ON** or **OFF**.

NOTE

There are several types of network device naming standards used to identify network devices with persistent names such as **em1** or **wl3sp0**. For information about these standards, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

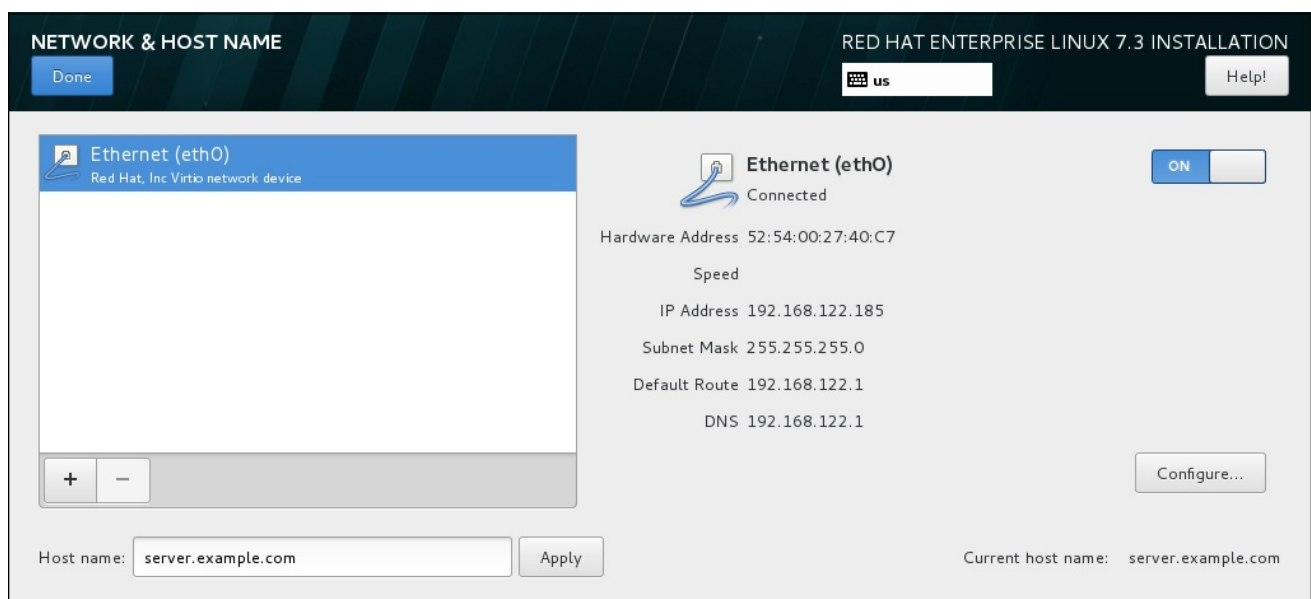
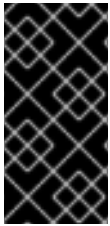


Figure 13.10. Network & Hostname Configuration Screen

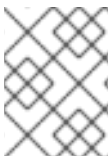
Below the list of connections, enter a host name for this computer in the **Hostname** input field. The host name can be either a *fully-qualified domain name* (FQDN) in the format *hostname.domainname* or a

short host name in the format *hostname*. Many networks have a *Dynamic Host Configuration Protocol* (DHCP) service that automatically supplies connected systems with a domain name. To allow the DHCP service to assign the domain name to this machine, only specify the short host name. The value **localhost.localdomain** means that no specific static host name for target system is configured, and the actual host name of installed system will be configured during process of network configuration (for example, by NetworkManager using DHCP or DNS).



IMPORTANT

If you want to manually assign the host name, make sure you do not use a domain name that is not delegated to you, as this can result in network resources becoming unavailable. For more information, see the recommended naming practices in the [Red Hat Enterprise Linux 7 Networking Guide](#).



NOTE

You can use the **Network** section of the system **Settings** dialog to change your network configuration after you have completed the installation.

Once you have finished network configuration, click **Done** to return to the **Installation Summary** screen.

13.13.1. Edit Network Connections

This section only details the most important settings for a typical wired connection used during installation. Many of the available options do not have to be changed in most installation scenarios and are not carried over to the installed system. Configuration of other types of network is broadly similar, although the specific configuration parameters are necessarily different. To learn more about network configuration after installation, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

To configure a network connection manually, click the **Configure** button in the lower right corner of the screen. A dialog appears that allows you to configure the selected connection. The configuration options presented depends on whether the connection is wired, wireless, mobile broadband, VPN, or DSL. If required, see the *Networking Guide* for more detailed information on network settings.

The most useful network configuration options to consider during installation are:

- Mark the **Automatically connect to this network when it is available** check box if you want to use the connection every time the system boots. You can use more than one connection that will connect automatically. This setting will carry over to the installed system.

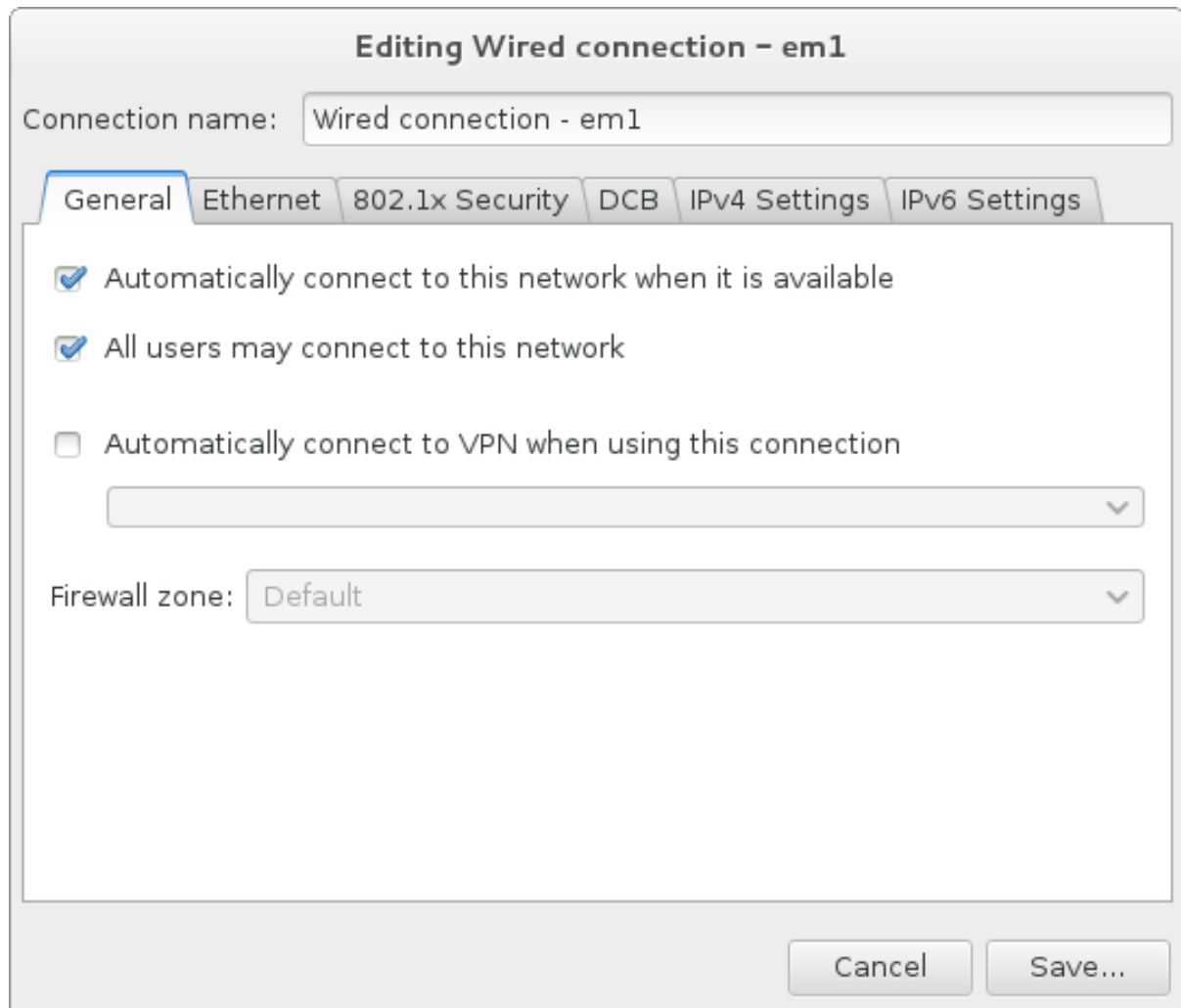


Figure 13.11. Network Auto-Connection Feature

- By default, IPv4 parameters are configured automatically by the DHCP service on the network. At the same time, the IPv6 configuration is set to the **Automatic** method. This combination is suitable for most installation scenarios and usually does not require any changes.

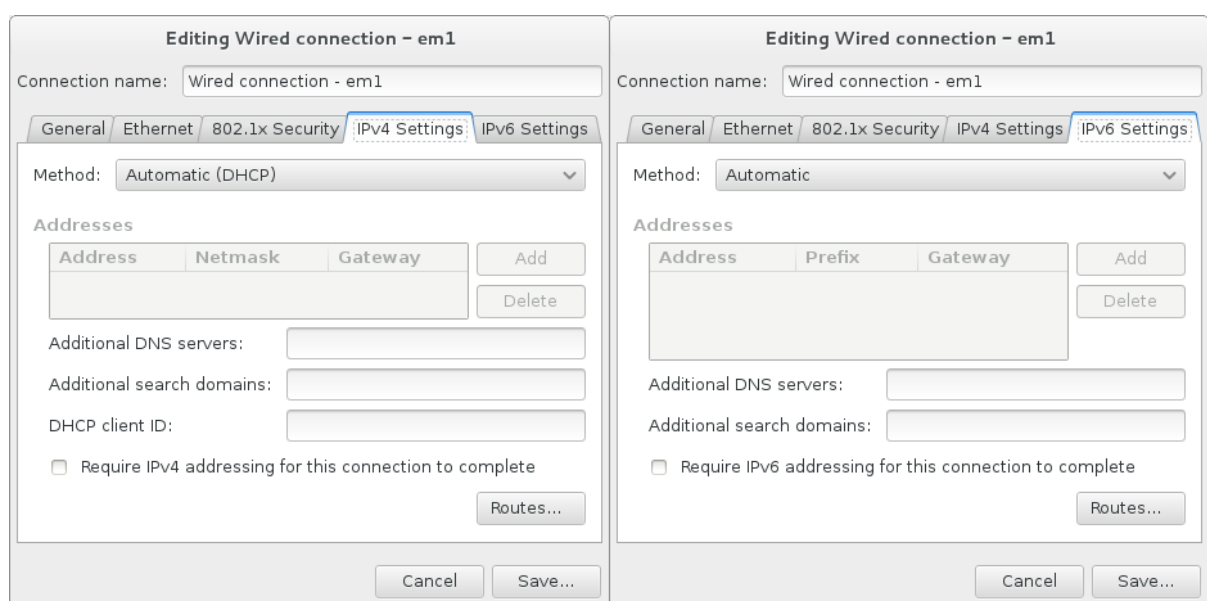


Figure 13.12. IP Protocol Settings

When you have finished editing network settings, click **Save** to save the new configuration. If you

reconfigured a device that was already active during installation, you must restart the device in order to use the new configuration in the installation environment. Use the **ON/OFF** switch on the **Network & Host Name** screen to restart the device.

13.13.2. Advanced Network Interfaces

Advanced network interfaces are also available for installation. This includes virtual local area networks (VLANs) and three methods to use aggregated links. Detailed description of these interfaces is beyond the scope of this document; read the [Red Hat Enterprise Linux 7 Networking Guide](#) for more information.

To create an advanced network interface, click the **+** button in the lower left corner of the **Network & Hostname** screen.

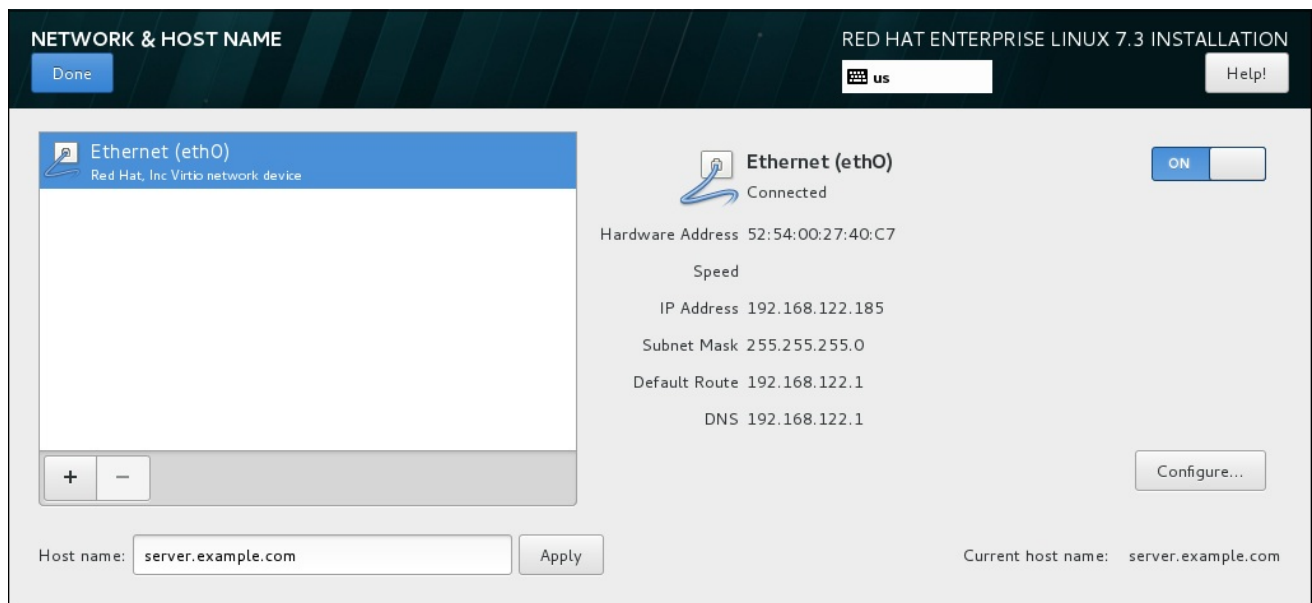


Figure 13.13. Network & Hostname Configuration Screen

A dialog appears with a drop-down menu with the following options:

- **Bond** - represents NIC (*Network Interface Controller*) Bonding, a method to bind multiple network interfaces together into a single, bonded, channel.
- **Bridge** - represents NIC Bridging, a method to connect multiple separate network into one aggregate network.
- **Team** - represents NIC Teaming, a new implementation to aggregate links, designed to provide a small kernel driver to implement the fast handling of packet flows, and various applications to do everything else in user space.
- **VLAN** - represents a method to create multiple distinct broadcast domains, which are mutually isolated.

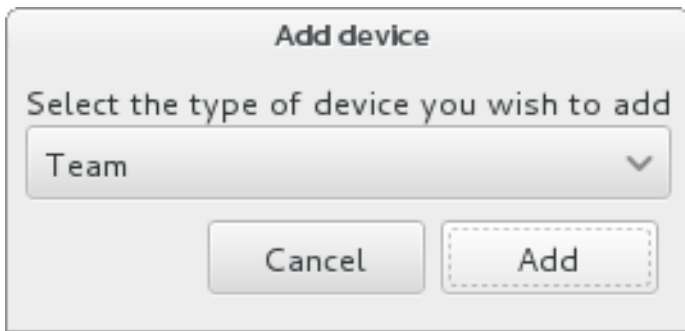


Figure 13.14. Advanced Network Interface Dialog



NOTE

Note that locally accessible interfaces, wired or wireless, are automatically detected by the installation program and cannot be manually added or deleted by using these controls.

Once you have selected an option and clicked the **Add** button, another dialog appears for you to configure the new interface. See the respective chapters in the [Red Hat Enterprise Linux 7 Networking Guide](#) for detailed instructions. To edit configuration on an existing advanced interface, click the **Configure** button in the lower right corner of the screen. You can also remove a manually-added interface by clicking the - button.

13.14. SOFTWARE SELECTION

To specify which packages will be installed, select **Software Selection** at the **Installation Summary** screen. The package groups are organized into *Base Environments*. These environments are pre-defined sets of packages with a specific purpose; for example, the **Virtualization Host** environment contains a set of software packages needed for running virtual machines on the system. Only one software environment can be selected at installation time.

For each environment, there are additional packages available in the form of *Add-ons*. Add-ons are presented in the right part of the screen and the list of them is refreshed when a new environment is selected. You can select multiple add-ons for your installation environment.

A horizontal line separates the list of add-ons into two areas:

- Add-ons listed *above* the horizontal line are specific to the environment you selected. If you select any add-ons in this part of the list and then select a different environment, your selection will be lost.
- Add-ons listed *below* the horizontal line are available for all environments. Selecting a different environment will not impact the selections made in this part of the list.

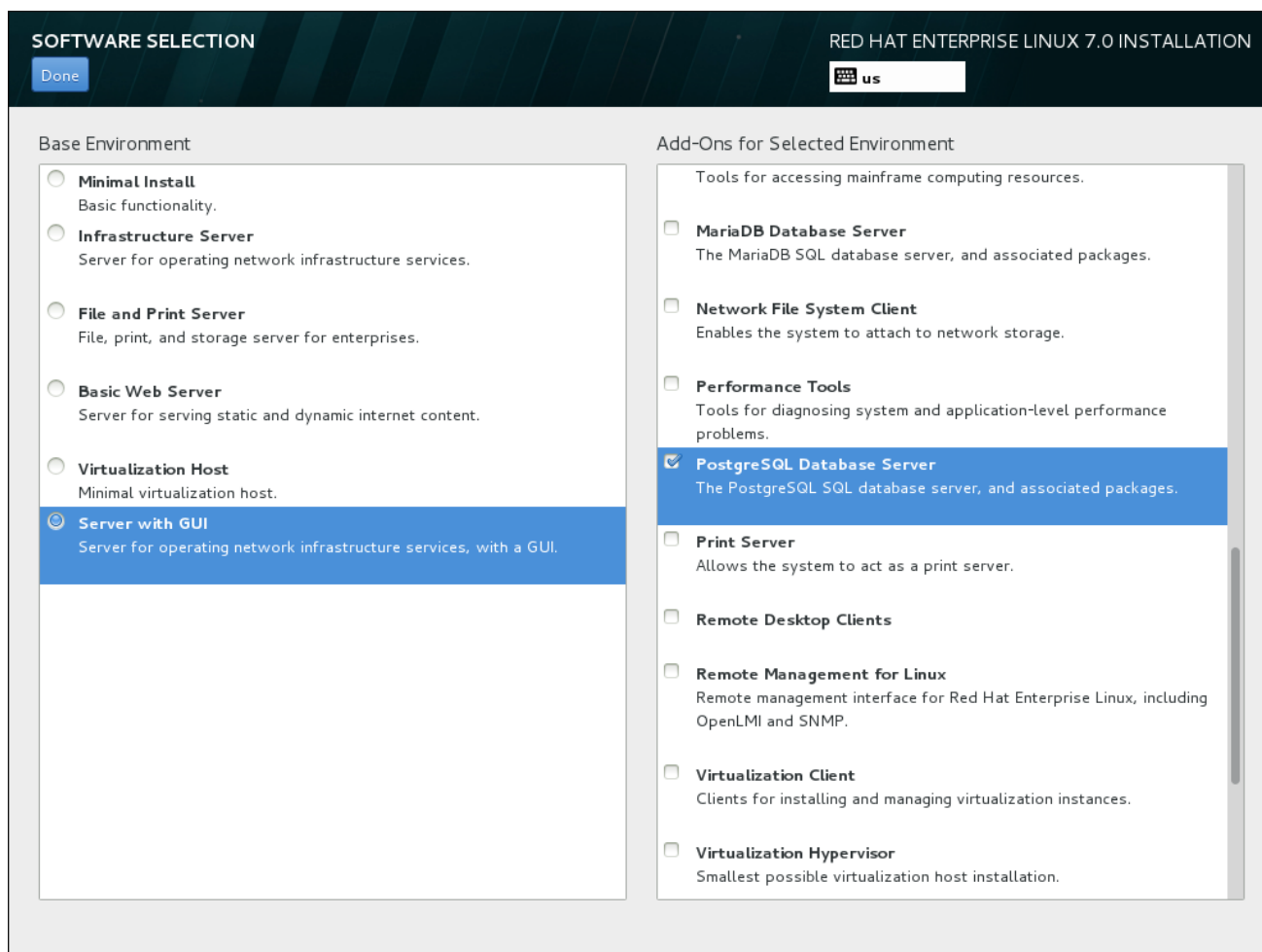


Figure 13.15. Example of a Software Selection for a Server Installation

The availability of base environments and add-ons depends on the variant of the installation ISO image which you are using as the installation source. For example, the **server** variant provides environments designed for servers, while the **workstation** variant has several choices for deployment as a developer workstation, and so on.

The installation program does not show which packages are contained in the available environments. To see which packages are contained in a specific environment or add-on, see the **repodata/*-comps-variant.architecture.xml** file on the Red Hat Enterprise Linux Installation DVD which you are using as the installation source. This file contains a structure describing available environments (marked by the **<environment>** tag) and add-ons (the **<group>** tag).



IMPORTANT

The pre-defined environments and add-ons allow you to customize your system, but in a manual installation, there is no way to select individual packages to install. If you are not sure what package should be installed, Red Hat recommends you to select the **Minimal Install** environment. **Minimal install** only installs a basic version of Red Hat Enterprise Linux with only a minimal amount of additional software. This will substantially reduce the chance of the system being affected by a vulnerability. After the system finishes installing and you log in for the first time, you can use the **Yum** package manager to install any additional software you need. For more details on **Minimal install**, see the [Installing the Minimum Amount of Packages Required](#) section of the Red Hat Enterprise Linux 7 Security Guide.

Alternatively, automating the installation with a Kickstart file allows for a much higher degree of control

over installed packages. You can specify environments, groups and individual packages in the **%packages** section of the Kickstart file. See [Section 27.3.2, “Package Selection”](#) for instructions on selecting packages to install in a Kickstart file, and [Chapter 27, Kickstart Installations](#) for general information about automating the installation with Kickstart.

Once you have selected an environment and add-ons to be installed, click **Done** to return to the **Installation Summary** screen.

13.14.1. Core Network Services

All Red Hat Enterprise Linux installations include the following network services:

- centralized logging through the **rsyslog** service
- email through SMTP (Simple Mail Transfer Protocol)
- network file sharing through NFS (Network File System)
- remote access through SSH (Secure SHell)
- resource advertising through mDNS (multicast DNS)

Some automated processes on your Red Hat Enterprise Linux system use the email service to send reports and messages to the system administrator. By default, the email, logging, and printing services do not accept connections from other systems.

You can configure your Red Hat Enterprise Linux system after installation to offer email, file sharing, logging, printing, and remote desktop access services. The SSH service is enabled by default. You can also use NFS to access files on other systems without enabling the NFS sharing service.

13.15. INSTALLATION DESTINATION

To select the disks and partition the storage space on which you will install Red Hat Enterprise Linux, select **Installation Destination** in the **Installation Summary** screen. If you are unfamiliar with disk partitions, see [Appendix A, An Introduction to Disk Partitions](#) for more information.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.



IMPORTANT

If you install Red Hat Enterprise Linux in text mode, you can only use the default partitioning schemes described in this section. You cannot add or remove partitions or file systems beyond those that the installation program automatically adds or removes.

IMPORTANT

Special cases

- If you have a RAID card, be aware that some BIOS types do not support booting from the RAID card. In such a case, the **/boot** partition must be created on a partition outside of the RAID array, such as on a separate hard drive. It is necessary to use an internal hard drive for partition creation with problematic RAID cards. A **/boot** partition is also necessary for software RAID setups. If you have chosen to automatically partition your system, you should manually edit your **/boot** partition; see [Section 13.15.4, “Manual Partitioning”](#) for more details.
- When you install Red Hat Enterprise Linux on a system with both multipath and non-multipath storage devices, the automatic partitioning layout in the installation program might create volume groups that contain a mix of multipath and non-multipath devices. This defeats the purpose of multipath storage. We advise that you select only multipath or only non-multipath devices on the **Installation Destination** screen. Alternatively, proceed to manual partitioning.

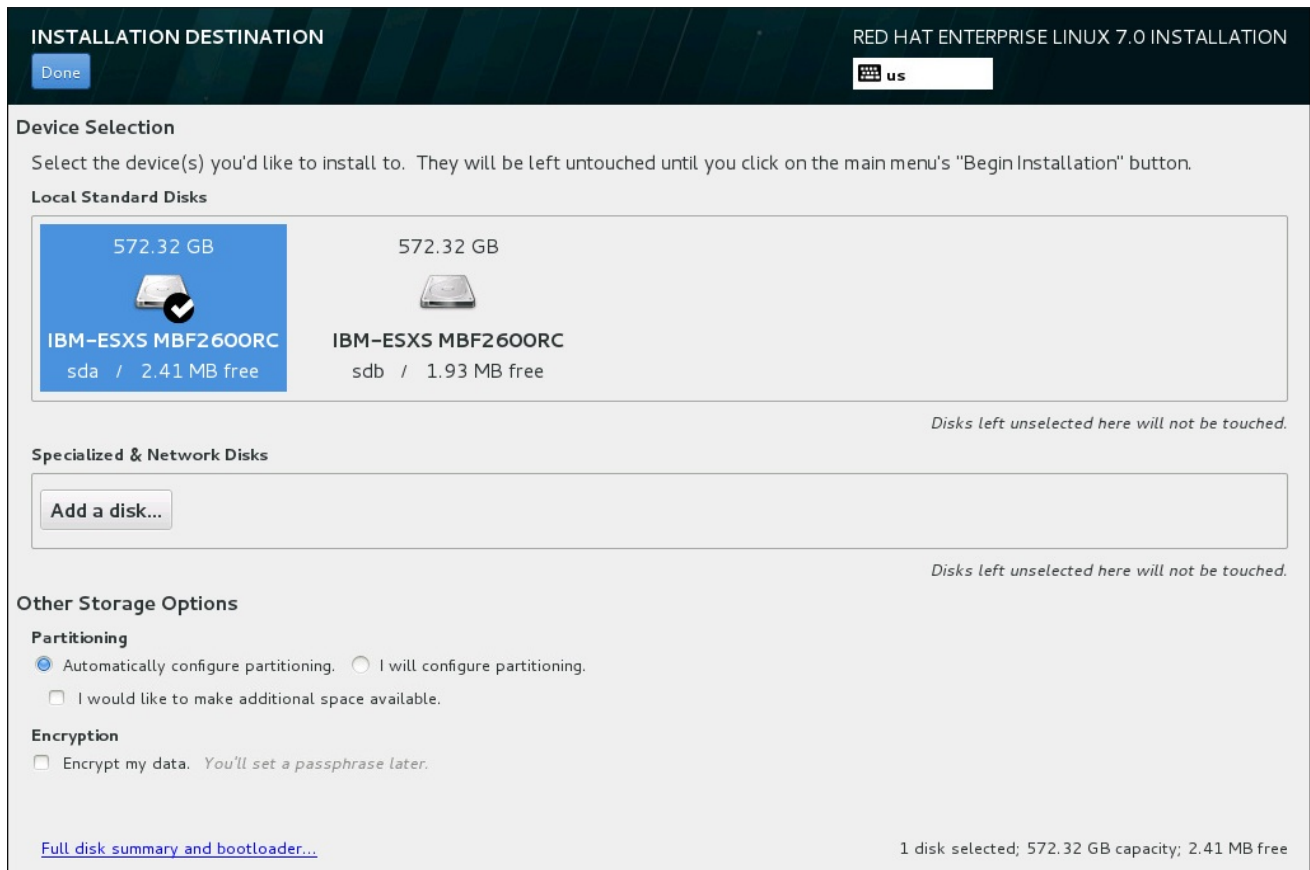


Figure 13.16. Storage Space Overview

On this screen, you can see storage devices available locally on your computer. You can also add additional specialized or network devices by clicking the **Add a disk** button. To learn more about these devices see [Section 13.16, “Storage Devices”](#).

If you do not feel comfortable with partitioning your system, leave the default selection of the **Automatically configure partitioning** radio button to let the installation program partition the storage devices for you.

Below the panes for storage devices is a form of additional controls labeled **Other Storage Options**:

- In the **Partitioning** section, you can select how storage devices are partitioned and how volumes are created. You can configure the partitions, manually or allow the installation program to do it automatically.

Automatic partitioning is recommended if you are doing a clean installation on previously unused storage or do not need to keep any data that might be present on the storage. To proceed this way, leave the default selection of the **Automatically configure partitioning** radio button, and the installation program will create the necessary partitions and volumes on the storage space for you.

For automatic partitioning, you can also select the **I would like to make additional space available** check box to choose how to reassign space from other file systems to this installation. After you click **Done**, there are two dialogs that can appear. If you selected automatic partitioning but there is not enough storage space to complete the installation using the recommended partitioning configuration, a dialog will appear:

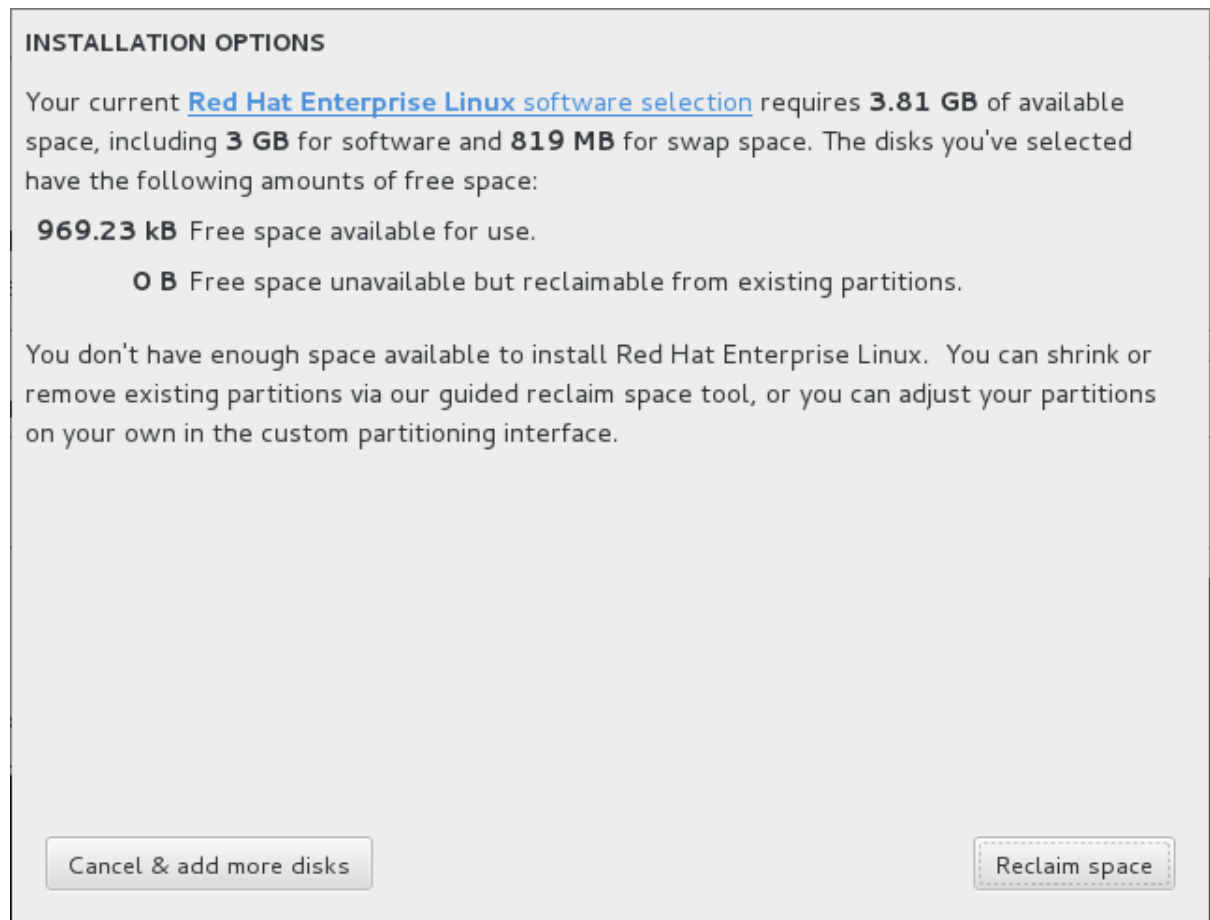


Figure 13.17. Installation Options Dialog with Option to Reclaim Space

You can click to the **Red Hat Enterprise Linux software selection** link. The link will navigate you to the **Software selection** section, where you can change what software you want to install, and free some additional storage space.

Alternatively, you can click **Cancel & add more disks** to return to the **Installation Destination** screen, where it is possible to add more storage devices or choose to configure partitioning manually. Click **Reclaim space** to free some storage space from existing file systems. See [Section 13.15.3, “Reclaim Disk Space”](#) for details.

The second dialog appears if you cannot create enough free space. In this case, you have to add more disks on the initial storage screen or quit the installation.

If you select the **I will configure partitioning** radio button for manual setup, you will be brought to the **Manual Partitioning** screen after clicking **Done**. See [Section 13.15.4, “Manual Partitioning”](#) for details.

- In the **Encryption** section, you can select the **Encrypt my data** check box to encrypt all partitions except for the **/boot** partition. See the [Red Hat Enterprise Linux 7 Security Guide](#) for information on encryption.

At the bottom of the screen is the **Full disk summary and bootloader** button for you to configure a disk on which a boot loader will be installed.

See [Section 13.15.1, “Boot Loader Installation”](#) for more information.

Click the **Done** button once you have made your selections to either return to the **Installation Summary** screen or to proceed to the **Manual Partitioning** screen.

13.15.1. Boot Loader Installation

Red Hat Enterprise Linux uses GRUB2 (GRand Unified Bootloader version 2) as its boot loader. The boot loader is the first program that runs when the computer starts and is responsible for loading and transferring control to an operating system. GRUB2 can boot any compatible operating system and can also use *chain loading* to transfer control to other boot loaders for unsupported operating systems.



WARNING

Installing GRUB2 can overwrite your existing boot loader.

If you have other operating systems already installed, Red Hat Enterprise Linux attempts to automatically detect and configure GRUB2 to boot them. You can manually configure any additional operating systems if they are not detected properly.

To specify which device the boot loader should be installed on, click the **Full disk summary and bootloader** link at the bottom of the **Installation Destination** screen. The **Selected Disks** dialog will appear. If you are partitioning the drive manually, this dialog can be reached by clicking **Storage device/s selected** on the **Manual Partitioning** screen.

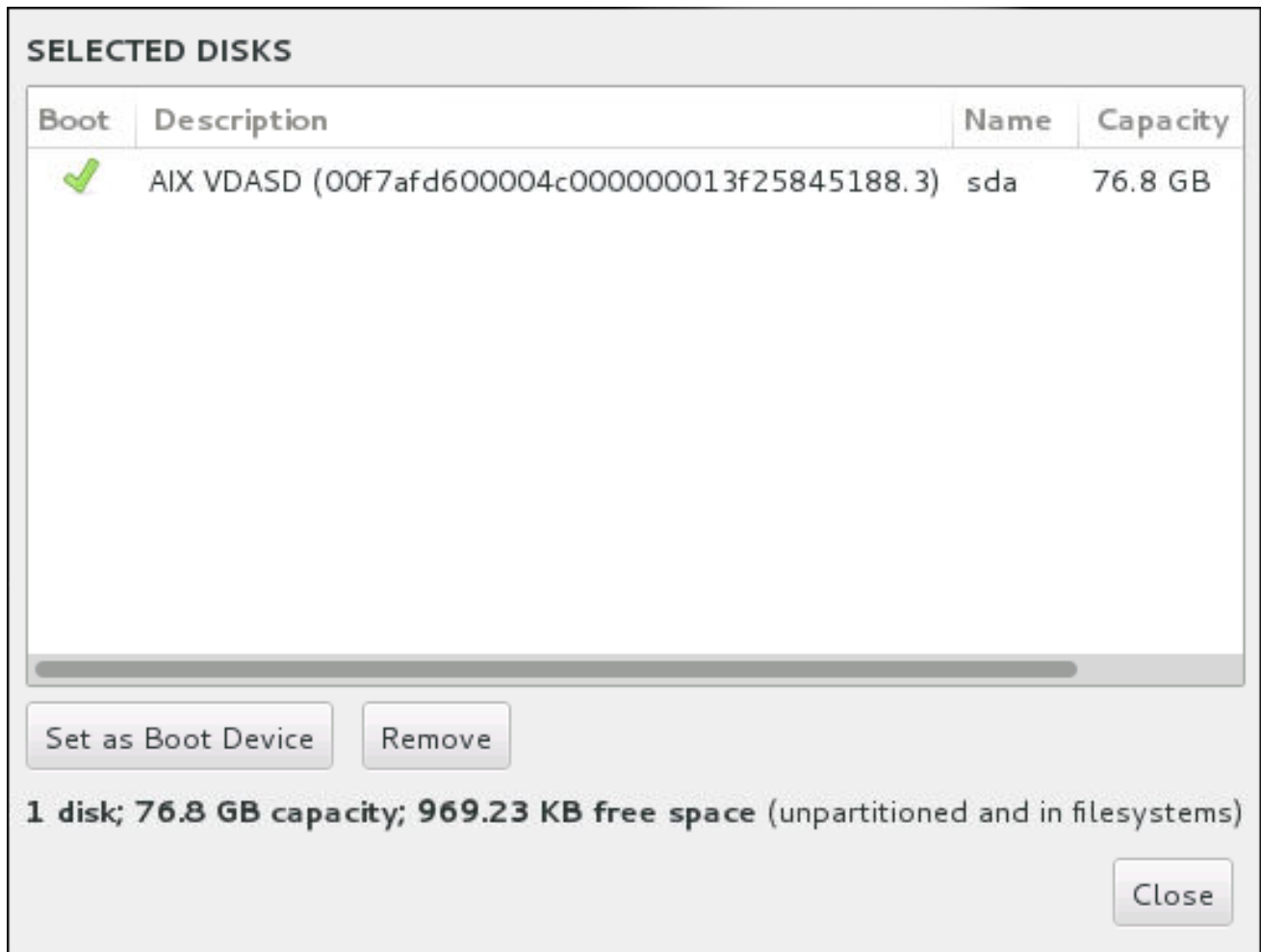


Figure 13.18. Summary of Selected Disks

In the **Boot** column, a green tick icon marks one of the devices as the intended boot device. To change the boot device, select a device from the list and click the **Set as Boot Device** button to install the boot loader there instead.

To decline installation of a new boot loader, select the marked device and click the **Do not install bootloader** button. This will remove the tick and ensure GRUB2 is not installed on any device.



WARNING

If you choose not to install a boot loader for any reason, you will not be able to boot the system directly, and you must use another boot method, such as a commercial boot loader application. Use this option only if you are sure you have another way to boot your system.

13.15.2. Encrypt Partitions

If you selected the **Encrypt my data** option, when you click to proceed to the next screen the installation program will prompt you for a passphrase with which to encrypt the partitions on the system.

Partitions are encrypted using the *Linux Unified Key Setup* – see the [Red Hat Enterprise Linux 7 Security Guide](#) for more information.

DISK ENCRYPTION PASSPHRASE

You have chosen to encrypt some of your data. You will need to create a passphrase that you will use to access your data when you start your computer.

Passphrase:

 Strong 

Confirm:

 Warning: You won't be able to switch between keyboard layouts (from the default one) when you decrypt your disks after install.

Figure 13.19. Enter Passphrase for an Encrypted Partition

Choose a passphrase and type it into each of the two fields in the dialog box. Note that you need to use the same keyboard layout for setting up this passphrase that you will use to unlock partitions later. Use the language layout icon to ensure the correct layout is selected. You must provide this passphrase every time that the system boots. Press **Tab** while in the **Passphrase** input field to retype it. If the passphrase is too weak, a warning icon appears in the field and you will not be allowed to type in the second field. Hover your mouse cursor over the warning icon to learn how to improve the passphrase.



WARNING

If you lose this passphrase, any encrypted partitions and the data on them will become completely inaccessible. There is no way to recover a lost passphrase.

Note that if you perform a Kickstart installation, you can save encryption passphrases and create backup encryption passphrases during installation. See the [Red Hat Enterprise Linux 7 Security Guide](#) for more information about disk encryption.

13.15.3. Reclaim Disk Space

If there is insufficient space to install Red Hat Enterprise Linux on the disks selected in **Installation Destination** and you selected **Reclaim Space** at the **Installation Options** dialog, the **Reclaim Disk Space** dialog appears.

**WARNING**

Unless you select to shrink a partition, reclaiming space on a partition involves deleting all the data on it and you should always verify that any data you need to keep was backed up.

RECLAIM DISK SPACE

You can remove existing filesystems you no longer need to free up space for this installation. Removing a filesystem will permanently delete all of the data it contains.

Disk	Name	Filesystem	Reclaimable Space	Action
▼ 76.8 GB AIX VDASD	sda		76.79 GB total	Preserve
└─ PPC PReP Boot	sda1	prepboot	Not resizeable	Preserve
└─ /boot (Red Hat Enterprise Linux Server Linux 7.0 for ppc64)	sda2	xf	Not resizeable	Preserve
└─ rhel_ibm-p730-06-lp3	sda3	lvmpv	Not resizeable	Preserve

Preserve
Delete
Shrink

Delete all

1 disk; 76.79 GB reclaimable space (in filesystems)

Total selected space to reclaim: 0 B

Installation requires a total of 1.24 GB for system data.

Cancel

Reclaim space

Figure 13.20. Reclaim Disk Space from Existing File Systems

The existing file systems Red Hat Enterprise Linux has detected are listed in a table as part of their respective disks. The **Reclaimable Space** column lists the space that could be reassigned to this installation. The **Action** column lists what action will be taken with the file system to reclaim space.

Beneath the table are four buttons:

- **Preserve** - leaves the file system untouched and no data will be deleted. This is the default action.
- **Delete** - removes the file system entirely. All the space it takes up on the disk will be made available for the installation.
- **Shrink** - recovers free space from the file system and makes it available for this installation. Use the slider to set a new size for the selected partition. Can only be used on resizable partitions where LVM or RAID is not used.

- **Delete all/Preserve all** - this button, located on the right, marks all file systems for deletion by default. Upon clicking, it changes the label and allows you to mark all file systems to be preserved again.

Select a file system or a whole disk in the table with your mouse and click one of the buttons. The label in the **Action** column will change to match your selection and the amount of **Total selected space to reclaim** displayed beneath the table will adjust accordingly. Beneath this value is the amount of space the installation requires based on the packages you have selected to install.

When enough space has been reclaimed for the installation to proceed, the **Reclaim Space** button will become available. Click this button to return to the Installation Summary screen and proceed with the installation.

13.15.4. Manual Partitioning

The **Manual Partitioning** screen is displayed when you click **Done** from Installation Destination if you selected the **I will configure partitioning** option. On this screen you configure your disk partitions and mount points. This defines the file system that Red Hat Enterprise Linux will be installed on.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.

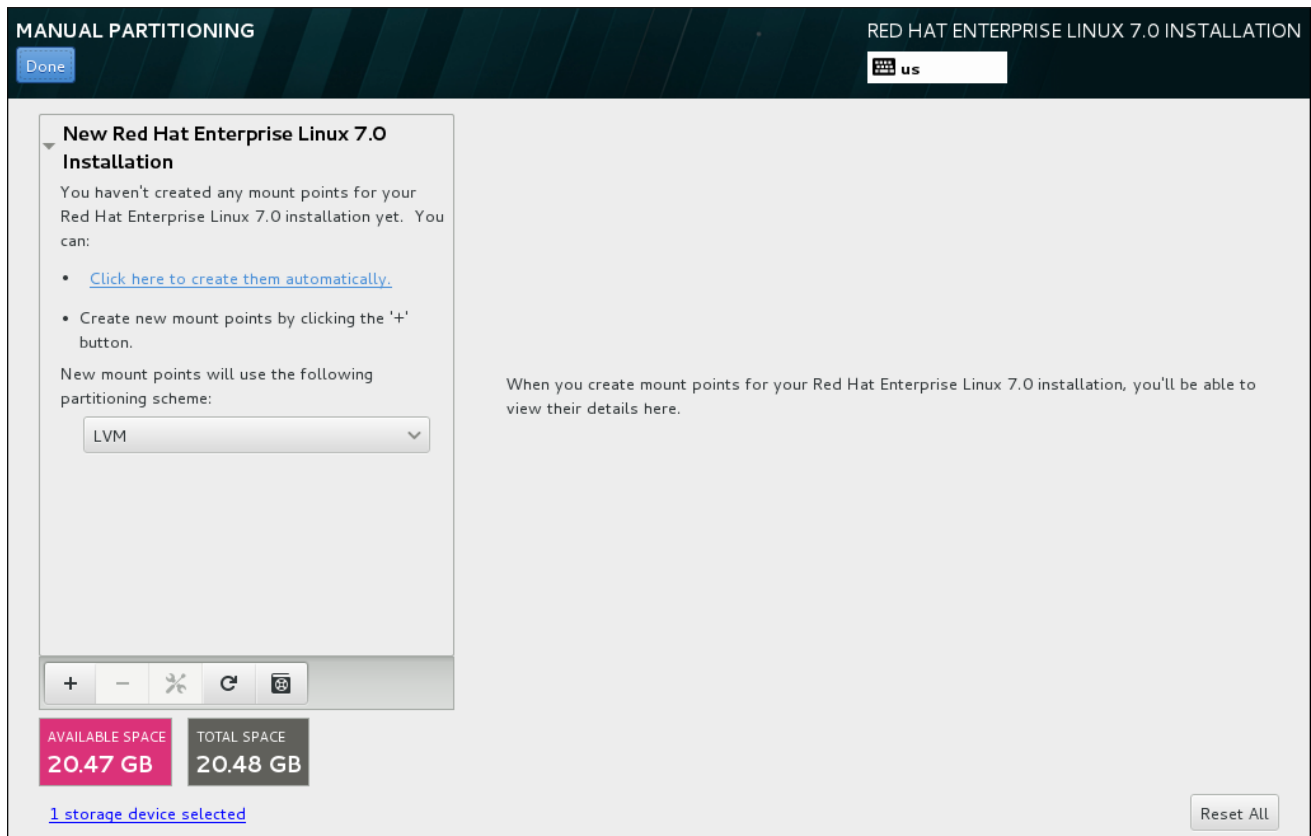


Figure 13.21. The Manual Partitioning Screen

The **Manual Partitioning** screen initially features a single pane on the left for the mount points. The pane is either empty except for information about creating mount points, or it displays existing mount points that the installation program has detected. These mount points are organized by detected operating system installations. Therefore, some file systems might be displayed multiple times if a partition is shared among several installations. The total space and available space on selected storage devices are displayed beneath this pane.

If your system contains existing file systems, ensure that enough space will be available for the installation. Use the - button to remove unneeded partitions.

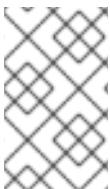


NOTE

For recommendations and additional information about disk partitions, see [Appendix A, *An Introduction to Disk Partitions*](#) and [Section 13.15.4.4, “Recommended Partitioning Scheme”](#). At a bare minimum, you need an appropriately sized root partition, and usually a swap partition appropriate to the amount of RAM you have on your system.

13.15.4.1. Adding File Systems and Configuring Partitions

An installation of Red Hat Enterprise Linux requires a PReP boot partition and one other partition but Red Hat recommends at least five: **PReP**, **/**, **/home**, **/boot**, and **swap**. You can also create additional partitions you require. See [Section 13.15.4.4, “Recommended Partitioning Scheme”](#) for further details.



NOTE

If you have any specific requirements for some partitions (for example, requiring that a particular partition be on a specific disk) and less specific requirements for other partitions, create the partitions first which have more specific requirements.

Adding a file system is a two-step process. You first create a mount point in a certain partitioning scheme. The mount point appears in the left pane. Next, you can customize it using the options in the right pane, where you can change the mount point, capacity, the device type, file system type, label, and whether to encrypt or reformat the corresponding partition.

If you have no existing file systems and want the installation program to create the required file systems and their mount points for you, select your preferred partitioning scheme from the drop-down menu in the left pane (default for Red Hat Enterprise Linux is LVM), then click the link on top of the pane for creating mount points automatically. This will generate a **/boot** partition, a **/** (root) volume, and a swap volume proportionate to the size of the available storage. These are the recommended file systems for a typical installation, but you can add additional file systems and mount points if you need to.

Alternatively, create individual mount points using the **+** button at the bottom of the pane. The **Add a New Mount Point** dialog then opens. Either select one of the preset paths from the **Mount Point** drop-down menu or type your own; for example, select **/** for the root partition or **/boot** for the boot partition. Then enter the size of the file system in the **Desired Capacity** text field; for example, **2GiB**. If you leave the field empty or if you specify a size bigger than available space, all remaining free space is used instead. After entering these details, click the **Add mount point** button to create the partition.



NOTE

To avoid problems with space allocation, first create small partitions with known fixed sizes, such as **/boot**, and then create the rest of the partitions, letting the installation program allocate the remaining capacity to them.

Similarly, if you have multiple disks that the system is to reside on, they differ in size, and a particular partition must be created on the first disk detected by BIOS, be sure to start by creating such a partition.

For each new mount point you create manually, you can set its partitioning scheme from the drop-down menu located in the left pane. The available options are **Standard Partition**, **BTRFS**, **LVM**, and **LVM Thin Provisioning**. Note that the **/boot** partition will always be located on a standard partition, regardless of the value selected in this menu.

To change on which devices a single non-LVM mount point should be located, select the mount point and click the **Modify...** button in the right pane to open the **Configure Mount Point** dialog. Select one or more devices and click **Select**. After the dialog closes, note that you also need to confirm this setting by clicking the **Update Settings** button on the right side of the **Manual Partitioning** screen.

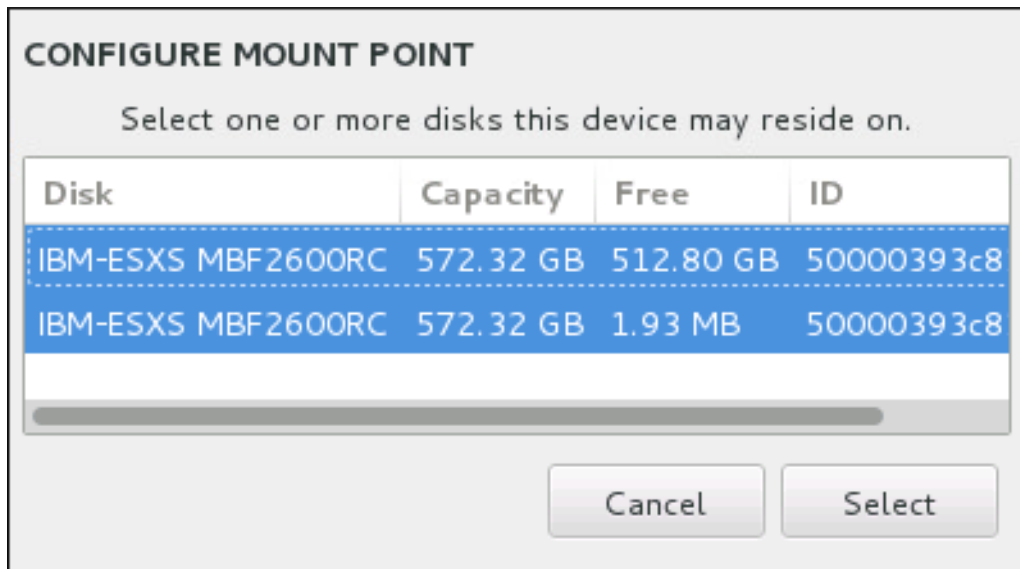


Figure 13.22. Configuring Mount Points

To refresh information about all local disks and partitions on them, click the **Rescan** button (with the circular arrow icon on it) in the toolbar. You only need to do this action after performing advanced partition configuration outside the installation program. Note that if you click the **Rescan Disks** button, all configuration changes you previously made in the installation program will be lost.

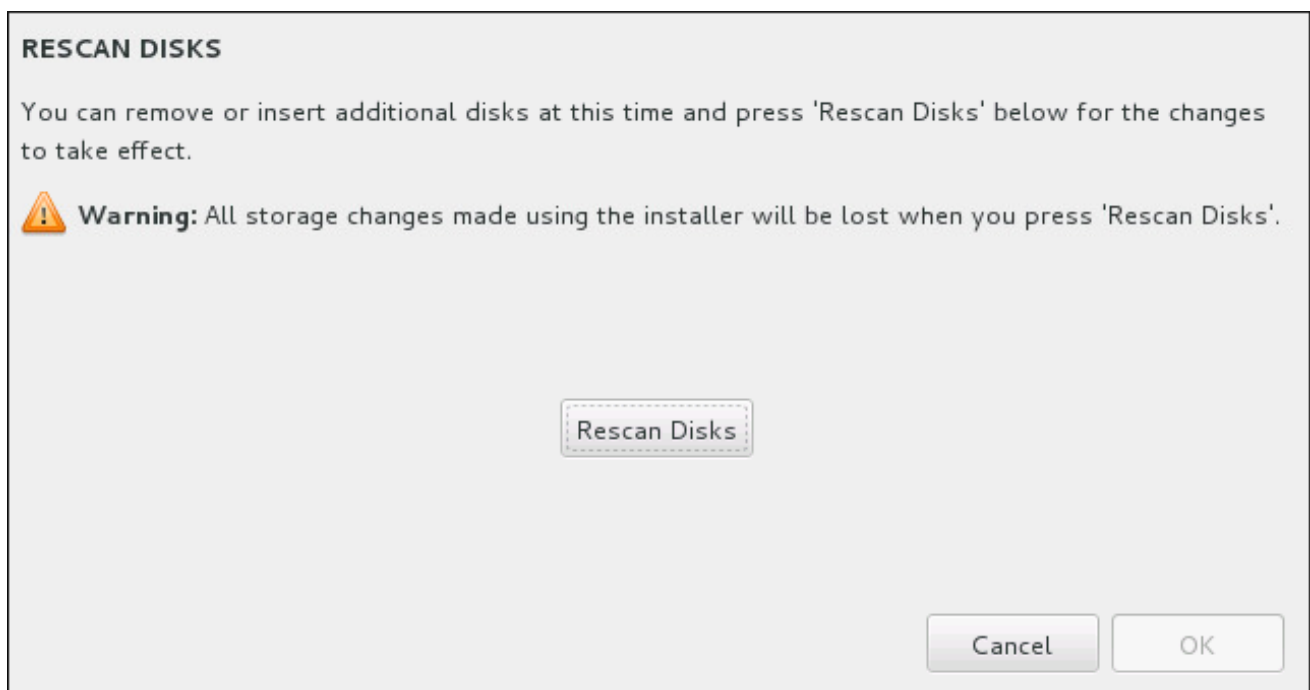


Figure 13.23. Rescanning Disks

At the bottom of the screen, a link states how many storage devices have been selected in **Installation Destination** (see [Section 13.15, "Installation Destination"](#)). Clicking on this link opens the **Selected Disks** dialog, where you review the information about the disks. See [Section 13.15.1, "Boot Loader Installation"](#) for more information.

To customize a partition or a volume, select its mount point in the left pane and the following customizable features then appear to the right:

MANUAL PARTITIONING RED HAT ENTERPRISE LINUX 7.1 INSTALLATION

Done us Help!

New Red Hat Enterprise Linux 7.1 Installation

DATA

SYSTEM

/boot 476 MiB

sdal

/ 4743.34 MiB >

rhel-root

swap 1020 MiB

rhel-swap

rhel-root

Mount Point: /

Device(s): ATA QEMU HARDDISK (sda) and 1 other

Desired Capacity: 4743.34 MiB

Device Type: LVM ☐ Encrypt

File System: xfs ☒ Reformat

Volume Group: rhel (0 B free)

Label:

Name: root

Update Settings

Note: The settings you make on this screen will not be applied until you click on the main menu's 'Begin Installation' button.

Reset All

AVAILABLE SPACE 2395 KiB TOTAL SPACE 6246.4 MiB

[2 storage devices selected](#)

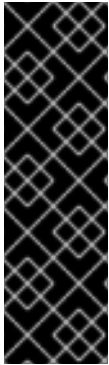
Figure 13.24. Customizing Partitions

- **Mount Point** – enter the file system's mount point. For example, if a file system should be the root file system, enter `/`; enter `/boot` for the `/boot` file system, and so on. For a swap file system, the mount point should not be set – setting the file system type to **swap** is sufficient.
- **Desired Capacity** – enter the desired size of the file system. You can use common size units such as KiB or GiB. The default is MiB if no other unit is specified.
- **Device type** – choose one of these types: **Standard Partition**, **LVM**, **RAID**, **LVM Thin Provisioning**, or **BTRFS**. Check the adjacent **Encrypt** box to encrypt the partition or volume. You will be prompted to set a password later. **RAID** is only available if two or more disks are selected for partitioning, and if you choose this type, you can also set the **RAID Level**. Similarly, if you select **LVM**, you can specify the **Volume Group**.
- **File system** – in the drop-down menu, select the appropriate file system type for this partition or volume. Check the adjacent **Reformat** box to format an existing partition, or leave it unchecked to retain your data. Note that newly created partitions and volumes must be reformatted, and the check box cannot be unchecked in this case.
- **Label** – assign a label to the partition. Labels are used for you to easily recognize and address individual partitions.
- **Name** – assign a name to an LVM or Btrfs volume. Note that standard partitions are named automatically when they are created and their name cannot be edited, such as `/home` being assigned the name `sda1`.

See [Section 13.15.4.1.1, "File System Types"](#) for more information about file system and device types.

Click the **Update Settings** button to save your changes and select another partition to customize. Note that the changes will not be applied until you actually start the installation from the Installation summary page. Click the **Reset All** button to discard all changes to all partitions and start over.

When all file systems and mount points have been created and customized, click the **Done** button. If you chose to encrypt any file system, you will now be prompted to create a passphrase. Then, a dialog appears, showing a summary of all actions related to storage that the installation program will take. This includes creating, resizing, or deleting partitions and file systems. You can review all the changes and click **Cancel & Return to Custom Partitioning** to go back. To confirm your changes, click **Accept Changes** to return to the Installation Summary page. To partition additional devices, select them in the **Installation Destination** screen, return to the **Manual Partitioning** screen, repeat the steps outlined in this section for the additional devices.



IMPORTANT

If **/usr** or **/var** is partitioned separately from the rest of the root volume, the boot process becomes much more complex because these directories contain components critical to it. In some situations, such as when these directories are placed on an iSCSI drive or an FCoE location, the system can either be unable to boot, or hang with a **Device is busy** error when powering off or rebooting.

This limitation only applies to **/usr** or **/var**, not to directories below them. For example, a separate partition for **/var/www** will work without issues.

13.15.4.1.1. File System Types

Red Hat Enterprise Linux allows you to create different device types and file systems. The following is a brief description of the different device types and file systems available, and how they can be used.

Device Types

- **standard partition** - A standard partition can contain a file system or swap space, or it can provide a container for software RAID or an LVM physical volume.
- **logical volume (LVM)** - Creating an LVM partition automatically generates an LVM logical volume. LVM can improve performance when using physical disks. For information on how to create a logical volume, see [Section 13.15.4.3, "Create LVM Logical Volume"](#). For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.
- **LVM thin provisioning** - Using thin provisioning, you can manage a storage pool of free space, known as a thin pool, which can be allocated to an arbitrary number of devices when needed by applications. The thin pool can be expanded dynamically when needed for cost-effective allocation of storage space. For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.



NOTE

The installer will automatically reserve 20% of any requested space for an LVM thin pool logical volume in the volume group containing it. This is a safety measure to ensure that you can extend either the metadata volume or the data volume of your thinly provisioned logical volume.

- **software RAID** - Creating two or more software RAID partitions allows you to create a RAID

device. One RAID partition is assigned to each disk on the system. To create a RAID device, see [Section 13.15.4.2, "Create Software RAID"](#). For more information regarding RAID, see the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

File Systems

- **xfs** – XFS is a highly scalable, high-performance file system that supports file systems up to 16 EiB (approximately 16 billion GiB), files up to 8 EiB (approximately 8 billion GiB), and directory structures containing tens of millions of entries. XFS supports metadata journaling, which facilitates quicker crash recovery. The XFS file system can also be defragmented and resized while mounted and active. This file system is selected by default and is highly recommended. For information on how to translate common commands from previously used ext4 file system to XFS, see [Appendix F, Reference Table for ext4 and XFS Commands](#).

The maximum supported size of an XFS file system in Red Hat Enterprise Linux is currently 500 TiB.

- **ext4** – The ext4 file system is based on the ext3 file system and features a number of improvements. These include support for larger file systems and larger files, faster and more efficient allocation of disk space, no limit on the number of subdirectories within a directory, faster file system checking, and more robust journaling.

The maximum supported size of an ext4 file system in Red Hat Enterprise Linux is currently 50 TiB.

- **ext3** – The ext3 file system is based on the ext2 file system and has one main advantage – journaling. Using a journaling file system reduces time spent recovering a file system after a crash as there is no need to check the file system for metadata consistency by running the **fsck** utility every time a crash occurs.
- **ext2** – An ext2 file system supports standard Unix file types, including regular files, directories, or symbolic links. It provides the ability to assign long file names, up to 255 characters.
- **vfat** – The VFAT file system is a Linux file system that is compatible with Microsoft Windows long file names on the FAT file system.
- **swap** – Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- **PReP** – this small boot partition is located on the first partition of the hard drive. The PReP boot partition contains the GRUB2 boot loader, which allows other IBM Power Systems servers to boot Red Hat Enterprise Linux.

Each file system has different size limits for the file system itself as well as individual files contained within. For a list of maximum supported file and file system sizes, see the Red Hat Enterprise Linux technology capabilities and limits page, available on the Customer Portal at <https://access.redhat.com/site/articles/rhel-limits>.

13.15.4.2. Create Software RAID

Redundant arrays of independent disks (RAIDs) are constructed from multiple storage devices that are arranged to provide increased performance and, in some configurations, greater fault tolerance. See below for a description of different kinds of RAID.

A RAID device is created in one step and disks are added or removed as necessary. One RAID partition per physical disk is allowed for each device, so the number of disks available to the installation program

determines which levels of RAID device are available to you. For example, if your system has two hard drives, the installation program will not allow you to create a RAID10 device, which requires 4 separate partitions.

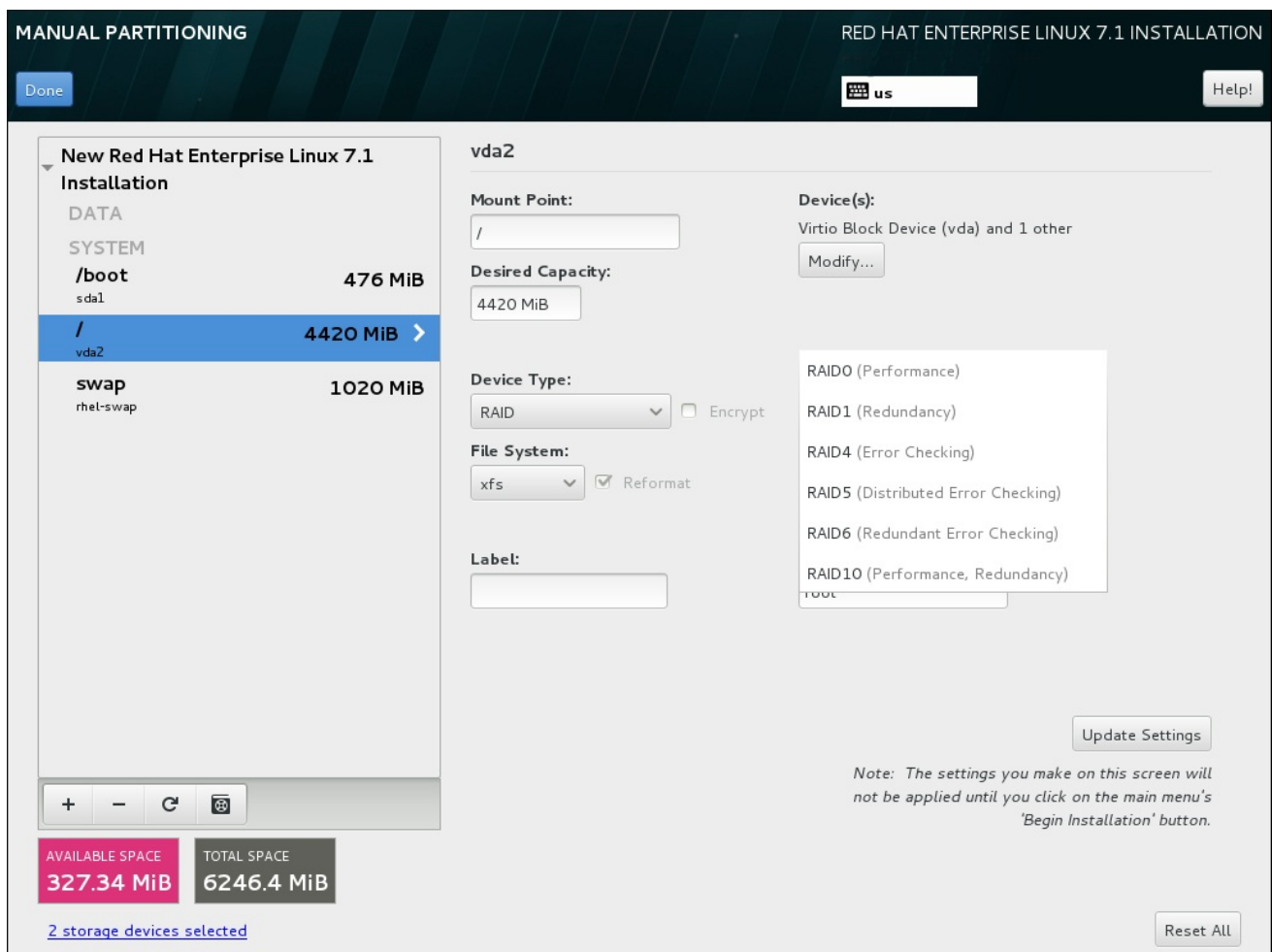


Figure 13.25. Creating a Software RAID Partition - the Device Type Menu Expanded

RAID configuration options are only visible if you have selected two or more disks for installation. At least two disks are required to create a RAID device.

To create a RAID device:

1. Create a mount point as described in [Section 13.15.4.1, "Adding File Systems and Configuring Partitions"](#). By configuring this mount point, you configure the RAID device.
2. Keeping the partition selected in the left pane, select the configuration button below the pane to open the **Configure Mount Point** dialog. Select which disks will be included in the RAID device and click **Select**.
3. Click the **Device Type** drop-down menu and select **RAID**.
4. Click the **File System** drop-down menu and select your preferred file system type (see [Section 8.14.4.1.1, "File System Types"](#)).
5. Click the **RAID Level** drop-down menu and select your preferred level of RAID.

The available RAID levels are:

RAID0 - Optimized performance (stripe)

Distributes data across multiple disks. Level 0 RAID arrays offer increased performance over standard partitions, and can be used to pool the storage of multiple disks into one large virtual device. Note that Level 0 RAID arrays offer no redundancy, and that the failure of one device in the array destroys data in the entire array. RAID 0 requires at least two RAID partitions.

RAID1 - Redundancy (mirror)

Mirrors all data on one disk onto one or more other disks. Additional devices in the array provide increasing levels of redundancy. RAID 1 requires at least two RAID partitions.

RAID4 - Error detection (parity)

Distributes data across multiple disks, and uses one disk in the array to store parity information that safeguards the array in case any disk within the array fails. Because all parity information is stored on one disk, access to this disk creates a bottleneck in the performance of the array. RAID 4 requires at least three RAID partitions.

RAID5 - Distributed error detection

Distributes data *and* parity information across multiple disks. Level 5 RAID arrays therefore offer the performance advantages of distributing data across multiple disks, but do not share the performance bottleneck of level 4 RAID arrays because the parity information is also distributed through the array. RAID 5 requires at least three RAID partitions.

RAID6 - Redundant

Level 6 RAID arrays are similar to level 5 RAID arrays, but instead of storing only one set of parity data, they store two sets. RAID 6 requires at least four RAID partitions.

RAID10 - Redundancy (mirror) and Optimized performance (stripe)

Level 10 RAID arrays are *nested RAID arrays* or *hybrid RAID arrays*. They are constructed by distributing data over mirrored sets of disks. For example, a level 10 RAID array constructed from four RAID partitions consists of two mirrored pairs of striped partitions. RAID 10 requires at least four RAID partitions.

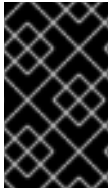
6. Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.

If fewer disks are included than the specified RAID level requires, a message will be displayed at the bottom of the window, informing you how many disks are actually required for your selected configuration.

13.15.4.3. Create LVM Logical Volume

Logical Volume Management (LVM) presents a simple logical view of underlying physical storage space, such as hard drives or LUNs. Partitions on physical storage are represented as *physical volumes* that can be grouped together into *volume groups*. Each volume group can be divided into multiple *logical volumes*, each of which is analogous to a standard disk partition. Therefore, LVM logical volumes function as partitions that can span multiple physical disks.

To learn more about LVM, see [Appendix D, Understanding LVM](#) or read the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide. Note that LVM configuration is only available in the graphical installation program.



IMPORTANT

During text-mode installation, LVM configuration is not available. If you need to create an LVM configuration from scratch, press **Ctrl+Alt+F2** to use a different virtual console, and run the **lvm** command. To return to the text-mode installation, press **Ctrl+Alt+F1**.

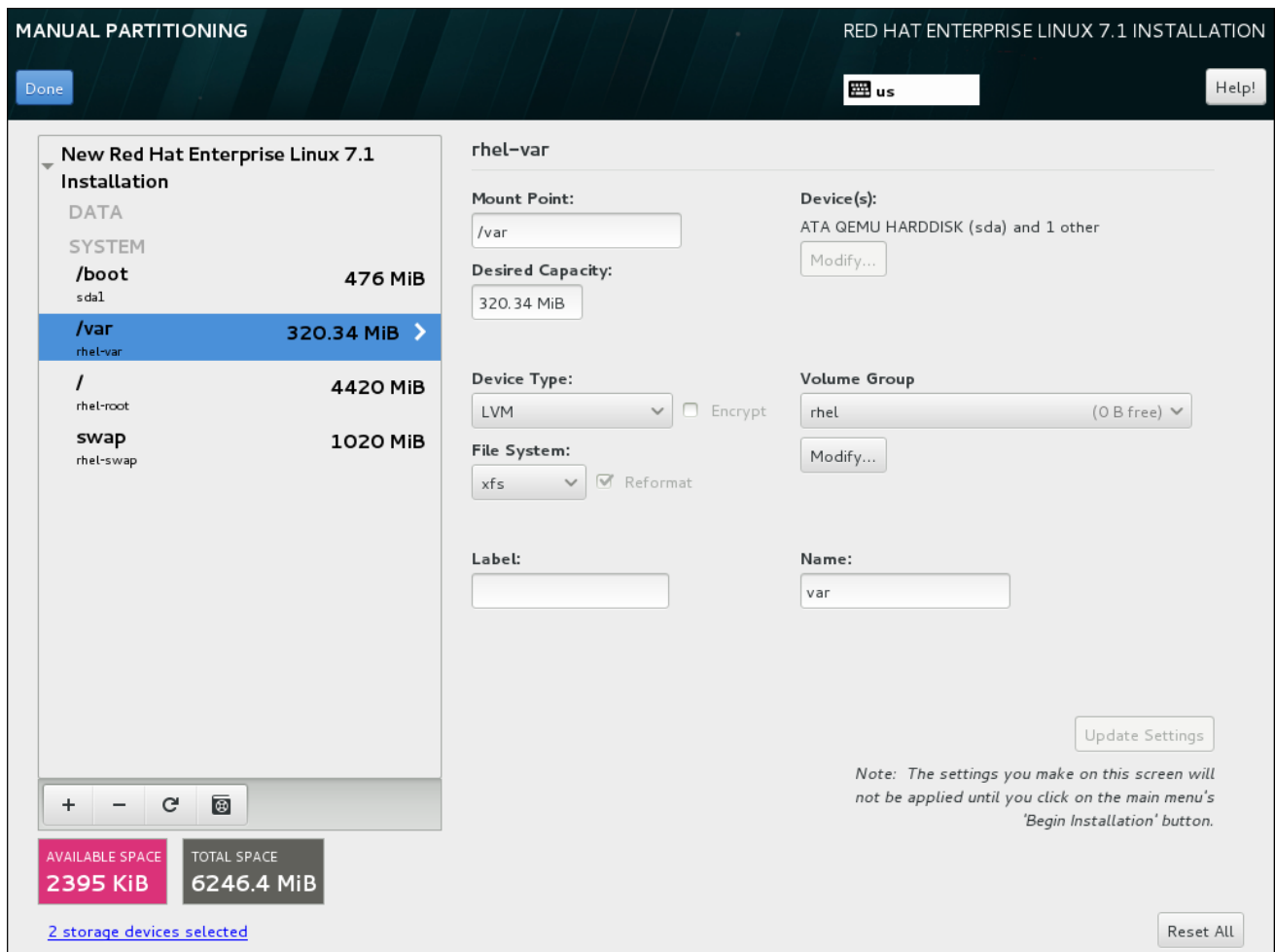


Figure 13.26. Configuring a Logical Volume

To create a logical volume and add it to a new or existing volume group:

1. Create a mount point for the LVM volume as described in [Section 13.15.4.1, “Adding File Systems and Configuring Partitions”](#).
2. Click the **Device Type** drop-down menu and select **LVM**. The **Volume Group** drop-down menu appears and displays the newly-created volume group name.
3. Optionally, either click the menu and select **Create a new volume group** or click **Modify** to configure the newly-created volume group, if you need to. Both the **Create a new volume group** option and the **Modify** button lead to the **Configure Volume Group** dialog, where you can rename the logical volume group and select which disks will be included.



NOTE

The configuration dialog does not allow you to specify the size of the volume group's physical extents. The size will always be set to the default value of 4 MiB. If you want to create a volume group with different physical extents, create it manually by switching to an interactive shell and using the **vgcreate** command, or use a Kickstart file with the **volgroup --pesize=size** command.

CONFIGURE VOLUME GROUP

Please create a name for this volume group and select at least one disk below.

Name:

Disk	Capacity	Free	ID
IBM-ESXS MBF2600RC	572.32 GB	512.80 GB	50000393c8187890
IBM-ESXS MBF2600RC	572.32 GB	1.93 MB	50000393c8187a18

RAID Level: ☐ Encrypt

Size policy:

Figure 13.27. Customizing an LVM Volume Group

The available RAID levels are the same as with actual RAID devices. See [Section 13.15.4.2, “Create Software RAID”](#) for more information. You can also mark the volume group for encryption and set the size policy for it. The available policy options are:

- **Automatic** - the size of the volume group is set automatically so that it is just large enough to contain the configured logical volumes. This is optimal if you do not need free space within the volume group.
- **As large as possible** - the volume group is created with maximum size, regardless of the size of the configured logical volumes it contains. This is optimal if you plan to keep most of your data on LVM and later need to increase the size of some existing logical volumes, or if you need to create additional logical volumes within this group.
- **Fixed** - with this option, you can set an exact size of the volume group. Any configured logical volumes must then fit within this fixed size. This is useful if you know exactly how large you would like the volume group to be.

Click **Save** when the group is configured.

- Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.



WARNING

Placing the **/boot** partition on an LVM volume is not supported.

13.15.4.4. Recommended Partitioning Scheme

Unless you have a reason for doing otherwise, Red Hat recommends that you create the following partitions:

A PReP boot partition - recommended size of 4 to 8 MiB

The first partition of the hard drive should include a PReP boot partition. This contains the **GRUB2** boot loader, which allows other IBM Power Systems servers to boot Red Hat Enterprise Linux.

/boot partition - recommended size at least 1 GiB

The partition mounted on **/boot** contains the operating system kernel, which allows your system to boot Red Hat Enterprise Linux, along with files used during the bootstrap process. Due to the limitations of most firmwares, creating a small partition to hold these is recommended. In most scenarios, a 1 GiB boot partition is adequate. Unlike other mount points, using an LVM volume for **/boot** is not possible - **/boot** must be located on a separate disk partition.



NOTE

If you have a RAID card, be aware that some BIOS types do not support booting from the RAID card. In such a case, the **/boot** partition must be created on a partition outside of the RAID array, such as on a separate hard drive.



WARNING

If you have a RAID card, be aware that Red Hat Enterprise Linux does not support setting up hardware RAID on an IPR card. You can boot the standalone diagnostics CD prior to installation to create a RAID array and then install to that RAID array.

root - recommended size of 10 GiB

This is where **/**, or the root directory, is located. The root directory is the top-level of the directory structure. By default, all files are written to this file system unless a different file system is mounted in the path being written to (for example, **/boot** or **/home**).

While a 5 GiB root file system allows you to install a minimal installation, it is recommended to allocate at least 10 GiB so that you can install as many package groups as you want.



IMPORTANT

Do not confuse the `/` directory with the `/root` directory. The `/root` directory is the home directory of the root user. The `/root` directory is sometimes referred to as *slash root* to distinguish it from the root directory.

`/home` - recommended size at least 1 GiB

To store user data separately from system data, create a dedicated file system for the `/home` directory. This file system should be sized based on the amount of data that will be stored locally, number of users, and so on. This will enable you to upgrade or reinstall Red Hat Enterprise Linux without erasing user data files. If you select automatic partitioning, it is recommended to have at least 55GiB of disk space available for the installation, to ensure that the `/home` file system is created.

swap partition - recommended size at least 1 GB

Swap file systems support virtual memory; data is written to a swap file system when there is not enough RAM to store the data your system is processing. Swap size is a function of system memory workload, not total system memory and therefore is not equal to the total system memory size. Therefore, it is important to analyze what applications a system will be running and the load those applications will serve in order to determine the system memory workload. Application providers and developers should be able to provide some guidance.

When the system runs out of swap space, the kernel terminates processes as the system RAM memory is exhausted. Configuring too much swap space results in storage devices being allocated but idle and is a poor use of resources. Too much swap space can also hide memory leaks. The maximum size for a swap partition and other additional information can be found in the **mkswap(8)** manual page.

The following table provides the recommended size of a swap partition depending on the amount of RAM in your system. If you let the installation program partition your system automatically, the swap partition size will be established using these guidelines. Automatic partitioning setup assumes that the maximum size of the swap partition is limited to 10% of the total size of the hard drive, and the installer cannot create swap partitions more than 128GB in size. If you want to set the swap partition size to more than 10% of the system's storage space, or more than 128GB, you must edit the partitioning layout manually.

Table 13.3. Recommended System Swap Space

Amount of RAM in the system	Recommended swap space	Recommended swap space if allowing for hibernation
less than 2 GB	2 times the amount of RAM	3 times the amount of RAM
2 GB - 8 GB	Equal to the amount of RAM	2 times the amount of RAM
8 GB - 64 GB	4GB to 0.5 times the amount of RAM	1.5 times the amount of RAM
more than 64 GB	workload dependent (at least 4GB)	hibernation not recommended

At the border between each range listed above (for example, a system with 2 GB, 8 GB, or 64 GB of system RAM), discretion can be exercised with regard to chosen swap space. If your system resources allow for it, increasing the swap space can lead to better performance.

Distributing swap space over multiple storage devices – particularly on systems with fast drives, controllers and interfaces – also improves swap space performance.



WARNING

The **PackageKit** update software downloads updated packages to `/var/cache/yum/` by default. If you create a separate partition or volume for `/var`, ensure that it is at least 3GB in size to accommodate downloaded package updates.

13.16. STORAGE DEVICES

You can install Red Hat Enterprise Linux on a large variety of storage devices. You can see basic, locally accessible, storage devices in the **Installation Destination** page, as described in [Section 13.15, “Installation Destination”](#). To add a specialized storage device, click the **Add a disk** button in the **Specialized & Network Disks** section of the screen.

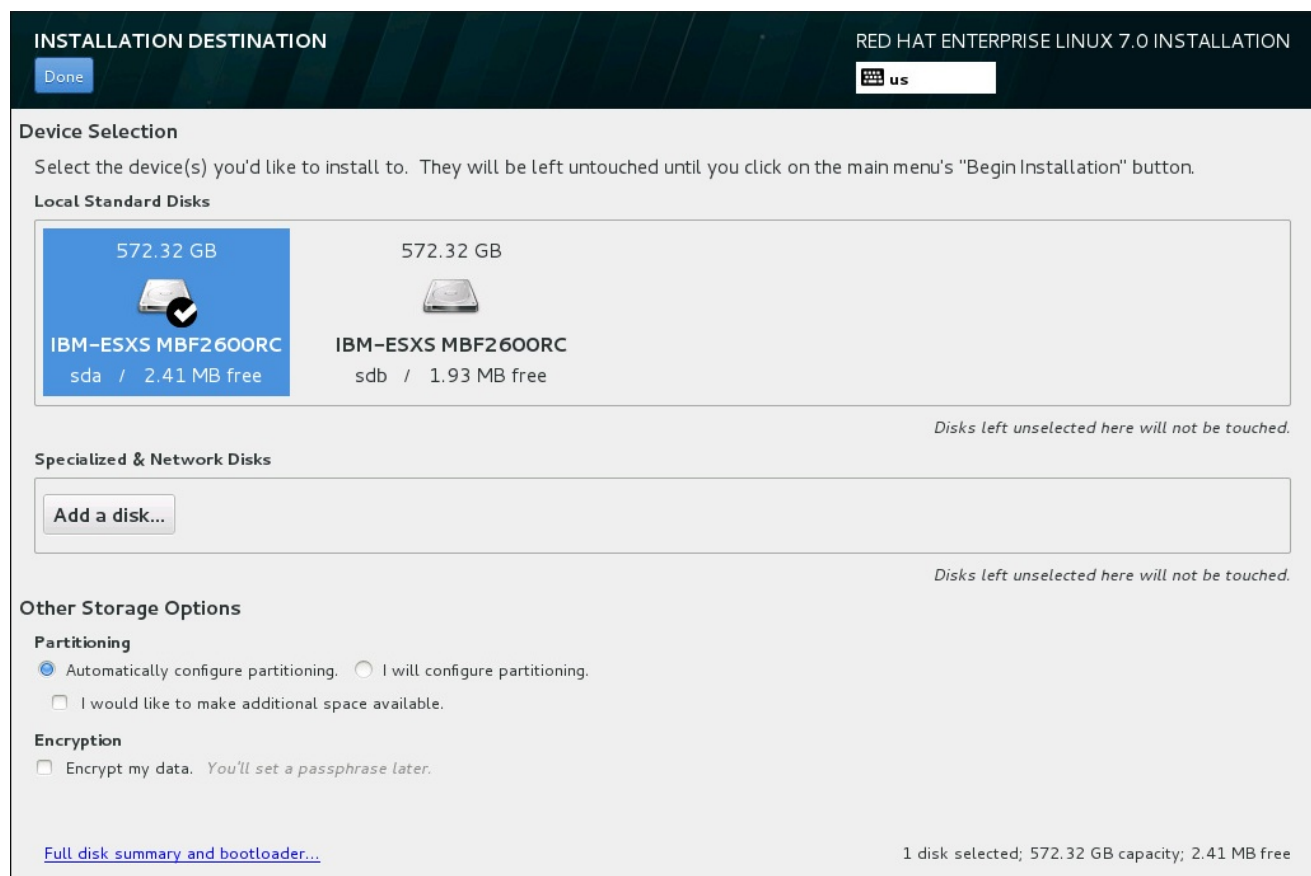


Figure 13.28. Storage Space Overview

13.16.1. The Storage Devices Selection Screen

The storage device selection screen displays all storage devices to which the **Anaconda** installation program has access.

The devices are grouped under the following tabs:

Multipath Devices

Storage devices accessible through more than one path, such as through multiple SCSI controllers or Fiber Channel ports on the same system.

The installation program only detects multipath storage devices with serial numbers that are 16 or 32 characters long.

Other SAN Devices

Devices available on a Storage Area Network (SAN).

Firmware RAID

Storage devices attached to a firmware RAID controller.

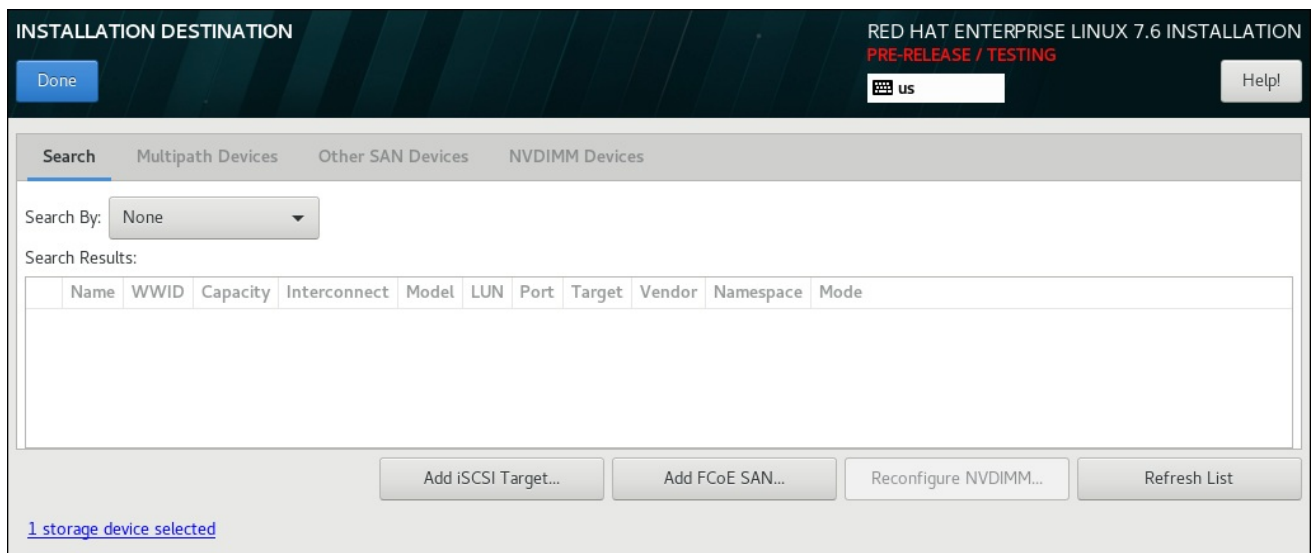


Figure 13.29. Tabbed Overview of Specialized Storage Devices

A set of buttons is available in the bottom right corner of the screen. Use these buttons to add additional storage devices. The available buttons are:

- **Add iSCSI Target** - use to attach iSCSI devices; continue with [Section 13.16.1.1.1, “Configure iSCSI Parameters”](#)
- **Add FCoE SAN** - use to configure a Fibre Channel Over Internet storage device; continue with [Section 13.16.1.1.2, “Configure FCoE Parameters”](#)

The overview page also contains the **Search** tab that allows you to filter storage devices either by their *World Wide Identifier* (WWID) or by the port, target, or *logical unit number* (LUN) at which they are accessed.

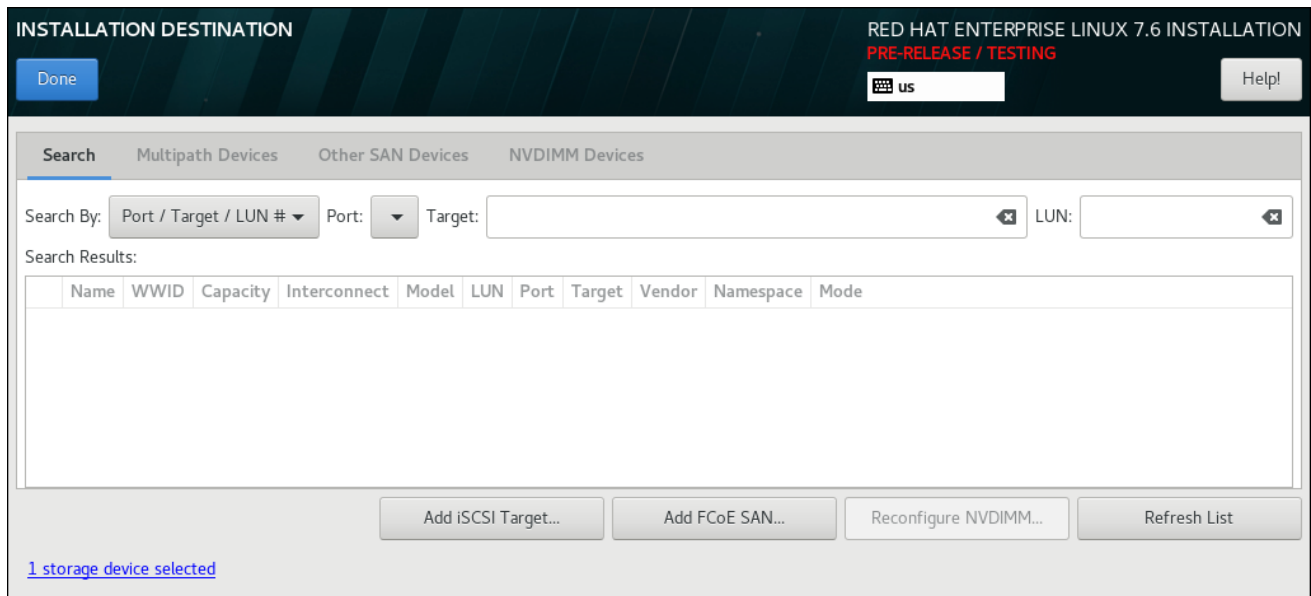


Figure 13.30. The Storage Devices Search Tab

The Search tab contains the **Search By** drop-down menu to select searching by port, target, LUN, or WWID. Searching by WWID or LUN requires additional values in the corresponding input text fields. Click the **Find** button to start the search.

Each device is presented on a separate row, with a check box to its left. Click the check box to make the device available during the installation process. Later in the installation process, you can choose to install Red Hat Enterprise Linux onto any of the devices selected here, and can choose to automatically mount any of the other devices selected here as part of the installed system.

Note that the devices that you select here are not automatically erased by the installation process. Selecting a device on this screen does not, in itself, place data stored on the device at risk. Also note that any devices that you do not select here to form part of the installed system can be added to the system after installation by modifying the **/etc/fstab** file.



IMPORTANT

Any storage devices that you do not select on this screen are hidden from **Anaconda** entirely. To *chain load* the Red Hat Enterprise Linux boot loader from a different boot loader, select all the devices presented in this screen.

When you have selected the storage devices to make available during installation, click **Done** to return to the Installation Destination screen.

13.16.1.1. Advanced Storage Options

To use an advanced storage device, you can configure an *iSCSI* (SCSI over TCP/IP) target or *FCoE* (Fibre Channel over Ethernet) *SAN* (Storage Area Network) by clicking the appropriate button in the lower right corner of the Installation Destination screen. See [Appendix B, *iSCSI Disks*](#) for an introduction to iSCSI.

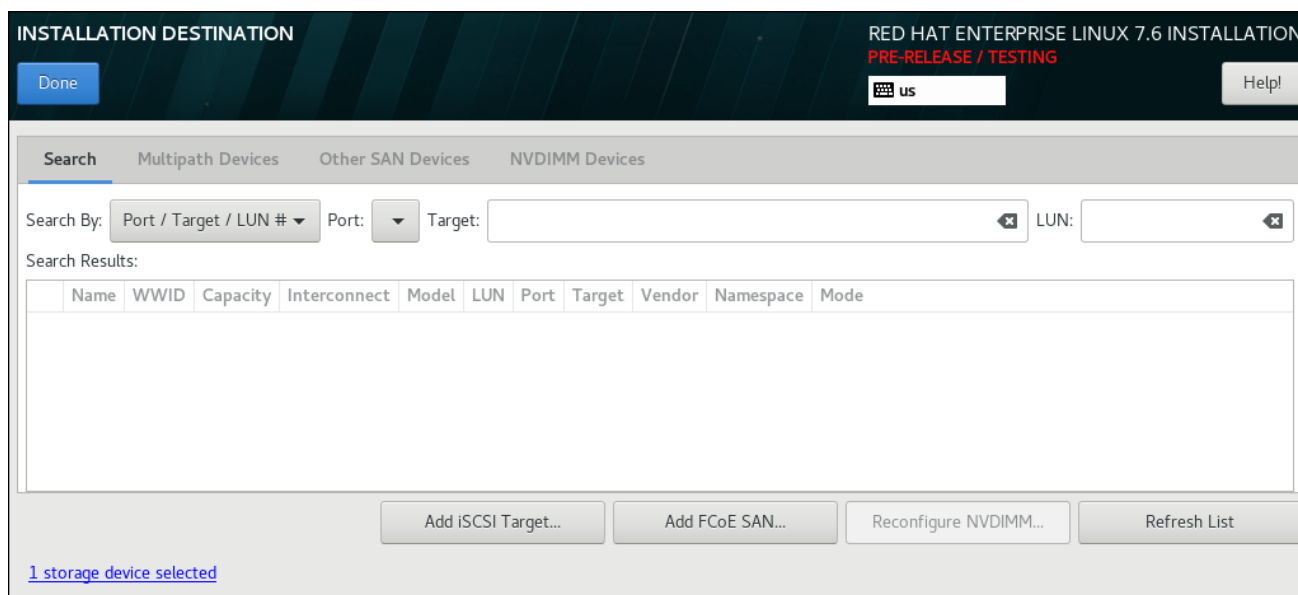


Figure 13.31. Advanced Storage Options

13.16.1.1.1. Configure iSCSI Parameters

When you click the **Add iSCSI target...** button, the **Add iSCSI Storage Target** dialog appears.

ADD iSCSI STORAGE TARGET

To use iSCSI disks, you must provide the address of your iSCSI target and the iSCSI initiator name you've configured for your host.

Target IP Address:

iSCSI Initiator Name:

 Example: iqn.2012-09.com.example:diskarrays-sn-a8675309

Discovery Authentication Type: CHAP pair and a reverse pair

CHAP Username:

CHAP Password:

Reverse CHAP Username:

Reverse CHAP Password:

☐ Bind targets to network interfaces

Figure 13.32. The iSCSI Discovery Details Dialog

To use iSCSI storage devices for the installation, **Anaconda** must be able to *discover* them as iSCSI targets and be able to create an iSCSI *session* to access them. Each of these steps might require a user name and password for *CHAP* (Challenge Handshake Authentication Protocol) authentication. Additionally, you can configure an iSCSI target to authenticate the iSCSI initiator on the system to which the target is attached (*reverse CHAP*), both for discovery and for the session. Used together, CHAP and reverse CHAP are called *mutual CHAP* or *two-way CHAP*. Mutual CHAP provides the greatest level of security for iSCSI connections, particularly if the user name and password are different for CHAP authentication and reverse CHAP authentication.



NOTE

Repeat the iSCSI discovery and iSCSI login steps as many times as necessary to add all required iSCSI storage. However, you cannot change the name of the iSCSI initiator after you attempt discovery for the first time. To change the iSCSI initiator name, you must restart the installation.

Procedure 13.1. iSCSI Discovery and Starting an iSCSI Session

Use the **Add iSCSI Storage Target** dialog to provide **Anaconda** with the information necessary to discover the iSCSI target.

1. Enter the IP address of the iSCSI target in the **Target IP Address** field.
2. Provide a name in the **iSCSI Initiator Name** field for the iSCSI initiator in *iSCSI qualified name* (IQN) format. A valid IQN entry contains:
 - the string **iqn.** (note the period)
 - a date code that specifies the year and month in which your organization's Internet domain or subdomain name was registered, represented as four digits for the year, a dash, and two digits for the month, followed by a period. For example, represent September 2010 as **2010-09**.
 - your organization's Internet domain or subdomain name, presented in reverse order with the top-level domain first. For example, represent the subdomain **storage.example.com** as **com.example.storage**
 - a colon followed by a string that uniquely identifies this particular iSCSI initiator within your domain or subdomain. For example, **:diskarrays-sn-a8675309**

A complete IQN can therefore look as follows: **iqn.2010-09.storage.example.com:diskarrays-sn-a8675309**. **Anaconda** prepopulates the **iSCSI Initiator Name** field with a name in this format to help you with the structure.

For more information on IQNs, see 3.2.6. *iSCSI Names* in *RFC 3720 - Internet Small Computer Systems Interface (iSCSI)* available from <http://tools.ietf.org/html/rfc3720#section-3.2.6> and 1. *iSCSI Names and Addresses* in *RFC 3721 - Internet Small Computer Systems Interface (iSCSI) Naming and Discovery* available from <http://tools.ietf.org/html/rfc3721#section-1>.

3. Use the **Discovery Authentication Type** drop-down menu to specify the type of authentication to use for iSCSI discovery. The following options are available:
 - no credentials
 - CHAP pair
 - CHAP pair and a reverse pair
4.
 - If you selected **CHAP pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** fields.
 - If you selected **CHAP pair and a reverse pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** field and the user name and password for the iSCSI initiator in the **Reverse CHAP Username** and **Reverse CHAP Password** fields.
5. Optionally check the box labeled **Bind targets to network interfaces**.
6. Click the **Start Discovery** button. **Anaconda** attempts to discover an iSCSI target based on the information that you provided. If discovery succeeds, the dialog displays a list of all iSCSI nodes discovered on the target.
7. Each node is presented with a check box beside it. Click the check boxes to select the nodes to use for installation.

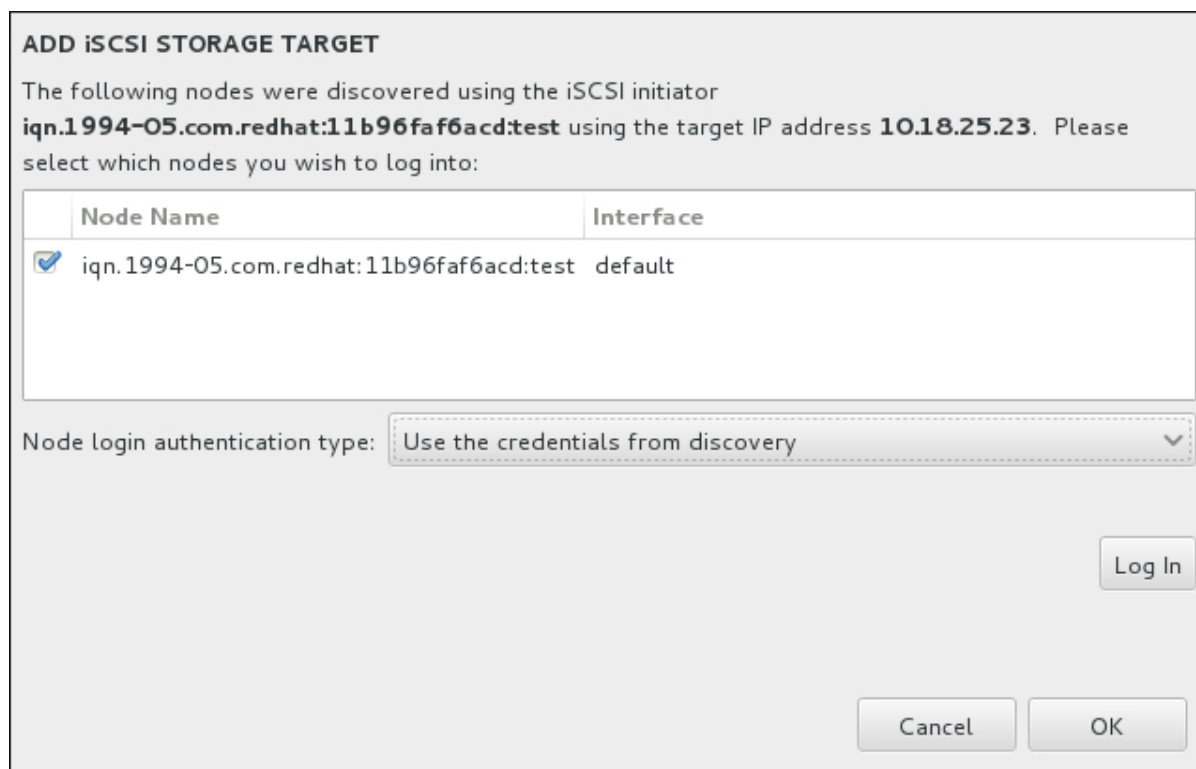


Figure 13.33. The Dialog of Discovered iSCSI Nodes

8. The **Node login authentication type** menu provides the same options as the **Discovery Authentication Type** menu described in step 3. However, if you needed credentials for discovery authentication, it is typical to use the same credentials to log into a discovered node. To do that, use the additional **Use the credentials from discovery** option from the menu. When the proper credentials have been provided, the **Log In** button becomes available.
9. Click **Log In** to initiate an iSCSI session.

13.16.1.1.2. Configure FCoE Parameters

When you click the **Add FCoE SAN...** button, a dialog appears for you to configure network interfaces for discovering FCoE storage devices.

First, select a network interface that is connected to a FCoE switch in the **NIC** drop-down menu and click the **Add FCoE disk(s)** button to scan the network for SAN devices.

Please select the network interface which is connected to your FCoE switch.

NIC: eth0 - 52:54:00:D9:65:B4

☐ Use DCB

☒ Use auto vlan

Add FCoE Disk(s)

Cancel

Figure 13.34. Configure FCoE Parameters

There are check boxes with additional options to consider:

Use DCB

Data Center Bridging (DCB) is a set of enhancements to the Ethernet protocols designed to increase the efficiency of Ethernet connections in storage networks and clusters. Enable or disable the installation program's awareness of DCB with the check box in this dialog. This option should only be enabled for network interfaces that require a host-based DCBX client. Configurations on interfaces that implement a hardware DCBX client should leave this check box empty.

Use auto vlan

Auto VLAN indicates whether VLAN discovery should be performed. If this box is checked, then the FIP (FCoE Initiation Protocol) VLAN discovery protocol will run on the Ethernet interface once the link configuration has been validated. If they are not already configured, network interfaces for any discovered FCoE VLANs will be automatically created and FCoE instances will be created on the VLAN interfaces. This option is enabled by default.

Discovered FCoE devices will be displayed under the **Other SAN Devices** tab in the Installation Destination screen.

13.17. KDUMP

Use this screen to select whether or not to use **Kdump** on this system. **Kdump** is a kernel crash dumping mechanism which, in the event of a system crash, captures information that can be invaluable in determining the cause of the crash.

Note that if you enable **Kdump**, you must reserve a certain amount of system memory for it. As a result, less memory is available for your processes.

IBM Power System LPARs support firmware-assisted dump (**fadump**), an alternate dump capture mechanism to **Kdump**. With **fadump**, dump capture takes place from a fully reset system that is loaded with a fresh copy of the kernel. In particular, PCI and I/O devices are reinitialized and are in a clean, consistent state making it a reliable alternative to **Kdump**. Note that although **fadump** is an alternative to **Kdump**, **fadump** requires **Kdump** to be enabled. You can enable **fadump** on this screen.

If you do not want to use **Kdump** on this system, uncheck **Enable kdump**. Otherwise, set the amount of

memory to reserve for **Kdump**. You can let the installer reserve a reasonable amount automatically, or you can set any amount manually. When you are satisfied with the settings, click **Done** to save the configuration and return to the previous screen.

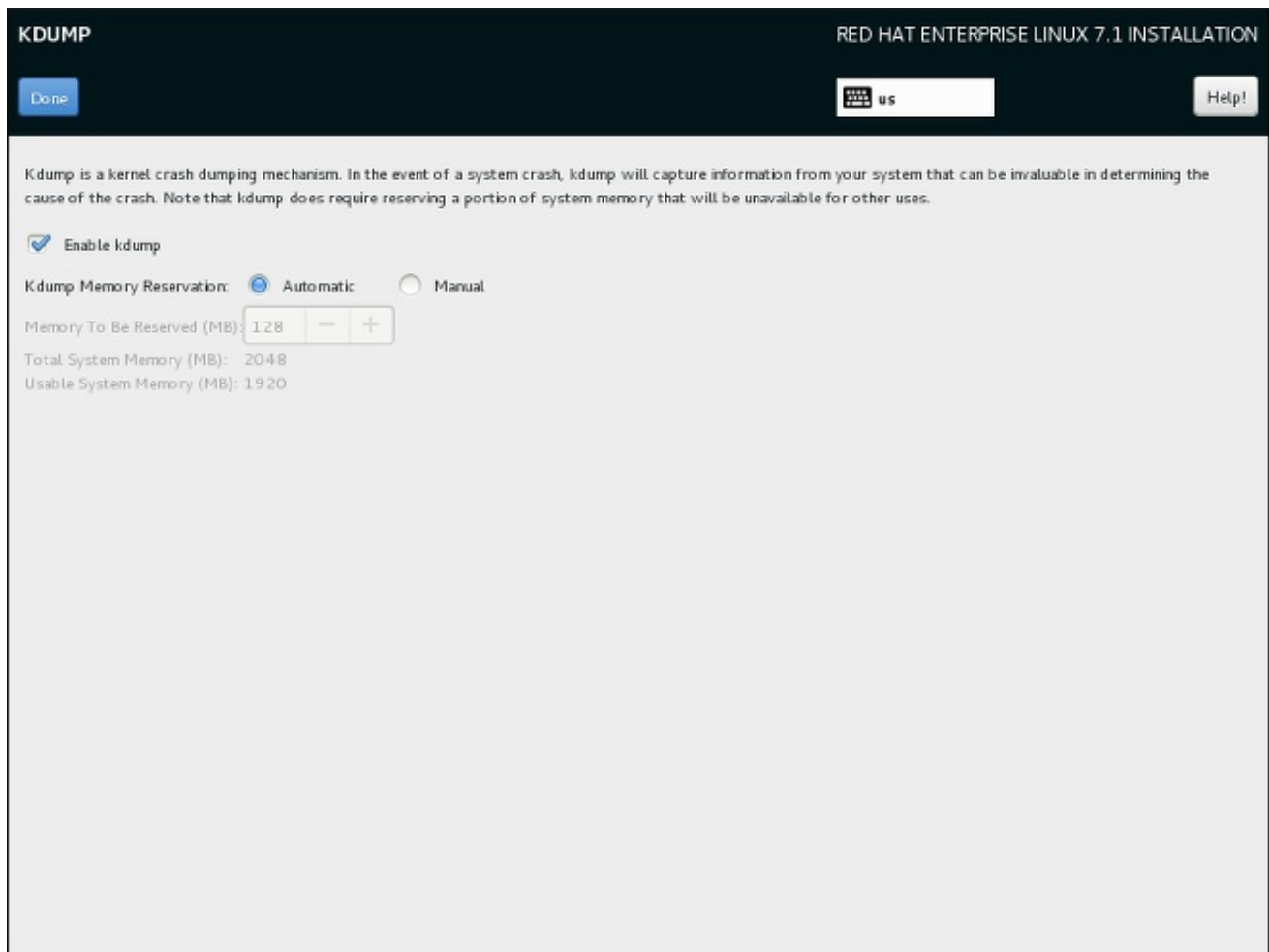


Figure 13.35. Kdump Enablement and Configuration

13.18. BEGIN INSTALLATION

When all required sections of the **Installation Summary** screen have been completed, the admonition at the bottom of the menu screen disappears and the **Begin Installation** button becomes available.

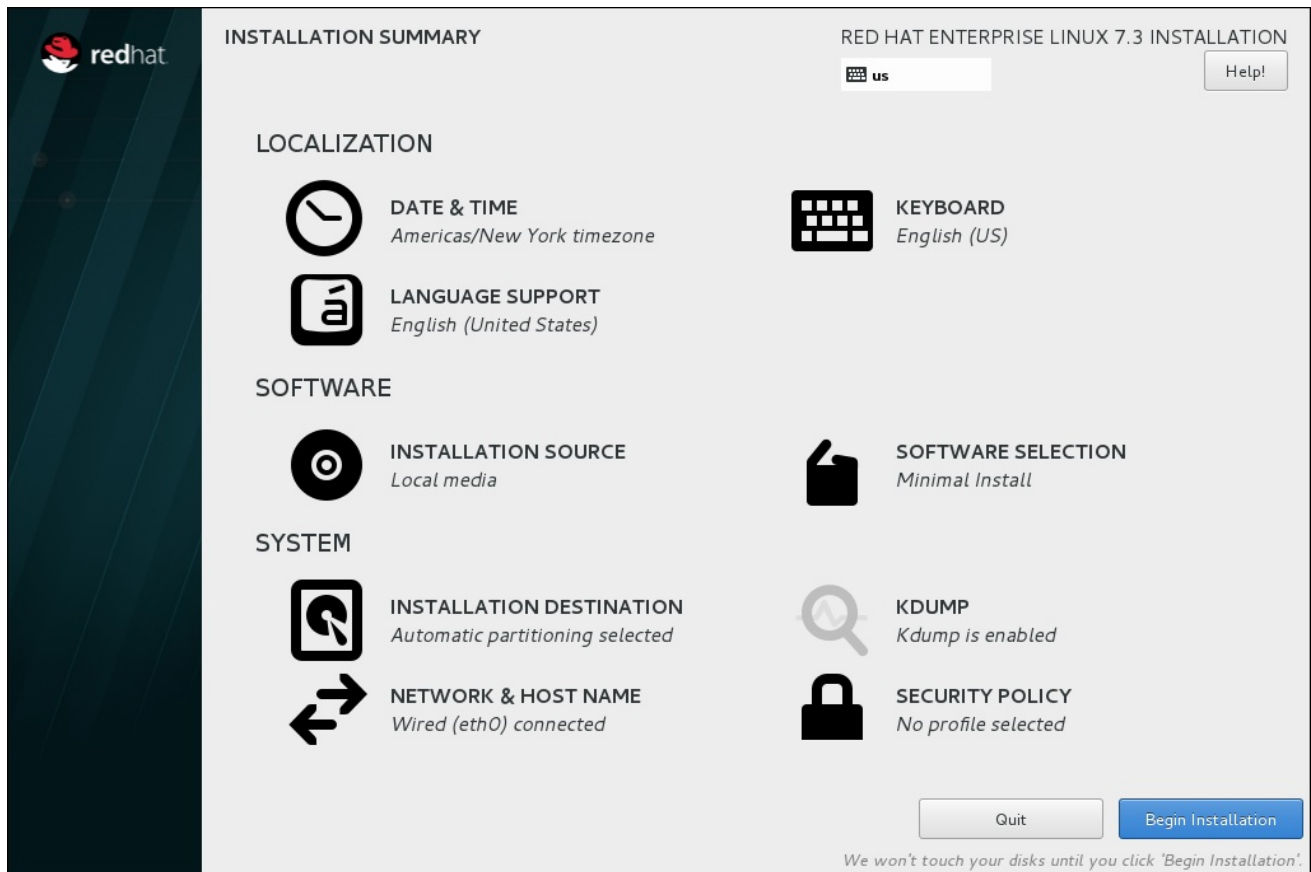


Figure 13.36. Ready to Install

**WARNING**

Up to this point in the installation process, no lasting changes have been made on your computer. When you click **Begin Installation**, the installation program will allocate space on your hard drive and start to transfer Red Hat Enterprise Linux into this space. Depending on the partitioning option that you chose, this process might include erasing data that already exists on your computer.

To revise any of the choices that you made up to this point, return to the relevant section of the **Installation Summary** screen. To cancel installation completely, click **Quit** or switch off your computer. To switch off most computers at this stage, press the power button and hold it down for a few seconds.

If you have finished customizing your installation and are certain that you want to proceed, click **Begin Installation**.

After you click **Begin Installation**, allow the installation process to complete. If the process is interrupted, for example, by you switching off or resetting the computer, or by a power outage, you will probably not be able to use your computer until you restart and complete the Red Hat Enterprise Linux installation process, or install a different operating system.

13.19. THE CONFIGURATION MENU AND PROGRESS SCREEN

Once you click **Begin Installation** at the **Installation Summary** screen, the progress screen appears. Red Hat Enterprise Linux reports the installation progress on the screen as it writes the selected packages to your system.

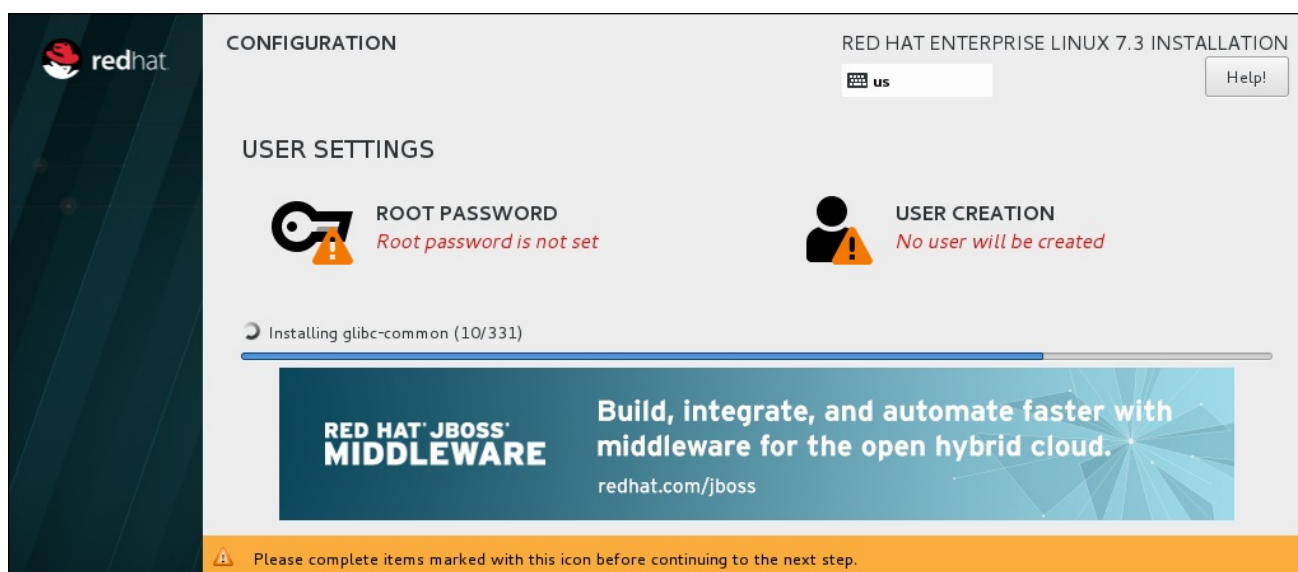


Figure 13.37. Installing Packages

For your reference, a complete log of your installation can be found in the `/var/log/anaconda/anaconda.packaging.log` file, once you reboot your system.

If you chose to encrypt one or more partitions during partitioning setup, a dialog window with a progress bar will be displayed during the early stage of the installation process. This window informs that the installer is attempting to gather enough entropy (random data) to ensure that the encryption is secure. This window will disappear after 256 bits of entropy are gathered, or after 10 minutes. You can speed up the gathering process by moving your mouse or randomly typing on the keyboard. After the window disappears, the installation process will continue.

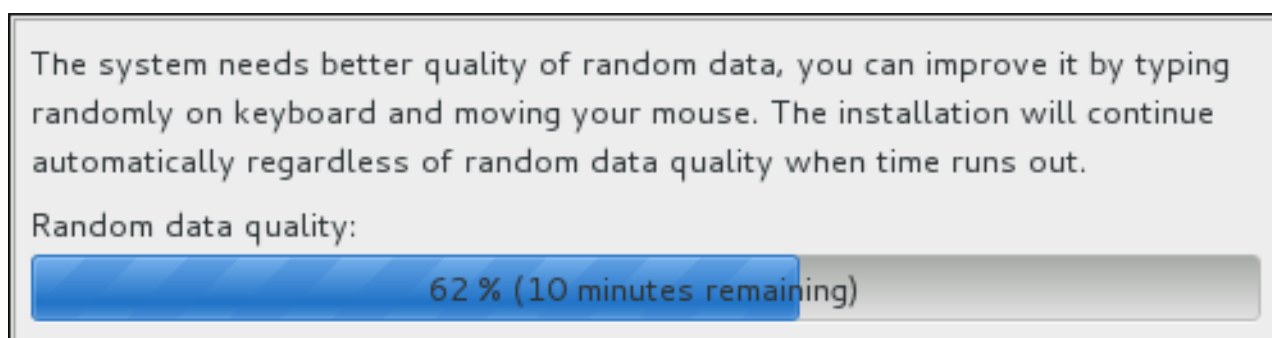


Figure 13.38. Gathering Entropy for Encryption

While the packages are being installed, more configuration is required. Above the installation progress bar are the **Root Password** and **User Creation** menu items.

The **Root Password** screen is used to configure the system's **root** account. This account can be used to perform critical system management and administration tasks. The same tasks can also be performed with a user account with the **wheel** group membership; if such an user account is created during installation, setting up a **root** password is not mandatory.

Creating a user account is optional and can be done after installation, but it is recommended to do it on this screen. A user account is used for normal work and to access the system. Best practice suggests that you always access the system through a user account, not the root account.

It is possible to disable access to the **Root Password** or **Create User** screens. To do so, use a Kickstart file which includes the **rootpw --lock** or **user --lock** commands. See [Section 27.3.1, “Kickstart Commands and Options”](#) for more information these commands.

13.19.1. Set the Root Password

Setting up a root account and password is an important step during your installation. The root account (also known as the superuser) is used to install packages, upgrade RPM packages, and perform most system maintenance. The root account gives you complete control over your system. For this reason, the root account is best used *only* to perform system maintenance or administration. See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about becoming root.

Figure 13.39. Root Password Screen



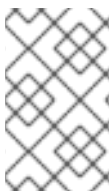
NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

Click the **Root Password** menu item and enter your new password into the **Root Password** field. Red Hat Enterprise Linux displays the characters as asterisks for security. Type the same password into the **Confirm** field to ensure it is set correctly. After you set the root password, click **Done** to return to the User Settings screen.

The following are the requirements and recommendations for creating a strong root password:

- *must* be at least eight characters long
- may contain numbers, letters (upper and lower case) and symbols
- is case-sensitive and should contain a mix of cases
- something you can remember but that is not easily guessed
- should not be a word, abbreviation, or number associated with you, your organization, or found in a dictionary (including foreign languages)
- should not be written down; if you must write it down keep it secure



NOTE

To change your root password after you have completed the installation, run the **passwd** command as **root**. If you forget the root password, see [Section 32.1.3, “Resetting the Root Password”](#) for instructions on how to use the rescue mode to set a new one.

13.19.2. Create a User Account

To create a regular (non-root) user account during the installation, click **User Settings** on the progress screen. The **Create User** screen appears, allowing you to set up the regular user account and configure its parameters. Though recommended to do during installation, this step is optional and can be performed after the installation is complete.



NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

To leave the user creation screen after you have entered it, without creating a user, leave all the fields empty and click **Done**.

Figure 13.40. User Account Configuration Screen

Enter the full name and the user name in their respective fields. Note that the system user name must be shorter than 32 characters and cannot contain spaces. It is highly recommended to set up a password for the new account.

When setting up a strong password even for a non-root user, follow the guidelines described in [Section 13.19.1, “Set the Root Password”](#).

Click the **Advanced** button to open a new dialog with additional settings.

ADVANCED USER CONFIGURATION

Home Directory

☒ Create a home directory for this user.

Home directory:

User and Group IDs

☐ Specify a user ID manually:
☐ Specify a group ID manually:

Group Membership

Add user to the following groups:

Tip: You may input a comma-separated list of group names and group IDs here.
Groups that do not already exist will be created; specify their GID in parentheses.

Figure 13.41. Advanced User Account Configuration

By default, each user gets a home directory corresponding to their user name. In most scenarios, there is no need to change this setting.

You can also manually define a system identification number for the new user and their default group by selecting the check boxes. The range for regular user IDs starts at the number **1000**. At the bottom of the dialog, you can enter the comma-separated list of additional groups, to which the new user shall belong. The new groups will be created in the system. To customize group IDs, specify the numbers in parenthesis.



NOTE

Consider setting IDs of regular users and their default groups at range starting at **5000** instead of **1000**. That is because the range reserved for system users and groups, **0-999**, might increase in the future and thus overlap with IDs of regular users.

For creating users with custom IDs using kickstart, see [user \(optional\)](#).

For changing the minimum UID and GID limits after the installation, which ensures that your chosen UID and GID ranges are applied automatically on user creation, see [the Users and Groups chapter of the System Administrator's Guide](#).

Once you have customized the user account, click **Save Changes** to return to the **User Settings** screen.

13.20. INSTALLATION COMPLETE

Congratulations! Your Red Hat Enterprise Linux installation is now complete!

Click the **Reboot** button to reboot your system and begin using Red Hat Enterprise Linux. Remember to remove any installation media if it is not ejected automatically upon reboot.

After your computer's normal power-up sequence has completed, Red Hat Enterprise Linux loads and starts. By default, the start process is hidden behind a graphical screen that displays a progress bar. Eventually, a GUI login screen (or if the X Window System is not installed, a **login:** prompt) appears.

If your system was installed with the X Window System during this installation process, the first time you start your Red Hat Enterprise Linux system, applications to set up your system are launched. These applications guide you through initial configuration of Red Hat Enterprise Linux and allow you to set your system time and date, register your machine with Red Hat Network, and more.

See [Chapter 30, *Initial Setup*](#) for information about the configuration process.

CHAPTER 14. TROUBLESHOOTING INSTALLATION ON IBM POWER SYSTEMS

This chapter discusses some common installation problems and their solutions.

For debugging purposes, **Anaconda** logs installation actions into files in the **/tmp** directory. These files are listed in the following table.

Table 14.1. Log Files Generated During the Installation

Log file	Contents
/tmp/anaconda.log	general Anaconda messages
/tmp/program.log	all external programs run during the installation
/tmp/storage.log	extensive storage module information
/tmp/packaging.log	yum and rpm package installation messages
/tmp/syslog	hardware-related system messages

If the installation fails, the messages from these files are consolidated into **/tmp/anaconda-tb-identifier**, where *identifier* is a random string.

After successful installation, by default, these files will be copied to the installed system under the directory **/var/log/anaconda/**. However, if installation is unsuccessful, or if the **inst.nosave=all** or **inst.nosave=logs** options are used when booting the installation system, these logs will only exist in the installation program's RAM disk. This means they are not saved permanently and will be lost once the system is powered down. To store them permanently, copy those files to another system on the network by using **scp** on the system running the installation program, or copy them to a mounted storage device (such as an USB flash drive). Details on how to transfer the log files over the network are below.



NOTE

The following procedure requires the installation system to be able to access the network and the target system to be able to receive files over the **ssh** protocol.

Procedure 14.1. Transferring Log Files Over the Network

1. On the system you are installing, press **Ctrl+Alt+F2** to access a shell prompt. You will be logged into a root account and you will have access to the installation program's temporary file system.
2. Switch to the **/tmp** directory where the log files are located:

```
# cd /tmp
```

3. Copy the log files onto another system on the network using the **scp** command:

```
# scp *log user@address:path
```

Replace *user* with a valid user name on the target system, *address* with the target system's address or host name, and *path* with the path to the directory you want to save the log files into. For example, if you want to log in as **john** to a system with an IP address of **192.168.0.122** and place the log files into the **/home/john/logs/** directory on that system, the command will have the following form:

```
# scp *log john@192.168.0.122:/home/john/logs/
```

When connecting to the target system for the first time, the SSH client asks you to confirm that the fingerprint of the remote system is correct and that you want to continue:

```
The authenticity of host '192.168.0.122 (192.168.0.122)' can't be established.  
ECDSA key fingerprint is a4:60:76:eb:b2:d0:aa:23:af:3d:59:5c:de:bb:c4:42.  
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press **Enter** to continue. Then, provide a valid password when prompted. The files will start transferring to the specified directory on the target system.

The log files from the installation are now permanently saved on the target system and available for review.

14.1. TROUBLE BEGINNING THE INSTALLATION

14.1.1. Problems with Booting into the Graphical Installation

Systems with some video cards have trouble booting into the graphical installation program. If the installation program does not run using its default settings, it attempts to run in a lower resolution mode. If that still fails, the installation program attempts to run in text mode.

There are several possible solutions to display issues, most of which involve specifying custom boot options. For more information, see [Section 23.1, “Configuring the Installation System at the Boot Menu”](#).

Use the basic graphics mode

You can attempt to perform the installation using the basic graphics driver. To do this, edit the installation program's options at the **boot:** prompt and append **inst.xdriver=vesa** at the end of the command line.

Specify the display resolution manually

If the installation program fails to detect your screen resolution, you can override the automatic detection and specify it manually. To do this, append the **inst.resolution=x** option at the boot menu, where *x* is your display's resolution (for example, **1024x768**).

14.1.2. Serial Console Not Detected

In some cases, attempting to install in text mode using a serial console will result in no output on the console. This happens on systems which have a graphics card, but no monitor connected. If **Anaconda** detects a graphics card, it will attempt to use it for a display, even if no display is connected.

If you want to perform a text-based installation on a serial console, use the **inst.text** and **console=** boot options. See [Chapter 23, Boot Options](#) for more details.

14.2. TROUBLE DURING THE INSTALLATION

14.2.1. No Disks Detected

In the **Installation Destination** screen, the following error message can appear at the bottom: **No disks detected. Please shut down the computer, connect at least one disk, and restart to complete installation.**

The message indicates that **Anaconda** did not find any writable storage devices to install to. In that case, first make sure that your system does have at least one storage device attached.

If your system uses a hardware RAID controller, verify that the controller is properly configured and working. See your controller's documentation for instructions.

If you are installing into one or more iSCSI devices and there is no local storage present on the system, make sure that all required LUNs (*Logical Unit Numbers*) are being presented to the appropriate HBA (*Host Bus Adapter*). For additional information about iSCSI, see [Appendix B, iSCSI Disks](#).

If you made sure you have a connected and properly configured storage device and the message still appears after you reboot the system and start the installation again, it means that the installation program failed to detect the storage. In most cases this message appears when you attempt to install on an SCSI device which has not been recognized by the installation program.

In that case, you will have to perform a driver update before starting the installation. Check your hardware vendor's website to determine if a driver update is available that fixes your problem. For more general information on driver updates, see [Chapter 11, Updating Drivers During Installation on IBM Power Systems](#).

You can also consult the *Red Hat Hardware Compatibility List*, available online at <https://hardware.redhat.com>.

14.2.2. Reporting Traceback Messages

If the graphical installation program encounters an error, it presents you with a crash reporting dialog box. You can then choose to send information about the problem you encountered to Red Hat. To send a crash report, you will need to enter your Customer Portal credentials. If you do not have a Customer Portal account, you can register at <https://www.redhat.com/wapps/ugc/register.html>. Automated crash reporting also requires a working network connection.

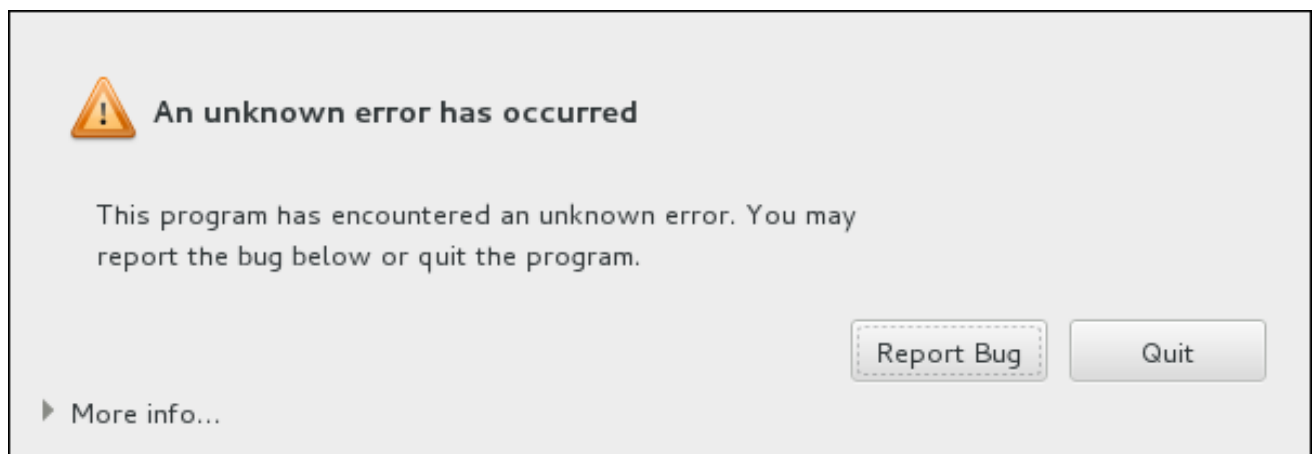


Figure 14.1. The Crash Reporting Dialog Box

When the dialog appears, select **Report Bug** to report the problem, or **Quit** to exit the installation.

Optionally, click **More Info** to display detailed output that can help determine the cause of the error. If you are familiar with debugging, click **Debug**. This will take you to virtual terminal **tty1**, where you can request more precise information that will enhance the bug report. To return to the graphical interface from **tty1**, use the **continue** command.

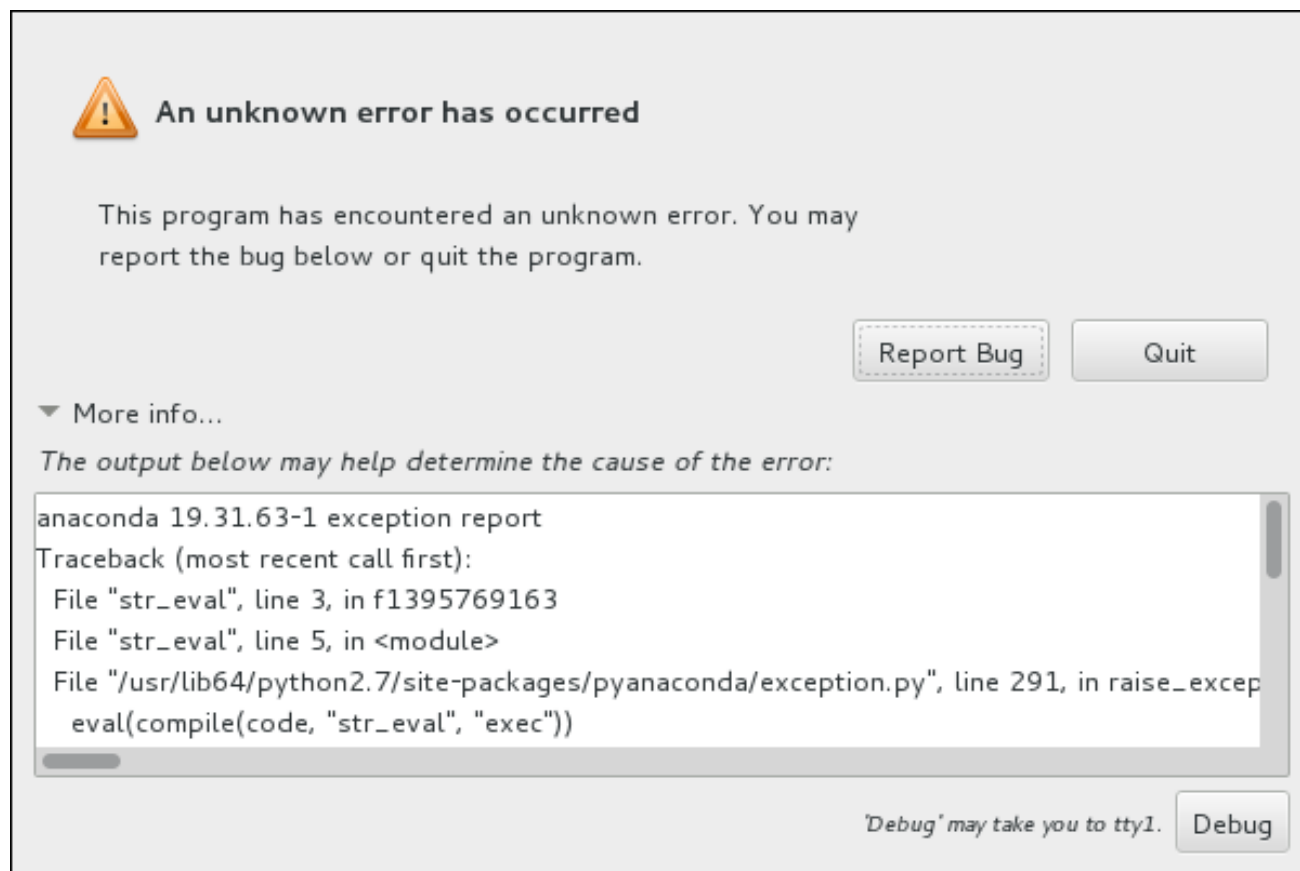


Figure 14.2. The Expanded Crash Reporting Dialog Box

If you want to report the bug to the customer portal, follow the procedure below.

Procedure 14.2. Reporting Errors to Red Hat Customer Support

1. In the menu that appears, select **Report a bug to Red Hat Customer Portal**.
2. To report the bug to Red Hat, you first need to provide your Customer Portal credentials. Click **Configure Red Hat Customer Support**.

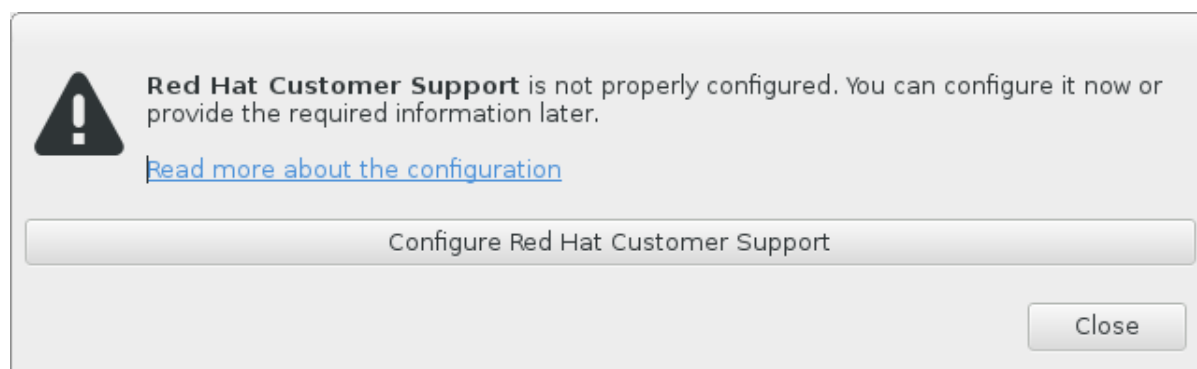
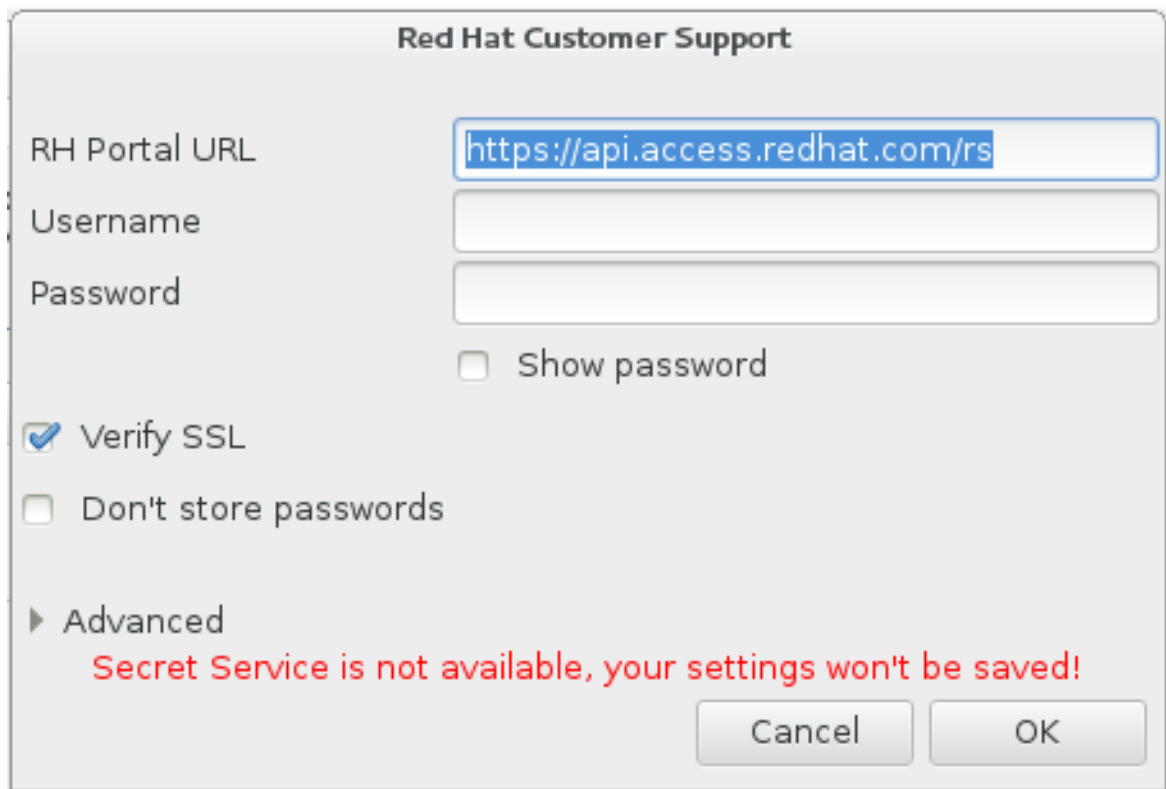


Figure 14.3. Customer Portal Credentials

3. A new window is now open, prompting you to enter your Customer Portal user name and password. Enter your Red Hat Customer Portal credentials.

A screenshot of a 'Red Hat Customer Support' configuration dialog box. The title bar reads 'Red Hat Customer Support'. Inside, there are three text input fields: 'RH Portal URL' (containing 'https://api.access.redhat.com/rs'), 'Username', and 'Password'. Below the 'Password' field is a checkbox labeled 'Show password'. Further down are two more checkboxes: 'Verify SSL' (checked) and 'Don't store passwords' (unchecked). A section titled 'Advanced' is expanded, showing a red error message: 'Secret Service is not available, your settings won't be saved!'. At the bottom right are 'Cancel' and 'OK' buttons.

Red Hat Customer Support

RH Portal URL

Username

Password

☐ Show password

☒ Verify SSL

☐ Don't store passwords

► Advanced

Secret Service is not available, your settings won't be saved!

Cancel OK

Figure 14.4. Configure Red Hat Customer Support

If your network settings require you to use a **HTTP** or **HTTPS** proxy, you can configure it by expanding the **Advanced** menu and entering the address of the proxy server.

When you put in all required credentials, click **OK** to proceed.

4. A new window appears, containing a text field. Write down any useful information and comments here. Describe how the error can be reproduced by explaining each step you took before the crash reporting dialog appeared. Provide as much relevant detail as possible, including any information you acquired when debugging. Be aware that the information you provide here can become publicly visible on the Customer Portal.

If you do not know what caused the error, check the box labeled **I don't know what caused this problem** at the bottom of the dialog.

Then, click **Forward**.

How did this problem happen (step-by-step)? How can it be reproduced? Any additional comments useful for diagnosing the problem? Please use English if possible.

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (stage1 on first disk). First disk is not used in partitioning section.

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the Kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to the second volume.

Your comments are not private. They may be included into publicly visible problem reports.

If you don't know how to describe it, you can [add a screencast](#)

☐ I don't know what caused this problem

Close Forward

Figure 14.5. Describe the Problem

- Next, review the information that will be sent to the Customer Portal. The explanation you provided is in the **comment** tab. Other tabs include such information as your system's host name and other details about the installation environment. You can remove any items you do not want sent to Red Hat, but be aware that providing less detail can affect the investigation of the issue.

Click **Forward** when you finish reviewing the information to be sent.

Please review the data before it gets reported. Depending on reporter chosen, it may end up publicly visible.

environ cmdline backtrace hostname **comment** reason

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the Kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to t

Close Forward

Figure 14.6. Review the Data to Be Sent

- Review the list of files that will be sent and included in the bug report as individual attachments.

These files provide system information that will assist the investigation. If you do not want to send certain files, uncheck the box next to each one. To provide additional files that can help find the problem, click **Attach a file**.

Once you have reviewed the files to be sent, check the box labeled **I have reviewed the data and agree with submitting it**. Then, click **Forward** to send the report and attachments to the Customer Portal.

Include	Name	Value
☒	anaconda.log	(click here to view/edit)
☒	packaging.log	(click here to view/edit)
☒	os_info	(click here to view/edit)
☒	environ	(click here to view/edit)
☒	last_occurrence	1395769224
☒	anaconda-tb	(click here to view/edit)
☒	version	7.0
☒	cmdline	/usr/bin/python /sbin/anaconda
☒	package	anaconda-19.31.63-1
☒	duphash	aeb4ed3992aab62d1cd03d3b1e0a89c79753f34da115a739be021d2873d0a131
☒	lsblk_output	(click here to view/edit)
☒	pkg_version	19.31.63
☒	product	Red Hat Enterprise Linux
☒	release	Red Hat Enterprise Linux Workstation release 7.0 Beta (Maipo)
☒	pkg_arch	x86_64

Attach a file

☐ I reviewed the data and agree with submitting it

Close Forward

Figure 14.7. Review the Files to Be Sent

- When the dialog reports that processing has finished, you can click **Show log** to view details of the reporting process or **Close** to return to the initial crash reporting dialog box. There, click **Quit** to exit the installation.

14.2.3. Creating Pre-installation Log Files

To debug installation problems you can set the **inst.debug** option to create log files from the environment before the installation starts. These log files contain, for example, the current storage configuration.

To set the option in the Red Hat Enterprise Linux installation boot menu:

- Select the **Install Red Hat Enterprise Linux 7.3** entry.
- Press the **Tab** key to edit the boot options.
- Append **inst.debug** to the options. For example:

```
> vmlinuz ... inst.debug
```

For further details, see [Chapter 23, Boot Options](#).

- Press **Enter** to start the setup.

The system stores the pre-installation log files in the `/tmp/pre-anaconda-logs/` directory before Anaconda starts. To access the log files:

1. Switch to the console. See [Section 8.2.1, “Accessing Consoles”](#).
2. Change into the `/tmp/pre-anaconda-logs/` directory:

```
# cd /tmp/pre-anaconda-logs/
```

14.2.4. Other Partitioning Problems for IBM Power Systems Users

If you create partitions manually, but cannot move to the next screen, you probably have not created all the partitions necessary for installation to proceed.

You must have the following partitions as a bare minimum:

- A `/` (root) partition
- A **PPReP Boot** partition
- A `/boot` partition (only if the root partition is a LVM logical volume or Btrfs subvolume)

See [Section 13.15.4.4, “Recommended Partitioning Scheme”](#) for more information.

14.3. PROBLEMS AFTER INSTALLATION

14.3.1. Trouble With the Graphical Boot Sequence

After you finish the installation and reboot your system for the first time, it is possible that the system stops responding during the graphical boot sequence, requiring a reset. In this case, the boot loader is displayed successfully, but selecting any entry and attempting to boot the system results in a halt. This usually means a problem with the graphical boot sequence; to solve this issue, you must disable graphical boot. To do this, temporarily alter the setting at boot time before changing it permanently.

Procedure 14.3. Disabling Graphical Boot Temporarily

1. Start your computer and wait until the boot loader menu appears. If you set your boot loader timeout period to 0, hold down the **Esc** key to access it.
2. When the boot loader menu appears, use your cursor keys to highlight the entry you want to boot and press the **e** key to edit this entry's options.
3. In the list of options, find the kernel line – that is, the line beginning with the keyword **linux**. On this line, locate the **rhgb** option and delete it. The option might not be immediately visible; use the cursor keys to scroll up and down.
4. Press **F10** or **Ctrl+X** to boot your system with the edited options.

If the system started successfully, you can log in normally. Then you will need to disable the graphical boot permanently – otherwise you will have to perform the previous procedure every time the system boots. To permanently change boot options, do the following.

Procedure 14.4. Disabling Graphical Boot Permanently

1. Log in to the **root** account using the **su -** command:

```
$ su -
```

2. Use the **grubby** tool to find the default **GRUB2** kernel:

```
# grubby --default-kernel  
/boot/vmlinuz-3.10.0-229.4.2.el7.ppc64
```

3. Use the **grubby** tool to remove the **rhgb** boot option from the default kernel, identified in the last step, in your **GRUB2** configuration. For example:

```
# grubby --remove-args="rhgb" --update-kernel /boot/vmlinuz-3.10.0-229.4.2.el7.ppc64
```

After you finish this procedure, you can reboot your computer. Red Hat Enterprise Linux will not use the graphical boot sequence any more. If you want to enable graphical boot in the future, follow the same procedure, replacing the **--remove-args="rhgb"** parameter with the **--args="rhgb"** parameter. This will restore the **rhgb** boot option to the default kernel in your **GRUB2** configuration.

See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about working with the **GRUB2** boot loader.

14.3.2. Booting into a Graphical Environment

If you have installed the **X Window System** but are not seeing a graphical desktop environment once you log into your system, you can start it manually using the **startx** command. Note, however, that this is just a one-time fix and does not change the log in process for future log ins.

To set up your system so that you can log in at a graphical login screen, you must change the default **systemd** target to **graphical.target**. When you are finished, reboot the computer. You will be presented with a graphical login prompt after the system restarts.

Procedure 14.5. Setting Graphical Login as Default

1. Open a shell prompt. If you are in your user account, become root by typing the **su -** command.
2. Change the default target to **graphical.target**. To do this, execute the following command:

```
# systemctl set-default graphical.target
```

Graphical login is now enabled by default – you will be presented with a graphical login prompt after the next reboot. If you want to reverse this change and keep using the text-based login prompt, execute the following command as **root**:

```
# systemctl set-default multi-user.target
```

For more information about targets in **systemd**, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

14.3.3. No Graphical User Interface Present

If you are having trouble getting **X** (the **X Window System**) to start, it is possible that it has not been installed. Some of the preset base environments you can select during the installation, such as **Minimal install** or **Web Server**, do not include a graphical interface – it has to be installed manually.

If you want **X**, you can install the necessary packages afterwards. See the Knowledgebase article at <https://access.redhat.com/site/solutions/5238> for information on installing a graphical desktop environment.

14.3.4. X Server Crashing After User Logs In

If you are having trouble with the **X** server crashing when a user logs in, one or more of your file systems can be full or nearly full. To verify that this is the problem you are experiencing, execute the following command:

```
$ df -h
```

The output will help you diagnose which partition is full – in most cases, the problem will be on the **/home** partition. The following is a sample output of the **df** command:

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/vg_rhel-root	20G	6.0G	13G	32%	/
devtmpfs	1.8G	0	1.8G	0%	/dev
tmpfs	1.8G	2.7M	1.8G	1%	/dev/shm
tmpfs	1.8G	1012K	1.8G	1%	/run
tmpfs	1.8G	0	1.8G	0%	/sys/fs/cgroup
tmpfs	1.8G	2.6M	1.8G	1%	/tmp
/dev/sda1	976M	150M	760M	17%	/boot
/dev/dm-4	90G	90G	0	100%	/home

In the above example, you can see that the **/home** partition is full, which causes the crash. You can make some room on the partition by removing unneeded files. After you free up some disk space, start **X** using the **startx** command.

For additional information about **df** and an explanation of the options available (such as the **-h** option used in this example), see the **df(1)** man page.

14.3.5. Is Your System Displaying Signal 11 Errors?

A signal 11 error, commonly known as a *segmentation fault*, means that a program accessed a memory location that was not assigned to it. A signal 11 error can occur due to a bug in one of the software programs that is installed, or faulty hardware.

If you receive a fatal signal 11 error during the installation, first make sure you are using the most recent installation images, and let **Anaconda** verify them to make sure they are not corrupted. Bad installation media (such as an improperly burned or scratched optical disk) are a common cause of signal 11 errors. Verifying the integrity of the installation media is recommended before every installation.

For information about obtaining the most recent installation media, see [Chapter 2, Downloading Red Hat Enterprise Linux](#). To perform a media check before the installation starts, append the **rd.live.check** boot option at the boot menu. See [Section 23.2.2, “Verifying Boot Media”](#) for details.

Other possible causes are beyond this document's scope. Consult your hardware manufacturer's documentation for more information.

14.3.6. Unable to IPL from Network Storage Space (*NWSSTG)

If you are experiencing difficulties when trying to IPL from Network Storage Space (*NWSSTG), in most cases the reason is a missing **PReP** partition. In this case, you must reinstall the system and make sure to create this partition during the partitioning phase or in the Kickstart file.

14.3.7. The GRUB2 `next_entry` variable can behave unexpectedly in a virtualized environment

IBM Power System users booting their virtual environment with SLOF firmware must manually unset the **`next_entry`** grub environment variable after a system reboot. The SLOF firmware does not support block writes at boot time by design thus the bootloader is unable to clear this variable at boot time.

PART III. IBM Z ARCHITECTURE - INSTALLATION AND BOOTING

This part discusses booting, or *initial program load* (IPL), and installation of Red Hat Enterprise Linux on IBM Z.

CHAPTER 15. PLANNING FOR INSTALLATION ON IBM Z

15.1. PRE-INSTALLATION

Red Hat Enterprise Linux 7 runs on zEnterprise 196 or later IBM mainframe systems.

The installation process assumes that you are familiar with the IBM Z and can set up *logical partitions* (LPARs) and z/VM guest virtual machines. For additional information on IBM Z, see <http://www.ibm.com/systems/z>.

For installation of Red Hat Enterprise Linux on IBM Z, Red Hat supports DASD (Direct Access Storage Device) and FCP (Fiber Channel Protocol) storage devices.

Before you install Red Hat Enterprise Linux, you must decide on the following:

- Decide whether you want to run the operating system on an LPAR or as a z/VM guest operating system.
- Decide if you need swap space and if so, how much. Although it is possible (and recommended) to assign enough memory to a z/VM guest virtual machine and let z/VM do the necessary swapping, there are cases where the amount of required RAM is hard to predict. Such instances should be examined on a case-by-case basis. See [Section 18.15.3.4, “Recommended Partitioning Scheme”](#).
- Decide on a network configuration. Red Hat Enterprise Linux 7 for IBM Z supports the following network devices:
 - Real and virtual *Open Systems Adapter* (OSA)
 - Real and virtual HiperSockets
 - *LAN channel station* (LCS) for real OSA

You require the following hardware:

- Disk space. Calculate how much disk space you need and allocate sufficient disk space on DASDs^[2] or SCSI^[3] disks. You require at least 10 GB for a server installation, and 20 GB if you want to install all packages. You also require disk space for any application data. After the installation, you can add or delete more DASD or SCSI disk partitions.

The disk space used by the newly installed Red Hat Enterprise Linux system (the Linux instance) must be separate from the disk space used by other operating systems you have installed on your system.

For more information about disks and partition configuration, see [Section 18.15.3.4, “Recommended Partitioning Scheme”](#).

- RAM. Acquire 1 GB (recommended) for the Linux instance. With some tuning, an instance might run with as little as 512 MB RAM.



NOTE

When initializing swap space on an FBA (*Fixed Block Architecture*) DASD using the **SWAPGEN** utility, the **FBAPART** option must be used.

15.2. OVERVIEW OF THE IBM Z INSTALLATION PROCEDURE

You can install Red Hat Enterprise Linux on IBM Z interactively or in unattended mode. Installation on IBM Z differs from installation on other architectures in that it is typically performed over a network and not from a local media. The installation consists of three phases:

1. Booting the Installation

Connect with the mainframe, then perform an *initial program load* (IPL), or boot, from the medium containing the installation program. See [Chapter 16, Booting the Installation on IBM Z](#) for details.

2. Connecting to the installation system

From a local machine, connect to the remote IBM Z system to continue with the installation process. See [Chapter 17, Connecting to the installation system](#) for details.

3. Anaconda

Use the **Anaconda** installation program to configure network, specify language support, installation source, software packages to be installed, and to perform the rest of the installation. See [Chapter 18, Installing Using Anaconda](#) for more information.

15.2.1. Booting the Installation

After establishing a connection with the mainframe, you need to perform an initial program load (IPL), or boot, from the medium containing the installation program. This document describes the most common methods of installing Red Hat Enterprise Linux on IBM Z. In general, you can use any method to boot the Linux installation system, which consists of a kernel (**kernel.img**) and initial RAM disk (**initrd.img**) with at least the parameters in the **generic.prm** file. Additionally, a **generic.ins** file is loaded which determines file names and memory addresses for the initrd, kernel and generic.prm.

The Linux installation system is also called the *installation program* in this book.

The control point from where you can start the IPL process depends on the environment where your Linux is to run. If your Linux is to run as a z/VM guest operating system, the control point is the *control program* (CP) of the hosting z/VM. If your Linux is to run in LPAR mode, the control point is the mainframe's *Support Element* (SE) or an attached IBM Z *Hardware Management Console* (HMC).

You can use the following boot media only if Linux is to run as a guest operating system under z/VM:

- z/VM reader – see [Section 16.3.1, “Using the z/VM Reader”](#) for details.

You can use the following boot media only if Linux is to run in LPAR mode:

- SE or HMC through a remote FTP server – see [Section 16.4.1, “Using an FTP Server”](#) for details.
- SE or HMC DVD – see [Section 16.4.4, “Using an FCP-attached SCSI DVD Drive”](#) for details.

You can use the following boot media for both z/VM and LPAR:

- DASD – see [Section 16.3.2, “Using a Prepared DASD”](#) for z/VM or [Section 16.4.2, “Using a Prepared DASD”](#) for LPAR.
- SCSI device that is attached through an FCP channel – see [Section 16.3.3, “Using a Prepared FCP-attached SCSI Disk”](#) for z/VM or [Section 16.4.3, “Using a Prepared FCP-attached SCSI Disk”](#) for LPAR.

- FCP-attached SCSI DVD – see [Section 16.3.4, “Using an FCP-attached SCSI DVD Drive”](#) for z/VM or [Section 16.4.4, “Using an FCP-attached SCSI DVD Drive”](#) for LPAR

If you use DASD and FCP-attached SCSI devices (except SCSI DVDs) as boot media, you must have a configured **zipl** boot loader.

15.2.2. Connecting to the installation system

From a local machine, connect to the remote IBM Z system to continue with the installation process. See [Chapter 17, *Connecting to the installation system*](#) for details.

15.2.3. Installation using Anaconda

In the second installation phase, you will use the **Anaconda** installation program in graphical, text-based, or command-line mode:

Graphical Mode

Graphical installation is done through a VNC client. You can use your mouse and keyboard to navigate through the screens, click buttons, and type into text fields. For more information on performing a graphical installation using VNC, see [Chapter 25, *Using VNC*](#).

Text-based Mode

This interface does not offer all interface elements of the GUI and does not support all settings. Use this for interactive installations if you cannot use a VNC client. For more information about text-based installations, see [Section 18.4, “Installing in Text Mode”](#).

Command-line Mode

This is intended for automated and non-interactive installations on IBM Z. Note that if the installation program encounters an invalid or missing kickstart command, the system will reboot. For more information about automated installation, see [Chapter 27, *Kickstart Installations*](#).

In Red Hat Enterprise Linux 7 the text-based installation has been reduced to minimize user interaction. Features like installation on FCP-attached SCSI devices, customizing partition layout, or package add-on selection are only available with the graphical user interface installation. Use the graphical installation whenever possible. See [Chapter 18, *Installing Using Anaconda*](#) for more details.

[2] *Direct Access Storage Devices* (DASDs) are hard disks that allow a maximum of three partitions per device. For example, **dasda** can have partitions **dasda1**, **dasda2**, and **dasda3**.

[3] Using the SCSI-over-Fibre Channel device driver (the **zfcp** device driver) and a switch, SCSI LUNs can be presented to Linux on IBM Z as if they were locally attached SCSI drives.

CHAPTER 16. BOOTING THE INSTALLATION ON IBM Z

The steps to perform the initial program boot (IPL) of the **Anaconda** installation program depend on the environment (either z/VM or LPAR) in which Red Hat Enterprise Linux will run.

16.1. CUSTOMIZING BOOT PARAMETERS

Before you can begin the installation, you must configure some mandatory boot parameters. When installing through z/VM, these parameters must be configured before you boot in the **generic.prm** file. When installing on an LPAR, the **rd.cmdline** parameter is set to **ask** by default, meaning that you will be given a prompt on which you can enter these boot parameters. In both cases, the required parameters are the same.



NOTE

Unlike Red Hat Enterprise Linux 6, which featured an interactive utility to assist network configuration, all network configuration must now be specified by the use of the following parameters, either by using a parameter file, or at the prompt.

Installation source

An installation source must always be configured. Use the **inst.repo=** option to specify the package source for the installation. See [Specifying the Installation Source](#) for details and syntax.

Network devices

Network configuration must be provided if network access will be required during the installation. If you plan to perform an unattended (Kickstart-based) installation using only local media such as a hard drive, network configuration can be omitted.

Use the **ip=** option for basic network configuration, and other options listed in [Network Boot Options](#) as required.

Also use the **rd.znet=** kernel option, which takes a network protocol type, a comma delimited list of sub-channels, and, optionally, comma delimited **sysfs** parameter and value pairs. This parameter can be specified multiple times to activate multiple network devices.

For example:

```
rd.znet=qeth,0.0.0600,0.0.0601,0.0.0602,layer2=1,portname=foo
```

Storage devices

At least one storage device must always be configured.

The **rd.dasd=** option takes a Direct Access Storage Device (DASD) adapter device bus identifier. For multiple DASDs, specify the parameter multiple times, or use a comma separated list of bus IDs. To specify a range of DASDs, specify the first and the last bus ID. Example:

```
rd.dasd=0.0.0200 rd.dasd=0.0.0202(ro),0.0.0203(ro:failfast),0.0.0205-0.0.0207
```

The **rd.zfcp=** option takes a SCSI over FCP (zFCP) adapter device bus identifier, a world wide port name (WWPN), and a FCP LUN, then activates the device. This parameter can be specified multiple times to activate multiple zFCP devices. Example:

```
rd.zfcp=0.0.4000,0x5005076300C213e9,0x5022000000000000
```

Kickstart options

If you are using a Kickstart file to perform an automatic installation, you must always specify the location of the Kickstart file using the **inst.ks=** option. For an unattended, fully automatic Kickstart installation, the **inst.cmdline** option is also useful. See [Section 21.4, “Parameters for Kickstart Installations”](#) for additional information.

An example customized **generic.prm** file containing all mandatory parameters look similar to the following example:

Example 16.1. Customized generic.prm file

```
ro ramdisk_size=40000 cio_ignore=all,!condev
inst.repo=http://example.com/path/to/repository
rd.znet=qeth,0.0.0600,0.0.0601,0.0.0602,layer2=1,portno=0,portname=foo
ip=192.168.17.115::192.168.17.254:24:foobar.systemz.example.com:enccw0.0.0600:none
nameserver=192.168.17.1
rd.dasd=0.0.0200 rd.dasd=0.0.0202
rd.zfcp=0.0.4000,0x5005076300C213e9,0x5022000000000000
inst.ks=http://example.com/path/to/kickstart
```

Some installation methods also require a file with a mapping of the location of installation data in the file system of the DVD or FTP server and the memory locations where the data is to be copied. The file is typically named **generic.ins**, and contains file names for the initial RAM disk, kernel image, and parameter file (**generic.prm**) and a memory location for each file. An example **generic.ins** will look similar to the following example:

Example 16.2. Sample generic.ins file

```
images/kernel.img 0x00000000
images/initrd.img 0x02000000
images/genericdvd.prm 0x00010480
images/initrd.addrsize 0x00010408
```

A valid **generic.ins** file is provided by Red Hat along with all other files required to boot the installer. Modify this file only if you want to, for example, load a different kernel version than default.

16.2. CONSIDERATIONS FOR HARD DRIVE INSTALLATION ON IBM Z

If you want to boot the installation program from a hard drive, you can optionally install the **zipl** boot loader on the same (or a different) disk. Be aware that **zipl** only supports one boot record per disk. If you have multiple partitions on a disk, they all "share" the disk's single boot record.

To prepare a hard drive to boot the installation program, install the **zipl** boot loader on the hard drive by entering the following command:

```
# zipl -V -t /mnt/ -i /mnt/images/kernel.img -r /mnt/images/initrd.img -p /mnt/images/generic.prm
```

See [Section 16.1, “Customizing boot parameters”](#) for details on customizing boot parameters in the **generic.prm** configuration file.

16.3. INSTALLING UNDER Z/VM

When installing under z/VM, you can boot from:

- the z/VM virtual reader
- a DASD or an FCP-attached SCSI device prepared with the **zipl** boot loader
- an FCP-attached SCSI DVD drive

Log on to the z/VM guest virtual machine chosen for the Linux installation. You can use the **x3270** or **c3270** terminal emulator, available in the `x3270-text` package in Red Hat Enterprise Linux, to log in to z/VM from other Linux systems. Alternatively, use the IBM 3270 terminal emulator on the IBM Z Hardware Management Console (HMC). If you are working from a machine with a Microsoft Windows operating system, Jolly Giant (<http://www.jollygiant.com/>) offers an SSL-enabled 3270 emulator. A free native Windows port of **c3270** called **wc3270** also exists.



NOTE

If your 3270 connection is interrupted and you cannot log in again because the previous session is still active, you can replace the old session with a new one by entering the following command on the z/VM logon screen:

```
logon user here
```

Replace *user* with the name of the z/VM guest virtual machine. Depending on whether an external security manager, for example RACF, is used, the logon command might vary.

If you are not already running **CMS** (single-user operating system shipped with z/VM) in your guest, boot it now by entering the command:

```
cp ipl cms
```

Be sure not to use CMS disks such as your A disk (often device number 0191) as installation targets. To find out which disks are in use by CMS, use the following query:

```
query disk
```

You can use the following CP (z/VM Control Program, which is the z/VM hypervisor) query commands to find out about the device configuration of your z/VM guest virtual machine:

- Query the available main memory, which is called *storage* in IBM Z terminology. Your guest should have at least 1 GB of main memory.

```
cp query virtual storage
```

- Query available network devices by type:

osa

OSA - CHPID type OSD, real or virtual (VSWITCH or GuestLAN), both in QDIO mode

hsi

HiperSockets - CHPID type IQD, real or virtual (GuestLAN type Hipers)

lcs

LCS - CHPID type OSE

For example, to query all of the network device types mentioned above, run:

```
cp query virtual osa
```

- Query available DASDs. Only those that are flagged **RW** for read-write mode can be used as installation targets:

```
cp query virtual dasd
```

- Query available FCP channels:

```
cp query virtual fcp
```

16.3.1. Using the z/VM Reader

Perform the following steps to boot from the z/VM reader:

1. If necessary, add the device containing the z/VM TCP/IP tools to your CMS disk list. For example:

```
cp link tcpmaint 592 592  
acc 592 fm
```

Replace *fm* with any **FILEMODE** letter.

2. Execute the command:

```
ftp host
```

Where *host* is the host name or IP address of the FTP server that hosts the boot images (**kernel.img** and **initrd.img**).

3. Log in and execute the following commands. Use the **(repl** option if you are overwriting existing **kernel.img**, **initrd.img**, **generic.prm**, or **redhat.exec** files:

```
cd /location/of/install-tree/images/  
ascii  
get generic.prm (repl  
get redhat.exec (repl  
locsite fix 80  
binary  
get kernel.img (repl  
get initrd.img (repl  
quit
```


- Optionally, check whether the files were transferred correctly by using the CMS command **filelist** to show the received files and their format. It is important that **kernel.img** and **initrd.img** have a fixed record length format denoted by **F** in the Format column and a record length of 80 in the **Lrecl** column. For example:

```
VMUSER FILELIST A0 V 169 Trunc=169 Size=6 Line=1 Col=1 Alt=0
Cmd Filename Filetype Fm Format Lrecl Records Blocks Date Time
REDHAT EXEC B1 V 22 1 1 4/15/10 9:30:40
GENERIC PRM B1 V 44 1 1 4/15/10 9:30:32
INITRD IMG B1 F 80 118545 2316 4/15/10 9:30:25
KERNEL IMG B1 F 80 74541 912 4/15/10 9:30:17
```

Press **PF3** to quit **filelist** and return to the CMS prompt.

- Customize boot parameters in **generic.prm** as necessary. See [Section 16.1, “Customizing boot parameters”](#) for details.

Another way to configure storage and network devices is by using a CMS configuration file. In such a case, add the **CMSDASD=** and **CMSCONFFILE=** parameters to **generic.prm**. See [Section 21.2, “The z/VM Configuration File”](#) for more details.

- Finally, execute the REXX script **redhat.exec** to boot the installation program:

```
redhat
```

16.3.2. Using a Prepared DASD

Boot from the prepared DASD and select the **zipl** boot menu entry referring to the Red Hat Enterprise Linux installation program. Use a command of the following form:

```
cp ipl DASD_device_number loadparm boot_entry_number
```

Replace *DASD_device_number* with the device number of the boot device, and *boot_entry_number* with the **zipl** configuration menu for this device. For example:

```
cp ipl eb1c loadparm 0
```

16.3.3. Using a Prepared FCP-attached SCSI Disk

Perform the following steps to boot from a prepared FCP-attached SCSI disk:

- Configure the SCSI boot loader of z/VM to access the prepared SCSI disk in the FCP Storage Area Network. Select the prepared **zipl** boot menu entry referring to the Red Hat Enterprise Linux installation program. Use a command of the following form:

```
cp set loaddev portname WWPN lun LUN bootprog boot_entry_number
```

Replace *WWPN* with the World Wide Port Name of the storage system and *LUN* with the Logical Unit Number of the disk. The 16-digit hexadecimal numbers must be split into two pairs of eight digits each. For example:

```
cp set loaddev portname 50050763 050b073d lun 40204011 00000000 bootprog 0
```

2. Optionally, confirm your settings with the command:

```
cp query loaddev
```

3. Boot the FCP device connected with the storage system containing the disk with the following command:

```
cp ipl FCP_device
```

For example:

```
cp ipl fc00
```

16.3.4. Using an FCP-attached SCSI DVD Drive

This requires a SCSI DVD drive attached to an FCP-to-SCSI bridge which is in turn connected to an FCP adapter in your IBM Z. The FCP adapter must be configured and available under z/VM.

1. Insert your Red Hat Enterprise Linux for IBM Z DVD into the DVD drive.
2. Configure the SCSI boot loader of z/VM to access the DVD drive in the FCP Storage Area Network and specify **1** for the boot entry on the Red Hat Enterprise Linux for IBM Z DVD. Use a command of the following form:

```
cp set loaddev portname WWPN lun FCP_LUN bootprog 1
```

Replace *WWPN* with the WWPN of the FCP-to-SCSI bridge and *FCP_LUN* with the LUN of the DVD drive. The 16-digit hexadecimal numbers must be split into two pairs of eight characters each. For example:

```
cp set loaddev portname 20010060 eb1c0103 lun 00010000 00000000 bootprog 1
```

3. Optionally, confirm your settings with the command:

```
cp query loaddev
```

4. IPL on the FCP device connected with the FCP-to-SCSI bridge.

```
cp ipl FCP_device
```

For example:

```
cp ipl fc00
```

16.4. INSTALLING IN AN LPAR

When installing in a *logical partition* (LPAR), you can boot from:

- an FTP server
- a DASD or an FCP-attached SCSI drive prepared with the **zipl** boot loader

- an FCP-attached SCSI DVD drive

Perform these common steps first:

1. Log in on the IBM Z *Hardware Management Console* (HMC) or the *Support Element* (SE) as a user with sufficient privileges to install a new operating system to an LPAR. The **SYSPROG** user is recommended.
2. Select **Images**, then select the LPAR to which you want to install. Use the arrows in the frame on the right side to navigate to the **CPC Recovery** menu.
3. Double-click **Operating System Messages** to show the text console on which Linux boot messages will appear.

Continue with the procedure for your installation source.



NOTE

Once you finish this procedure and one of the following ones depending on your installation source, the installation will begin. The installer will then prompt you to provide additional boot parameters. Required parameters are described in [Section 16.1, "Customizing boot parameters"](#).

16.4.1. Using an FTP Server

1. Double-click **Load from CD-ROM, DVD, or Server**.
2. In the dialog box that follows, select **FTP Source**, and enter the following information:
 - **Host Computer** - Host name or IP address of the FTP server you want to install from, for example **ftp.redhat.com**
 - **User ID** - Your user name on the FTP server. Or, specify **anonymous**.
 - **Password** - Your password. Use your email address if you are logging in as **anonymous**.
 - **Account (optional)** - Leave this field empty.
 - **File location (optional)** - Directory on the FTP server holding the Red Hat Enterprise Linux for IBM Z, for example **/rhel/s390x/**.
3. Click **Continue**.
4. In the dialog that follows, keep the default selection of **generic.ins** and click **Continue**.

16.4.2. Using a Prepared DASD

1. Double-click **Load**.
2. In the dialog box that follows, select **Normal** as the **Load type**.
3. As **Load address**, fill in the device number of the DASD.
4. As **Load parameter**, fill in the number corresponding the **zipl** boot menu entry that you prepared for booting the Red Hat Enterprise Linux installation program.

5. Click the **OK** button.

16.4.3. Using a Prepared FCP-attached SCSI Disk

1. Double-click **Load**.
2. In the dialog box that follows, select **SCSI** as the **Load type**.
3. As **Load address**, fill in the device number of the FCP channel connected with the SCSI disk.
4. As **World wide port name**, fill in the WWPN of the storage system containing the disk as a 16-digit hexadecimal number.
5. As **Logical unit number**, fill in the LUN of the disk as a 16-digit hexadecimal number.
6. As **Boot program selector**, fill in the number corresponding the **zipl** boot menu entry that you prepared for booting the Red Hat Enterprise Linux installation program.
7. Leave the **Boot record logical block address** as **0** and the **Operating system specific load parameters** empty.
8. Click the **OK** button.

16.4.4. Using an FCP-attached SCSI DVD Drive

This requires a SCSI DVD drive attached to an FCP-to-SCSI bridge which is in turn connected to an FCP adapter in your IBM Z machine. The FCP adapter must be configured and available in your LPAR.

1. Insert your Red Hat Enterprise Linux for IBM Z DVD into the DVD drive.
2. Double-click **Load**.
3. In the dialog box that follows, select **SCSI** as the **Load type**.
4. As **Load address**, fill in the device number of the FCP channel connected with the FCP-to-SCSI bridge.
5. As **World wide port name**, fill in the WWPN of the FCP-to-SCSI bridge as a 16-digit hexadecimal number.
6. As **Logical unit number**, fill in the LUN of the DVD drive as a 16-digit hexadecimal number.
7. As **Boot program selector**, fill in the number **1** to select the boot entry on the Red Hat Enterprise Linux for IBM Z DVD.
8. Leave the **Boot record logical block address** as **0** and the **Operating system specific load parameters** empty.
9. Click the **OK** button.

CHAPTER 17. CONNECTING TO THE INSTALLATION SYSTEM

After the Initial Program Load (IPL) of the Anaconda installation program is complete, connect to the IBM Z system from a local machine, as an 'install' user, using an ssh connection.

You need to connect to the installation system to continue the installation process. Use a VNC mode to run a GUI-based installation or use the established connection to run a text mode installation.

Additional resources:

For more information about installing VNC and various VNC modes for GUI-based installation, see [Chapter 25, Using VNC](#)

17.1. SETTING UP REMOTE CONNECTION USING VNC

From a local machine, run the steps below to set up a remote connection with the IBM Z system.

Prerequisite:

- The initial program boot is complete on the IBM Z system, and the command prompt displays:

```
Starting installer, one moment...
Please ssh install@my-z-system (system ip address) to begin the install.
```

- If you want to restrict VNC access to the installation system, then ensure **inst.vncpassword=PASSWORD** boot parameter is configured.

1. On the command prompt, run the following command:

```
$ssh install@my-z-system-domain-name
```

OR

```
$ssh install@my-z-system-IP-address
```

2. Depending on whether or not have you configured the **inst.vnc** parameter, the ssh session displays the following output:

When **inst.vnc** parameter is configured:

```
Starting installer, one moment...
Please manually connect your vnc client to my-z-system:1 (system-ip-address:1) to begin the
install.
```

When **inst.vnc** parameter is not configured:

```
Starting installer, one moment...
Graphical installation is not available. Starting text mode.
=====
Text mode provides a limited set of installation options.
It does not offer custom partitioning for full control
over the disk layout. Would you like to use VNC mode instead?
```

1) Start VNC

2) Use text mode

Please make your choice from above ['q' to quit | 'c' to continue | 'r' to refresh]:

If you have configured the **inst.vnc** parameter, proceed to step 5.

3. Enter 1 to start VNC.
4. Enter a password, if you have not set the **inst.vncpassword=** boot option, but want to secure the server connection.
5. From a new command prompt, connect to the VNC server.

\$vncviewer *my-z-system-ip-address:display_number*

If you have secured the connection, use the password that you have entered in the previous step or the one that you had set for **inst.vncpassword=** boot option.

The RHEL installer is launched in the VNC client.

CHAPTER 18. INSTALLING USING ANACONDA

This chapter provides step-by-step instructions for installing Red Hat Enterprise Linux using the **Anaconda** installer. The bulk of this chapter describes installation using the graphical user interface; on IBM Z, the graphical interface is accessed over the VNC protocol from another system. A text mode is also available for systems with no graphical display, but this mode is limited in certain aspects (for example, custom partitioning is not possible in text mode).

If you cannot use VNC mode with a graphical interface, consider using Kickstart to automate the installation. See [Chapter 27, Kickstart Installations](#) for information about Kickstart.

18.1. INTRODUCTION TO ANACONDA

The Red Hat Enterprise Linux installer, **Anaconda**, is different from most other operating system installation programs due to its parallel nature. Most installers follow a fixed path: you must choose your language first, then you configure network, then installation type, then partitioning, and so on. There is usually only one way to proceed at any given time.

In **Anaconda** you are only required to select your language and locale first, and then you are presented with a central screen, where you can configure most aspects of the installation in any order you like. This does not apply to all parts of the installation process, however – for example, when installing from a network location, you must configure the network before you can select which packages to install.

Some screens will be automatically configured depending on your hardware and the type of media you used to start the installation. You can still change the detected settings in any screen. Screens which have not been automatically configured, and therefore require your attention before you begin the installation, are marked by an exclamation mark. You cannot start the actual installation process before you finish configuring these settings.

Additional differences appear in certain screens; notably the custom partitioning process is very different from other Linux distributions. These differences are described in each screen's subsection.

18.2. CONSOLES AND LOGGING DURING THE INSTALLATION

The following sections describe how to access logs and an interactive shell during the installation. This is useful when troubleshooting problems, but should not be necessary in most cases.

18.2.1. Accessing Consoles

The Red Hat Enterprise Linux installer uses the **tmux** terminal multiplexer to display and control several windows you can use in addition to the main interface. Each of these windows serves a different purpose – they display several different logs, which can be used to troubleshoot any issues during the installation, and one of the windows provides an interactive shell prompt with **root** privileges, unless this prompt was specifically disabled using a boot option or a Kickstart command.



NOTE

In general, there is no reason to leave the default graphical installation environment unless you need to diagnose an installation problem.

The terminal multiplexer is running in virtual console 1. To switch from the graphical installation environment to **tmux**, press **Ctrl+Alt+F1**. To go back to the main installation interface which runs in virtual console 6, press **Ctrl+Alt+F6**.

**NOTE**

If you choose text mode installation, you will start in virtual console 1 (**tmux**), and switching to console 6 will open a shell prompt instead of a graphical interface.

The console running **tmux** has 5 available windows; their contents are described in the table below, along with keyboard shortcuts used to access them. Note that the keyboard shortcuts are two-part: first press **Ctrl+b**, then release both keys, and press the number key for the window you want to use.

You can also use **Ctrl+b n** and **Ctrl+b p** to switch to the next or previous **tmux** window, respectively.

Table 18.1. Available tmux Windows

Shortcut	Contents
Ctrl+b 1	Main installation program window. Contains text-based prompts (during text mode installation or if you use VNC Direct Mode), and also some debugging information.
Ctrl+b 2	Interactive shell prompt with root privileges.
Ctrl+b 3	Installation log; displays messages stored in /tmp/anaconda.log .
Ctrl+b 4	Storage log; displays messages related storage devices from kernel and system services, stored in /tmp/storage.log .
Ctrl+b 5	Program log; displays messages from other system utilities, stored in /tmp/program.log .

In addition to displaying diagnostic information in **tmux** windows, **Anaconda** also generates several log files, which can be transferred from the installation system. These log files are described in [Table 19.1, “Log Files Generated During the Installation”](#), and directions for transferring them from the installation system are available in [Chapter 19, Troubleshooting Installation on IBM Z](#).

18.2.2. Saving Screenshots

You can press **Shift+Print Screen** at any time during the graphical installation to capture the current screen. These screenshots are saved to **/tmp/anaconda-screenshots/**.

Additionally, you can use the **autostep --autoscreenshot** command in a Kickstart file to capture and save each step of the installation automatically. See [Section 27.3.1, “Kickstart Commands and Options”](#) for details.

18.3. INSTALLATION IN NON-INTERACTIVE LINE MODE

If the **inst.cmdline** option was specified as a boot option in your parameter file (see [Section 21.4, “Parameters for Kickstart Installations”](#)) or the **cmdline** option was specified in your Kickstart file (see [Chapter 27, Kickstart Installations](#)), **Anaconda** starts with non-interactive text line mode. In this mode, all necessary information must be provided in the Kickstart file. The installation program will not allow user interaction and it will stop if any required commands are missing.

18.4. INSTALLING IN TEXT MODE

Text mode installation offers an interactive, non-graphical interface for installing Red Hat Enterprise Linux. This can be useful on systems with no graphical capabilities; however, you should always consider the available alternatives (an automated Kickstart installation or using the graphical user interface over VNC) before starting a text-based installation. Text mode is limited in the amount of choices you can make during the installation.

```

Installation

1) [!] Timezone settings          2) [x] Language settings
   (Timezone is not set.)        (English (United States))
3) [!] Software selection        4) [!] Installation source
   (Processing...)              (Processing...)
5) [x] Network settings          6) [!] Install Destination
   (Wired (eth0) connected)     (No disks selected)
7) [x] Kdump                     8) [!] Set root password
   (Kdump is enabled)           (Password is not set.)
9) [!] Create user
   (No user will be created)

Please make your choice from above ['q' to quit | 'b' to begin installation |
'r' to refresh]: _

```

Figure 18.1. Text Mode Installation

Installation in text mode follows a pattern similar to the graphical installation: There is no single fixed progression; you can configure many settings in any order you want using the main status screen. Screens which have already been configured, either automatically or by you, are marked as **[x]**, and screens which require your attention before the installation can begin are marked with **[!]**. Available commands are displayed below the list of available options.



NOTE

When related background tasks are being run, certain menu items can be temporarily unavailable or display the **Processing...** label. To refresh to the current status of text menu items, use the **r** option at the text mode prompt.

At the bottom of the screen in text mode, a green bar is displayed showing five menu options. These options represent different screens in the **tmux** terminal multiplexer; by default you start in screen 1, and you can use keyboard shortcuts to switch to other screens which contain logs and an interactive command prompt. For information about available screens and shortcuts to switch to them, see [Section 18.2.1, “Accessing Consoles”](#).

Limits of interactive text mode installation include:

- The installer will always use the English language and the US English keyboard layout. You can configure your language and keyboard settings, but these settings will only apply to the installed system, not to the installation.
- You cannot configure any advanced storage methods (LVM, software RAID, FCoE, zFCP and iSCSI).
- It is not possible to configure custom partitioning; you must use one of the automatic partitioning settings. You also cannot configure where the boot loader will be installed.

- You cannot select any package add-ons to be installed; they must be added after the installation finishes using the **Yum** package manager.

To start a text mode installation, boot the installation with the **inst.text** boot option used in the parameter file (**generic.prm**). See [Chapter 21, Parameter and Configuration Files on IBM Z](#) for information about the parameter file.

18.5. INSTALLING IN THE GRAPHICAL USER INTERFACE

The graphical installation interface is the preferred method of manually installing Red Hat Enterprise Linux. It allows you full control over all available settings, including custom partitioning and advanced storage configuration, and it is also localized to many languages other than English, allowing you to perform the entire installation in a different language. The graphical mode is used by default when you boot the system from local media (a CD, DVD or a USB flash drive).

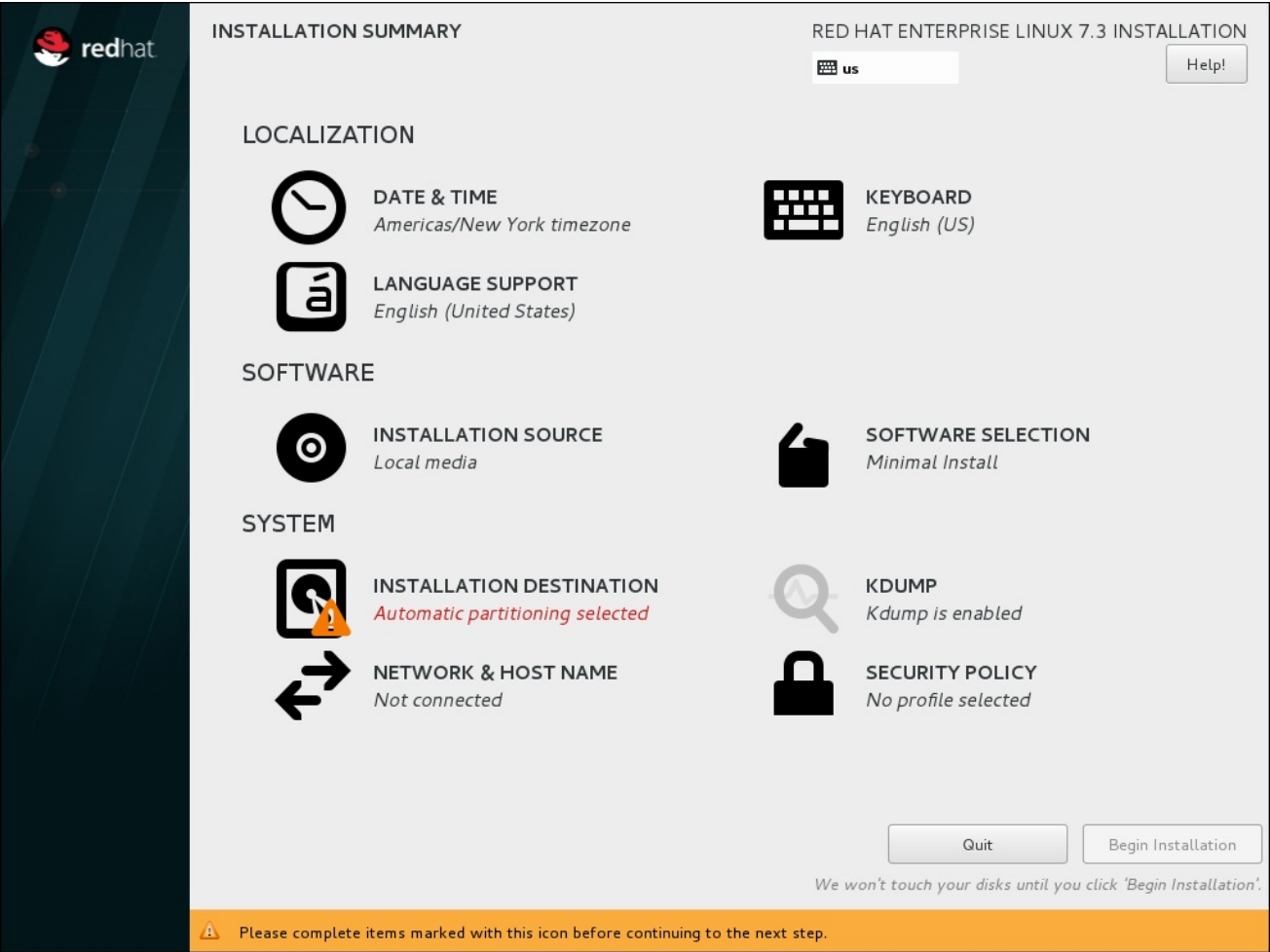


Figure 18.2. The Installation Summary Screen

The sections below discuss each screen available in the installation process. Note that due to the installer's parallel nature, most of the screens do not have to be completed in the order in which they are described here.

Each screen in the graphical interface contains a **Help** button. This button opens the **Yelp** help browser displaying the section of the *Red Hat Enterprise Linux Installation Guide* relevant to the current screen.

You can also control the graphical installer with your keyboard. Following table shows you the shortcuts you can use.

Table 18.2. Graphical installer keyboard shortcuts

Shortcut keys	Usage
Tab and Shift+Tab	Cycle through active control elements (buttons, check boxes, and so on.) on the current screen
Up and Down	Scroll through lists
Left and Right	Scroll through horizontal toolbars and table entries
Space and Enter	Select or remove a highlighted item from selection and expand and collapse drop-down menus

Additionally, elements in each screen can be toggled using their respective shortcuts. These shortcuts are highlighted (underlined) when you hold down the **Alt** key; to toggle that element, press **Alt+X**, where **X** is the highlighted letter.

Your current keyboard layout is displayed in the top right hand corner. Only one layout is configured by default; if you configure more than layout in the **Keyboard Layout** screen ([Section 18.10, “Keyboard Configuration”](#)), you can switch between them by clicking the layout indicator.

18.6. WELCOME SCREEN AND LANGUAGE SELECTION

The first screen of the installation program is the **Welcome to Red Hat Enterprise Linux** screen. Here you select the language that **Anaconda** will use for the rest of the installation. This selection will also become the default for the installed system, unless changed later. In the left panel, select your language of choice, for example **English**. Then you can select a locale specific to your region in the right panel, for example **English (United States)**.



NOTE

One language is pre-selected by default on top of the list. If network access is configured at this point (for example, if you booted from a network server instead of local media), the pre-selected language will be determined based on automatic location detection using the **GeoIP** module.

Alternatively, type your preferred language into the search box as shown below.

Once you have made your selection, click the **Continue** button to proceed to the **Installation Summary** screen.

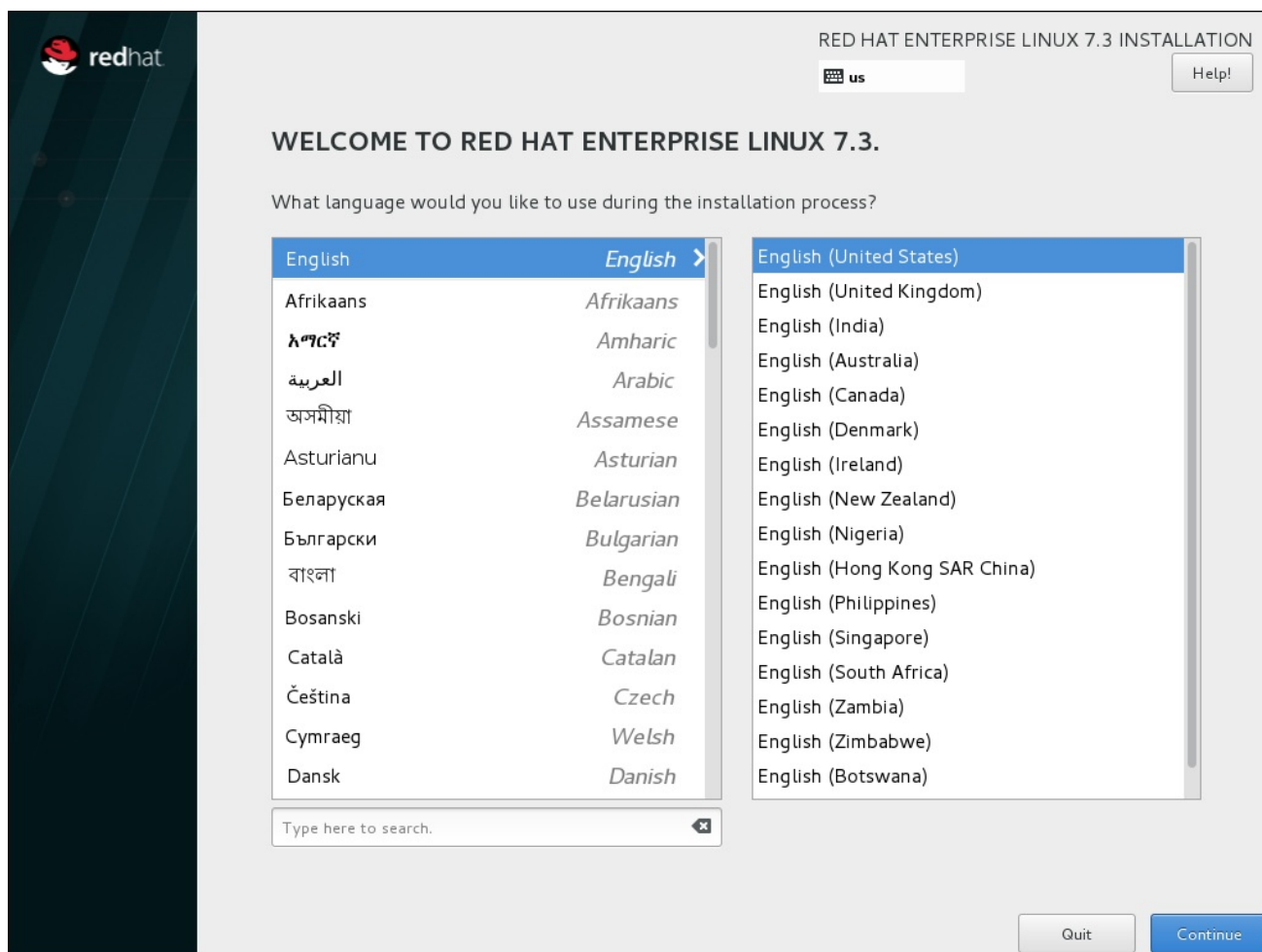


Figure 18.3. Language Configuration

After you click the **Continue** button, the unsupported hardware dialog may appear. This happens if you are using hardware that the kernel does not support.

18.7. THE INSTALLATION SUMMARY SCREEN

The **Installation Summary** screen is the central location for setting up an installation.

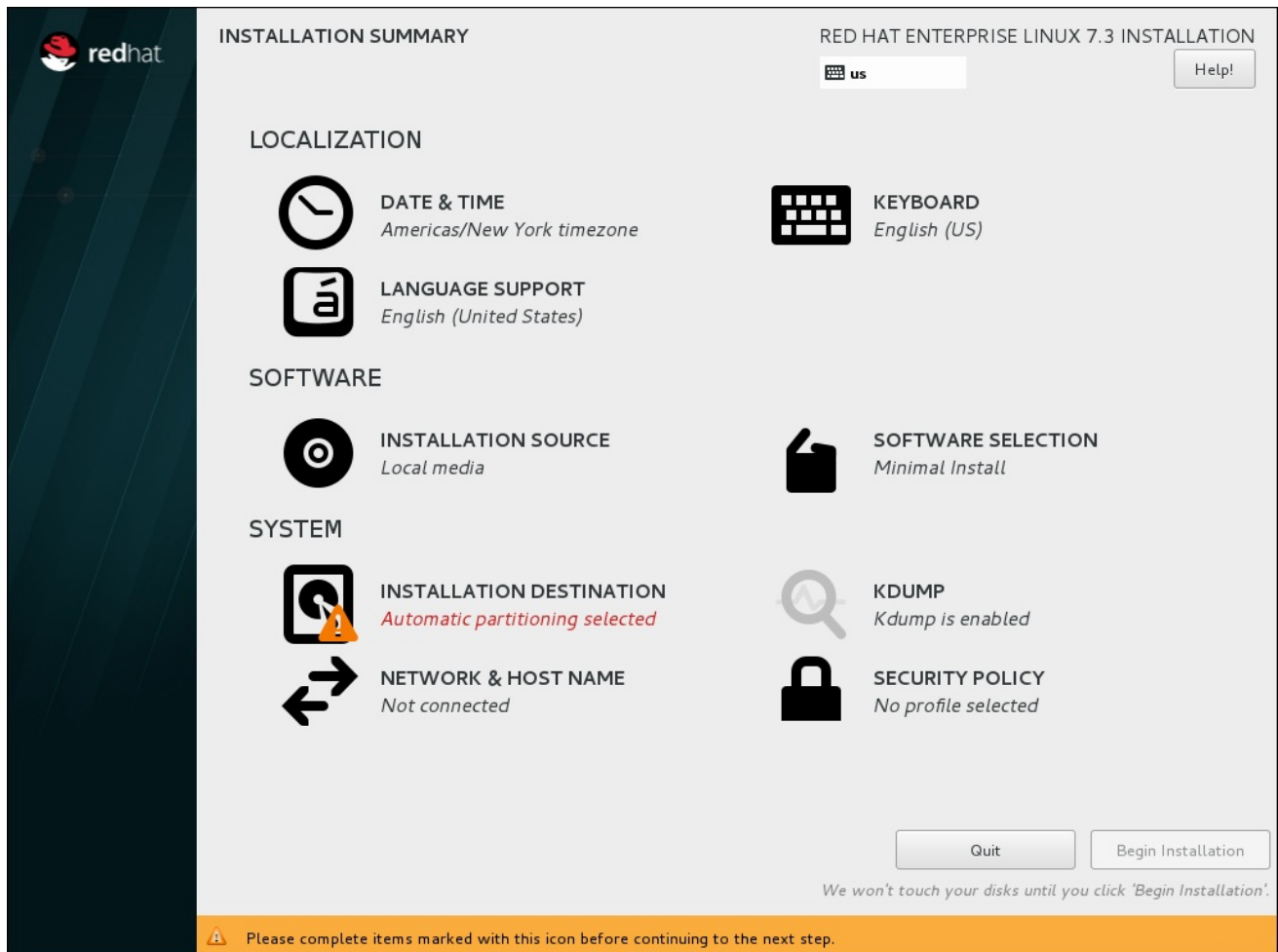


Figure 18.4. The Installation Summary Screen

Instead of directing you through consecutive screens, the Red Hat Enterprise Linux installation program allows you to configure your installation in the order you choose.

Use your mouse to select a menu item to configure a section of the installation. When you have completed configuring a section, or if you would like to complete that section later, click the **Done** button located in the upper left corner of the screen.

Only sections marked with a warning symbol are mandatory. A note at the bottom of the screen warns you that these sections must be completed before the installation can begin. The remaining sections are optional. Beneath each section's title, the current configuration is summarized. Using this you can determine whether you need to visit the section to configure it further.

Once all required sections are complete, click the **Begin Installation** button. Also see [Section 18.18, "Begin Installation"](#).

To cancel the installation, click the **Quit** button.



NOTE

When related background tasks are running, certain menu items might be temporarily unavailable.

18.8. DATE & TIME

To configure time zone, date, and optionally settings for network time, select **Date & Time** at the **Installation Summary** screen.

There are three ways for you to select a time zone:

- Using your mouse, click on the interactive map to select a specific city. A red pin appears indicating your selection.
- You can also scroll through the **Region** and **City** drop-down menus at the top of the screen to select your time zone.
- Select **Etc** at the bottom of the **Region** drop-down menu, then select your time zone in the next menu adjusted to GMT/UTC, for example **GMT+1**.

If your city is not available on the map or in the drop-down menu, select the nearest major city in the same time zone. Alternatively you can use a Kickstart file, which will allow you to specify some additional time zones which are not available in the graphical interface. See the **timezone** command in [timezone \(required\)](#) for details.



NOTE

The list of available cities and regions comes from the Time Zone Database (tzdata) public domain, which is maintained by the Internet Assigned Numbers Authority (IANA). Red Hat cannot add cities or regions into this database. You can find more information at the official website, available at <http://www.iana.org/time-zones>.

Specify a time zone even if you plan to use NTP (Network Time Protocol) to maintain the accuracy of the system clock.

If you are connected to the network, the **Network Time** switch will be enabled. To set the date and time using NTP, leave the **Network Time** switch in the **ON** position and click the configuration icon to select which NTP servers Red Hat Enterprise Linux should use. To set the date and time manually, move the switch to the **OFF** position. The system clock should use your time zone selection to display the correct date and time at the bottom of the screen. If they are still incorrect, adjust them manually.

Note that NTP servers might be unavailable at the time of installation. In such a case, enabling them will not set the time automatically. When the servers become available, the date and time will update.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your time zone configuration after you have completed the installation, visit the **Date & Time** section of the **Settings** dialog window.

18.9. LANGUAGE SUPPORT

To install support for additional locales and language dialects, select **Language Support** from the **Installation Summary** screen.

Use your mouse to select the language for which you would like to install support. In the left panel, select your language of choice, for example **Español**. Then you can select a locale specific to your region in the right panel, for example **Español (Costa Rica)**. You can select multiple languages and multiple locales. The selected languages are highlighted in bold in the left panel.

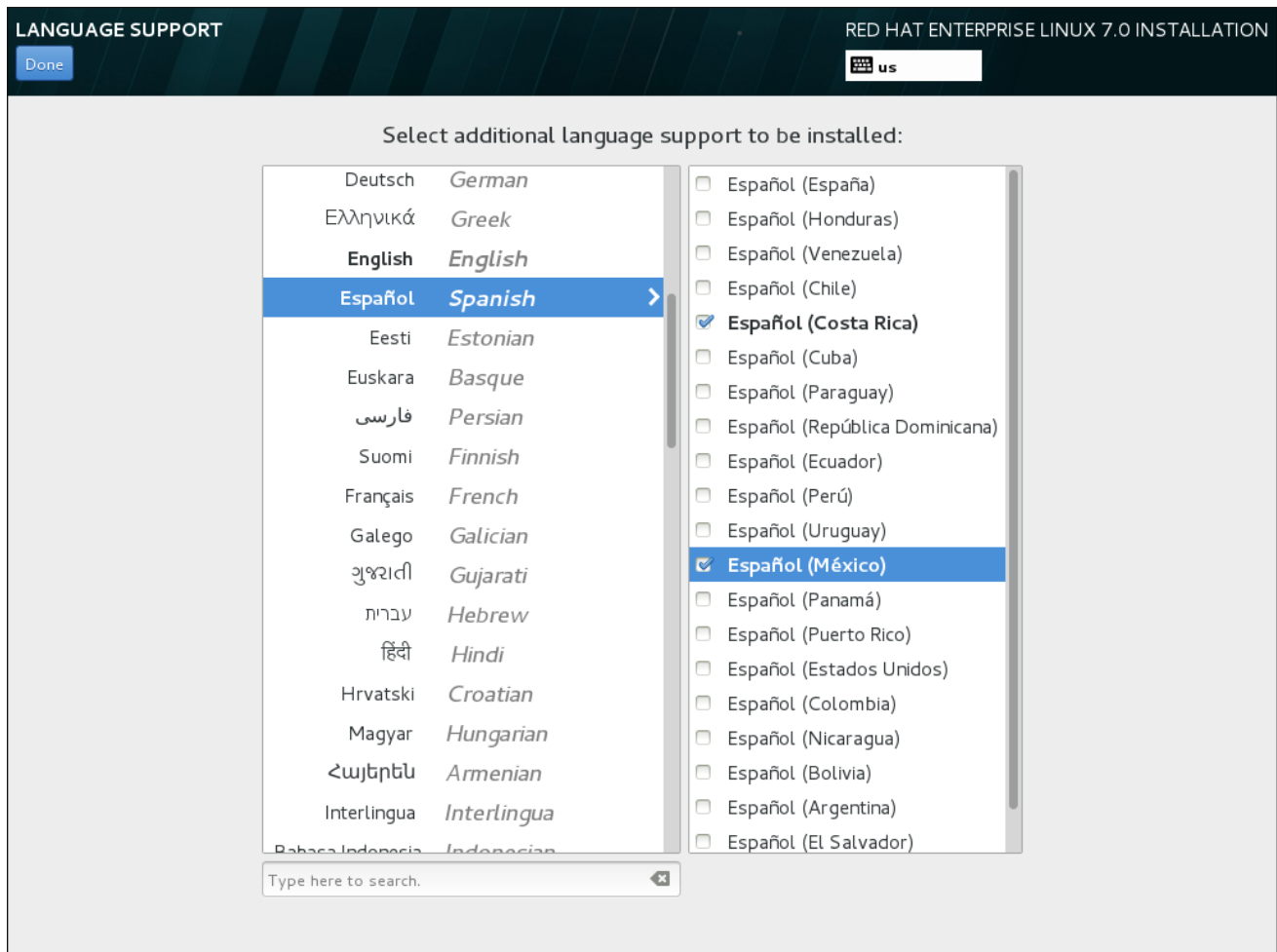


Figure 18.5. Configuring Language Support

Once you have made your selections, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your language support configuration after you have completed the installation, visit the **Region & Language** section of the **Settings** dialog window.

18.10. KEYBOARD CONFIGURATION

To add multiple keyboard layouts to your system, select **Keyboard** from the **Installation Summary** screen. Upon saving, the keyboard layouts are immediately available in the installation program and you can switch between them by using the keyboard icon located at all times in the upper right corner of the screen.

Initially, only the language you selected in the welcome screen is listed as the keyboard layout in the left pane. You can either replace the initial layout or add more layouts. However, if your language does not use ASCII characters, you might need to add a keyboard layout that does, to be able to properly set a password for an encrypted disk partition or the root user, among other things.

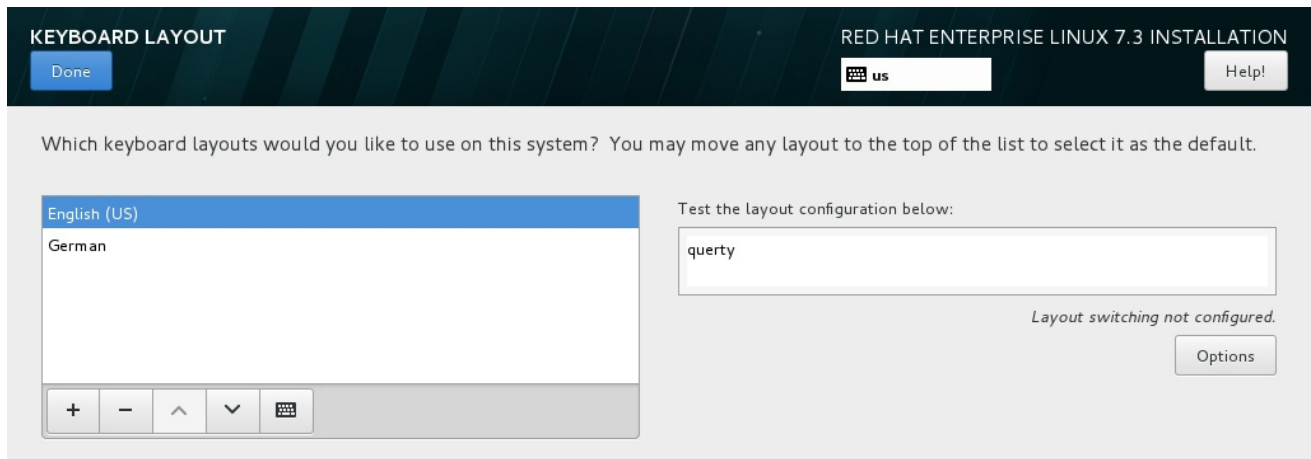
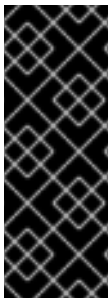


Figure 18.6. Keyboard Configuration

To add an additional layout, click the **+** button, select it from the list, and click **Add**. To delete a layout, select it and click the **-** button. Use the arrow buttons to arrange the layouts in order of preference. For a visual preview of the keyboard layout, select it and click the keyboard button.

To test a layout, use the mouse to click inside the text box on the right. Type some text to confirm that your selection functions correctly.

To test additional layouts, you can click the language selector at the top on the screen to switch them. However, it is recommended to set up a keyboard combination for switching layout. Click the **Options** button at the right to open the **Layout Switching Options** dialog and choose a combination from the list by selecting its check box. The combination will then be displayed above the **Options** button. This combination applies both during the installation and on the installed system, so you must configure a combination here in order to use one after installation. You can also select more than one combination to switch between layouts.



IMPORTANT

If you use a layout that cannot accept Latin characters, such as **Russian**, Red Hat recommends additionally adding the **English (United States)** layout and configuring a keyboard combination to switch between the two layouts. If you only select a layout without Latin characters, you might be unable to enter a valid root password and user credentials later in the installation process. This can prevent you from completing the installation.

Once you have made your selection, click **Done** to return to the **Installation Summary** screen.



NOTE

To change your keyboard configuration after you have completed the installation, visit the **Keyboard** section of the **Settings** dialogue window.

18.11. SECURITY POLICY

The **Security Policy** spoke allows you to configure the installed system following restrictions and recommendations (*compliance policies*) defined by the Security Content Automation Protocol (SCAP) standard. This functionality is provided by an add-on which has been enabled by default since Red Hat

Enterprise Linux 7.2. When enabled, the packages necessary to provide this functionality will automatically be installed. However, by default, no policies are enforced, meaning that no checks are performed during or after installation unless specifically configured.

The [Red Hat Enterprise Linux 7 Security Guide](#) provides detailed information about security compliance including background information, practical examples, and additional resources.



IMPORTANT

Applying a security policy is not necessary on all systems. This screen should only be used when a specific policy is mandated by your organization rules or government regulations.

If you apply a security policy to the system, it will be installed using restrictions and recommendations defined in the selected profile. The `openscap-scanner` package will also be added to your package selection, providing a preinstalled tool for compliance and vulnerability scanning. After the installation finishes, the system will be automatically scanned to verify compliance. The results of this scan will be saved to the `/root/openscap_data` directory on the installed system.

Pre-defined policies which are available in this screen are provided by **SCAP Security Guide**. See the [OpenSCAP Portal](#) for links to detailed information about each available profile.

You can also load additional profiles from an HTTP, HTTPS or FTP server.

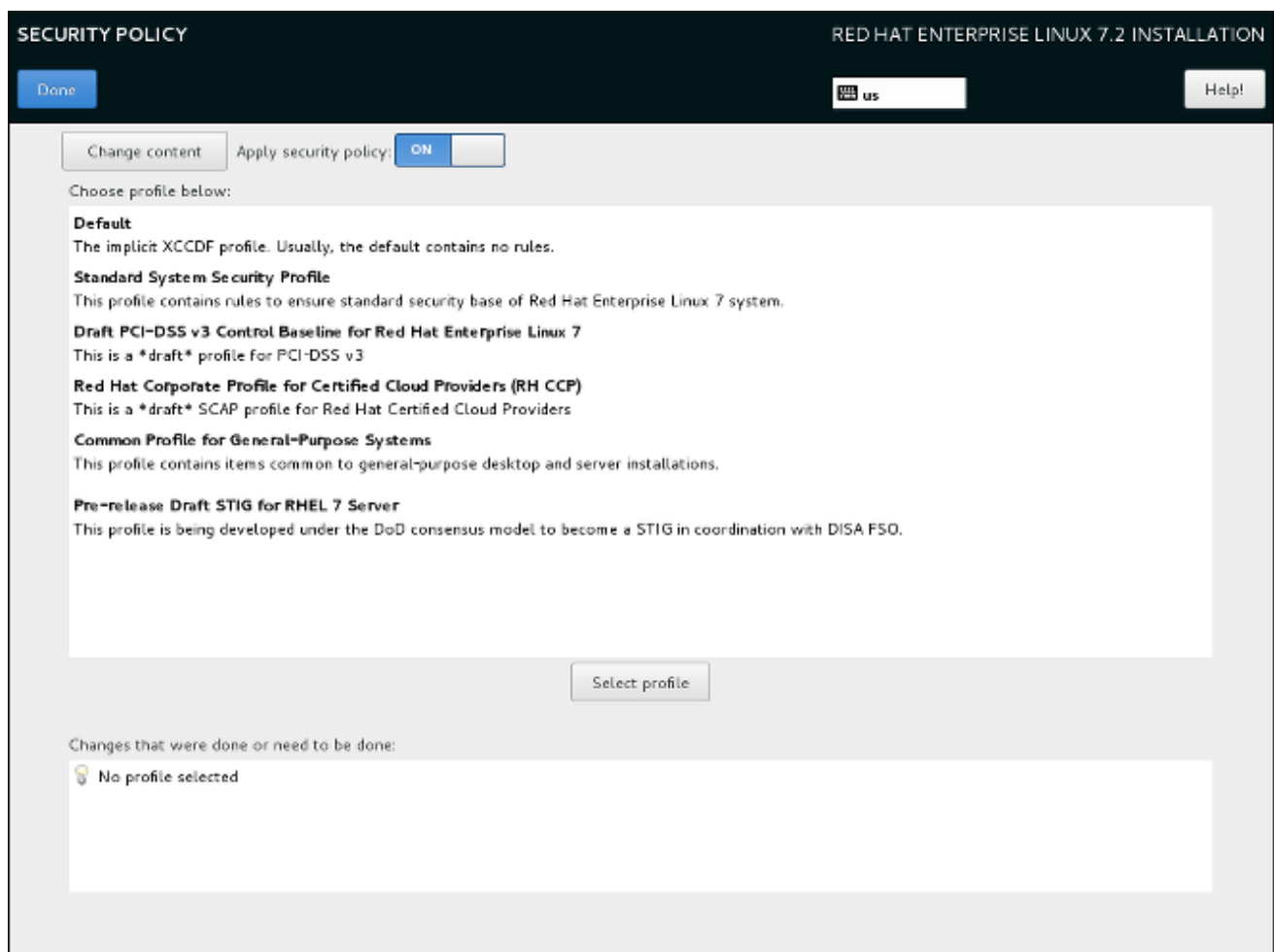


Figure 18.7. Security policy selection screen

To configure the use of security policies on the system, first enable configuration by setting the **Apply security policy** switch to **ON**. If the switch is in the **OFF** position, controls in the rest of this screen have no effect.

After enabling security policy configuration using the switch, select one of the profiles listed in the top window of the screen, and click the **Select profile** below. When a profile is selected, a green check mark will appear on the right side, and the bottom field will display whether any changes will be made before beginning the installation.



NOTE

None of the profiles available by default perform any changes before the installation begins. However, loading a custom profile as described below can require some pre-installation actions.

To use a custom profile, click the **Change content** button in the top left corner. This will open another screen where you can enter an URL of a valid security content. To go back to the default security content selection screen, click **Use SCAP Security Guide** in the top left corner.

Custom profiles can be loaded from an **HTTP**, **HTTPS** or **FTP** server. Use the full address of the content, including the protocol (such as **http://**). A network connection must be active (enabled in [Section 18.13, "Network & Hostname"](#)) before you can load a custom profile. The content type will be detected automatically by the installer.

After you select a profile, or if you want to leave the screen, click **Done** in the top left corner to return to [Section 18.7, "The Installation Summary Screen"](#).

18.12. INSTALLATION SOURCE

To specify a file or a location to install Red Hat Enterprise Linux from, select **Installation Source** from the **Installation Summary** screen. On this screen, you can choose between locally available installation media such an ISO file, or a network location.

Figure 18.8. Installation Source Screen

Select one of the following options:

ISO file

This option will appear if the installation program detected a partitioned hard drive with mountable file systems. Select this option, click the **Choose an ISO** button, and browse to the installation ISO file's location on your system. Then click **Verify** to ensure that the file is suitable for installation.

On the network

To specify a network location, select this option and choose from the following options in the drop-down menu:

- **http://**
- **https://**
- **ftp://**
- **nfs**

Using your selection as the start of the location URL, type the rest into the address box. If you choose NFS, another box will appear for you to specify any NFS mount options.



IMPORTANT

When selecting an NFS-based installation source, you must specify the address with a colon (:) character separating the host name from the path. For example:

server.example.com:/path/to/directory

To configure a proxy for an HTTP or HTTPS source, click the **Proxy setup** button. Check **Enable HTTP proxy** and type the URL into the **Proxy URL** box. If your proxy requires authentication, check **Use Authentication** and enter a user name and password. Click **Add**.

If your HTTP or HTTP URL refers to a repository mirror list, mark the check box under the input field.

You can also specify additional repositories to gain access to more installation environments and software add-ons. See [Section 18.14, "Software Selection"](#) for more information.

To add a repository, click the **+** button. To delete a repository, click the **-** button. Click the arrow icon to revert to the previous list of repositories, that is, to replace current entries with those that were present at the time you entered the **Installation Source** screen. To activate or deactivate a repository, click the check box in the **Enabled** column at each entry in the list.

In the right part of the form, you can name your additional repository and configure it the same way as the primary repository on the network.

Once you have selected your installation source, click **Done** to return to the **Installation Summary** screen.

18.13. NETWORK & HOSTNAME

To configure essential networking features for your system, select **Network & Hostname** at the **Installation Summary** screen.

Locally accessible interfaces are automatically detected by the installation program and cannot be manually added or deleted. The detected interfaces are listed in the left pane. Click an interface in the list to display more details about it on the right. To activate or deactivate a network interface, move the switch in the top right corner of the screen to either **ON** or **OFF**.



NOTE

There are several types of network device naming standards used to identify network devices with persistent names such as **em1** or **wl3sp0**. For information about these standards, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

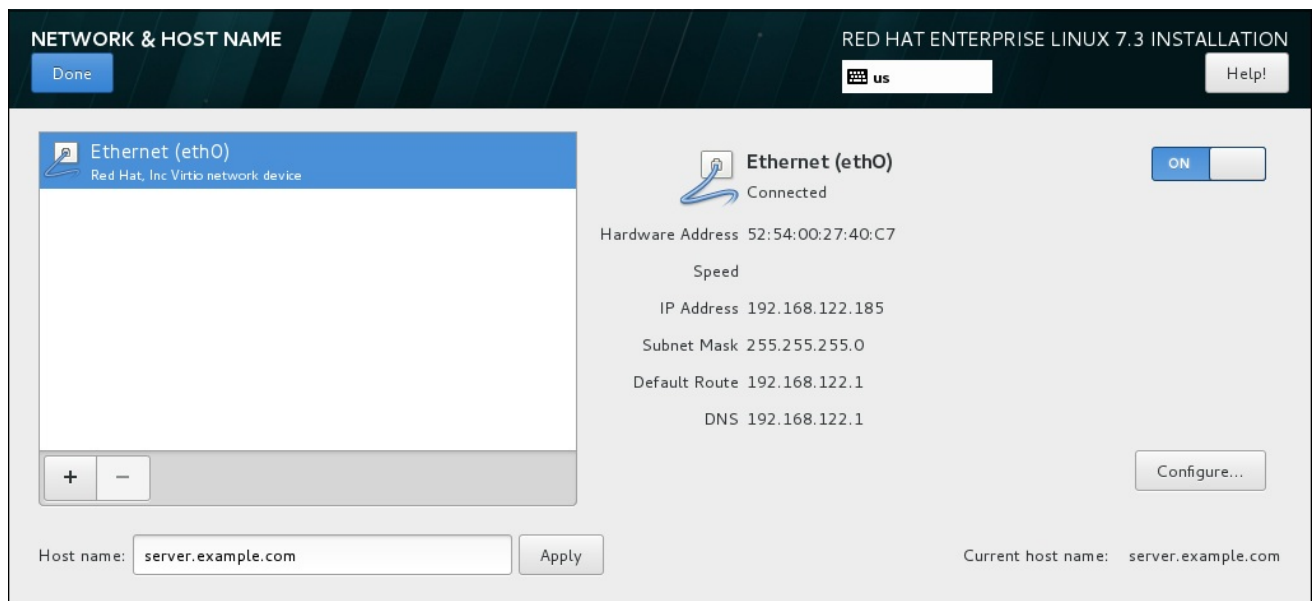
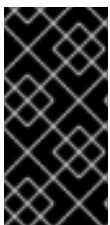


Figure 18.9. Network & Hostname Configuration Screen

Below the list of connections, enter a host name for this computer in the **Hostname** input field. The host name can be either a *fully-qualified domain name* (FQDN) in the format *hostname.domainname* or a *short host name* in the format *hostname*. Many networks have a *Dynamic Host Configuration Protocol* (DHCP) service that automatically supplies connected systems with a domain name. To allow the DHCP service to assign the domain name to this machine, only specify the short host name. The value **localhost.localdomain** means that no specific static host name for target system is configured, and the actual host name of installed system will be configured during process of network configuration (for example, by NetworkManager using DHCP or DNS).



IMPORTANT

If you want to manually assign the host name, make sure you do not use a domain name that is not delegated to you, as this can result in network resources becoming unavailable. For more information, see the recommended naming practices in the [Red Hat Enterprise Linux 7 Networking Guide](#).

Change the default setting *localhost.localdomain* to a unique host name for each of your Linux instances.

Once you have finished network configuration, click **Done** to return to the **Installation Summary** screen.

18.13.1. Edit Network Connections

All network connections on IBM Z are listed in the **Network & Hostname** screen. By default, the list contains the connection configured earlier in the booting phase and is either OSA, LCS, or HiperSockets. All of these interface types use names in the form of **enccw***device_id*, for example **enccw0.0.0a00**. Note that on IBM Z, you cannot add a new connection because the network subchannels need to be grouped and set online beforehand, and this is currently only done in the booting phase. See [Chapter 16, Booting the Installation on IBM Z](#) for details.

Usually, the network connection configured earlier in the booting phase does not need to be modified during the rest of the installation. However, if you do need to modify the existing connection, click the **Configure** button. A **NetworkManager** dialog appears with a set of tabs appropriate to wired connections, as described below. Here, you can configure network connections for the system, not all of which are relevant to IBM Z.

This section only details the most important settings for a typical wired connection used during installation. Many of the available options do not have to be changed in most installation scenarios and are not carried over to the installed system. Configuration of other types of network is broadly similar, although the specific configuration parameters are necessarily different. To learn more about network configuration after installation, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

To configure a network connection manually, click the **Configure** button in the lower right corner of the screen. A dialog appears that allows you to configure the selected connection. If required, see the *Networking Guide* for more detailed information on network settings.

The most useful network configuration options to consider during installation are:

- Mark the **Automatically connect to this network when it is available** check box if you want to use the connection every time the system boots. You can use more than one connection that will connect automatically. This setting will carry over to the installed system.

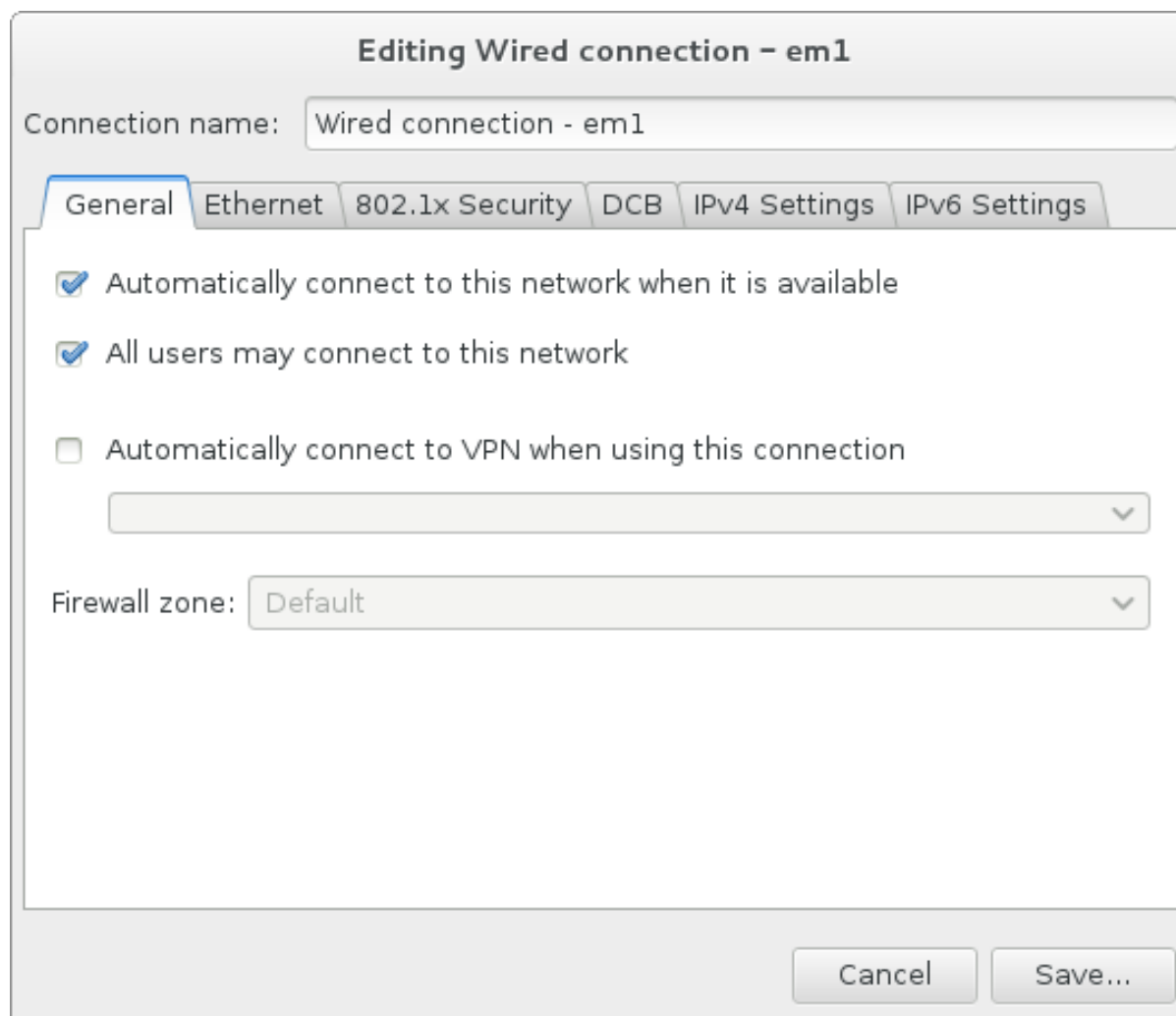


Figure 18.10. Network Auto-Connection Feature

- By default, IPv4 parameters are configured automatically by the DHCP service on the network. At the same time, the IPv6 configuration is set to the **Automatic** method. This combination is suitable for most installation scenarios and usually does not require any changes.

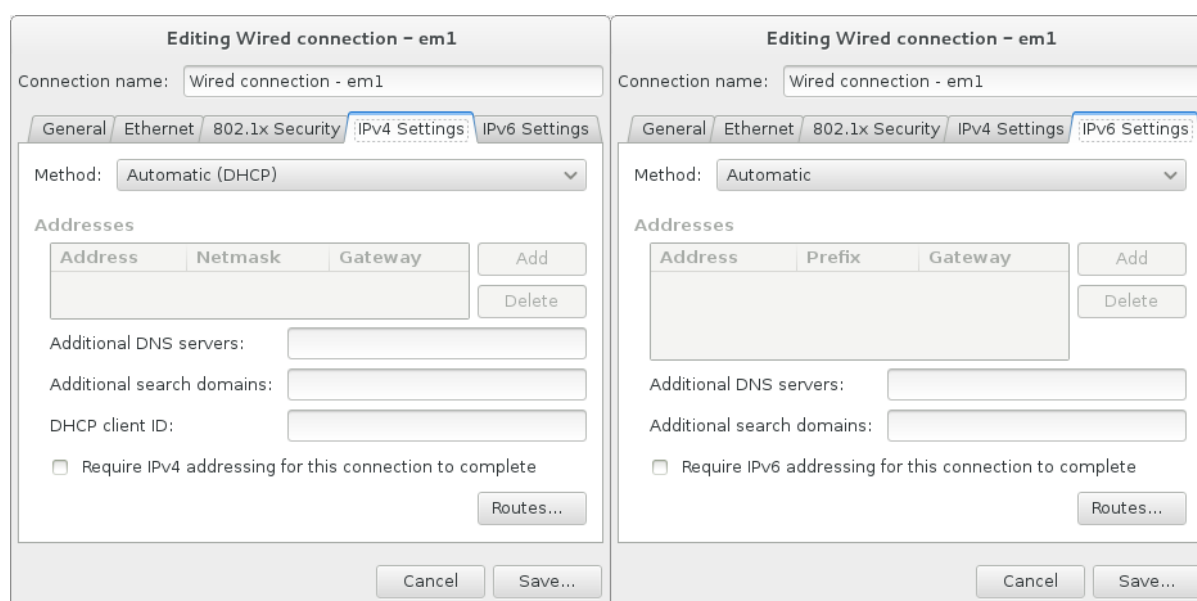


Figure 18.11. IP Protocol Settings

When you have finished editing network settings, click **Save** to save the new configuration. If you

reconfigured a device that was already active during installation, you must restart the device in order to use the new configuration in the installation environment. Use the **ON/OFF** switch on the **Network & Host Name** screen to restart the device.

18.13.2. Advanced Network Interfaces

Advanced network interfaces are also available for installation. This includes virtual local area networks (VLANs) and three methods to use aggregated links. Detailed description of these interfaces is beyond the scope of this document; read the [Red Hat Enterprise Linux 7 Networking Guide](#) for more information.

To create an advanced network interface, click the **+** button in the lower left corner of the **Network & Hostname** screen.

A dialog appears with a drop-down menu with the following options:

- **Bond** - represents NIC (*Network Interface Controller*) Bonding, a method to bind multiple network interfaces together into a single, bonded, channel.
- **Bridge** - represents NIC Bridging, a method to connect multiple separate network into one aggregate network.
- **Team** - represents NIC Teaming, a new implementation to aggregate links, designed to provide a small kernel driver to implement the fast handling of packet flows, and various applications to do everything else in user space.
- **VLAN** - represents a method to create multiple distinct broadcast domains, which are mutually isolated.

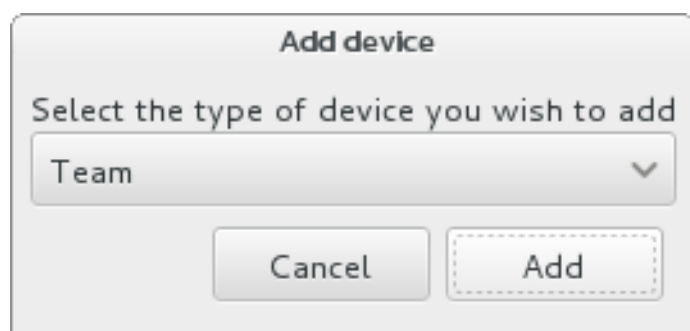


Figure 18.12. Advanced Network Interface Dialog



NOTE

Note that locally accessible interfaces, wired or wireless, are automatically detected by the installation program and cannot be manually added or deleted by using these controls.

Once you have selected an option and clicked the **Add** button, another dialog appears for you to configure the new interface. See the respective chapters in the [Red Hat Enterprise Linux 7 Networking Guide](#) for detailed instructions. To edit configuration on an existing advanced interface, click the **Configure** button in the lower right corner of the screen. You can also remove a manually-added interface by clicking the **-** button.

18.14. SOFTWARE SELECTION

To specify which packages will be installed, select **Software Selection** at the **Installation Summary** screen. The package groups are organized into *Base Environments*. These environments are pre-defined sets of packages with a specific purpose; for example, the **Virtualization Host** environment contains a set of software packages needed for running virtual machines on the system. Only one software environment can be selected at installation time.

For each environment, there are additional packages available in the form of *Add-ons*. Add-ons are presented in the right part of the screen and the list of them is refreshed when a new environment is selected. You can select multiple add-ons for your installation environment.

A horizontal line separates the list of add-ons into two areas:

- Add-ons listed *above* the horizontal line are specific to the environment you selected. If you select any add-ons in this part of the list and then select a different environment, your selection will be lost.
- Add-ons listed *below* the horizontal line are available for all environments. Selecting a different environment will not impact the selections made in this part of the list.

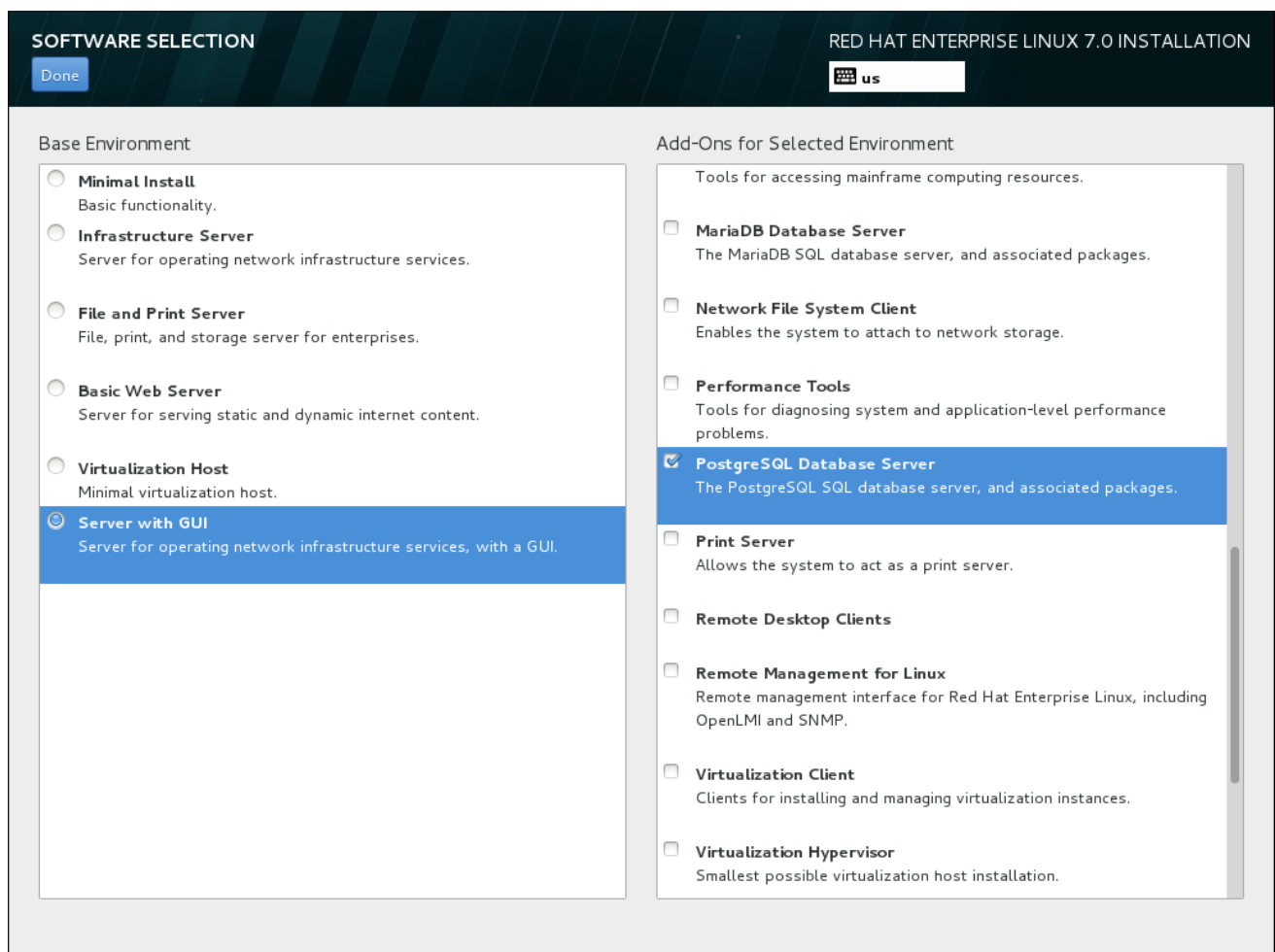
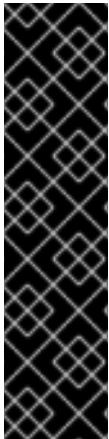


Figure 18.13. Example of a Software Selection for a Server Installation

The availability of base environments and add-ons depends on the variant of the installation ISO image which you are using as the installation source. For example, the **server** variant provides environments designed for servers, while the **workstation** variant has several choices for deployment as a developer workstation, and so on.

The installation program does not show which packages are contained in the available environments. To see which packages are contained in a specific environment or add-on, see the **repodata/*-**

comps-variant.architecture.xml file on the Red Hat Enterprise Linux Installation DVD which you are using as the installation source. This file contains a structure describing available environments (marked by the **<environment>** tag) and add-ons (the **<group>** tag).



IMPORTANT

The pre-defined environments and add-ons allow you to customize your system, but in a manual installation, there is no way to select individual packages to install. If you are not sure what package should be installed, Red Hat recommends you to select the **Minimal Install** environment. **Minimal install** only installs a basic version of Red Hat Enterprise Linux with only a minimal amount of additional software. This will substantially reduce the chance of the system being affected by a vulnerability. After the system finishes installing and you log in for the first time, you can use the **Yum** package manager to install any additional software you need. For more details on **Minimal install**, see the [Installing the Minimum Amount of Packages Required](#) section of the Red Hat Enterprise Linux 7 Security Guide.

Alternatively, automating the installation with a Kickstart file allows for a much higher degree of control over installed packages. You can specify environments, groups and individual packages in the **%packages** section of the Kickstart file. See [Section 27.3.2, “Package Selection”](#) for instructions on selecting packages to install in a Kickstart file, and [Chapter 27, Kickstart Installations](#) for general information about automating the installation with Kickstart.

Once you have selected an environment and add-ons to be installed, click **Done** to return to the **Installation Summary** screen.

18.14.1. Core Network Services

All Red Hat Enterprise Linux installations include the following network services:

- centralized logging through the **rsyslog** service
- email through SMTP (Simple Mail Transfer Protocol)
- network file sharing through NFS (Network File System)
- remote access through SSH (Secure SHell)
- resource advertising through mDNS (multicast DNS)

Some automated processes on your Red Hat Enterprise Linux system use the email service to send reports and messages to the system administrator. By default, the email, logging, and printing services do not accept connections from other systems.

You can configure your Red Hat Enterprise Linux system after installation to offer email, file sharing, logging, printing, and remote desktop access services. The SSH service is enabled by default. You can also use NFS to access files on other systems without enabling the NFS sharing service.

18.15. INSTALLATION DESTINATION

To select the disks and partition the storage space on which you will install Red Hat Enterprise Linux, select **Installation Destination** in the **Installation Summary** screen. If you are unfamiliar with disk partitions, see [Appendix A, An Introduction to Disk Partitions](#) for more information.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.



IMPORTANT

If you install Red Hat Enterprise Linux in text mode, you can only use the default partitioning schemes described in this section. You cannot add or remove partitions or file systems beyond those that the installation program automatically adds or removes.

INSTALLATION DESTINATION

Done

RED HAT ENTERPRISE LINUX 7.0 INSTALLATION

us

Device Selection

Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.

Local Standard Disks

2.34 GB DASD device 0.0.0200 dasda / 908.62 MB free	2.34 GB DASD device 0.0.0201 dasdb / 0 B free	2.34 GB DASD device 0.0.0202 dasdc / 2.34 GB free	2.34 GB DASD device 0.0.0203 dasdd / 2.34 GB free
--	--	--	--

Disks left unselected here will not be touched.

Specialized & Network Disks

Add a disk...

Disks left unselected here will not be touched.

Other Storage Options

Partitioning

☒ Automatically configure partitioning.
 ☐ I will configure partitioning.

☐ I would like to make additional space available.

Encryption

☐ Encrypt my data. You'll set a passphrase later.

[Full disk summary and bootloader...](#)

3 disks selected; 7.04 GB capacity; 3.25 GB free

Figure 18.14. Storage Space Overview

On this screen, you can see storage devices available locally on your computer. You can also add additional specialized or network devices by clicking the **Add a disk** button. To learn more about these devices see [Section 18.16, "Storage Devices"](#).

**WARNING**

A known issue prevents DASDs configured as HyperPAV aliases to be automatically attached to the system after the installation finishes. These storage devices will be available on this screen during the installation, but will not be immediately accessible after you finish installing and reboot. To attach HyperPAV alias devices, add them manually to the system's `/etc/dasd.conf` configuration file as described in [Section 20.1.3, “Persistently Setting DASDs Online”](#).

If you do not feel comfortable with partitioning your system, leave the default selection of the **Automatically configure partitioning** radio button to let the installation program partition the storage devices for you.

Below the panes for storage devices is a form of additional controls labeled **Other Storage Options**:

- In the **Partitioning** section, you can select how storage devices are partitioned and how volumes are created. You can configure the partitions, manually or allow the installation program to do it automatically.

Automatic partitioning is recommended if you are doing a clean installation on previously unused storage or do not need to keep any data that might be present on the storage. To proceed this way, leave the default selection of the **Automatically configure partitioning** radio button, and the installation program will create the necessary partitions and volumes on the storage space for you.

For automatic partitioning, you can also select the **I would like to make additional space available** check box to choose how to reassign space from other file systems to this installation. After you click **Done**, there are two dialogs that can appear. If you selected automatic partitioning but there is not enough storage space to complete the installation using the recommended partitioning configuration, a dialog will appear:

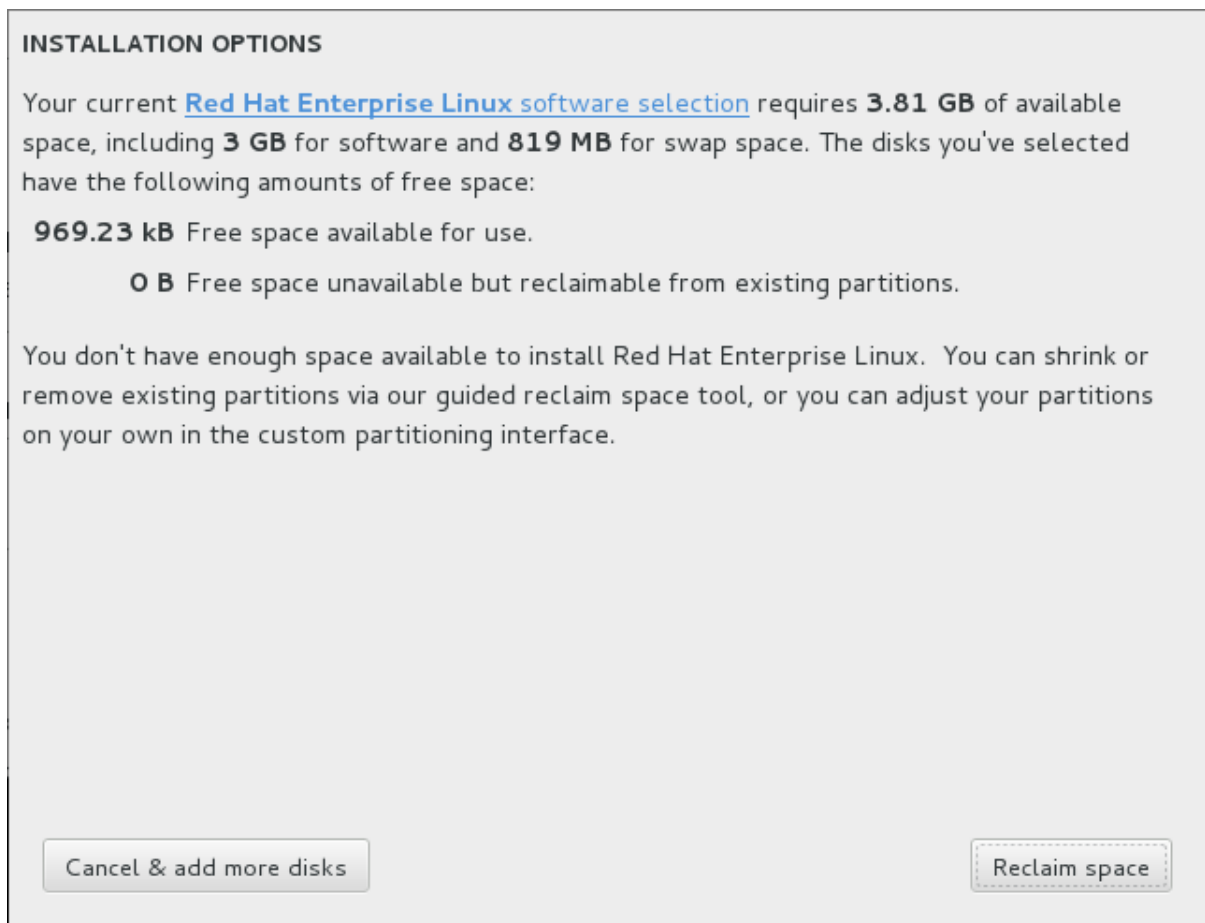


Figure 18.15. Installation Options Dialog with Option to Reclaim Space

You can click to the **Red Hat Enterprise Linux software selection** link. The link will navigate you to the **Software selection** section, where you can change what software you want to install, and free some additional storage space.

Alternatively, you can click **Cancel & add more disks** to return to the **Installation Destination** screen, where it is possible to add more storage devices, or to choose to configure partitioning manually. Click **Reclaim space** to free some storage space from existing file systems. See [Section 18.15.2, "Reclaim Disk Space"](#) for details.

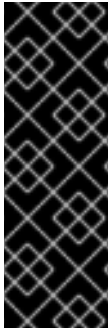
The second dialog appears if you cannot create enough free space. In this case, you have to add more disks on the initial storage screen or quit the installation.

If you select the **I will configure partitioning** radio button for manual setup, you will be brought to the **Manual Partitioning** screen after clicking **Done**. See [Section 18.15.3, "Manual Partitioning"](#) for details.

- In the **Encryption** section, you can select the **Encrypt my data** check box to encrypt all partitions except for the **/boot** partition. See the [Red Hat Enterprise Linux 7 Security Guide](#) for information on encryption.

At the bottom of the screen is the **Full disk summary and bootloader** button for you to configure a disk on which a boot loader will be installed.

Click the **Done** button once you have made your selections to either return to the **Installation Summary** screen or to proceed to the **Manual Partitioning** screen.



IMPORTANT

When you install Red Hat Enterprise Linux on a system with both multipath and non-multipath storage devices, the automatic partitioning layout in the installation program might create volume groups that contain a mix of multipath and non-multipath devices. This defeats the purpose of multipath storage.

We advise that you select only multipath or only non-multipath devices on the **Installation Destination** screen. Alternatively, proceed to manual partitioning.

18.15.1. Encrypt Partitions

If you selected the **Encrypt my data** option, when you click to proceed to the next screen the installation program will prompt you for a passphrase with which to encrypt the partitions on the system.

Partitions are encrypted using the *Linux Unified Key Setup* – see the [Red Hat Enterprise Linux 7 Security Guide](#) for more information.

DISK ENCRYPTION PASSPHRASE

You have chosen to encrypt some of your data. You will need to create a passphrase that you will use to access your data when you start your computer.

Passphrase:

 Strong

Confirm:

 Warning: You won't be able to switch between keyboard layouts (from the default one) when you decrypt your disks after install.

Figure 18.16. Enter Passphrase for an Encrypted Partition

Choose a passphrase and type it into each of the two fields in the dialog box. Note that you need to use the same keyboard layout for setting up this passphrase that you will use to unlock partitions later. Use the language layout icon to ensure the correct layout is selected. You must provide this passphrase every time that the system boots. Press **Tab** while in the **Passphrase** input field to retype it. If the passphrase is too weak, a warning icon appears in the field and you will not be allowed to type in the second field. Hover your mouse cursor over the warning icon to learn how to improve the passphrase.

**WARNING**

If you lose this passphrase, any encrypted partitions and the data on them will become completely inaccessible. There is no way to recover a lost passphrase.

Note that if you perform a Kickstart installation, you can save encryption passphrases and create backup encryption passphrases during installation. See the [Red Hat Enterprise Linux 7 Security Guide](#) for more information about disk encryption.

18.15.2. Reclaim Disk Space

If there is insufficient space to install Red Hat Enterprise Linux on the disks selected in **Installation Destination** and you selected **Reclaim Space** at the **Installation Options** dialog, the **Reclaim Disk Space** dialog appears.

**WARNING**

Unless you select to shrink a partition, reclaiming space on a partition involves deleting all the data on it and you should always verify that any data you need to keep was backed up.

RECLAIM DISK SPACE

You can remove existing filesystems you no longer need to free up space for this installation. Removing a filesystem will permanently delete all of the data it contains.

Disk	Name	Filesystem	Reclaimable Space	Action
▼ 2.34 GB DASD device 0.0.3527	dasda		2.29 GB total	Delete
└─ rhel_rtt7	dasda1	lvmpv	Not resizeable	Delete
└─ Free space			55.59 MB	
▼ 2.34 GB DASD device 0.0.3727	dasdb		2.34 GB total	Preserve
└─ /boot (Red Hat Enterprise Linux Server Linux 7.0 for s390x)	dasdb1	xfs	Not resizeable	Preserve
└─ rhel_rtt7	dasdb2	lvmpv	Not resizeable	Preserve
▼ 2.34 GB DASD device 0.0.3627	dasdc		2.12 GB total	Preserve
└─ rhel_rtt7	dasdc1	lvmpv	Not resizeable	Preserve
└─ Free space			222.23 MB	

3 disks; 6.76 GB reclaimable space (in filesystems)

Total selected space to reclaim: **2.29 GB**

Installation requires a total of **919.82 MB** for system data.

Figure 18.17. Reclaim Disk Space from Existing File Systems

The existing file systems Red Hat Enterprise Linux has detected are listed in a table as part of their respective disks. The **Reclaimable Space** column lists the space that could be reassigned to this installation. The **Action** column lists what action will be taken with the file system to reclaim space.

Beneath the table are four buttons:

- **Preserve** - leaves the file system untouched and no data will be deleted. This is the default action.
- **Delete** - removes the file system entirely. All the space it takes up on the disk will be made available for the installation.
- **Shrink** - recovers free space from the file system and makes it available for this installation. Use the slider to set a new size for the selected partition. Can only be used on resizable partitions where LVM or RAID is not used.
- **Delete all/Preserve all** - this button, located on the right, marks all file systems for deletion by default. Upon clicking, it changes the label and allows you to mark all file systems to be preserved again.

Select a file system or a whole disk in the table with your mouse and click one of the buttons. The label in the **Action** column will change to match your selection and the amount of **Total selected space to reclaim** displayed beneath the table will adjust accordingly. Beneath this value is the amount of space the installation requires based on the packages you have selected to install.

When enough space has been reclaimed for the installation to proceed, the **Reclaim Space** button will become available. Click this button to return to the Installation Summary screen and proceed with the installation.

18.15.3. Manual Partitioning

The **Manual Partitioning** screen is displayed when you click **Done** from Installation Destination if you selected the **I will configure partitioning** option. On this screen you configure your disk partitions and mount points. This defines the file system that Red Hat Enterprise Linux will be installed on.



WARNING

Red Hat recommends that you always back up any data that you have on your systems. For example, if you are upgrading or creating a dual-boot system, you should back up any data you want to keep on your storage devices. Unforeseen circumstances can result in loss of all your data.

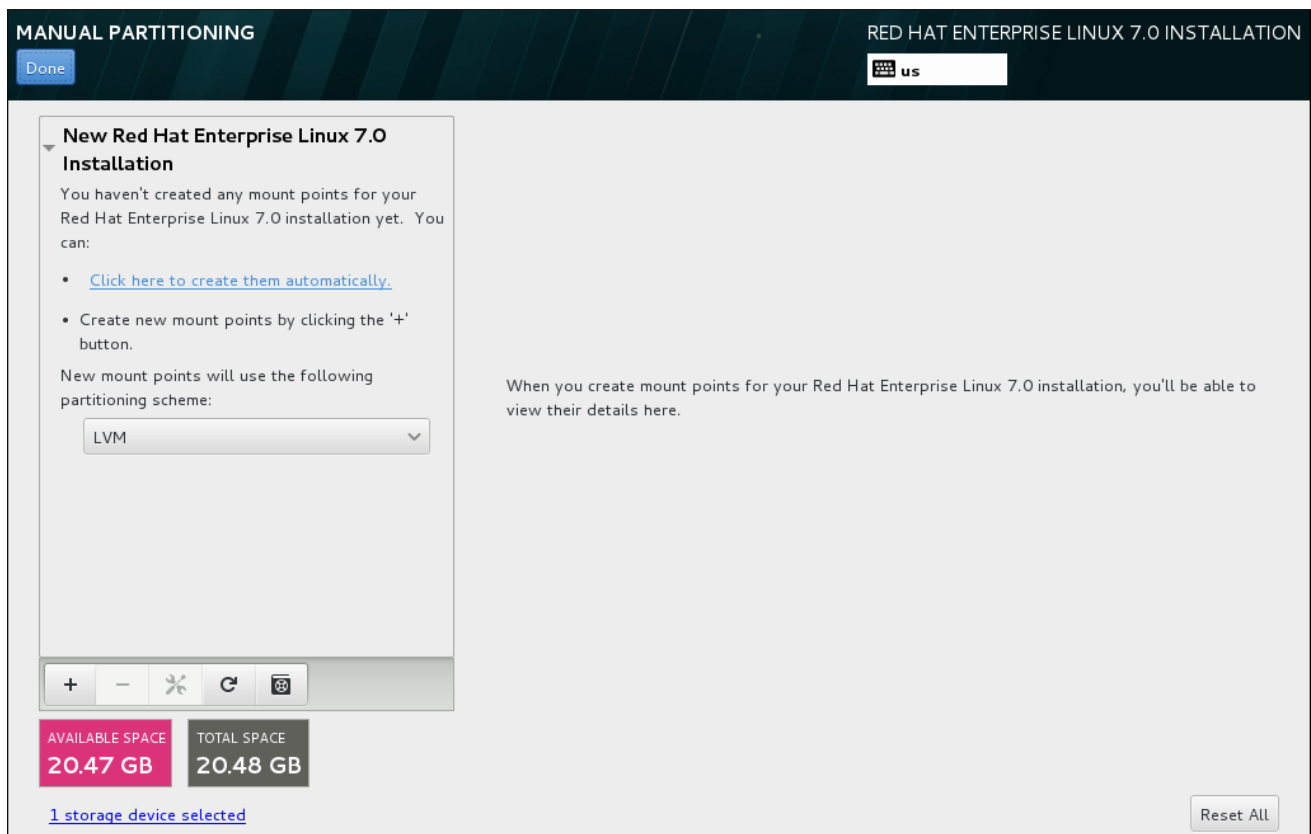


Figure 18.18. The Manual Partitioning Screen

The **Manual Partitioning** screen initially features a single pane on the left for the mount points. The pane is either empty except for information about creating mount points, or it displays existing mount points that the installation program has detected. These mount points are organized by detected operating system installations. Therefore, some file systems might be displayed multiple times if a partition is shared among several installations. The total space and available space on selected storage devices are displayed beneath this pane.

If your system contains existing file systems, ensure that enough space will be available for the installation. Use the **-** button to remove unneeded partitions.



NOTE

For recommendations and additional information about disk partitions, see [Appendix A, *An Introduction to Disk Partitions*](#) and [Section 18.15.3.4, “Recommended Partitioning Scheme”](#). At a bare minimum, you need an appropriately sized root partition, and usually a swap partition appropriate to the amount of RAM you have on your system.

Note which device is associated with **/boot**. The kernel files and boot loader sector will be associated with this device. The first DASD or SCSI LUN will be used, and the device number will be used when re-IPLing the post-installed system.

18.15.3.1. Adding File Systems and Configuring Partitions

An installation of Red Hat Enterprise Linux requires a minimum of one partition but Red Hat recommends using at least the following partitions or volumes: **/**, **/home**, **/boot**, and **swap**. You can also create additional partitions and volumes as you require. See [Section 18.15.3.4, “Recommended Partitioning Scheme”](#) for further details.



NOTE

If you have any specific requirements for some partitions (for example, requiring that a particular partition be on a specific disk) and less specific requirements for other partitions, create the partitions first which have more specific requirements.

Adding a file system is a two-step process. You first create a mount point in a certain partitioning scheme. The mount point appears in the left pane. Next, you can customize it using the options in the right pane, where you can change the mount point, capacity, the device type, file system type, label, and whether to encrypt or reformat the corresponding partition.

If you have no existing file systems and want the installation program to create the required file systems and their mount points for you, select your preferred partitioning scheme from the drop-down menu in the left pane (default for Red Hat Enterprise Linux is LVM), then click the link on top of the pane for creating mount points automatically. This will generate a **/boot** partition, a **/** (root) volume, and a swap volume proportionate to the size of the available storage. These are the recommended file systems for a typical installation, but you can add additional file systems and mount points if you need to.

Alternatively, create individual mount points using the **+** button at the bottom of the pane. The **Add a New Mount Point** dialog then opens. Either select one of the preset paths from the **Mount Point** drop-down menu or type your own; for example, select **/** for the root partition or **/boot** for the boot partition. Then enter the size of the file system in the **Desired Capacity** text field; for example, **2GiB**. If you leave the field empty or if you specify a size bigger than available space, all remaining free space is used instead. After entering these details, click the **Add mount point** button to create the partition.



NOTE

To avoid problems with space allocation, first create small partitions with known fixed sizes, such as **/boot**, and then create the rest of the partitions, letting the installation program allocate the remaining capacity to them.

Similarly, if you have multiple disks that the system is to reside on, they differ in size, and a particular partition must be created on the first disk detected by BIOS, be sure to start by creating such a partition.

For each new mount point you create manually, you can set its partitioning scheme from the drop-down menu located in the left pane. The available options are **Standard Partition**, **BTRFS**, **LVM**, and **LVM Thin Provisioning**. Note that the **/boot** partition will always be located on a standard partition, regardless of the value selected in this menu.

To change on which devices a single non-LVM mount point should be located, select the mount point and click the **Modify...** button in the right pane to open the **Configure Mount Point** dialog. Select one or more devices and click **Select**. After the dialog closes, note that you also need to confirm this setting by clicking the **Update Settings** button on the right side of the **Manual Partitioning** screen.

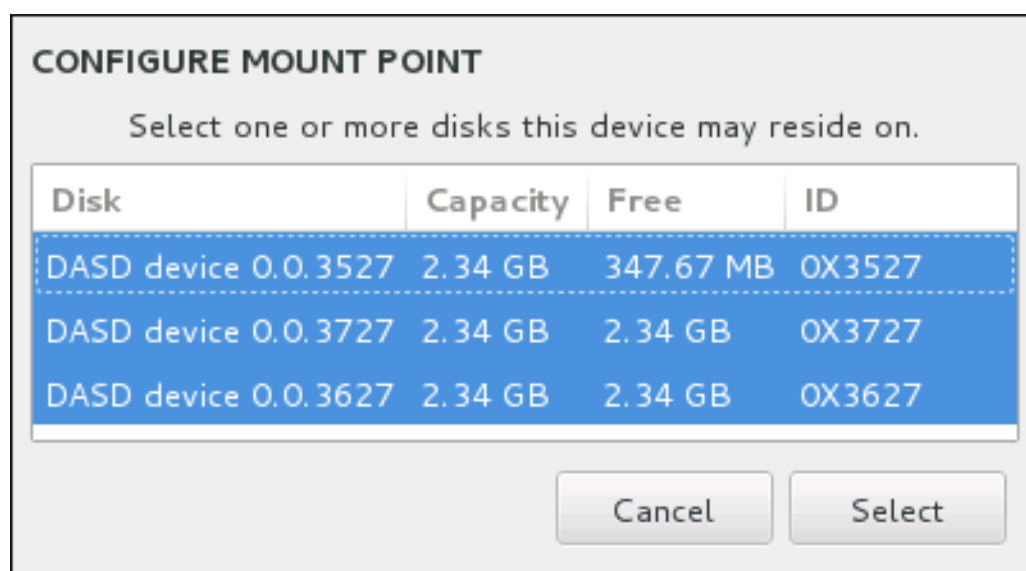


Figure 18.19. Configuring Mount Points

To refresh information about all local disks and partitions on them, click the **Rescan** button (with the circular arrow icon on it) in the toolbar. You only need to do this action after performing advanced partition configuration outside the installation program. Note that if you click the **Rescan Disks** button, all configuration changes you previously made in the installation program will be lost.



Figure 18.20. Rescanning Disks

At the bottom of the screen, a link states how many storage devices have been selected in **Installation Destination** (see [Section 18.15, "Installation Destination"](#)). Clicking on this link opens the **Selected Disks** dialog, where you review the information about the disks.

To customize a partition or a volume, select its mount point in the left pane and the following customizable features then appear to the right:

MANUAL PARTITIONING RED HAT ENTERPRISE LINUX 7.1 INSTALLATION

Done us Help!

New Red Hat Enterprise Linux 7.1 Installation

DATA

SYSTEM

/boot 476 MiB

sda1

/ 4743.34 MiB >

rhel-root

swap 1020 MiB

rhel-swap

rhel-root

Mount Point: /

Device(s): ATA QEMU HARDISK (sda) and 1 other

Desired Capacity: 4743.34 MiB

Device Type: LVM ☐ Encrypt

File System: xfs ☒ Reformat

Volume Group: rhel (0 B free)

Label:

Name: root

Update Settings

Note: The settings you make on this screen will not be applied until you click on the main menu's 'Begin Installation' button.

Reset All

AVAILABLE SPACE 2395 KiB TOTAL SPACE 6246.4 MiB

[2 storage devices selected](#)

Figure 18.21. Customizing Partitions

- **Mount Point** – enter the file system's mount point. For example, if a file system should be the root file system, enter `/`; enter `/boot` for the `/boot` file system, and so on. For a swap file system, the mount point should not be set – setting the file system type to **swap** is sufficient.
- **Desired Capacity** – enter the desired size of the file system. You can use common size units such as KiB or GiB. The default is MiB if no other unit is specified.
- **Device type** – choose one of these types: **Standard Partition**, **LVM**, **RAID**, **LVM Thin Provisioning**, or **BTRFS**. Check the adjacent **Encrypt** box to encrypt the partition or volume. You will be prompted to set a password later. **RAID** is only available if two or more disks are selected for partitioning, and if you choose this type, you can also set the **RAID Level**. Similarly, if you select **LVM**, you can specify the **Volume Group**.
- **File system** – in the drop-down menu, select the appropriate file system type for this partition or volume. Check the adjacent **Reformat** box to format an existing partition, or leave it unchecked to retain your data. Note that newly created partitions and volumes must be reformatted, and the check box cannot be unchecked in this case.
- **Label** – assign a label to the partition. Labels are used for you to easily recognize and address individual partitions.
- **Name** – assign a name to an LVM or Btrfs volume. Note that standard partitions are named automatically when they are created and their name cannot be edited, such as `/home` being assigned the name `sda1`.

See [Section 18.15.3.1.1, “File System Types”](#) for more information about file system and device types.

Click the **Update Settings** button to save your changes and select another partition to customize. Note that the changes will not be applied until you actually start the installation from the Installation summary page. Click the **Reset All** button to discard all changes to all partitions and start over.

When all file systems and mount points have been created and customized, click the **Done** button. If you chose to encrypt any file system, you will now be prompted to create a passphrase. Then, a dialog appears, showing a summary of all actions related to storage that the installation program will take. This includes creating, resizing, or deleting partitions and file systems. You can review all the changes and click **Cancel & Return to Custom Partitioning** to go back. To confirm your changes, click **Accept Changes** to return to the Installation Summary page. To partition additional devices, select them in the **Installation Destination** screen, return to the **Manual Partitioning** screen, repeat the steps outlined in this section for the additional devices.



IMPORTANT

If **/usr** or **/var** is partitioned separately from the rest of the root volume, the boot process becomes much more complex because these directories contain components critical to it. In some situations, such as when these directories are placed on an iSCSI drive or an FCoE location, the system can either be unable to boot, or hang with a **Device is busy** error when powering off or rebooting.

This limitation only applies to **/usr** or **/var**, not to directories below them. For example, a separate partition for **/var/www** will work without issues.

18.15.3.1.1. File System Types

Red Hat Enterprise Linux allows you to create different device types and file systems. The following is a brief description of the different device types and file systems available, and how they can be used.

Device Types

- **standard partition** - A standard partition can contain a file system or swap space, or it can provide a container for software RAID or an LVM physical volume.
- **logical volume (LVM)** - Creating an LVM partition automatically generates an LVM logical volume. LVM can improve performance when using physical disks. For information on how to create a logical volume, see [Section 18.15.3.3, "Create LVM Logical Volume"](#). For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.
- **LVM thin provisioning** - Using thin provisioning, you can manage a storage pool of free space, known as a thin pool, which can be allocated to an arbitrary number of devices when needed by applications. The thin pool can be expanded dynamically when needed for cost-effective allocation of storage space. For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide.



NOTE

The installer will automatically reserve 20% of any requested space for an LVM thin pool logical volume in the volume group containing it. This is a safety measure to ensure that you can extend either the metadata volume or the data volume of your thinly provisioned logical volume.

- **software RAID** - Creating two or more software RAID partitions allows you to create a RAID

device. One RAID partition is assigned to each disk on the system. To create a RAID device, see [Section 18.15.3.2, “Create Software RAID”](#). For more information regarding RAID, see the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

File Systems

- **xfs** – XFS is a highly scalable, high-performance file system that supports file systems up to 16 EiB (approximately 16 billion GiB), files up to 8 EiB (approximately 8 billion GiB), and directory structures containing tens of millions of entries. XFS supports metadata journaling, which facilitates quicker crash recovery. The XFS file system can also be defragmented and resized while mounted and active. This file system is selected by default and is highly recommended. For information on how to translate common commands from previously used ext4 file system to XFS, see [Appendix F, Reference Table for ext4 and XFS Commands](#).

The maximum supported size of an XFS file system in Red Hat Enterprise Linux is currently 500 TiB.

- **ext4** – The ext4 file system is based on the ext3 file system and features a number of improvements. These include support for larger file systems and larger files, faster and more efficient allocation of disk space, no limit on the number of subdirectories within a directory, faster file system checking, and more robust journaling.

The maximum supported size of an ext4 file system in Red Hat Enterprise Linux is currently 50 TiB.

- **ext3** – The ext3 file system is based on the ext2 file system and has one main advantage – journaling. Using a journaling file system reduces time spent recovering a file system after a crash as there is no need to check the file system for metadata consistency by running the **fsck** utility every time a crash occurs.
- **ext2** – An ext2 file system supports standard Unix file types, including regular files, directories, or symbolic links. It provides the ability to assign long file names, up to 255 characters.
- **vfat** – The VFAT file system is a Linux file system that is compatible with Microsoft Windows long file names on the FAT file system.
- **swap** – Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.

Each file system has different size limits for the file system itself as well as individual files contained within. For a list of maximum supported file and file system sizes, see the Red Hat Enterprise Linux technology capabilities and limits page, available on the Customer Portal at <https://access.redhat.com/site/articles/rhel-limits>.

18.15.3.2. Create Software RAID



NOTE

On IBM Z, the storage subsystem uses RAID transparently. There is no need to set up a software RAID manually.

Redundant arrays of independent disks (RAIDs) are constructed from multiple storage devices that are arranged to provide increased performance and, in some configurations, greater fault tolerance. See below for a description of different kinds of RAID.

A RAID device is created in one step and disks are added or removed as necessary. One RAID partition per physical disk is allowed for each device, so the number of disks available to the installation program determines which levels of RAID device are available to you. For example, if your system has two hard drives, the installation program will not allow you to create a RAID10 device, which requires 4 separate partitions.

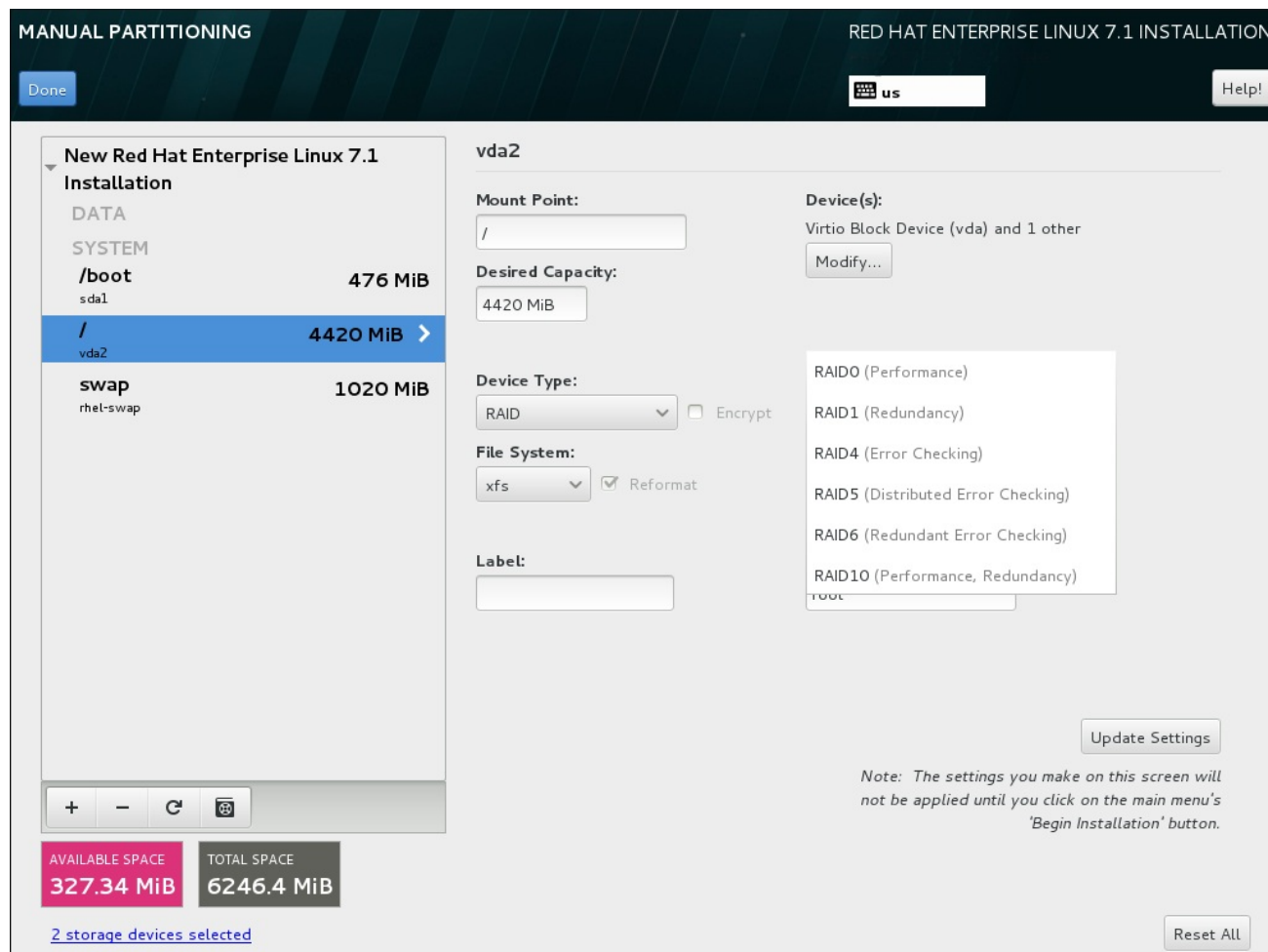


Figure 18.22. Creating a Software RAID Partition - the **Device Type** Menu Expanded

RAID configuration options are only visible if you have selected two or more disks for installation. At least two disks are required to create a RAID device.

To create a RAID device:

1. Create a mount point as described in [Section 18.15.3.1, “Adding File Systems and Configuring Partitions”](#). By configuring this mount point, you configure the RAID device.
2. Keeping the partition selected in the left pane, select the configuration button below the pane to open the **Configure Mount Point** dialog. Select which disks will be included in the RAID device and click **Select**.
3. Click the **Device Type** drop-down menu and select **RAID**.
4. Click the **File System** drop-down menu and select your preferred file system type (see [Section 8.14.4.1.1, “File System Types”](#)).
5. Click the **RAID Level** drop-down menu and select your preferred level of RAID.

The available RAID levels are:

RAID0 - Optimized performance (stripe)

Distributes data across multiple disks. Level 0 RAID offers increased performance over standard partitions, and can be used to pool the storage of multiple disks into one large virtual device. Note that Level 0 RAID offers no redundancy, and that the failure of one device in the array destroys data in the entire array. RAID 0 requires at least two RAID partitions.

RAID1 - Redundancy (mirror)

Mirrors all data on one disk onto one or more other disks. Additional devices in the array provide increasing levels of redundancy. RAID 1 requires at least two RAID partitions.

RAID4 - Error detection (parity)

Distributes data across multiple disks, and uses one disk in the array to store parity information that safeguards the array in case any disk within the array fails. Because all parity information is stored on one disk, access to this disk creates a bottleneck in the performance of the array. RAID 4 requires at least three RAID partitions.

RAID5 - Distributed error detection

Distributes data *and* parity information across multiple disks. Level 5 RAID therefore offers the performance advantages of distributing data across multiple disks, but does not share the performance bottleneck of level 4 RAID because the parity information is also distributed through the array. RAID 5 requires at least three RAID partitions.

RAID6 - Redundant

Level 6 RAID is similar to level 5 RAID, but instead of storing only one set of parity data, they store two sets. RAID 6 requires at least four RAID partitions.

RAID10 - Redundancy (mirror) and Optimized performance (stripe)

Level 10 RAID is a *nested RAID* or *hybrid RAID*. They are constructed by distributing data over mirrored sets of disks. For example, a level 10 RAID array constructed from four RAID partitions consists of two mirrored pairs of striped partitions. RAID 10 requires at least four RAID partitions.

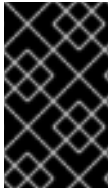
6. Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.

If fewer disks are included than the specified RAID level requires, a message will be displayed at the bottom of the window, informing you how many disks are actually required for your selected configuration.

18.15.3.3. Create LVM Logical Volume

Logical Volume Management (LVM) presents a simple logical view of underlying physical storage space, such as hard drives or LUNs. Partitions on physical storage are represented as *physical volumes* that can be grouped together into *volume groups*. Each volume group can be divided into multiple *logical volumes*, each of which is analogous to a standard disk partition. Therefore, LVM logical volumes function as partitions that can span multiple physical disks.

To learn more about LVM, see [Appendix D, Understanding LVM](#) or read the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide. Note that LVM configuration is only available in the graphical installation program.



IMPORTANT

During text-mode installation, LVM configuration is not available. If you need to create an LVM configuration from scratch, press **Ctrl+Alt+F2** to use a different virtual console, and run the **lvm** command. To return to the text-mode installation, press **Ctrl+Alt+F1**.

MANUAL PARTITIONING RED HAT ENTERPRISE LINUX 7.1 INSTALLATION

[Done](#) [Help!](#)

New Red Hat Enterprise Linux 7.1 Installation

DATA

SYSTEM

/boot 476 MiB
sda1

/var 320.34 MiB [▶](#)
rhel-var

/ 4420 MiB
rhel-root

swap 1020 MiB
rhel-swap

rhel-var

Mount Point: /var

Device(s): ATA QEMU HARDDISK (sda) and 1 other
[Modify...](#)

Desired Capacity: 320.34 MiB

Device Type: LVM ☐ Encrypt

File System: xfs ☒ Reformat

Volume Group rhel (0 B free) [Modify...](#)

Label:

Name: var

[Update Settings](#)

Note: The settings you make on this screen will not be applied until you click on the main menu's 'Begin Installation' button.

AVAILABLE SPACE 2395 KiB **TOTAL SPACE** 6246.4 MiB

[2 storage devices selected](#)

[Reset All](#)

Figure 18.23. Configuring a Logical Volume

To create a logical volume and add it to a new or existing volume group:

1. Create a mount point for the LVM volume as described in [Section 18.15.3.1, “Adding File Systems and Configuring Partitions”](#).
2. Click the **Device Type** drop-down menu and select **LVM**. The **Volume Group** drop-down menu appears and displays the newly-created volume group name.
3. Optionally, either click the menu and select **Create a new volume group** or click **Modify** to configure the newly-created volume group, if you need to. Both the **Create a new volume group** option and the **Modify** button lead to the **Configure Volume Group** dialog, where you can rename the logical volume group and select which disks will be included.



NOTE

The configuration dialog does not allow you to specify the size of the volume group's physical extents. The size will always be set to the default value of 4 MiB. If you want to create a volume group with different physical extents, create it manually by switching to an interactive shell and using the **vgcreate** command, or use a Kickstart file with the **volgroup --pesize=size** command.

CONFIGURE VOLUME GROUP

Please create a name for this volume group and select at least one disk below.

Name:

Disk	Capacity	Free	ID
DASD device 0.0.3527	2.34 GB	0 B	0X3527
DASD device 0.0.3727	2.34 GB	0 B	0X3727
DASD device 0.0.3627	2.34 GB	0 B	0X3627

RAID Level:
☐ Encrypt

Size policy:

Figure 18.24. Customizing an LVM Volume Group

The available RAID levels are the same as with actual RAID devices. See [Section 18.15.3.2, “Create Software RAID”](#) for more information. You can also mark the volume group for encryption and set the size policy for it. The available policy options are:

- **Automatic** – the size of the volume group is set automatically so that it is just large enough to contain the configured logical volumes. This is optimal if you do not need free space within the volume group.
- **As large as possible** – the volume group is created with maximum size, regardless of the size of the configured logical volumes it contains. This is optimal if you plan to keep most of your data on LVM and later need to increase the size of some existing logical volumes, or if you need to create additional logical volumes within this group.
- **Fixed** – with this option, you can set an exact size of the volume group. Any configured logical volumes must then fit within this fixed size. This is useful if you know exactly how large you would like the volume group to be.

Click **Save** when the group is configured.

- Click **Update Settings** to save your changes, and either continue with another partition or click **Done** to return to the **Installation Summary** screen.

**WARNING**

Placing the **/boot** partition on an LVM volume is not supported.

18.15.3.4. Recommended Partitioning Scheme

Configuring efficient swap space for Linux on IBM Z is a complex task. It very much depends on the specific environment and should be tuned to the actual system load.

Consult the following resources for more information and to guide your decision:

- 'Chapter 7. Linux Swapping' in the IBM Redbooks publication *Linux on IBM System z: Performance Measurement and Tuning* [IBM Form Number SG24-6926-01], [ISBN 0738485586], available from <http://www.redbooks.ibm.com/abstracts/sg246926.html>
- *Linux Performance when running under VM*, available from <http://www.vm.ibm.com/perf/tips/linuxper.html>

18.16. STORAGE DEVICES

You can install Red Hat Enterprise Linux on a large variety of storage devices. You can see basic, locally accessible, storage devices in the **Installation Destination** page, as described in [Section 18.15, "Installation Destination"](#). To add a specialized storage device, click the **Add a disk** button in the **Specialized & Network Disks** section of the screen.

Basic storage devices directly connected to the local system, such as hard disk drives and solid-state drives, are seen in the **Local Standard Disks** section of the screen. On IBM Z, this contains activated *Direct Access Storage Devices* (DASDs).

**WARNING**

A known issue prevents DASDs configured as HyperPAV aliases to be automatically attached to the system after the installation finishes. These storage devices will be available on this screen during the installation, but will not be immediately accessible after you finish installing and reboot. To attach HyperPAV alias devices, add them manually to the system's **/etc/dasd.conf** configuration file as described in [Section 20.1.3, "Persistently Setting DASDs Online"](#).

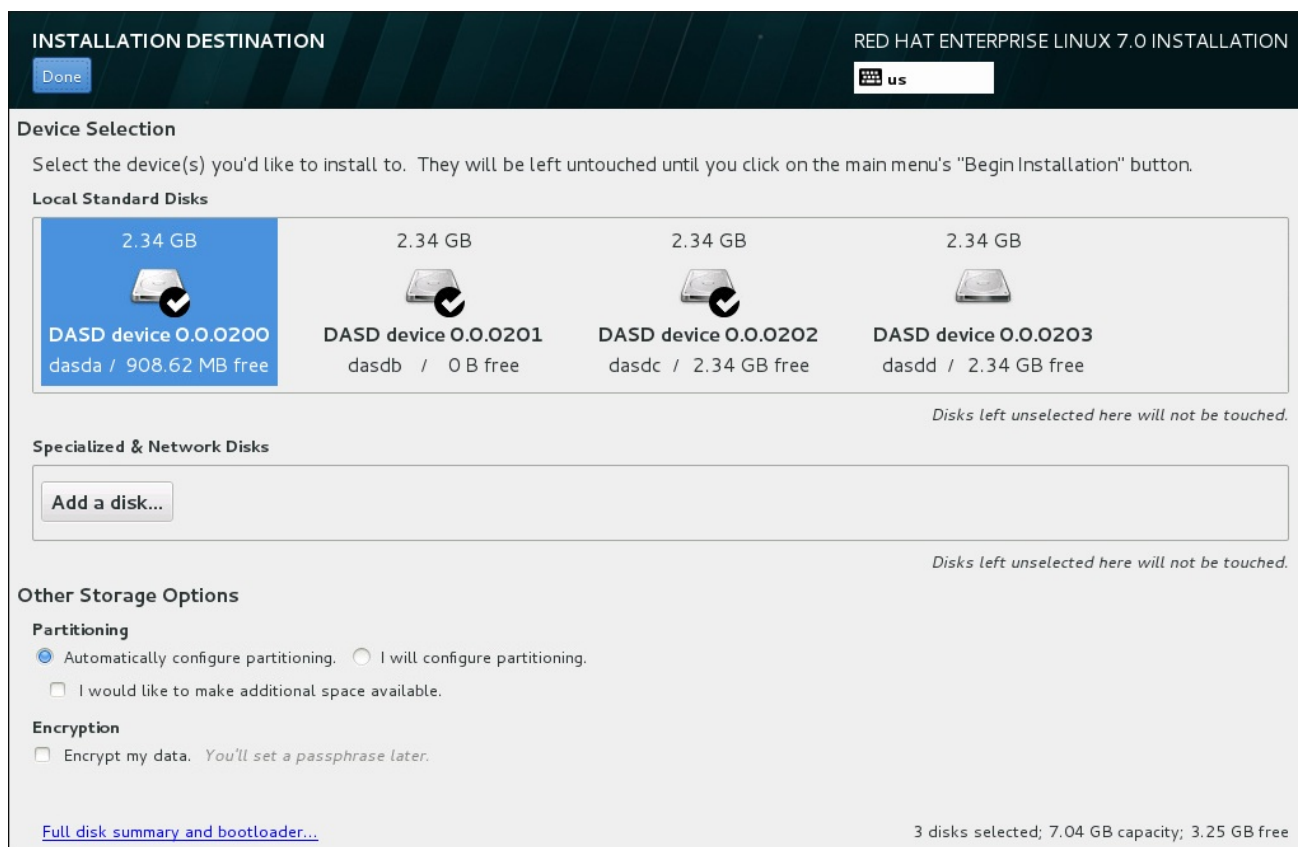


Figure 18.25. Storage Space Overview

18.16.1. The Storage Devices Selection Screen

The storage device selection screen displays all storage devices to which the **Anaconda** installation program has access.

The devices are grouped under the following tabs:

Multipath Devices

Storage devices accessible through more than one path, such as through multiple SCSI controllers or Fiber Channel ports on the same system.



IMPORTANT

The installation program only detects multipath storage devices with serial numbers that are 16 or 32 characters long.

Other SAN Devices

Any other devices available on a Storage Area Network (SAN) such as FCP LUNs attached over one single path.

Firmware RAID

Storage devices attached to a firmware RAID controller. This tab does not apply to IBM Z.

System z Devices

This tab contains storage devices, or Logical Units (LUNs), attached through the zSeries Linux FCP (Fiber Channel Protocol) driver.

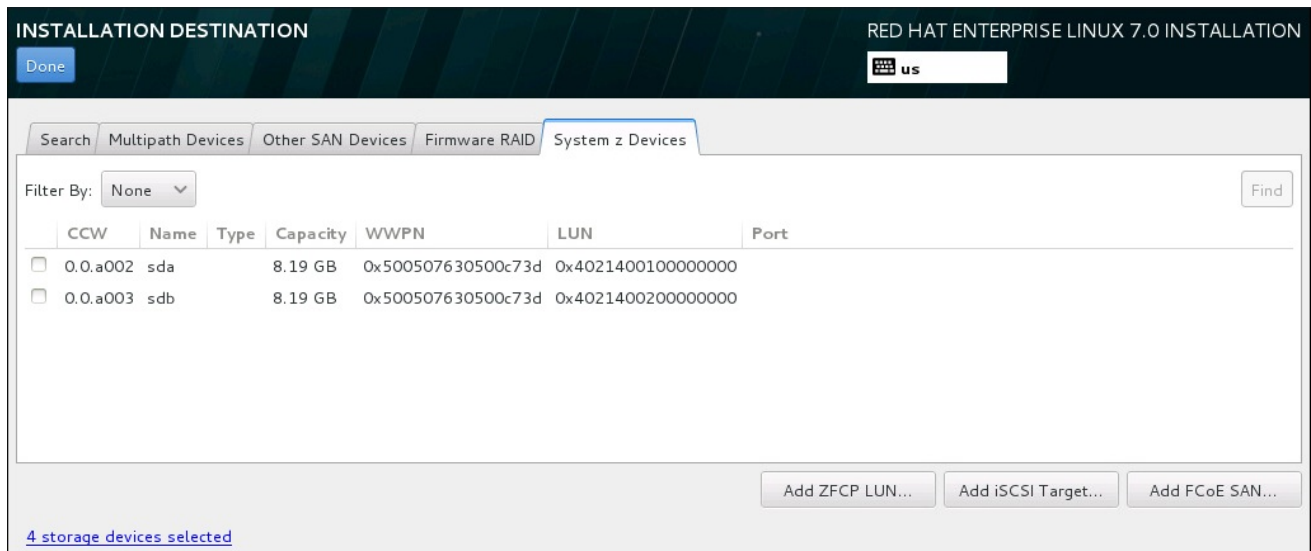


Figure 18.26. Tabbed Overview of Specialized Storage Devices

A set of buttons is available in the bottom right corner of the screen. Use these buttons to add additional storage devices. The available buttons are:

- **Add ZFCP LUN** - press this button to add a zFCP storage device, and continue with [Section 18.16.1.2.3, "FCP Devices"](#)
- **Add DASD** - press this to add additional DASD devices, and continue with [Section 18.16.1.2.2, "DASD storage devices"](#)
- **Add iSCSI Target** - use to attach iSCSI devices; continue with [Section 18.16.1.2.1, "Configuring iSCSI Parameters"](#)
- **Add FCoE SAN** - use to configure a Fibre Channel Over Internet storage device; continue with [Section 18.16.1.2.4, "Configure FCoE Parameters"](#)

The overview page also contains the **Search** tab that allows you to filter storage devices either by their *World Wide Identifier* (WWID) or by the port, target, or *logical unit number* (LUN) at which they are accessed.

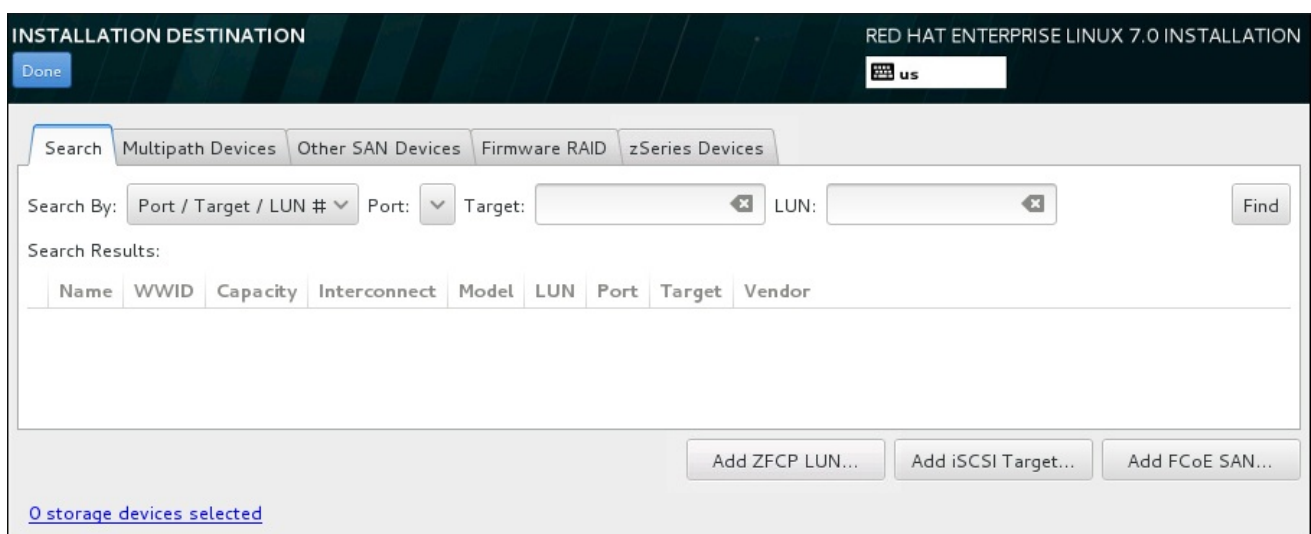


Figure 18.27. The Storage Devices Search Tab

The Search tab contains the **Search By** drop-down menu to select searching by port, target, LUN, or WWID. Searching by WWID or LUN requires additional values in the corresponding input text fields. Click the **Find** button to start the search.

Each device is presented on a separate row, with a check box to its left. Click the check box to make the device available during the installation process. Later in the installation process, you can choose to install Red Hat Enterprise Linux onto any of the devices selected here, and can choose to automatically mount any of the other devices selected here as part of the installed system.

Note that the devices that you select here are not automatically erased by the installation process. Selecting a device on this screen does not, in itself, place data stored on the device at risk. Also note that any devices that you do not select here to form part of the installed system can be added to the system after installation by modifying the `/etc/fstab` file.

When you have selected the storage devices to make available during installation, click **Done** to return to the Installation Destination screen.

18.16.1.1. DASD Low-level Formatting

For installation on a DASD, formatting on a low level in the Compatible Disk Layout (CDL) format is recommended. However, it is also possible to use a FBA DASD formatted in the CMS format. When you select DASDs in the **Installation Destination** screen and click **Done**, the installation program detects any unformatted or incompatibly formatted disks, and the following dialog appears:

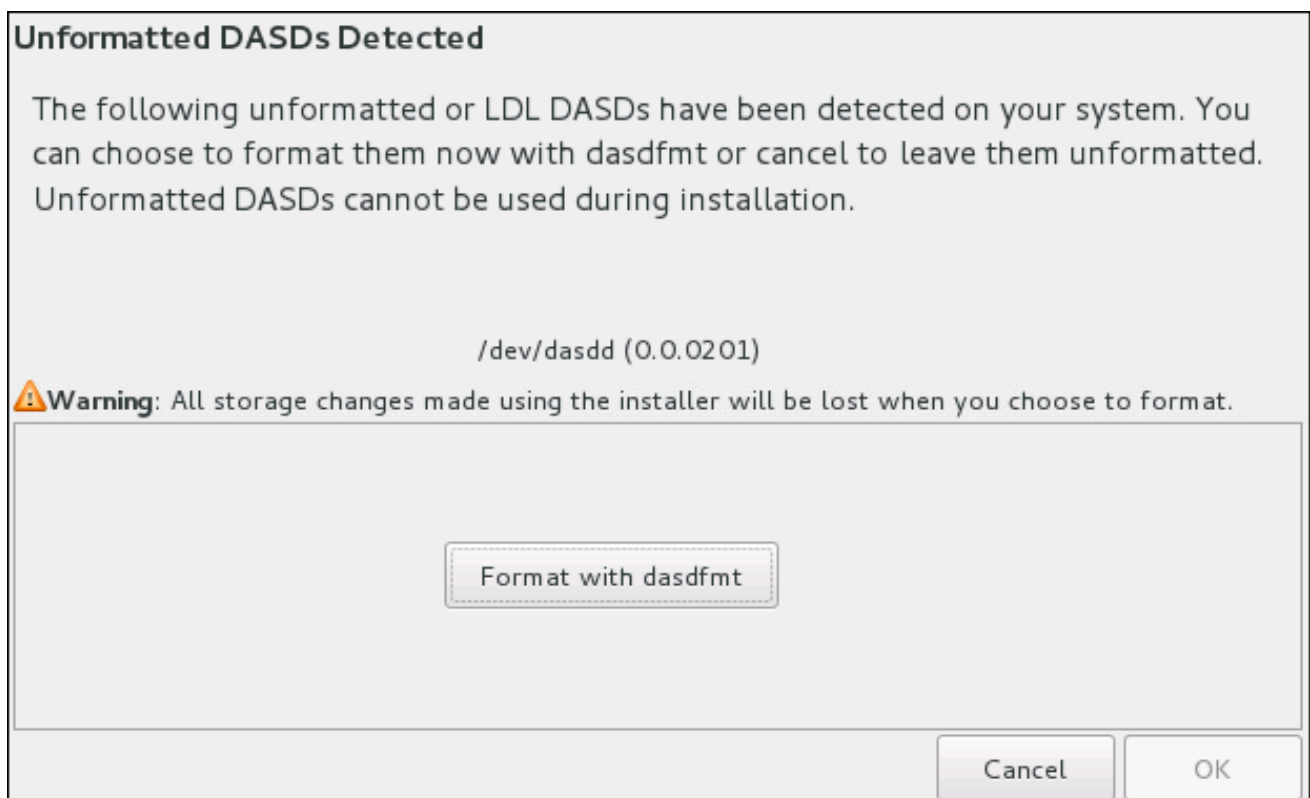


Figure 18.28. Dialog for Formatting DASD Devices

In the dialog, you can click **Cancel** to return to the **Installation Destination** screen and edit disk selection. If the selection is correct, click the **Format with dasdfmt** button to launch the `dasdfmt` utility on all unformatted DASDs.

When the formatting process is complete, clicking the **OK** button brings you back to the **Installation Destination** screen where the list of DASDs will be refreshed. You then need to re-select your disks for the installation to proceed.

To automatically allow low-level formatting of unformatted online DASDs, specify the Kickstart command **zerombr**. See [zerombr \(optional\)](#) for more details.

When installing Red Hat Enterprise Linux on IBM Z with the FBA DASD CMS disk layout as the target, only three partitions are allowed. The installer first creates an **msdos** partition table on the target DASD. The table does not allow more than three partitions on one DASD device. You can create the partitions manually or use the **autopart --nohome** Kickstart option, which ensures that the installer does not create a separate **/home/** partition. The absence of the **home** directory keeps the number of partitions at three.

18.16.1.2. Advanced Storage Options

To use an advanced storage device, you can configure an *iSCSI* (SCSI over TCP/IP) target or *zFCP* (zSeries Fibre Channel Protocol) *LUN* (logical unit) by clicking the appropriate button in the lower right corner of the Installation Destination screen. See [Appendix B, iSCSI Disks](#) for an introduction to iSCSI.

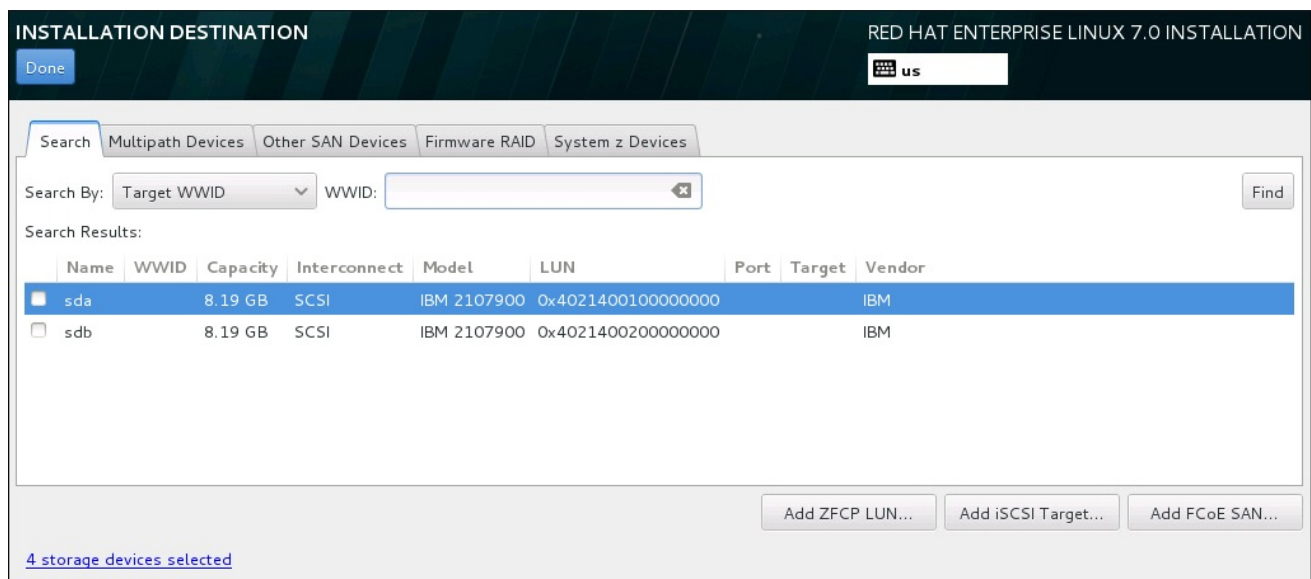


Figure 18.29. Advanced Storage Options

18.16.1.2.1. Configuring iSCSI Parameters

When you click the **Add iSCSI target...** button, the **Add iSCSI Storage Target** dialog appears.

ADD iSCSI STORAGE TARGET

To use iSCSI disks, you must provide the address of your iSCSI target and the iSCSI initiator name you've configured for your host.

Target IP Address:

iSCSI Initiator Name:

 Example: iqn.2012-09.com.example:diskarrays-sn-a8675309

Discovery Authentication Type: CHAP pair and a reverse pair ▾

CHAP Username:

CHAP Password:

Reverse CHAP Username:

Reverse CHAP Password:

☐ Bind targets to network interfaces

Figure 18.30. The iSCSI Discovery Details Dialog

To use iSCSI storage devices for the installation, **Anaconda** must be able to *discover* them as iSCSI targets and be able to create an iSCSI *session* to access them. Each of these steps might require a user name and password for *CHAP* (Challenge Handshake Authentication Protocol) authentication. Additionally, you can configure an iSCSI target to authenticate the iSCSI initiator on the system to which the target is attached (*reverse CHAP*), both for discovery and for the session. Used together, CHAP and reverse CHAP are called *mutual CHAP* or *two-way CHAP*. Mutual CHAP provides the greatest level of security for iSCSI connections, particularly if the user name and password are different for CHAP authentication and reverse CHAP authentication.



NOTE

Repeat the iSCSI discovery and iSCSI login steps as many times as necessary to add all required iSCSI storage. However, you cannot change the name of the iSCSI initiator after you attempt discovery for the first time. To change the iSCSI initiator name, you must restart the installation.

Procedure 18.1. iSCSI Discovery and Starting an iSCSI Session

Use the **Add iSCSI Storage Target** dialog to provide **Anaconda** with the information necessary to discover the iSCSI target.

1. Enter the IP address of the iSCSI target in the **Target IP Address** field.
2. Provide a name in the **iSCSI Initiator Name** field for the iSCSI initiator in *iSCSI qualified name* (IQN) format. A valid IQN entry contains:
 - the string **iqn.** (note the period)
 - a date code that specifies the year and month in which your organization's Internet domain or subdomain name was registered, represented as four digits for the year, a dash, and two digits for the month, followed by a period. For example, represent September 2010 as **2010-09.**
 - your organization's Internet domain or subdomain name, presented in reverse order with the top-level domain first. For example, represent the subdomain **storage.example.com** as **com.example.storage**
 - a colon followed by a string that uniquely identifies this particular iSCSI initiator within your domain or subdomain. For example, **:diskarrays-sn-a8675309**

A complete IQN can therefore look as follows: **iqn.2010-09.storage.example.com:diskarrays-sn-a8675309**. **Anaconda** prepopulates the **iSCSI Initiator Name** field with a name in this format to help you with the structure.

For more information on IQNs, see 3.2.6. *iSCSI Names* in *RFC 3720 - Internet Small Computer Systems Interface (iSCSI)* available from <http://tools.ietf.org/html/rfc3720#section-3.2.6> and 1. *iSCSI Names and Addresses* in *RFC 3721 - Internet Small Computer Systems Interface (iSCSI) Naming and Discovery* available from <http://tools.ietf.org/html/rfc3721#section-1>.

3. Use the **Discovery Authentication Type** drop-down menu to specify the type of authentication to use for iSCSI discovery. The following options are available:
 - no credentials
 - CHAP pair
 - CHAP pair and a reverse pair
4.
 - If you selected **CHAP pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** fields.
 - If you selected **CHAP pair and a reverse pair** as the authentication type, provide the user name and password for the iSCSI target in the **CHAP Username** and **CHAP Password** field and the user name and password for the iSCSI initiator in the **Reverse CHAP Username** and **Reverse CHAP Password** fields.
5. Optionally check the box labeled **Bind targets to network interfaces**.
6. Click the **Start Discovery** button. **Anaconda** attempts to discover an iSCSI target based on the information that you provided. If discovery succeeds, the dialog displays a list of all iSCSI nodes discovered on the target.
7. Each node is presented with a check box beside it. Click the check boxes to select the nodes to use for installation.

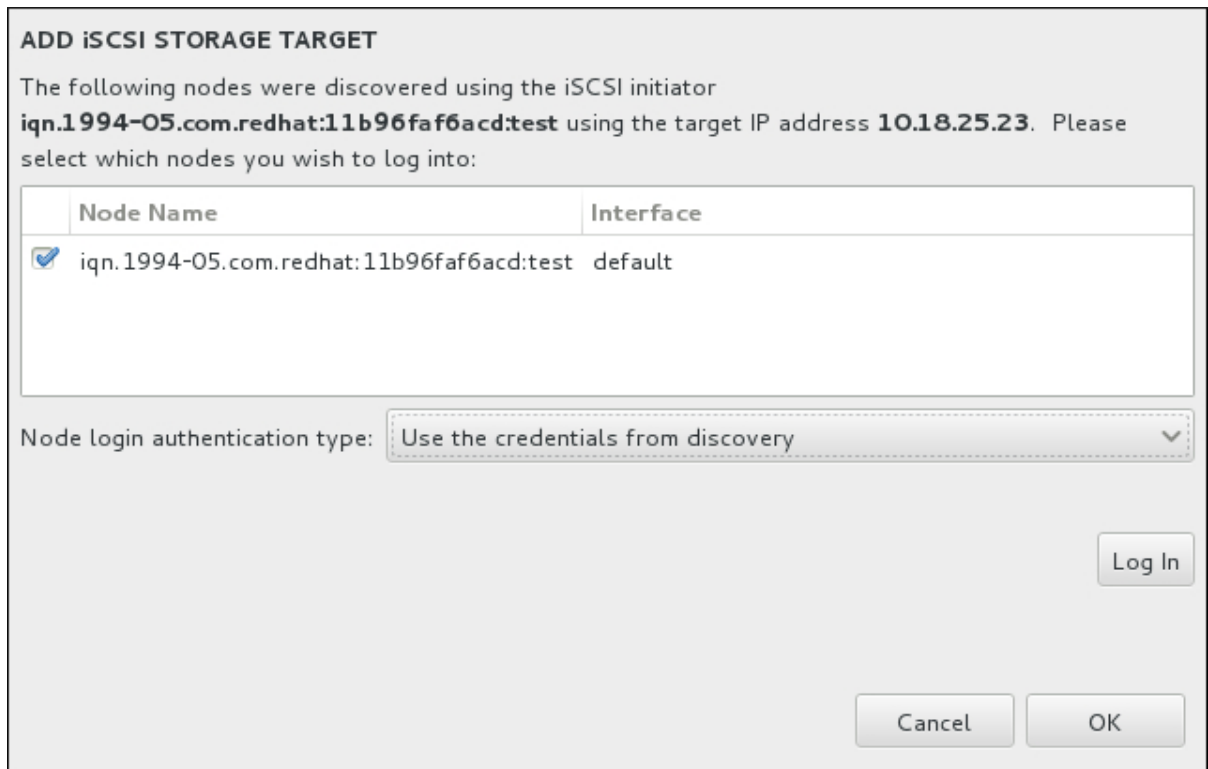


Figure 18.31. The Dialog of Discovered iSCSI Nodes

8. The **Node login authentication type** menu provides the same options as the **Discovery Authentication Type** menu described in step 3. However, if you needed credentials for discovery authentication, it is typical to use the same credentials to log into a discovered node. To do that, use the additional **Use the credentials from discovery** option from the menu. When the proper credentials have been provided, the **Log In** button becomes available.
9. Click **Log In** to initiate an iSCSI session.

18.16.1.2.2. DASD storage devices

After clicking the **Add DASD** button at the **Installation Destination** screen, a dialog appears for you to add a DASD (Direct Access Storage Device) storage device. This dialog allows you to attach additional DASDs which were not detected when the installation started.



Figure 18.32. Add DASD Storage Target

The **Add DASD Storage Target** dialog prompts you to specify a device number, such as **0.0.0204**. Enter the device number of the DASD you want to attach, and click **Start Discovery**.

If a DASD with the specified device number is found, and if it is not already attached, the dialog window will close and the newly discovered drives will appear in the list of drives in [Section 18.16.1, “The Storage Devices Selection Screen”](#). There, you can use the check boxes on the left side of the screen to select which of the drives should be made available; after you do so, press **Done** in the top left corner to return to [Section 18.15, “Installation Destination”](#). The new DASDs will then be available for selection (marked as **DASD device 0.0.xxxx**) in the **Local Standard Disks** section of the screen.

If you entered an invalid device number, or if the DASD with the specified device number is already attached to the system, an error message will appear within the dialog window, explaining the error and prompting you to try again with a different device number.

18.16.1.2.3. FCP Devices

When you click the **Add ZFCP LUN** button, a dialog appears for you to add a FCP (Fibre Channel Protocol) storage device.

FCP devices enable IBM Z to use SCSI devices rather than, or in addition to, Direct Access Storage Device (DASD) devices. FCP devices provide a switched fabric topology that enables IBM Z systems to use SCSI LUNs as disk devices in addition to traditional DASD devices.

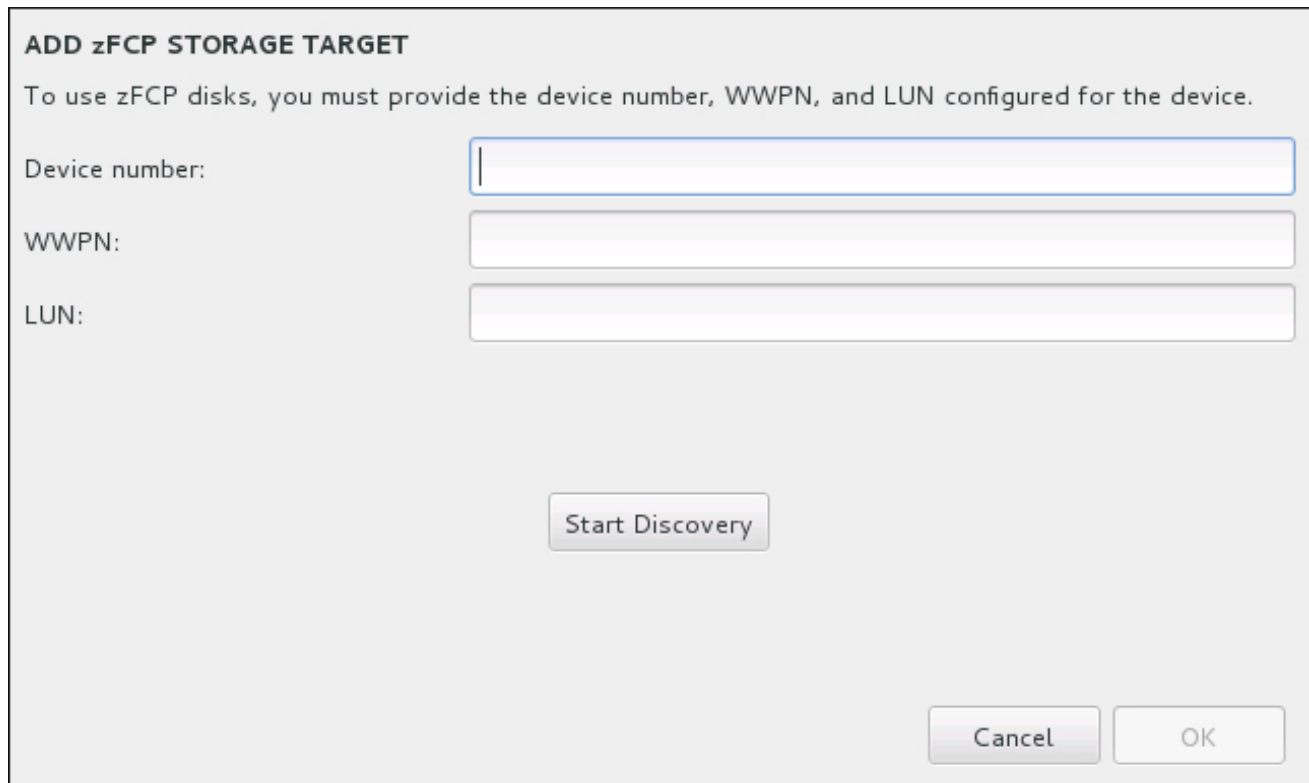
IBM Z requires that any FCP device is entered manually for the installation program to activate FCP LUNs. This can be done either in **Anaconda** interactively, or specified as a unique parameter entry in the parameter or CMS configuration file. The values entered here are unique to each site in which they are set up.

Notes

- Interactive creation of an FCP device is only possible in graphical mode. It is not possible to interactively configure an FCP device in a text mode installation.

- Use only lower-case letters in hex values. If you enter an incorrect value and hit the **Start discovery** button, the installation program will display a warning and allow you to edit the configuration information and retry the discovery attempt.
- For more information on these values, consult the hardware documentation and check with the system administrator who set up the network for this system.

To configure a Fiber Channel Protocol SCSI device, fill in the 16-bit device number, 64-bit World Wide Port Number (WWPN), and 64-bit FCP LUN identifier. Click the **Start Discovery** button to connect to the FCP device using this information.



ADD zFCP STORAGE TARGET

To use zFCP disks, you must provide the device number, WWPN, and LUN configured for the device.

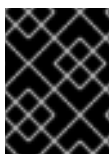
Device number:

WWPN:

LUN:

Figure 18.33. Add FCP Device

The newly added devices are displayed in the **System z Devices** tab of the Installation Destination screen.



IMPORTANT

For a SCSI-only installation, remove the **DASD=** from the parameter or CMS configuration file to indicate no DASD is present.

18.16.1.2.4. Configure FCoE Parameters

When you click the **Add FCoE SAN...** button, a dialog appears for you to configure network interfaces for discovering FCoE storage devices.

First, select a network interface that is connected to a FCoE switch in the **NIC** drop-down menu and click the **Add FCoE disk(s)** button to scan the network for SAN devices.

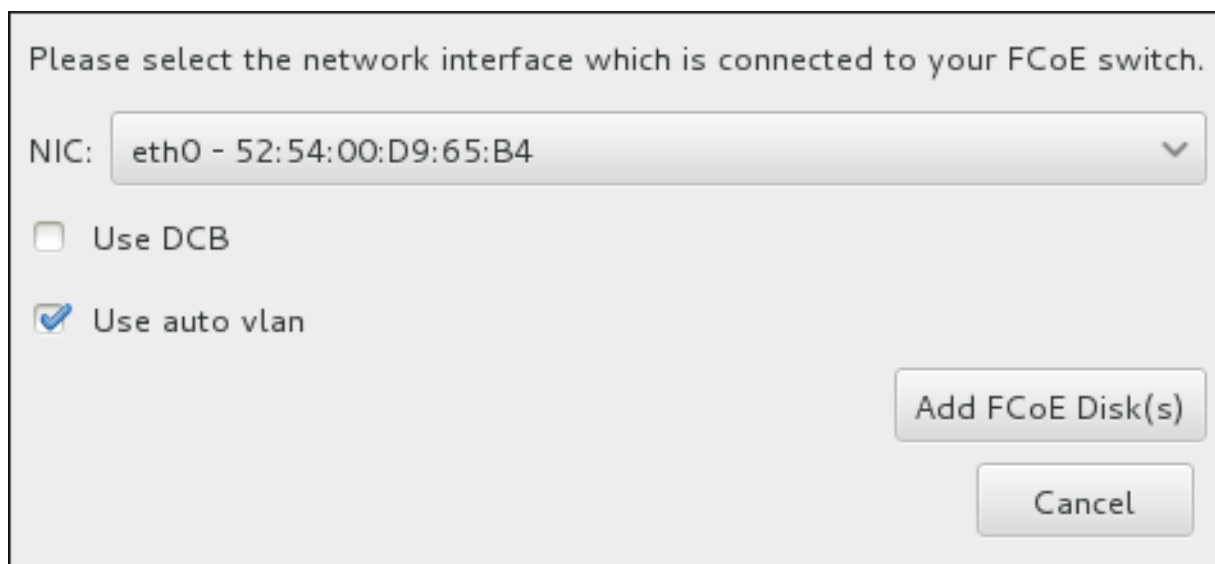


Figure 18.34. Configure FCoE Parameters

There are check boxes with additional options to consider:

Use DCB

Data Center Bridging (DCB) is a set of enhancements to the Ethernet protocols designed to increase the efficiency of Ethernet connections in storage networks and clusters. Enable or disable the installation program's awareness of DCB with the check box in this dialog. This option should only be enabled for network interfaces that require a host-based DCBX client. Configurations on interfaces that implement a hardware DCBX client should leave this check box empty.

Use auto vlan

Auto VLAN indicates whether VLAN discovery should be performed. If this box is checked, then the FIP (FCoE Initiation Protocol) VLAN discovery protocol will run on the Ethernet interface once the link configuration has been validated. If they are not already configured, network interfaces for any discovered FCoE VLANs will be automatically created and FCoE instances will be created on the VLAN interfaces. This option is enabled by default.

Discovered FCoE devices will be displayed under the **Other SAN Devices** tab in the Installation Destination screen.

18.17. KDUMP

Use this screen to select whether or not to use **Kdump** on this system. **Kdump** is a kernel crash dumping mechanism which, in the event of a system crash, captures information that can be invaluable in determining the cause of the crash.

Note that if you enable **Kdump**, you must reserve a certain amount of system memory for it. As a result, less memory is available for your processes.

If you do not want to use **Kdump** on this system, uncheck **Enable kdump**. Otherwise, set the amount of memory to reserve for **Kdump**. You can let the installer reserve a reasonable amount automatically, or you can set any amount manually. When you are satisfied with the settings, click **Done** to save the configuration and return to the previous screen.

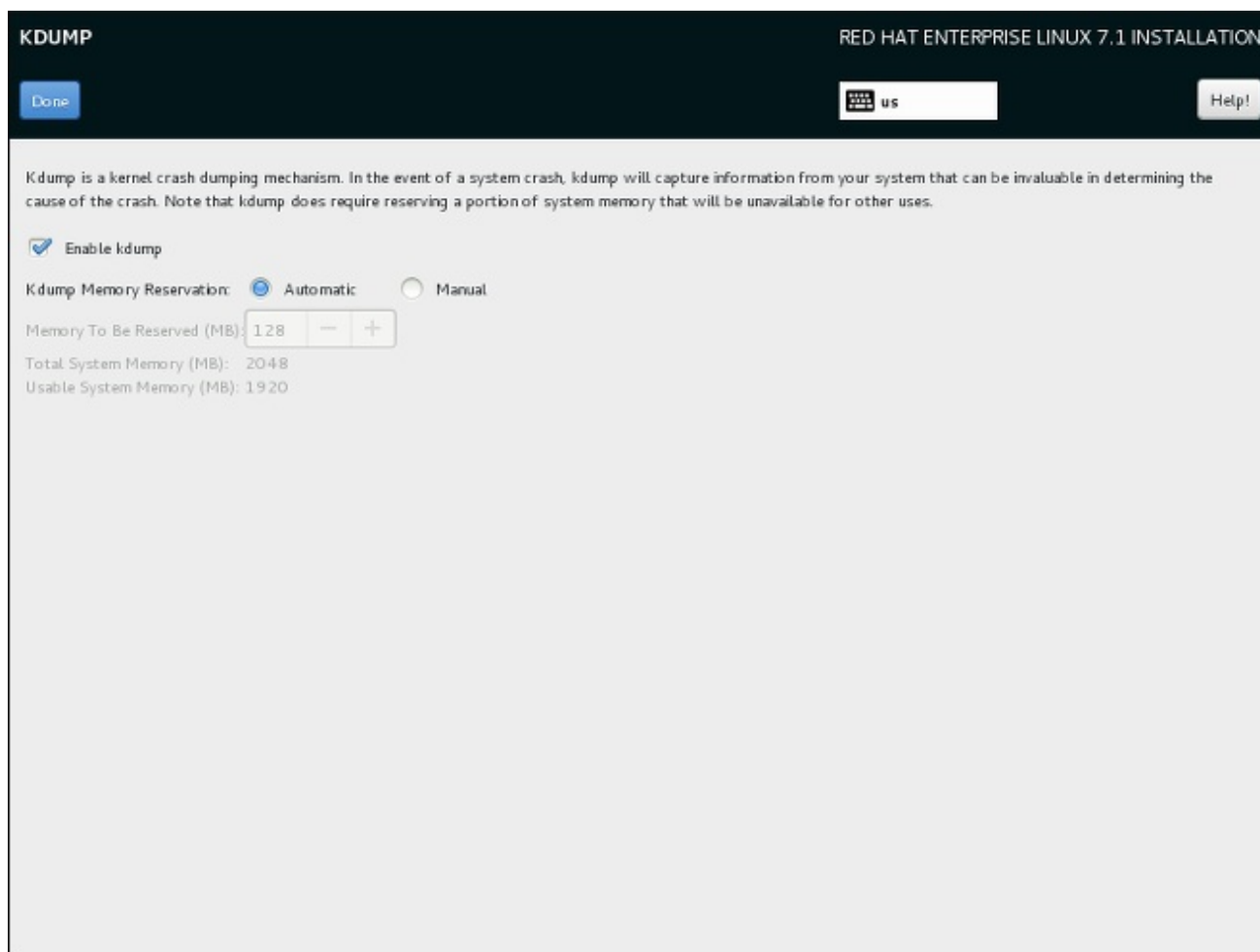


Figure 18.35. Kdump Enablement and Configuration

18.18. BEGIN INSTALLATION

When all required sections of the **Installation Summary** screen have been completed, the admonition at the bottom of the menu screen disappears and the **Begin Installation** button becomes available.



Figure 18.36. Ready to Install

**WARNING**

Up to this point in the installation process, no lasting changes have been made on your computer. When you click **Begin Installation**, the installation program will allocate space on your hard drive and start to transfer Red Hat Enterprise Linux into this space. Depending on the partitioning option that you chose, this process might include erasing data that already exists on your computer.

To revise any of the choices that you made up to this point, return to the relevant section of the **Installation Summary** screen. To cancel installation completely, click **Quit** or switch off your computer. To switch off most computers at this stage, press the power button and hold it down for a few seconds.

If you have finished customizing your installation and are certain that you want to proceed, click **Begin Installation**.

After you click **Begin Installation**, allow the installation process to complete. If the process is interrupted, for example, by you switching off or resetting the computer, or by a power outage, you will probably not be able to use your computer until you restart and complete the Red Hat Enterprise Linux installation process, or install a different operating system.

18.19. THE CONFIGURATION MENU AND PROGRESS SCREEN

Once you click **Begin Installation** at the **Installation Summary** screen, the progress screen appears. Red Hat Enterprise Linux reports the installation progress on the screen as it writes the selected packages to your system.

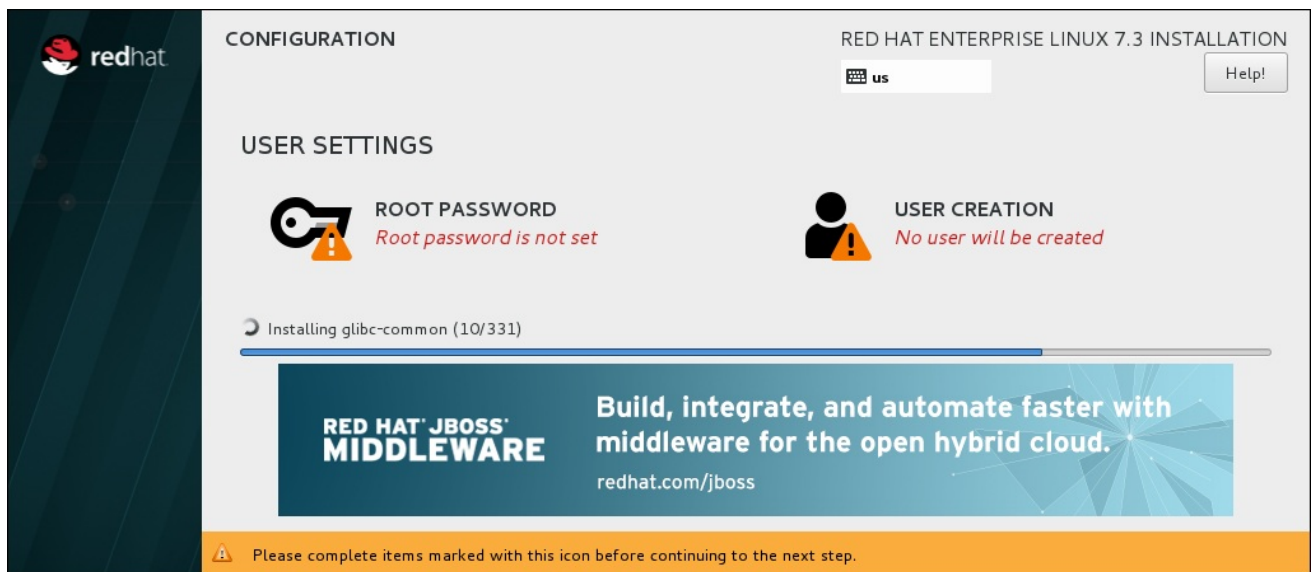


Figure 18.37. Installing Packages

For your reference, a complete log of your installation can be found in the `/var/log/anaconda/anaconda.packaging.log` file, once you reboot your system.

If you chose to encrypt one or more partitions during partitioning setup, a dialog window with a progress bar will be displayed during the early stage of the installation process. This window informs that the installer is attempting to gather enough entropy (random data) to ensure that the encryption is secure. This window will disappear after 256 bits of entropy are gathered, or after 10 minutes. You can speed up the gathering process by moving your mouse or randomly typing on the keyboard. After the window disappears, the installation process will continue.

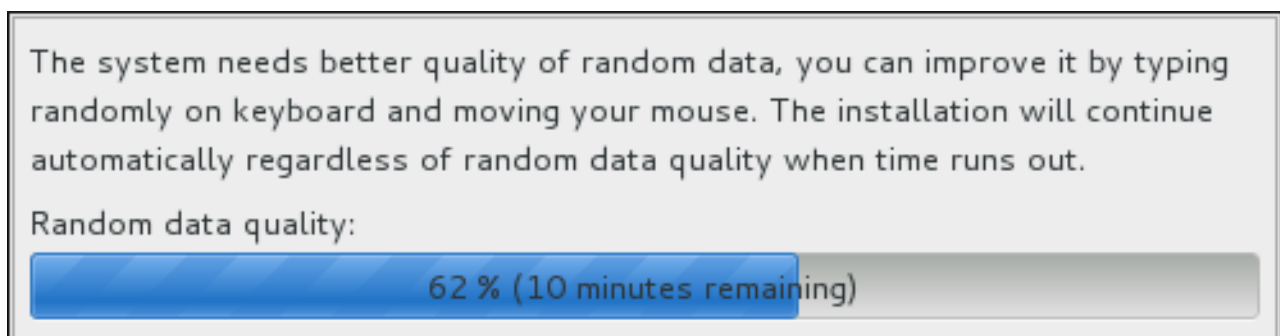


Figure 18.38. Gathering Entropy for Encryption

While the packages are being installed, more configuration is required. Above the installation progress bar are the **Root Password** and **User Creation** menu items.

The **Root Password** screen is used to configure the system's **root** account. This account can be used to perform critical system management and administration tasks. The same tasks can also be performed with a user account with the **wheel** group membership; if such an user account is created during installation, setting up a **root** password is not mandatory.

Creating a user account is optional and can be done after installation, but it is recommended to do it on this screen. A user account is used for normal work and to access the system. Best practice suggests that you always access the system through a user account, not the root account.

It is possible to disable access to the **Root Password** or **Create User** screens. To do so, use a Kickstart file which includes the **rootpw --lock** or **user --lock** commands. See [Section 27.3.1, “Kickstart Commands and Options”](#) for more information these commands.

18.19.1. Set the Root Password

Setting up a root account and password is an important step during your installation. The root account (also known as the superuser) is used to install packages, upgrade RPM packages, and perform most system maintenance. The root account gives you complete control over your system. For this reason, the root account is best used *only* to perform system maintenance or administration. See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information about becoming root.

Figure 18.39. Root Password Screen



NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

Click the **Root Password** menu item and enter your new password into the **Root Password** field. Red Hat Enterprise Linux displays the characters as asterisks for security. Type the same password into the **Confirm** field to ensure it is set correctly. After you set the root password, click **Done** to return to the User Settings screen.

The following are the requirements and recommendations for creating a strong root password:

- *must* be at least eight characters long
- may contain numbers, letters (upper and lower case) and symbols
- is case-sensitive and should contain a mix of cases
- something you can remember but that is not easily guessed
- should not be a word, abbreviation, or number associated with you, your organization, or found in a dictionary (including foreign languages)
- should not be written down; if you must write it down keep it secure



NOTE

To change your root password after you have completed the installation, run the **passwd** command as **root**. If you forget the root password, see [Section 32.1.3, “Resetting the Root Password”](#) for instructions on how to use the rescue mode to set a new one.

18.19.2. Create a User Account

To create a regular (non-root) user account during the installation, click **User Settings** on the progress screen. The **Create User** screen appears, allowing you to set up the regular user account and configure its parameters. Though recommended to do during installation, this step is optional and can be performed after the installation is complete.



NOTE

You must always set up at least one way to gain root privileges to the installed system: either using a **root** account, or by creating a user account with administrative privileges (member of the **wheel** group), or both.

To leave the user creation screen after you have entered it, without creating a user, leave all the fields empty and click **Done**.

Figure 18.40. User Account Configuration Screen

Enter the full name and the user name in their respective fields. Note that the system user name must be shorter than 32 characters and cannot contain spaces. It is highly recommended to set up a password for the new account.

When setting up a strong password even for a non-root user, follow the guidelines described in [Section 18.19.1, “Set the Root Password”](#).

Click the **Advanced** button to open a new dialog with additional settings.

ADVANCED USER CONFIGURATION

Home Directory

☒ Create a home directory for this user.

Home directory:

User and Group IDs

☐ Specify a user ID manually:
☐ Specify a group ID manually:

Group Membership

Add user to the following groups:

Example: wheel, my-team (1245), project-x (29935)

Tip: You may input a comma-separated list of group names and group IDs here.
Groups that do not already exist will be created; specify their GID in parentheses.

Figure 18.41. Advanced User Account Configuration

By default, each user gets a home directory corresponding to their user name. In most scenarios, there is no need to change this setting.

You can also manually define a system identification number for the new user and their default group by selecting the check boxes. The range for regular user IDs starts at the number **1000**. At the bottom of the dialog, you can enter the comma-separated list of additional groups, to which the new user shall belong. The new groups will be created in the system. To customize group IDs, specify the numbers in parenthesis.



NOTE

Consider setting IDs of regular users and their default groups at range starting at **5000** instead of **1000**. That is because the range reserved for system users and groups, **0-999**, might increase in the future and thus overlap with IDs of regular users.

For creating users with custom IDs using kickstart, see [user \(optional\)](#).

For changing the minimum UID and GID limits after the installation, which ensures that your chosen UID and GID ranges are applied automatically on user creation, see [the Users and Groups chapter of the System Administrator's Guide](#).

Once you have customized the user account, click **Save Changes** to return to the **User Settings** screen.

18.20. INSTALLATION COMPLETE

Congratulations! Your Red Hat Enterprise Linux installation is now complete!

The installation program prompts you to prepare your system for reboot.

The installation program automatically reboots into the installed system.

Should the installation program not reboot, the installation program shows information from which device to do an IPL (boot). Accept the shutdown option and after shutdown, IPL from the DASD or SCSI LUN where the **/boot** partition for Red Hat Enterprise Linux has been installed.

18.20.1. IPL under z/VM

To IPL from a DASD, for example using the DASD device 200 on the 3270 console, issue the command:

```
#cp i 200
```

In DASD only environments where automatic partitioning (clearing data from all partitions) was used, the first activated DASD is where the **/boot** partition is typically located.

Using **/boot** on an FCP LUN, you must provide the WWPN and LUN for the FCP-attached device from which to IPL.

To IPL from an FCP-attached device:

1. Provide FCP routing information to an FCP-attached device, for example, where **0x50050763050B073D** is the WWPN, and **0x4020400100000000** is the FCP LUN:

```
#cp set loaddev portname 50050763 050B073D lun 40204001 00000000
```

2. IPL the FCP adapter, for example **FC00**:

```
#cp ipl FC00
```



NOTE

To disconnect from the 3270 terminal without stopping the Linux running in your virtual machine, use **#cp disconnect** instead of **#cp logoff**. When your virtual machine is re-connected using the usual logon procedure, it might be placed in CP console function mode (**CP READ**). If so, to resume execution on your virtual machine, enter the **BEGIN** command.

18.20.2. IPL on an LPAR

For LPAR-based installations, on the HMC, issue a load command to the LPAR, specifying the particular DASD, or the FCP adapter, WWPN, and FCP LUN where the **/boot** partition is located.

18.20.3. Continuing after Reboot (re-IPL)

Following the automatic reboot or the manual IPL of the installed Red Hat Enterprise Linux operating system, you can log on to the system through **ssh**. Note that the only place from which you can log in as root is from the 3270 terminal or from other terminal devices listed in **/etc/securetty**.

The first time you start your Red Hat Enterprise Linux system in a graphical environment, you can use

Initial Setup to guide you through Red Hat Enterprise Linux configuration. **Initial Setup** lets you configure your environment at the beginning, so that you can get started using your Red Hat Enterprise Linux system quickly.

See [Chapter 30, *Initial Setup*](#) for information about the configuration process.

CHAPTER 19. TROUBLESHOOTING INSTALLATION ON IBM Z

This chapter discusses some common installation problems and their solutions.

For debugging purposes, **Anaconda** logs installation actions into files in the **/tmp** directory. These files are listed in the following table.

Table 19.1. Log Files Generated During the Installation

Log file	Contents
/tmp/anaconda.log	general Anaconda messages
/tmp/program.log	all external programs run during the installation
/tmp/storage.log	extensive storage module information
/tmp/packaging.log	yum and rpm package installation messages
/tmp/syslog	hardware-related system messages

If the installation fails, the messages from these files are consolidated into **/tmp/anaconda-tb-identifier**, where *identifier* is a random string.

After successful installation, by default, these files will be copied to the installed system under the directory **/var/log/anaconda/**. However, if installation is unsuccessful, or if the **inst.nosave=all** or **inst.nosave=logs** options are used when booting the installation system, these logs will only exist in the installation program's RAM disk. This means they are not saved permanently and will be lost once the system is powered down. To store them permanently, copy those files to another system on the network by using **scp** on the system running the installation program, or copy them to a mounted storage device (such as a USB flash drive). Details on how to transfer the log files over the network are below.



NOTE

The following procedure requires the installation system to be able to access the network and the target system to be able to receive files over the **ssh** protocol.

Procedure 19.1. Transferring Log Files Over the Network

1. Access the shell prompt on the installation system. This can be done in the following ways:
 - In a running **tmux** session on the installation system, press **Ctrl+b p** and **Ctrl+b n** to switch to the previous or next terminal, respectively, and find the terminal with a root shell.
 - Connect to the installation system over **ssh**.

In both cases, you will be able to use the installation system's shell as **root**.

2. Switch to the **/tmp** directory where the log files are located:

```
# cd /tmp
```

3. Copy the log files onto another system on the network using the **scp** command:

```
# scp *log user@address:path
```

Replace *user* with a valid user name on the target system, *address* with the target system's address or host name, and *path* with the path to the directory you want to save the log files into. For example, if you want to log in as **john** to a system with an IP address of **192.168.0.122** and place the log files into the **/home/john/logs/** directory on that system, the command will have the following form:

```
# scp *log john@192.168.0.122:/home/john/logs/
```

When connecting to the target system for the first time, the SSH client asks you to confirm that the fingerprint of the remote system is correct and that you want to continue:

```
The authenticity of host '192.168.0.122 (192.168.0.122)' can't be established.  
ECDSA key fingerprint is a4:60:76:eb:b2:d0:aa:23:af:3d:59:5c:de:bb:c4:42.  
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press **Enter** to continue. Then, provide a valid password when prompted. The files will start transferring to the specified directory on the target system.

The log files from the installation are now permanently saved on the target system and available for review.

19.1. TROUBLE DURING THE INSTALLATION

19.1.1. No Disks Detected

In the **Installation Destination** screen, the following error message can appear at the bottom: **No disks detected. Please shut down the computer, connect at least one disk, and restart to complete installation.**

This message usually indicates that there is an issue with your DASD (*Direct Access Storage Device*) devices. If you encounter this error, add the **DASD=<disks>** parameter to your parameter file or CMS configuration file (where *disks* is the DASD range reserved for installation) and start the installation again.

Additionally, make sure you format the DASDs using the **dasdfmt** command within a Linux root shell, instead of formatting the DASDs using CMS. **Anaconda** automatically detects any DASD devices that are not yet formatted and asks you whether to format the devices.

If you are installing into one or more iSCSI devices and there is no local storage present on the system, make sure that all required LUNs (*Logical Unit Numbers*) are being presented to the appropriate HBA (*Host Bus Adapter*). For additional information about iSCSI, see [Appendix B, iSCSI Disks](#).

19.1.2. Reporting Traceback Messages

If the graphical installation program encounters an error, it presents you with a crash reporting dialog box. You can then choose to send information about the problem you encountered to Red Hat. To send a crash report, you will need to enter your Customer Portal credentials. If you do not have a Customer Portal account, you can register at <https://www.redhat.com/wapps/ugc/register.html>. Automated crash reporting also requires a working network connection.



Figure 19.1. The Crash Reporting Dialog Box

When the dialog appears, select **Report Bug** to report the problem, or **Quit** to exit the installation.

Optionally, click **More Info** to display detailed output that can help determine the cause of the error. If you are familiar with debugging, click **Debug**. This will take you to virtual terminal **tty1**, where you can request more precise information that will enhance the bug report. To return to the graphical interface from **tty1**, use the **continue** command.

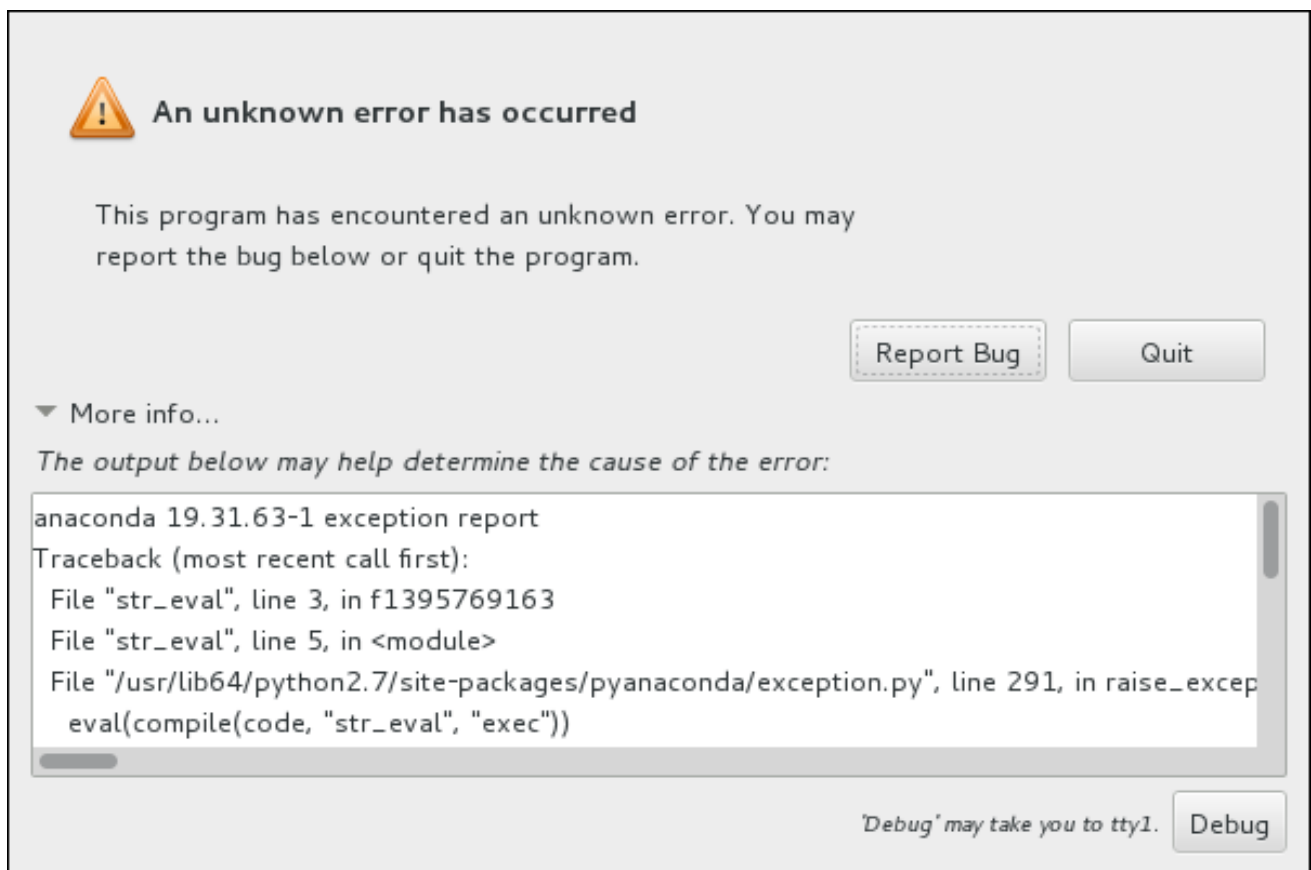


Figure 19.2. The Expanded Crash Reporting Dialog Box

If you want to report the bug to the customer portal, follow the procedure below.

Procedure 19.2. Reporting Errors to Red Hat Customer Support

1. In the menu that appears, select **Report a bug to Red Hat Customer Portal**.

2. To report the bug to Red Hat, you first need to provide your Customer Portal credentials. Click **Configure Red Hat Customer Support**.



Figure 19.3. Customer Portal Credentials

3. A new window is now open, prompting you to enter your Customer Portal user name and password. Enter your Red Hat Customer Portal credentials.

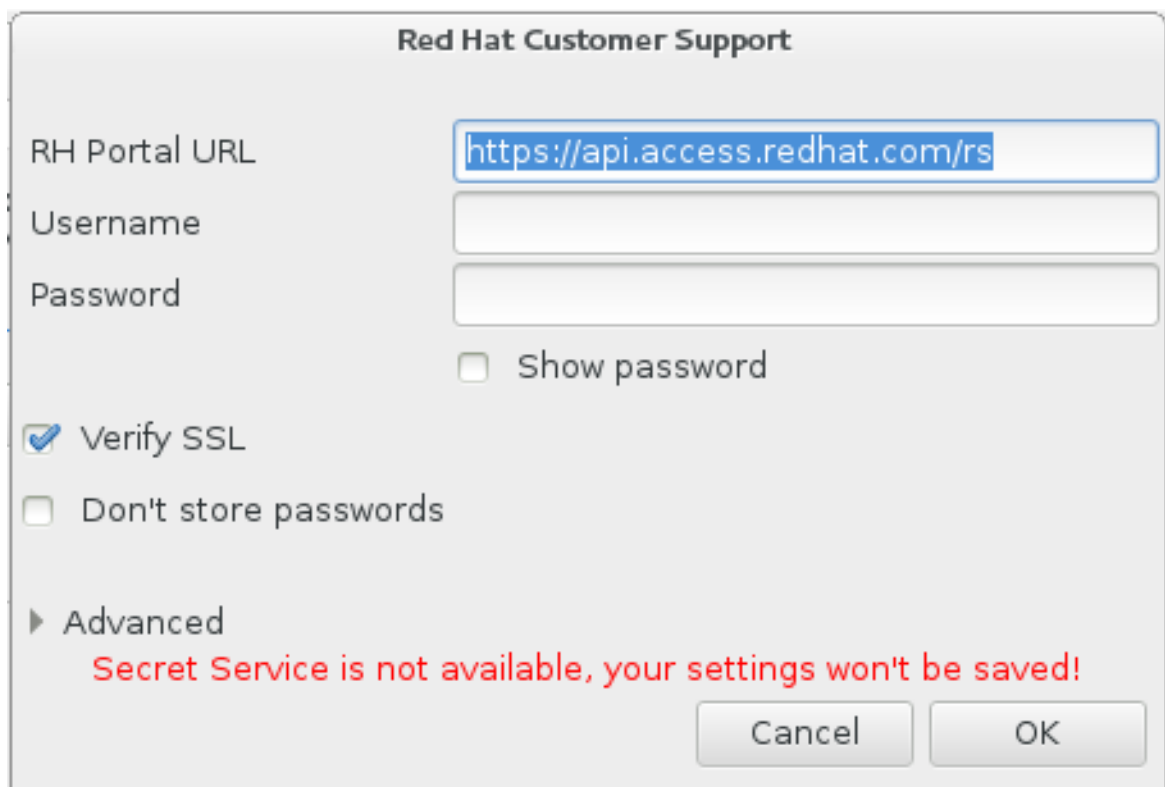


Figure 19.4. Configure Red Hat Customer Support

If your network settings require you to use a **HTTP** or **HTTPS** proxy, you can configure it by expanding the **Advanced** menu and entering the address of the proxy server.

When you put in all required credentials, click **OK** to proceed.

4. A new window appears, containing a text field. Write down any useful information and comments here. Describe how the error can be reproduced by explaining each step you took before the crash reporting dialog appeared. Provide as much relevant detail as possible, including any information you acquired when debugging. Be aware that the information you provide here can become publicly visible on the Customer Portal.

If you do not know what caused the error, check the box labeled **I don't know what caused this problem** at the bottom of the dialog.

Then, click **Forward**.

How did this problem happen (step-by-step)? How can it be reproduced? Any additional comments useful for diagnosing the problem? Please use English if possible.

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (stage1 on first disk). First disk is not used in partitioning section.

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the Kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to the second volume.

Your comments are not private. They may be included into publicly visible problem reports.

If you don't know how to describe it, you can [add a screencast](#)

☐ I don't know what caused this problem

Close Forward

Figure 19.5. Describe the Problem

- Next, review the information that will be sent to the Customer Portal. The explanation you provided is in the **comment** tab. Other tabs include such information as your system's host name and other details about the installation environment. You can remove any items you do not want sent to Red Hat, but be aware that providing less detail can affect the investigation of the issue.

Click **Forward** when you finish reviewing the information to be sent.

Please review the data before it gets reported. Depending on reporter chosen, it may end up publicly visible.

environ cmdline backtrace hostname **comment** reason

Description of problem:
Installation of Red Hat Enterprise Linux on second disk crashes during boot loader installation (

How reproducible: always

Steps to reproduce:
1. Attach 2 disks to platform
2. Run Kickstart installation on second disk with the following in the kickstart file:

```
bootloader --location=mbr --driveorder=sda,sdb
clearpart --all --initlabel
part / --fstype ext4 --size=1 --grow --ondisk=sdb
part swap --fstype swap --recommended --ondisk=sdb
part /boot --fstype ext4 --size=1000 --ondisk=sdb
```

Actual results: Installation crashes

Expected results: Installation finishes properly

Additional info:
This issue can also be reproduced using two RAID volumes, when the system is being installed to t

Close Forward

Figure 19.6. Review the Data to Be Sent

- Review the list of files that will be sent and included in the bug report as individual attachments. These files provide system information that will assist the investigation. If you do not want to send certain files, uncheck the box next to each one. To provide additional files that can help find the problem, click **Attach a file**.

Once you have reviewed the files to be sent, check the box labeled **I have reviewed the data and agree with submitting it**. Then, click **Forward** to send the report and attachments to the Customer Portal.

Size: 618133 bytes, 41 files

Include	Name	Value
☒	anaconda.log	(click here to view/edit)
☒	packaging.log	(click here to view/edit)
☒	os_info	(click here to view/edit)
☒	environ	(click here to view/edit)
☒	last_occurrence	1395769224
☒	anaconda-tb	(click here to view/edit)
☒	version	7.0
☒	cmdline	/usr/bin/python /sbin/anaconda
☒	package	anaconda-19.31.63-1
☒	duphash	aeb4ed3992aab62d1cd03d3b1e0a89c79753f34da115a739be021d2873d0a131
☒	lsblk_output	(click here to view/edit)
☒	pkg_version	19.31.63
☒	product	Red Hat Enterprise Linux
☒	release	Red Hat Enterprise Linux Workstation release 7.0 Beta (Maipo)
☒	pkg_arch	x86_64

Attach a file

☐ I reviewed the data and agree with submitting it

Close Forward

Figure 19.7. Review the Files to Be Sent

- When the dialog reports that processing has finished, you can click **Show log** to view details of the reporting process or **Close** to return to the initial crash reporting dialog box. There, click **Quit** to exit the installation.

19.1.3. Creating Pre-installation Log Files

To debug installation problems you can set the **inst.debug** option to create log files from the environment before the installation starts. These log files contain, for example, the current storage configuration.

To set the option in the Red Hat Enterprise Linux installation boot menu:

- Select the **Install Red Hat Enterprise Linux 7.3** entry.
- Press the **Tab** key to edit the boot options.
- Append **inst.debug** to the options. For example:

```
> vmlinuz ... inst.debug
```

For further details, see [Chapter 23, Boot Options](#).

- Press **Enter** to start the setup.

The system stores the pre-installation log files in the **/tmp/pre-anaconda-logs/** directory before Anaconda starts. To access the log files:

- Switch to the console. See [Section 8.2.1, "Accessing Consoles"](#).
- Change into the **/tmp/pre-anaconda-logs/** directory:

```
# cd /tmp/pre-anaconda-logs/
```

19.2. PROBLEMS AFTER INSTALLATION

19.2.1. Remote Graphical Desktops and XDMCP

If you have installed the **X Window System** and would like to log in to your Red Hat Enterprise Linux system using a graphical login manager, enable the *X Display Manager Control Protocol* (XDMCP). This protocol allows users to remotely log in to a desktop environment from any X-compatible client, such as a network-connected workstation or X11 terminal. The procedure below explains how to enable XDMCP.

Procedure 19.3. Enabling XDMCP on IBM Z

- Open the **/etc/gdm/custom.conf** configuration file in a plain text editor such as **vi** or **nano**.
- In the **custom.conf** file, locate the section starting with **[xdmcp]**. In this section, add the following line:

```
Enable=true
```

- Save the file, and exit the text editor.

4. Restart the **X Window System**. To do this, either reboot the whole system, or restart the **GNOME Display Manager** using the following command as **root**:

```
# systemctl restart gdm.service
```

Wait for the login prompt to appear again, and log in using your normal user name and password.

The IBM Z server is now configured for XDMCP. You can connect to it from another workstation (client) by starting a remote **X** session using the **X** command on the client workstation. For example:

```
$ X :1 -query address
```

Replace *address* with the host name of the remote X11 server. The command connects to the remote X11 server using XDMCP and displays the remote graphical login screen on display **:1** of the X11 server system (usually accessible by pressing **Ctrl-Alt-F8**).

You can also access remote desktop sessions using a *nested* X11 server, which opens the remote desktop as a window in your current X11 session. **Xnest** allows users to open a remote desktop nested within their local X11 session. For example, run **Xnest** using the following command, replacing *address* with the host name of the remote X11 server:

```
$ Xnest :1 -query address
```

For more information about XDMCP, see the **X Window System** documentation at <http://www.x.org/releases/X11R7.6/doc/libXdmcp/xdmcp.html>.

19.2.2. Is Your System Displaying Signal 11 Errors?

A signal 11 error, commonly known as a *segmentation fault*, means that a program accessed a memory location that was not assigned to it. A signal 11 error can occur due to a bug in one of the software programs that is installed, or faulty hardware.

If you receive a fatal signal 11 error during the installation, first make sure you are using the most recent installation images, and let **Anaconda** verify them to make sure they are not corrupted. Bad installation media (such as an improperly burned or scratched optical disk) are a common cause of signal 11 errors. Verifying the integrity of the installation media is recommended before every installation.

For information about obtaining the most recent installation media, see [Chapter 2, Downloading Red Hat Enterprise Linux](#). To perform a media check before the installation starts, append the **rd.live.check** boot option at the boot menu. See [Section 23.2.2, “Verifying Boot Media”](#) for details.

Other possible causes are beyond this document's scope. Consult your hardware manufacturer's documentation for more information.

CHAPTER 20. CONFIGURING AN INSTALLED LINUX ON IBM Z INSTANCE

For more information about Linux on IBM Z, see the publications listed in [Chapter 22, IBM Z References](#). Some of the most common tasks are described here.

20.1. ADDING DASDS

DASDs (*Direct Access Storage Devices*) are a type of storage commonly used with IBM Z. Additional information about working with these storage devices can be found at the IBM Knowledge Center at http://www-01.ibm.com/support/knowledgecenter/linuxonibm/com.ibm.linux.z.lgdd/lgdd_t_dasd_wrk.html.

The following is an example of how to set a DASD online, format it, and make the change persistent.



NOTE

Make sure the device is attached or linked to the Linux system if running under z/VM.

CP ATTACH EB1C TO *

To link a mini disk to which you have access, issue, for example:

CP LINK RHEL7X 4B2E 4B2E MR
DASD 4B2E LINKED R/W

See *z/VM: CP Commands and Utilities Reference, SC24-6175* for details about the commands.

20.1.1. Dynamically Setting DASDs Online

To set a DASD online, follow these steps:

1. Use the **cio_ignore** utility to remove the DASD from the list of ignored devices and make it visible to Linux:

```
# cio_ignore -r device_number
```

Replace *device_number* with the device number of the DASD. For example:

```
# cio_ignore -r 4b2e
```

2. Set the device online. Use a command of the following form:

```
# chccwdev -e device_number
```

Replace *device_number* with the device number of the DASD. For example:

```
# chccwdev -e 4b2e
```

As an alternative, you can set the device online using sysfs attributes:

1. Use the **cd** command to change to the `/sys/` directory that represents that volume:

```
# cd /sys/bus/ccw/drivers/dasd-eckd/0.0.4b2e/
# ls -l
total 0
-r--r--r-- 1 root root 4096 Aug 25 17:04 availability
-rw-r--r-- 1 root root 4096 Aug 25 17:04 cmb_enable
-r--r--r-- 1 root root 4096 Aug 25 17:04 cutype
-rw-r--r-- 1 root root 4096 Aug 25 17:04 detach_state
-r--r--r-- 1 root root 4096 Aug 25 17:04 devtype
-r--r--r-- 1 root root 4096 Aug 25 17:04 discipline
-rw-r--r-- 1 root root 4096 Aug 25 17:04 online
-rw-r--r-- 1 root root 4096 Aug 25 17:04 readonly
-rw-r--r-- 1 root root 4096 Aug 25 17:04 use_diag
```

2. Check to see if the device is already online:

```
# cat online
0
```

3. If it is not online, enter the following command to bring it online:

```
# echo 1 > online
# cat online
1
```

3. Verify which block devnode it is being accessed as:

```
# ls -l
total 0
-r--r--r-- 1 root root 4096 Aug 25 17:04 availability
lrwxrwxrwx 1 root root  0 Aug 25 17:07 block -> ../../../../block/dasdb
-rw-r--r-- 1 root root 4096 Aug 25 17:04 cmb_enable
-r--r--r-- 1 root root 4096 Aug 25 17:04 cutype
-rw-r--r-- 1 root root 4096 Aug 25 17:04 detach_state
-r--r--r-- 1 root root 4096 Aug 25 17:04 devtype
-r--r--r-- 1 root root 4096 Aug 25 17:04 discipline
-rw-r--r-- 1 root root  0 Aug 25 17:04 online
-rw-r--r-- 1 root root 4096 Aug 25 17:04 readonly
-rw-r--r-- 1 root root 4096 Aug 25 17:04 use_diag
```

As shown in this example, device 4B2E is being accessed as `/dev/dasdb`.

These instructions set a DASD online for the current session, but this is not persistent across reboots. For instructions on how to set a DASD online persistently, see [Section 20.1.3, “Persistently Setting DASDs Online”](#). When you work with DASDs, use the persistent device symbolic links under `/dev/disk/by-path/`. See the chapter about persistent storage device naming in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#) for more in-depth information about different ways to consistently refer to storage devices.

20.1.2. Preparing a New DASD with Low-level Formatting

Once the disk is online, change back to the `/root` directory and low-level format the device. This is only required once for a DASD during its entire lifetime:

```
# cd /root
# dasdfmt -b 4096 -d cdl -p /dev/disk/by-path/ccw-0.0.4b2e
Drive Geometry: 10017 Cylinders * 15 Heads = 150255 Tracks
```

I am going to format the device `/dev/disk/by-path/ccw-0.0.4b2e` in the following way:

```
Device number of device : 0x4b2e
Labelling device       : yes
Disk label             : VOL1
Disk identifier        : 0X4B2E
Extent start (trk no)  : 0
Extent end (trk no)    : 150254
Compatible Disk Layout : yes
Blocksize              : 4096
```

```
--->> ATTENTION! <<---
```

All data of that device will be lost.

Type "yes" to continue, no will leave the disk untouched: yes

```
cyl  97 of 3338 |#-----| 2%
```

When the progress bar reaches the end and the format is complete, **dasdfmt** prints the following output:

```
Rereading the partition table...
Exiting...
```

Now, use **fdasd** to partition the DASD. You can create up to three partitions on a DASD. In our example here, we create one partition spanning the whole disk:

```
# fdasd -a /dev/disk/by-path/ccw-0.0.4b2e
auto-creating one partition for the whole disk...
writing volume label...
writing VTOC...
checking !
wrote NATIVE!
rereading partition table...
```

After a (low-level formatted) DASD is online, it can be used like any other disk under Linux. For instance, you can create file systems, LVM physical volumes, or swap space on its partitions, for example `/dev/disk/by-path/ccw-0.0.4b2e-part1`. Never use the full DASD device (**dev/dasdb**) for anything but the commands **dasdfmt** and **fdasd**. If you want to use the entire DASD, create one partition spanning the entire drive as in the **fdasd** example above.

To add additional disks later without breaking existing disk entries in, for example, `/etc/fstab`, use the persistent device symbolic links under `/dev/disk/by-path/`.

20.1.3. Persistently Setting DASDs Online

The above instructions described how to activate DASDs dynamically in a running system. However, such changes are not persistent and do not survive a reboot. Making changes to the DASD configuration persistent in your Linux system depends on whether the DASDs belong to the root file system. Those DASDs required for the root file system need to be activated very early during the boot process by the **initramfs** to be able to mount the root file system.

The **cio_ignore** commands are handled transparently for persistent device configurations and you do not need to free devices from the ignore list manually.

20.1.3.1. DASDs That Are Part of the Root File System

The only file you have to modify to add DASDs that are part of the root file system is **/etc/zipl.conf**. Then run the **zipl** boot loader tool. There is no need to recreate the **initramfs**.

There is one boot option to activate DASDs early in the boot process: **rd.dasd=**. This option takes a Direct Access Storage Device (DASD) adapter device bus identifier. For multiple DASDs, specify the parameter multiple times, or use a comma separated list of bus IDs. To specify a range of DASDs, specify the first and the last bus ID.

Below is an example **zipl.conf** for a system that uses physical volumes on partitions of two DASDs for an LVM volume group **vg_devel1** that contains a logical volume **lv_root** for the root file system.

```
[defaultboot]
default=linux
target=/boot/

[linux]
image=/boot/vmlinuz-2.6.32-19.el7.s390x
ramdisk=/boot/initramfs-2.6.32-19.el7.s390x.img
parameters="root=/dev/mapper/vg_devel1-lv_root
rd.dasd=0.0.0200,use_diag=0,readonly=0,erplog=0,failfast=0
rd.dasd=0.0.0207,use_diag=0,readonly=0,erplog=0,failfast=0 rd_LVM_LV=vg_devel1/lv_root
rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8 SYSFONT=latarcyrheb-sun16
KEYTABLE=us cio_ignore=all,!condev"
```

Suppose that you want to add another physical volume on a partition of a third DASD with device bus ID **0.0.202b**. To do this, add **rd.dasd=0.0.202b** to the parameters line of your boot kernel in **zipl.conf**:

```
[defaultboot]
default=linux
target=/boot/

[linux]
image=/boot/vmlinuz-2.6.32-19.el7.s390x
ramdisk=/boot/initramfs-2.6.32-19.el7.s390x.img
parameters="root=/dev/mapper/vg_devel1-lv_root
rd.dasd=0.0.0200,use_diag=0,readonly=0,erplog=0,failfast=0
rd.dasd=0.0.0207,use_diag=0,readonly=0,erplog=0,failfast=0 rd.dasd=0.0.202b
rd_LVM_LV=vg_devel1/lv_root rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8
SYSFONT=latarcyrheb-sun16 KEYTABLE=us cio_ignore=all,!condev"
```



WARNING

Make sure the length of the kernel command line in **/etc/zipl.conf** does not exceed 896 bytes. Otherwise, the boot loader cannot be saved, and the installation fails.

Run **zipl** to apply the changes of **/etc/zipl.conf** for the next IPL:

```
# zipl -V
```

```

Using config file '/etc/zipl.conf'
Target device information
Device.....: 5e:00
Partition.....: 5e:01
Device name.....: dasda
DASD device number.....: 0201
Type.....: disk partition
Disk layout.....: ECKD/compatible disk layout
Geometry - heads.....: 15
Geometry - sectors.....: 12
Geometry - cylinders.....: 3308
Geometry - start.....: 24
File system block size.....: 4096
Physical block size.....: 4096
Device size in physical blocks...: 595416
Building bootmap in '/boot/'
Building menu 'rh-automatic-menu'
Adding #1: IPL section 'linux' (default)
kernel image.....: /boot/vmlinuz-2.6.32-19.el7.s390x
kernel parmline...: 'root=/dev/mapper/vg_devel1-lv_root
rd.dasd=0.0.0200,use_diag=0,readonly=0,erplog=0,failfast=0
rd.dasd=0.0.0207,use_diag=0,readonly=0,erplog=0,failfast=0 rd.dasd=0.0.202b
rd_LVM_LV=vg_devel1/lv_root rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8
SYSFONT=latacyrheb-sun16 KEYTABLE=us cio_ignore=all,!condev'
initial ramdisk...: /boot/initramfs-2.6.32-19.el7.s390x.img
component address:
kernel image.....: 0x00010000-0x00a70fff
parmline.....: 0x00001000-0x00001fff
initial ramdisk.: 0x02000000-0x022d2fff
internal loader.: 0x0000a000-0x0000afff
Preparing boot device: dasda (0201).
Preparing boot menu
Interactive prompt.....: enabled
Menu timeout.....: 15 seconds
Default configuration...: 'linux'
Syncing disks...
Done.

```

20.1.3.2. DASDs That Are Not Part of the Root File System

DASDs that are not part of the root file system, that is, *data disks*, are persistently configured in the file **/etc/dasd.conf**. It contains one DASD per line. Each line begins with the device bus ID of a DASD. Optionally, each line can continue with options separated by space or tab characters. Options consist of key-value-pairs, where the key and value are separated by an equals sign.

The key corresponds to any valid **sysfs** attribute a DASD can have. The value will be written to the key's **sysfs** attribute. Entries in **/etc/dasd.conf** are activated and configured by udev when a DASD is added to the system. At boot time, all DASDs visible to the system get added and trigger **udev**.

Example content of **/etc/dasd.conf**:

```

0.0.0207
0.0.0200 use_diag=1 readonly=1

```

Modifications of **/etc/dasd.conf** only become effective after a reboot of the system or after the

dynamic addition of a new DASD by changing the system's I/O configuration (that is, the DASD is attached under z/VM). Alternatively, you can trigger the activation of a new entry in **/etc/dasd.conf** for a DASD which was previously not active, by executing the following commands:

1. Use the **cio_ignore** utility to remove the DASD from the list of ignored devices and make it visible to Linux:

```
# cio_ignore -r device_number
```

For example:

```
# cio_ignore -r 021a
```

2. Trigger the activation by writing to the **uevent** attribute of the device:

```
# echo add > /sys/bus/ccw/devices/device-bus-ID/uevent
```

For example:

```
# echo add > /sys/bus/ccw/devices/0.0.021a/uevent
```

20.2. ADDING FCP-ATTACHED LOGICAL UNITS (LUNS)

The following is an example of how to add an FCP LUN.



NOTE

If running under z/VM, make sure the FCP adapter is attached to the z/VM guest virtual machine. For multipathing in production environments there would be at least two FCP devices on two different physical adapters (CHPIDs). For example:

```
CP ATTACH FC00 TO *
CP ATTACH FCD0 TO *
```

20.2.1. Dynamically Activating an FCP LUN

Follow these steps to activate a LUN:

1. Use the **cio_ignore** utility to remove the FCP adapter from the list of ignored devices and make it visible to Linux:

```
# cio_ignore -r device_number
```

Replace *device_number* with the device number of the FCP adapter. For example:

2. To bring the FCP adapter device online, use the following command:

```
# chccwdev -e fc00
```

3. Verify that the required WWPN was found by the automatic port scanning of the zfcpc device driver:

```
# ls -l /sys/bus/ccw/drivers/zfcp/0.0.fc00/
drwxr-xr-x. 3 root root  0 Apr 28 18:19 0x500507630040710b
drwxr-xr-x. 3 root root  0 Apr 28 18:19 0x50050763050b073d
drwxr-xr-x. 3 root root  0 Apr 28 18:19 0x500507630e060521
drwxr-xr-x. 3 root root  0 Apr 28 18:19 0x500507630e860521
-r--r--r--. 1 root root 4096 Apr 28 18:17 availability
-r--r--r--. 1 root root 4096 Apr 28 18:19 card_version
-rw-r--r--. 1 root root 4096 Apr 28 18:17 cmb_enable
-r--r--r--. 1 root root 4096 Apr 28 18:17 ctype
-r--r--r--. 1 root root 4096 Apr 28 18:17 devtype
lrwxrwxrwx. 1 root root  0 Apr 28 18:17 driver -> ../../../../bus/ccw/drivers/zfcp
-rw-r--r--. 1 root root 4096 Apr 28 18:17 failed
-r--r--r--. 1 root root 4096 Apr 28 18:19 hardware_version
drwxr-xr-x. 35 root root  0 Apr 28 18:17 host0
-r--r--r--. 1 root root 4096 Apr 28 18:17 in_recovery
-r--r--r--. 1 root root 4096 Apr 28 18:19 lic_version
-r--r--r--. 1 root root 4096 Apr 28 18:17 modalias
-rw-r--r--. 1 root root 4096 Apr 28 18:17 online
-r--r--r--. 1 root root 4096 Apr 28 18:19 peer_d_id
-r--r--r--. 1 root root 4096 Apr 28 18:19 peer_wwnn
-r--r--r--. 1 root root 4096 Apr 28 18:19 peer_wwpn
--w-----. 1 root root 4096 Apr 28 18:19 port_remove
--w-----. 1 root root 4096 Apr 28 18:19 port_rescan
drwxr-xr-x. 2 root root  0 Apr 28 18:19 power
-r--r--r--. 1 root root 4096 Apr 28 18:19 status
lrwxrwxrwx. 1 root root  0 Apr 28 18:17 subsystem -> ../../../../bus/ccw
-rw-r--r--. 1 root root 4096 Apr 28 18:17 uevent
```

4. Activate the FCP LUN by adding it to the port (WWPN) through which you would like to access the LUN:

```
# echo 0x4020400100000000 >
/sys/bus/ccw/drivers/zfcp/0.0.fc00/0x50050763050b073d/unit_add
```

5. Find out the assigned SCSI device name:

```
# lszfcp -DV
/sys/devices/css0/0.0.0015/0.0.fc00/0x50050763050b073d/0x4020400100000000
/sys/bus/ccw/drivers/zfcp/0.0.fc00/host0/rport-0:0-21/target0:0:21/0:0:21:1089355792
```

20.2.2. Persistently activating FCP LUNs

The above instructions described how to activate FCP LUNs dynamically in a running system. However, such changes are not persistent and do not survive a reboot. How you make the changes to the FCP configuration persistent in your Linux system depends on whether the FCP LUNs belong to the root file system. Those required for the root file system need to be activated very early during the boot process by the **initramfs** to be able to mount the root file system. The **cio_ignore** commands are handled transparently for persistent device configurations and you do not need to free devices from the ignore list manually.

20.2.2.1. FCP LUNs That Are Part of the Root File System

The only file you have to modify for adding FCP LUNs that are part of the root file system is **/etc/zipl.conf** followed by a run of the **zipl** boot loader tool. There is no more need to recreate the **initramfs**.

Red Hat Enterprise Linux provides a parameter to activate FCP LUNs early in the boot process: **rd.zfcp=**. The value is a comma-separated list containing the device bus ID, the WWPN as 16 digit hexadecimal number prefixed with **0x**, and the FCP LUN prefixed with **0x** and padded with zeroes to the right to have 16 hexadecimal digits.

The following example **zipl.conf** is for a system that uses physical volumes on partitions of two FCP LUNs for an LVM volume group **vg_devel1** that contains a logical volume **lv_root** for the root file system. For simplicity, the example shows a configuration without multipathing.

```
[defaultboot]
default=linux
target=/boot/

[linux]
image=/boot/vmlinuz-2.6.32-19.el7.s390x
ramdisk=/boot/initramfs-2.6.32-19.el7.s390x.img
parameters="root=/dev/mapper/vg_devel1-lv_root
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a000000000
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a100000000
rd_LVM_LV=vg_devel1/lv_root rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8
SYSEFONT=latarcyrheb-sun16 KEYTABLE=us cio_ignore=all,!condev"
```

To add another physical volume on a partition of a third FCP LUN with device bus ID 0.0.fc00, WWPN 0x5105074308c212e9 and FCP LUN 0x401040a300000000, add

rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a300000000 to the parameters line of your boot kernel in **zipl.conf**. For example:

```
[defaultboot]
default=linux
target=/boot/

[linux]
image=/boot/vmlinuz-2.6.32-19.el7.s390x
ramdisk=/boot/initramfs-2.6.32-19.el7.s390x.img
parameters="root=/dev/mapper/vg_devel1-lv_root
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a000000000
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a100000000
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a300000000
rd_LVM_LV=vg_devel1/lv_root rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8
SYSEFONT=latarcyrheb-sun16 KEYTABLE=us cio_ignore=all,!condev"
```



WARNING

Make sure the length of the kernel command line in **/etc/zipl.conf** does not exceed 896 bytes. Otherwise, the boot loader cannot be saved, and the installation fails.

Run **zipl** to apply the changes of **/etc/zipl.conf** for the next IPL:

```
# zipl -V
Using config file '/etc/zipl.conf'
Target device information
Device.....: 08:00
Partition.....: 08:01
Device name.....: sda
Device driver name.....: sd
Type.....: disk partition
Disk layout.....: SCSI disk layout
Geometry - start.....: 2048
File system block size.....: 4096
Physical block size.....: 512
Device size in physical blocks...: 10074112
Building bootmap in '/boot/'
Building menu 'rh-automatic-menu'
Adding #1: IPL section 'linux' (default)
kernel image.....: /boot/vmlinuz-2.6.32-19.el7.s390x
kernel parmline....: 'root=/dev/mapper/vg_devel1-lv_root
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a000000000
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a100000000
rd.zfcp=0.0.fc00,0x5105074308c212e9,0x401040a300000000 rd_LVM_LV=vg_devel1/lv_root
rd_NO_LUKS rd_NO_MD rd_NO_DM LANG=en_US.UTF-8 SYSFONT=latarcyrheb-sun16
KEYTABLE=us cio_ignore=all,!condev'
initial ramdisk...: /boot/initramfs-2.6.32-19.el7.s390x.img
component address:
kernel image.....: 0x00010000-0x007a21ff
parmline.....: 0x00001000-0x000011ff
initial ramdisk.: 0x02000000-0x028f63ff
internal loader.: 0x0000a000-0x0000a3ff
Preparing boot device: sda.
Detected SCSI PCBIOS disk layout.
Writing SCSI master boot record.
Syncing disks...
Done.
```

20.2.2.2. FCP LUNs That Are Not Part of the Root File System

FCP LUNs that are not part of the root file system, such as data disks, are persistently configured in the file **/etc/zfcp.conf**. It contains one FCP LUN per line. Each line contains the device bus ID of the FCP adapter, the WWPN as 16 digit hexadecimal number prefixed with **0x**, and the FCP LUN prefixed with **0x** and padded with zeroes to the right to have 16 hexadecimal digits, separated by a space or tab. Entries in **/etc/zfcp.conf** are activated and configured by udev when an FCP adapter is added to the system. At boot time, all FCP adapters visible to the system are added and trigger **udev**.

Example content of **/etc/zfcp.conf**:

```
0.0.fc00 0x5105074308c212e9 0x401040a000000000
0.0.fc00 0x5105074308c212e9 0x401040a100000000
0.0.fc00 0x5105074308c212e9 0x401040a300000000
0.0.fcd0 0x5105074308c2aee9 0x401040a000000000
0.0.fcd0 0x5105074308c2aee9 0x401040a100000000
0.0.fcd0 0x5105074308c2aee9 0x401040a300000000
```

Modifications of **/etc/zfcp.conf** only become effective after a reboot of the system or after the dynamic addition of a new FCP channel by changing the system's I/O configuration (for example, a channel is attached under z/VM). Alternatively, you can trigger the activation of a new entry in **/etc/zfcp.conf** for an FCP adapter which was previously not active, by executing the following commands:

1. Use the **cio_ignore** utility to remove the FCP adapter from the list of ignored devices and make it visible to Linux:

```
# cio_ignore -r device_number
```

Replace *device_number* with the device number of the FCP adapter. For example:

```
# cio_ignore -r fcfc
```

2. To trigger the uevent that activates the change, issue:

```
# echo add > /sys/bus/ccw/devices/device-bus-ID/uevent
```

For example:

```
# echo add > /sys/bus/ccw/devices/0.0.fcfc/uevent
```

20.3. ADDING A NETWORK DEVICE

Network device driver modules are loaded automatically by **udev**.

You can add a network interface on IBM Z dynamically or persistently.

- Dynamically
 1. Load the device driver
 2. Remove the network devices from the list of ignored devices.
 3. Create the group device.
 4. Configure the device.
 5. Set the device online.
- Persistently
 1. Create a configuration script.
 2. Activate the interface.

The following sections provide basic information for each task of each IBM Z network device driver. [Section 20.3.1, “Adding a qeth Device”](#) describes how to add a qeth device to an existing instance of Red Hat Enterprise Linux. [Section 20.3.2, “Adding an LCS Device”](#) describes how to add an lcs device to an existing instance of Red Hat Enterprise Linux.

20.3.1. Adding a qeth Device

The **qeth** network device driver supports IBM Z OSA-Express features in QDIO mode, HiperSockets, z/VM guest LAN, and z/VM VSWITCH.

The **qeth** device driver assigns the same interface name for Ethernet and Hipersockets devices: **enccwbus_ID**. The bus ID is composed of the channel subsystem ID, subchannel set ID, and device number, for example **enccw0.0.0a00**.

20.3.1.1. Dynamically Adding a qeth Device

To add a **qeth** device dynamically, follow these steps:

1. Determine whether the **qeth** device driver modules are loaded. The following example shows loaded **qeth** modules:

```
# lsmod | grep qeth
qeth_l3          127056  9
qeth_l2          73008  3
ipv6             492872  155ip6t_REJECT,nf_conntrack_ipv6,qeth_l3
qeth            115808  2 qeth_l3,qeth_l2
qdio             68240  1 qeth
ccwgroup         12112  2 qeth
```

If the output of the **lsmod** command shows that the **qeth** modules are not loaded, run the **modprobe** command to load them:

```
# modprobe qeth
```

2. Use the **cio_ignore** utility to remove the network channels from the list of ignored devices and make them visible to Linux:

```
# cio_ignore -r read_device_bus_id,write_device_bus_id,data_device_bus_id
```

Replace *read_device_bus_id*, *write_device_bus_id*, *data_device_bus_id* with the three device bus IDs representing a network device. For example, if the *read_device_bus_id* is **0.0.f500**, the *write_device_bus_id* is **0.0.f501**, and the *data_device_bus_id* is **0.0.f502**:

```
# cio_ignore -r 0.0.f500,0.0.f501,0.0.f502
```

3. Use the **znetconf** utility to sense and list candidate configurations for network devices:

```
# znetconf -u
Scanning for network devices...
Device IDs          Type   Card Type   CHPID Drv.
-----
0.0.f500,0.0.f501,0.0.f502 1731/01 OSA (QDIO)    00 qeth
0.0.f503,0.0.f504,0.0.f505 1731/01 OSA (QDIO)    01 qeth
0.0.0400,0.0.0401,0.0.0402 1731/05 HiperSockets  02 qeth
```

4. Select the configuration you want to work with and use **znetconf** to apply the configuration and to bring the configured group device online as network device.


```
# znetconf -a f500
Scanning for network devices...
Successfully configured device 0.0.f500 (enccw0.0.f500)
```

- Optionally, you can also pass arguments that are configured on the group device before it is set online:

```
# znetconf -a f500 -o portname=myname
Scanning for network devices...
Successfully configured device 0.0.f500 (enccw0.0.f500)
```

Now you can continue to configure the **enccw0.0.f500** network interface.

Alternatively, you can use **sysfs** attributes to set the device online as follows:

- Create a **qeth** group device:

```
# echo read_device_bus_id,write_device_bus_id,data_device_bus_id >
/sys/bus/ccwgroup/drivers/qeth/group
```

For example:

```
# echo 0.0.f500,0.0.f501,0.0.f502 > /sys/bus/ccwgroup/drivers/qeth/group
```

- Next, verify that the **qeth** group device was created properly by looking for the read channel:

```
# ls /sys/bus/ccwgroup/drivers/qeth/0.0.f500
```

You can optionally set additional parameters and features, depending on the way you are setting up your system and the features you require, such as:

- **portno**
- **layer2**
- **portname**

- Bring the device online by writing **1** to the online **sysfs** attribute:

```
# echo 1 > /sys/bus/ccwgroup/drivers/qeth/0.0.f500/online
```

- Then verify the state of the device:

```
# cat /sys/bus/ccwgroup/drivers/qeth/0.0.f500/online
1
```

A return value of **1** indicates that the device is online, while a return value **0** indicates that the device is offline.

- Find the interface name that was assigned to the device:

```
# cat /sys/bus/ccwgroup/drivers/qeth/0.0.f500/if_name
enccw0.0.f500
```

Now you can continue to configure the **enccw0.0.f500** network interface.

The following command from the **s390utils** package shows the most important settings of your **qeth** device:

```
# lsqeth enccw0.0.f500
Device name           : enccw0.0.f500
-----
card_type             : OSD_1000
cdev0                 : 0.0.f500
cdev1                 : 0.0.f501
cdev2                 : 0.0.f502
chpid                 : 76
online                : 1
portname              : OSAPORT
portno                : 0
state                 : UP (LAN ONLINE)
priority_queueing     : always queue 0
buffer_count          : 16
layer2                : 1
isolation              : none
```

20.3.1.2. Dynamically Removing a qeth Device

To remove a **qeth** device, use the **znetconf** utility. For example:

1. Use the **znetconf** utility to show you all configured network devices:

```
# znetconf -c
Device IDs           Type   Card Type   CHPID Drv. Name   State
-----
0.0.8036,0.0.8037,0.0.8038 1731/05 HiperSockets   FB qeth hsi1      online
0.0.f5f0,0.0.f5f1,0.0.f5f2 1731/01 OSD_1000     76 qeth enccw0.0.09a0 online
0.0.f500,0.0.f501,0.0.f502 1731/01 GuestLAN QDIO  00 qeth enccw0.0.f500 online
```

2. Select the network device to be removed and run **znetconf** to set the device offline and ungroup the **ccw>** group device.

```
# znetconf -r f500
Remove network device 0.0.f500 (0.0.f500,0.0.f501,0.0.f502)?
Warning: this may affect network connectivity!
Do you want to continue (y/n)?y
Successfully removed device 0.0.f500 (enccw0.0.f500)
```

3. Verify the success of the removal:

```
# znetconf -c
Device IDs           Type   Card Type   CHPID Drv. Name   State
-----
0.0.8036,0.0.8037,0.0.8038 1731/05 HiperSockets   FB qeth hsi1      online
0.0.f5f0,0.0.f5f1,0.0.f5f2 1731/01 OSD_1000     76 qeth enccw0.0.09a0 online
```

20.3.1.3. Persistently Adding a qeth Device

To make your new **qeth** device persistent, you need to create the configuration file for your new interface. The network interface configuration files are placed in the **/etc/sysconfig/network-scripts/** directory.

The network configuration files use the naming convention **ifcfg-*device***, where *device* is the value found in the **if_name** file in the **qeth** group device that was created earlier, for example **enccw0.0.09a0**. The **cio_ignore** commands are handled transparently for persistent device configurations and you do not need to free devices from the ignore list manually.

If a configuration file for another device of the same type already exists, the simplest way is to copy it to the new name and then edit it:

```
# cd /etc/sysconfig/network-scripts
# cp ifcfg-enccw0.0.09a0 ifcfg-enccw0.0.0600
```

To learn IDs of your network devices, use the **lsqeth** utility:

```
# lsqeth -p
devices          CHPID interface  cardtype      port chksum prio-q'ing rtr4 rtr6 lay'2 cnt
-----
0.0.09a0/0.0.09a1/0.0.09a2 x00  enccw0.0.09a0  Virt.NIC QDIO 0   sw   always_q_2 n/a  n/a  1
64
0.0.0600/0.0.0601/0.0.0602 x00  enccw0.0.0600  Virt.NIC QDIO 0   sw   always_q_2 n/a  n/a  1
64
```

If you do not have a similar device defined, you must create a new file. Use this example of **/etc/sysconfig/network-scripts/ifcfg-0.0.09a0** as a template:

```
# IBM QETH
DEVICE=enccw0.0.09a0
BOOTPROTO=static
IPADDR=10.12.20.136
NETMASK=255.255.255.0
ONBOOT=yes
NETTYPE=qeth
SUBCHANNELS=0.0.09a0,0.0.09a1,0.0.09a2
PORTNAME=OSAPORT
OPTIONS='layer2=1 portno=0'
MACADDR=02:00:00:23:65:1a
TYPE=Ethernet
```

Edit the new **ifcfg-0.0.0600** file as follows:

1. Modify the **DEVICE** statement to reflect the contents of the **if_name** file from your **ccw** group.
2. Modify the **IPADDR** statement to reflect the IP address of your new interface.
3. Modify the **NETMASK** statement as needed.
4. If the new interface is to be activated at boot time, then make sure **ONBOOT** is set to **yes**.
5. Make sure the **SUBCHANNELS** statement matches the hardware addresses for your qeth device. Please, note that the IDs must be specified in lowercase.
6. Modify the **PORTNAME** statement or leave it out if it is not necessary in your environment.

- You can add any valid **sysfs** attribute and its value to the **OPTIONS** parameter. The Red Hat Enterprise Linux installation program currently uses this to configure the layer mode (**layer2**) and the relative port number (**portno**) of **qeth** devices.

The **qeth** device driver default for OSA devices is now layer 2 mode. To continue using old **ifcfg** definitions that rely on the previous default of layer 3 mode, add **layer2=0** to the **OPTIONS** parameter.

/etc/sysconfig/network-scripts/ifcfg-0.0.0600

```
# IBM QETH
DEVICE=enccw0.0.0600
BOOTPROTO=static
IPADDR=192.168.70.87
NETMASK=255.255.255.0
ONBOOT=yes
NETTYPE=qeth
SUBCHANNELS=0.0.0600,0.0.0601,0.0.0602
PORTNAME=OSAPORT
OPTIONS='layer2=1 portno=0'
MACADDR=02:00:00:b3:84:ef
TYPE=Ethernet
```

Changes to an **ifcfg** file only become effective after rebooting the system or after the dynamic addition of new network device channels by changing the system's I/O configuration (for example, attaching under z/VM). Alternatively, you can trigger the activation of a **ifcfg** file for network channels which were previously not active yet, by executing the following commands:

- Use the **cio_ignore** utility to remove the network channels from the list of ignored devices and make them visible to Linux:

```
# cio_ignore -r read_device_bus_id,write_device_bus_id,data_device_bus_id
```

Replace *read_device_bus_id*, *write_device_bus_id*, *data_device_bus_id* with the three device bus IDs representing a network device. For example, if the *read_device_bus_id* is **0.0.0600**, the *write_device_bus_id* is **0.0.0601**, and the *data_device_bus_id* is **0.0.0602**:

```
# cio_ignore -r 0.0.0600,0.0.0601,0.0.0602
```

- To trigger the uevent that activates the change, issue:

```
# echo add > /sys/bus/ccw/devices/read-channel/uevent
```

For example:

```
# echo add > /sys/bus/ccw/devices/0.0.0600/uevent
```

- Check the status of the network device:

```
# lsqeth
```

- Now start the new interface:

```
# ifup encw0.0.0600
```

5. Check the status of the interface:

```
# ip addr show encw0.0.0600
3: encw0.0.0600: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast
state UP group default qlen 1000
link/ether 3c:97:0e:51:38:17 brd ff:ff:ff:ff:ff:ff
inet 10.85.1.245/24 brd 10.34.3.255 scope global dynamic encw0.0.0600
valid_lft 81487sec preferred_lft 81487sec
inet6 1574:12:5:1185:3e97:eff:fe51:3817/64 scope global noprefixroute dynamic
valid_lft 2591994sec preferred_lft 604794sec
inet6 fe45::a455:eff:d078:3847/64 scope link
valid_lft forever preferred_lft forever
```

6. Check the routing for the new interface:

```
# ip route
default via 10.85.1.245 dev encw0.0.0600 proto static metric 1024
12.34.4.95/24 dev enp0s25 proto kernel scope link src 12.34.4.201
12.38.4.128 via 12.38.19.254 dev enp0s25 proto dhcp metric 1
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1
```

7. Verify your changes by using the **ping** utility to ping the gateway or another host on the subnet of the new device:

```
# ping -c 1 192.168.70.8
PING 192.168.70.8 (192.168.70.8) 56(84) bytes of data.
64 bytes from 192.168.70.8: icmp_seq=0 ttl=63 time=8.07 ms
```

8. If the default route information has changed, you must also update `/etc/sysconfig/network` accordingly.

20.3.2. Adding an LCS Device

The *LAN channel station* (LCS) device driver supports 1000Base-T Ethernet on the OSA-Express2 and OSA-Express 3 features.

The **LCS** device driver assigns the following interface name for OSA-Express Fast Ethernet and Gigabit Ethernet devices: **encwbus_ID**. The bus ID is composed of the channel subsystem ID, subchannel set ID, and device number, for example **encw0.0.0a00**.

20.3.2.1. Dynamically Adding an LCS Device

1. Load the device driver:

```
# modprobe lcs
```

2. Use the **cio_ignore** utility to remove the network channels from the list of ignored devices and make them visible to Linux:

```
# cio_ignore -r read_device_bus_id,write_device_bus_id
```

Replace *read_device_bus_id* and *write_device_bus_id* with the two device bus IDs representing a network device. For example:

```
# cio_ignore -r 0.0.09a0,0.0.09a1
```

3. Create the group device:

```
# echo read_device_bus_id,write_device_bus_id > /sys/bus/ccwgroup/drivers/lcs/group
```

4. Configure the device. OSA cards can provide up to 16 ports for a single CHPID. By default, the LCS group device uses port **0**. To use a different port, issue a command similar to the following:

```
# echo portno > /sys/bus/ccwgroup/drivers/lcs/device_bus_id/portno
```

Replace *portno* with the port number you want to use.

5. Set the device online:

```
# echo 1 > /sys/bus/ccwgroup/drivers/lcs/read_device_bus_id/online
```

6. To find out what network device name has been assigned, enter the command:

```
# ls -l /sys/bus/ccwgroup/drivers/lcs/read_device_bus_ID/net/  
drwxr-xr-x 4 root root 0 2010-04-22 16:54 enc cw0.0.0600
```

20.3.2.2. Persistently Adding an LCS Device

The **cio_ignore** commands are handled transparently for persistent device configurations and you do not need to free devices from the ignore list manually.

To add an LCS device persistently, follow these steps:

1. Create a configuration script as file in **/etc/sysconfig/network-scripts/** with a name like **ifcfg-device**, where *device* is the value found in the **if_name** file in the **qeth** group device that was created earlier, for example **enc cw0.0.09a0**. The file should look similar to the following:

```
# IBM LCS  
DEVICE=enc cw0.0.09a0  
BOOTPROTO=static  
IPADDR=10.12.20.136  
NETMASK=255.255.255.0  
ONBOOT=yes  
NETTYPE=lcs  
SUBCHANNELS=0.0.09a0,0.0.09a1  
PORTNAME=0  
OPTIONS="  
TYPE=Ethernet
```

2. Modify the value of **PORTNAME** to reflect the LCS port number (**portno**) you would like to use. You can add any valid lcs sysfs attribute and its value to the optional **OPTIONS** parameter. See [Section 20.3.1.3, "Persistently Adding a qeth Device"](#) for the syntax.
3. Set the **DEVICE** parameter as follows:

```
DEVICE=enccwbus_ID
```

4. Issue an **ifup** command to activate the device:

```
# ifup enccwbus_ID
```

Changes to an **ifcfg** file only become effective after rebooting the system. You can trigger the activation of a **ifcfg** file for network channels by executing the following commands:

1. Use the **cio_ignore** utility to remove the LCS device adapter from the list of ignored devices and make it visible to Linux:

```
# cio_ignore -r read_device_bus_id,write_device_bus_id
```

Replace *read_device_bus_id* and *write_device_bus_id* with the device bus IDs of the LCS device. For example:

```
# cio_ignore -r 0.0.09a0,0.0.09a1
```

2. To trigger the uevent that activates the change, issue:

```
# echo add > /sys/bus/ccw/devices/read-channel/uevent
```

For example:

```
# echo add > /sys/bus/ccw/devices/0.0.09a0/uevent
```

20.3.3. Configuring a IBM Z Network Device for Network Root File System

To add a network device that is required to access the root file system, you only have to change the boot options. The boot options can be in a parameter file (see [Chapter 21, Parameter and Configuration Files on IBM Z](#)) or part of a **zipl.conf** on a DASD or FCP-attached SCSI LUN prepared with the **zipl** boot loader. There is no need to recreate the initramfs.

Dracut, the **mkinitrd** successor that provides the functionality in the initramfs that in turn replaces **initrd**, provides a boot parameter to activate network devices on IBM Z early in the boot process: **rd.znet=**.

As input, this parameter takes a comma-separated list of the **NETTYPE** (qeth, lcs, etc), two (lcs, etc) or three (qeth) device bus IDs, and optional additional parameters consisting of key-value pairs corresponding to network device sysfs attributes. This parameter configures and activates the IBM Z network hardware. The configuration of IP addresses and other network specifics works the same as for other platforms. See the **dracut** documentation for more details.

The **cio_ignore** commands for the network channels are handled transparently on boot.

Example boot options for a root file system accessed over the network through NFS:

```
root=10.16.105.196:/nfs/nfs_root cio_ignore=all,!condev
rd.znet=qeth,0.0.0a00,0.0.0a01,0.0.0a02,layer2=1,portno=0,portname=OSAPORT
ip=10.16.105.197:10.16.105.196:10.16.111.254:255.255.248.0:nfs-server.subdomain.domain:enccw0.0.09a0:none rd_NO_LUKS rd_NO_LVM rd_NO_MD rd_NO_DM LANG=en_US.UTF-8
SYSEFONT=latacyrheb-sun16 KEYTABLE=us
```

■

CHAPTER 21. PARAMETER AND CONFIGURATION FILES ON IBM Z

The IBM Z architecture can use a customized parameter file to pass boot parameters to the kernel and the installation program. This section describes the contents of this parameter file.

You need only read this section if you intend to change the shipped parameter file. You need to change the parameter file if you want to:

- install unattended with Kickstart.
- choose non-default installation settings that are not accessible through the installation program's interactive user interface, such as rescue mode.

The parameter file can be used to set up networking non-interactively before the installation program (loader and **Anaconda**) starts.

The kernel parameter file is limited to 895 characters plus an end-of-line character. The parameter file can be variable or fixed record format. Fixed record format increases the file size by padding each line up to the record length. Should you encounter problems with the installation program not recognizing all specified parameters in LPAR environments, you can try to put all parameters in one single line or start and end each line with a space character.

The parameter file contains kernel parameters, such as **ro**, and parameters for the installation process, such as **vncpassword=test** or **vnc**.

21.1. REQUIRED PARAMETERS

The following parameters are required and must be included in the parameter file. They are also provided in the file **generic.prm** in directory **images/** of the installation DVD:

ro

mounts the root file system, which is a RAM disk, read-only.

ramdisk_size=size

modifies the memory size reserved for the RAM disk to ensure that the Red Hat Enterprise Linux installation program fits within it. For example: **ramdisk_size=40000**.

The **generic.prm** file also contains the additional parameter **cio_ignore=all,!condev**. This setting speeds up boot and device detection on systems with many devices. The installation program transparently handles the activation of ignored devices.



IMPORTANT

To avoid installation problems arising from **cio_ignore** support not being implemented throughout the entire stack, adapt the **cio_ignore=** parameter value to your system or remove the parameter entirely from your parameter file used for booting (IPL) the installation program.

21.2. THE Z/VM CONFIGURATION FILE

This applies only if installing under z/VM. Under z/VM, you can use a configuration file on a CMS-

formatted disk. The purpose of the CMS configuration file is to save space in the parameter file by moving the parameters that configure the initial network setup, the DASD, and the FCP specification out of the parameter file (see [Section 21.3, "Installation Network Parameters"](#)).

Each line of the CMS configuration file contains a single variable and its associated value, in the following shell-style syntax: ***variable=value***.

You must also add the ***CMSDASD*** and ***CMSCONFFILE*** parameters to the parameter file. These parameters point the installation program to the configuration file:

CMSDASD=cmsdasd_address

Where *cmsdasd_address* is the device number of a CMS-formatted disk that contains the configuration file. This is usually the CMS user's **A** disk.

For example: ***CMSDASD=191***

CMSCONFFILE=configuration_file

Where *configuration_file* is the name of the configuration file. **This value must be specified in lower case.** It is specified in a Linux file name format: ***CMS_file_name.CMS_file_type***.

The CMS file **REDHAT CONF** is specified as **redhat.conf**. The CMS file name and the file type can each be from one to eight characters that follow the CMS conventions.

For example: ***CMSCONFFILE=redhat.conf***

21.3. INSTALLATION NETWORK PARAMETERS

The following parameters can be used to set up the preliminary network automatically and can be defined in the CMS configuration file. The parameters in this section are the only parameters that can also be used in a CMS configuration file. All other parameters in other sections must be specified in the parameter file.

NETTYPE="type"

Where *type* must be one of the following: **qeth**, **lcs**, or **ctc**. The default is **qeth**.

Choose **lcs** for:

- OSA-2 Ethernet/Token Ring
- OSA-Express Fast Ethernet in non-QDIO mode
- OSA-Express High Speed Token Ring in non-QDIO mode
- Gigabit Ethernet in non-QDIO mode

Choose **qeth** for:

- OSA-Express Fast Ethernet
- Gigabit Ethernet (including 1000Base-T)
- High Speed Token Ring
- HiperSockets

- ATM (running Ethernet LAN emulation)

SUBCHANNELS="device_bus_IDs"

Where *device_bus_IDs* is a comma-separated list of two or three device bus IDs. The IDs must be specified in lowercase.

Provides required device bus IDs for the various network interfaces:

```
qeth: SUBCHANNELS="read_device_bus_id,write_device_bus_id,data_device_bus_id"
lcs or ctc: SUBCHANNELS="read_device_bus_id,write_device_bus_id"
```

For example (a sample qeth SUBCHANNEL statement):

```
SUBCHANNELS="0.0.f5f0,0.0.f5f1,0.0.f5f2"
```

PORTNAME="osa_portname", PORTNAME="lcs_portnumber"

This variable supports OSA devices operating in qdio mode or in non-qdio mode.

When using qdio mode (**NETTYPE="qeth"**), *osa_portname* is the portname specified on the OSA device when operating in qeth mode.

When using non-qdio mode (**NETTYPE="lcs"**), *lcs_portnumber* is used to pass the relative port number as a decimal integer in the range of 0 through 15.

PORTNO="portnumber"

You can add either **PORTNO="0"** (to use port 0) or **PORTNO="1"** (to use port 1 of OSA features with two ports per CHPID) to the CMS configuration file to avoid being prompted for the mode.

LAYER2="value"

Where *value* can be **0** or **1**.

Use **LAYER2="0"** to operate an OSA or HiperSockets device in layer 3 mode (**NETTYPE="qeth"**). Use **LAYER2="1"** for layer 2 mode. For virtual network devices under z/VM this setting must match the definition of the GuestLAN or VSWITCH to which the device is coupled.

To use network services that operate on layer 2 (the Data Link Layer or its MAC sublayer) such as DHCP, layer 2 mode is a good choice.

The qeth device driver default for OSA devices is now layer 2 mode. To continue using the previous default of layer 3 mode, set **LAYER2="0"** explicitly.

VSWITCH="value"

Where *value* can be **0** or **1**.

Specify **VSWITCH="1"** when connecting to a z/VM VSWITCH or GuestLAN, or **VSWITCH="0"** (or nothing at all) when using directly attached real OSA or directly attached real HiperSockets.

MACADDR="MAC_address"

If you specify **LAYER2="1"** and **VSWITCH="0"**, you can optionally use this parameter to specify a MAC address. Linux requires six colon-separated octets as pairs lower case hex digits - for example, **MACADDR=62:a3:18:e7:bc:5f**. Note that this is different from the notation used by z/VM.

If you specify **LAYER2="1"** and **VSWITCH="1"**, you must not specify the **MACADDR**, because z/VM assigns a unique MAC address to virtual network devices in layer 2 mode.

CTCProt="value"

Where *value* can be **0**, **1**, or **3**.

Specifies the CTC protocol for **NETTYPE="ctc"**. The default is **0**.

HOSTNAME="string"

Where *string* is the host name of the newly-installed Linux instance.

IPADDR="IP"

Where *IP* is the IP address of the new Linux instance.

NETMASK="netmask"

Where *netmask* is the netmask.

The netmask supports the syntax of a prefix integer (from 1 to 32) as specified in IPv4 *classless interdomain routing* (CIDR). For example, you can specify **24** instead of **255.255.255.0**, or **20** instead of **255.255.240.0**.

GATEWAY="gw"

Where *gw* is the gateway IP address for this network device.

MTU="mtu"

Where *mtu* is the *Maximum Transmission Unit* (MTU) for this network device.

DNS="server1:server2:additional_server_terms:serverN"

Where *"server1:server2:additional_server_terms:serverN"* is a list of DNS servers, separated by colons. For example:

```
DNS="10.1.2.3:10.3.2.1"
```

SEARCHDNS="domain1:domain2:additional_dns_terms:domainN"

Where *"domain1:domain2:additional_dns_terms:domainN"* is a list of the search domains, separated by colons. For example:

```
SEARCHDNS="subdomain.domain:domain"
```

You only need to specify **SEARCHDNS=** if you specify the **DNS=** parameter.

DASD=

Defines the DASD or range of DASDs to configure for the installation.

The installation program supports a comma-separated list of device bus IDs or of ranges of device bus IDs with the optional attributes **ro**, **diag**, **erplog**, and **failfast**. Optionally, you can abbreviate device bus IDs to device numbers with leading zeros stripped. Any optional attributes should be separated by colons and enclosed in parentheses. Optional attributes follow a device bus ID or a range of device bus IDs.

The only supported global option is **autodetect**. This does not support the specification of non-existent DASDs to reserve kernel device names for later addition of DASDs. Use persistent DASD device names (for example **/dev/disk/by-path/...**) to enable transparent addition of disks later. Other global options such as **probeonly**, **nopav**, or **nofcx** are not supported by the installation program.

Only specify those DASDs that you really need to install your system. All unformatted DASDs specified here must be formatted after a confirmation later on in the installation program (see [Section 18.16.1.1, “DASD Low-level Formatting”](#)). Add any data DASDs that are not needed for the root file system or the **/boot** partition after installation as described in [Section 20.1.3.2, “DASDs That Are Not Part of the Root File System”](#).

For example:

```
DASD="eb1c,0.0.a000-0.0.a003,eb10-eb14(diag),0.0.ab1c(ro:diag)"
```

For FCP-only environments, remove the **DASD=** option from the CMS configuration file to indicate no DASD is present.

FCP_n="device_bus_ID WWPN FCP_LUN"

Where:

- *n* is typically an integer value (for example **FCP_1** or **FCP_2**) but could be any string with alphabetic or numeric characters or underscores.
- *device_bus_ID* specifies the device bus ID of the FCP device representing the *host bus adapter* (HBA) (for example **0.0.fc00** for device fc00).
- *WWPN* is the world wide port name used for routing (often in conjunction with multipathing) and is as a 16-digit hex value (for example **0x50050763050b073d**).
- *FCP_LUN* refers to the storage logical unit identifier and is specified as a 16-digit hexadecimal value padded with zeroes to the right (for example **0x4020400100000000**).

These variables can be used on systems with FCP devices to activate FCP LUNs such as SCSI disks. Additional FCP LUNs can be activated during the installation interactively or by means of a Kickstart file. An example value looks similar to the following:

```
FCP_1="0.0.fc00 0x50050763050b073d 0x4020400100000000"
```



IMPORTANT

Each of the values used in the FCP parameters (for example **FCP_1** or **FCP_2**) are site-specific and are normally supplied by the FCP storage administrator.

The installation program prompts you for any required parameters not specified in the parameter or configuration file except for FCP_n.

21.4. PARAMETERS FOR KICKSTART INSTALLATIONS

The following parameters can be defined in a parameter file but do not work in a CMS configuration file.

inst.ks=URL

References a Kickstart file, which usually resides on the network for Linux installations on IBM Z. Replace *URL* with the full path including the file name of the Kickstart file. This parameter activates automatic installation with Kickstart. See [Kickstart Boot Options](#) and [Section 27.2.5, “Starting the Kickstart Installation”](#) for more details.

RUNKS=value



IMPORTANT

This parameter is deprecated. If you use it in a Kickstart file, it will be ignored. Only the ***inst.ks=*** parameter is necessary to start a Kickstart installation on IBM Z.

Where *value* is defined as *1* if you want to run the loader automatically on the Linux console without having to log in over the network with SSH. To use ***RUNKS=1***, the console must either support full-screen or the ***inst.cmdline*** option (below) should be used. The latter applies for the 3270 terminal under z/VM or the operating system messages console for LPAR. We recommend ***RUNKS=1*** for fully automatic installations with Kickstart. When ***RUNKS=1*** is set, the installation program automatically continues in case of parameter errors and does not interrupt unattended installations by prompting for user interaction.

Leave out the parameter or specify ***RUNKS=0*** otherwise.

inst.cmdline

When this option is specified, output on line-mode terminals (such as 3270 under z/VM or operating system messages for LPAR) becomes readable, as the installation program disables escape terminal sequences that are only applicable to UNIX-like consoles. This requires installation with a Kickstart file that answers all questions, because the installation program does not support interactive user input in cmdline mode.

Ensure that your Kickstart file contains all required parameters before you use the ***inst.cmdline*** option. If a required command is missing, the installation will fail. See [Chapter 27, Kickstart Installations](#) for details.

21.5. MISCELLANEOUS PARAMETERS

The following parameters can be defined in a parameter file but do not work in a CMS configuration file.

rd.live.check

Turns on testing of an ISO-based installation source; for example, when booted from an FCP-attached DVD or using ***inst.repo=*** with an ISO on local hard disk or mounted with NFS.

nompath

Disables support for multipath devices.

proxy=[protocol://][username[:password]@]host[:port]

Specify a proxy to use with installation over HTTP, HTTPS, or FTP.

inst.rescue

Boot into a rescue system running from a RAM disk that can be used to fix and restore an installed system.

inst.stage2=URL

Specifies a path to an **install.img** file instead of to an installation source. Otherwise, follows the same syntax as **inst.repo=**. If **inst.stage2** is specified, it typically takes precedence over other methods of finding **install.img**. However, if **Anaconda** finds **install.img** on local media, the **inst.stage2** URL will be ignored.

If **inst.stage2** is not specified and **install.img** cannot be found locally, **Anaconda** looks to the location given by **inst.repo=** or **method=**.

If only **inst.stage2=** is given without **inst.repo=** or **method=**, **Anaconda** uses whatever repos the installed system would have enabled by default for installation.

Use the option multiple times to specify multiple HTTP, HTTPS or FTP sources. The HTTP, HTTPS or FTP paths are then tried sequentially until one succeeds:

```
inst.stage2=host1/install.img inst.stage2=host2/install.img inst.stage3=host3/install.img
```

inst.syslog=IP/hostname[:port]

Sends log messages to a remote syslog server.

The boot parameters described here are the most useful for installations and trouble shooting on IBM Z, but only a subset of those that influence the installation program. See [Chapter 23, Boot Options](#) for a more complete list of available boot parameters.

21.6. SAMPLE PARAMETER FILE AND CMS CONFIGURATION FILE

To change the parameter file, begin by extending the shipped **generic.prm** file.

Example of **generic.prm** file:

```
ro ramdisk_size=40000 cio_ignore=all,!condev
CMSDASD="191" CMSCONFFILE="redhat.conf"
vnc
inst.repo=http://example.com/path/to/repository
```

Example of **redhat.conf** file configuring a QETH network device (pointed to by **CMSCONFFILE** in **generic.prm**):

```
NETTYPE="qeth"
SUBCHANNELS="0.0.0600,0.0.0601,0.0.0602"
PORTNAME="FOOBAR"
PORTNO="0"
LAYER2="1"
MACADDR="02:00:be:3a:01:f3"
HOSTNAME="foobar.systemz.example.com"
IPADDR="192.168.17.115"
NETMASK="255.255.255.0"
GATEWAY="192.168.17.254"
DNS="192.168.17.1"
SEARCHDNS="systemz.example.com:example.com"
DASD="200-203"
```

CHAPTER 22. IBM Z REFERENCES

22.1. IBM Z PUBLICATIONS

Current versions of the Linux on IBM Z publications can be found at http://www.ibm.com/developerworks/linux/linux390/documentation_red_hat.html. They include:

Linux on System z - How to use FC-attached SCSI devices with Linux on System z9 and zSeries IBM . 2008. SC33-8413.

Linux on System z - How to Improve Performance with PAV IBM . 2008. SC33-8414.

z/VM - Getting Started with Linux on System z IBM . 2009. SC24-6194.

22.2. IBM REDBOOKS PUBLICATIONS FOR IBM Z

Current versions of IBM Redbooks publications can be found at <http://www.redbooks.ibm.com/>. They include:

Introductory publications

Introduction to the New Mainframe: z/VM Basics IBM Redbooks . 2007. SG24-7316.

Practical Migration to Linux on System z. IBM Redbooks . 2009. SG24-7727.

Performance and high availability

Linux on IBM System z: Performance Measurement and Tuning IBM Redbooks . 2011. SG24-6926.

Achieving High Availability on Linux for System z with Linux-HA Release 2 IBM Redbooks . 2009. SG24-7711.

Security

Security for Linux on System z. IBM Redbooks . 2013. SG24-7728.

Networking

IBM System z Connectivity Handbook. IBM Redbooks . 2013. SG24-5444.

OSA Express Implementation Guide IBM Redbooks . 2009. SG24-5948.

HiperSockets Implementation Guide. IBM Redbooks . 2007. SG24-6816.

Fibre Channel Protocol for Linux and z/VM on IBM System z IBM Redbooks . 2007. SG24-7266.

22.3. ONLINE RESOURCES

For z/VM publications, refer to <http://www.vm.ibm.com/library/> .

For IBM Z I/O connectivity information, refer to <http://www.ibm.com/systems/z/hardware/connectivity/index.html> .

For IBM Z cryptographic coprocessor information, refer to <http://www.ibm.com/security/cryptocards/> .

For IBM Z DASD storage information, refer to http://www-01.ibm.com/support/knowledgecenter/linuxonibm/com.ibm.linux.z.lgdd/lgdd_t_dasd_wrk.html.

PART IV. ADVANCED INSTALLATION OPTIONS

This part of the *Red Hat Enterprise Linux Installation Guide* covers more advanced or uncommon methods of installing Red Hat Enterprise Linux, including:

- customizing the installation program's behavior by specifying boot options
- setting up a PXE server to boot the installation program over a network
- installing with remote access through VNC
- using a Kickstart file to automate the installation process
- installing into a disk image instead of a physical drive
- upgrading a previous release of Red Hat Enterprise Linux to the current version

CHAPTER 23. BOOT OPTIONS

The Red Hat Enterprise Linux installation system includes a range of boot options for administrators, which modify the default behavior of the installation program by enabling (or disabling) certain functions. To use boot options, append them to the boot command line, as described in [Section 23.1, “Configuring the Installation System at the Boot Menu”](#). Multiple options added to the boot line need to be separated by a single space.

There are two basic types of options described in this chapter:

- Options presented as ending with an “equals” sign (=) require a value to be specified – they cannot be used on their own. For example, the **inst.vncpassword=** option must also contain a value (in this case, a password). The correct form is therefore **inst.vncpassword=password**. On its own, without a password specified, the option is invalid.
- Options presented without the “=” sign do not accept any values or parameters. For example, the **rd.live.check** option forces **Anaconda** to verify the installation media before starting the installation; if this option is present, the check will be performed, and if it is not present, the check will be skipped.

23.1. CONFIGURING THE INSTALLATION SYSTEM AT THE BOOT MENU



NOTE

The exact way to specify custom boot options is different on every system architecture. For architecture-specific instructions about editing boot options, see:

- [Section 7.2, “The Boot Menu”](#) for 64-bit AMD, Intel and ARM systems
- [Section 12.1, “The Boot Menu”](#) for IBM Power Systems servers
- [Chapter 21, *Parameter and Configuration Files on IBM Z*](#) for IBM Z

There are several different ways to edit boot options at the boot menu (that is, the menu which appears after you boot the installation media):

- The **boot:** prompt, accessed by pressing the **Esc** key anywhere in the boot menu. When using this prompt, the first option must always specify the installation program image file to be loaded. In most cases, the image can be specified using the **linux** keyword. After that, additional options can be specified as needed.

Pressing the **Tab** key at this prompt will display help in the form of usable commands where applicable. To start the installation with your options, press the **Enter** key. To return from the **boot:** prompt to the boot menu, restart the computer and boot from the installation media again.

- The **>** prompt on BIOS-based AMD64 and Intel 64 systems, accessed by highlighting an entry in the boot menu and pressing the **Tab** key. Unlike the **boot:** prompt, this prompt allows you to edit a predefined set of boot options. For example, if you highlight the entry labeled **Test this media & install Red Hat Enterprise Linux 7.5**, a full set of options used by this menu entry will be displayed on the prompt, allowing you to add your own options.

Pressing **Enter** will start the installation using the options you specified. To cancel editing and return to the boot menu, press the **Esc** key at any time.

- The **GRUB2** menu on UEFI-based 64-bit AMD, Intel and ARM systems. If your system uses UEFI, you can edit boot options by highlighting an entry and pressing the **e** key. When you finish editing, press **F10** or **Ctrl+X** to start the installation using the options you specified.

In addition to the options described in this chapter, the boot prompt also accepts **dracut** kernel options. A list of these options is available as the **dracut.cmdline(7)** man page.



NOTE

Boot options specific to the installation program always start with **inst.** in this guide. Currently, this prefix is optional, for example, **resolution=1024x768** will work exactly the same as **inst.resolution=1024x768**. However, it is expected that the **inst.** prefix will be mandatory in future releases.

Specifying the Installation Source

inst.repo=

Specifies the installation source – that is, a location where the installation program can find the images and packages it requires. For example:

```
inst.repo=cdrom
```

The target must be either:

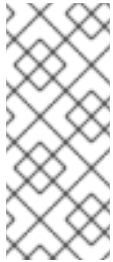
- an installable tree, which is a directory structure containing the installation program's images, packages and repodata as well as a valid **.treeinfo** file
- a DVD (a physical disk present in the system's DVD drive)
- an ISO image of the full Red Hat Enterprise Linux installation DVD, placed on a hard drive or a network location accessible from the installation system (requires specifying **NFS Server** as the installation source)

This option allows for the configuration of different installation methods using different formats. The syntax is described in the table below.

Table 23.1. Installation Sources

Installation source	Option format
Any CD/DVD drive	inst.repo=cdrom
Specific CD/DVD drive	inst.repo=cdrom:device
Hard Drive	inst.repo=hd:device:/path
HMC	inst.repo=hmc
HTTP Server	inst.repo=http://host/path

Installation source	Option format
HTTPS Server	inst.repo=https://<i>host/path</i>
FTP Server	inst.repo=ftp://<i>username:password@host/path</i>
NFS Server	inst.repo=nfs:[<i>options</i>]:<i>server:/path</i> ^[a]
[a] This option uses NFS protocol version 3 by default. To use a different version, add nfsvers=<i>X</i> to <i>options</i> , replacing <i>X</i> with the version number that you want to use.	



NOTE

In previous releases of Red Hat Enterprise Linux, there were separate options for an installable tree accessible by NFS (the **nfs** option) and an ISO image located on an NFS source (the **nfsiso** option). In Red Hat Enterprise Linux 7, the installation program can automatically detect whether the source is an installable tree or a directory containing an ISO image, and the **nfsiso** option is deprecated.

Disk device names can be set using the following formats:

- Kernel device name, for example **/dev/sda1** or **sdb2**
- File system label, for example **LABEL=Flash** or **LABEL=RHEL7**
- File system UUID, for example **UUID=8176c7bf-04ff-403a-a832-9557f94e61db**

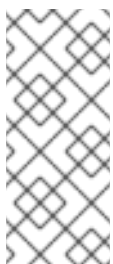
Non-alphanumeric characters must be represented as **\xNN**, where *NN* is the hexadecimal representation of the character. For example, **\x20** is a white space (" ").

inst.stage2=

Specifies the location of the installation program runtime image to be loaded. The syntax is the same as in [Specifying the Installation Source](#). This option expects a path to a directory containing a valid **.treeinfo** file; the location of the runtime image will be read from this file if found. If a **.treeinfo** file is not available, **Anaconda** will try to load the image from **LiveOS/squashfs.img**.

Use the option multiple times to specify multiple HTTP, HTTPS or FTP sources.

```
inst.stage2=host1/install.img inst.stage2=host2/install.img inst.stage2=host3/install.img
```



NOTE

By default, the **inst.stage2=** boot option is used on the installation media and set to a specific label (for example, **inst.stage2=hd:LABEL=RHEL7\x20Server.x86_64**). If you modify the default label of the file system containing the runtime image, or if you use a customized procedure to boot the installation system, you must ensure this option is set to the correct value.

inst.dd=

If you need to perform a driver update during the installation, use the **inst.dd=** option. It can be used multiple times. The location of a driver RPM package can be specified using any of the formats detailed in [Specifying the Installation Source](#). With the exception of the **inst.dd=cdrom** option, the device name must always be specified. For example:

```
inst.dd=/dev/sdb1
```

Using this option without any parameters (only as **inst.dd**) will prompt the installation program to ask you for a driver update disk with an interactive menu.

Driver disks can also be loaded from a hard disk drive or a similar device instead of being loaded over the network or from **initrd**. Follow this procedure:

1. Load the driver disk on a hard disk drive, a USB or any similar device.
2. Set the label, for example, *DD*, to this device.
3. Start the installation with:

```
inst.dd=hd:LABEL=DD:/dd.rpm
```

as the boot argument.

Replace *DD* with a specific label and replace *dd.rpm* with a specific name. Use anything supported by the **inst.repo** command instead of *LABEL* to specify your hard disk drive.

For more information about driver updates during the installation, see [Chapter 6, Updating Drivers During Installation on AMD64 and Intel 64 Systems](#) for AMD64 and Intel 64 systems and [Chapter 11, Updating Drivers During Installation on IBM Power Systems](#) for IBM Power Systems servers.

Kickstart Boot Options

inst.ks=

Gives the location of a Kickstart file to be used to automate the installation. Locations can be specified using any of the formats valid for **inst.repo**. See [Specifying the Installation Source](#) for details.

Use the option multiple times to specify multiple HTTP, HTTPS and FTP sources. If multiple HTTP, HTTPS and FTP locations are specified, the locations are tried sequentially until one succeeds:

```
inst.ks=host1/directory/ks.cfg inst.ks=host2/directory/ks.cfg inst.ks=host3/directory/ks.cfg
```

If you only specify a device and not a path, the installation program will look for the Kickstart file in **/ks.cfg** on the specified device. If you use this option without specifying a device, the installation program will use the following:

```
inst.ks=nfs:next-server:/filename
```

In the above example, *next-server* is the DHCP **next-server** option or the IP address of the DHCP server itself, and *filename* is the DHCP **filename** option, or **/kickstart/**. If the given file name ends with the */* character, **ip-kickstart** is appended. For example:

Table 23.2. Default Kickstart File Location

DHCP server address	Client address	Kickstart file location
192.168.122.1	192.168.122.100	192.168.122.1:/kickstart/192.168.122.100-kickstart

Additionally, starting with Red Hat Enterprise Linux 7.2, the installer will attempt to load a Kickstart file named **ks.cfg** from a volume with a label of **OEMDRV** if present. If your Kickstart file is in this location, you do not need to use the **inst.ks=** boot option at all.

inst.ks.sendmac

Adds headers to outgoing **HTTP** requests with the MAC addresses of all network interfaces. For example:

```
X-RHN-Provisioning-MAC-0: eth0 01:23:45:67:89:ab
```

This can be useful when using **inst.ks=http** to provision systems.

inst.ks.sendsn

Adds a header to outgoing **HTTP** requests. This header will contain the system's serial number, read from **/sys/class/dmi/id/product_serial**. The header has the following syntax:

```
X-System-Serial-Number: R8VA23D
```

Console, Environment and Display Options

console=

This kernel option specifies a device to be used as the primary console. For example, to use a console on the first serial port, use **console=ttyS0**. This option should be used along with the **inst.text** option.

You can use this option multiple times. In that case, the boot message will be displayed on all specified consoles, but only the last one will be used by the installation program afterwards. For example, if you specify **console=ttyS0 console=ttyS1**, the installation program will use **ttyS1**.

noshell

Disables access to the root shell during the installation. This is useful with automated (Kickstart) installations - if you use this option, a user can watch the installation progress, but they cannot interfere with it by accessing the root shell by pressing **Ctrl+Alt+F2**.

inst.lang=

Sets the language to be used during the installation. Language codes are the same as the ones used in the **lang** Kickstart command as described in [Section 27.3.1, "Kickstart Commands and Options"](#). On systems where the **system-config-language** package is installed, a list of valid values can also be found in **/usr/share/system-config-language/locale-list**.

inst.geoloc=

Configures geolocation usage in the installation program. Geolocation is used to preset the language and time zone, and uses the following syntax: **inst.geoloc=value**

The *value* parameter can be any of the following:

Table 23.3. Valid Values for the `inst.geoloc` Option

Disable geolocation	<code>inst.geoloc=0</code>
Use the Fedora GeoIP API	<code>inst.geoloc=provider_fedora_geoip</code>
Use the Hostip.info GeoIP API	<code>inst.geoloc=provider_hostip</code>

If this option is not specified, **Anaconda** will use **`provider_fedora_geoip`**.

`inst.keymap=`

Specifies the keyboard layout to be used by the installation program. Layout codes are the same as the ones used in the **`keyboard`** Kickstart command as described in [Section 27.3.1, “Kickstart Commands and Options”](#).

`inst.text`

Forces the installation program to run in text mode instead of graphical mode. The text user interface is limited, for example, it does not allow you to modify the partition layout or set up LVM. When installing a system on a machine with a limited graphical capabilities, it is recommended to use VNC as described in [Enabling Remote Access](#).

`inst.cmdline`

Forces the installation program to run in command line mode. This mode does not allow any interaction, all options must be specified in a Kickstart file or on the command line.

`inst.graphical`

Forces the installation program to run in graphical mode. This mode is the default.

`inst.resolution=`

Specifies the screen resolution in graphical mode. The format is *NxM*, where *N* is the screen width and *M* is the screen height (in pixels). The lowest supported resolution is **`800x600`**.

`inst.headless`

Specifies that the machine being installed onto does not have any display hardware. In other words, this option prevents the installation program from trying to detect a screen.

`inst.xdriver=`

Specifies the name of the **X** driver to be used both during the installation and on the installed system.

`inst.usefbx`

Tells the installation program to use the frame buffer **X** driver instead of a hardware-specific driver. This option is equivalent to **`inst.xdriver=fbdev`**.

`modprobe.blacklist=`

Blacklists (completely disables) one or more drivers. Drivers (mods) disabled using this option will be prevented from loading when the installation starts, and after the installation finishes, the installed

system will keep these settings. The blacklisted drivers can then be found in the `/etc/modprobe.d/` directory.

Use a comma-separated list to disable multiple drivers. For example:

```
modprobe.blacklist=ahci,firewire_ohci
```

inst.sshd=0

By default, **sshd** is only automatically started on IBM Z, and on other architectures, **sshd** is not started unless the **inst.sshd** option is used. This option prevents **sshd** from starting automatically on IBM Z.

inst.sshd

Starts the **sshd** service during the installation, which allows you to connect to the system during the installation using **SSH** and monitor its progress. For more information on SSH, see the **ssh(1)** man page and the corresponding chapter in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#). By default, **sshd** is only automatically started on IBM Z, and on other architectures, **sshd** is not started unless the **inst.sshd** option is used.



NOTE

During the installation, the **root** account has no password by default. You can set a root password to be used during the installation with the **sshpw** Kickstart command as described in [Section 27.3.1, "Kickstart Commands and Options"](#).

inst.kdump_addon=

Enables or disables the **Kdump** configuration screen (add-on) in the installer. This screen is enabled by default; use **inst.kdump_addon=off** to disable it. Note that disabling the add-on will disable the **Kdump** screens in both the graphical and text-based interface as well as the **%addon com_redhat_kdump** Kickstart command.

Network Boot Options

Initial network initialization is handled by **dracut**. This section only lists some of the more commonly used options; for a complete list, see the **dracut.cmdline(7)** man page. Additional information on networking is also available in [Red Hat Enterprise Linux 7 Networking Guide](#).

ip=

Configures one or more network interfaces. To configure multiple interfaces, you can use the **ip** option multiple times - once for each interface. If multiple interfaces are configured, you must also use the option **rd.neednet=1**, and you must specify a primary boot interface using the **bootdev** option, described below. Alternatively, you can use the **ip** option once, and then use Kickstart to set up further interfaces.

This option accepts several different formats. The most common are described in [Table 23.4, "Network Interface Configuration Formats"](#).

Table 23.4. Network Interface Configuration Formats

Configuration Method	Option format
Automatic configuration of any interface	ip=method
Automatic configuration of a specific interface	ip=interface:method
Static configuration	ip=ip::gateway:netmask:hostname:interface:none
Automatic configuration of a specific interface with an override [a]	ip=ip::gateway:netmask:hostname:interface:method:mtu
[a] Brings up the specified interface using the specified method of automatic configuration, such as dhcp, but overrides the automatically obtained IP address, gateway, netmask, host name or other specified parameter. All parameters are optional; only specify the ones you want to override and automatically obtained values will be used for the others.	

The **method** parameter can be any the following:

Table 23.5. Automatic Interface Configuration Methods

Automatic configuration method	Value
DHCP	dhcp
IPv6 DHCP	dhcp6
IPv6 automatic configuration	auto6
iBFT (iSCSI Boot Firmware Table)	ibft



NOTE

If you use a boot option which requires network access, such as **inst.ks=http://host/path**, without specifying the **ip** option, the installation program will use **ip=dhcp**.



IMPORTANT

To connect automatically to an iSCSI target, a network device for accessing the target needs to be activated. The recommended way to do so is to use **ip=ibft** boot option.

In the above tables, the *ip* parameter specifies the client's IP address. **IPv6** addresses can be specified by putting them in square brackets, for example, **[2001:DB8::1]**.

The *gateway* parameter is the default gateway. IPv6 addresses are accepted here as well.

The *netmask* parameter is the netmask to be used. This can either be a full netmask for IPv4 (for example **255.255.255.0**) or a prefix for IPv6 (for example **64**).

The *hostname* parameter is the host name of the client system. This parameter is optional.

nameserver=

Specifies the address of the name server. This option can be used multiple times.

rd.neednet=

You must use the option **rd.neednet=1** if you use more than one **ip** option. Alternatively, to set up multiple network interfaces you can use the **ip** once, and then set up further interfaces using Kickstart.

bootdev=

Specifies the boot interface. This option is mandatory if you use more than one **ip** option.

ifname=

Assigns a given interface name to a network device with a given MAC address. Can be used multiple times. The syntax is **ifname=interface:MAC**. For example:

```
ifname=eth0:01:23:45:67:89:ab
```



NOTE

Using the **ifname=** option is the only supported way to set custom network interface names during installation.

inst.dhcpclass=

Specifies the DHCP vendor class identifier. The **dhcpcd** service will see this value as **vendor-class-identifier**. The default value is **anaconda-\$(uname -srm)**.

inst.waitfornet=

Using the **inst.waitfornet=SECONDS** boot option causes the installation system to wait for network connectivity before installation. The value given in the *SECONDS* argument specifies maximum amount of time to wait for network connectivity before timing out and continuing the installation process even if network connectivity is not present.

vlan=

Sets up a Virtual LAN (VLAN) device on a specified interface with a given name. The syntax is **vlan=name:interface**. For example:

```
vlan=vlan5:em1
```

The above will set up a VLAN device named **vlan5** on the **em1** interface. The *name* can take the following forms:

Table 23.6. VLAN Device Naming Conventions

Naming scheme	Example
VLAN_PLUS_VID	vlan0005
VLAN_PLUS_VID_NO_PAD	vlan5
DEV_PLUS_VID	em1.0005.
DEV_PLUS_VID_NO_PAD	em1.5.

bond=

Set up a bonding device with the following syntax: **bond=*name*[:*slaves*][:*options*]**. Replace *name* with the bonding device name, *slaves* with a comma-separated list of physical (ethernet) interfaces, and *options* with a comma-separated list of bonding options. For example:

```
bond=bond0:em1,em2:mode=active-backup,tx_queues=32,downdelay=5000
```

For a list of available options, execute the **modinfo bonding** command.

Using this option without any parameters will assume **bond=bond0:eth0,eth1:mode=balance-rr**.

team=

Set up a team device with the following syntax: **team=*master*:*slaves***. Replace *master* with the name of the master team device and *slaves* with a comma-separated list of physical (ethernet) devices to be used as slaves in the team device. For example:

```
team=team0:em1,em2
```

Advanced Installation Options**inst.kexec**

If this option is specified, the installer will use the **kexec** system call at the end of the installation, instead of performing a reboot. This loads the new system immediately, and bypasses the hardware initialization normally performed by the BIOS or firmware.

**IMPORTANT**

Due to the complexities involved with booting systems using **kexec**, it cannot be explicitly tested and guaranteed to function in every situation.

When **kexec** is used, device registers (which would normally be cleared during a full system reboot) might stay filled with data, which could potentially create issues for some device drivers.

inst.gpt

Force the installation program to install partition information into a GUID Partition Table (GPT) instead of a Master Boot Record (MBR). This option is meaningless on UEFI-based systems, unless they are in BIOS compatibility mode.

Normally, BIOS-based systems and UEFI-based systems in BIOS compatibility mode will attempt to use the MBR schema for storing partitioning information, unless the disk is 2^{32} sectors in size or larger. Most commonly, disk sectors are 512 bytes in size, meaning that this is usually equivalent to 2 TiB. Using this option will change this behavior, allowing a GPT to be written to disks smaller than this.

See [Section 8.14.1.1, “MBR and GPT Considerations”](#) for more information about GPT and MBR, and [Section A.1.4, “GUID Partition Table \(GPT\)”](#) for more general information about GPT, MBR and disk partitioning in general.

inst.multilib

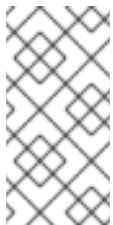
Configure the system for multilib packages (that is, to allow installing 32-bit packages on a 64-bit AMD64 or Intel 64 system) and install packages specified in this section as such.

Normally, on an AMD64 or Intel 64 system, only packages for this architecture (marked as **x86_64**) and packages for all architectures (marked as **noarch**) would be installed. When you use this option, packages for 32-bit AMD or Intel systems (marked as **i686**) will be automatically installed as well if available.

This only applies to packages directly specified in the **%packages** section. If a package is only installed as a dependency, only the exact specified dependency will be installed. For example, if you are installing package `bash` which depends on package `glibc`, the former will be installed in multiple variants, while the latter will only be installed in variants specifically required.

selinux=0

By default, SELinux operates in **permissive** mode in the installer, and in **enforcing** mode in the installed system. This option disables the use of SELinux in the installer and the installed system entirely.



NOTE

The **selinux=0** and **inst.selinux=0** options are not the same. The **selinux=0** option disables the use of SELinux in the installer and the installed system, whereas **inst.selinux=0** disables SELinux only in the installer. By default, SELinux is set to operate in **permissive** mode in the installer, so disabling it has little effect.

inst.nosave=

This option, introduced in Red Hat Enterprise Linux 7.3, controls which Kickstart files and installation logs are saved to the installed system. It can be especially useful to disable saving such data when performing OEM operating system installations, or when generating images using sensitive resources (such as internal repository URLs), as these resources might otherwise be mentioned in kickstart files, or in logs on the image, or both. Possible values for this option are:

input_ks - disables saving of the input Kickstart file (if any).

output_ks - disables saving of the output Kickstart file generated by Anaconda.

all_ks - disables saving of both input and output Kickstart files.

logs - disables saving of all installation logs.

all - disables saving of all Kickstart files and all installation logs.

Multiple values can be combined as a comma separated list, for example: **input_ks,logs**

inst.zram

This option controls the usage of *zRAM* swap during the installation. It creates a compressed block device inside the system RAM and uses it for swap space instead of the hard drive. This allows the installer to essentially increase the amount of memory available, which makes the installation faster on systems with low memory.

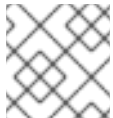
By default, swap on zRAM is enabled on systems with 2 GiB or less RAM, and disabled on systems with more than 2 GiB of memory. You can use this option to change this behavior – on a system with more than 2 GiB RAM, use **inst.zram=1** to enable it, and on systems with 2 GiB or less memory, use **inst.zram=0** to disable this feature.

Enabling Remote Access

The following options are necessary to configure **Anaconda** for remote graphical installation. See [Chapter 25, Using VNC](#) for more details.

inst.vnc

Specifies that the installation program's graphical interface should be run in a **VNC** session. If you specify this option, you will need to connect to the system using a VNC client application to be able to interact with the installation program. VNC sharing is enabled, so multiple clients can connect to the system at the same time.



NOTE

A system installed using VNC will start in text mode by default.

inst.vncpassword=

Sets a password on the VNC server used by the installation program. Any VNC client attempting to connecting to the system will have to provide the correct password to gain access. For example, **inst.vncpassword=testpwd** will set the password to **testpwd**. The VNC password must be between 6 and 8 characters long.



NOTE

If you specify an invalid password (one that is too short or too long), you will be prompted to specify a new one by a message from the installation program:

VNC password must be six to eight characters long.
Please enter a new one, or leave blank for no password.

Password:

inst.vncconnect=

Connect to a listening VNC client at a specified host and port once the installation starts. The correct syntax is **inst.vncconnect=host:port**, where *host* is the address to the VNC client's host, and *port* specifies which port to use. The *port* parameter is optional, if you do not specify one, the installation program will use **5900**.

Debugging and Troubleshooting

inst.updates=

Specifies the location of the **updates.img** file to be applied to the installation program runtime. The syntax is the same as in the **inst.repo** option - see [Table 23.1, "Installation Sources"](#) for details. In all formats, if you do not specify a file name but only a directory, the installation program will look for a file named **updates.img**.

inst.loglevel=

Specifies the minimum level for messages to be logged on a terminal. This only concerns terminal logging; log files will always contain messages of all levels.

Possible values for this option from the lowest to highest level are: **debug**, **info**, **warning**, **error** and **critical**. The default value is **info**, which means that by default, the logging terminal will display messages ranging from **info** to **critical**.

inst.syslog=

Once the installation starts, this option sends log messages to the syslog process on the specified host. The remote syslog process must be configured to accept incoming connections. For information on how to configure a syslog service to accept incoming connections, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

inst.virtio-log=

Specifies a **virtio** port (a character device at **/dev/virtio-ports/name**) to be used for forwarding logs. The default value is **org.fedoraproject.anaconda.log.0**; if this port is present, it will be used.

rd.live.ram

If this option is specified, the stage 2 image will be copied into RAM. When the stage2 image on NFS repository is used, this option may make the installation proceed smoothly, since the installation is sometimes affected by reconfiguring network in an environment built upon the stage 2 image on NFS.

Note that using this option when the stage 2 image is on an NFS server will increase the minimum required memory by the size of the image - roughly 500 MiB.

inst.nokill

A debugging option that prevents anaconda from and rebooting when a fatal error occurs or at the end of the installation process. This allows you to capture installation logs which would be lost upon reboot.

23.1.1. Deprecated and Removed Boot Options

Deprecated Boot Options

Options in this list are *deprecated*. They will still work, but there are other options which offer the same functionality. Using deprecated options is not recommended and they are expected to be removed in future releases.

**NOTE**

Note that as [Section 23.1, “Configuring the Installation System at the Boot Menu”](#) describes, options specific to the installation program now use the **inst.** prefix. For example, the **vnc=** option is considered deprecated and replaced by the **inst.vnc=** option. These changes are not listed here.

method=

Configured the installation method. Use the **inst.repo=** option instead.

repo=nfsiso:server:/path

In NFS installations, specified that the target is an ISO image located on an NFS server instead of an installable tree. The difference is now detected automatically, which means this option is the same as **inst.repo=nfs:server:/path**.

dns=

Configured the Domain Name Server (DNS). Use the **nameserver=** option instead.

netmask=, gateway=, hostname=, ip=, ipv6=

These options have been consolidated under the **ip=** option.

ksdevice=

Select network device to be used at early stage of installation. Different values have been replaced with different options; see the table below.

Table 23.7. Automatic Interface Configuration Methods

Value	Current behavior
Not present	Activation of all devices is attempted using dhcp , unless the desired device and configuration is specified by the ip= option or the BOOTIF option.
ksdevice=link	Similar to the above, with the difference that network will always be activated in the initramfs, whether it is needed or not. The supported rd.neednet dracut option should be used to achieve the same result.
ksdevice=bootif	Ignored (the BOOTIF= option is used by default when specified)
ksdevice=ibft	Replaced with the ip=ibft dracut option
ksdevice=MAC	Replaced with BOOTIF=MAC
ksdevice=device	Replaced by specifying the device name using the ip= dracut option.

blacklist=

Used to disable specified drivers. This is now handled by the **modprobe.blacklist=** option.

nofirewire=

Disabled support for the FireWire interface. You can disable the FireWire driver (**firewire_ohci**) by using the **modprobe.blacklist=** option instead:

```
modprobe.blacklist=firewire_ohci
```

nicdelay=

Used to indicate the delay after which the network was considered active; the system waited until either the gateway was successfully pinged, or until the amount of seconds specified in this parameter passed. In RHEL 7, network devices are configured and activated during the early stage of installation by the **dracut** modules which ensure that the gateway is accessible before proceeding. For more information about **dracut**, see the **dracut.cmdline(7)** man page.

linksleep=

Used to configure how long anaconda should wait for a link on a device before activating it. This functionality is now available in the **dracut** modules where specific **rd.net.timeout.*** options can be configured to handle issues caused by slow network device initialization. For more information about **dracut**, see the **dracut.cmdline(7)** man page.

Removed Boot Options

The following options are removed. They were present in previous releases of Red Hat Enterprise Linux, but they cannot be used anymore.

askmethod, asknetwork

The installation program's **initramfs** is now completely non-interactive, which means that these options are not available anymore. Instead, use the **inst.repo=** to specify the installation method and **ip=** to configure network settings.

serial

This option forced **Anaconda** to use the **/dev/ttyS0** console as the output. Use the **console=/dev/ttyS0** (or similar) instead.

updates=

Specified the location of updates for the installation program. Use the **inst.updates=** option instead.

essid=, wepkey=, wpakey=

Configured wireless network access. Network configuration is now being handled by **dracut**, which does not support wireless networking, rendering these options useless.

ethtool=

Used in the past to configure additional low-level network settings. All network settings are now handled by the **ip=** option.

gdb

Allowed you to debug the loader. Use **rd.debug** instead.

mediacheck

Verified the installation media before starting the installation. Replaced with the **rd.live.check** option.

ks=floppy

Specified a 3.5 inch diskette as the Kickstart file source. These drives are not supported anymore.

display=

Configured a remote display. Replaced with the **inst.vnc** option.

utf8

Added UTF8 support when installing in text mode. UTF8 support now works automatically.

noipv6

Used to disable IPv6 support in the installation program. IPv6 is now built into the kernel so the driver cannot be blacklisted; however, it is possible to disable IPv6 using the **ipv6.disable dracut** option.

upgradeany

Upgrades are done in a different way in Red Hat Enterprise Linux 7. For more information about upgrading your system, see [Chapter 29, Upgrading Your Current System](#).

vlanid=

Used to configure Virtual LAN (802.1q tag) devices. Use the **vlan= dracut** option instead.

23.2. USING THE MAINTENANCE BOOT MODES

23.2.1. Loading the Memory (RAM) Testing Mode

Faults in memory (RAM) modules can cause your system to freeze or crash unpredictably. In certain situations, memory faults might only cause errors with particular combinations of software. For this reason, you should test the memory of a computer before you install Red Hat Enterprise Linux for the first time, even if it has previously run other operating systems.

Red Hat Enterprise Linux includes the **Memtest86+** memory testing application. To start memory testing mode, choose **Troubleshooting > Memory test** at the boot menu. Testing will begin immediately. By default, **Memtest86+** carries out ten tests in every pass; a different configuration can be specified by accessing the configuration screen using the **c** key. After the first pass completes, a message will appear at the bottom informing you of the current status, and another pass will start automatically.

**NOTE**

Memtest86+ only works on BIOS systems. Support for UEFI systems is currently unavailable.

```

Memtest86+ v4.20 | Pass 3% #
2894 MHz | Test 46% #####
L1 Cache: 32K 115740 MB/s | Test #3 [Moving inversions, 8 bit pattern]
L2 Cache: 2048K 51669 MB/s | Testing: 196K - 1024M 1024M
L3 Cache: None | Pattern: efefefef
Memory : 1024M 9425 MB/s |-----
Chipset : Intel i440FX

WallTime   Cached   RsvdMem   MemMap   Cache   ECC   Test   Pass   Errors   ECC   Errs
-----
0:00:14   1024M       0K      e820     on    off   Std     0       0

(ESC)Reboot (c)configuration (SP)scroll_lock (CR)scroll_unlock

```

Figure 23.1. Memory Check Using Memtest86+

The main screen displayed while testing is in progress is divided into three main areas:

- The upper left corner shows information about your system's memory configuration - the amount of detected memory and processor cache and their throughputs and processor and chipset information. This information is detected when **Memtest86+** starts.
- The upper right corner displays information about the tests - progress of the current pass and the currently running test in that pass as well as a description of the test.
- The central part of the screen is used to display information about the entire set of tests from the moment when the tool has started, such as the total time, the number of completed passes, number of detected errors and your test selection. On some systems, detailed information about the installed memory (such as the number of installed modules, their manufacturer, frequency and latency) will be also displayed here. After the each pass completes, a short summary will appear in this location. For example:

```

** Pass complete, no errors, press Esc to exit **

```

If **Memtest86+** detects an error, it will also be displayed in this area and highlighted red. The message will include detailed information such as which test detected a problem, the memory location which is failing, and others.

In most cases, a single successful pass (that is, a single run of all 10 tests) is sufficient to verify that your RAM is in good condition. In some rare circumstances, however, errors that went undetected on the first pass might appear on subsequent passes. To perform a thorough test on an important system, leave the tests running overnight or even for a few days in order to complete multiple passes.

**NOTE**

The amount of time it takes to complete a single full pass of **Memtest86+** varies depending on your system's configuration (notably the RAM size and speed). For example, on a system with 2 GiB of DDR2 memory at 667 MHz, a single pass will take roughly 20 minutes to complete.

To halt the tests and reboot your computer, press the **Esc** key at any time.

For more information about using **Memtest86+**, see the official website at <http://www.memtest.org/>. A **README** file is also located in `/usr/share/doc/memtest86+-version/` on Red Hat Enterprise Linux systems with the `memtest86+` package installed.

23.2.2. Verifying Boot Media

You can test the integrity of an ISO-based installation source before using it to install Red Hat Enterprise Linux. These sources include DVD, and ISO images stored on a hard drive or NFS server. Verifying that the ISO images are intact before you attempt an installation helps to avoid problems that are often encountered during installation.

To test the checksum integrity of an ISO image, append the **rd.live.check** to the boot loader command line. Note that this option is used automatically if you select the default installation option from the boot menu (**Test this media & install Red Hat Enterprise Linux 7.0**).

23.2.3. Booting Your Computer in Rescue Mode

You can boot a command-line Linux system from an installation disc without actually installing Red Hat Enterprise Linux on the computer. This enables you to use the utilities and functions of a running Linux system to modify or repair already installed operating systems.

To load the rescue system with the installation disk or USB drive, choose **Rescue a Red Hat Enterprise Linux system** from the **Troubleshooting** submenu in the boot menu, or use the **inst.rescue** boot option.

Specify the language, keyboard layout and network settings for the rescue system with the screens that follow. The final setup screen configures access to the existing system on your computer.

By default, rescue mode attaches an existing operating system to the rescue system under the directory **/mnt/sysimage/**.

For additional information about rescue mode and other maintenance modes, see [Chapter 32, Basic System Recovery](#).

CHAPTER 24. PREPARING FOR A NETWORK INSTALLATION

A network installation using an installation server allows you to install Red Hat Enterprise Linux on multiple systems using a *network boot* server. This way, all systems configured to do so will boot using an image provided by this server and start the installation program automatically.



NOTE

Red Hat Satellite has the ability to automate the setup of a PXE server. See the [Red Hat Satellite documentation](#) for more information.

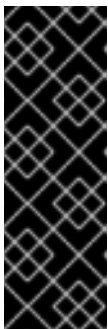
A minimum of two systems is required for a network installation:

- A *server* – a system running a DHCP server, a TFTP server to provide boot files, and an HTTP, FTP or NFS server which hosts the installation image. Theoretically, each of the servers can run on a different physical system; procedures in this section assume a single system runs all of them for simplicity.
- A *client* – the system which you are installing Red Hat Enterprise Linux to. When the installation begins, the client will query the DHCP server, obtain boot files from the TFTP server, and download the installation image from the HTTP, FTP or NFS server.

Unlike most other means of installation, no physical boot media is required to be plugged in the client (that is, the system you are installing into) in order to begin the installation. This chapter describes the steps you must take to prepare for network installations.

The following steps must be performed to prepare for a network installation:

1. Configure the network server (**NFS**, **HTTPS**, **HTTP**, or **FTP**) to export the installation tree or the installation ISO image. For procedures describing the configuration, see [Section 3.3.3, “Installation Source on a Network”](#).
2. Configure the files on the **tftp** server necessary for network boot, configure **DHCP**, and start the **tftp** service on the PXE server. See [Section 24.1, “Configuring Network Boot Services”](#) for details.



IMPORTANT

The **GRUB2** boot loader supports network boot from **HTTP** in addition to a **tftp** server. However, obtaining boot files (the kernel and initial ram disk for the installer) over this protocol is very slow and suffers a risk of timeout failures. Using a **tftp** server to provide the boot files is recommended.

This warning only applies to the kernel and initial ram disk (**vmlinuz** and **initrd**). Obtaining the *installation source* from an **HTTP** server does not carry this risk.

3. Boot the client (the system you want to install Red Hat Enterprise Linux on) and start the installation.



NOTE

The procedures in this chapter describe setting up a network boot server on a Red Hat Enterprise Linux 7 system. For details about configuring network boot on earlier releases of Red Hat Enterprise Linux, see the appropriate *Installation Guide* for that release.

24.1. CONFIGURING NETWORK BOOT SERVICES

After setting up a network server containing the package repositories to be used in the installation, the next step is to configure the PXE server itself. This server will contain files necessary to boot the Red Hat Enterprise Linux and start the installation. Additionally, a **DHCP** server must be configured, and all necessary services must be enabled and started.



NOTE

The network boot configuration procedure differs based on whether the AMD64/Intel 64 system you want to install Red Hat Enterprise Linux on uses BIOS or UEFI. Consult your hardware's documentation to see which system is used on your hardware, and then follow the appropriate procedure in this chapter.

A separate procedure is provided for booting IBM Power Systems from a network location with the **GRUB2** boot loader. See [Section 24.1.3, "Configuring Network Boot for IBM Power Systems Using GRUB2"](#) for details.

For more information on configuring a network boot server for use with headless systems (systems without a directly connected display, keyboard and mouse), see [Chapter 26, Headless Systems](#).

24.1.1. Configuring a TFTP Server for BIOS-based AMD64 and Intel 64 Clients

The following procedure will prepare the PXE server for booting BIOS-based AMD64 and Intel 64 systems. For information on UEFI-based systems, see [Section 24.1.2, "Configuring a TFTP Server for UEFI-based AMD64/Intel 64 and ARM Clients"](#).

Procedure 24.1. Configuring a TFTP Boot Server for BIOS-based Systems

1. Install the `tftp-server` package. To do this, enter the following command as **root**:

```
# yum install tftp-server
```

2. Allow incoming connections to the **tftp** service in the firewall:

```
# firewall-cmd --add-service=tftp
```



NOTE

The above command only enables access until the next server reboot. To allow access permanently, add the **--permanent** option. For more information about firewall configuration, see the [Red Hat Enterprise Linux 7 Security Guide](#).

3. Configure your **DHCP** server to use the boot images packaged with **SYSLINUX**. If you do not have one installed, see the [Red Hat Enterprise Linux 7 Networking Guide](#) for instructions.

A sample configuration in the `/etc/dhcp/dhcpd.conf` file might look like:

```
option space pxelinux;
option pxelinux.magic code 208 = string;
option pxelinux.configfile code 209 = text;
option pxelinux.pathprefix code 210 = text;
```

```

option pxelinux.reboottime code 211 = unsigned integer 32;
option architecture-type code 93 = unsigned integer 16;

subnet 10.0.0.0 netmask 255.255.255.0 {
    option routers 10.0.0.254;
    range 10.0.0.2 10.0.0.253;

    class "pxeclients" {
        match if substring (option vendor-class-identifier, 0, 9) = "PXEClient";
        next-server 10.0.0.1;

        if option architecture-type = 00:07 {
            filename "uefi/shim.efi";
        } else {
            filename "pxelinux/pxelinux.0";
        }
    }
}

```

4. You now need the **pxelinux.0** file from the SYSLINUX package in the ISO image file of the full installation DVD. To access it, enter the following commands as root:

```
# mount -t iso9660 /path_to_image/name_of_image.iso /mount_point -o loop,ro
```

```
# cp -pr /mount_point/Packages/syslinux-version-architecture.rpm
/publicly_available_directory
```

```
# umount /mount_point
```

Extract the package:

```
# rpm2cpio syslinux-version-architecture.rpm | cpio -dimv
```

5. Create a **pxelinux/** directory within **tftpboot/** and copy the **pxelinux.0** file into it:

```
# mkdir /var/lib/tftpboot/pxelinux
```

```
# cp /publicly_available_directory/usr/share/syslinux/pxelinux.0 /var/lib/tftpboot/pxelinux
```

6. Create the directory **pxelinux.cfg/** in the **pxelinux/** directory:

```
# mkdir /var/lib/tftpboot/pxelinux/pxelinux.cfg
```

Add a configuration file named **default** to the **pxelinux.cfg/** directory.

A sample configuration file at **/var/lib/tftpboot/pxelinux/pxelinux.cfg/default** might look like:

```

default vesamenu.c32
prompt 1
timeout 600

display boot.msg

```

```

label linux
  menu label ^Install system
  menu default
  kernel images/RHEL-7.1/vmlinuz
  append initrd=images/RHEL-7.1/initrd.img ip=dhcp
inst.repo=http://10.32.5.1/mnt/archive/RHEL-7/7.x/Server/x86_64/os/
label vesa
  menu label Install system with ^basic video driver
  kernel images/RHEL-7.1/vmlinuz
  append initrd=images/RHEL-7.1/initrd.img ip=dhcp inst.xdriver=vesa nomodeset
inst.repo=http://10.32.5.1/mnt/archive/RHEL-7/7.x/Server/x86_64/os/
label rescue
  menu label ^Rescue installed system
  kernel images/RHEL-7.1/vmlinuz
  append initrd=images/RHEL-7.1/initrd.img rescue
label local
  menu label Boot from ^local drive
  localboot 0xffff

```



IMPORTANT

The **inst.repo= Anaconda** option, shown in the example above, must always be used to specify the installation program's image as well as the installation source. Without this option, the installation program will be unable to boot. For more information about boot options for **Anaconda**, see [Section 23.1, “Configuring the Installation System at the Boot Menu”](#).

7. Create a subdirectory to store the boot image files within the **/var/lib/tftpboot/** directory, and copy the boot image files to it. In this example, we use the directory **/var/lib/tftpboot/pxelinux/images/RHEL-release_number/**:

```

# mkdir -p /var/lib/tftpboot/pxelinux/images/RHEL-7.1/
# cp /path_to_x86_64_images/pxeboot/{vmlinuz,initrd.img}
/var/lib/tftpboot/pxelinux/images/RHEL-release_number/

```

8. Finally, start and enable the services:

- The **dhcpd** service:

```

# systemctl start dhcpd
# systemctl enable dhcpd

```

- The **xinetd** service, that manages the **tftpd** service:

```

# systemctl start xinetd
# systemctl enable xinetd

```

After finishing this procedure, the PXE boot server is ready to serve PXE clients. You can now start the system you want to install Red Hat Enterprise Linux on, select PXE Boot when prompted to specify a boot source, and start the network installation. For more information, see [Section 7.1.2, “Booting from the Network Using PXE”](#).

24.1.2. Configuring a TFTP Server for UEFI-based AMD64/Intel 64 and ARM Clients

The following procedure will prepare the PXE server for booting UEFI-based AMD64/Intel 64 and ARM systems. For information on BIOS-based systems, see [Section 24.1.1, "Configuring a TFTP Server for BIOS-based AMD64 and Intel 64 Clients"](#).

Procedure 24.2. Configuring TFTP Boot for UEFI-based Systems



NOTE

Red Hat Enterprise Linux 7 UEFI PXE boot supports a lowercase file format for a MAC-based grub menu file. For example, the MAC address file format for grub2 is **grub.cfg-01-aa-bb-cc-dd-ee-ff**

1. Install the `tftp-server` package. To do this, enter the following command as **root**:

```
# yum install tftp-server
```

2. Allow incoming connections to the **tftp** service in the firewall:

```
# firewall-cmd --add-service=tftp
```



NOTE

The above command only enables access until the next server reboot. To allow access permanently, add the **--permanent** option. For more information about firewall configuration, see the [Red Hat Enterprise Linux 7 Security Guide](#).

3. Configure your **DHCP** server to use the EFI boot images packaged with `shim`. If you do not have one installed, see the [Red Hat Enterprise Linux 7 Networking Guide](#) for instructions.

A sample configuration in the `/etc/dhcp/dhcpd.conf` file might look like:

```
option space pxelinux;
option pxelinux.magic code 208 = string;
option pxelinux.configfile code 209 = text;
option pxelinux.pathprefix code 210 = text;
option pxelinux.reboottime code 211 = unsigned integer 32;
option architecture-type code 93 = unsigned integer 16;

subnet 10.0.0.0 netmask 255.255.255.0 {
    option routers 10.0.0.254;
    range 10.0.0.2 10.0.0.253;

    class "pxeclients" {
        match if substring (option vendor-class-identifier, 0, 9) = "PXEClient";
        next-server 10.0.0.1;

        if option architecture-type = 00:07 {
            filename "shim.efi";
        } else {
            filename "pxelinux/pxelinux.0";
        }
    }
}
```

```
}
}
}
```

- You now need the **shim.efi** file from the **shim** package and the **grubx64.efi** file from the **grub2-efi** package in the ISO image file. To access them, enter the following commands as root:

```
# mount -t iso9660 /path_to_image/name_of_image.iso /mount_point -o loop,ro
```

```
# cp -pr /mount_point/Packages/shim-version-architecture.rpm /publicly_available_directory
```

```
# cp -pr /mount_point/Packages/grub2-efi-version-architecture.rpm
/publicly_available_directory
```

```
# umount /mount_point
```

Extract the packages:

```
# rpm2cpio shim-version-architecture.rpm | cpio -dimv
```

```
# rpm2cpio grub2-efi-version-architecture.rpm | cpio -dimv
```

- Copy the EFI boot images from your boot directory:

```
# cp /publicly_available_directory/boot/efi/EFI/redhat/shim.efi /var/lib/tftpboot/
```

```
# cp /publicly_available_directory/boot/efi/EFI/redhat/grubx64.efi /var/lib/tftpboot/
```

- Add a configuration file named **grub.cfg** to the **tftpboot/** directory. A sample configuration file at **/var/lib/tftpboot/grub.cfg** might look like:

```
set timeout=60
menuentry 'RHEL 7' {
    linuxefi images/RHEL-7.1/vmlinuz ip=dhcp inst.repo=http://10.32.5.1/mnt/archive/RHEL-
7/7.1/Server/x86_64/os/
    initrdefi images/RHEL-7.1/initrd.img
}
```

IMPORTANT

The **inst.repo= Anaconda** option, shown in the example above, must always be used to specify the installation program's image as well as the installation source. Without this option, the installation program will be unable to boot. For more information about boot options for **Anaconda**, see [Section 23.1, "Configuring the Installation System at the Boot Menu"](#).

- Create a subdirectory to store the boot image files within the **/var/lib/tftpboot/** directory, and copy the boot image files to it. In this example, we use the directory **/var/lib/tftpboot/images/RHEL-7.1/**:

```
# mkdir -p /var/lib/tftpboot/images/RHEL-7.1/# cp
/path_to_x86_64_images/pxeboot/{vmlinuz,initrd.img} /var/lib/tftpboot/images/RHEL-7.1/
```

8. Finally, start and enable the services:

- The **dhcpcd** service:

```
# systemctl start dhcpcd
# systemctl enable dhcpcd
```

- The **xinetd** service, that manages the **tftpd** service:

```
# systemctl start xinetd
# systemctl enable xinetd
```

After finishing this procedure, the PXE boot server is ready to serve PXE clients. You can now start the system you want to install Red Hat Enterprise Linux on, select PXE Boot when prompted to specify a boot source, and start the network installation. For more information, see [Section 7.1.2, “Booting from the Network Using PXE”](#).

24.1.3. Configuring Network Boot for IBM Power Systems Using GRUB2

Procedure 24.3. Configuring a Network Boot Server for IBM Power Systems Using GRUB2

1. Install the **tftp-server** package. To do this, enter the following command as **root**:

```
# yum install tftp-server
```

2. Allow incoming connections to the **tftp** service in the firewall:

```
# firewall-cmd --add-service=tftp
```



NOTE

The above command only enables access until the next server reboot. To allow access permanently, add the **--permanent** option. For more information about firewall configuration, see the [Red Hat Enterprise Linux 7 Security Guide](#).

3. Create a **GRUB2** network boot directory inside the **tftp** root:

```
# grub2-mknetdir --net-directory=/var/lib/tftpboot
Netboot directory for powerpc-ieee1275 created. Configure your DHCP server to point to
/boot/grub2/powerpc-ieee1275/core.elf
```

Note the command's output, which informs you about which file needs to be configured as the **filename** in your **DHCP** configuration. This will become important further in the procedure.

4. Create a **GRUB2** configuration file: **/var/lib/tftpboot/boot/grub2/grub.cfg**. The **grub.cfg** syntax is described in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Below is an example configuration file:

```

set default=0
set timeout=5

echo -e "\nWelcome to the Red Hat Enterprise Linux 7 installer!\n\n"

menuentry 'Red Hat Enterprise Linux 7' {
    linux grub2-ppc64/vmlinuz ro ip=dhcp inst.repo=http://10.32.5.1/mnt/archive/RHEL-7/7.6-
Beta/Server/ppc64/os/
    initrd grub2-ppc64/initrd.img
}

```



IMPORTANT

The **inst.repo= Anaconda** option, shown in the example above, must always be used to specify the installation program's image as well as the installation source. Without this option, the installation program will be unable to boot. For more information about boot options for **Anaconda**, see [Section 23.1, “Configuring the Installation System at the Boot Menu”](#).

5. Configure your **DHCP** server to use the boot images packaged with **GRUB2**. If you do not have one installed, see the [Red Hat Enterprise Linux 7 Networking Guide](#) for instructions.

A sample configuration in the **/etc/dhcp/dhcpd.conf** file might look like:

```

subnet 192.168.0.1 netmask 255.255.255.0 {
    allow bootp;
    option routers 192.168.0.5;
    group { #BOOTP POWER clients
        filename "boot/grub2/powerpc-ieee1275/core.elf";
        host client1 {
            hardware ethernet 01:23:45:67:89:ab;
            fixed-address 192.168.0.112;
        }
    }
}

```

Adjust the sample parameters (**subnet**, **netmask**, **routers**, **fixed-address** and **hardware ethernet**) to fit your network configuration. Also note the **filename** parameter; this is the file name which was output by the **grub2-mknetdir** command earlier in the procedure.

6. Finally, start and enable the services:

- The **dhcpd** service:

```

# systemctl start dhcpd
# systemctl enable dhcpd

```

- The **xinetd** service that manages the **tftpd** service:

```

# systemctl start xinetd
# systemctl enable xinetd

```

After finishing this procedure, the PXE boot server is ready to serve PXE clients. You can now follow the steps described in [Chapter 12, *Booting the Installation on IBM Power Systems*](#) to boot your Power Systems client from this server.

Additional information about setting up network boot for IBM Power Systems clients can be found in the [Netbooting on POWER - An Introduction](#) at the IBM DeveloperWorks website.

CHAPTER 25. USING VNC

The graphical installation interface is the recommended method of installing Red Hat Enterprise Linux. However, in some cases, accessing the graphical interface directly is difficult or impossible. Many enterprise systems, notably servers (IBM Power Systems and IBM Z), lack the capability to connect a display and a keyboard, making VNC a necessity for manual (non-Kickstart) installations.

To allow manual installations on *headless systems* (systems without a directly connected display, keyboard and mouse), the **Anaconda** installation program includes a *Virtual Network Computing* (VNC) installation which allows the graphical mode of the installation program to run locally, but display on a system connected to the network. The VNC installation provides you with the full range of installation options, even in situations where the system lacks a display or input devices.

This chapter provides instructions on activating VNC mode on the installation system and connecting to it using a VNC viewer.

25.1. INSTALLING A VNC VIEWER

Performing a VNC installation requires a VNC viewer running on your workstation or another terminal computer. VNC viewers are available in the repositories of most Linux distributions; free VNC viewers are also available for other operating systems such as Windows. On Linux systems, use your package manager to search for a viewer for your distribution.

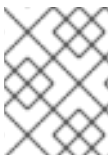
The following VNC viewers are available in Red Hat Enterprise Linux:

- **TigerVNC** - A basic viewer independent of your desktop environment. Installed as the `tigervnc` package.
- **Vinagre** - A viewer for the **GNOME** desktop environment. Installed as the `vinagre` package.
- **KRDC** - A viewer integrated with the **KDE** desktop environment. Installed as the `kdenetwork-krdc` package.

To install any of the viewers listed above, execute the following command as **root**:

```
# yum install package
```

Replace *package* with the package name of the viewer you want to use (for example, `tigervnc`).



NOTE

Procedures in this chapter assume you are using **TigerVNC** as your VNC viewer. Specific instructions for other viewers can differ, but the general principles still apply.

25.2. PERFORMING A VNC INSTALLATION

The **Anaconda** installation program offers two modes for VNC installation. The modes are *Direct Mode* and *Connect Mode*. Direct Mode requires the VNC viewer to initiate the connection to the system being installed. Connect Mode requires the system being installed to initiate the connection to the VNC viewer. Once the connection is established, the two modes do not differ. The mode you select depends on the configuration in your environment.

Direct Mode

In this mode, **Anaconda** is configured to start the installation and wait for a VNC viewer before

proceeding. The IP address and port are displayed on the system being installed. Using this information, you can connect to the installation system from a different computer. For this reason you must have visual and interactive access to the system being installed.

Connect Mode

In this mode, the VNC viewer is started on the remote system in *listening mode*. The VNC viewer waits for an incoming connection on a specified port. Then, **Anaconda** is started and the host name and port number are provided using a boot option or a Kickstart command. When the installation begins, the installation program establishes a connection with the listening VNC viewer using the specified host name and port number. For this reason, your remote system must be able to accept incoming network connections.

Considerations for choosing a VNC installation mode

- Visual and Interactive access to the system
 - If visual and interactive access to the system being installed is not available, then you must use Connect Mode.
- Network Connection Rules and Firewalls
 - If the system being installed is not allowed inbound connections by a firewall, then you must use Connect Mode or disable the firewall. Disabling a firewall can have security implications.
 - If the remote system running the VNC viewer is not allowed incoming connections by a firewall, then you must use Direct Mode, or disable the firewall. Disabling a firewall can have security implications. See the [Red Hat Enterprise Linux 7 Security Guide](#) for information about configuring the firewall.



NOTE

You must specify custom boot options to start a VNC installation. The exact way to do this differs depending on the system architecture. For architecture-specific instructions about editing boot options, see:

- [Section 7.2, “The Boot Menu”](#) for 64-bit AMD, Intel, and ARM systems
- [Section 12.1, “The Boot Menu”](#) for IBM Power Systems servers
- [Chapter 21, *Parameter and Configuration Files on IBM Z*](#) for IBM Z

25.2.1. Installing in VNC Direct Mode

The Direct Mode expects the VNC viewer to initiate a connection to the system being installed. **Anaconda** asks you to initiate this connection.

Procedure 25.1. Starting VNC in Direct Mode

1. Run the VNC viewer of your choice on the workstation you are using to connect to the system being installed. For example, if you use **TigerVNC**:

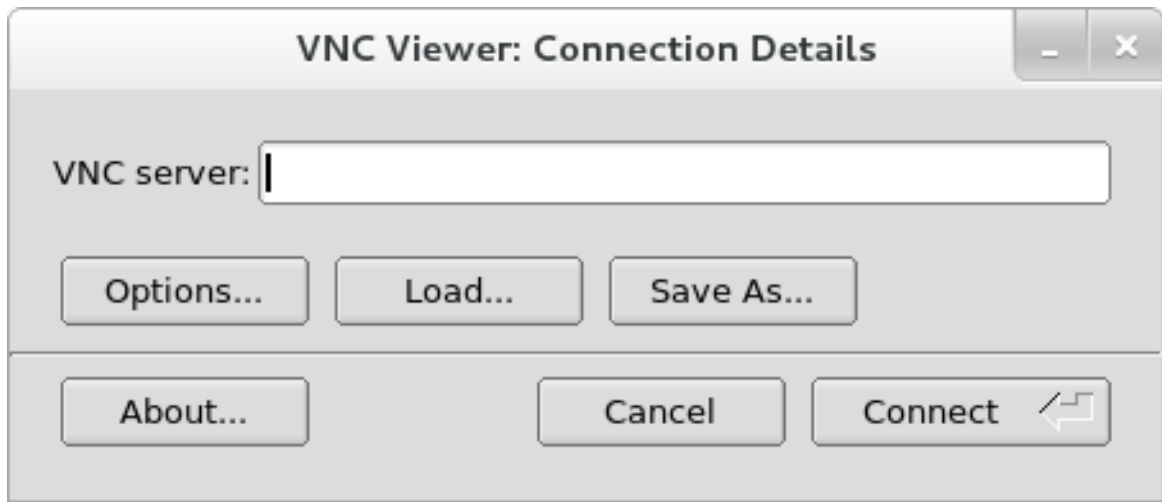
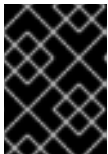


Figure 25.1. TigerVNC Connection Details

2. Boot the installation system and wait for the boot menu to appear. In the menu, press the **Tab** key to edit boot options. Append the **inst.vnc** option to the end of the command line.

Optionally, if you want to restrict VNC access to the installation system, add the **inst.vncpassword=PASSWORD** boot option as well. Replace *PASSWORD* with the password you want to use for the installation. The VNC password must be between 6 and 8 characters long.



IMPORTANT

Use a temporary password for the **inst.vncpassword=** option. It should not be a real or root password you use on any system.

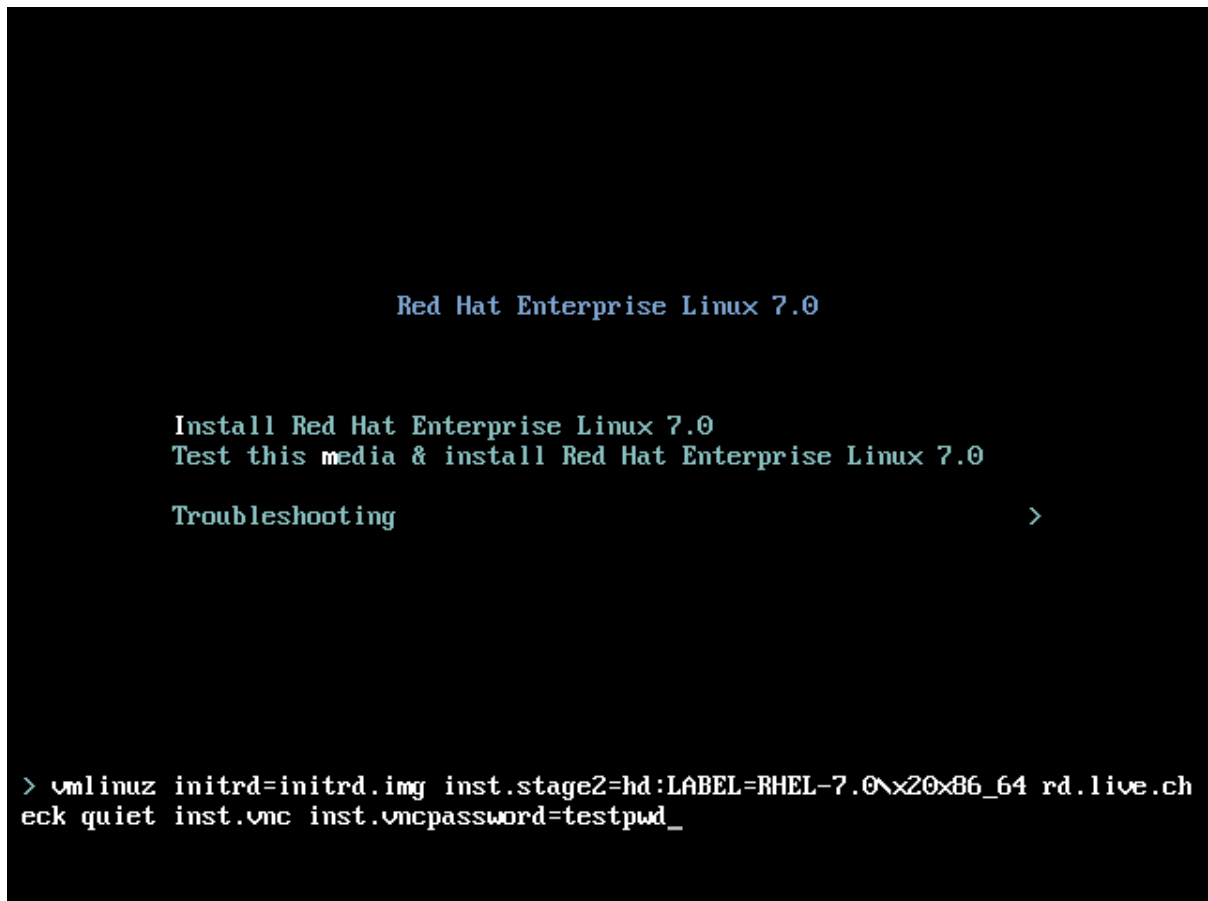


Figure 25.2. Adding VNC Boot Options on AMD, Intel, and ARM Systems

3. Press **Enter** to start the installation. The system initializes the installation program and starts the necessary services. When the system is ready, you get a message on the screen similar to the following:

```
13:14:47 Please manually connect your VNC viewer to 192.168.100.131:1 to begin the
install.
```

Note the IP address and port number (in the above example, **192.168.100.131:1**).

4. On the system running the VNC Viewer, enter the IP address and port number obtained in the previous step into the **Connection Details** dialog in the same format as it was displayed on the screen by Anaconda. Then, click **Connect**. The VNC viewer connects to the installation system. If you set up a VNC password, enter it when prompted and click **OK**.

For further details about using a VNC client, see the corresponding section in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

After you finish the procedure, a new window opens with the VNC connection established, displaying the installation menu. In this window, you can use the **Anaconda** graphical interface the same way you would use it when installing directly on the system.

You can proceed with:

- [Chapter 8, Installing Using Anaconda](#) for 64-bit AMD, Intel, and ARM systems
- [Chapter 13, Installing Using Anaconda](#) for IBM Power Systems servers
- [Chapter 18, Installing Using Anaconda](#) for IBM Z

25.2.2. Installing in VNC Connect Mode

In Connect Mode, the system being installed initiates a connection to the VNC viewer running on a remote system. Before you start, make sure the remote system is configured to accept an incoming connection on the port you want to use for VNC. The exact way to make sure the connection is not blocked depends on your network and on your workstation's configuration. Information about configuring the firewall is available in the [Red Hat Enterprise Linux 7 Security Guide](#).

Procedure 25.2. Starting VNC in Connect Mode

1. Start the VNC viewer on the client system in listening mode. For example, on Red Hat Enterprise Linux using **TigerVNC**, execute the following command:

```
$ vncviewer -listen PORT
```

Replace *PORT* with the port number you want to use for the connection.

The terminal displays a message similar to the following example:

Example 25.1. TigerVNC Viewer Listening

```
TigerVNC Viewer 64-bit v1.3.0 (20130924)
Built on Sep 24 2013 at 16:32:56
Copyright (C) 1999-2011 TigerVNC Team and many others (see README.txt)
See http://www.tigervnc.org for information on TigerVNC.
```

```
Thu Feb 20 15:23:54 2014
main:      Listening on port 5901
```

The VNC viewer is now ready and waiting for an incoming connection from the installation system.

2. Boot the system being installed and wait for the boot menu to appear. In the menu, press the **Tab** key to edit boot options. Append the following options to the command line:

```
inst.vnc inst.vncconnect=HOST:PORT
```

Replace *HOST* with the IP address of the system running the listening VNC viewer, and *PORT* with the port number that the VNC viewer is listening on.

3. Press **Enter** to start the installation. The system initializes the installation program and starts the necessary services. Once the initialization is finished, **Anaconda** attempts to connect to the IP address and port you provided in the previous step.

When the connection is successfully established, a new window opens on the system running the VNC viewer, displaying the installation menu. In this window, you can use the **Anaconda** graphical interface the same way you would use it when installing directly on the system.

After you finish the procedure, you can proceed with:

- [Chapter 8, Installing Using Anaconda](#) for 64-bit AMD, Intel, and ARM systems
- [Chapter 13, Installing Using Anaconda](#) for IBM Power Systems servers

- [Chapter 18, *Installing Using Anaconda*](#) for IBM Z

25.3. KICKSTART CONSIDERATIONS

Commands for using VNC are also available in Kickstart installations. Using only the **vnc** command results in an installation using Direct Mode. Additional options are available to set up an installation using Connect Mode. For more information about the **vnc** command and options used in Kickstart files, see [Section 27.3.1, “Kickstart Commands and Options”](#).

CHAPTER 26. HEADLESS SYSTEMS

When installing headless systems, you can only choose between an automated Kickstart installation and an interactive VNC installation using Connect Mode. For more information about automated Kickstart installation, see [Section 27.3.1, “Kickstart Commands and Options”](#). The general process for interactive VNC installation is described below.

1. Set up a network boot server to start the installation. Information about installing and performing basic configuring of a network boot server can be found in [Chapter 24, *Preparing for a Network Installation*](#).
2. Configure the server to use the boot options for a Connect Mode VNC installation. For information on these boot options, see [Section 25.2.2, “Installing in VNC Connect Mode”](#).
3. Follow the procedure for VNC Installation using Connect Mode as described in the procedure [Procedure 25.2, “Starting VNC in Connect Mode”](#). However, when directed to boot the system, boot it from the network server instead of local media.

CHAPTER 27. KICKSTART INSTALLATIONS

27.1. WHAT ARE KICKSTART INSTALLATIONS?

Kickstart installations offer a means to automate the installation process, either partially or fully. Kickstart files contain answers to all questions normally asked by the installation program, such as what time zone you want the system to use, how the drives should be partitioned, or which packages should be installed. Providing a prepared Kickstart file when the installation begins therefore allows you to perform the installation automatically, without need for any intervention from the user. This is especially useful when deploying Red Hat Enterprise Linux on a large number of systems at once.

Kickstart files can be kept on a single server system and read by individual computers during the installation. This installation method can support the use of a single Kickstart file to install Red Hat Enterprise Linux on multiple machines, making it ideal for network and system administrators.

All Kickstart scripts and the log files of their execution are stored in the **/tmp** directory to assist with debugging installation failures.



NOTE

In previous versions of Red Hat Enterprise Linux, Kickstart allowed for upgrading the system as well. In Red Hat Enterprise Linux 7, this functionality has been removed and system upgrades are instead handled by specialized tools. See [Chapter 29, Upgrading Your Current System](#) for details.

27.2. HOW DO YOU PERFORM A KICKSTART INSTALLATION?

Kickstart installations can be performed using a local DVD, a local hard drive, NFS, FTP, HTTP, or HTTPS.

To use Kickstart, you must:

1. Create a Kickstart file.
2. Make the Kickstart file available on removable media, a hard drive or a network location.
3. Create boot media, which will be used to begin the installation.
4. Make the installation source available.
5. Start the Kickstart installation.

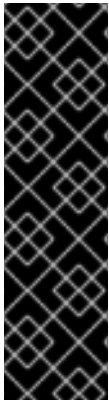
This chapter explains these steps in detail.

27.2.1. Creating a Kickstart File

The Kickstart file itself is a plain text file, containing keywords listed in [Section 27.3, “Kickstart Syntax Reference”](#), which serve as directions for the installation. Any text editor able to save files as ASCII text, such as **Gedit** or **vim** on Linux systems or **Notepad** on Windows systems, can be used to create and edit Kickstart files. The file name of your Kickstart configuration does not matter; however, it is recommended to use a simple name as you will need to specify this name later in other configuration files or dialogs.

The recommended approach to creating Kickstart files is to perform a manual installation on one system

first. After the installation completes, all choices made during the installation are saved into a file named **anaconda-ks.cfg**, located in the **/root/** directory on the installed system. You can then copy this file, make any changes you need, and use the resulting configuration file in further installations.



IMPORTANT

If you have a Red Hat Customer Portal account, you can use the **Kickstart Configuration Tool** available at <https://access.redhat.com/labs/kickstartconfig/> in the Customer Portal Labs. This tool will walk you through basic configuration and allows you to download the resulting Kickstart file. However, the tool currently does not support any advanced partitioning.

Kickstart Configurator, the graphical tool for creating Kickstart files, is still available. However, it is no longer being updated and it does not reflect changes in Kickstart syntax between Red Hat Enterprise Linux 6 and 7.

When creating a Kickstart file, keep in mind the following:

- Sections must be specified *in order*. Items within the sections do not have to be in a specific order unless otherwise specified. The section order is:
 - Command section - See [Section 27.3.1, “Kickstart Commands and Options”](#) for a list of Kickstart options. You must include the required options.

Add-ons for **Anaconda** which expand the functionality of the installer can also be used in the command section by using the **%addon *addon_name*** command. See [Section 27.3.7, “Kickstart Add-ons”](#) for details.

- The **%packages** section - See [Section 27.3.2, “Package Selection”](#) for details.
- The **%pre** and **%post** sections, and the **%onerror** section - these sections can be in any order and are not required. See [Section 27.3.3, “Pre-installation Script”](#), [Section 27.3.5, “Post-installation Script”](#) and [Section 27.3.6, “Kickstart error handling”](#) for details.



IMPORTANT

Sections **%addon**, **%packages**, **%onerror**, **%pre** and **%post** must end with **%end**, otherwise the installation program will refuse the Kickstart file.

- Items that are not required can be omitted.
- Omitting any required item results in the installation program prompting the user for an answer to the related item, just as the user would be prompted during a typical installation. Once the answer is given, the installation continues unattended (unless it finds another missing item).
- Lines starting with a pound (also known as number) sign (**#**) are treated as comments and are ignored.

27.2.2. Maintaining the Kickstart File

27.2.2.1. Verifying the Kickstart File

When creating or customizing your Kickstart file, it is useful to verify that it is valid before attempting to use it in an installation. Red Hat Enterprise Linux 7 includes the **ksvalidator** command line utility which

can be used to do this. This tool is a part of the `pykickstart` package. To install this package, execute the following command as **root**:

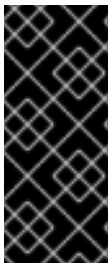
```
# yum install pykickstart
```

After installing the package, you can validate a Kickstart file using the following command:

```
$ ksvalidator /path/to/kickstart.ks
```

Replace `/path/to/kickstart.ks` with the path to the Kickstart file you want to verify.

For more information about this tool, see the **ksvalidator(1)** man page.



IMPORTANT

Keep in mind that the validation tool has its limitations. The Kickstart file can be very complicated; **ksvalidator** can make sure the syntax is correct and that the file does not include deprecated options, but it cannot guarantee the installation will be successful. It also does not attempt to validate the **%pre**, **%post** and **%packages** sections of the Kickstart file.

27.2.2.2. Changes in Kickstart Syntax

While the general principles of Kickstart installations tend to stay the same, the commands and options can change between major releases of Red Hat Enterprise Linux. You can use the **ksverdiff** command to display the differences between two versions of the Kickstart syntax. This is useful when updating an existing Kickstart file to be used with a new release. To display a list of changes in syntax between Red Hat Enterprise Linux 6 and 7, use the following command:

```
$ ksverdiff -f RHEL6 -t RHEL7
```

The **-f** option specifies the release to start the comparison with, and the **-t** option to specify the release to end with. For additional information, see the **ksverdiff(1)** man page.

27.2.3. Making the Kickstart File Available

A Kickstart file must be placed in one of the following locations:

- On *removable media*, such as a DVD or USB flash drive
- On a *hard drive* connected to the installation system
- On a *network share* reachable from the installation system

Normally, a Kickstart file is copied to removable media or a hard drive, or made available on the network. Placing the file in a network location complements the usual approach to Kickstart installations, which is also network-based: the system is booted using a PXE server, the Kickstart file is downloaded from a network share, and software packages specified in the file are downloaded from remote repositories.

Making the Kickstart file available and reachable from the installation system is exactly the same as making the installation source available, only with the Kickstart file instead of the installation ISO image or tree. For full procedures, see [Section 3.3, “Preparing Installation Sources”](#).

27.2.4. Making the Installation Source Available

The Kickstart installation must access an installation source in order to install the packages needed by your system. The source can be either the full Red Hat Enterprise Linux installation DVD ISO image, or an *installation tree*. An installation tree is a copy of the binary Red Hat Enterprise Linux DVD with the same directory structure.

If you are performing a DVD-based installation, insert the Red Hat Enterprise Linux installation DVD into the computer before starting the Kickstart installation. See [Section 3.3.1, “Installation Source on a DVD”](#) for information about using a Red Hat Enterprise Linux DVD as the installation source.

If you are performing a hard drive installation (using either a hard drive or an USB flash drive), make sure the ISO images of the binary Red Hat Enterprise Linux DVD are on a hard drive in the computer. See [Section 3.3.2, “Installation Source on a Hard Drive”](#) for details about using a hard drive as the installation source.

If you are performing a network-based (NFS, FTP or HTTP) installation, you must make the installation tree or the binary DVD ISO image (depending on the protocol used) available over the network. See [Section 3.3.3, “Installation Source on a Network”](#) for details.

27.2.5. Starting the Kickstart Installation



NOTE

To load your Kickstart file automatically without having to specify the **inst.ks=** boot option, name the file **ks.cfg** and place it on a storage volume labeled **OEMDRV**.

To start a Kickstart installation, use the boot option **inst.ks=location** when booting the installation system, replacing *location* with the location of your Kickstart file. The exact way to specify the boot option depends on your system's architecture – see [Chapter 23, Boot Options](#) for details.

64-bit AMD, Intel and ARM systems and IBM Power Systems servers have the ability to boot using a PXE server. When you configure the PXE server, you can add the boot option into the boot loader configuration file, which in turn allows you to start the installation automatically. Using this approach, it is possible to automate the installation completely, including the boot process. For information about setting up a PXE server, see [Chapter 24, Preparing for a Network Installation](#).

Procedures in this section assume that you already have a Kickstart file ready in a location accessible from the installation system, as well as boot media or a PXE server which can be used to boot the system and begin the installation. The procedures are intended as a general reference; some steps differ based on your system's architecture, and not all options are available on all architectures (for example, you cannot use PXE boot on IBM Z).

27.2.5.1. Starting the Kickstart Installation Manually

This section explains how to start a Kickstart installation manually, which means some user interaction (adding boot options at the **boot:** prompt) will be required.

Procedure 27.1. Starting the Kickstart Installation Using a Boot Option

1. Boot the system using either local media (a CD, DVD, or a USB flash drive). For architecture-specific instructions, see:
 - [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) for 64-bit AMD, Intel, and ARM systems

- [Chapter 12, Booting the Installation on IBM Power Systems](#) for IBM Power Systems servers
 - [Chapter 16, Booting the Installation on IBM Z](#) for IBM Z
2. At the boot prompt, specify the **inst.ks=** boot option and the location of the Kickstart file. If the Kickstart file is in a network location, you must also configure the network using the **ip=** option. In some cases, the **inst.repo=** option is also necessary in order to access a software source from which necessary packages will be installed.

For details about boot options and valid syntax, see [Chapter 23, Boot Options](#).

3. Start the installation by confirming your added boot options.

The installation begins now, using the options specified in the Kickstart file. If the Kickstart file is valid and contains all required commands, the installation is completely automated from this point forward.

27.2.5.2. Starting the Kickstart Installation Automatically

The following procedure explains how to completely automate the Kickstart installation, using a network boot (PXE) server and a properly configured boot loader. If you follow this procedure, you only need to turn on the system; no other interaction will be required from that moment until the installation finishes.



NOTE

PXE installations are not available on IBM Z.

Procedure 27.2. Starting the Kickstart Installation by Editing Boot Loader Configuration

1. Open the boot loader configuration file on your PXE server, and add the **inst.ks=** boot option to the appropriate line. The name of the file and its syntax depends on your system's architecture and hardware:
 - On AMD64 and Intel 64 systems with *BIOS*, the file name can be either **default** or based on your system's IP address. In this case, add the **inst.ks=** option to the **append** line in the installation entry. A sample **append** line in the configuration file looks similar to the following:

```
append initrd=initrd.img inst.ks=http://10.32.5.1/mnt/archive/RHEL-
7/7.x/Server/x86_64/kickstarts/ks.cfg
```

- On systems using the **GRUB2** boot loader (64-bit AMD, Intel, and ARM systems with UEFI firmware and IBM Power Systems servers), the file name will be **grub.cfg**. In this file, append the **inst.ks=** option to the **kernel** line in the installation entry. A sample **kernel** line in the configuration file will look similar to the following:

```
kernel vmlinuz inst.ks=http://10.32.5.1/mnt/archive/RHEL-
7/7.x/Server/x86_64/kickstarts/ks.cfg
```

2. Boot the installation from the network server. For architecture-specific instructions, see:
 - [Section 7.1.2, "Booting from the Network Using PXE"](#) for 64-bit AMD, Intel, and ARM systems

- [Section 12.3, “Booting from the Network Using an Installation Server”](#) for IBM Power Systems servers

The installation begins now, using the installation options specified in the Kickstart file. If the Kickstart file is valid and contains all required commands, the installation is completely automated.

27.3. KICKSTART SYNTAX REFERENCE

27.3.1. Kickstart Commands and Options



NOTE

If an option is followed by an equals mark (=), a value must be specified after it. In the example commands, options in square brackets ([]) are optional arguments for the command.

auth or **authconfig** (optional)

Sets up the authentication options for the system using the **authconfig** command, which can also be run on the command line after the installation finishes. See the **authconfig(8)** manual page and the **authconfig --help** command for more details. Passwords are shadowed by default.



WARNING

When using OpenLDAP with the **SSL** protocol for security, make sure that the **SSLv2** and **SSLv3** protocols are disabled in the server configuration. This is due to the POODLE SSL vulnerability (CVE-2014-3566). See <https://access.redhat.com/solutions/1234843> for details.

- **--enablenis** - Turns on NIS support. By default, **--enablenis** uses whatever domain it finds on the network. A domain should almost always be set by hand with the **--nisdomain=** option.
- **--nisdomain=** - NIS domain name to use for NIS services.
- **--nisserver=** - Server to use for NIS services (broadcasts by default).
- **--useshadow** or **--enableshadow** - Use shadow passwords.
- **--enableldap** - Turns on LDAP support in **/etc/nsswitch.conf**, allowing your system to retrieve information about users (for example, their UIDs, home directories, and shells) from an LDAP directory. To use this option, you must install the **nss-pam-ldapd** package. You must also specify a server and a base *DN* (distinguished name) with **--ldapserver=** and **--ldapbasedn=**.
- **--enableldappauth** - Use LDAP as an authentication method. This enables the **pam_ldap** module for authentication and changing passwords, using an LDAP directory. To use this option, you must have the **nss-pam-ldapd** package installed. You must also specify a server

and a base DN with **--ldapserver=** and **--ldapbasedn=**. If your environment does not use *TLS* (Transport Layer Security), use the **--disableldaptls** switch to ensure that the resulting configuration file works.

- **--ldapserver=** - If you specified either **--enableldap** or **--enableldapauth**, use this option to specify the name of the LDAP server to use. This option is set in the **/etc/ldap.conf** file.
- **--ldapbasedn=** - If you specified either **--enableldap** or **--enableldapauth**, use this option to specify the DN in your LDAP directory tree under which user information is stored. This option is set in the **/etc/ldap.conf** file.
- **--enableldaptls** - Use TLS (Transport Layer Security) lookups. This option allows LDAP to send encrypted user names and passwords to an LDAP server before authentication.
- **--disableldaptls** - Do not use TLS (Transport Layer Security) lookups in an environment that uses LDAP for authentication.
- **--enablekrb5** - Use Kerberos 5 for authenticating users. Kerberos itself does not know about home directories, UIDs, or shells. If you enable Kerberos, you must make users' accounts known to this workstation by enabling LDAP, NIS, or Hesiod or by using the **useradd** command. If you use this option, you must have the **pam_krb5** package installed.
- **--krb5realm=** - The Kerberos 5 realm to which your workstation belongs.
- **--krb5kdc=** - The KDC (or KDCs) that serve requests for the realm. If you have multiple KDCs in your realm, use a comma-separated list without spaces.
- **--krb5adminserver=** - The KDC in your realm that is also running **kadmind**. This server handles password changing and other administrative requests. This server must be run on the master KDC if you have more than one KDC.
- **--enablehesiod** - Enables Hesiod support for looking up user home directories, UIDs, and shells. More information on setting up and using Hesiod on your network is in **/usr/share/doc/glibc-2.x.x/README.hesiod**, which is included in the **glibc** package. Hesiod is an extension of DNS that uses DNS records to store information about users, groups, and various other items.
- **--hesiodlhs** and **--hesiodrhs** - The **Hesiod** LHS (left-hand side) and RHS (right-hand side) values, set in **/etc/hesiod.conf**. The **Hesiod** library uses these values to search DNS for a name, similar to the way that **LDAP** uses a base DN.

To look up user information for the user name **jim**, the Hesiod library looks up **jim.passwdLHSRHS**, which should resolve to a TXT record that contains a string identical to an entry for that user in the **passwd** file: **jim:*:1001:1001:Jungle Jim:/home/jim:/bin/bash**. To look up groups, the Hesiod library looks up **jim.groupLHSRHS** instead.

To look up users and groups by number, make **1001.uid** a CNAME for **jim.passwd**, and **1001.gid** a CNAME for **jim.group**. Note that the library does not place a period (**.**) in front of the LHS and RHS values when performing a search. Therefore, if the LHS and RHS values need to have a period placed in front of them, you must include the period in the values you set for **--hesiodlhs** and **--hesiodrhs**.

- **--enablesmbauth** - Enables authentication of users against an SMB server (typically a Samba or Windows server). SMB authentication support does not know about home directories, UIDs, or shells. If you enable SMB, you must make users' accounts known to the workstation by enabling LDAP, NIS, or Hesiod or by using the **useradd** command.

- **--smbservers=** - The name of the servers to use for SMB authentication. To specify more than one server, separate the names with commas (,).
- **--smbworkgroup=** - The name of the workgroup for the SMB servers.
- **--enablecache** - Enables the **nscd** service. The **nscd** service caches information about users, groups, and various other types of information. Caching is especially helpful if you choose to distribute information about users and groups over your network using **NIS**, **LDAP**, or **Hesiod**.
- **--passalgo=** - Specify **sha256** to set up the SHA-256 hashing algorithm or **sha512** to set up the SHA-512 hashing algorithm.

autopart (optional)

Automatically creates partitions: a root (/) partition (1 GB or larger), a **swap** partition, and an appropriate **/boot** partition for the architecture. On large enough drives (50 GB and larger), this also creates a **/home** partition.



IMPORTANT

The **autopart** option cannot be used together with the **part/partition**, **raid**, **logvol**, or **volgroup** options in the same Kickstart file.

- **--type=** - Selects one of the predefined automatic partitioning schemes you want to use. Accepts the following values:
 - **lvm**: The LVM partitioning scheme.
 - **btrfs**: The Btrfs partitioning scheme.
 - **plain**: Regular partitions with no LVM or Btrfs.
 - **thinp**: The LVM Thin Provisioning partitioning scheme.

For a description of the available partition schemes, see [Section 8.14.4.1.1, "File System Types"](#).

- **--fstype=** - Selects one of the available file system types. The available values are **ext2**, **ext3**, **ext4**, **xfs**, and **vfat**. The default file system is **xfs**. For information about these file systems, see [Section 8.14.4.1.1, "File System Types"](#).
- **--nohome** - Disables automatic creation of the **/home** partition.
- **--nolvm** - Do not use LVM or Btrfs for automatic partitioning. This option is equal to **--type=plain**.
- **--encrypted** - Encrypts all partitions. This is equivalent to checking the **Encrypt partitions** check box on the initial partitioning screen during a manual graphical installation.

**NOTE**

When encrypting one or more partitions, **Anaconda** attempts to gather 256 bits of entropy to ensure the partitions are encrypted securely. Gathering entropy can take some time - the process will stop after a maximum of 10 minutes, regardless of whether sufficient entropy has been gathered.

The process can be sped up by interacting with the installation system (typing on the keyboard or moving the mouse). If you are installing in a virtual machine, you can also attach a virtio-rng device (a virtual random number generator) to the guest as described in the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#).

- **--passphrase=** - Provides a default system-wide passphrase for all encrypted devices.
- **--escrowcert=URL_of_X.509_certificate** - Stores data encryption keys of all encrypted volumes as files in **/root**, encrypted using the X.509 certificate from the URL specified with **URL_of_X.509_certificate**. The keys are stored as a separate file for each encrypted volume. This option is only meaningful if **--encrypted** is specified.
- **--backuppassphrase** - Adds a randomly-generated passphrase to each encrypted volume. Store these passphrases in separate files in **/root**, encrypted using the X.509 certificate specified with **--escrowcert**. This option is only meaningful if **--escrowcert** is specified.
- **--cipher=** - Specifies the type of encryption to use if the **Anaconda** default **aes-xts-plain64** is not satisfactory. You must use this option together with the **--encrypted** option; by itself it has no effect. Available types of encryption are listed in the [Red Hat Enterprise Linux 7 Security Guide](#), but Red Hat strongly recommends using either **aes-xts-plain64** or **aes-cbc-essiv:sha256**.

**NOTE**

It is recommended to use the **autopart --nohome** Kickstart option when installing on a single FBA DASD of the CMS type. This ensures that the installer does not create a separate **/home** partition. The installation then proceeds successfully.

autostep (optional)

Normally, Kickstart installations skip unnecessary screens. This option makes the installation program step through every screen, displaying each briefly. This option should not be used when deploying a system because it can disrupt package installation.

- **--autoscreenshot** - Take a screenshot at every step during installation. These screenshots are stored in **/tmp/anaconda-screenshots/** during the installation, and after the installation finishes you can find them in **/root/anaconda-screenshots**.

Each screen is only captured right before the installer switches to the next one. This is important, because if you do not use all required Kickstart options and the installation therefore does not begin automatically, you can go to the screens which were not automatically configured, perform any configuration you want. Then, when you click **Done** to continue, the screen is captured including the configuration you just provided.

bootloader (required)

Specifies how the boot loader should be installed.



IMPORTANT

Red Hat recommends setting up a boot loader password on every system. An unprotected boot loader can allow a potential attacker to modify the system's boot options and gain unauthorized access to the system.



IMPORTANT

Device names in the **sdX** (or **/dev/sdX**) format are not guaranteed to be consistent across reboots, which can complicate usage of some Kickstart commands. When a command calls for a device node name, you can instead use any item from **/dev/disk**. For example, instead of:

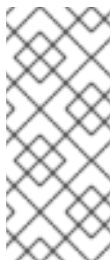
```
part / --fstype=xfs --onpart=sda1
```

You could use an entry similar to one of the following:

```
part / --fstype=xfs --onpart=/dev/disk/by-path/pci-0000:00:05.0-scsi-0:0:0:0-part1
```

```
part / --fstype=xfs --onpart=/dev/disk/by-id/ata-ST3160815AS_6RA0C882-part1
```

This way the command will always target the same storage device. This is especially useful in large storage environments. See the chapter about persistent storage device naming in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#) for more in-depth information about different ways to consistently refer to storage devices.



NOTE

In some cases, a special partition is required to install the boot loader on 64-bit AMD, Intel, and ARM systems. The type and size of this partition depends on whether the disk you are installing the boot loader to uses the *Master Boot Record* (MBR) or a *GUID Partition Table* (GPT) schema. For more information, see [Section 8.14.1, "Boot Loader Installation"](#).

- **--append=** - Specifies additional kernel parameters. To specify multiple parameters, separate them with spaces. For example:

```
bootloader --location=mbr --append="hdd=ide-scsi ide=nodma"
```

The **rhgb** and **quiet** parameters are always used, even if you do not specify them here or do not use the **--append=** command at all.

This option is useful for disabling mechanisms which were implemented to mitigate the Meltdown and Spectre speculative execution vulnerabilities found in most modern processors (CVE-2017-5754, CVE-2017-5753, and CVE-2017-5715). In some cases, these mechanisms may be unnecessary, and keeping them enabled causes decreased performance with no improvement in security. To disable these mechanisms, add the options to do so into your Kickstart file - for example, **bootloader --append="nopti noibrs noibpb"** on AMD64/Intel 64 systems.

**WARNING**

Ensure your system is not at risk of attack before disabling any of the vulnerability mitigation mechanisms. See the [Red Hat vulnerability response article](#) for information about the Meltdown and Spectre vulnerabilities.

- **--boot-drive=** - Specifies which drive the boot loader should be written to, and therefore which drive the computer will boot from. If you use a multipath device as the boot drive, specify the device using its disk/by-id/dm-uuid-mpath-WWID name.

**IMPORTANT**

The **--boot-drive=** option is currently being ignored in Red Hat Enterprise Linux installations on IBM Z systems using the **zipl** boot loader. When **zipl** is installed, it determines the boot drive on its own.

- **--leavebootorder** - The installer will add Red Hat Enterprise Linux 7 to the top of the list of installed systems in the boot loader, and preserve all existing entries as well as their order.
- **--driveorder=** - Specifies which drive is first in the BIOS boot order. For example:

```
bootloader --driveorder=sda,hda
```

- **--location=** - Specifies where the boot record is written. Valid values are the following:
 - **mbr** - The default option. Depends on whether the drive uses the Master Boot Record (MBR) or GUID Partition Table (GPT) scheme:
 - On a GPT-formatted disk, this option installs stage 1.5 of the boot loader into the BIOS boot partition.
 - On an MBR-formatted disk, stage 1.5 is installed into the empty space between the MBR and the first partition.
 - **partition** - Install the boot loader on the first sector of the partition containing the kernel.
 - **none** - Do not install the boot loader.

In most cases, this option does not need to be specified.

- **--password=** - If using **GRUB2**, sets the boot loader password to the one specified with this option. This should be used to restrict access to the **GRUB2** shell, where arbitrary kernel options can be passed.

If a password is specified, **GRUB2** also asks for a user name. The user name is always **root**.

- **--iscrypted** – Normally, when you specify a boot loader password using the **--password=** option, it is stored in the Kickstart file in plain text. If you want to encrypt the password, use this option and an encrypted password.

To generate an encrypted password, use the **grub2-mkpasswd-pbkdf2** command, enter the password you want to use, and copy the command's output (the hash starting with **grub.pbkdf2**) into the Kickstart file. An example **bootloader** Kickstart entry with an encrypted password looks similar to the following:

```
bootloader --iscrypted --
password=grub.pbkdf2.sha512.10000.5520C6C9832F3AC3D149AC0B24BE69E2D4FB0D
BEEDBD29CA1D30A044DE2645C4C7A291E585D4DC43F8A4D82479F8B95CA4BA4381
F8550510B75E8E0BB2938990.C688B6F0EF935701FF9BD1A8EC7FE5BD2333799C98F
28420C5CC8F1A2A233DE22C83705BB614EA17F3FDFDF4AC2161CEA3384E56EB38A
2E39102F5334C47405E
```

- **--timeout=** – Specifies the amount of time the boot loader waits before booting the default option (in seconds).
- **--default=** – Sets the default boot image in the boot loader configuration.
- **--extlinux** – Use the **extlinux** boot loader instead of **GRUB2**. This option only works on systems supported by **extlinux**.
- **--disabled** – This option is a stronger version of **--location=none**. While **--location=none** simply disables boot loader installation, **--disabled** disables boot loader installation and also disables installation of the package containing the boot loader, thus saving space.

btrfs (optional)

Create a Btrfs volume or subvolume. For a volume, the syntax is:

```
btrfs mntpoint --data=level --metadata=level --label=label partitions
```

One or more partitions can be specified in *partitions*. When specifying more than one partitions, the entries must be separated by a single space. See [Example 27.1, "Creating Btrfs Volumes and Subvolumes"](#) for a demonstration.

For a subvolume, the syntax is:

```
btrfs mntpoint --subvol --name=path parent
```

parent should be the identifier of the subvolume's parent volume and *mntpoint* is the location where the file system is mounted.

- **--data=** – RAID level to use for file system data (such as 0, 1, or 10). This parameter is optional, has no meaning for subvolumes, and requires more than one physical disk.
- **--metadata=** – RAID level to use for file system/volume metadata (such as **0, 1, or 10**). Optional. This option has no meaning for subvolumes and requires more than one physical disk.
- **--label=** – Specify a label for the Btrfs file system. If the given label is already in use by another file system, a new label is created. This option has no meaning for subvolumes.

- **--noformat** or **--useexisting** - Use an existing Btrfs volume (or subvolume) and do not reformat the file system.
- **--mkfsoptions=** - Specifies additional parameters to be passed to the program that makes a filesystem on this partition. No processing is done on the list of arguments, so they must be supplied in a format that can be passed directly to the **mkfs** program. This means multiple options should be comma-separated or surrounded by double quotes, depending on the filesystem.

The following example shows how to create a Btrfs volume from member partitions on three disks with subvolumes for **/** and **/home**. The main volume is not mounted or used directly in this example.

Example 27.1. Creating Btrfs Volumes and Subvolumes

```
part btrfs.01 --size=6000 --ondisk=sda
part btrfs.02 --size=6000 --ondisk=sdb
part btrfs.03 --size=6000 --ondisk=sdz

btrfs none --data=0 --metadata=1 --label=rhel7 btrfs.01 btrfs.02 btrfs.03
btrfs / --subvol --name=root LABEL=rhel7
btrfs /home --subvol --name=home rhel7
```

clearpart (optional)

Removes partitions from the system, prior to creation of new partitions. By default, no partitions are removed.

IMPORTANT

Device names in the **sdX** (or **/dev/sdX**) format are not guaranteed to be consistent across reboots, which can complicate usage of some Kickstart commands. When a command calls for a device node name, you can instead use any item from **/dev/disk**. For example, instead of:

```
part / --fstype=xfs --onpart=sda1
```

You could use an entry similar to one of the following:

```
part / --fstype=xfs --onpart=/dev/disk/by-path/pci-0000:00:05.0-scsi-0:0:0:0-part1
```

```
part / --fstype=xfs --onpart=/dev/disk/by-id/ata-ST3160815AS_6RA0C882-part1
```

This way the command will always target the same storage device. This is especially useful in large storage environments. See the chapter about persistent storage device naming in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#) for more in-depth information about different ways to consistently refer to storage devices.

NOTE

If the **clearpart** command is used, then the **part --onpart** command cannot be used on a logical partition.

For a detailed example of partitioning including the **clearpart** command, see [Section 27.4.1, “Advanced Partitioning Example”](#).

- **--all** - Erases all partitions from the system.



WARNING

This option will erase all disks which can be reached by the installer, including any attached network storage. Use this option with caution.

You can prevent **clearpart** from wiping storage you want to preserve by using the **--drives=** option and specifying only the drives you want to clear, by attaching network storage later (for example, in the **%post** section of the Kickstart file), or by blacklisting the kernel modules used to access network storage.

- **--drives=** - Specifies which drives to clear partitions from. For example, the following clears all the partitions on the first two drives on the primary IDE controller:

```
clearpart --drives=hda,hdb --all
```

To clear a multipath device, use the format **disk/by-id/scsi-*WWID***, where *WWID* is the *world-wide identifier* for the device. For example, to clear a disk with WWID **58095BEC5510947BE8C0360F604351918**, use:

```
clearpart --drives=disk/by-id/scsi-58095BEC5510947BE8C0360F604351918
```

This format is preferable for all multipath devices, but if errors arise, multipath devices that do not use *logical volume management* (LVM) can also be cleared using the format **disk/by-id/dm-uuid-mpath-*WWID***, where *WWID* is the *world-wide identifier* for the device. For example, to clear a disk with WWID **2416CD96995134CA5D787F00A5AA11017**, use:

```
clearpart --drives=disk/by-id/dm-uuid-mpath-2416CD96995134CA5D787F00A5AA11017
```



WARNING

Never specify multipath devices by device names like **mpatha**. Device names such as this are not specific to a particular disk. The disk named **/dev/mpatha** during installation might not be the one that you expect it to be. Therefore, the **clearpart** command could target the wrong disk.

- **--initlabel** - Initializes a disk (or disks) by creating a default disk label for all disks in their respective architecture that have been designated for formatting (for example, msdos for x86). Because **--initlabel** can see all disks, it is important to ensure only those drives that are

to be formatted are connected.

```
clearpart --initlabel --drives=names_of_disks
```

For example:

```
clearpart --initlabel --drives=dasda,dasdb,dasdc
```

- **--list=** - Specifies which partitions to clear. This option overrides the **--all** and **--linux** options if used. Can be used across different drives. For example:

```
clearpart --list=sda2,sda3,sdb1
```

- **--linux** - Erases all Linux partitions.
- **--none** (default) - Do not remove any partitions.



NOTE

Using the **clearpart --all** command in a Kickstart file to remove all existing partitions during the installation can cause **Anaconda** to pause and prompt you for a confirmation in certain cases. If you need to perform the installation automatically with no interaction, add the **zerombr** command to your Kickstart file.

cmdline (optional)

Perform the installation in a completely non-interactive command line mode. Any prompt for interaction halts the installation. This mode is useful on IBM Z systems with the x3270 terminal. See [Section 21.4, "Parameters for Kickstart Installations"](#).



IMPORTANT

For a fully automatic installation, you must either specify one of the available modes (**graphical**, **text**, or **cmdline**) in the Kickstart file, or you must use the **console=** boot option as described in [Console, Environment and Display Options](#). If no mode is specified, the system will prompt you to choose one before continuing.

device (optional)

On most PCI systems, the installation program automatically detects Ethernet and SCSI cards. However, on older systems and some PCI systems, Kickstart requires a hint to find the proper devices. The **device** command, which tells the installation program to install extra modules, uses the following format:

```
device moduleName --opts=options
```

- *moduleName* - Replace with the name of the kernel module which should be installed.
- **--opts=** - Options to pass to the kernel module. For example:

```
device --opts="aic152x=0x340 io=11"
```

driverdisk (optional)

Driver disks can be used during Kickstart installations to provide additional drivers not included by default. You must copy the driver disks's contents to the root directory of a partition on the system's hard drive. Then, you must use the **driverdisk** command to specify that the installation program should look for a driver disk and its location.

```
driverdisk [partition]--source=url--biospart=biospart]
```

Alternatively, a network location can be specified for the driver disk:

```
driverdisk --source=ftp://path/to/dd.img  
driverdisk --source=http://path/to/dd.img  
driverdisk --source=nfs:host:/path/to/img
```

- *partition* - Partition containing the driver disk. Note that the partition must be specified as a full path (for example, **/dev/sdb1**), *not* just the partition name (for example, **sdb1**).
- **--source=** - URL for the driver disk. NFS locations can be given in the form of **nfs:host:/path/to/img**.
- **--biospart=** - BIOS partition containing the driver disk (for example, **82p2**).

Driver disks can also be loaded from a hard disk drive or a similar device instead of being loaded over the network or from **initrd**. Follow this procedure:

1. Load the driver disk on a hard disk drive, a USB or any similar device.
2. Set the label, for example, *DD*, to this device.
3. Add the following line to your Kickstart file:

```
driverdisk LABEL=DD:e1000.rpm
```

Replace *DD* with a specific label and replace *dd.rpm* with a specific name. Use anything supported by the **inst.repo** command instead of *LABEL* to specify your hard disk drive.

eula (optional)

Use this option to accept the *End User License Agreement* (EULA) without user interaction. Specifying this option prevents **Initial Setup** from prompting you to accept the license agreement after you finish the installation and reboot the system for the first time. See [Chapter 30, Initial Setup](#) for more information.

- **--agreed** (required) - Accept the EULA. This option must always be used, otherwise the **eula** command is meaningless.

fcoe (optional)

Specify which FCoE devices should be activated automatically in addition to those discovered by *Enhanced Disk Drive Services* (EDD).

```
fcoe --nic=name [options]
```

- **--nic=** (required) - The name of the device to be activated.

- **--dcb=** - Establish *Data Center Bridging* (DCB) settings.
- **--autovlan** - Discover VLANs automatically.

firewall (optional)

Specify the firewall configuration for the installed system.

firewall **--enabled|--disabled** *device* [*options*]

- **--enabled** or **--enable** - Reject incoming connections that are not in response to outbound requests, such as DNS replies or DHCP requests. If access to services running on this machine is needed, you can choose to allow specific services through the firewall.
- **--remove-service** - Do not allow services through the firewall.
- **--disabled** or **--disable** - Do not configure any iptables rules.
- **--trust=** - Listing a device here, such as `em1`, allows all traffic coming to and from that device to go through the firewall. To list more than one device, use **--trust em1 --trust em2**. Do NOT use a comma-separated format such as **--trust em1, em2**.
- *incoming* - Replace with one or more of the following to allow the specified services through the firewall.
 - **--ssh**
 - **--smtp**
 - **--http**
 - **--ftp**
- **--port=** - You can specify that ports be allowed through the firewall using the port:protocol format. For example, to allow IMAP access through your firewall, specify **imap:tcp**. Numeric ports can also be specified explicitly; for example, to allow UDP packets on port 1234 through, specify **1234:udp**. To specify multiple ports, separate them by commas.
- **--service=** - This option provides a higher-level way to allow services through the firewall. Some services (like **cups**, **avahi**, and so on.) require multiple ports to be open or other special configuration in order for the service to work. You can specify each individual port with the **--port** option, or specify **--service=** and open them all at once.

Valid options are anything recognized by the **firewall-offline-cmd** program in the **firewalld** package. If **firewalld** is running, **firewall-cmd --get-services** provides a list of known service names.

firstboot (optional)

Determine whether the **Initial Setup** application starts the first time the system is booted. If enabled, the initial-setup package must be installed. If not specified, this option is disabled by default.

- **--enable** or **--enabled** - **Initial Setup** is started the first time the system boots.
- **--disable** or **--disabled** - **Initial Setup** is not started the first time the system boots.

- **--reconfig** - Enable the **Initial Setup** to start at boot time in reconfiguration mode. This mode enables the language, mouse, keyboard, root password, security level, time zone and networking configuration options in addition to the default ones.

group (optional)

Creates a new user group on the system. If a group with the given name or GID already exists, this command fails. In addition, the **user** command can be used to create a new group for the newly created user.

```
group --name=name [--gid=gid]
```

- **--name=** - Provides the name of the group.
- **--gid=** - The group's GID. If not provided, defaults to the next available non-system GID.

graphical (optional)

Perform the installation in graphical mode. This is the default.



IMPORTANT

For a fully automatic installation, you must either specify one of the available modes (**graphical**, **text**, or **cmdline**) in the Kickstart file, or you must use the **console=** boot option as described in [Console, Environment and Display Options](#). If no mode is specified, the system will prompt you to choose one before continuing.

halt (optional)

Halt the system after the installation has successfully completed. This is similar to a manual installation, where **Anaconda** displays a message and waits for the user to press a key before rebooting. During a Kickstart installation, if no completion method is specified, this option is used as the default.

The **halt** command is equivalent to the **shutdown -h** command.

For other completion methods, see the **poweroff**, **reboot**, and **shutdown** commands.

ignoredisk (optional)

Causes the installation program to ignore the specified disks. This is useful if you use automatic partitioning and want to be sure that some disks are ignored. For example, without **ignoredisk**, attempting to deploy on a SAN-cluster the Kickstart would fail, as the installation program detects passive paths to the SAN that return no partition table.

```
ignoredisk --drives=drive1,drive2,...
```

where *driveN* is one of **sda**, **sdb**,..., **hda**,... and so on.

IMPORTANT

Device names in the **sdX** (or **/dev/sdX**) format are not guaranteed to be consistent across reboots, which can complicate usage of some Kickstart commands. When a command calls for a device node name, you can instead use any item from **/dev/disk**. For example, instead of:

```
part / --fstype=xfs --onpart=sda1
```

You could use an entry similar to one of the following:

```
part / --fstype=xfs --onpart=/dev/disk/by-path/pci-0000:00:05.0-scsi-0:0:0:0-part1
```

```
part / --fstype=xfs --onpart=/dev/disk/by-id/ata-ST3160815AS_6RA0C882-part1
```

This way the command will always target the same storage device. This is especially useful in large storage environments. See the chapter about persistent storage device naming in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#) for more in-depth information about different ways to consistently refer to storage devices.

To ignore a multipath device that does not use *logical volume management* (LVM), use the format **disk/by-id/dm-uuid-mpath-*WWID***, where *WWID* is the *world-wide identifier* for the device. For example, to ignore a disk with WWID **2416CD96995134CA5D787F00A5AA11017**, use:

```
ignoredisk --drives=disk/by-id/dm-uuid-mpath-2416CD96995134CA5D787F00A5AA11017
```



WARNING

Never specify multipath devices by device names like **mpatha**. Device names such as this are not specific to a particular disk. The disk named **/dev/mpatha** during installation might not be the one that you expect it to be. Therefore, the **clearpart** command could target the wrong disk.

- **--only-use** - Specifies a list of disks for the installation program to use. All other disks are ignored. For example, to use disk **sda** during installation and ignore all other disks:

```
ignoredisk --only-use=sda
```

To include a multipath device that does not use LVM:

```
ignoredisk --only-use=disk/by-id/dm-uuid-mpath-2416CD96995134CA5D787F00A5AA11017
```

To include a multipath device that uses LVM:

```
ignoredisk --only-use=/dev/disk/by-id/dm-uuid-mpath-
```

```
bootloader --location=mbr
```

- **--interactive** - Allows you to manually navigate the advanced storage screen.

install (optional)

The default installation mode. You must specify the type of installation from **cdrom**, **harddrive**, **nfs**, **liveimg**, or **url** (for FTP, HTTP, or HTTPS installations). The **install** command and the installation method command must be on separate lines. For example:

```
install
liveimg --url=file:///images/install/squashfs.img --noverifyssl
```

- **cdrom** - Install from the first optical drive on the system.
- **harddrive** - Install from a Red Hat installation tree or full installation ISO image on a local drive. The drive must contain a file system the installation program can mount: **ext2**, **ext3**, **ext4**, **vfat**, or **xfs**.
 - **--biospart=** - BIOS partition to install from (such as **82**).
 - **--partition=** - Partition to install from (such as **sdb2**).
 - **--dir=** - Directory containing the **variant** directory of the installation tree, or the ISO image of the full installation DVD.

For example:

```
harddrive --partition=hdb2 --dir=/tmp/install-tree
```

- **liveimg** - Install from a disk image instead of packages. The image can be the **squashfs.img** file from a live ISO image, a compressed tar file (**.tar**, **.tbz**, **.tgz**, **.txz**, **.tar.bz2**, **.tar.gz**, or **.tar.xz**), or any file system that the installation media can mount. Supported file systems are **ext2**, **ext3**, **ext4**, **vfat**, and **xfs**.



NOTE

When using the **liveimg** installation mode with a driver disk, drivers on the disk will not automatically be included in the installed system. If necessary, these drivers should be installed manually, or in the **%post** section of a kickstart script.

- **--url=** - The location to install from. Supported protocols are **HTTP**, **HTTPS**, **FTP**, and **file**.
- **--proxy=** - Specify an **HTTP**, **HTTPS** or **FTP** proxy to use while performing the installation.
- **--checksum=** - An optional argument with the **SHA256** checksum of the image file, used for verification.
- **--noverifyssl** - Disable SSL verification when connecting to an **HTTPS** server.

For example:


```
liveimg --url=file:///images/install/squashfs.img --
checksum=03825f567f17705100de3308a20354b4d81ac9d8bed4bb4692b2381045e56197
--noverifyssl
```

- **nfs** - Install from the NFS server specified.
 - **--server=** - Server from which to install (host name or IP).
 - **--dir=** - Directory containing the **variant** directory of the installation tree.
 - **--opts=** - Mount options to use for mounting the NFS export. (optional)

For example:

```
nfs --server=nfsserver.example.com --dir=/tmp/install-tree
```

- **url** - Install from an installation tree on a remote server using the **FTP**, **HTTP**, or **HTTPS** protocol. You can only specify one URL.
 - **--url=** - Specifies the **HTTP**, **HTTPS**, **FTP**, or **file** location to install from.
 - **--mirrorlist=** - Specifies the mirror URL to install from.
 - **--proxy=** - Specifies an **HTTP**, **HTTPS**, or **FTP** proxy to use during the installation.
 - **--noverifyssl** - Disables SSL verification when connecting to an **HTTPS** server.

For example:

```
url --url http://server/path
```

or:

```
url --url ftp://username:password@server/path
```

iscsi (optional)

```
iscsi --ipaddr=address [options]
```

Specifies additional iSCSI storage to be attached during installation. If you use the **iscsi** command, you must also assign a name to the iSCSI node, using the **iscsiname** command. The **iscsiname** command must appear before the **iscsi** command in the Kickstart file.

We recommend that wherever possible you configure iSCSI storage in the system BIOS or firmware (iBFT for Intel systems) rather than use the **iscsi** command. **Anaconda** automatically detects and uses disks configured in BIOS or firmware and no special configuration is necessary in the Kickstart file.

If you must use the **iscsi** command, ensure that networking is activated at the beginning of the installation, and that the **iscsi** command appears in the Kickstart file *before* you refer to iSCSI disks with commands such as **clearpart** or **ignoredisk**.

- **--ipaddr=** (required) - the IP address of the target to connect to.

- **--port=** (required) - the port number (typically, **--port=3260**)
- **--target=** - the target *IQN* (iSCSI Qualified Name).
- **--iface=** - bind the connection to a specific network interface instead of using the default one determined by the network layer. Once used, it must be specified in all instances of the **iscsi** command in the entire Kickstart file.
- **--user=** - the user name required to authenticate with the target
- **--password=** - the password that corresponds with the user name specified for the target
- **--reverse-user=** - the user name required to authenticate with the initiator from a target that uses reverse CHAP authentication
- **--reverse-password=** - the password that corresponds with the user name specified for the initiator

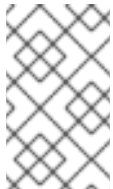
iscsiname (optional)

Assigns a name to an iSCSI node specified by the **iscsi** parameter. If you use the **iscsi** parameter in your Kickstart file, you must specify **iscsiname** *earlier* in the Kickstart file.

 `iscsiname iqn`

%addon com_redhat_kdump (optional)

This command configures the **kdump** kernel crash dumping mechanism.



NOTE

The syntax for this command is unusual because it is an add-on rather than a built-in Kickstart command. For more information about add-ons, see [Section 27.3.7, “Kickstart Add-ons”](#).

Kdump is a kernel crash dumping mechanism that allows you to save the contents of the system's memory for later analysis. It relies on **kexec**, which can be used to boot a Linux kernel from the context of another kernel without rebooting the system, and preserve the contents of the first kernel's memory that would otherwise be lost.

In case of a system crash, **kexec** boots into a second kernel (a *capture kernel*). This capture kernel resides in a reserved part of the system memory that is inaccessible to the first kernel. **Kdump** then captures the contents of the crashed kernel's memory (a *crash dump*) and saves it to a specified location. The location cannot be configured using this Kickstart command; it must be configured after the installation by editing the **/etc/kdump.conf** configuration file.

For more information about **Kdump**, see the [Red Hat Enterprise Linux 7 Kernel Crash Dump Guide](#).

Available options are:

- **--enable** - Enable kdump on the installed system.
- **--disable** - Disable kdump on the installed system.
- **--reserve-mb=** - The amount of memory you want to reserve for kdump, in MiB. For example:

```
%addon com_redhat_kdump --enable --reserve-mb=128
%end
```

You can also specify **auto** instead of a numeric value. In that case, the installer will determine the amount of memory automatically based on the criteria described in the [Red Hat Enterprise Linux 7 Kernel Crash Dump Guide](#).

If you enable **kdump** and do not specify a **--reserve-mb=** option, the value **auto** will be used.

- **--enablefadump** - Enable firmware-assisted dumping on systems which allow it (notably, IBM Power Systems servers).

keyboard (required)

Sets one or more available keyboard layouts for the system.

- **--vckeymap=** - Specify a **VConsole** keymap which should be used. Valid names correspond to the list of files in the `/usr/lib/kbd/keymaps/` directory, without the **.map.gz** extension.
- **--xlayouts=** - Specify a list of **X** layouts that should be used as a comma-separated list without spaces. Accepts values in the same format as **setxkbmap(1)**, either in the **layout** format (such as **cz**), or in the **layout (variant)** format (such as **cz (qwerty)**).

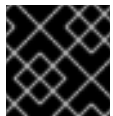
All available layouts can be viewed on the **xkeyboard-config(7)** man page under **Layouts**.

- **--switch=** - Specify a list of layout-switching options (shortcuts for switching between multiple keyboard layouts). Multiple options must be separated by commas without spaces. Accepts values in the same format as **setxkbmap(1)**.

Available switching options can be viewed on the **xkeyboard-config(7)** man page under **Options**.

The following example sets up two keyboard layouts (**English (US)** and **Czech (qwerty)**) using the **--xlayouts=** option, and allows to switch between them using **Alt+Shift**:

```
keyboard --xlayouts=us,'cz (qwerty)' --switch=grp:alt_shift_toggle
```



IMPORTANT

Either the **--vckeymap=** or the **--xlayouts=** option must be used.

lang (required)

Sets the language to use during installation and the default language to use on the installed system. For example, to set the language to English, the Kickstart file should contain the following line:

```
lang en_US
```

The file `/usr/share/system-config-language/locale-list` provides a list of the valid language codes in the first column of each line and is part of the `system-config-language` package.

Certain languages (for example, Chinese, Japanese, Korean, and Indic languages) are not supported during text-mode installation. If you specify one of these languages with the **lang** command, the installation process continues in English, but the installed system uses your selection as its default language.

- **--addsupport=** - Add support for additional languages. Takes the form of comma-separated list without spaces. For example:

```
lang en_US --addsupport=cs_CZ,de_DE,en_UK
```

logging (optional)

Controls the error logging of **Anaconda** during installation. It has no effect on the installed system.

```
logging [--host=host] [--port=port] [--level=debug|info|error|critical]
```

- **--host=** - Send logging information to the given remote host, which must be running a syslogd process configured to accept remote logging.
- **--port=** - If the remote syslogd process uses a port other than the default, set it using this option.
- **--level=** - Specify the minimum level of messages that appear on tty3. All messages are still sent to the log file regardless of this level, however. Possible values are **debug**, **info**, **warning**, **error**, or **critical**.

logvol (optional)

Create a logical volume for Logical Volume Management (LVM). For more information regarding LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration](#) guide. This command uses the following syntax:

```
logvol mntpoint --vgname=name --name=name [options]
```

NOTE

Do not use the dash (-) character in logical volume and volume group names when installing Red Hat Enterprise Linux using Kickstart. If this character is used, the installation finishes normally, but the **/dev/mapper/** directory will list these volumes and volume groups with every dash doubled. For example, a volume group named **volgrp-01** containing a logical volume named **logvol-01** will be listed as **/dev/mapper/volgrp--01-logvol--01**.

This limitation only applies to newly created logical volume and volume group names. If you are reusing existing ones using the **--noformat** option, their names will not be changed.

For a detailed example of **logvol** in action, see [Section 27.4.1, “Advanced Partitioning Example”](#).

- The *mntpoint* is where the partition is mounted and must be of one of the following forms:

- **/path**

For example, **/** or **/home**

- **swap**

The partition is used as swap space.

To determine the size of the swap partition automatically, use the **--recommended** option:

```
swap --recommended
```

To determine the size of the swap partition automatically and also allow extra space for your system to hibernate, use the **--hibernation** option:

```
swap --hibernation
```

The size assigned will be equivalent to the swap space assigned by **--recommended** plus the amount of RAM on your system.

For the swap sizes assigned by these commands, see [Section 8.14.4.4, “Recommended Partitioning Scheme”](#) for 64-bit AMD, Intel, and ARM systems, [Section 13.15.4.4, “Recommended Partitioning Scheme”](#) for IBM Power Systems servers, and [Section 18.15.3.4, “Recommended Partitioning Scheme”](#) for IBM Z.

The options are as follows:

- **--noformat** - Use an existing logical volume and do not format it.
- **--useexisting** - Use an existing logical volume and reformat it.
- **--fstype=** - Sets the file system type for the logical volume. Valid values are **xfs**, **ext2**, **ext3**, **ext4**, **swap**, and **vfat**.
- **--fsoptions=** - Specifies a free form string of options to be used when mounting the filesystem. This string will be copied into the **/etc/fstab** file of the installed system and should be enclosed in quotes.
- **--mkfsoptions=** - Specifies additional parameters to be passed to the program that makes a filesystem on this partition. No processing is done on the list of arguments, so they must be supplied in a format that can be passed directly to the **mkfs** program. This means multiple options should be comma-separated or surrounded by double quotes, depending on the filesystem.
- **--label=** - Sets a label for the logical volume.
- **--grow** - Tells the logical volume to grow to fill available space (if any), or up to the maximum size setting, if one is specified. A minimum size must be given, using either the **--percent=** option or the **--size=** option.



NOTE

For more information about allocating space when writing to an image file, see [section 3.5.4.Host Storage](#) describing **Image files** in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

- **--size=** - The size of the logical volume in MiB. This option cannot be used together with the **--percent=** option.
- **--percent=** - The size of the logical volume, as a percentage of the free space in the volume group after any statically-sized logical volumes are taken into account. This option cannot be used together with the **--size=** option.



IMPORTANT

When creating a new logical volume, you must either specify its size statically using the **--size=** option, or as a percentage of remaining free space using the **--percent=** option. You cannot use both of these options on the same logical volume.

Note that this behavior is only applies to Red Hat Enterprise Linux 7.1 and later. In Red Hat Enterprise Linux 7.0, these two options interacted differently.

- **--maxsize=** - The maximum size in MiB when the logical volume is set to grow. Specify an integer value here such as **500** (do not include the unit).
- **--recommended** - Use this option when creating a **swap** logical volume to determine the size of this volume automatically, based on your system's hardware. For details about the recommended scheme, see [Section 8.14.4.4, "Recommended Partitioning Scheme"](#) for 64-bit AMD, Intel, and ARM systems, [Section 13.15.4.4, "Recommended Partitioning Scheme"](#) for IBM Power Systems, and [Section 18.15.3.4, "Recommended Partitioning Scheme"](#) for IBM Z.
- **--resize** - Resize a logical volume. If you use this option, you must also specify **--useexisting** and **--size**.
- **--encrypted** - Specifies that this logical volume should be encrypted, using the passphrase provided in the **--passphrase=** option. If you do not specify a passphrase, the installation program uses the default, system-wide passphrase set with the **autopart --passphrase** command, or stops the installation and prompts you to provide a passphrase if no default is set.



NOTE

When encrypting one or more partitions, **Anaconda** attempts to gather 256 bits of entropy to ensure the partitions are encrypted securely. Gathering entropy can take some time - the process will stop after a maximum of 10 minutes, regardless of whether sufficient entropy has been gathered.

The process can be sped up by interacting with the installation system (typing on the keyboard or moving the mouse). If you are installing in a virtual machine, you can also attach a virtio-rng device (a virtual random number generator) to the guest as described in the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#).

- **--passphrase=** - Specifies the passphrase to use when encrypting this logical volume. You must use this option together with the **--encrypted** option; it has no effect by itself.
- **--cipher=** - Specifies which type of encryption will be used if the **Anaconda** default aes-xts-plain64 is not satisfactory. You must use this option together with the **--encrypted** option; by itself it has no effect. Available types of encryption are listed in the [Red Hat Enterprise Linux 7 Security Guide](#), but Red Hat strongly recommends using either aes-xts-plain64 or aes-cbc-essiv:sha256.
- **--escrowcert=URL_of_X.509_certificate** - Store data encryption keys of all encrypted volumes as files in **/root**, encrypted using the X.509 certificate from the URL specified with **URL_of_X.509_certificate**. The keys are stored as a separate file for each encrypted volume. This option is only meaningful if **--encrypted** is specified.

- **--backupperphrase** - Add a randomly-generated passphrase to each encrypted volume. Store these passphrases in separate files in **/root**, encrypted using the X.509 certificate specified with **--escrowcert**. This option is only meaningful if **--escrowcert** is specified.
- **--thinpool** - Creates a thin pool logical volume. (Use a mount point of **none**)
- **--metadatasize=size** - Specify the metadata area size (in MiB) for a new thin pool device.
- **--chunksize=size** - Specify the chunk size (in KiB) for a new thin pool device.
- **--thin** - Create a thin logical volume. (Requires use of **--poolname**)
- **--poolname=name** - Specify the name of the thin pool in which to create a thin logical volume. Requires the **--thin** option.
- **--profile=name** - Specify the configuration profile name to use with thin logical volumes. If used, the name will also be included in the metadata for the given logical volume. By default, the available profiles are **default** and **thin-performance** and are defined in the **/etc/lvm/profile/** directory. See the **lvm(8)** man page for additional information.
- **--cachepvs=** - A comma-separated list of physical volumes which should be used as a cache for this volume.
- **--cachemode=** - Specify which mode should be used to cache this logical volume - either **writeback** or **writethrough**.



NOTE

For more information about cached logical volumes and their modes, see the **lvmcache(7)** man page.

- **--cachesize=** - Size of cache attached to the logical volume, specified in MiB. This option requires the **--cachepvs=** option.

Create the partition first, create the logical volume group, and then create the logical volume. For example:

```
part pv.01 --size 3000
volgroup myvg pv.01
logvol / --vgname=myvg --size=2000 --name=rootvol
```

Create the partition first, create the logical volume group, and then create the logical volume to occupy 90% of the remaining space in the volume group. For example:

```
part pv.01 --size 1 --grow
volgroup myvg pv.01
logvol / --vgname=myvg --name=rootvol --percent=90
```

mediacheck (optional)

If given, this command forces the installation program to perform a media check (**rd.live.check**) before starting the installation. This command requires that installations be attended, so it is disabled by default.

mount (optional)

Assigns a mount point to an existing block device, and optionally reformats it to a given format.

```
mount [--reformat [REFORMAT]] [--mkfsoptions MKFS_OPTS] [--mountoptions MOUNT_OPTS]
device mntpoint
```

Unlike most other storage configuration commands in Kickstart, **mount** does not require you to describe the entire storage configuration in the Kickstart file. You only need to ensure that the described block device exists on the system. However, if you want to *create* the storage stack with all the devices mounted, you must use other commands such as **part** to do so.



IMPORTANT

You can not use **mount** together with other storage-related commands such as **part**, **logvol**, or **autopart** in the same Kickstart file.

Mandatory arguments:

- **device** - The block device to mount.
- **mntpoint** - Where to mount the **device**. It must be a valid mount point, such as `/` or `/usr`, or **none** if the device is unmountable (for example **swap**).

Optional arguments:

- **--reformat=** - Specifies a new format (such as **ext4**) to which the device should be reformatted.
- **--mkfsoptions=** - Specifies additional argument to be passed to the command which creates the new file system specified in **--reformat=**. The list of arguments provided here is not processed, so they must be specified in a format that can be passed directly to the **mkfs** program. The list of options should be either comma-separated or surrounded by double quotes, depending on the file system. Check the **mkfs** man page for the file system you want to create (for example **mkfs.ext4(8)** or **mkfs.xfs(8)**) for specific details.
- **--mountoptions=** - Specifies a free form string that contains options to be used when mounting the file system. The string will be copied to the **/etc/fstab** file on the installed system and should be enclosed in double quotes. See the **mount(8)** man page for a full list of mount options, and **fstab(5)** for basics.

network (optional)

Configures network information for the target system and activates network devices in the installation environment. The device specified in the first **network** command is activated automatically. Activation of the device can be also explicitly required by the **--activate** option.



NOTE

There are several types of network device naming standards used to identify network devices with persistent names such as **em1** or **wl3sp0**. For information about these standards, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

- **--activate** - activate this device in the installation environment.

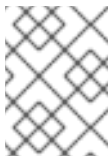
If you use the **--activate** option on a device that has already been activated (for example, an interface you configured with boot options so that the system could retrieve the Kickstart file) the device is reactivated to use the details specified in the Kickstart file.

Use the **--nodefroute** option to prevent the device from using the default route.

- **--no-activate** - do not activate this device in the installation environment.

By default, Anaconda activates the first network device in the Kickstart file regardless of the **--activate** option. You can disable the default setting by using the **--no-activate** option.

- **--bootproto=** - One of **dhcp**, **bootp**, **ibft**, or **static**. The default option is **dhcp**; the **dhcp** and **bootp** options are treated the same. To disable **ipv4** configuration of the device, use **--noipv4** option.



NOTE

This option configures ipv4 configuration of the device. For ipv6 configuration use **--ipv6** and **--ipv6gateway** options.

The DHCP method uses a DHCP server system to obtain its networking configuration. The BOOTP method is similar, requiring a BOOTP server to supply the networking configuration. To direct a system to use DHCP:

```
network --bootproto=dhcp
```

To direct a machine to use BOOTP to obtain its networking configuration, use the following line in the Kickstart file:

```
network --bootproto=bootp
```

To direct a machine to use the configuration specified in iBFT, use:

```
network --bootproto=ibft
```

The **static** method requires that you specify at least the IP address and netmask in the Kickstart file. This information is static and is used during and after the installation.

All static networking configuration information must be specified on *one* line; you cannot wrap lines using a backslash (\) as you can on a command line.

```
network --bootproto=static --ip=10.0.2.15 --netmask=255.255.255.0 --  
gateway=10.0.2.254 --nameserver=10.0.2.1
```

You can also configure multiple nameservers at the same time. To do so, use the **--nameserver=** option once, and specify each of their IP addresses, separated by commas.

```
network --bootproto=static --ip=10.0.2.15 --netmask=255.255.255.0 --  
gateway=10.0.2.254 --nameserver=192.168.2.1,192.168.3.1
```

- **--device=** - specifies the device to be configured (and eventually activated in **Anaconda**) with the **network** command.

If the **--device=** option is missing on the *first* use of the **network** command, the value of the

ksdevice= Anaconda boot option is used, if available. Note that this is considered deprecated behavior; in most cases, you should always specify a **--device=** for every **network** command.

The behavior of any subsequent **network** command in the same Kickstart file is unspecified if its **--device=** option is missing. Make sure you specify this option for any **network** command beyond the first.

You can specify a device to be activated in any of the following ways:

- the device name of the interface, for example, **em1**
- the MAC address of the interface, for example, **01:23:45:67:89:ab**
- the keyword **link**, which specifies the first interface with its link in the **up** state
- the keyword **bootif**, which uses the MAC address that **pxelinux** set in the **BOOTIF** variable. Set **IPAPPEND 2** in your **pxelinux.cfg** file to have **pxelinux** set the **BOOTIF** variable.

For example:

```
network --bootproto=dhcp --device=em1
```

- **--ip=** - IP address of the device.
- **--ipv6=** - IPv6 address of the device, in the form of *address[/prefix length]* - for example, **3ffe:ffff:0:1::1/128**. If *prefix* is omitted, **64** is used. You can also use **auto** for automatic configuration, or **dhcp** for DHCPv6-only configuration (no router advertisements).
- **--gateway=** - Default gateway as a single IPv4 address.
- **--ipv6gateway=** - Default gateway as a single IPv6 address.
- **--nodefroute** - Prevents the interface being set as the default route. Use this option when you activate additional devices with the **--activate=** option, for example, a NIC on a separate subnet for an iSCSI target.
- **--nameserver=** - DNS name server, as an IP address. To specify more than one name server, use this option once, and separate each IP address with a comma.
- **--nodns** - Do not configure any DNS server.
- **--netmask=** - Network mask for the installed system.
- **--hostname=** - The host name for the installed system. The host name can either be a *fully-qualified domain name* (FQDN) in the format **host_name.domainname**, or a short host name with no domain. Many networks have a *Dynamic Host Configuration Protocol* (DHCP) service which automatically supplies connected systems with a domain name; to allow DHCP to assign the domain name, only specify a short host name.



IMPORTANT

If your network does *not* provide a DHCP service, always use the FQDN as the system's host name.

- **--ethtool=** - Specifies additional low-level settings for the network device which will be passed to the **ethtool** program.
- **--essid=** - The network ID for wireless networks.
- **--wepkey=** - The WEP encryption key for wireless networks.
- **--wpakey=** - The WPA encryption key for wireless networks.
- **--onboot=** - Whether or not to enable the device at boot time.
- **--dhcpclass=** - The DHCP class.
- **--mtu=** - The MTU of the device.
- **--noipv4** - Disable IPv4 on this device.
- **--noipv6** - Disable IPv6 on this device.
- **--bondslaves=** - When this option is used, the network device specified in the **--device=** option is created using slaves defined in the **--bondslaves=** option. For example:

```
network --device=mynetwork --bondslaves=em1,em2
```

The above command creates a bond device named **mynetwork** using the **em1** and **em2** interfaces as its slaves.

- **--bondopts=** - a list of optional parameters for a bonded interface, which is specified using the **--bondslaves=** and **--device=** options. Options in this list must be separated by commas (",") or semicolons (";"). If an option itself contains a comma, use a semicolon to separate the options. For example:

```
network --bondopts=mode=active-backup,balance-rr;primary=eth1
```

Available optional parameters are listed in the *Working with Kernel Modules* chapter of the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).



IMPORTANT

The **--bondopts=mode=** parameter only supports full mode names such as **balance-rr** or **broadcast**, not their numerical representations such as **0** or **3**.

- **--vlanid=** - Specifies virtual LAN (VLAN) ID number (802.1q tag) for the device created using the device specified in **--device=** as a parent. For example, **network --device=em1 --vlanid=171** creates a virtual LAN device **em1.171**.
- **--interfacename=** - Specify a custom interface name for a virtual LAN device. This option should be used when the default name generated by the **--vlanid=** option is not desirable. This option must be used along with **--vlanid=**. For example:

```
network --device=em1 --vlanid=171 --interfacename=vlan171
```

The above command creates a virtual LAN interface named **vlan171** on the **em1** device with an ID of **171**.

The interface name can be arbitrary (for example, **my-vlan**), but in specific cases, the following conventions must be followed:

- If the name contains a dot (.), it must take the form of **NAME.ID**. The *NAME* is arbitrary, but the *ID* must be the VLAN ID. For example: **em1.171** or **my-vlan.171**.
- Names starting with **vlan** must take the form of **vlan/ID** – for example, **vlan171**.
- **--teamslaves=** – Team device specified by the **--device=** option will be created using slaves specified in this option. Slaves are separated by commas. A slave can be followed by its configuration, which is a single-quoted JSON string with double quotes escaped by the `\` character. For example:

```
network --teamslaves="p3p1'{"prio": -10, "sticky": true}',p3p2'{"prio": 100}'"
```

See also the **--teamconfig=** option.

- **--teamconfig=** – Double-quoted team device configuration which is a single-quoted JSON string with double quotes escaped by the `\` character. The device name is specified by **--device=** option and its slaves and their configuration by **--teamslaves=** option. For example:

```
network --device team0 --activate --bootproto static --ip=10.34.102.222 --
netmask=255.255.255.0 --gateway=10.34.102.254 --nameserver=10.34.39.2 --
teamslaves="p3p1'{"prio": -10, "sticky": true}',p3p2'{"prio": 100}'" --teamconfig="
{"runner": {"name": "activebackup"}}
```

- **--bridgeslaves=** – When this option is used, the network bridge with device name specified using the **--device=** option will be created and devices defined in the **--bridgeslaves=** option will be added to the bridge. For example:

```
network --device=bridge0 --bridgeslaves=em1
```

- **--bridgeopts=** – An optional comma-separated list of parameters for the bridged interface. Available values are **stp**, **priority**, **forward-delay**, **hello-time**, **max-age**, and **ageing-time**. For information about these parameters, see the *bridge setting* table in the **nm-settings(5)** man page or at <https://developer.gnome.org/NetworkManager/0.9/ref-settings.html>.

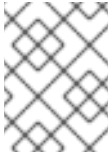
Also see the [Red Hat Enterprise Linux 7 Networking Guide](#) for general information about network bridging.

- **--bindto=mac** – Bind the device configuration (**ifcfg**) file on the installed system to the device MAC address (**HWADDR**) instead of the default binding to the interface name (**DEVICE**). Note that this option is independent of the **--device=** option – **--bindto=mac** will be applied even if the same **network** command also specifies a device name, **link**, or **bootif**.

nvdimm (optional)

Performs an action on Non-Volatile Dual In-line Memory Module (NVDIMM) devices. This command uses the following format:

```
nvdimm action [options]
```



NOTE

By default, all NVDIMM devices are ignored by the installer. You must use the **nvdimm** command to enable installation on these devices.

The following actions are available:

- **reconfigure** – Reconfigure a specific NVDIMM device into a given mode. Additionally, the specified device is implicitly marked as to be used, so a subsequent **nvdimm use** command for the same device is redundant. This action uses the following format:

```
nvdimm reconfigure [--namespace=NAMESPACE] [--mode=MODE] [--sectorsize=SECTORSIZE]
```

- **--namespace=** – The device specification by namespace. For example:

```
nvdimm reconfigure --namespace=namespace0.0 --mode=sector --sectorsize=512
```

- **--mode=** – The mode specification. Currently, only the value **sector** is available.

- **--sectorsize=** – Size of a sector for sector mode. For example:

```
nvdimm reconfigure --namespace=namespace0.0 --mode=sector --sectorsize=512
```

The supported sector sizes are 512 and 4096 bytes.

- **use** – Specify a NVDIMM device as a target for installation. The device must be already configured to the sector mode by the **nvdimm reconfigure** command. This action uses the following format:

```
nvdimm use [--namespace=NAMESPACE]--blockdevs=DEVICES]
```

- **--namespace=** – Specifies the device by namespace. For example:

```
nvdimm use --namespace=namespace0.0
```

- **--blockdevs=** – Specifies a comma-separated list of block devices corresponding to the NVDIMM devices to be used. The asterisk ***** wildcard is supported. For example:

```
nvdimm use --blockdevs=pmem0s,pmem1s
nvdimm use --blockdevs=pmem*
```

%addon org_fedora_oscap (optional)

The OpenSCAP installer add-on is used to apply SCAP (Security Content Automation Protocol) content – security policies – on the installed system. This add-on has been enabled by default since Red Hat Enterprise Linux 7.2. When enabled, the packages necessary to provide this functionality will automatically be installed. However, by default, no policies are enforced, meaning that no checks are performed during or after installation unless specifically configured.



IMPORTANT

Applying a security policy is not necessary on all systems. This screen should only be used when a specific policy is mandated by your organization rules or government regulations.

Unlike most other commands, this add-on does not accept regular options, but uses key-value pairs in the body of the **%addon** definition instead. These pairs are whitespace-agnostic. Values can be optionally enclosed in single quotes (') or double quotes (").

The following keys are recognized by the add-on:

- **content-type** - Type of the security content. Possible values are **datastream**, **archive**, **rpm**, and **scap-security-guide**.

If the **content-type** is **scap-security-guide**, the add-on will use content provided by the **scap-security-guide** package, which is present on the boot media. This means that all other keys except **profile** will have no effect.

- **content-url** - Location of the security content. The content must be accessible using HTTP, HTTPS, or FTP; local storage is currently not supported. A network connection must be available to reach content definitions in a remote location.
- **datastream-id** - ID of the data stream referenced in the **content-url** value. Used only if **content-type** is **datastream**.
- **xccdf-id** - ID of the benchmark you want to use.
- **xccdf-path** - Path to the XCCDF file which should be used; given as a relative path in the archive.
- **profile** - ID of the profile to be applied. Use **default** to apply the default profile.
- **fingerprint** - A MD5, SHA1 or SHA2 checksum of the content referenced by **content-url**.
- **tailoring-path** - Path to a tailoring file which should be used, given as a relative path in the archive.

The following is an example **%addon org_fedora_oscap** section which uses content from the **scap-security-guide** on the installation media:

Example 27.2. Sample OpenSCAP Add-on Definition Using SCAP Security Guide

```
%addon org_fedora_oscap
content-type = scap-security-guide
profile = xccdf_org.ssgproject.content_profile_pci-dss
%end
```

The following is a more complex example which loads a custom profile from a web server:

Example 27.3. Sample OpenSCAP Add-on Definition Using a Datastream

```
%addon org_fedora_oscap
content-type = datastream
```

```

content-url = http://www.example.com/scap/testing_ds.xml
datastream-id = scap_example.com_datastream_testing
xccdf-id = scap_example.com_cref_xccdf.xml
profile = xccdf_example.com_profile_my_profile
fingerprint = 240f2f18222faa98856c3b4fc50c4195
%end

```

For more information about the profiles available in the SCAP Security Guide and what they do, see [Red Hat Enterprise Linux 7 Security Guide](#).

part or partition (required)

Creates a partition on the system.



WARNING

All partitions created are formatted as part of the installation process unless **--noformat** and **--onpart** are used.

IMPORTANT

Device names in the **sdX** (or **/dev/sdX**) format are not guaranteed to be consistent across reboots, which can complicate usage of some Kickstart commands. When a command calls for a device node name, you can instead use any item from **/dev/disk**. For example, instead of:

```
part / --fstype=xfs --onpart=sda1
```

You could use an entry similar to one of the following:

```
part / --fstype=xfs --onpart=/dev/disk/by-path/pci-0000:00:05.0-scsi-0:0:0:0-part1
```

```
part / --fstype=xfs --onpart=/dev/disk/by-id/ata-ST3160815AS_6RA0C882-part1
```

This way the command will always target the same storage device. This is especially useful in large storage environments. See the chapter about persistent storage device naming in the [Red Hat Enterprise Linux 7 Storage Administration Guide](#) for more in-depth information about different ways to consistently refer to storage devices.

For a detailed example of **part** in action, see [Section 27.4.1, "Advanced Partitioning Example"](#).

```
part|partition mntpoint --name=name --device=device --rule=rule [options]
```

- *mntpoint* - Where the partition is mounted. The value must be of one of the following forms:
 - */path*

For example, `/`, `/usr`, `/home`

- **swap**

The partition is used as swap space.

To determine the size of the swap partition automatically, use the **--recommended** option:

```
swap --recommended
```

The size assigned will be effective but not precisely calibrated for your system.

To determine the size of the swap partition automatically but also allow extra space for your system to hibernate, use the **--hibernation** option:

```
swap --hibernation
```

The size assigned will be equivalent to the swap space assigned by **--recommended** plus the amount of RAM on your system.

For the swap sizes assigned by these commands, see [Section 8.14.4.4, "Recommended Partitioning Scheme"](#) for 64-bit AMD, Intel, and ARM systems, [Section 13.15.4.4, "Recommended Partitioning Scheme"](#) for IBM Power Systems servers, and [Section 18.15.3.4, "Recommended Partitioning Scheme"](#) for IBM Z.

- **raid.id**

The partition is used for software RAID (see **raid**).

- **pv.id**

The partition is used for LVM (see **logvol**).

- **biosboot**

The partition will be used for a BIOS Boot partition. A 1 MiB BIOS boot partition is necessary on BIOS-based AMD64 and Intel 64 systems using a *GUID Partition Table* (GPT); the boot loader will be installed into it. It is not necessary on UEFI systems. See also the **bootloader** command.

- **/boot/efi**

An EFI System Partition. A 50 MiB EFI partition is necessary on UEFI-based AMD, Intel, and ARM; the recommended size is 200 MiB. It is not necessary on BIOS systems. See also the **bootloader** command.

- **--size=** – The minimum partition size in MiB. Specify an integer value here such as **500** (do not include the unit).



IMPORTANT

If the **--size** value is too small, the installation fails. Set the **--size** value as the minimum amount of space you require. For size recommendations, see [Section 8.14.4.4, "Recommended Partitioning Scheme"](#).

- **--grow** - Tells the partition to grow to fill available space (if any), or up to the maximum size setting, if one is specified.



NOTE

If you use **--grow=** without setting **--maxsize=** on a swap partition, **Anaconda** limits the maximum size of the swap partition. For systems that have less than 2 GB of physical memory, the imposed limit is twice the amount of physical memory. For systems with more than 2 GB, the imposed limit is the size of physical memory plus 2GB.

- **--maxsize=** - The maximum partition size in MiB when the partition is set to grow. Specify an integer value here such as **500** (do not include the unit).
- **--noformat** - Specifies that the partition should not be formatted, for use with the **--onpart** command.
- **--onpart=** or **--usepart=** - Uses existing blank device and format it to the new specified type. For example:

```
partition /home --onpart=hda1
```

puts **/home** on **/dev/hda1**.

These options can also add a partition to a logical volume. For example:

```
partition pv.1 --onpart=hda2
```

The device must already exist on the system; the **--onpart** option will not create it.

It is also possible to specify an entire drive, rather than a partition, in which case **Anaconda** will format and use the drive without creating a partition table. Note, however, that installation of GRUB2 is not supported on a device formatted in this way, and must be placed on a drive with a partition table.

- **--ondisk=** or **--ondrive=** - Creates a partition (specified by the part command) on an existing disk. This command always creates a partition. For example, **--ondisk=sdb** puts the partition on the second SCSI disk on the system.

To specify a multipath device that does not use *logical volume management* (LVM), use the format **disk/by-id/dm-uuid-mpath-*WWID***, where *WWID* is the *world-wide identifier* for the device. For example, to specify a disk with WWID **2416CD96995134CA5D787F00A5AA11017**, use:

```
part / --fstype=xfst --grow --asprimary --size=8192 --ondisk=disk/by-id/dm-uuid-mpath-2416CD96995134CA5D787F00A5AA11017
```

**WARNING**

Never specify multipath devices by device names like **mpatha**. Device names such as this are not specific to a particular disk. The disk named **/dev/mpatha** during installation might not be the one that you expect it to be. Therefore, the **clearpart** command could target the wrong disk.

- **--asprimary** - Forces the partition to be allocated as a *primary* partition. If the partition cannot be allocated as primary (usually due to too many primary partitions being already allocated), the partitioning process fails. This option only makes sense when the disk uses a Master Boot Record (MBR); for GUID Partition Table (GPT)-labeled disks this option has no meaning. For information about primary (and extended) partitions, see [Section A.1.2, "Partitions: Turning One Drive Into Many"](#).
- **--fsprofile=** - Specifies a *usage type* to be passed to the program that makes a filesystem on this partition. A usage type defines a variety of tuning parameters to be used when making a filesystem. For this option to work, the filesystem must support the concept of usage types and there must be a configuration file that lists valid types. For **ext2**, **ext3**, **ext4**, this configuration file is **/etc/mke2fs.conf**.
- **--mkfsoptions=** - Specifies additional parameters to be passed to the program that makes a filesystem on this partition. This is similar to **--fsprofile** but works for all filesystems, not just the ones that support the profile concept. No processing is done on the list of arguments, so they must be supplied in a format that can be passed directly to the **mkfs** program. This means multiple options should be comma-separated or surrounded by double quotes, depending on the filesystem.
- **--fstype=** - Sets the file system type for the partition. Valid values are **xfs**, **ext2**, **ext3**, **ext4**, **swap**, **vfat**, **efi** and **biosboot**.
- **--fsoptions** - Specifies a free form string of options to be used when mounting the filesystem. This string will be copied into the **/etc/fstab** file of the installed system and should be enclosed in quotes.
- **--label=** - assign a label to an individual partition.
- **--recommended** - Determine the size of the partition automatically. For details about the recommended scheme, see [Section 8.14.4.4, "Recommended Partitioning Scheme"](#) for 64-bit AMD, Intel, and ARM, [Section 13.15.4.4, "Recommended Partitioning Scheme"](#) for IBM Power Systems, and [Section 18.15.3.4, "Recommended Partitioning Scheme"](#) for IBM Z.

**IMPORTANT**

This option can only be used for partitions which result in a file system such as the **/boot** partition and **swap** space. It cannot be used to create LVM physical volumes or RAID members.

- **--onbiosdisk** - Forces the partition to be created on a particular disk as discovered by the BIOS.
- **--encrypted** - Specifies that this partition should be encrypted, using the passphrase

provided in the **--passphrase** option. If you do not specify a passphrase, **Anaconda** uses the default, system-wide passphrase set with the **autopart --passphrase** command, or stops the installation and prompts you to provide a passphrase if no default is set.

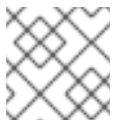


NOTE

When encrypting one or more partitions, **Anaconda** attempts to gather 256 bits of entropy to ensure the partitions are encrypted securely. Gathering entropy can take some time - the process will stop after a maximum of 10 minutes, regardless of whether sufficient entropy has been gathered.

The process can be sped up by interacting with the installation system (typing on the keyboard or moving the mouse). If you are installing in a virtual machine, you can also attach a virtio-rng device (a virtual random number generator) to the guest as described in the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#).

- **--passphrase=** - Specifies the passphrase to use when encrypting this partition. You must use this option together with the **--encrypted** option; by itself it has no effect.
- **--cipher=** - Specifies which type of encryption will be used if the **Anaconda** default **aes-xts-plain64** is not satisfactory. You must use this option together with the **--encrypted** option; by itself it has no effect. Available types of encryption are listed in the [Red Hat Enterprise Linux 7 Security Guide](#), but Red Hat strongly recommends using either **aes-xts-plain64** or **aes-cbc-essiv:sha256**.
- **--escrowcert=URL_of_X.509_certificate** - Store data encryption keys of all encrypted partitions as files in **/root**, encrypted using the X.509 certificate from the URL specified with **URL_of_X.509_certificate**. The keys are stored as a separate file for each encrypted partition. This option is only meaningful if **--encrypted** is specified.
- **--backuppssphrase** - Add a randomly-generated passphrase to each encrypted partition. Store these passphrases in separate files in **/root**, encrypted using the X.509 certificate specified with **--escrowcert**. This option is only meaningful if **--escrowcert** is specified.
- **--resize=** - Resize an existing partition. When using this option, specify the target size (in MiB) using the **--size=** option and the target partition using the **--onpart=** option.



NOTE

If partitioning fails for any reason, diagnostic messages appear on virtual console 3.

poweroff (optional)

Shut down and power off the system after the installation has successfully completed. Normally during a manual installation, **Anaconda** displays a message and waits for the user to press a key before rebooting. During a Kickstart installation, if no completion method is specified, the **halt** option is used as default.

The **poweroff** option is equivalent to the **shutdown -p** command.



NOTE

The **poweroff** command is highly dependent on the system hardware in use. Specifically, certain hardware components such as the BIOS, APM (advanced power management), and ACPI (advanced configuration and power interface) must be able to interact with the system kernel. Consult your hardware documentation for more information on your system's APM/ACPI abilities.

For other completion methods, see the **halt**, **reboot**, and **shutdown** Kickstart commands.

raid (optional)

Assembles a software RAID device. This command is of the form:

```
raid mntpoint --level=level --device=device-name partitions*
```

- *mntpoint* – Location where the RAID file system is mounted. If it is */*, the RAID level must be 1 unless a boot partition (**/boot**) is present. If a boot partition is present, the **/boot** partition must be level 1 and the root (*/*) partition can be any of the available types. The *partitions** (which denotes that multiple partitions can be listed) lists the RAID identifiers to add to the RAID array.



IMPORTANT

On IBM Power Systems, if a RAID device has been prepared and has not been reformatted during the installation, ensure that the RAID metadata version is **0.90** if you intend to put the **/boot** and **PReP** partitions on the RAID device.

The default Red Hat Enterprise Linux 7 **mdadm** metadata version is not supported for the boot device.

For a detailed example of **raid** in action, see [Section 27.4.1, “Advanced Partitioning Example”](#).

- **--level=** – RAID level to use (0, 1, 4, 5, 6, or 10). See [Section 8.14.4.2, “Create Software RAID”](#) for information about various available RAID levels.
- **--device=** – Name of the RAID device to use – for example, **--device=root**.



IMPORTANT

Do not use **mdraid** names in the form of **md0** – these names are not guaranteed to be persistent. Instead, use meaningful names such as **root** or **swap**. Using meaningful names creates a symbolic link from **/dev/md/name** to whichever **/dev/mdX** node is assigned to the array.

If you have an old (v0.90 metadata) array that you cannot assign a name to, you can specify the array by a filesystem label or UUID (for example, **--device=rhel7-root --label=rhel7-root**).

- **--chunksize=** – Sets the chunk size of a RAID storage in KiB. In certain situations, using a different chunk size than the default (**512 Kib**) can improve the performance of the RAID.
- **--spares=** – Specifies the number of spare drives allocated for the RAID array. Spare drives are used to rebuild the array in case of drive failure.

- **--fsprofile=** - Specifies a *usage type* to be passed to the program that makes a filesystem on this partition. A usage type defines a variety of tuning parameters to be used when making a filesystem. For this option to work, the filesystem must support the concept of usage types and there must be a configuration file that lists valid types. For ext2, ext3, and ext4, this configuration file is **/etc/mke2fs.conf**.
- **--fstype=** - Sets the file system type for the RAID array. Valid values are **xfs**, **ext2**, **ext3**, **ext4**, **swap**, and **vfat**.
- **--fsoptions=** - Specifies a free form string of options to be used when mounting the filesystem. This string will be copied into the **/etc/fstab** file of the installed system and should be enclosed in quotes.
- **--mkfsoptions=** - Specifies additional parameters to be passed to the program that makes a filesystem on this partition. No processing is done on the list of arguments, so they must be supplied in a format that can be passed directly to the **mkfs** program. This means multiple options should be comma-separated or surrounded by double quotes, depending on the filesystem.
- **--label=** - Specify the label to give to the filesystem to be made. If the given label is already in use by another filesystem, a new label will be created.
- **--noformat** - Use an existing RAID device and do not format the RAID array.
- **--useexisting** - Use an existing RAID device and reformat it.
- **--encrypted** - Specifies that this RAID device should be encrypted, using the passphrase provided in the **--passphrase** option. If you do not specify a passphrase, **Anaconda** uses the default, system-wide passphrase set with the **autopart --passphrase** command, or stops the installation and prompts you to provide a passphrase if no default is set.



NOTE

When encrypting one or more partitions, **Anaconda** attempts to gather 256 bits of entropy to ensure the partitions are encrypted securely. Gathering entropy can take some time - the process will stop after a maximum of 10 minutes, regardless of whether sufficient entropy has been gathered.

The process can be sped up by interacting with the installation system (typing on the keyboard or moving the mouse). If you are installing in a virtual machine, you can also attach a virtio-rng device (a virtual random number generator) to the guest as described in the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#).

- **--cipher=** - Specifies which type of encryption will be used if the **Anaconda** default aes-xts-plain64 is not satisfactory. You must use this option together with the **--encrypted** option; by itself it has no effect. Available types of encryption are listed in the [Red Hat Enterprise Linux 7 Security Guide](#), but Red Hat strongly recommends using either aes-xts-plain64 or aes-cbc-essiv:sha256.
- **--passphrase=** - Specifies the passphrase to use when encrypting this RAID device. You must use this option together with the **--encrypted** option; by itself it has no effect.
- **--escrowcert=URL_of_X.509_certificate** - Store the data encryption key for this device in a file in **/root**, encrypted using the X.509 certificate from the URL specified with **URL_of_X.509_certificate**. This option is only meaningful if **--encrypted** is specified.

- **--backupperphrase** - Add a randomly-generated passphrase to this device. Store the passphrase in a file in **/root**, encrypted using the X.509 certificate specified with **--escrowcert**. This option is only meaningful if **--escrowcert** is specified.

The following example shows how to create a RAID level 1 partition for **/**, and a RAID level 5 for **/home**, assuming there are three SCSI disks on the system. It also creates three swap partitions, one on each drive.

Example 27.4. Using the raid Kickstart command

```
part raid.01 --size=6000 --ondisk=sda
part raid.02 --size=6000 --ondisk=sdb
part raid.03 --size=6000 --ondisk=sd

part swap --size=512 --ondisk=sda
part swap --size=512 --ondisk=sdb
part swap --size=512 --ondisk=sd

part raid.11 --size=1 --grow --ondisk=sda
part raid.12 --size=1 --grow --ondisk=sdb
part raid.13 --size=1 --grow --ondisk=sd

raid / --level=1 --device=rhel7-root --label=rhel7-root raid.01 raid.02 raid.03
raid /home --level=5 --device=rhel7-home --label=rhel7-home raid.11 raid.12 raid.13
```

realm (optional)

Join an Active Directory or IPA domain. For more information about this command, see the **join** section of the **realm(8)** man page.

```
realm join [options] domain
```

- **--computer-ou=OU=** - Provide the distinguished name of an organizational unit in order to create the computer account. The exact format of the distinguished name depends on the client software and membership software. The root DSE portion of the distinguished name can usually be left out.
- **--no-password** - Join automatically without a password.
- **--one-time-password=** - Join using a one-time password. This is not possible with all types of realm.
- **--client-software=** - Only join realms which can run this client software. Valid values include **sssd** and **winbind**. Not all realms support all values. By default, the client software is chosen automatically.
- **--server-software=** - Only join realms which can run this server software. Possible values include **active-directory** or **freeipa**.
- **--membership-software=** - Use this software when joining the realm. Valid values include **samba** and **adcli**. Not all realms support all values. By default, the membership software is chosen automatically.

reboot (optional)

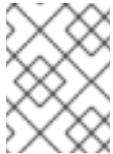
Reboot after the installation is successfully completed (no arguments). Normally, Kickstart displays a message and waits for the user to press a key before rebooting.

The **reboot** option is equivalent to the **shutdown -r** command.

Specify **reboot** to automate installation fully when installing in command line mode on IBM Z.

For other completion methods, see the **halt**, **poweroff**, and **shutdown** Kickstart options.

The **halt** option is the default completion method if no other methods are explicitly specified in the Kickstart file.



NOTE

Use of the **reboot** option *might* result in an endless installation loop, depending on the installation media and method.

- **--eject** - Attempt to eject the bootable media (DVD, USB, or other media) before rebooting.
- **--kexec** - Uses the **kexec** system call instead of performing a full reboot, which immediately loads the installed system into memory, bypassing the hardware initialization normally performed by the BIOS or firmware.



IMPORTANT

Due to the complexities involved with booting systems using **kexec**, it cannot be explicitly tested and guaranteed to function in every situation.

When **kexec** is used, device registers (which would normally be cleared during a full system reboot) might stay filled with data, which could potentially create issues for some device drivers.

repo (optional)

Configures additional **yum** repositories that can be used as sources for package installation. You can add multiple **repo** lines.

```
repo --name=repoid [--baseurl=<url>|--mirrorlist=url] [options]
```

- **--name=** - The repository id. This option is required. If a repository has a name which conflicts with another previously added repository, it is ignored. Because the installation program uses a list of preset repositories, this means that you cannot add repositories with the same names as the preset ones.
- **--baseurl=** - The URL to the repository. The variables that can be used in yum repo config files are not supported here. This option cannot be used together with the **--mirrorlist** option in the same repository definition.
- **--mirrorlist=** - The URL pointing at a list of mirrors for the repository. The variables that can be used in yum repository configuration files are not supported here. This option cannot be used together with the **--baseurl** option in the same repository definition.
- **--install** - Save the provided repository configuration on the installed system in the **/etc/yum.repos.d/** directory. Without using this option, a repository configured in a Kickstart file will only be available during the installation process, not on the installed system.

- **--cost=** - An integer value to assign a cost to this repository. If multiple repositories provide the same packages, this number is used to prioritize which repository will be used before another. Repositories with a lower cost take priority over repositories with higher cost.
- **--excludepkgs=** - A comma-separated list of package names that must *not* be pulled from this repository. This is useful if multiple repositories provide the same package and you want to make sure it comes from a particular repository. Both full package names (such as **publican**) and globs (such as **gnome-***) are accepted.
- **--includepkgs=** - A comma-separated list of package names and globs that must be pulled from this repository. This is useful if multiple repositories provide the same package and you want to make sure it comes from this repository.
- **--proxy=[protocol://][username[:password]@]host[:port]** - Specify an HTTP/HTTPS/FTP proxy to use just for this repository. This setting does not affect any other repositories, nor how the **install.img** is fetched on HTTP installations.
- **--ignoregroups=true** - This option is used when composing installation trees and has no effect on the installation process itself. It tells the compose tools to not look at the package group information when mirroring trees so as to avoid mirroring large amounts of unnecessary data.
- **--noverifyssl** - Disable SSL verification when connecting to an **HTTPS** server.



IMPORTANT

Repositories used for installation must be stable. The installation can fail if a repository is modified before the installation concludes.

rescue (optional)

Automatically enters the installation program's rescue mode. This gives you a chance to repair the system in case of any problems.

```
rescue [--nomount|--romount]
```

- **--nomount** or **--romount** - Controls how the installed system is mounted in the rescue environment. By default, the installation program finds your system and mount it in read-write mode, telling you where it has performed this mount. You can optionally select to not mount anything (the **--nomount** option) or mount in read-only mode (the **--romount** option). Only one of these two options can be used.

reqpart (optional)

Automatically creates partitions required by your hardware platform. These include a **/boot/efi** partition for systems with UEFI firmware, a **biosboot** partition for systems with BIOS firmware and GPT, and a **PRePBoot** partition for IBM Power Systems.

```
reqpart [--add-boot]
```

- **--add-boot** - Creates a separate **/boot** partition in addition to the platform-specific partition created by the base command.



NOTE

This command cannot be used together with **autopart**, because **autopart** does everything the **reqpart** command does and, in addition, creates other partitions or logical volumes such as **/** and **swap**. In contrast with **autopart**, this command only creates platform-specific partitions and leaves the rest of the drive empty, allowing you to create a custom layout.

rootpw (required)

Sets the system's root password to the *password* argument.

```
rootpw [--iscrypted|--plaintext] [--lock] password
```

- **--iscrypted** - If this option is present, the password argument is assumed to already be encrypted. This option is mutually exclusive with **--plaintext**. To create an encrypted password, you can use **python**:

```
$ python -c 'import crypt,getpass;pw=getpass.getpass();print(crypt.crypt(pw) if
(pw==getpass.getpass("Confirm: ")) else exit())'
```

This generates a sha512 crypt-compatible hash of your password using a random salt.

- **--plaintext** - If this option is present, the password argument is assumed to be in plain text. This option is mutually exclusive with **--iscrypted**.
- **--lock** - If this option is present, the root account is locked by default. This means that the root user will not be able to log in from the console. This option will also disable the **Root Password** screens in both the graphical and text-based manual installation.

selinux (optional)

Sets the state of SELinux on the installed system. The default SELinux policy is **enforcing**.

```
selinux [--disabled|--enforcing|--permissive]
```

- **--enforcing** - Enables SELinux with the default targeted policy being **enforcing**.
- **--permissive** - Outputs warnings based on the SELinux policy, but does not actually enforce the policy.
- **--disabled** - Disables SELinux completely on the system.

For more information regarding SELinux, see the [Red Hat Enterprise Linux 7 SELinux User's and Administrator's Guide](#).

services (optional)

Modifies the default set of services that will run under the default **systemd** target. The list of disabled services is processed before the list of enabled services. Therefore, if a service appears on both lists, it will be enabled.

```
services [--disabled=list] [--enabled=list]
```

- **--disabled=** - Disable the services given in the comma separated list.

- **--enabled=** - Enable the services given in the comma separated list.



IMPORTANT

Do not include spaces in the list of services. If you do, Kickstart will enable or disable only the services up to the first space. For example:

```
services --disabled=auditd, cups,smartd, nfslock
```

disables only the **auditd** service. To disable all four services, this entry should include no spaces:

```
services --disabled=auditd,cups,smartd,nfslock
```

shutdown (optional)

Shut down the system after the installation has successfully completed. During a Kickstart installation, if no completion method is specified, the **halt** command is used.

The **shutdown** Kickstart option is equivalent to the **shutdown** command.

For other completion methods, see the **halt**, **poweroff**, and **reboot** Kickstart options.

skipx (optional)

If present, **X** is not configured on the installed system.



IMPORTANT

If you install a display manager among your package selection options, this package creates an **X** configuration, and the installed system defaults to **graphical.target**. The effect of the **skipx** option is overridden.

snapshot (optional)

Using the **snapshot** command, you can create LVM thin volume snapshots during the installation process. This enables you to back up a logical volume before or after the installation.

To create multiple snapshots, add the **snaphost** Kickstart command multiple times.

```
snapshots vg_name/lv_name --name=snapshot_name --when=pre-install|post-install
```

- **vg_name/lv_name** - Sets the name of the volume group and logical volume to create the snapshot from.
- **--name=snapshot_name** - Sets the name of the snapshot. This name must be unique within the volume group.
- **--when=pre-install|post-install** - Sets if the snapshot is created before the installation begins or after the installation is completed.

sshpw (optional)

During the installation, you can interact with the installation program and monitor its progress over

an **SSH** connection. Use the **sshpw** command to create temporary accounts through which to log on. Each instance of the command creates a separate account that exists only in the installation environment. These accounts are not transferred to the installed system.

```
sshpw --username=name password [--iscrypted|--plaintext] [--lock]
```

- **--username** - Provides the name of the user. This option is required.
- **--iscrypted** - If this option is present, the password argument is assumed to already be encrypted. This option is mutually exclusive with **--plaintext**. To create an encrypted password, you can use **python**:

```
$ python -c 'import crypt,getpass;pw=getpass.getpass();print(crypt.crypt(pw) if
(pw==getpass.getpass("Confirm: ")) else exit())'
```

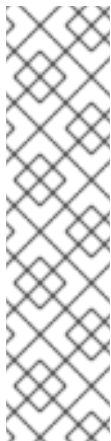
This generates a sha512 crypt-compatible hash of your password using a random salt.

- **--plaintext** - If this option is present, the password argument is assumed to be in plain text. This option is mutually exclusive with **--iscrypted**
- **--lock** - If this option is present, this account is locked by default. This means that the user will not be able to log in from the console.
- **--ssh** - If this option is present, then the *password* string is interpreted as an ssh key value.



IMPORTANT

By default, the **ssh** server is not started during the installation. To make **ssh** available during the installation, boot the system with the kernel boot option **inst.sshd**. See [Console, Environment and Display Options](#) for details.



NOTE

If you want to disable root **ssh** access, while allowing another user **ssh** access, use the following:

```
sshpw --username=example_username example_password --plaintext
sshpw --username=root example_password --lock
```

To simply disable root **ssh** access, use the following:

```
sshpw --username=root example_password --lock
```

text (optional)

Perform the Kickstart installation in text mode. Kickstart installations are performed in graphical mode by default.



IMPORTANT

For a fully automatic installation, you must either specify one of the available modes (**graphical**, **text**, or **cmdline**) in the Kickstart file, or you must use the **console=** boot option as described in [Console, Environment and Display Options](#). If no mode is specified, the system will prompt you to choose one before continuing.

timezone (required)

Sets the system time zone to *timezone*.

timezone *timezone* [*options*]

- **--utc** - If present, the system assumes the hardware clock is set to UTC (Greenwich Mean) time.
- **--nntp** - Disable the NTP service automatic starting.
- **--ntpervers=** - Specify a list of NTP servers to be used as a comma-separated list without spaces.

Starting with Red Hat Enterprise Linux 7.5, time zone names are validated using the **pytz.all_timezones** list, provided by the `pytz` package. In previous releases, the names were validated against **pytz.common_timezones**, which is a subset of the currently used list. Note that the graphical and text mode interfaces still use the more restricted **pytz.common_timezones** list; you must use a Kickstart file to use additional time zone definitions.

unsupported_hardware (optional)

Tells the installation program to suppress the **Unsupported Hardware Detected** alert. If this command is not included and unsupported hardware is detected, the installation stalls at this alert.

user (optional)

Creates a new user on the system.

user **--name=***username* [*options*]

- **--name=** - Provides the name of the user. This option is required.
- **--gecos=** - Provides the GECOS information for the user. This is a string of various system-specific fields separated by a comma. It is frequently used to specify the user's full name, office number, and so on. See the **passwd(5)** man page for more details.
- **--groups=** - In addition to the default group, a comma separated list of group names the user should belong to. The groups must exist before the user account is created. See the **group** command.
- **--homedir=** - The home directory for the user. If not provided, this defaults to **/home/username**.
- **--lock** - If this option is present, this account is locked by default. This means that the user will not be able to log in from the console. This option will also disable the **Create User** screens in both the graphical and text-based manual installation.

- **--password=** - The new user's password. If not provided, the account will be locked by default.
- **--iscrypted** - If this option is present, the password argument is assumed to already be encrypted. This option is mutually exclusive with **--plaintext**. To create an encrypted password, you can use **python**:

```
$ python -c 'import crypt,getpass;pw=getpass.getpass();print(crypt.crypt(pw) if
(pw==getpass.getpass("Confirm: ")) else exit())'
```

This generates a sha512 crypt-compatible hash of your password using a random salt.

- **--plaintext** - If this option is present, the password argument is assumed to be in plain text. This option is mutually exclusive with **--iscrypted**
- **--shell=** - The user's login shell. If not provided, the system default is used.
- **--uid=** - The user's *UID* (User ID). If not provided, this defaults to the next available non-system UID.
- **--gid=** - The *GID* (Group ID) to be used for the user's group. If not provided, this defaults to the next available non-system group ID.

NOTE

Consider using the **--uid** and **--gid** options to set IDs of regular users and their default groups at range starting at **5000** instead of **1000**. That is because the range reserved for system users and groups, **0-999**, might increase in the future and thus overlap with IDs of regular users.

For changing the minimum UID and GID limits after the installation, which ensures that your chosen UID and GID ranges are applied automatically on user creation, see [the Users and Groups chapter of the System Administrator's Guide](#).

NOTE

Files and directories are created with various permissions, dictated by the application used to create the file or directory. For example, the **mkdir** command creates directories with all permissions enabled. However, applications are prevented from granting certain permissions to newly created files, as specified by the **user file-creation mask** setting.

The **user file-creation mask** can be controlled with the **umask** command. The default setting of the **user file-creation mask** for new users is defined by the **UMASK** variable in the **/etc/login.defs** configuration file on the installed system. If unset, it defaults to **022**. This means that by default when an application creates a file, it is prevented from granting write permission to users other than the owner of the file. However, this can be overridden by other settings or scripts. More information can be found in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

vnc (optional)

Allows the graphical installation to be viewed remotely through VNC. This method is usually preferred over text mode, as there are some size and language limitations in text installations. With no

additional options, this command starts a VNC server on the installation system with no password and displays the details required to connect to it.

```
vnc [--host=host_name] [--port=port] [--password=password]
```

- **--host=** - Connect to the VNC viewer process listening on the given host name.
- **--port=** - Provide a port that the remote VNC viewer process is listening on. If not provided, **Anaconda** uses the VNC default port of 5900.
- **--password=** - Set a password which must be provided to connect to the VNC session. This is optional, but recommended.

For more information about VNC installations, including instructions on how to connect to the installation system, see [Chapter 25, Using VNC](#).

volgroup (optional)

Creates a Logical Volume Management (LVM) group.

```
volgroup name partition [options]
```

IMPORTANT

Do not use the dash (-) character in logical volume and volume group names when installing Red Hat Enterprise Linux using Kickstart. If this character is used, the installation finishes normally, but the **/dev/mapper/** directory will list these volumes and volume groups with every dash doubled. For example, a volume group named **volgrp-01** containing a logical volume named **logvol-01** will be listed as **/dev/mapper/volgrp--01-logvol--01**.

This limitation only applies to newly created logical volume and volume group names. If you are reusing existing ones using the **--noformat** option, their names will not be changed.

For a detailed partitioning example including **volgroup**, see [Section 27.4.1, "Advanced Partitioning Example"](#).

The options are as follows:

- **--noformat** - Use an existing volume group and do not format it.
- **--useexisting** - Use an existing volume group and reformat it. If you use this option, do not specify a *partition*. For example:

```
volgroup rhel00 --useexisting --noformat
```

- **--pesize=** - Set the size of the volume group's physical extents in KiB. The default value is 4096 (4 MiB), and the minimum value is 1024 (1 MiB).
- **--reserved-space=** - Specify an amount of space to leave unused in a volume group in MiB. Applicable only to newly created volume groups.

- **--reserved-percent=** – Specify a percentage of total volume group space to leave unused. Applicable only to newly created volume groups.

Create the partition first, then create the logical volume group, and then create the logical volume. For example:

```
part pv.01 --size 10000
volgroup volgrp pv.01
logvol / --vgname=volgrp --size=2000 --name=root
```

xconfig (optional)

Configures the **X Window System**. If you install the **X Window System** with a Kickstart file that does not include the **xconfig** command, you must provide the **X** configuration manually during installation.

Do not use this command in a Kickstart file that does not install the **X Window System**.

- **--defaultdesktop=** – Specify either **GNOME** or **KDE** to set the default desktop (assumes that the chosen environment, either the **GNOME Desktop Environment** or the **KDE Desktop Environment**, has been installed in the **%packages** section).



IMPORTANT

It is currently not possible to specify **KDE** as your default desktop environment using this option. This is a known issue. See <https://access.redhat.com/solutions/1125833> for a workaround. The workaround can be used as a Kickstart post-installation script as described in [Section 27.3.5, “Post-installation Script”](#).

- **--startxonboot** – Use a graphical login on the installed system.

zerombr (optional)

The **zerombr** initializes any invalid partition tables that are found on disks and destroys all of the contents of disks with invalid partition tables. This command is required when performing an installation on an IBM Z system with unformatted Direct Access Storage Device (DASD) disks, otherwise the unformatted disks are not formatted and used during the installation.



WARNING

On IBM Z, if **zerombr** is specified, any *Direct Access Storage Device* (DASD) visible to the installation program which is not already low-level formatted is automatically low-level formatted with **dasdfmt**. The command also prevents user choice during interactive installations.

If **zerombr** is not specified and there is at least one unformatted DASD visible to the installation program, a non-interactive Kickstart installation exits unsuccessfully.

If **zerombr** is not specified and there is at least one unformatted DASD visible to the installation program, an interactive installation exits if the user does not agree to format all visible and unformatted DASDs. To circumvent this, only activate those DASDs that you will use during installation. You can always add more DASDs after installation is complete.

zfcpl (optional)

Define a Fibre channel device. This option only applies on IBM Z. All of the options described below must be specified.

```
zfcpl --devnum=devnum --wwpn=wwpn --fcplun=lun
```

- **--devnum** - The device number (zFCP adapter device bus ID).
- **--wwpn** - The device's World Wide Port Name (WWPN). Takes the form of a 16-digit number, preceded by **0x**.
- **--fcplun** - The device's Logical Unit Number (LUN). Takes the form of a 16-digit number, preceded by **0x**.

For example:

```
zfcpl --devnum=0.0.4000 --wwpn=0x5005076300C213e9 --fcplun=0x5022000000000000
```

%include (optional)

Use the **%include** */path/to/file* command to include the contents of another file in the Kickstart file as though the contents were at the location of the **%include** command in the Kickstart file.

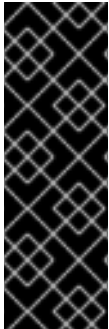
27.3.2. Package Selection

Use the **%packages** command to begin a Kickstart section which describes the software packages to be installed.

You can specify packages by *environment*, *group*, or by their package names. Several environments and groups that contain related packages are defined. See the **repodata/*-comps-variant.architecture.xml** file on the Red Hat Enterprise Linux 7 Installation DVD for a list of environments and groups.

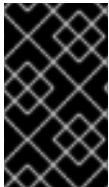
The ***-comps-variant.architecture.xml** file contains a structure describing available environments (marked by the **<environment>** tag) and groups (the **<group>** tag). Each entry has an ID, user visibility value, name, description, and package list. If the group is selected for installation, the packages marked **mandatory** in the package list are always installed, the packages marked **default** are installed if they are not specifically excluded elsewhere, and the packages marked **optional** must be specifically included elsewhere even when the group is selected.

You can specify a package group or environment using either its ID (the **<id>** tag) or name (the **<name>** tag).



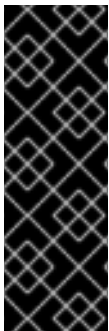
IMPORTANT

If you are not sure what package should be installed, Red Hat recommends you to select the **Minimal Install** environment. **Minimal install** provides only the packages which are essential for running Red Hat Enterprise Linux 7. This will substantially reduce the chance of the system being affected by a vulnerability. If necessary, additional packages can be added later after the installation. For more details on **Minimal install**, see the [Installing the Minimum Amount of Packages Required](#) section of the Red Hat Enterprise Linux 7 Security Guide.



IMPORTANT

To install a 32-bit package on a 64-bit system, append the package name with the 32-bit architecture for which the package was built; for example, `glibc.i686`. The **--multilib** option also must be specified in the Kickstart file; see the available options below.



IMPORTANT

Initial Setup does not run after a system is installed from a Kickstart file unless a desktop environment and the **X Window System** were included in the installation and graphical login was enabled. This means that by default, no users except for **root** are created. You can either create a user with the **user** option in the Kickstart file before installing additional systems from it (see [Section 27.3.1, “Kickstart Commands and Options”](#) for details) or log into the installed system with a virtual console as **root** and add users with the **useradd** command.

The **%packages** section must end with the **%end** command.

Specifying an Environment

In addition to groups, you specify an entire environment to be installed:

```
%packages
@^Infrastructure Server
%end
```

This command installs all packages which are part of the **Infrastructure Server** environment. All available environments are described in the **repodata/*-comps-variant.architecture.xml** file on the Red Hat Enterprise Linux 7 Installation DVD. Only a single environment can be specified in the Kickstart file.

Specifying Groups

Specify groups, one entry to a line, starting with an **@** symbol, and then the full group name or group id as given in the ***-comps-variant.architecture.xml** file. For example:

```
%packages
@X Window System
@Desktop
@Sound and Video
%end
```

The **Core** group is always selected – it is not necessary to specify it in the **%packages** section.

The ***-comps-variant.architecture.xml** file also defines groups called **Conflicts (variant)** for each variant of Red Hat Enterprise Linux. This group contains all packages which are known to cause file conflicts, and is intended to be excluded.

Specifying Individual Packages

Specify individual packages by name, one entry to a line. You can use the asterisk character (*) as a *wildcard* in package names. For example:

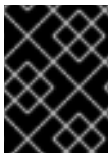
```
%packages
sqlite
curl
aspell
docbook*
%end
```

The **docbook*** entry includes the packages `docbook-dtds`, `docbook-simple`, `docbook-slides` and others that match the pattern represented with the wildcard.

Excluding Environments, Groups, or Packages

Use a leading dash (-) to specify packages or groups to exclude from the installation. For example:

```
%packages
-@Graphical Internet
-autofs
-ipa*fonts
%end
```



IMPORTANT

Installing all available packages using only ***** in a Kickstart file is not supported, even if you exclude the **@Conflicts (variant)** group.

You can change the default behavior of the **%packages** section by using several options. Some options work for the entire package selection, others are used with only specific groups.

Common Package Selection Options

The following options are available for the **%packages**. To use an option, append it to the start of the package selection section. For example:

```
%packages --multilib --ignoremissing
```

--default

Install the default set of packages. This corresponds to the package set which would be installed if no other selections were made in the **Package Selection** screen during an interactive installation.

--excludedocs

Do not install any documentation contained within packages. In most cases, this excludes any files normally installed in the **/usr/share/doc** directory, but the specific files to be excluded depend on individual packages.

--ignoremissing

Ignore any packages, groups and environments missing in the installation source, instead of halting the installation to ask if the installation should be aborted or continued.

--instLangs=

Specify a list of languages to install. Note that this is different from package group level selections. This option does not describe which package groups should be installed; instead, it sets RPM macros controlling which translation files from individual packages should be installed.

Red Hat Enterprise Linux packages use the ISO 639 language codes. Running the **locale -a** command provides you with a comprehensive list of language codes applicable with **%packages --instLangs=** arguments.

Example 27.5. Specifying a list of languages to install

- A colon separated list of language codes installs specified locales:

```
%packages --instLangs=es:fr:it
```

- Omitting the **--instLangs=** option installs all locales:

```
%packages
```

- Omitting language codes installs no locales:

```
%packages --instLangs=
```

--multilib

Configure the installed system for multilib packages, to allow installing 32-bit packages on a 64-bit system, and install packages specified in this section as such.

Normally, on an AMD64 and Intel 64 system, you can install only the **x86_64** and the **noarch** packages. However, with the **--multilib** option, you can automatically install the 32-bit AMD and the i686 Intel system packages available, if any.

This only applies to packages explicitly specified in the **%packages** section. Packages which are only being installed as dependencies without being specified in the Kickstart file are only installed in architecture versions in which they are needed, even if they are available for more architectures.

This option works both during the installation and on the running system using the ``yum`` command.

--nocore

Disables installation of the **@Core** package group which is otherwise always installed by default. Disabling the **@Core** package group with **--nocore** should be only used for creating lightweight containers; installing a desktop or server system with **--nocore** will result in an unusable system.



NOTE

- Using **-@Core** to exclude packages in the **@Core** package group does not work. The only way to exclude the **@Core** package group is with the **--nocore** option.
- The **@Core** package group is defined as a minimal set of packages needed for installing a working system. It is not related in any way to core packages as defined in the [Package Manifest](#) and [Scope of Coverage Details](#).

--retries=

Sets the number of times **Yum** will attempt to download packages (retries). The default value is **10**. This option only applies during the installation, and will not affect **Yum** configuration on the installed system.

--timeout=

Sets the **Yum** timeout in seconds. The default value is **30**. This option only applies during the installation, and will not affect **Yum** configuration on the installed system.

Options for Specific Package Groups

The options in this list only apply to a single package group. Instead of using them at the **%packages** command in the Kickstart file, append them to the group name. For example:

```
%packages
@Graphical Internet --optional
%end
```

--nodefaults

Only install the group's mandatory packages, not the default selections.

--optional

Install packages marked as optional in the group definition in the ***-comps-variant.architecture.xml** file, in addition to installing the default selections.

Note that some package groups, such as **Scientific Support**, do not have any mandatory or default packages specified – only optional packages. In this case the **--optional** option must always be used, otherwise no packages from this group will be installed.

27.3.3. Pre-installation Script

The **%pre** script is run on the system immediately after the Kickstart file has been parsed, but before installation begins. This section must be placed towards the end of the Kickstart file, after the Kickstart commands described in [Section 27.3.1, “Kickstart Commands and Options”](#), and must start with **%pre** and end with **%end**. If your Kickstart file also includes a **%post** section, the order in which the **%pre** and **%post** sections are included does not matter.

The **%pre** script can be used for activation and configuration of networking and storage devices. It is also possible to run scripts, using interpreters available in the installation environment. Adding a **%pre** script can be useful if you have networking and storage that needs special configuration before proceeding with the installation, or have a script that, for example, sets up additional logging parameters or environment variables. Debugging problems with **%pre** scripts can be difficult, so it is recommended only to use a **%pre** script when necessary.



IMPORTANT

The **%pre** section of Kickstart is executed at the stage of installation which happens after the installer image (**inst.stage2**) is fetched: it means *after* root switches to the installer environment (the installer image) and *after* the **Anaconda** installer itself starts. Then the configuration in **%pre** is applied and can be used to fetch packages from installation repositories configured, for example, by URL in Kickstart. However, it *cannot* be used to configure network to fetch the image (**inst.stage2**) from network.

Commands related to networking, storage, and file systems are available to use in the **%pre** script, in addition to most of the utilities in the installation environment's **/sbin** and **/bin** directories.

You can access the network in the **%pre** section. However, the *name* service has not been configured at this point, so only IP addresses work, not URLs.

The pre-installation script section of Kickstart *cannot* manage multiple install trees or source media. This information must be included for each created Kickstart file, as the pre-installation script occurs during the second stage of the installation process.



NOTE

Unlike the post-installation script, the pre-installation script is not run in the **chroot** environment.

The following options can be used to change the behavior of pre-installation scripts. To use an option, append it to the **%pre** line at the beginning of the script. For example:

```
%pre --interpreter=/usr/bin/python
--- Python script omitted --
%end
```

--interpreter=

Allows you to specify a different scripting language, such as Python. Any scripting language available on the system can be used; in most cases, these are **/usr/bin/sh**, **/usr/bin/bash**, and **/usr/bin/python**.

--erroronfail

Display an error and halt the installation if the script fails. The error message will direct you to where the cause of the failure is logged.

--log=

Logs the script's output into the specified log file. For example:

```
%pre --log=/mnt/sysimage/root/ks-pre.log
```

The following is an example **%pre** section:

Example 27.6. Sample %pre Script

```
%pre
#!/bin/sh
hds=""
mymedia=""
for file in /proc/ide/h* do
  mymedia=`cat $file/media`
  if [ $mymedia == "disk" ] ; then
    hds="$hds `basename $file`"
  fi
done
set $hds
numhd=`echo $#`
drive1=`echo $hds | cut -d' ' -f1`
drive2=`echo $hds | cut -d' ' -f2`

#Write out partition scheme based on whether there are 1 or 2 hard drives
if [ $numhd == "2" ] ; then
  #2 drives
  echo "#partitioning scheme generated in %pre for 2 drives" > /tmp/part-include
  echo "clearpart --all" >> /tmp/part-include
  echo "part /boot --fstype xfs --size 75 --ondisk hda" >> /tmp/part-include
  echo "part / --fstype xfs --size 1 --grow --ondisk hda" >> /tmp/part-include
  echo "part swap --recommended --ondisk $drive1" >> /tmp/part-include
  echo "part /home --fstype xfs --size 1 --grow --ondisk hdb" >> /tmp/part-include
else
  #1 drive
  echo "#partitioning scheme generated in %pre for 1 drive" > /tmp/part-include
  echo "clearpart --all" >> /tmp/part-include
  echo "part /boot --fstype xfs --size 75" >> /tmp/part-include
  echo "part swap --recommended" >> /tmp/part-include
  echo "part / --fstype xfs --size 2048" >> /tmp/part-include
  echo "part /home --fstype xfs --size 2048 --grow" >> /tmp/part-include
fi
%end
```

This script determines the number of hard drives in the system and writes a text file with a different partitioning scheme depending on whether it has one or two drives. Instead of having a set of partitioning commands in the Kickstart file, include the following line:

```
%include /tmp/part-include
```

The partitioning commands selected in the script will be used.

27.3.4. Anaconda configuration

Additional installation options can be configured in the **%anaconda** section of your Kickstart file. This section controls the behavior of the user interface of the installation system.

This section must be placed towards the end of the Kickstart file, after the Kickstart commands described in [Section 27.3.1, “Kickstart Commands and Options”](#), and must start with **%anaconda** and end with **%end**.

Currently, the only command that can be used in the **%anaconda** section is **pwpolicy**. See [Section 27.3.1, “Kickstart Commands and Options”](#) for more details.

The following is an example **%anaconda** section:

Example 27.7. Sample %anaconda Script

```
%anaconda
pwpolicy root --minlen=10 --strict
%end
```

This example **%anaconda** section sets a password policy which requires that the root password be at least 10 characters long, and strictly forbids passwords which do not match this requirement.

27.3.5. Post-installation Script

You have the option of adding commands to run on the system once the installation is complete, but before the system is rebooted for the first time. This section must be placed towards the end of the Kickstart file, after the Kickstart commands described in [Section 27.3.1, “Kickstart Commands and Options”](#), and must start with **%post** and end with **%end**. If your Kickstart file also includes a **%pre** section, the order of the **%pre** and **%post** sections does not matter.

This section is useful for functions such as installing additional software or configuring an additional name server. The post-install script is run in a chroot environment, therefore, performing tasks such as copying scripts or RPM packages from the installation media do not work by default. You can change this behavior using the **--nochroot** option as described below.

Because post-install script runs in a chroot environment, most **systemctl** commands will refuse to perform any action. For more information, see the [“Behavior of systemctl in a chroot Environment”](#) section of System Administrator's Guide for Red Hat Enterprise Linux 7.



IMPORTANT

If you configured the network with static IP information, including a name server, you can access the network and resolve IP addresses in the **%post** section. If you configured the network for **DHCP**, the **/etc/resolv.conf** file has not been completed when the installation executes the **%post** section. You can access the network, but you cannot resolve IP addresses. Thus, if you are using **DHCP**, you must specify IP addresses in the **%post** section.

The following options can be used to change the behavior of post-installation scripts. To use an option, append it to the **%post** line at the beginning of the script. For example:

```
%post --interpreter=/usr/bin/python
--- Python script omitted --
%end
```

--interpreter=

Allows you to specify a different scripting language, such as Python. For example:

```
%post --interpreter=/usr/bin/python
```

Any scripting language available on the system can be used; in most cases, these are **/usr/bin/sh**, **/usr/bin/bash**, and **/usr/bin/python**.

--nochroot

Allows you to specify commands that you would like to run outside of the chroot environment.

The following example copies the file **/etc/resolv.conf** to the file system that was just installed.

```
%post --nochroot
cp /etc/resolv.conf /mnt/sysimage/etc/resolv.conf
%end
```

--erroronfail

Display an error and halt the installation if the script fails. The error message will direct you to where the cause of the failure is logged.

--log=

Logs the script's output into the specified log file. Note that the path of the log file must take into account whether or not you use the **--nochroot** option. For example, without **--nochroot**:

```
%post --log=/root/ks-post.log
```

with **--nochroot**:

```
%post --nochroot --log=/mnt/sysimage/root/ks-post.log
```

The following is an example **%post** section:

Example 27.8. Sample %post Script

```
# Start of the %post section with logging into /root/ks-post.log
%post --log=/root/ks-post.log

# Mount an NFS share
mkdir /mnt/temp
mount -o nolock 10.10.0.2:/usr/new-machines /mnt/temp
openvt -s -w -- /mnt/temp/runme
umount /mnt/temp

# End of the %post section
%end
```

The above example mounts an NFS share and executes a script named **runme** located at **/usr/new-machines/** on the share. Note that NFS file locking is *not* supported while in Kickstart mode, therefore the **-o nolock** option is required.

One of the most common uses of post-installation scripts in Kickstart installations is automatic registration of the installed system using Red Hat Subscription Manager. The following is an example of automatic subscription in a **%post** script:

Example 27.9. Running subscription-manager as a Post-Install Script

```
%post --log=/root/ks-post.log
/usr/sbin/subscription-manager register --username=admin@example.com --password=secret --
serverurl=sam-server.example.com --org="Admin Group" --environment="Dev" --
servicelevel=standard --release="7.0"
%end
```

The **subscription-manager** command-line script registers a system to a Red Hat Subscription Management server (Customer Portal Subscription Management, Subscription Asset Manager, or CloudForms System Engine). This script can also be used to assign or attach subscriptions automatically to the system that best-match that system.

When registering to the Customer Portal, use the Red Hat Network login credentials. When registering to Subscription Asset Manager or CloudForms System Engine, use a user account created by the local administrator.

Additional options can be used with the registration command to set a preferred service level for the system and to restrict updates and errata to a specific operating system version.

Also see the [How do I use subscription-manager in a kickstart file?](#) article on the Red Hat Customer Portal for additional information about using **subscription-manager** in a Kickstart **%post** section.

27.3.6. Kickstart error handling

Starting with Red Hat Enterprise Linux 7, Kickstart installations can contain custom scripts which are run when the installer encounters a fatal error – for example, an error in a package that has been requested for installation, failure to start VNC when specified, or an error when scanning storage devices. Installation cannot continue after such an error has occurred. The installer will run all **%onerror** scripts in the order they are provided in the Kickstart file. In addition, **%onerror** scripts will be run in the event of a traceback.

Each **%onerror** script is required to end with **%end**.

--erroronfail

Display an error and halt the installation if the script fails. The error message will direct you to where the cause of the failure is logged.

--interpreter=

Allows you to specify a different scripting language, such as Python. For example:

```
%post --interpreter=/usr/bin/python
```

Any scripting language available on the system can be used; in most cases, these are **/usr/bin/sh**, **/usr/bin/bash**, and **/usr/bin/python**.

--log=

Logs the script's output into the specified log file.

27.3.7. Kickstart Add-ons

Starting with Red Hat Enterprise Linux 7, Kickstart installations support add-ons. These add-ons can expand the basic Kickstart (and Anaconda) functionality in many ways.

To use an add-on in your Kickstart file, use the **%addon *addon_name options*** command, and finish the command with an **%end** statement, similar to pre-installation and post-installation scripts described in previous sections. For example, if you want to use the **Kdump** add-on, which is distributed with **Anaconda** by default, use the following commands:

```
%addon com_redhat_kdump --enable --reserve-mb=auto
%end
```

The **%addon** command does not include any options of its own – all options are dependent on the actual add-on. For more information about add-ons, see the [Red Hat Enterprise Linux 7 Anaconda Customization Guide](#).

27.4. SAMPLE KICKSTART CONFIGURATIONS

27.4.1. Advanced Partitioning Example

The following is an integrated example showing the **clearpart**, **zerombr**, **part**, **raid**, **volgroup**, and **logvol** Kickstart options in action:

Example 27.10. Advanced Partitioning Example

```
clearpart --drives=hda,hdc
zerombr
# Raid 1 IDE config
part raid.11 --size 1000 --asprimary --ondrive=hda
part raid.12 --size 1000 --asprimary --ondrive=hda
part raid.13 --size 2000 --asprimary --ondrive=hda
part raid.14 --size 8000 --ondrive=hda
part raid.15 --size 16384 --grow --ondrive=hda
part raid.21 --size 1000 --asprimary --ondrive=hdc
part raid.22 --size 1000 --asprimary --ondrive=hdc
part raid.23 --size 2000 --asprimary --ondrive=hdc
part raid.24 --size 8000 --ondrive=hdc
part raid.25 --size 16384 --grow --ondrive=hdc

# You can add --spares=x
raid / --fstype xfs --device root --level=RAID1 raid.11 raid.21
raid /safe --fstype xfs --device safe --level=RAID1 raid.12 raid.22
raid swap --fstype swap --device swap --level=RAID1 raid.13 raid.23
raid /usr --fstype xfs --device usr --level=RAID1 raid.14 raid.24
raid pv.01 --fstype xfs --device pv.01 --level=RAID1 raid.15 raid.25

# LVM configuration so that we can resize /var and /usr/local later
volgroup sysvg pv.01
logvol /var --vgname=sysvg --size=8000 --name=var
logvol /var/freespace --vgname=sysvg --size=8000 --name=freespacetouse
logvol /usr/local --vgname=sysvg --size=1 --grow --name=usrlocal
```

This advanced example implements LVM over RAID, as well as the ability to resize various directories for future growth.

First, the **clearpart** command is used on drives **hda** and **hdc** to wipe them. The **zerombr** command initializes unused partition tables.

Then, the two drives are partitioned to prepare them for RAID configuration. Each drive is divided into five partitions, and each drive is partitioned into an identical layout.

The next part uses these pairs of physical partitions to create a software RAID device with RAID1 level (mirroring). The first four RAID devices are used for **/** (root), **/safe**, **swap** and **/usr**. The fifth, largest pair of partitions is named **pv.01** and will be used in the following part as a physical volume for LVM.

Finally, the last set of commands first creates a volume group named **sysvg** on the **pv.01** physical volume. Then, three logical volumes (**/var**, **/var/freespace** and **/usr/local**) are created and added to the **sysvg** volume group. The **/var** and **/var/freespace** volumes have a set size of 8 GB, and the **/usr/local** volume uses the **--grow** option to fill all remaining available space.

27.4.2. User Input Example

The following is an example showing how to prompt the user for input, and then read that input and save it as a variable, using bash:

Example 27.11. User Input Example

```
%pre
exec < /dev/tty6 > /dev/tty6 2> /dev/tty6
chvt 6
IFS=$'\n'
echo -n "Enter input: "
read USERINPUT
echo
echo -n "You entered:" "$USERINPUT"
echo
chvt 1
exec < /dev/tty1 > /dev/tty1 2> /dev/tty1
%end
```

Due to the way Kickstart operates, the script must switch to a new virtual terminal before reading input from the user. This is accomplished by the **exec < /dev/tty6 > /dev/tty6 2> /dev/tty6** and **chvt 6** commands. The **read USERINPUT** reads input from the user until enter is pressed, and stores it in the variable **USERINPUT**. The **echo -n "You entered:" "\$USERINPUT"** command displays the text **You entered:** followed by the user's input. Finally, the **chvt 1** and **exec < /dev/tty1 > /dev/tty1 2> /dev/tty1** commands switch back to the original terminal and allow Kickstart to continue installation.

27.4.3. Example Kickstart file for installing and starting the RNG daemon

The following is an example Kickstart file which demonstrates how to install and enable a service, in this case the Random Number Generator (RNG) daemon, which supplies entropy to the system kernel:

Example 27.12. Example Kickstart file for installing and starting the RNG daemon

-

```
services --enabled=rngd
```

```
%packages
```

```
rng-tools
```

```
%end
```

The **services --enabled=rngd** command instructs the installed system to start the RNG daemon each time the system starts. The `rng-tools` package, which contains the RNG daemon, is then designated for installation.

CHAPTER 28. INSTALLING INTO A DISK IMAGE

This chapter describes the process of creating custom, bootable images of several different types, and other related topics. The image creation and installation process can be either performed manually in a procedure similar to a normal hard drive installation, or it can be automated using a Kickstart file and the **livemedia-creator** tool.



NOTE

Creating custom images using **livemedia-creator** is currently supported only on AMD64 and Intel 64 (x86_64) and IBM POWER (big endian) systems.

Additionally, Red Hat only supports creating custom images of Red Hat Enterprise Linux 7.

If you choose the manual approach, you will be able to perform the installation interactively, using the graphical installation program. The process is similar to installing using Red Hat Enterprise Linux bootable media and the graphical installation program; however, before you begin the installation, you must create one or more empty image files manually.

Automated disk image installations using **livemedia-creator** are somewhat similar to Kickstart installations with network boot. To use this approach, you must prepare a valid Kickstart file, which will be used by **livemedia-creator** to perform the installation. The disk image file will be created automatically.

Both approaches to disk image installations require a separate installation source. In most cases, the best approach is to use an ISO image of the binary Red Hat Enterprise Linux DVD. See [Chapter 2, Downloading Red Hat Enterprise Linux](#) for information about obtaining installation ISO images.



IMPORTANT

It is not currently possible to use an installation ISO image of Red Hat Enterprise Linux without any additional preparation. The installation source for a disk image installation must be prepared the same way it would be prepared when performing a normal installation. See [Section 3.3, "Preparing Installation Sources"](#) for information about preparing installation sources.

28.1. MANUAL DISK IMAGE INSTALLATION

A manual installation into a disk image is performed by executing the **Anaconda** installation program on an existing system and specifying one or more disk image files as installation targets. Additional options can also be used to configure **Anaconda** further. A list of available options can be obtained by using the **anaconda -h** command.

**WARNING**

Image installation using **Anaconda** is potentially dangerous, because it uses the installation program on an already installed system. While no bugs are known at this moment which could cause any problems, it is possible that this process could render the entire system unusable. Installation into disk images should always be performed on systems or virtual machines specifically reserved for this purpose, and not on systems containing any valuable data.

This section provides information about creating empty disk images and using the **Anaconda** installation program to install Red Hat Enterprise Linux into these images.

28.1.1. Preparing a Disk Image

The first step in manual disk image installation is creating one or more image files, which will later be used as installation targets similar to physical storage devices. On Red Hat Enterprise Linux, a disk image file can be created using the following command:

```
$ fallocate -l size name
```

Replace *size* with a value representing the size of the image (such as **10G** or **5000M**), and *name* with the file name of the image to be created. For example, to create a disk image file named **myimage.raw** with the size of 30GB, use the following command:

```
$ fallocate -l 30G myimage.raw
```

**NOTE**

The **fallocate** command allows you to specify the size of the file to be created in different ways, depending on the suffix used. For details about specifying the size, see the **fallocate(1)** man page.

The size of the disk image file you create will limit the maximum capacity of file systems created during the installation. The image must always have a minimum size of 3GB, but in most cases, the space requirements will be larger. The exact size you will need for your installation will vary depending on the software you want to install, the amount of swap space, and the amount of space you will need to be available after the installation. More details about partitioning are available in:

- [Section 8.14.4.4, “Recommended Partitioning Scheme”](#) for 64-bit AMD, Intel, and ARM systems
- [Section 13.15.4.4, “Recommended Partitioning Scheme”](#) for IBM Power Systems servers

After you create one or more empty disk image files, continue with [Section 28.1.2, “Installing Red Hat Enterprise Linux into a Disk Image”](#).

28.1.2. Installing Red Hat Enterprise Linux into a Disk Image



IMPORTANT

Set Security Enhanced Linux (**SELinux**) to permissive (or disabled) mode before creating custom images with **Anaconda**. See [Red Hat Enterprise Linux 7 SELinux User's and Administrator's Guide](#) for information on setting **SELinux** modes.

To start the installation into a disk image file, execute the following command as **root**:

```
# anaconda --image=/path/to/image/file
```

Replace */path/to/image/file* with the *full* path to the image file you created earlier.

After executing this command, **Anaconda** will start on your system. The installation interface will be the same as if you performed the installation normally (booting the system from Red Hat Enterprise Linux media), but the graphical installation will start directly, skipping the boot menu. This means that boot options must be specified as additional arguments to the **anaconda** command. You can view the full list of supported commands by executing **anaconda -h** on a command line.

One of the most important options is **--repo=**, which allows you to specify an installation source. This option uses the same syntax as the **inst.repo=** boot option. See [Section 23.1, "Configuring the Installation System at the Boot Menu"](#) for more information.

When you use the **--image=** option, *only* the disk image file specified will be available as the installation target. No other devices will be visible in the **Installation Destination** dialog. If you want to use multiple disk images, you must specify the **--image=** option separately for each image file separately. For example:

```
# anaconda --image=/home/testuser/diskinstall/image1.raw --  
image=/home/testuser/diskinstall/image2.raw
```

The above command will start **Anaconda**, and in the **Installation Destination** screen, both image files specified will be available as installation targets.

Optionally, you can also assign custom names to the disk image files used in the installation. To assign a name to a disk image file, append **:name** to the end of the disk image file name. For example, to use a disk image file located in **/home/testuser/diskinstall/image1.raw** and assign the name **myimage** to it, execute the following command:

```
# anaconda --image=/home/testuser/diskinstall/image1.raw:myimage
```

28.2. AUTOMATIC DISK IMAGE INSTALLATION

Creation of disk images and the installation into them can be automated using **livemedia-creator**. To perform an automatic installation, you will need an installed Red Hat Enterprise Linux system and a Kickstart file. The disk images themselves do not need to be created manually. For information about creating and using Kickstart files, see [Chapter 27, Kickstart Installations](#).

28.2.1. Overview of livemedia-creator

Creating custom images using **livemedia-creator** is usually a two stage process. In the first stage, a temporary disk image file is created and **Anaconda**, the Red Hat Enterprise Linux installation program, installs a system on this image based on the parameters provided in a Kickstart file. Then, in the second stage, **livemedia-creator** uses this temporary system to create the final, bootable image.

This behavior can be changed by specifying additional options. For example, it is possible to go through the first stage only, with the result being a disk image file, or to skip the first stage and use an existing disk or file system image to create the final bootable ISO image.

Sample usage of **livemedia-creator** is described in [Section 28.2.4, “Creating Custom Images”](#). On a system where the **lorax** package is installed, a list of all available options can be displayed using the **livemedia-creator --help** command. Additional documentation is also installed along with the **lorax** package: the **livemedia-creator(1)** man page and the **README.livemedia-creator** file located in the **/usr/share/doc/lorax-version/** directory, where *version* is the version of the **lorax** package you have installed.

28.2.2. Installing livemedia-creator

The **livemedia-creator** tool is a part of the **lorax** package. To install the package, execute the following command as **root**:

```
# yum install lorax
```

You will also need to install several other packages in addition to **lorax** itself. These packages are not dependencies of **lorax** and therefore they are not installed automatically, but you might need them depending on what exactly are you using **livemedia-creator** for. Among these packages are:

- **virt-install**: a package providing tools to build new virtual machines, used in the first stage of live media creation unless the **--no-virt** option is specified.
- **libvirt**, **qemu-kvm**, **libvirt-client** and other virtualization tools: when using **virt-install**, your system must be prepared to create, run and manage a virtual machine. See the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#) for information on virtualization in Red Hat Enterprise Linux and for documentation about installing and working with virtualization tools.
- **anaconda**: the Red Hat Enterprise Linux installation program, used in the first stage instead of **virt-install** if the **--no-virt** option is used.

Other applications, which are beyond the scope of this chapter, might be necessary. If you attempt to execute **livemedia-creator** and a package required with the options you specified is missing, the program will stop and an error message will be displayed informing you of packages you need to install before proceeding.

28.2.3. Sample Kickstart Files

To successfully create a custom live image, you will need a valid Kickstart configuration file. Two samples are automatically installed along with **lorax**. You can use these samples as a reference when creating your own custom images, or you can copy them and modify them to suit your intended usage. Both provided samples are located in the **/usr/share/doc/lorax-version/** directory, where *version* is the version number of the **lorax** package installed on your system.

The available samples are:

- **rhel7-minimal.ks**: a configuration file which provides only a minimal installation (the **@core** group) and other essentials such as the kernel and the **GRUB2** boot loader. No users apart from **root** are created, and no graphical interface or additional packages are installed.
- **rhel7-livemedia.ks**: a more advanced configuration file which creates a live system with a graphical interface. A user named **liveuser** is created along with **root**.

Both sample configurations need to be modified to use a valid location as the installation source. To do this edit the file and update the url command to refer to a valid installation source. See [Chapter 27, Kickstart Installations](#) for more information about Kickstart. No other changes are necessary for these samples to work.



IMPORTANT

Do not modify the samples in their original location. Copy them to another directory and modify the copies instead.



NOTE

Red Hat only supports the repositories provided by Red Hat as installation source.



NOTE

When modifying the provided samples or making a Kickstart file from scratch, do not use the **%include** statement as they are not supported by **livemedia-creator**. All Kickstart files must be flattened either manually or using the **ksflatten** tool before they can be used.

28.2.4. Creating Custom Images

This section describes several common usage patterns for tools in the lorax package. It is not intended to be a complete list of available options. To view every available option for **livemedia-creator**, execute **livemedia-creator --help** or see the **livemedia-creator(1)** man page. For more information about lorax, see <https://weldr.io/lorax/>.

28.2.4.1. Using lorax to create a boot.iso file

The installation system of Red Hat Enterprise Linux is created by a tool named lorax. The lorax tool can also be used to create your own installation media, for example containing an updated kernel or additional packages.



NOTE

It is recommended to use lorax on a system that is the same release of Red Hat Enterprise Linux as the image being created.

After running the following as root, an installation image named **boot.iso** will exist in the **results/images** directory. The **--repo=** command loads a Yum repository configuration file, in this case the default one, and use it to download package from Red Hat's content delivery network the same way they would be downloaded during a system update.

```
# yum install lorax
# setenforce 0
# lorax -p RHEL -v 7.4 -r 7.4 --repo=/etc/yum.repos.d/redhat.repo ./results/
# setenforce 1
```

**NOTE**

The above commands require that the system they are executed on has been registered and subscribed with Red Hat Subscription Manager. For more information, see [Red Hat Subscription Management](#).

28.2.4.2. Creating a Live Image Using virt-install

Perhaps the most common use of **livemedia-creator** involves using **virt-install** to create a temporary virtual machine to be used for the live image creation process. To create a live ISO using **virt-install**, you will need a valid Kickstart file and a bootable ISO image which contains the **Anaconda** installation program. Such images are provided by Red Hat as "minimal boot media"; see [Section 3.2, "Making Installation USB Media"](#) for details.

The following command is the bare minimum you need to create a live image using **virt-install**:

```
# livemedia-creator --make-iso --iso=/path/to/boot.iso --ks=/path/to/valid/kickstart.ks
```

Replace */path/to/boot.iso* with a path to a minimal boot image and */path/to/valid/kickstart.ks* with a path to a valid Kickstart file to be used in the image creation process.

Other frequently used options in this particular use case are:

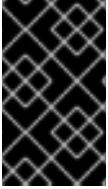
- **--vnc vnc**: this option allows you to watch the installation process using a VNC client such as **TigerVNC**. The option is passed to **virt-install**'s **--graphics** option. See [Chapter 25, Using VNC](#) for more information.
- **--ram x**: allows you to specify the amount of RAM for the temporary virtual machine in MiB.
- **--vcpus x**: the amount of the virtual machine's processors.

28.2.4.3. Creating a Live Image Using Anaconda's Image Install

The **livemedia-creator** application can create a live image using the **Anaconda** package if **virt-install** is not available. In this case, no image containing the installation program is needed, but the **Anaconda** package must be installed on the system. Again, the process has two stages: first, a temporary disk image is created and a system is installed into it, and then this image is used to create the final bootable ISO.

**WARNING**

Live image creation using **Anaconda** is potentially dangerous, because it uses the installation program on the system itself instead of inside a virtual machine. While no bugs are known at this moment that would cause any problems, it is possible that this process could render the entire system unusable. Running **livemedia-creator** with the **--no-virt** option is therefore only recommended on virtual machines (guests) specifically reserved for this purpose.



IMPORTANT

Set Security Enhanced Linux (**SELinux**) to permissive (or disabled) mode before creating custom images with **Anaconda**. See [Red Hat Enterprise Linux 7 SELinux User's and Administrator's Guide](#) for information on setting **SELinux** modes.

To create a live image using **Anaconda**, use the **--no-virt** option. For example:

```
# livemedia-creator --make-iso --ks=/path/to/valid/kickstart.ks --no-virt
```

28.2.4.4. Creating a Disk or File System Image

You can also use **livemedia-creator** to create a disk or file system image. This means running only the first stage of the image creation process. The final ISO will not be created, the program will stop after finishing the installation process on the temporary disk or file system image file. You can then mount and inspect this image for errors, which can be useful when troubleshooting a modified Kickstart file, and you can also keep it for future use to save time when creating images in the future.



NOTE

It is also possible to use the **--no-virt** option in all examples in this section.

There are several ways to stop the creation process after the first stage. You can use the **--image-only** option as illustrated in the following example:

```
# livemedia-creator --make-iso --ks=/path/to/valid/kickstart.ks --iso=/path/to/boot.iso --image-only
```

Alternatively, you can use the **--make-disk** option instead of **--make-iso**:

```
# livemedia-creator --make-disk --ks=/path/to/valid/kickstart.ks --iso=/path/to/boot.iso
```

You can also create a file system image instead of partitioned disk image using the **--make-fsimage** option:

```
# livemedia-creator --make-fsimage --ks=/path/to/valid/kickstart.ks --iso=/path/to/boot.iso
```

In all cases, the result will be a partitioned disk image or a file system image, located in the **/var/tmp/** directory by default. To change the location of the result, use the **--tmp /path/to/temporary/directory/** option, where **/path/to/temporary/directory/** is the path to the target directory.

28.2.4.5. Using a Previously Created Disk or File System Image

If you already have a disk or file system image (see [Section 28.2.4.4, "Creating a Disk or File System Image"](#)), you can supply it to **livemedia-creator** to produce the final bootable ISO image. In this case no Kickstart File or **Anaconda** installation image is necessary; these are only needed in the first stage of the image creation process, which is skipped in this case.

To create a final image from an existing partitioned disk image file, use the **--disk-image** option. For example:

```
# livemedia-creator --make-iso --disk-image=/path/to/disk/image.img
```

If you want to use a file system image instead of a disk image, use the **--fs-image** option instead:

```
# livemedia-creator --make-iso --fs-image=/path/to/filesystem/image.img
```

28.2.4.6. Creating an Appliance

The **livemedia-creator** utility can be used to create appliance images (a partitioned disk image), including an XML file containing its description, generated using a template. Virtual machine installations as well as image installations are supported in this case. To create an appliance image and description, use the **--make-appliance** option instead of **--make-iso**. For example:

```
# livemedia-creator --make-appliance --ks=/path/to/valid/kickstart.ks --iso=/path/to/boot.iso
```

Both the image and the description XML file will be stored in the **/var/tmp/** directory unless a different one is specified using the **--resultdir** option.

Additional options specific to appliance creation are:

- **--app-name *name***: specifies the name of the appliance, which will appear in the XML description file marked by the **<name>** tag. The default value is **None**.
- **--app-template */path/to/template.tmpl***: specifies the template to be used. The default is **/usr/share/lorax/appliance/libvirt.tmpl**.
- **--app-file */path/to/app/file.xml***: specifies name of the generated description XML file. The default value is **appliance.xml**.

28.2.4.7. Creating an Amazon Machine Image (AMI)

To create an Amazon Machine Image (AMI) for use within the Amazon Elastic Compute Cloud (EC2), use the **--make-ami** option. Virtualized and image installations are both supported.

```
# livemedia-creator --make-ami --ks=/path/to/valid/kickstart.ks --iso=/path/to/boot.iso
```

The result will be an image file named **ami-root.img**, located in the **/var/tmp/** directory, unless you used the **--resultdir** option to specify a different one.

28.2.4.8. Additional Arguments

The following options can be used with all use cases listed above (virtual installations, **Anaconda** image installations and others).

- **--keep-image**: when you specify this option, the temporary disk image file used in the first stage of the installation will not be deleted. It will be located in the **/var/tmp/** directory and it will have a randomly generated name such as **diskgU42Cq.img**.
- **--image-only**: using this option means that only the first stage of the image creation process will be executed. Instead of producing the final bootable ISO image, **livemedia-creator** will only create the temporary disk image file and perform an installation on it. This option allows you to save time when testing modifications to your Kickstart file, because you can skip the time-consuming second stage and inspect the temporary disk image file.
- **--image-name *name***: allows you to specify a custom name for the temporary disk image file. The default name is randomly generated (for example, **disk1Fac8G.img**).

- **--tmp /path/to/temporary/directory/**: specifies the top level temporary directory. The default value is **/var/tmp/**. When using this option, you must specify a directory which already exists.
- **--resultdir /path/to/results/directory/**: specifies the directory where the results (the bootable ISO image) will appear after **livemedia-creator** finishes. An already existing directory cannot be specified. The default is **/var/tmp/**. This option only applies to the final ISO image; if you are creating a disk or file system image and want it to be saved at a specific location, use the **--tmp** option.
- **--logfile /path/to/log/file/**: specifies the location of the program's log file.

28.2.5. Troubleshooting livemedia-creator Problems

This section offers suggestions on solving various issues commonly encountered when using **livemedia-creator**. If you encounter a problem not described here, you can look into the program's log files, which are automatically generated during every run and saved into the directory from which you executed the tool, unless you specify a different directory using the **--logfile** option. The log files will be different based on the options you used – for example, **virt-install.log** will not be generated when you use the **--no-virt** option (instead, you will get log files from **Anaconda**, located in the **anaconda/** directory). Other files, namely **livemedia.log** and **program.log**, are generated every time.

Another way to find and solve problems is using the **--image-only** option when running the utility. This option will stop the program after the first stage, so only a disk image file will be generated instead of the final bootable ISO. You can then mount the disk image file and examine its contents without having to wait for the second stage to finish. Alternatively, you can use the **--keep-image** option, which will execute both stages, but keep the temporary disk image for later analysis.

Using the **--vnc** option is recommended when testing changes to the Kickstart file. This option will allow you to use a VNC client to connect to the virtual machine and watch the installation progress. See [Chapter 25, Using VNC](#) for details.

28.2.5.1. Stuck Virtual Machine Installation

If the installation program stops for any reason during the first stage of a virtual installation, **livemedia-creator** will stop as well, waiting for the installation to finish. You can either interrupt the program directly, or you can solve this problem by stopping the temporary virtual machine. **Livemedia-creator** will detect that the guest operating system has been stopped, delete all temporary files and exit.

To stop the temporary virtual machine, follow this procedure:

Procedure 28.1. Stopping the Temporary Virtual Machine

1. Use **virsh** to list all virtual machines (guests) currently available on the system. The output will be similar to the following:

```
# virsh list --all
Id   Name                                     State
-----
93   LiveOS-2a198971-ba97-454e-a056-799f453e1bd7 running
-    RHEL7                                   shut off
```

Identify the temporary virtual machine. Its name will always start with **LiveOS**, followed by a string of random numbers and characters.

2. Once you have identified the temporary virtual machine, stop it using the **virsh destroy *name*** command, where *name* is the virtual machine's name:

```
# virsh destroy LiveOS-2a198971-ba97-454e-a056-799f453e1bd7
Domain LiveOS-2a198971-ba97-454e-a056-799f453e1bd7 destroyed
```

28.2.5.2. Failed installation using a Virtual Machine

If you were performing a virtual installation and the process was interrupted for any reason (such as hardware failure, power outage or a keyboard interrupt) during the first stage, virt-install will not be able to start again until the previously created temporary disk image and virtual machine have been removed. The following procedure explains how to do this.

Not all steps might be necessary every time. For example, if you are recovering after a system crash, you will not have to stop the temporary virtual machine, instead you can just use the **virsh undefine *name*** command. You can also use steps 4 and 5 if you only want to clean up temporary files created by **livemedia-creator** and nothing else.

Procedure 28.2. Removing Temporary Guests And Disk Image Files

1. Use **virsh** to list all virtual machines (guests) currently available on the system. The output will be similar to the following:

```
# virsh list --all
Id   Name                                     State
-----
93   LiveOS-2a198971-ba97-454e-a056-799f453e1bd7 running
-    RHEL7                                   shut off
```

Identify the temporary virtual machine. Its name will always start with **LiveOS**, followed by a string of random numbers and characters.

2. Once you have identified the temporary virtual machine, stop it using the **virsh destroy *name*** command, where *name* is the virtual machine's name:

```
# virsh destroy LiveOS-2a198971-ba97-454e-a056-799f453e1bd7
Domain LiveOS-2a198971-ba97-454e-a056-799f453e1bd7 destroyed
```

3. Delete the temporary virtual machine using **virsh undefine *name***, using the same *name* as in the previous step.

```
# virsh undefine LiveOS-2a198971-ba97-454e-a056-799f453e1bd7
Domain LiveOS-2a198971-ba97-454e-a056-799f453e1bd7 has been undefined
```

4. Find the temporary file system's mount. It will be targeted to the **/var/tmp/** directory and its name will be **lorax.imgutils** followed by six random numbers or characters.

```
# findmnt -T /var/tmp/lorax.imgutils*
TARGET                SOURCE  FSTYPE  OPTIONS
/var/tmp/lorax.imgutils.bg6iPJ /dev/loop1 iso9660 ro,relatime
```

Then, unmount it using the **umount** command:

```
# umount /var/tmp/lorax.imgutils.bg6iPJ
```

- Find the temporary disk image created by `virt-install` in the `/var/tmp/` directory. The name of this file is printed to command line at the beginning of the installation process and is randomly generated, unless you specify a name using the `--image-name` option. For example:

```
2013-10-30 09:53:03,161: disk_size = 5GB
2013-10-30 09:53:03,161: disk_img = /var/tmp/diskQBkzRz.img
2013-10-30 09:53:03,161: install_log = /home/pbokoc/lorax/virt-install.log
mount: /dev/loop1 is write-protected, mounting read-only
```

In the above example, the temporary disk image is `/var/tmp/diskQBkzRz.img`.

If you cannot find the initial messages, you can identify the temporary files manually. List all contents of the `/var/tmp/` directory using the `ls` command and filter the output for files containing **disk** in their names:

```
# ls /var/tmp/ | grep disk
diskQBkzRz.img
```

Then, delete the temporary disk image:

```
# rm -f /var/tmp/diskQBkzRz.img
```

If you followed all steps in this procedure, you are now able to start a new installation with `virt-install`.

28.2.5.3. Failed Installation Using Anaconda

Recovery from an interrupted installation using **Anaconda** image install feature (the `--no-virt` option) can be achieved by executing the **anaconda-cleanup** script, which is installed along with the `anaconda` package. This script is located in the `/usr/bin/` directory.

Use the following command to execute the cleanup script. You will need root privileges to do this.

```
# anaconda-cleanup
```

CHAPTER 29. UPGRADING YOUR CURRENT SYSTEM

The procedure for performing an in-place upgrade on your current system is handled by the following utilities:

- The **Preupgrade Assistant**, which is a diagnostics utility that assesses your current system and identifies potential problems you might encounter during or after the upgrade.
- The **Red Hat Upgrade Tool** utility, which is used to upgrade a system from Red Hat Enterprise Linux version 6 to version 7.



NOTE

In-place upgrades are currently only supported on AMD64 and Intel 64 (**x86_64**) systems and on IBM Z (**s390x**). Additionally, only the **Server** variant can be upgraded with **Red Hat Upgrade Tool**.

Full documentation covering the process of upgrading from an earlier release of Red Hat Enterprise Linux to Red Hat Enterprise Linux 7 is available in the [Red Hat Enterprise Linux 7 Migration Planning Guide](#).

You can also use the [Red Hat Enterprise Linux Upgrade Helper](#) to guide you through migration from Red Hat Enterprise Linux 6 to 7.

PART V. AFTER INSTALLATION

This part of the *Red Hat Enterprise Linux Installation Guide* covers finalizing the installation, as well as some installation tasks related to installation that you might perform at some time in the future. These include:

- performing common post-installation tasks, such as registering the system to Red Hat Subscription Management services
- using a Red Hat Enterprise Linux installation disc to rescue a damaged system
- removing Red Hat Enterprise Linux from your computer

CHAPTER 30. INITIAL SETUP

After you started a new Red Hat Enterprise Linux system the first time, the **Initial Setup** application launches, if you installed the server:

- Using the **Server with GUI** base environment in the Red Hat Enterprise Linux setup.
- Using a Kickstart file that contains one of the following entries in the **%packages** section:
 - the initial-setup-gui package for graphical mode
 - the initial-setup package for text mode
 - the x11 group
 - the gnome-desktop group
 - the kde-desktop group



NOTE

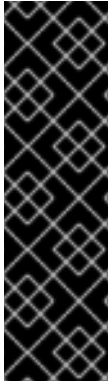
The Initial Setup application in graphical mode only launches for a system with a graphical output. If you install Red Hat Enterprise Linux system with an environment that includes Initial Setup (like "Server with a GUI", "GNOME Desktop" or "KDE Plasma Workspaces") in a system that has no graphic card, the Initial Setup application launches in text mode.

Options in the Initial Setup

The **Initial Setup** application can display the following options:

Option	Graphical User Interface	Text User Interface
License agreement [a]	Yes	Yes
Language settings [b]	No	Yes
Date & time [b]	Yes	Yes [c]
Subscription Manager	Yes	No
Network & host name [b]	Yes	No
Root password [b]	Yes	Yes [c]
User creation [b]	Yes	Yes

Option	Graphical User Interface	Text User Interface
[a] This option is only displayed if you did not previously agree to the license.		
[b] This option is only displayed if you did not previously set it.		
[c] This option is only available if you run Initial Setup in reconfiguration mode		



IMPORTANT

Options set during the setup are not displayed in the **Initial Setup**. To display all options in **Initial Setup**, you must install Red Hat Enterprise Linux using a Kickstart file that contains the following command:

```
firstboot --enable --reconfig
```

The **--reconfig** option specifies that all options should be displayed. For further details about Kickstart installations, see [Chapter 27, Kickstart Installations](#).

30.1. GRAPHICAL MODE

In graphical mode, the **Initial Setup** displays the following screen:

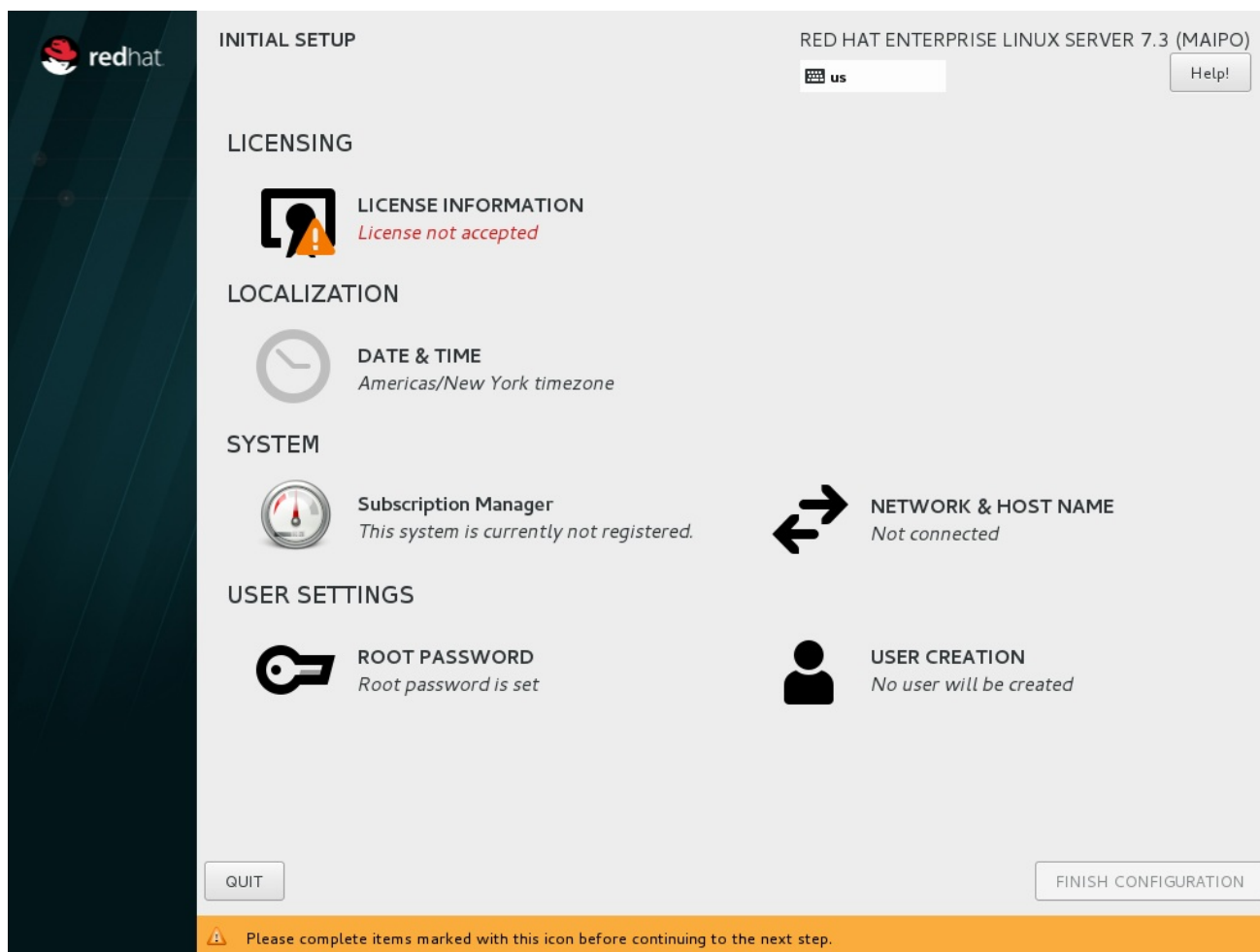


Figure 30.1. Main Initial Setup Screen

The **License Agreement** screen displays the overall licensing terms for Red Hat Enterprise Linux.

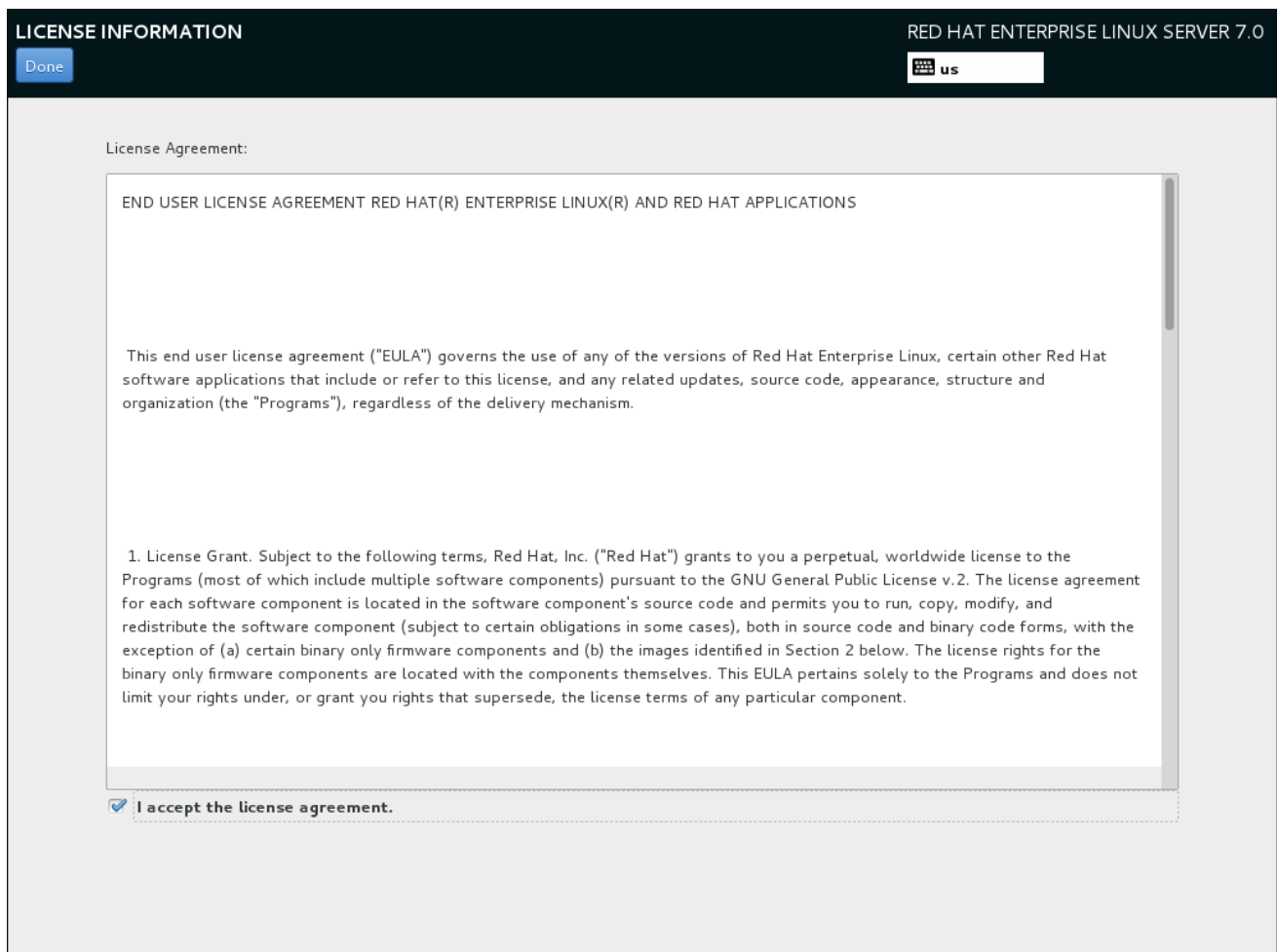


Figure 30.2. License Information Screen

In order to continue with the configuration process, the license agreement must be accepted. Exiting **Initial Setup** without completing this step will cause the system to restart, and once the system finishes rebooting, you will be prompted to accept the agreement again.

Review the license agreement. Then, select **I accept the license agreement.** and click **Done** to continue.

The **User Creation** screen is the same as the one used when creating an account during the installation. See [Section 8.18.2, "Create a User Account"](#) for detailed information.

Similarly, the **Network & Host Name** screen is the same as the one used when setting up network. See [Section 8.12, "Network & Hostname"](#) for information.

The **Subscription Manager** screen allows you to register your system with Red Hat to receive updates and install additional packages from repositories provided by Red Hat. For information about how to register your system, see [Section 30.1.1, "Subscription Manager"](#).

Once ready, click the **FINISH CONFIGURATION** button to register your system, before completing the **Initial Setup** configuration process.

To start the **Initial Setup** again, see [Section 30.3, "Starting Initial Setup Manually"](#).

30.1.1. Subscription Manager

The **Subscription Manager** screen allows you to register your system with Red Hat in order to receive updates and access to package repositories.

**NOTE**

The **Subscription Manager** screen in **Initial Setup** replaces the **Firstboot** tool, which was used to register systems in Red Hat Enterprise Linux 7.1 and earlier.

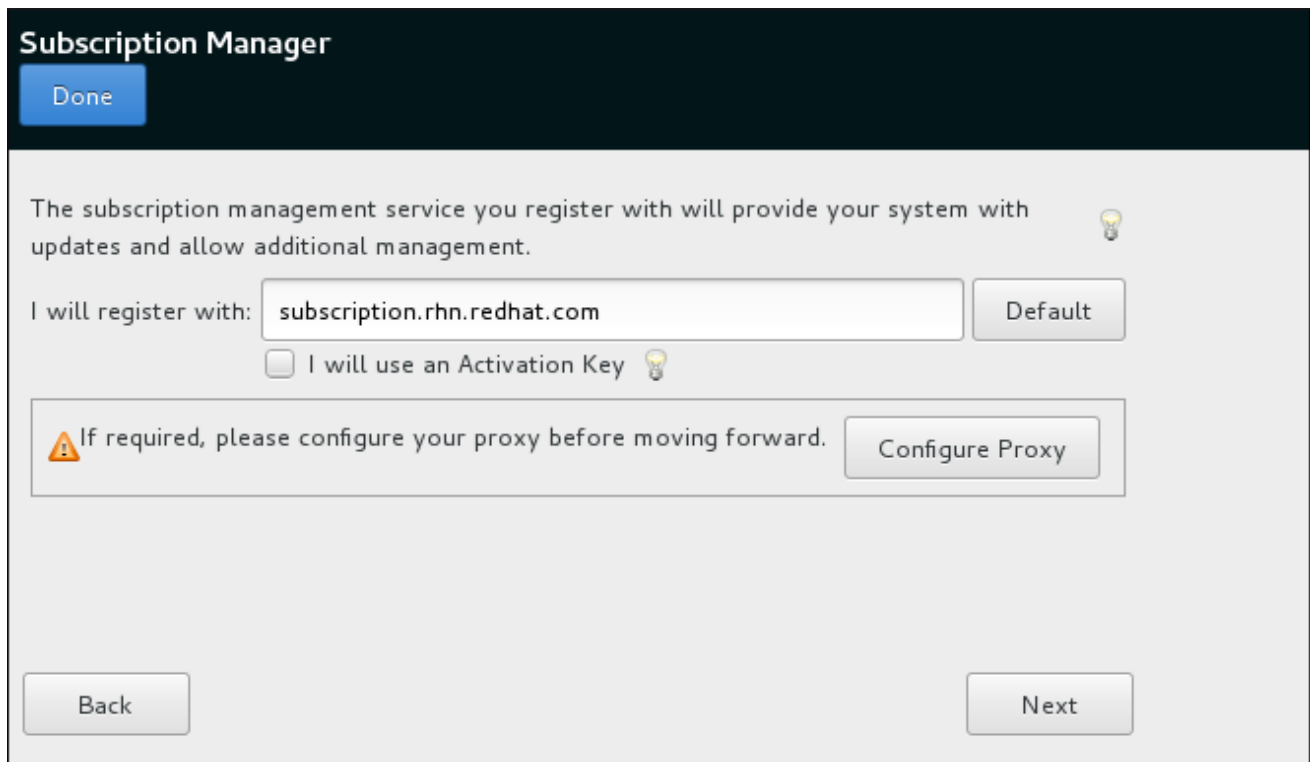


Figure 30.3. Subscription Manager Screen

The products installed on a system (including the operating system itself) are covered by *subscriptions*. A subscription service is used to track registered systems, the products installed on those systems, and the subscriptions *attached* to the system to cover those products. Red Hat provides several different subscription services which a system can registered with:

- Customer Portal Subscription Management, hosted services from Red Hat (the default)
- Subscription Asset Manager, an on-premise subscription server which proxies content delivery back to the Customer Portal's services
- CloudForms System Engine, an on-premise service which handles both subscription services and content delivery

The **Subscription Manager** screen provides a basic interface which is suitable for most use cases. In some scenarios, you require options which are not present in **Initial Setup**; in that case, you can skip the post-installation registration process and use **Subscription Manager** from the command line or using the `subscription-manager-gui` package which provides a graphical interface.

Also note that some registration scenarios, such as registering using CloudForms System Engine, require additional setup steps - you must have a registration server ready before registering your system.

To register your system, follow on-screen instructions, providing your credentials when prompted. Note that if you want to leave the **Subscription Manager** screen and return to the main **Initial Setup** screen, you must use the **Done** button in the top left corner of the screen, not the **Back** or **Next** buttons in the main window.

For full documentation concerning various tools for system registration and management, see the [Red Hat Subscription Management](#) section of the Red Hat Customer Portal. Additionally, you can use the [Registration Assistant](#) for an interactive guide through the registration process.

30.2. TEXT MODE

If you installed Red Hat Enterprise Linux without the **X Window System**, the **Initial Setup** starts in text mode:

```
Initial setup of Red Hat Enterprise Linux Server 7.3 (Maipo)

1) [!] License information          2) [x] Language settings
   (License not accepted)         (English (United States))
3) [x] Time settings              4) [x] Root password
   (America/New_York timezone)    (Password is set.)
5) [ ] User creation
   (No user will be created)
Please make your choice from above ['q' to quit | 'c' to continue |
'r' to refresh]: _
```

Figure 30.4. Initial Setup in text mode

To configure an entry, enter the menu number and press **Enter**. Additionally, you can press the following keys:

- **q** to close the application. Until you accepted the license agreement, closing the application causes the system to reboot.
- **c** to continue. Pressing this key in a submenu returns you to the main menu. In the main menu, pressing the **c** key stores the settings and closes the application. Note that you cannot continue without accepting the license agreement.
- **r** to refresh the menu.

Menu entries can have different statuses:

- **[x]**: This setting is already configured. However, you can change the setting.
- **[!]**: This setting is mandatory but not yet set.
- **[]**: This setting is optional and not yet set.

To start the **Initial Setup** again, see [Section 30.3, “Starting Initial Setup Manually”](#).

30.3. STARTING INITIAL SETUP MANUALLY

Once you once finished the **Initial Setup**, the application does not start again when you boot the system. To manually start the **Initial Setup** on system boot:

1. Enable the service:

```
# systemctl enable initial-setup.service
```

2. Optionally, to display all menu options, including the ones previously set, create an empty **/.unconfigured** file to start **Initial Setup** in reconfiguration mode:

```
# touch /.unconfigured
```

Note that, regardless of this setting, the license agreement entry is not displayed again if you previously accepted it.

3. Reboot the system.



NOTE

The initial-setup-gui (graphical mode) or initial-setup (text mode) packages must be installed for to run **Initial Setup**.

CHAPTER 31. YOUR NEXT STEPS

This chapter lists common steps that might be required after installation. Not all steps listed here are always necessary. You can use this list to find other manuals, describing how to perform the tasks you need.

Find help, answers, and utilize diagnostic services

Red Hat Access is a GUI application which allows you to conveniently access Red Hat knowledge and solutions. It can be used to search for error codes, messages, or any topic of interest, and view related knowledge from the Red Hat Customer Portal. More information about **Red Hat Access** can be found in the [Red Hat Access GUI](#) article on the Red Hat Customer Portal.

Recover a lost root password

The root password, which is configured during the installation, is required for accessing the system as the root user. Without the root password you will not be able to configure your system or install additional software. If you lost or forgot your root password, you can reset it by following the steps described in [Section 32.1.3, “Resetting the Root Password”](#).

Install driver updates

Usually, drivers for system devices are already supported in the kernel provided by Red Hat Enterprise Linux. However, occasionally, support for devices that have been released recently can be missing. In these cases, a driver update enabling your device might be available.

Devices necessary to complete the installation can have driver updates provided before the installation begins. If a device is missing a driver, but it is not essential during the installation, it is recommended to wait until after the installation completes, and install additional drivers afterwards. For instructions on installing and enabling additional drivers on the installed system using **RPM** and **Yum**, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Configure the network

In most cases network access is configured during the installation process, either in the installation program or in a Kickstart file. For information on configuring the network after the installation, see the [Red Hat Enterprise Linux 7 Networking Guide](#).

Set up Kdump

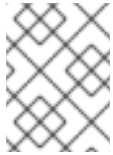
Kdump is a kernel crash dumping mechanism. If your system encounters a significant error, **Kdump** can save the contents of the system's memory into a *kernel crash dump*, which can then be analyzed to find the cause of the error.

Kdump can be enabled during the installation process (see [Section 8.16, “Kdump”](#)). It can also be configured at any time afterwards. [Red Hat Enterprise Linux 7 Kernel Crash Dump Guide](#) provides all information necessary to understand how **Kdump** works and how to configure it on your system.

Register the system

The products installed on a system (including the operating system itself) are covered by subscriptions. A subscription service is used to track registered systems, the products installed on those systems, and the subscriptions attached to those products. Registration is a part of the **Initial Setup** configuration process (see [Section 30.1.1, “Subscription Manager”](#)).

However, if you have not registered your system during **Initial Setup**, you can register it afterwards. See [Using and Configuring Red Hat Subscription Manager](#) and [Red Hat Satellite Quick Start Guide](#) for more information.

**NOTE**

You can also use the [Registration Assistant](#) application to guide you through the registration process.

Automate the initial configuration of cloud instances using cloud-init

For the initial configuration of cloud instances, you can use the cloud-init package. On a new cloud instance, **cloud-init** can automatically:

- set the default locale
- configure the host name
- configure network interfaces
- generate private SSH keys
- add SSH keys to the user's **.ssh/authorized_keys** file
- set up ephemeral mount points

Cloud-init is used with Red Hat's cloud products. See documentation on using **cloud-init** with Red Hat products:

- Red Hat Enterprise Linux Atomic Host 7 [Installation and Configuration Guide](#)
- Red Hat OpenStack Platform 8 [Instances and Images Guide](#)
- Red Hat Enterprise Virtualization [Virtual Machine Management Guide](#)
- Red Hat CloudForms [Provisioning Virtual Machines and Hosts Guide](#)

See also [upstream cloud-init documentation](#)

Perform an initial system update

After the installation is complete, Red Hat recommends that you perform an initial system update. During this process, all installed packages are updated to their latest available versions. Updates to packages provide security fixes, bug fixes and enhancements.

In Red Hat Enterprise Linux, the **Yum** package manager is used for updating the installed packages. For more information about updating your system with **Yum**, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Configure additional repositories

New software is installed from *package repositories*. Package repositories are organized sets of software and metadata that can be accessed by the **Yum** package manager. If you registered your system with Red Hat, update repositories are configured automatically and you can install updates and additional software from those. However, if you want to set up additional repositories, for example containing your own software, some extra steps are needed.

For information about configuring additional software repositories, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Install additional packages

You can control which packages will be installed by selecting an environment in the **Software Selection** dialog in the graphical installation. This dialog does not provide a way to choose individual packages, only predefined sets. However, you can use the **Yum** packages manager to install additional packages after the installation. See the [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information.

Switch to a graphical login

Depending on the options you chose during the installation process, it is possible that your system does not have a graphical interface, instead offering only a text-based prompt. If this is the case and you want to enable a graphical desktop after the installation, you must install the **X Window System** and your preferred desktop environment (either **GNOME** or **KDE**).

As with all other software, these packages can be installed using the **Yum** package manager. For information about using **Yum** to install new packages, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#). For information on how to enable graphical login by default, see [Section 9.3.3, "Booting into a Graphical Environment"](#).

Enable or disable GNOME 3 extensions

The default desktop environment in Red Hat Enterprise Linux 7 is **GNOME** 3 which provides **GNOME Shell** and **GNOME Classic** user interfaces. It is possible to customize these interfaces by enabling and disabling **GNOME** 3 extensions. See the [Red Hat Enterprise Linux 7 Desktop Migration and Administration Guide](#) for more information.

CHAPTER 32. BASIC SYSTEM RECOVERY

When things go wrong, there are ways to fix problems. However, these methods require that you understand the system well. This chapter contains information on common problems you might face and it also describes *installation program rescue mode*, which can be used to fix these problems.

32.1. COMMON PROBLEMS

You might need to boot into installation program rescue mode for any of the following reasons:

- You are unable to boot into Red Hat Enterprise Linux normally.
- You are having hardware or software problems, and you want to recover data from your system's hard drive.
- You forgot the root password.

32.1.1. Unable to Boot into Red Hat Enterprise Linux

This problem is often caused by the installation of another operating system after you have installed Red Hat Enterprise Linux. Some other operating systems assume that you have no other operating system(s) on your computer. They overwrite the Master Boot Record (MBR) that originally contained the GRUB2 boot loader. If the boot loader is overwritten in this manner, you cannot boot Red Hat Enterprise Linux unless you can boot into installation program rescue mode and reconfigure the boot loader.

Another common problem occurs when using a partitioning tool to resize a partition or create a new partition from free space after installation, and it changes the order of your partitions. If the partition number of your `/` partition changes, the boot loader might not be able to find it to mount the partition. To fix this problem, you will need to reinstall the boot loader. See [Section 32.2.2, “Reinstalling the Boot Loader”](#) for instructions on how to do this.

32.1.2. Hardware and Software Problems

This category includes a wide variety of different situations. Two examples include failing hard drives and specifying an invalid root device or kernel in the boot loader configuration file. If either of these occur, you might not be able to reboot into Red Hat Enterprise Linux. However, if you boot into installation program rescue mode, you might be able to resolve the problem or at least get copies of your most important files.

32.1.3. Resetting the Root Password

See the corresponding section in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

32.2. ANACONDA RESCUE MODE

The **Anaconda** installation program's rescue mode is a minimal Linux environment that can be booted from the Red Hat Enterprise Linux 7 DVD or other boot media. It contains command-line utilities for repairing a wide variety of issues. This rescue mode can be accessed from the **Troubleshooting** submenu of the boot menu. In this mode, you can mount file systems as read-only or even to not mount them at all, blacklist or add a driver provided on a driver disc, install or upgrade system packages, or manage partitions.



NOTE

Anaconda rescue mode is different from *rescue mode* (an equivalent to *single-user mode*) and *emergency mode*, which are provided as parts of the **systemd** system and service manager. For more information about these modes, see [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

To boot into **Anaconda** rescue mode, you must be able to boot the system using one Red Hat Enterprise Linux boot media, such as a minimal boot disc or USB drive, or a full installation DVD.

For detailed information about booting the system using media provided by Red Hat, see the appropriate chapters:

- [Chapter 7, Booting the Installation on 64-bit AMD, Intel, and ARM systems](#) for 64-bit AMD, Intel, and ARM systems
- [Chapter 12, Booting the Installation on IBM Power Systems](#) for IBM Power Systems servers
- [Chapter 16, Booting the Installation on IBM Z](#) for IBM Z



IMPORTANT

Advanced storage, such as iSCSI or zFCP devices, must be configured either using **dracut** boot options (such as **rd.zfcp=** or **root=iscsi:options**), or in the CMS configuration file on IBM Z. It is not possible to configure these storage devices interactively after booting into rescue mode.

For information about **dracut** boot options, see the **dracut.cmdline(7)** man page. For information about the CMS configuration file, see [Chapter 21, Parameter and Configuration Files on IBM Z](#).

Procedure 32.1. Booting into Anaconda Rescue Mode

1. Boot the system from either minimal boot media, or a full installation DVD or USB drive, and wait for the boot menu to appear.
2. From the boot menu, either select the **Rescue a Red Hat Enterprise Linux system** option from the **Troubleshooting** submenu, or append the **inst.rescue** option to the boot command line. To enter the boot command line, press the **Tab** key on BIOS-based systems or the **e** key on the UEFI-based systems.
3. If your system requires a third-party driver provided on a *driver disc* to boot, append the **inst.dd=driver_name** to the boot command line:

```
inst.rescue inst.dd=driver_name
```

For more information on using a driver disc at boot time, see [Section 6.3.3, "Manual Driver Update"](#) for AMD64 and Intel 64 systems or [Section 11.2.3, "Manual Driver Update"](#) for IBM Power Systems servers.

4. If a driver that is part of the Red Hat Enterprise Linux 7 distribution prevents the system from booting, append the **modprobe.blacklist=** option to the boot command line:

```
inst.rescue modprobe.blacklist=driver_name
```

For more information about blacklisting drivers, see [Section 6.3.4, “Blacklisting a Driver”](#).

- When ready, press **Enter** (BIOS-based systems) or **Ctrl+X** (UEFI-based systems) to boot the modified option. Then wait until the following message is displayed:

The rescue environment will now attempt to find your Linux installation and mount it under the `/mnt/sysimage/` directory. You can then make any changes required to your system. If you want to proceed with this step choose 'Continue'. You can also choose to mount your file systems read-only instead of read-write by choosing 'Read-only'. If for some reason this process fails you can choose 'Skip' and this step will be skipped and you will go directly to a command line.

If you select **Continue**, it attempts to mount your file system under the directory `/mnt/sysimage/`. If it fails to mount a partition, you will be notified. If you select **Read-Only**, it attempts to mount your file system under the directory `/mnt/sysimage/`, but in read-only mode. If you select **Skip**, your file system is not mounted. Choose **Skip** if you think your file system is corrupted.

- Once you have your system in rescue mode, a prompt appears on VC (virtual console) 1 and VC 2 (use the **Ctrl+Alt+F1** key combination to access VC 1 and **Ctrl+Alt+F2** to access VC 2):

```
sh-4.2#
```

Even if your file system is mounted, the default root partition while in **Anaconda** rescue mode is a temporary root partition, not the root partition of the file system used during normal user mode (**multi-user.target** or **graphical.target**). If you selected to mount your file system and it mounted successfully, you can change the root partition of the **Anaconda** rescue mode environment to the root partition of your file system by executing the following command:

```
sh-4.2# chroot /mnt/sysimage
```

This is useful if you need to run commands, such as **rpm**, that require your root partition to be mounted as `/`. To exit the **chroot** environment, type **exit** to return to the prompt.

If you selected **Skip**, you can still try to mount a partition or LVM2 logical volume manually inside **Anaconda** rescue mode by creating a directory, such as `/directory`, and typing the following command:

```
sh-4.2# mount -t xfs /dev/mapper/VolGroup00-LogVol02 /directory
```

In the above command, `/directory` is a directory that you have created and `/dev/mapper/VolGroup00-LogVol02` is the LVM2 logical volume you want to mount. If the partition is a different type than XFS, replace the `xfs` string with the correct type (such as **ext4**).

If you do not know the names of all physical partitions, use the following command to list them:

```
sh-4.2# fdisk -l
```

If you do not know the names of all LVM2 physical volumes, volume groups, or logical volumes, use the **pvdisplay**, **vgdisplay** or **lvdisplay** commands, respectively.

From the prompt, you can run many useful commands, such as:

- **ssh**, **scp**, and **ping** if the network is started

For details, see the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

- **dump** and **restore** for users with tape drives

For details, see the [RHEL Backup and Restore Assistant](#)

- **parted** and **fdisk** for managing partitions

For details, see the [Red Hat Enterprise Linux 7 Storage Administration Guide](#).

- **yum** for installing or upgrading software

For details, see the [Red Hat Enterprise Linux 7 Administrator's Guide](#)

32.2.1. Capturing an **sosreport**

The **sosreport** command-line utility collects configuration and diagnostic information, such as the running kernel version, loaded modules, and system and service configuration files, from the system. The utility output is stored in a tar archive in the **/var/tmp/** directory.

The **sosreport** utility is useful for analyzing the system errors and can make troubleshooting easier. The following procedure describes how to capture an **sosreport** output in **Anaconda** rescue mode:

Procedure 32.2. Using **sosreport** in Anaconda Rescue Mode

1. Follow steps in [Procedure 32.1, "Booting into Anaconda Rescue Mode"](#) to boot into **Anaconda** rescue mode. Ensure that you mount the installed system / (root) partition in read-write mode.
2. Change the root directory to the **/mnt/sysimage/** directory:

```
sh-4.2# chroot /mnt/sysimage/
```

3. Execute **sosreport** to generate an archive with system configuration and diagnostic information:

```
sh-4.2# sosreport
```

IMPORTANT

When running, **sosreport** will prompt you to enter your name and case number that you get when you contact Red Hat Support service and open a new support case. Use only letters and numbers because adding any of the following characters or spaces could render the report unusable:

```
# % & { } \ < > * ? / $ ~ ' " : @ + ` | =
```

4. *Optional.* If you want to transfer the generated archive to a new location using the network, it is necessary to have a network interface configured. In case you use the dynamic IP addressing, there are no other steps required. However, when using the static addressing, enter the following command to assign an IP address (for example **10.13.153.64/23**) to a network interface (for example **dev eth0**):

```
bash-4.2# ip addr add 10.13.153.64/23 dev eth0
```

See the [Red Hat Enterprise Linux 7 Networking Guide](#) for additional information about static addressing.

5. Exit the chroot environment:

```
sh-4.2# exit
```

6. Store the generated archive in a new location, from where it can be easily accessible:

```
sh-4.2# cp /mnt/sysimage/var/tmp/sosreport new_location
```

For transferring the archive through the network, use the **scp** utility:

```
sh-4.2# scp /mnt/sysimage/var/tmp/sosreport username@hostname:sosreport
```

See the references below for further information:

- For general information about **sosreport**, see [What is a sosreport and how to create one in Red Hat Enterprise Linux 4.6 and later?](#).
- For information about using **sosreport** within **Anaconda** rescue mode, see [How to generate sosreport from the rescue environment](#).
- For information about generating an **sosreport** to a different location than **/tmp**, see [How do I make sosreport write to an alternative location?](#).
- For information about collecting an **sosreport** manually, see [Sosreport fails. What data should I provide in its place?](#).

32.2.2. Reinstalling the Boot Loader

In some cases, the GRUB2 boot loader can mistakenly be deleted, corrupted, or replaced by other operating systems. The following steps detail the process on how GRUB is reinstalled on the master boot record:

Procedure 32.3. Reinstalling the GRUB2 Boot Loader

1. Follow instructions in [Procedure 32.1, "Bootting into Anaconda Rescue Mode"](#) to boot into **Anaconda** rescue mode. Ensure that you mount the installed system's **/** (root) partition in read-write mode.
2. Change the root partition:

```
sh-4.2# chroot /mnt/sysimage/
```

3. Use the following command to reinstall the GRUB2 boot loader, where *install_device* is the boot device (typically, **/dev/sda**):

```
sh-4.2# /sbin/grub2-install install_device
```

4. Reboot the system.

32.2.3. Using RPM to Add, Remove, or Replace a Driver

Missing or malfunctioning drivers can cause problems when booting the system. **Anaconda** rescue mode provides an environment in which you can add, remove, or replace a driver even when the system fails to boot. Wherever possible, we recommend that you use the **RPM** package manager to remove malfunctioning drivers or to add updated or missing drivers.



NOTE

When you install a driver from a driver disc, the driver disc updates all initramfs images on the system to use this driver. If a problem with a driver prevents a system from booting, you cannot rely on booting the system from another initramfs image.

Procedure 32.4. Using RPM to Remove a Driver

1. Boot the system into **Anaconda** rescue mode. Follow the instructions in [Procedure 32.1, “Booting into Anaconda Rescue Mode”](#). Ensure that you mount the installed system in read-write mode.
2. Change the root directory to **/mnt/sysimage/**:

```
sh-4.2# chroot /mnt/sysimage/
```

3. Use the **rpm -e** command to remove the driver package. For example, to remove the `xorg-x11-drv-wacom` driver package, run:

```
sh-4.2# rpm -e xorg-x11-drv-wacom
```

4. Exit the chroot environment:

```
sh-4.2# exit
```

If you cannot remove a malfunctioning driver for some reason, you can instead *blacklist* the driver so that it does not load at boot time. See [Section 6.3.4, “Blacklisting a Driver”](#) and [Chapter 23, Boot Options](#) for more information about blacklisting drivers.

Installing a driver is a similar process but the RPM package must be available on the system:

Procedure 32.5. Installing a Driver from an RPM package

1. Boot the system into **Anaconda** rescue mode. Follow the instructions in [Procedure 32.1, “Booting into Anaconda Rescue Mode”](#). Do *not* choose to mount the installed system as read only.
2. Make the RPM package that contains the driver available. For example, mount a CD or USB flash drive and copy the RPM package to a location of your choice under **/mnt/sysimage/**, for example: **/mnt/sysimage/root/drivers/**
3. Change the root directory to **/mnt/sysimage/**:

```
sh-4.2# chroot /mnt/sysimage/
```

4. Use the **rpm -ivh** command to install the driver package. For example, to install the `xorg-x11-drv-wacom` driver package from **/root/drivers/**, run:

```
sh-4.2# rpm -ivh /root/drivers/xorg-x11-drv-wacom-0.23.0-6.el7.x86_64.rpm
```

**NOTE**

The **/root/drivers/** directory in this chroot environment is the **/mnt/sysimage/root/drivers/** directory in the original rescue environment.

5. Exit the chroot environment:

```
sh-4.2# exit
```

When you have finished removing and installing drivers, reboot the system.

CHAPTER 33. UNREGISTERING FROM RED HAT SUBSCRIPTION MANAGEMENT SERVICES

A system can only be registered with one subscription service. If you need to change which service your system is registered with or need to delete the registration in general, then the method to unregister depends on which type of subscription service the system was originally registered with.

33.1. SYSTEMS REGISTERED WITH RED HAT SUBSCRIPTION MANAGEMENT

Several different subscription services use the same, certificate-based framework to identify systems, installed products, and attached subscriptions. These services are Customer Portal Subscription Management (hosted), Subscription Asset Manager (on-premise subscription service), and CloudForms System Engine (on-premise subscription and content delivery services). These are all part of *Red Hat Subscription Management*.

For all services within Red Hat Subscription Management, the systems are managed with the Red Hat Subscription Manager client tools.

To unregister a system registered with a Red Hat Subscription Management server, use the **unregister** command as **root** without any additional parameters:

```
# subscription-manager unregister
```

For additional information, see [Using and Configuring Red Hat Subscription Manager](#).

33.2. SYSTEMS REGISTERED WITH RED HAT SATELLITE

For a Satellite registration on the server, locate the system in the **Systems** tab and delete the appropriate profile.

For additional information, see [Red Hat Satellite Quick Start Guide](#).

CHAPTER 34. UNINSTALLING RED HAT ENTERPRISE LINUX

34.1. REMOVING RED HAT ENTERPRISE LINUX FROM 64-BIT AMD, INTEL, AND ARM SYSTEMS

The method for removing Red Hat Enterprise Linux from your computer varies, depending on whether Red Hat Enterprise Linux is the only operating system installed on the computer.

Before proceeding be sure you have considered the following information:

- In certain situations you need the install media for any non-Red Hat Enterprise Linux operating system you are going to be using on the system after you complete this process.
- If you have multiple operating systems installed, ensure that you can boot each one separately and have all administrator passwords, including any passwords that might have been set automatically by your computer manufacturer or the manufacturer of the operating system.
- If you want to retain any data from the installation of Red Hat Enterprise Linux that you are going to remove, it will need to be backed up to a different location. If you are deleting an installation that contains sensitive data, ensure that you destroy the data according to your security policy. Ensure that any backup medium is readable on the operating system where you will restore the data. For example, without extra third-party software, Microsoft Windows cannot read an external hard drive that you have formatted with Red Hat Enterprise Linux to use the ext2, ext3, ext4 or XFS file system.



WARNING

As a precaution, back up all data from any operating systems, including Red Hat Enterprise Linux, that are installed on the same computer. Unforeseen circumstances can result in loss of all your data.

- If you are only uninstalling Red Hat Enterprise Linux and not reinstalling the entire computer, you should familiarize yourself with your partition layout. In particular, the output of the **mount** command can be helpful. It can also be helpful to note which **menuitem** is used to boot your Red Hat Enterprise Linux installation in **grub.cfg**.

In general, to uninstall Red Hat Enterprise Linux from your system, you perform two steps:

1. Remove the Red Hat Enterprise Linux boot loader information from your master boot record (MBR).
2. Remove any partitions that contain the Red Hat Enterprise Linux operating system.

These instructions cannot cover every possible computer configuration, common configurations are listed here.

- only Red Hat Enterprise Linux

See [Section 34.1.1, “Only Red Hat Enterprise Linux is Installed”](#).

- Red Hat Enterprise Linux and different Linux Distribution

See [Section 34.1.2, “Red Hat Enterprise Linux installed with a Different Linux Distribution”](#) .

- Red Hat Enterprise Linux and Windows 2000, Windows Server 2000, Windows XP, Windows Vista, Windows Server 2003 and Windows Server 2008

See [Section 34.1.3, “Red Hat Enterprise Linux installed with a Microsoft Windows Operating System”](#).

If your configuration is not listed or has a highly-customized partition scheme, use the following sections as a general guide. In these situations, you will also need to learn to configure your chosen boot loader. See [Red Hat Enterprise Linux 7 System Administrator's Guide](#) for more information on the **GRUB2** boot loader.

To keep neither Red Hat Enterprise Linux nor the other operating system, follow the steps described for a computer with only Red Hat Enterprise Linux installed.

34.1.1. Only Red Hat Enterprise Linux is Installed

The following procedure shows how to remove Red Hat Enterprise Linux on systems where it is the only operating system installed. You will use the installation media for the replacement operating system to remove Red Hat Enterprise Linux. Examples of installation media include the Windows XP installation CD, Windows Vista installation DVD, or the installation CD, CDs, or DVD of another Linux distribution.

Note that some manufacturers of factory-built computers pre-installed with Microsoft Windows do not supply the Windows installation CD or DVD with the computer. In this case the manufacturer usually supplied their own "system restore disc" instead, or have included software with the computer that allowed you to create your own "system restore disc" when you first started the computer. In some cases, the system restore software is stored on a separate partition on the system's hard drive. If you cannot identify the installation media for an operating system that was pre-installed on your computer, consult the documentation supplied with the machine, or contact the manufacturer.

When you have located the installation media for your chosen operating system:

1. Back up any data that you want to keep.
2. Shut down the computer.
3. Boot your computer with the installation disc for the replacement operating system.
4. Follow the prompts presented during the installation process. Windows, OS X, and most Linux installation discs allow you to manually partition your hard drive during the installation process, or will offer you the option to remove all partitions and start with a fresh partition scheme. At this point, remove any existing partitions that the installation software detects or allow the installation program to remove the partitions automatically. "System restore" media for computers pre-installed with Microsoft Windows might create a default partition layout automatically without input from you.

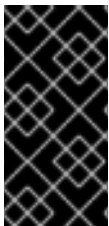
**WARNING**

If your computer has system restore software stored on a partition on a hard drive, take care when removing partitions while installing an operating system from other media. Under these circumstances, you could destroy the partition holding the system restore software.

34.1.2. Red Hat Enterprise Linux installed with a Different Linux Distribution

The following procedure shows how to remove Red Hat Enterprise Linux on systems also installed with another Linux distribution. You can use the other Linux distribution to remove the boot loader entry (or entries) and to remove any Red Hat Enterprise Linux partitions.

Because of the differences between the many different Linux distributions, these instructions are a general guide only. Specific details vary according to the configuration of your particular system and the Linux distribution that dual-boots with Red Hat Enterprise Linux.

**IMPORTANT**

These instructions assume that your system uses the **GRUB2** boot loader. If you use a different boot loader (such as **LILO**), consult the documentation for that software to identify and remove Red Hat Enterprise Linux entries from its list of boot targets and to ensure that your default operating system is correctly specified.

1. Remove Red Hat Enterprise Linux Entries from Your Boot Loader
 - a. Boot the Linux Distribution you are keeping on your computer, not Red Hat Enterprise Linux.
 - b. At the command line, type **su -** and press **Enter**. When the system prompts you for the root password, type the password and press **Enter**.
 - c. Use a text editor such as **vim** to open the **/boot/grub2/grub.cfg** configuration file. In this file, find the entry of the system you are removing. A typical Red Hat Enterprise Linux entry in the **grub.cfg** file looks similar to the following example:

Example 34.1. A Red Hat Enterprise Linux Entry in grub.cfg

```
menuentry 'Red Hat Enterprise Linux Server (3.10.0-57.el7.x86_64) 7 (Maipo)' --class
red --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-3.10.0-
53.el7.x86_64-advanced-9eecdce6-58ce-439b-bfa4-76a9ea6b0906' {
  load_video
  set gfxpayload=keep
  insmod gzio
  insmod part_msdos
  insmod xfs
  set root='hd0,msdos1'
  if [x$feature_platform_search_hint = xy ]; then
    search --no-floppy --fs-uuid --set=root --hint='hd0,msdos1' 0c70bc74-7675-4989-
9dc8-bbcf5418ddf1
```

```

else
    search --no-floppy --fs-uuid --set=root 0c70bc74-7675-4989-9dc8-bbcf5418ddf1
fi
linux16 /vmlinuz-3.10.0-57.el7.x86_64 root=/dev/mapper/rhel-root ro
rd.lvm.lv=rhel/root vconsole.font=latarcyrheb-sun16 rd.lvm.lv=rhel/swap
crashkernel=auto vconsole.keymap=us rhgb quiet LANG=en_US.UTF-8
initrd16 /initramfs-3.10.0-57.el7.x86_64.img
}

```

- d. Delete the entire entry, starting with the *menuentry* keyword and ending with *}*.

Depending on the configuration of your system, there might be multiple Red Hat Enterprise Linux entries in **grub.cfg**, each corresponding to a different version of the Linux kernel. Delete each of the Red Hat Enterprise Linux entries from the file.

- e. Save the updated **grub.cfg** file and close **vim**

2. Remove Red Hat Enterprise Linux partitions on multi-boot environments:



NOTE

Be careful not to remove partitions that are still in use by the other installations.

- a. Boot the Linux Distribution you are keeping on your computer, not Red Hat Enterprise Linux.
- b. Remove any unwanted and unnecessary partitions, for example, using; for example, with **fdisk** for standard partitions, or **lvremove** and **vgremove** to remove logical volumes and volume groups. Additional information on these utilities can be found in their respective man pages, or the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).
 - **fdisk** for standard partitions. For details, see [How do you use fdisk to delete a partition?](#)
 - **lvremove** and **vgremove** to remove logical volumes and volume groups. For details, see the [Red Hat Storage Administration Guide](#).

34.1.3. Red Hat Enterprise Linux installed with a Microsoft Windows Operating System

The following procedure shows how to remove Red Hat Enterprise Linux on systems also installed with Windows 2000, Windows Server 2000, Windows XP, Windows Server 2003, Windows Vista or Windows Server 2008. You can use the Microsoft Windows installation and its installation media to remove the boot loader and to remove any Red Hat Enterprise Linux partitions.

The removal of Red Hat Enterprise Linux on systems also installed with MS-DOS or versions of Microsoft Windows prior to Windows XP (except Windows 2000) is not covered in this document. These operating systems do not have robust partition management and cannot remove Linux partitions.

Because of the differences between each version of Microsoft Windows, these instructions need to be reviewed completely before being followed. It can be helpful to consult the documentation for your Microsoft Windows operating system as only utilities from that operating system are used in this procedure.



WARNING

This procedure relies on the **Windows Recovery Console** or the **Windows Recovery Environment** that loads from the Windows installation disk, therefore, you will not be able to complete the procedure without access to this disk. If you start this procedure and do not complete it, you could leave your computer in a condition where you cannot boot it. The "system restore disk" supplied with some factory-built computers that are sold with Windows pre-installed on them might not include the **Windows Recovery Console** or **Windows Recovery Environment**.

Users of Windows 2000, Windows Server 2000, Windows XP, and Windows Server 2003 following this procedure will be prompted for the Administrator password for their Windows system. Do not follow these instructions unless you know the Administrator password for your system or are certain that an Administrator password has never been created, even by the computer manufacturer.

1. Remove the Red Hat Enterprise Linux partitions

- a. Boot your computer into your Microsoft Windows environment.
- b. Click **Start>Run**, type **diskmgmt.msc** and press **Enter**. The **Disk Management** tool opens.

The tool displays a graphical representation of your disk, with bars representing each partition. The first partition is usually labeled **NTFS** and corresponds to your **C:** drive. At least two Red Hat Enterprise Linux partitions will be visible. Windows will not display a file system type for these partitions, but can allocate drive letters to some of them.

- c. Right-click on one of the Red Hat Enterprise Linux partitions, then click **Delete Partition** and click **Yes** to confirm the deletion. Repeat this process for the other Red Hat Enterprise Linux partitions on your system. As you delete partitions, Windows labels the space on the hard drive previously occupied by those partitions as **unallocated**.

You can add this unallocated space to an existing Windows partition or to use this space in some other way. Directions for doing this can be found in the manuals for your non-Red Hat Enterprise Linux operating system.

2. Restore the Windows boot loader

- a. On Windows 2000, Windows Server 2000, Windows XP, and Windows Server 2003:
 - i. Insert the Windows installation disk and restart your computer. As your computer starts, the following message will appear on the screen for a few seconds:

Press any key to boot from CD

Press any key while the message is still showing and the Windows installation software will load.

- ii. When the **Welcome to Setup** screen appears, you can start the **Windows Recovery Console**. The procedure is slightly different on different versions of Windows:

- A. On Windows 2000 and Windows Server 2000, press the **R** key, then the **C** key.
- B. On Windows XP and Windows Server 2003, press the **R** key.
- iii. The **Windows Recovery Console** scans your hard drives for Windows installations, and assigns a number to each one. It displays a list of Windows installations and prompts you to select one. Type the number corresponding to the Windows installation that you want to restore.
- iv. The **Windows Recovery Console** prompts you for the Administrator password for your Windows installation. Type the Administrator password and press the **Enter** key. If there is no administrator password for this system, press only the **Enter** key.
- v. At the prompt, type the command **fixmbr** and press the **Enter**. The **fixmbr** tool now restores the Master Boot Record for the system.
- vi. When the prompt reappears, type **exit** and press the **Enter** key.
- vii. Your computer will restart and boot your Windows operating system.
- b. On Windows Vista and Windows Server 2008:
 - i. Insert the Windows installation disk and restart your computer. As your computer starts, the following message will appear on the screen for a few seconds:

Press any key to boot from CD or DVD

Press any key while the message is still showing and the Windows installation software will load.
 - ii. In the **Install Windows** dialog, select a language, time and currency format, and keyboard type. Click **Next**
 - iii. Click **Repair your computer**.
 - iv. The **Windows Recovery Environment**(WRE) shows you the Windows installations that it can detect on your system. Select the installation that you want to restore, then click **Next**.
 - v. Click **Command prompt**. A command window will open.
 - vi. Type **bootrec /fixmbr** and press **Enter**.
 - vii. When the prompt reappears, close the command window, then click **Restart**.
 - viii. Your computer will restart and boot your Windows operating system.

34.2. REMOVING RED HAT ENTERPRISE LINUX FROM IBM Z

If you want to delete the existing operating system data, first, if any Linux disks contain sensitive data, ensure that you destroy the data according to your security policy. To proceed you can consider these options:

- Overwrite the disks with a new installation.
- Make the DASD or SCSI disk where Linux was installed visible from another system, then delete

the data. However, this might require special privileges. Ask your system administrator for advice. You can use Linux commands such as **dasdfmt** (DASD only), **parted**, **mke2fs** or **dd**. For more details about the commands, see the respective man pages.

34.2.1. Running a Different Operating System on Your z/VM Guest or LPAR

If you want to boot from a DASD or SCSI disk different from where the currently installed system resides under a z/VM guest virtual machine or an LPAR, shut down the Red Hat Enterprise Linux installed and use the desired disk, where another Linux instance is installed, to boot from. This leaves the contents of the installed system unchanged.

PART VI. TECHNICAL APPENDIXES

The appendixes in this section do not contain instructions on installing Red Hat Enterprise Linux. Instead, they provide technical background that you might find helpful to understand the options that Red Hat Enterprise Linux offers you at various points in the installation process.

APPENDIX A. AN INTRODUCTION TO DISK PARTITIONS



NOTE

This appendix is not necessarily applicable to architectures other than AMD64 and Intel 64. However, the general concepts mentioned here might apply.

This section discusses basic disk concepts, disk repartitioning strategies, the partition naming scheme used by Linux systems, and related topics.

If you are comfortable with disk partitions, you can skip ahead to [Section A.2, “Strategies for Disk Repartitioning”](#) for more information on the process of freeing up disk space to prepare for a Red Hat Enterprise Linux installation.



NOTE

Before installation, you should consider whether you want to use partitioned or unpartitioned disk devices. For more information, see the Knowledgebase article at <https://access.redhat.com/solutions/163853>.

A.1. HARD DISK BASIC CONCEPTS

Hard disks perform a very simple function – they store data and reliably retrieve it on command.

When discussing issues such as disk partitioning, it is important to have a understanding of the underlying hardware; however, since the theory is very complicated and expansive, only the basic concepts will be explained here. This appendix uses a set of simplified diagrams of a disk drive to help explain what is the process and theory behind partitions.

[Figure A.1, “An Unused Disk Drive”](#), shows a brand-new, unused disk drive.

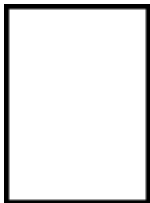


Figure A.1. An Unused Disk Drive

A.1.1. File Systems

To store data on a disk drive, it is necessary to *format* the disk drive first. Formatting (usually known as “making a *file system*”) writes information to the drive, creating order out of the empty space in an unformatted drive.

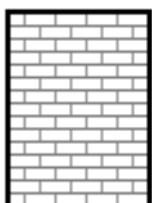


Figure A.2. Disk Drive with a File System

As the previous diagram implies, the order imposed by a file system involves some trade-offs:

- A small percentage of the driver's available space is used to store file system-related data and can be considered as overhead.
- A file system splits the remaining space into small, consistently-sized segments. For Linux, these segments are known as *blocks*.^[4]

Note that there is no single, universal file system. As the following diagram shows, a disk drive can have one of many different file systems written on it. Different file systems tend to be incompatible; that is, an operating system that supports one file system (or a handful of related file system types) might not support another. However, for example, Red Hat Enterprise Linux supports a wide variety of file systems (including many commonly used by other operating systems), making data interchange between different file systems easy.

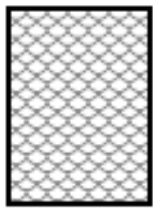


Figure A.3. Disk Drive with a Different File System

Writing a file system to disk is only the first step. The goal of this process is to actually *store* and *retrieve* data. The figure below shows a drive disk after some data have been written to it:

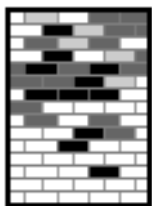


Figure A.4. Disk Drive with Data Written to It

As the previous diagram shows, some of the previously empty blocks are now holding data. However, by just looking at this picture, we cannot determine exactly how many files reside on this drive. There might only be one file or many, as all files use at least one block and some files use multiple blocks. Another important point to note is that the used blocks do not have to form a contiguous region; used and unused blocks can be interspersed. This is known as *fragmentation*. Fragmentation can play a part when attempting to resize an existing partition.

As with most computer-related technologies, disk drives changed over time after their introduction. In particular, they got bigger. Not larger in physical size, but bigger in their capacity to store information. And, this additional capacity drove a fundamental change in the way disk drives were used.

A.1.2. Partitions: Turning One Drive Into Many

Disk drives can be divided into *partitions*. Each partition can be accessed as if it was a separate disk. This is done through the addition of a *partition table*.

There are several reasons for allocating disk space into separate disk partitions, for example:

- Logical separation of the operating system data from the user data

- Ability to use different file systems
- Ability to run multiple operating systems on one machine

There are currently two partitioning layout standards for physical hard disks: Master Boot Record (MBR) and GUID Partition Table (GPT). MBR is an older method of disk partitioning used with BIOS-based computers. GPT is a newer partitioning layout that is a part of the Unified Extensible Firmware Interface (UEFI). This section and [Section A.1.3, "Partitions Within Partitions - An Overview of Extended Partitions"](#) mainly describe the *Master Boot Record* (MBR) disk partitioning scheme. For information about the *GUID Partition Table* (GPT) partitioning layout, see [Section A.1.4, "GUID Partition Table \(GPT\)"](#).



NOTE

While the diagrams in this chapter show the partition table as being separate from the actual disk drive, this is not entirely accurate. In reality, the partition table is stored at the very start of the disk, before any file system or user data. But for clarity, they are separate in our diagrams.

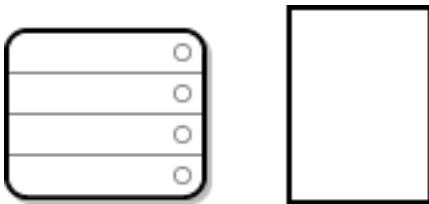


Figure A.5. Disk Drive with Partition Table

As the previous diagram shows, the partition table is divided into four sections or four *primary* partitions. A primary partition is a partition on a hard drive that can contain only one logical drive (or section). Each section can hold the information necessary to define a single partition, meaning that the partition table can define no more than four partitions.

Each partition table entry contains several important characteristics of the partition:

- The points on the disk where the partition starts and ends
- Whether the partition is "active"
- The partition's type

The starting and ending points define the partition's size and location on the disk. The "active" flag is used by some operating systems' boot loaders. In other words, the operating system in the partition that is marked "active" is booted.

The type is a number that identifies the partition's anticipated usage. Some operating systems use the partition type to denote a specific file system type, to flag the partition as being associated with a particular operating system, to indicate that the partition contains a bootable operating system, or some combination of the three.

The following shows an example of a disk drive with single partition:

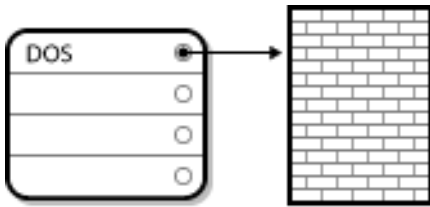


Figure A.6. Disk Drive With Single Partition

The single partition in this example is labeled as **DOS**. This label shows the *partition type*, with **DOS** being one of the most common ones. The table below shows a list of some of the commonly used partition types and hexadecimal numbers used to represent them.

Table A.1. Partition Types

Partition Type	Value	Partition Type	Value
Empty	00	Novell Netware 386	65
DOS 12-bit FAT	01	PIC/IX	75
XENIX root	02	Old MINIX	80
XENIX usr	03	Linux/MINUX	81
DOS 16-bit <=32M	04	Linux swap	82
Extended	05	Linux native	83
DOS 16-bit >=32	06	Linux extended	85
OS/2 HPFS	07	Amoeba	93
AIX	08	Amoeba BBT	94
AIX bootable	09	BSD/386	a5
OS/2 Boot Manager	0a	OpenBSD	a6
Win95 FAT32	0b	NEXTSTEP	a7
Win95 FAT32 (LBA)	0c	BSDI fs	b7
Win95 FAT16 (LBA)	0e	BSDI swap	b8
Win95 Extended (LBA)	0f	Syrinx	c7
Venix 80286	40	CP/M	db

Partition Type	Value	Partition Type	Value
Novell	51	DOS access	e1
PreP Boot	41	DOS R/O	e3
GNU HURD	63	DOS secondary	f2
Novell Netware 286	64	BBT	ff

A.1.3. Partitions Within Partitions - An Overview of Extended Partitions

In case four partitions are insufficient for your needs, you can use *extended partitions* to create up additional partitions. You do this by setting the type of a partition to "Extended".

An extended partition is like a disk drive in its own right - it has its own partition table which points to one or more partitions (now called *logical partitions*, as opposed to the four *primary partitions*) contained entirely within the extended partition itself. The following diagram shows a disk drive with one primary partition and one extended partition containing two logical partitions (along with some unpartitioned free space):

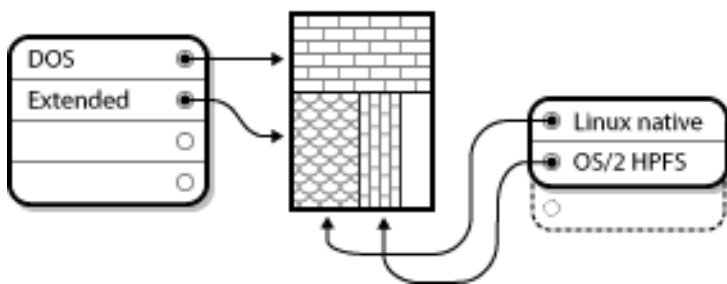


Figure A.7. Disk Drive With Extended Partition

As this figure implies, there is a difference between primary and logical partitions - there can only be four primary partitions, but there is no fixed limit to the number of logical partitions that can exist. However, due to the way in which partitions are accessed in Linux, no more than 12 logical partitions should be defined on a single disk drive.

A.1.4. GUID Partition Table (GPT)

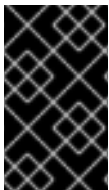
GUID Partition Table (GPT) is a newer partitioning scheme based on using Globally Unique Identifiers (GUID). GPT was developed to cope with limitations of the MBR partition table, especially with the limited maximum addressable storage space of a disk. Unlike MBR, which is unable to address storage space larger than 2 TiB (equivalent to approximately 2.2 TB), GPT can be used with hard disks larger than this; the maximum addressable disk size is 2.2 ZiB. In addition GPT, by default, supports creating up to 128 primary partitions. This number could be extended by allocating more space to the partition table.

GPT disks use logical block addressing (LBA) and the partition layout is as follows:

- To preserve backward compatibility with MBR disks, the first sector (LBA 0) of GPT is reserved for MBR data and it is called "protective MBR".
- The *primary GPT header* begins on the second logical block (LBA 1) of the device. The header contains the disk GUID, the location of the primary partition table, the location of the secondary

GPT header, and CRC32 checksums of itself and the primary partition table. It also specifies the number of partition entries of the table.

- The *primary GPT table* includes, by default, 128 partition entries, each with an entry size 128 bytes, its partition type GUID and unique partition GUID.
- The *secondary GPT table* is identical to the primary GPT table. It is used mainly as a backup table for recovery in case the primary partition table is corrupted.
- The *secondary GPT header* is located on the last logical sector of the disk and it can be used to recover GPT information in case the primary header is corrupted. It contains the disk GUID, the location of the secondary partition table and the primary GPT header, CRC32 checksums of itself and the secondary partition table, and the number of possible partition entries.



IMPORTANT

There must be a BIOS boot partition for the boot loader to be installed successfully onto a disk that contains a GPT (GUID Partition Table). This includes disks initialized by **Anaconda**. If the disk already contains a BIOS boot partition, it can be reused.

A.2. STRATEGIES FOR DISK REPARTITIONING

There are several different ways that a disk can be repartitioned. This section discusses the following possible approaches:

- Unpartitioned free space is available
- An unused partition is available
- Free space in an actively used partition is available

Note that this section discusses the aforementioned concepts only theoretically and it does not include any procedures showing how to perform disk repartitioning step-by-step. Such detailed information are beyond the scope of this document.



NOTE

Keep in mind that the following illustrations are simplified in the interest of clarity and do not reflect the exact partition layout that you encounter when actually installing Red Hat Enterprise Linux.

A.2.1. Using Unpartitioned Free Space

In this situation, the partitions already defined do not span the entire hard disk, leaving unallocated space that is not part of any defined partition. The following diagram shows what this might look like:

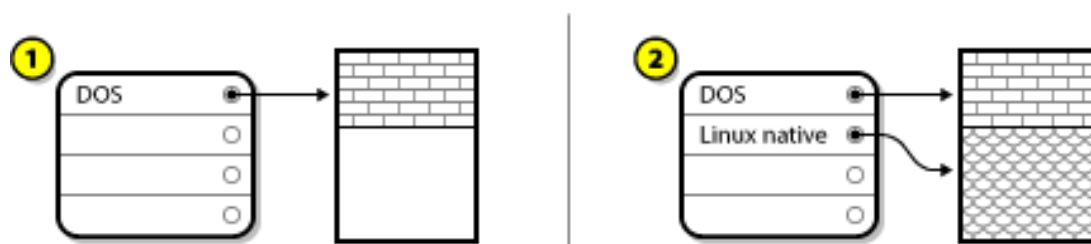


Figure A.8. Disk Drive with Unpartitioned Free Space

In the above example, 1 represents an undefined partition with unallocated space and 2 represents a defined partition with allocated space.

An unused hard disk also falls into this category. The only difference is that *all* the space is not part of any defined partition.

In any case, you can create the necessary partitions from the unused space. Unfortunately, this scenario, although very simple, is not very likely (unless you have just purchased a new disk just for Red Hat Enterprise Linux). Most pre-installed operating systems are configured to take up all available space on a disk drive (see [Section A.2.3, “Using Free Space from an Active Partition”](#)).

A.2.2. Using Space from an Unused Partition

In this case, maybe you have one or more partitions that you do not use any longer. The following diagram illustrates such a situation.

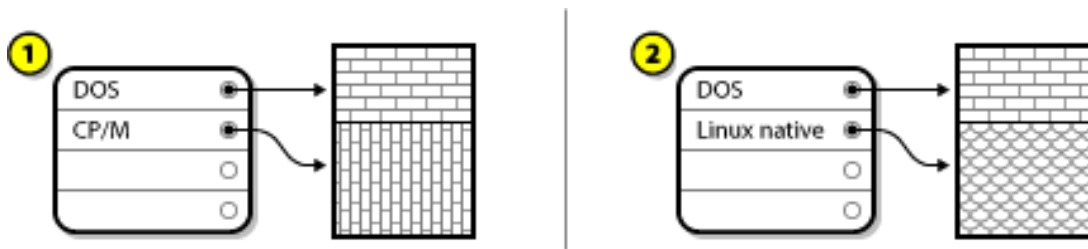


Figure A.9. Disk Drive with an Unused Partition

In the above example, 1 represents an unused partition and 2 represents reallocating an unused partition for Linux.

In this situation, you can use the space allocated to the unused partition. You first must delete the partition and then create the appropriate Linux partition(s) in its place. You can delete the unused partition and manually create new partitions during the installation process.

A.2.3. Using Free Space from an Active Partition

This is the most common situation. It is also, unfortunately, the hardest to handle. The main problem is that, even if you have enough free space, it is presently allocated to a partition that is already in use. If you purchased a computer with pre-installed software, the hard disk most likely has one massive partition holding the operating system and data.

Aside from adding a new hard drive to your system, you have two choices:

Destructive Repartitioning

In this case, the single large partition is deleted and several smaller ones are created instead. Any data held in the original partition is destroyed. This means that making a complete backup is necessary. It is highly recommended to make two backups, use verification (if available in your backup software), and try to read data from the backup *before* deleting the partition.

**WARNING**

If an operating system was installed on that partition, it must be reinstalled if you want to use that system as well. Be aware that some computers sold with pre-installed operating systems might not include the installation media to reinstall the original operating system. You should check whether this applies to your system is *before* you destroy your original partition and its operating system installation.

After creating a smaller partition for your existing operating system, you can reinstall software, restore your data, and start your Red Hat Enterprise Linux installation.

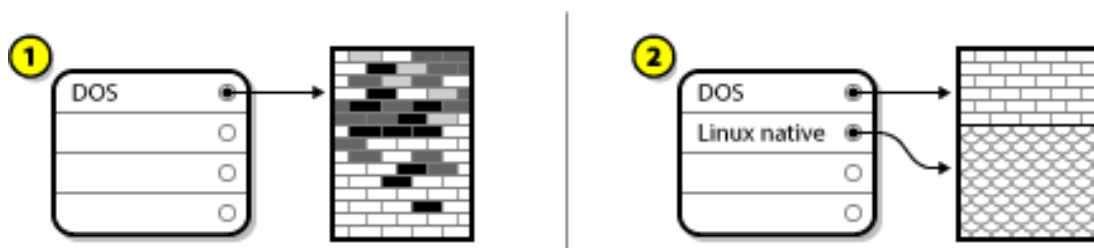


Figure A.10. Disk Drive Being Destructively Repartitioned

In the above example, 1 represents before and 2 represents after.

**WARNING**

Any data previously present in the original partition is lost.

Non-Destructive Repartitioning

With non-destructive repartitioning you execute a program that makes a big partition smaller without losing any of the files stored in that partition. This method is usually reliable, but can be very time-consuming on large drives.

While the process of non-destructive repartitioning is rather straightforward, there are three steps involved:

1. Compress and backup existing data
2. Resize the existing partition
3. Create new partition(s)

Each step is described further in more detail.

A.2.3.1. Compress Existing Data

As the following figure shows, the first step is to compress the data in your existing partition. The reason for doing this is to rearrange the data such that it maximizes the available free space at the "end" of the partition.

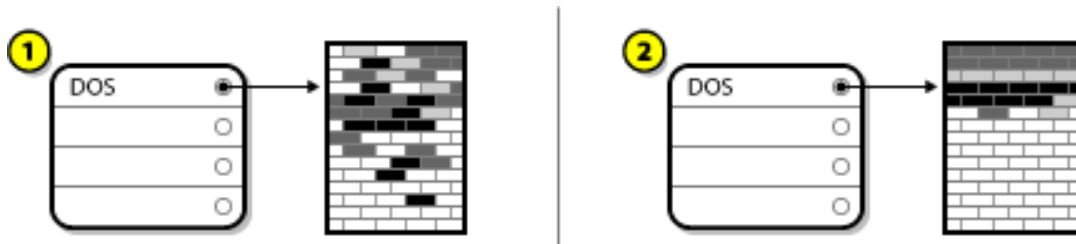


Figure A.11. Disk Drive Being Compressed

In the above example, 1 represents before and 2 represents after.

This step is crucial. Without it, the location of the data could prevent the partition from being resized to the extent desired. Note also that, for one reason or another, some data cannot be moved. If this is the case (and it severely restricts the size of your new partitions), you might be forced to destructively repartition your disk.

A.2.3.2. Resize the Existing Partition

Figure A.12, "Disk Drive with Partition Resized" shows the actual resizing process. While the actual result of the resizing operation varies depending on the software used, in most cases the newly freed space is used to create an unformatted partition of the same type as the original partition.

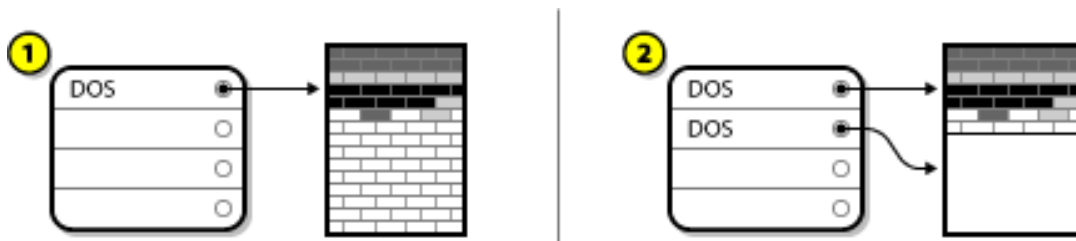


Figure A.12. Disk Drive with Partition Resized

In the above example, 1 represents before and 2 represents after.

It is important to understand what the resizing software you use does with the newly freed space, so that you can take the appropriate steps. In the case illustrated here, it would be best to delete the new DOS partition and create the appropriate Linux partition(s).

A.2.3.3. Create new partitions

As the previous step implied, it might or might not be necessary to create new partitions. However, unless your resizing software supports systems with Linux installed, it is likely that you must delete the partition that was created during the resizing process.

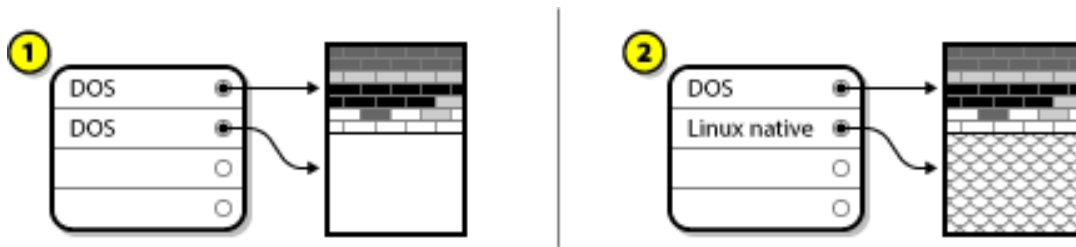


Figure A.13. Disk Drive with Final Partition Configuration

In the above example, 1 represents before and 2 represents after.

A.3. PARTITION NAMING SCHEMES AND MOUNT POINTS

A common source of confusion for users unfamiliar with Linux is the matter of how partitions are used and accessed by the Linux operating system. In DOS/Windows, it is relatively simple: Each partition gets a "drive letter." You then use the correct drive letter to refer to files and directories on its corresponding partition. This is entirely different from how Linux deals with partitions and, for that matter, with disk storage in general. This section describes the main principles of partition naming scheme and the way how partitions are accessed in Red Hat Enterprise Linux.

A.3.1. Partition Naming Scheme

Red Hat Enterprise Linux uses a naming scheme that is file-based, with file names in the form of **/dev/xxyn**.

Device and partition names consist of the following:

/dev/

This is the name of the directory in which all device files reside. Because partitions reside on hard disks, and hard disks are devices, the files representing all possible partitions reside in **/dev/**.

xx

The first two letters of the partition name indicate the type of device on which the partition resides, usually **sd**.

y

This letter indicates which device the partition is on. For example, **/dev/sda** for the first hard disk, **/dev/sdb** for the second, and so on.

N

The final number denotes the partition. The first four (primary or extended) partitions are numbered **1** through **4**. Logical partitions start at **5**. So, for example, **/dev/sda3** is the third primary or extended partition on the first hard disk, and **/dev/sdb6** is the second logical partition on the second hard disk.



NOTE

Even if Red Hat Enterprise Linux can identify and refer to *all* types of disk partitions, it might not be able to read the file system and therefore access stored data on every partition type. However, in many cases, it is possible to successfully access data on a partition dedicated to another operating system.

A.3.2. Disk Partitions and Mount Points

In Red Hat Enterprise Linux each partition is used to form part of the storage necessary to support a single set of files and directories. This is done by associating a partition with a directory through a process known as *mounting*. Mounting a partition makes its storage available starting at the specified directory (known as a *mount point*).

For example, if partition **/dev/sda5** is mounted on **/usr/**, that would mean that all files and directories under **/usr/** physically reside on **/dev/sda5**. So the file **/usr/share/doc/FAQ/txt/Linux-FAQ** would be stored on **/dev/sda5**, while the file **/etc/gdm/custom.conf** would not.

Continuing the example, it is also possible that one or more directories below **/usr/** would be mount points for other partitions. For instance, a partition (say, **/dev/sda7**) could be mounted on **/usr/local/**, meaning that **/usr/local/man/whatis** would then reside on **/dev/sda7** rather than **/dev/sda5**.

A.3.3. How Many Partitions?

At this point in the process of preparing to install Red Hat Enterprise Linux, you must give some consideration to the number and size of the partitions to be used by your new operating system. However, there is no one right answer to this question. It depends on your needs and requirements.

Keeping this in mind, Red Hat recommends that, unless you have a reason for doing otherwise, you should *at least* create the following partitions: **swap**, **/boot/**, and **/** (root).

For more information, see [Section 8.14.4.4, “Recommended Partitioning Scheme”](#) for AMD64 and Intel 64 systems, [Section 13.15.4.4, “Recommended Partitioning Scheme”](#) for IBM Power Systems servers, and [Section 18.15.3.4, “Recommended Partitioning Scheme”](#) for IBM Z.

[4] Blocks really *are* consistently sized, unlike our illustrations. Keep in mind, also, that an average disk drive contains thousands of blocks. The picture is simplified for the purposes of this discussion.

APPENDIX B. ISCSI DISKS

Internet Small Computer System Interface (iSCSI) is a protocol that allows computers to communicate with storage devices by SCSI requests and responses carried over TCP/IP. Because iSCSI is based on the standard SCSI protocols, it uses some terminology from SCSI. The device on the SCSI bus to which requests get sent, and which answers these requests, is known as the *target* and the device issuing requests is known as the *initiator*. In other words, an iSCSI disk is a target and the iSCSI software equivalent of a SCSI controller or SCSI Host Bus Adapter (HBA) is called an initiator. This appendix only covers Linux as an iSCSI initiator; how Linux uses iSCSI disks, but not how Linux hosts iSCSI disks.

Linux has a software iSCSI initiator in the kernel that takes the place and form of a SCSI HBA driver and therefore allows Linux to use iSCSI disks. However, as iSCSI is a fully network-based protocol, iSCSI initiator support requires more than just the ability to send SCSI packets over the network. Before Linux can use an iSCSI target, Linux must find the target on the network and make a connection to it. In some cases, Linux must send authentication information to gain access to the target. Linux must also detect any failure of the network connection and must establish a new connection, including logging in again if necessary.

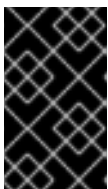
The discovery, connection, and logging in is handled in user space by the **iscsiadm** utility, while errors are handled, also in user space, by the **iscsid** utility.

Both **iscsiadm** and **iscsid** are part of the **iscsi-initiator-utils** package under Red Hat Enterprise Linux.

B.1. ISCSI DISKS IN ANACONDA

The **Anaconda** installation program can discover and log in to iSCSI disks in two ways:

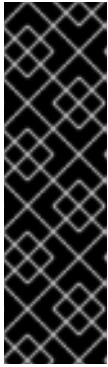
1. When **Anaconda** starts, it checks if the BIOS or add-on boot ROMs of the system support *iSCSI/Boot Firmware Table* (iBFT), a BIOS extension for systems which can boot from iSCSI. If the BIOS supports iBFT, **Anaconda** will read the iSCSI target information for the configured boot disk from the BIOS and log in to this target, making it available as an installation target.



IMPORTANT

To connect automatically to an iSCSI target, a network device for accessing the target needs to be activated. The recommended way to do so is to use **ip=ibft** boot option.

2. You can discover and add iSCSI targets manually in the graphical user interface in **anaconda**. From the main menu, the Installation Summary screen, click the Installation Destination option. Then click the **Add a disk** in the **Specialized & Network Disks** section of the screen. A tabbed list of available storage devices appears. In the lower right corner, click the **Add iSCSI Target** button and proceed with the discovery process. See [Section 8.15.1, "The Storage Devices Selection Screen"](#) for more information.



IMPORTANT

Restriction: The **/boot** partition cannot be placed on iSCSI targets that have been manually added using this method – an iSCSI target containing a **/boot** partition must be configured for use with iBFT.

However, in instances where the installed system is expected to boot from iSCSI with iBFT configuration provided by a method other than firmware iBFT, for example using iPXE, the **/boot** partition restriction can be disabled using the **inst.nonibftiscsiboot** installer boot option.

While **Anaconda** uses **iscsiadm** to find and log into iSCSI targets, **iscsiadm** automatically stores any information about these targets in the **iscsiadm** iSCSI database. **Anaconda** then copies this database to the installed system and marks any iSCSI targets not used for **/** so that the system will automatically log in to them when it starts. If **/** is placed on an iSCSI target, **initrd** will log into this target and **Anaconda** does not include this target in start up scripts to avoid multiple attempts to log into the same target.

B.2. ISCSI DISKS DURING START UP

Events related to iSCSI might occur at a number of points while the system is starting:

1. The init script in the **initrd** will log into iSCSI targets used for **/**, if any. This is done using the **iscsistart** utility, without requiring **iscsid** to run.



NOTE

If the root file system is on an iSCSI disk connected using IPv6, ensure that the installed system is using the correct **ip=** boot option, for example **ip=eth0:auto6**. If this option is not set, the installed system can spend up to 20 minutes at boot time attempting to establish a connection, before eventually succeeding. Using the correct **ip=** option eliminates this delay.

2. When the root file system has been mounted and the various service init scripts are running, the **iscsi** init script will get called. This script then starts the **iscsid** daemon if any iSCSI targets are used for **/**, or if any targets in the iSCSI database are marked to be logged into automatically.
3. After the classic network service script has been run, the **iscsi** init script will run. If the network is accessible, this will log into any targets in the iSCSI database that are marked to be logged into automatically. If the network is not accessible, this script will exit quietly.
4. When using **NetworkManager** to access the network, instead of the classic network service script, **NetworkManager** will call the **iscsi** init script. Also see the **/etc/NetworkManager/dispatcher.d/04-iscsi** file for further reference.



IMPORTANT

Because **NetworkManager** is installed in the **/usr** directory, you cannot use it to configure network access if **/usr** is on network-attached storage such as an iSCSI target.

If **iscsid** is not needed as the system starts, it will not start automatically. If you start **iscsiadm**, **iscsiadm** will start **iscsid** in turn.

APPENDIX C. ANACONDA UI SPECIFIC COMMANDS

C.1. COMMANDS USED IN ANACONDA

The "pwpolicy" command is an Anaconda UI specific command that can be used only in the %anaconda section of the kickstart file.

pwpolicy (optional)

This command can be used to enforce a custom password policy, which specifies requirements for passwords created during installation, based on factors such as password length and strength.

```
pwpolicy name [--minlen=length] [--minquality=quality] [--strict|--nostrict] [--emptyok|--noempty] [--changesok|--nochanges]
```

Replace *name* with either **root**, **user** or **luks** to enforce the policy for the **root** password, user passwords, or LUKS passphrase, respectively.

The **libpwquality** library is used to check minimum password requirements (length and quality). You can use the **pwscore** and **pwmake** commands provided by the **libpwquality** package to check the quality score of a password, or to create a random password with a given score. See the **pwscore(1)** and **pwmake(1)** man page for details about these commands.



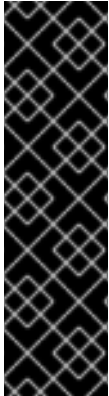
IMPORTANT

This command can only be used inside the %anaconda section.

- **--minlen=** - Sets the minimum allowed password length, in characters. The default is **6**.
- **--minquality=** - Sets the minimum allowed password quality as defined by the **libpwquality** library. The default value is **1**.
- **--strict** - Enables strict password enforcement. Passwords which do not meet the requirements specified in **--minquality=** and **--minlen=** will not be accepted. This option is disabled by default.
- **--notstrict** - Passwords which do *not* meet the minimum quality requirements specified by the **--minquality=** and **--minlen=** options will be allowed, after **Done** is clicked twice.
- **--emptyok** - Allows the use of empty passwords. Enabled by default for user passwords.
- **--notempty** - Disallows the use of empty passwords. Enabled by default for the root password and the LUKS passphrase.
- **--changesok** - Allows changing the password in the user interface, even if the Kickstart file already specifies a password. Disabled by default.
- **--nochanges** - Disallows changing passwords which are already set in the Kickstart file. Enabled by default.

APPENDIX D. UNDERSTANDING LVM

LVM (Logical Volume Management) partitions provide a number of advantages over standard partitions. LVM partitions are formatted as *physical volumes*. One or more physical volumes are combined to form a *volume group*. Each volume group's total storage is then divided into one or more *logical volumes*. The logical volumes function much like standard partitions. They have a file system type, such as **xfs**, and a mount point.



IMPORTANT

On AMD, Intel, and ARM systems and IBM Power Systems servers, the boot loader cannot read LVM volumes. You must make a standard, non-LVM disk partition for your **/boot** partition.

On IBM Z, the **zipl** boot loader supports **/boot** on LVM logical volumes with linear mapping.

By default, the installation process always creates the **/** and **swap** partitions within LVM volumes, with a separate **/boot** partition on a physical volume.

To understand LVM better, imagine the physical volume as a pile of *blocks*. A block is a storage unit used to store data. Several piles of blocks can be combined to make a much larger pile, just as physical volumes are combined to make a volume group. The resulting pile can be subdivided into several smaller piles of arbitrary size, just as a volume group is allocated to several logical volumes.

An administrator can grow or shrink logical volumes without destroying data, unlike standard disk partitions. If the physical volumes in a volume group are on separate drives or RAID arrays, then administrators can also spread a logical volume across the storage devices.

You can lose data if you shrink a logical volume to a smaller capacity than the data on the volume requires. To ensure maximum flexibility, create logical volumes to meet your current needs, and leave excess storage capacity unallocated. You can safely extend logical volumes to use unallocated space, depending on your needs.

APPENDIX E. OTHER TECHNICAL DOCUMENTATION

Both **Anaconda** and Red Hat Enterprise Linux systems use a common set of software components. For detailed information on key technologies, see the web sites listed below.

Boot Loader

Red Hat Enterprise Linux uses the **GRUB2** boot loader. See the corresponding chapter in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Storage Management

Logical Volume Management (LVM) provides administrators with a range of facilities to manage storage. By default, the Red Hat Enterprise Linux installation process formats drives as LVM volumes. For further details about LVM, see the [Red Hat Enterprise Linux 7 Logical Volume Manager Administration Guide](#).

Remote Displays

Red Hat Enterprise Linux and **Anaconda** include VNC (Virtual Network Computing) software to enable remote access to graphical displays. For more information about VNC, see the TigerVNC chapter in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

You can also use the [VNC Configurator](#) application on the Red Hat Customer Portal to guide you through VNC server and client setup.

Remote System Access

Red Hat Enterprise Linux incorporates the OpenSSH suite to provide remote access to the system. The SSH service enables a number of functions, which include access to the command-line from other systems, remote command execution, and network file transfers. During the installation process, **Anaconda** might use the **scp** feature of OpenSSH to transfer crash reports to remote systems. For further details about OpenSSH, see the corresponding section in the [Red Hat Enterprise Linux 7 System Administrator's Guide](#).

Access Control

SELinux provides Mandatory Access Control (MAC) capabilities that supplement the standard Linux security features. See the [Red Hat Enterprise Linux 7 SELinux User's and Administrator's Guide](#) for further information.

Firewall

Red Hat Enterprise Linux uses **firewalld** to provide firewall features. For further information, see the corresponding chapter in the [Red Hat Enterprise Linux 7 Security Guide](#).

Software Installation

Red Hat Enterprise Linux uses **yum** to manage the RPM packages that make up the system. See the corresponding chapter in the [Red Hat Enterprise Linux 7 Administrator's Guide](#) for further information.

Virtualization

Virtualization provides the capability to simultaneously run multiple operating systems on the same computer. Red Hat Enterprise Linux also includes tools to install and manage the secondary systems on a Red Hat Enterprise Linux host. You can select virtualization support during the installation process, or at any time thereafter. See the [Red Hat Enterprise Linux 7 Virtualization Deployment and Administration Guide](#) for more information.

APPENDIX F. REFERENCE TABLE FOR EXT4 AND XFS COMMANDS

XFS replaces ext4 as the default file system in Red Hat Enterprise Linux 7. This table serves as a cross reference listing common file system manipulation tasks and any changes in these commands between ext4 and XFS.

Table F.1. Reference Table for ext4 and XFS Commands

Task	ext4	XFS
Creating a file system	mkfs.ext4	mkfs.xfs
Mounting a file system	mount	mount
Resizing a file system	resize2fs	xfs_growfs ^[a]
Repairing a file system	e2fsck	xfs_repair
Changing the label on a file system	e2label	xfs_admin -L
Reporting on disk space and file usage	quota	quota
Debugging a file system	debugfs	xfs_db
Saving critical file system metadata to a file	e2image	xfs_metadump
[a] The size of XFS file systems cannot be reduced; the command is used only to increase the size.		

APPENDIX G. DATA SIZE TERMINOLOGY REFERENCE TABLE

Table G.1. Data Size Terminology Reference Table

Term	Abbreviation	Size (in bytes)
Binary (Bytes)		
Kibibyte	KiB	1024
Mebibyte	MiB	1024^2
Gibibyte	GiB	1024^3
Tebibyte	TiB	1024^4
Pebibyte	PiB	1024^5
Exbibyte	EiB	1024^6
Zebibyte	ZiB	1024^7
Yobibyte	YiB	1024^8
Decimal (Bytes)		
Kilobyte	KB	1000
Megabyte	MB	1000^2
Gigabyte	GB	1000^3
Terabyte	TB	1000^4
Petabyte	PB	1000^5
Exabyte	EB	1000^6
Zettabyte	ZB	1000^7
Yottabyte	YB	1000^8
Decimal (Bits)		
Kilobit	Kb	125

Term	Abbreviation	Size (in bytes)
Megabit	Mb	125,000
Gigabit	Gb	125,000,000

APPENDIX H. REVISION HISTORY

Note that revision numbers relate to the edition of this manual, not to version numbers of Red Hat Enterprise Linux.

Revision 1.5-6	Wed Aug 07 2019	Eliane Pereira
Revision 1.5-5	Thu Jul 11 2019	Eliane Pereira
Image Builder has been removed and split to its own Guide		
Revision 1.5-4	Fri May 24 2019	Sharon Moroney
Preparing document for 7.7 Beta publication.		
Revision 1.5-3	Tue Oct 30 2018	Vladimír Slávik
Preparing document for 7.6 GA publication.		
Revision 1.5-2	Tue Aug 21 2018	Vladimír Slávik
Preparing document for 7.6 Beta publication.		
Revision 1.5-1	Fri Apr 6 2018	Petr Bokoč
Preparing document for 7.5 GA publication.		
Revision 1.5-0	Fri Dec 15 2017	Petr Bokoč
Preparing document for 7.5 Beta publication.		
Revision 1.4-2	Thu Nov 23 2017	Petr Bokoč
Asynchronous update.		
Revision 1.4-1	Fri Oct 13 2017	Petr Bokoč
Asynchronous update.		
Revision 1.4-0	Tue Aug 1 2017	Petr Bokoč
Preparing document for 7.4 GA publication.		
Revision 1.3-9	Mon May 15 2017	Petr Bokoč
Preparing document for 7.4 Beta publication.		
Revision 1.3-8	Tue Apr 4 2017	Petr Bokoč
Asynchronous update.		
Revision 1.3-7	Sun Nov 6 2016	Robert Kratky
Version for 7.3 GA publication.		
Revision 1.3-4	Mon Nov 16 2015	Petr Bokoč
Version for 7.2 GA publication.		
Revision 1.2-2	Wed Feb 18 2015	Petr Bokoč
Version for 7.1 GA publication.		
Revision 1.0-0	Tue Jun 03 2014	Petr Bokoč
Version for 7.0 GA publication.		

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